GOOGLE | 288 QUESTIONS | INTERACTIVE MASTER BOOK

Part A: Medium | Part B: Easy + Hard

Study sections

Part A puts all 197 Medium questions first. It has Part 1 for question information, clarification, brute force and the short best solution, followed by Part 2 for matching follow-ups. Part B repeats the same Part 1 and Part 2 flow for the 38 Easy and 53 Hard questions. Every question keeps its original number and topic group.

38	197	53
EASY	MEDIUM	HARD

PART A CONTENTS | A - PART 1 | A - PART 2
PART B CONTENTS | B - PART 1 | B - PART 2

Part A Contents - Medium Questions

197 Medium questions | question opens Part A - Part 1 | FU opens Part A - Part 2

Groups keep their original 1-43 study-path numbers; tap a question for Part 1 or FU for Part 2

1. Frontier search (BFS or DFS), Tier 1: spread, count and measure distance (6) Go >

M * 2. * Number of Islands — 1Point3Acres * 2026-09-30 * [FU]
M * 3. * Rotten Oranges / Multi-Source BFS (taxis) — 1Point3Acres * 2026-09-08 * [FU]
M * 4. * Find shortest relationship path using BFS — PracHub * 2026-02-12 * [FU]
M * 5. * Compute distance to nearest taxi in grid — PracHub * 2026-01-10 * [FU]
M * 6. * Solve Grid and Counting Problems — PracHub * 2025-01-25 * [FU]
M * 7. * Farthest City for a New Data Center — ColorStack * 2025-10-10 * [FU]

2. Sweep over events in time order (6) Go >

M * 10. * Meeting Rooms / Interval Scheduling Variants — 1Point3Acres * 2026-09-29 * [FU]
M * 11. * Train / Flight Schedule Reachability — 1Point3Acres * 2026-10-08 * [FU]
M * 12. * Solve Rooms and Top-K Streams — PracHub * 2026-06-05 * [FU]
M * 13. * Gate Entry / Exit Time Resolution with Tie-Break Rules — 1Point3Acres * 2026-05-04 * [FU]
M * 14. * Compute servers needed for daily recurring jobs — PracHub * 2025-11-05 * [FU]
M * 15. * Meeting Rooms III Variant — HackTheRounds ND [FU]

3. Tree recursion, Tier 1: one function that visits a node, asks its children, and reports upward (4) Go >

M * 22. * Expression Tree Single-Leaf Mutation — 1Point3Acres * 2026-09-27 * [FU]
M * 23. * Longest Consecutive Path in Binary Tree — 1Point3Acres * 2026-07-02 * [FU]
M * 24. * Implement zigzag level-order traversal — PracHub * 2025-08-01 * [FU]
M * 25. * Given the root of a binary tree, return the length of the diameter of the tree. — Exponent ND [FU]

4. A heap of what is available: pop the best, put back the changed one (8) Go >

M * 28. * Streaming Median (LC 295) with Follow-ups — 1Point3Acres * 2026-08-03 * [FU]
M * 29. * Compute minimum number of rooms needed — PracHub * 2026-02-12 * [FU]
M * 30. * Implement a Reusable Locker Allocation System — PracHub * 2026-08-14 * [FU]
M * 31. * Simulate room assignments for scheduled meetings — PracHub * 2026-01-06 * [FU]
M * 32. * Minimum Makespan on Identical CPUs — ColorStack * 2025-09-23 * [FU]
M * 33. * Design server allocation for multi-type nodes — PracHub * 2025-09-06 * [FU]
M * 34. * Trip Expense Settlement — HackTheRounds ND [FU]
M * 35. * Playlist Shuffle No Adjacent Artists — HackTheRounds ND [FU]

5. Recurrence on a position or a range (memo or table) (6) Go >

M * 37. * Target Expressions from Ordered Digits — 1Point3Acres * 2026-08-13 * [FU]
M * 38. * Count Word Decompositions into Chemical Element Symbols — 1Point3Acres * 2026-07-14 * [FU]
M * 39. * Fence Painting Minimum Operations (Gemini Team) — 1Point3Acres * 2026-04-27 * [FU]
M * 40. * Maximize 1D deliveries within distance and minimize total distance — PracHub * 2025-12-27 * [FU]
M * 41. * Find the longest palindromic subsequence using dynamic programming. — Exponent ND [FU]
M * 42. * Jump Game with Profit — 1Point3Acres ND [FU]

6. Best-first search with a heap: Dijkstra and its minimax twin (5) Go >

M * 46. * Path Minimizing the Maximum Cell (Swim in Rising Water) — 1Point3Acres * 2026-09-29 * [FU]
M * 47. * Nearest Favorite City by Travel Time — ColorStack * 2025-10-17 * [FU]
M * 48. * Given a train schedule with departure city, arrival city, start time, end time, and duration, and given a source and destination, find the sequence of trains that gets you there at the earliest possible time. — Exponent ND [FU]
M * 49. * Find the shortest path in a graph. — Exponent ND [FU]
M * 50. * Given a matrix with a rat, bread, cat, and obstacles, find a path from the rat to the bread that maximizes the minimum distance from cat. — Exponent ND [FU]

7. Backtracking: choose, recurse, undo (8) Go >

M * 55. * Mahjong Winning Hand Detection (14 Tiles) — 1Point3Acres * 2026-10-09 * [FU]
M * 56. * Count Prefix-Matching Simple Paths in a Labeled Graph — PracHub * 2026-08-09 * [FU]
M * 57. * Determine if a 14-tile hand is winning — PracHub * 2026-03-09 * [FU]
M * 58. * American Football Score Combinations — ColorStack * 2025-08-25 * [FU]
M * 59. * Count Passcode Patterns on a Keypad — ColorStack * 2025-11-07 * [FU]
M * 60. * Match people to questions using tags and priorities — PracHub * 2025-12-27 * [FU]
M * 61. * Generate Parentheses — Exponent ND [FU]
M * 62. * Permutations — Exponent ND [FU]

8. Runs and groups: extend while it fits, restart or close when it does not (7) Go >

M * 66. * Longest Non-Decreasing Subarray with One Replacement — 1Point3Acres * 2026-10-04 * [FU]
M * 67. * Piano Hand Movement Grouping — 1Point3Acres * 2026-07-14 * [FU]
M * 68. * Find longest increasing contiguous subarray — PracHub * 2026-01-06 * [FU]
M * 69. * Sum of Good Subarrays (Adjacent Diff ±1) — 1Point3Acres * 2026-02-08 * [FU]
M * 70. * Sustained-Error Time Window — ColorStack * 2025-09-16 * [FU]
M * 71. * Longest Non-Decreasing Subarray From Two Arrays — ColorStack * 2025-09-24 * [FU]
M * 72. * Count Super Streaks — ColorStack * 2025-10-03 * [FU]

9. Union-find: merge the pairs, then ask who is together (7) Go >

M * 73. * Count Islands in a Grid and Track the Count as Land Cells Are Added — PracHub * 2026-10-02 * [FU]
M * 74. * 2D Points Connected Components by Radius — 1Point3Acres * 2026-08-04 * [FU]
M * 75. * Count Clusters of 2D Points Within a Radius — PracHub * 2026-06-18 * [FU]
M * 76. * Find A Threshold-Limited Path With Minimum Required Safety — PracHub * 2026-07-06 * [FU]
M * 77. * Determine Reporting Relationships — PracHub * 2026-02-22 * [FU]
M * 78. * Determine whether two nodes are related — PracHub * 2026-01-22 * [FU]
M * 79. * Movie Recommendations via Transitive Similarity — ColorStack * 2025-10-09 * [FU]

10. A dict of per-key state that you update, expire, or rank (7) Go >

M * 82. * Active Communication Top Users — 1Point3Acres * 2026-09-22 * [FU]
M * 83. * Filter Repeated Robot Status Messages — PracHub * 2026-07-05 * [FU]
M * 84. * Next-Word Predictor — 1Point3Acres * 2026-09-02 * [FU]
M * 85. * Batch and Streaming Log Deduplication — 1Point3Acres * 2026-07-23 * [FU]
M * 86. * Design a logger that returns top-K messages — PracHub * 2026-01-06 * [FU]
M * 87. * Website Activity Analytics: Unique Users + Session Time — 1Point3Acres * 2026-06-26 * [FU]
M * 88. * Hit Counter — HackTheRounds ND [FU]

11. Binary search on a yes/no condition (4) Go >

M * 90. * Cutting Ribbons (LC 1891) — 1Point3Acres * 2026-08-16 * [FU]
M * 91. * First Bad Version with Parallel Search Follow-up — 1Point3Acres * 2026-06-15 * [FU]
M * 92. * Solve Banana Speed and Interval Merge — PracHub * 2025-01-27 * [FU]
M * 93. * Given a list with response time for each server, calculate how long time it will take to process n requests. — Exponent ND [FU]

12. Greedy: scan once, carry the best choice so far, act only when forced (5) Go >

M * 100. * Visible People in a Line (Last Person, Then Everyone) — 1Point3Acres * 2026-10-07 * [FU]
M * 101. * Bounded Knapsack Variant — 1Point3Acres * 2026-10-16 * [FU]
M * 102. * Find the Largest Monotone Increasing Number — PracHub * 2026-07-04 * [FU]
M * 103. * Min boards to cover roof holes — PracHub * 2026-02-12 * [FU]
M * 104. * Jump Game II (Minimum Jumps) — ColorStack * 2025-08-19 * [FU]

13. Sets and counters: remember the first collection, check the second against it (3) Go >

M * 109. * Reconstruct original array from doubled shuffle — PracHub * 2026-01-04 * [FU]
M * 110. * Min deletions to avoid overlap in first k — PracHub * 2026-02-12 * [FU]
M * 111. * Filter Common Elements Per Window (Two Lists) — ColorStack * 2025-10-08 * [FU]

14. bisect on a sorted list: find a position, count a range, keep it sorted (5) Go >

M * 114. * Sliding-Window Mean After Removing the Largest K Values — PracHub * 2026-09-01 * [FU]

M * 115. * Streaming Insert + Find K-th Largest — 1Point3Acres * 2026-02-04 * [FU]
M * 116. * Count Sorted Strings With a Given Prefix — ColorStack * 2025-10-08 * [FU]
M * 117. * Russian Doll Envelopes — 1Point3Acres ND [FU]
M * 118. * Given positions of N kids and M ice cream sellers on a 1D line, find the smallest distance E such that every kid lies within the range [P - E, P + E] of at least one seller. — Exponent ND [FU]

15. Prefix sums, Tier 1: precompute once, answer ranges and balance points in O(1) (5) Go >

M * 124. * Range Duplicate Check with Constant-Time Queries — 1Point3Acres * 2026-09-29 * [FU]
M * 125. * Find a Horizontal Cut That Bisects Rectangle Area — PracHub * 2026-06-04 * [FU]
M * 126. * Check pivot after removing one element — PracHub * 2026-02-12 * [FU]
M * 127. * Gold Chain Split into Two Equal Halves — 1Point3Acres * 2026-04-21 * [FU]
M * 128. * Count the number of ways to split a string. — Exponent ND [FU]

16. Sorted intervals: merge while they touch, then read the gaps or search (5) Go >

M * 130. * Find Containing Range — PracHub * 2026-06-04 * [FU]
M * 131. * Union of Sorted Interval Lists — 1Point3Acres * 2026-08-16 * [FU]
M * 132. * Range Module (Insert and Point Query) — ColorStack * 2026-01-20 * [FU]
M * 133. * Meetings Minus a Do-Not-Schedule Block — ColorStack * 2025-10-03 * [FU]
M * 134. * Merge Intervals — Exponent ND [FU]

17. Trie: nested dicts, one character per level (7) Go >

M * 137. * URL Request Handler Router — 1Point3Acres * 2026-10-16 * [FU]
M * 138. * Trie Prefix Search / Autocomplete — 1Point3Acres * 2026-07-02 * [FU]
M * 139. * Build Prefix Lookup with a Trie — PracHub * 2026-07-03 * [FU]
M * 140. * Greedy Longest-Match Dictionary Tokenizer — 1Point3Acres * 2026-06-15 * [FU]
M * 141. * Check if all substrings are dictionary words — PracHub * 2026-03-11 * [FU]
M * 142. * Return dictionary words matching a prefix — PracHub * 2026-03-09 * [FU]
M * 143. * Implement Trie Insert and Exact-Word Search — PracHub * 2026-02-20 * [FU]

18. A dict for lookup plus an ordered container for who is next (lazy deletion) (6) Go >

M * 146. * Design a restaurant waitlist system — PracHub * 2026-03-01 * [FU]
M * 147. * Implement Python's itertools.tee — 1Point3Acres * 2026-08-20 * [FU]
M * 148. * Restaurant Waitlist: join / delete / find_first_match — 1Point3Acres * 2026-02-01 * [FU]
M * 149. * Store With Cheapest-Item Removal — ColorStack * 2025-09-29 * [FU]
M * 150. * Round-Robin Job Scheduler — ColorStack * 2025-08-19 * [FU]
M * 151. * Aggregate FizzBuzz Offer Matches — ColorStack * 2025-10-13 * [FU]

19. Rolling state: the next state comes only from the previous one (4) Go >

M * 155. * Maximize sum without choosing adjacent elements — PracHub * 2026-01-06 * [FU]
M * 156. * Paint House II — ColorStack * 2025-10-17 * [FU]
M * 157. * Given n houses in a line with money in each, find the maximum amount a robber can steal, without stealing from two adjacent houses. — Exponent ND [FU]
M * 158. * Write SQL code to publish the Fibonacci series. — Exponent ND [FU]

20. Two pointers: move i and j toward each other (or along two lists) (3) Go >

M * 164. * Find median of merged RLE arrays — PracHub * 2026-01-06 * [FU]
M * 165. * Run-Length Vector Storage + Dot Product — 1Point3Acres * 2026-01-21 * [FU]
M * 166. * Count Unique Triplets That Sum to Zero — ColorStack * 2025-09-20 * [FU]

21. Walk two strings at once with two indices (3) Go >

M * 170. * Can board states be transformed? — PracHub * 2026-04-01 * [FU]
M * 171. * Decide string transform with directional tokens and walls — PracHub * 2025-07-15 * [FU]
M * 172. * Determine if two sentences are similar. — Exponent ND [FU]

22. Simulate a machine: a class with state and one branch per kind of event (5) Go >

M * 175. * Tic-Tac-Toe System with K Players and 3-in-a-Row Win — 1Point3Acres * 2026-05-18 * [FU]
M * 176. * Design Tic-Tac-Toe With K Players — PracHub * 2026-02-12 * [FU]

M * 177. * Implement iterator merging lists with filtering — PracHub * 2026-01-06 * [FU]
M * 178. * Infinite Deck Card Duel — ColorStack * 2025-09-26 * [FU]
M * 179. * Design an elevator control system — PracHub * 2025-09-06 * [FU]

23. Top-K by frequency: Counter plus nsmallest (6) — Go >

M * 181. * Top-K Frequent in Large Logs — 1Point3Acres * 2026-07-19 * [FU]
M * 182. * Rearrange Tasks With Cooldown — PracHub * 2026-06-03 * [FU]
M * 183. * Implement frequency + distance top-K queries — PracHub * 2026-02-12 * [FU]
M * 184. * Find top/bottom-k words in list or stream — PracHub * 2026-01-11 * [FU]
M * 185. * Solve chat, grid paths, and car rentals — PracHub * 2025-12-02 * [FU]
M * 186. * Top-K Chatters by Word Count — ColorStack * 2025-09-21 * [FU]

24. Digits and characters: peel one at a time and carry (3) — Go >

M * 188. * Solve several coding interview problems — PracHub * 2026-03-04 * [FU]
M * 189. * Check Whether Digits Are Monotone Increasing — PracHub * 2026-07-04 * [FU]
M * 190. * Add two big integers from digit lists — PracHub * 2025-11-24 * [FU]

25. Look at the input once and keep only a small summary (and count your calls) (4) — Go >

M * 193. * Streaming Points: Emit K Within Distance — 1Point3Acres * 2026-06-23 * [FU]
M * 194. * Two-Day Log Intersection under Memory Limit — 1Point3Acres * 2025-11-27 * [FU]
M * 195. * Find shared objects across two log files — PracHub * 2025-11-05 * [FU]
M * 196. * Minimize calls to find all bad test pairs — PracHub * 2025-11-05 * [FU]

26. Grid DP: each cell from its neighbours above and to the left (3) — Go >

M * 199. * Grid Path Count with Diagonal Moves and Checkpoints — 1Point3Acres * 2026-09-29 * [FU]
M * 200. * Longest Same-Color Grid Sequence — 1Point3Acres * 2026-09-08 * [FU]
M * 201. * Count Same-Color Squares in Matrix — 1Point3Acres ND [FU]

27. Nested structures: a stack of contexts, or recursion with a cache (5) — Go >

M * 204. * Decode String — 1Point3Acres * 2026-07-29 * [FU]
M * 205. * Template Variable Expansion with Follow-Ups — 1Point3Acres * 2026-10-17 * [FU]
M * 206. * Expand Recursive Key-Value Placeholders in a Template — PracHub * 2026-08-04 * [FU]
M * 207. * Nested List Weight Sum (LC 339) with Follow-Ups — 1Point3Acres * 2026-08-16 * [FU]
M * 208. * Nested Function Expression Evaluator — 1Point3Acres * 2026-06-10 * [FU]

28. Frontier search (BFS or DFS), Tier 2: free steps, early exit, neighbours worked out on the fly (5) — Go >

M * 209. * Shortest Path Nodes and Escaping a Pursuer — 1Point3Acres * 2026-09-29 * [FU]
M * 210. * Radio Broadcast Reachability — ColorStack * 2026-01-30 * [FU]
M * 211. * Minimum Direction Violations in a Directed Graph — 1Point3Acres * 2026-07-19 * [FU]
M * 212. * Compute shortest paths with blocked nodes — PracHub * 2025-09-06 * [FU]
M * 213. * Shortest Directed Cycle Through a Node — 1Point3Acres ND [FU]

29. Topological sort (Kahn) and cycle detection (5) — Go >

M * 214. * Course Scheduling and Dependency Variants — 1Point3Acres * 2026-10-04 * [FU]
M * 215. * Validate Race Results (Cycle Detection) — ColorStack * 2025-10-01 * [FU]
M * 216. * Solve tree, graph, sliding-window problems — PracHub * 2025-08-04 * [FU]
M * 217. * Validate course catalog dependencies — PracHub * 2025-08-01 * [FU]
M * 218. * Course Schedule — Exponent ND [FU]

30. Counting pairs by looking up each partner in a Counter or a bucket (4) — Go >

M * 219. * Progressive Array Partitioning — 1Point3Acres * 2026-08-11 * [FU]
M * 220. * K-diff Pairs in an Array — ColorStack * 2025-09-16 * [FU]
M * 221. * Count Axis-Aligned Squares Among Grid Points — ColorStack * 2025-11-03 * [FU]
M * 222. * Sum of Two Squares in Two Ways — HackTheRounds ND [FU]

31. Tree DP: build a parent order once, then add up from the leaves (3) — Go >

M * 224. * Compute File System Total Size — 1Point3Acres * 2026-09-02 * [FU]

M *	225. * Minimum-Cost Tree Disconnection	1Point3Acres * 2026-09-09 * [FU]
M *	226. * Given a large n-ary tree where the root node can pass information to one child per iteration, and in subsequent iterations, nodes can pass information to one child while the root continues with others, find the minimum number of iterations to reach all nodes.	Exponent ND [FU]

32. Counting or reaching an amount: the knapsack loop (5) Go >

M *	229. * Maximize Shelf Space (Unbounded Knapsack)	ColorStack * 2025-10-17 * [FU]
M *	230. * Coin Change	Exponent ND [FU]
M *	231. * Partition an array into two sub-arrays with equal sum.	Exponent ND [FU]
M *	232. * Explain how to find a target sum in an array.	Exponent ND [FU]
M *	233. * Reverse Coin Change	HackTheRounds ND [FU]

33. Prefix sums, Tier 2: running total plus a dict of earlier totals (4) Go >

M *	234. * Solve array-sum and city-community problems	PracHub * 2026-02-20 * [FU]
M *	235. * Maximum-Sum Subarray With Equal Endpoints	ColorStack * 2025-09-29 * [FU]
M *	236. * Maximum Subarray Sum With a Minimum Length Constraint	ColorStack * 2025-08-19 * [FU]
M *	237. * Find the maximum subarray sum.	Exponent ND [FU]

34. Group by a canonical key (3) Go >

M *	240. * Check if all substrings are anagrams of words	PracHub * 2026-03-11 * [FU]
M *	241. * Group anagrams	Exponent ND [FU]
M *	242. * Given an array of strings, return groups of "buddy strings," where the strings have the same length and the distance between corresponding characters is the same.	Exponent ND [FU]

35. K-way merge: a heap of the next item from each source (2) Go >

M *	245. * K Largest Elements Across Sorted Lists	ColorStack * 2025-10-31 * [FU]
M *	246. * Sort a nearly sorted (or K sorted) array.	Exponent ND [FU]

36. Tree recursion, Tier 2: pass a rule down, return a node or a yes/no up (4) Go >

M *	250. * Find mode in a trinary search tree	PracHub * 2025-11-16 * [FU]
M *	251. * Next Upcoming Birthday in a Family Tree	ColorStack * 2025-10-02 * [FU]
M *	252. * Lowest Common Ancestor	HackTheRounds ND [FU]
M *	253. * Determine if a given binary tree is a binary search tree (BST).	Exponent ND [FU]

37. Random pick without repeats: swap the chosen item with the last one and pop (4) Go >

M *	254. * Design set with O(1) random access	PracHub * 2026-01-01 * [FU]
M *	255. * Random Generator Without Repeats	ColorStack * 2025-10-10 * [FU]
M *	256. * Random Song Shuffler With a No-Repeat Window	ColorStack * 2025-10-06 * [FU]
M *	257. * Unique Random Number Generator	ColorStack * 2025-10-03 * [FU]

38. Build and serialize: write a structure out, read it back with the same walk (3) Go >

M *	259. * Print a Directory Tree	ColorStack * 2025-09-21 * [FU]
M *	260. * Construct a binary tree given its preorder and inorder traversals.	Exponent ND [FU]
M *	261. * Write functions to serialize and deserialize a list of strings.	Exponent ND [FU]

39. Games: win or lose with a cache, and guessing with feedback (1) Go >

M *	264. * Choose optimal guesses for green-only Wordle	PracHub * 2025-10-14 * [FU]

40. Sliding window: grow on the right, shrink on the left while it is too bad (4) Go >

M *	269. * Temperature Monitoring Data Structure	1Point3Acres * 2026-09-10 * [FU]
M *	270. * Footpath Beauty After Repainting	ColorStack * 2025-09-22 * [FU]
M *	271. * Shortest Subarray with K Distinct Integers	HackTheRounds ND [FU]
M *	272. * Find the minimum window substring.	Exponent ND [FU]

41. Parent pointers: store who is above each node and climb (4) Go >

M *	275. * Employee Hierarchy Add / Delete	1Point3Acres * 2026-06-10 * [FU]
M *	276. * Validate parent array forms a tree	PracHub * 2026-02-11 * [FU]

M * 277. * Root a Tree So It Becomes a Binary Tree ColorStack * 2025-11-07 * [FU]
M * 278. * Lowest Common Ancestor of a Binary Tree III HackTheRounds ND [FU]

42. Build or transform a grid with index arithmetic (4) Go >

M * 280. * Connected Crop Arrangement on a Grid 1Point3Acres * 2026-10-04 * [FU]
M * 281. * Solve matrix groups and recipe inventory PracHub * 2026-01-22 * [FU]
M * 282. * Range Sum Query 2D (Immutable Matrix) ColorStack * 2025-08-19 * [FU]
M * 283. * Rotate Any m×n 2D Matrix 1Point3Acres ND [FU]

43. Split into pieces, fix each piece, join (and pack words into lines) (2) Go >

M * 285. * Snake Case to Camel Case with Repository-Wide Refactor 1Point3Acres * 2026-07-14 * [FU]
M * 286. * Google Doc Line Wrap (Word Wrap with Whitespace and Newlines) 1Point3Acres * 2026-02-01 * [FU]

PART A - PART 1 | 197 MEDIUM QUESTIONS

PART A - Part 1: question to short best solution

This section contains the Medium questions. Understand the prompt, clarify the rules, state the brute-force baseline, then code the shortest complete best solution. The purple button opens the matching PART A - Part 2 follow-ups.

GROUP 1 OF 43 | 6 QUESTIONS | PART A - PART 1

Frontier search (BFS or DFS), Tier 1: spread, count and measure distance

PATTERN CONTEXT

Every one of these keeps a frontier of cells or nodes still to visit, takes one out, looks at its neighbours, and adds the new ones. A dict of distances doubles as the seen set. Change what you start from, which neighbours are allowed, and what you do when a node comes out.

2 * Number of Islands * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: loop over every cell; each unseen land cell starts a new frontier_search (share one seen set); islands += 1 per start

QUESTION INFO

QUESTION

Given a grid of 0s (water) and 1s (land), count the islands. An island is a group of 1s connected up, down, left or right (diagonal cells are not connected).

Function signature

def num_islands(grid: list) -> int

EXAMPLES

Example 1	
Input	`[[1, 1, 0, 0], [1, 1, 0, 0], [0, 0, 1, 1], [0, 0, 1, 1]]`
Output	`2`
Why	`Two 2x2 blocks of land that only touch the water between them: 2 islands.`

Example 2	
Input	`[[1, 0, 1], [0, 1, 0], [1, 0, 1]]`
Output	`5`
Why	`Land that touches only diagonally is not connected, so every 1 is its own island.`

Example 3	
Input	`[[0, 0], [0, 0]]`
Output	`0`
Why	`No land, no islands.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

DIRECT APPROACH | Recursive flood fill

DIRECT IDEA

Scan; at each unvisited land cell count one island and recurse in four directions to mark it. This is the standard recursive approach, but deep recursion can overflow on a large snake-shaped island.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
seen = empty set; islands = 0
scan every cell in the grid
if it is unvisited land, increment islands
```

```
DFS or BFS from it; mark all 4-directional land as seen
return islands
```

DIRECT-APPROACH COMPLEXITY | Time O(m * n) | Space O(m * n) recursion

WHY THIS IS WEAKER | Time is already O(m * n); the weakness is the recursion depth, which can reach m * n on a snake-shaped island and overflow the stack. An explicit queue or stack avoids that.

HOW TO IMPROVE IT

loop over every cell; each unseen land cell starts a new frontier_search (share one seen set); islands += 1 per start

SHORT BEST SOLUTION

Each unvisited land cell starts one island. DFS marks its complete 4-directional component before the scan continues.

```python
# BASE QUESTION (optimal): Recursive DFS flood fill
# Idea: Every unvisited land cell starts one island. DFS
# marks its whole 4-directional component before scanning on.
def numIslands(grid):
    if not grid:
        return 0
    rows, cols = len(grid), len(grid[0])
    seen = set()

    def dfs(row, col):
        if not (0 <= row < rows and 0 <= col < cols):
            return
        if grid[row][col] != 1 or (row, col) in seen:
            return
        seen.add((row, col))
        dfs(row + 1, col)
        dfs(row - 1, col)
        dfs(row, col + 1)
        dfs(row, col - 1)

    islands = 0
    for row in range(rows):
        for col in range(cols):
            if grid[row][col] == 1 and (row, col) not in seen:
                islands += 1
                dfs(row, col)
    return islands
```

COMPLEXITY | Time O(m * n) | Space O(m * n) for seen and recursion

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 2

3 * Rotten Oranges / Multi-Source BFS (taxis) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: starts = every rotten orange; usable = fresh orange; answer = rounds taken, or -1 if a fresh orange stays unreached

QUESTION INFO

QUESTION

Each cell of a grid is 0 (empty), 1 (a fresh orange) or 2 (a rotten orange). Every minute, each fresh orange that is next to (up, down, left or right) a rotten one becomes rotten. Return the number of minutes until no fresh orange is left, or -1 if some fresh orange can never rot.

Function signature

def rotting_minutes(grid: list) -> int

EXAMPLES

QUESTION

Example 1	
Input	`[[2, 1, 1], [1, 1, 0], [0, 1, 1]]`
Output	`4`
Why	The rot spreads outward from the corner and reaches the last orange after 4 minutes.

Example 2	
Input	`[[2, 1, 1], [0, 1, 1], [1, 0, 1]]`
Output	`-1`
Why	The orange at the bottom left is cut off by empty cells, so it never rots.

Example 3	
Input	`[[0, 2]]`
Output	`0`
Why	Nothing fresh to rot: 0 minutes.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Re-scan the whole grid every minute

BRUTE-FORCE IDEA

Each minute, scan the grid and collect every fresh orange next to a rotten one. Rot them together, then repeat until no new orange changes. This is slow because every minute rescans the whole grid.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
minutes = 0
repeat: collect fresh oranges next to rotten oranges
if none were collected, stop
rot all collected oranges together; minutes += 1
return minutes if no fresh oranges remain, otherwise -1
```

BRUTE-FORCE COMPLEXITY | Time O(minutes * m * n) | Space O(m * n)

WHY THIS IS WEAKER | Each minute scans the entire grid, O(minutes * m * n). A queue of the newly rotten oranges touches each cell only once, O(m * n).

HOW TO IMPROVE IT

starts = every rotten orange; usable = fresh orange; answer = rounds taken, or -1 if a fresh orange stays unreached

SHORT BEST SOLUTION

Part 1: Start one BFS with every rotten orange; each level is one minute.

Part 2: Start one BFS with every taxi and store the first distance written to each open cell.

```python
from collections import deque


def rotting_minutes(grid):
    if not grid:
        return 0
    rows, cols = len(grid), len(grid[0])
    queue = deque((row, col) for row in range(rows) for col in range(cols)
                  if grid[row][col] == 2)
    fresh = sum(cell == 1 for row in grid for cell in row)
    minutes = 0
```

```
    while queue and fresh:
        minutes += 1
        for _ in range(len(queue)):
            row, col = queue.popleft()
            for row_step, col_step in ((1, 0), (-1, 0), (0, 1), (0, -1)):
                next_row, next_col = row + row_step, col + col_step
                if 0 <= next_row < rows and 0 <= next_col < cols and grid[next_row][next_col] == 1:
                    grid[next_row][next_col] = 2
                    fresh -= 1
                    queue.append((next_row, next_col))
    return minutes if fresh == 0 else -1


def nearest_taxi(grid):
    if not grid:
        return []
    rows, cols = len(grid), len(grid[0])
    distance = [[-1] * cols for _ in range(rows)]
    queue = deque()
    for row in range(rows):
        for col in range(cols):
            if grid[row][col] == "T":
                distance[row][col] = 0
                queue.append((row, col))
    while queue:
        row, col = queue.popleft()
        for row_step, col_step in ((1, 0), (-1, 0), (0, 1), (0, -1)):
            next_row, next_col = row + row_step, col + col_step
            if (0 <= next_row < rows and 0 <= next_col < cols
                    and grid[next_row][next_col] != "#" and distance[next_row][next_col] == -1):
                distance[next_row][next_col] = distance[row][col] + 1
                queue.append((next_row, next_col))
    return distance
```

COMPLEXITY | Part 1: Time O(m * n) | Space O(m * n)
Part 2: Time O(m * n) | Space O(m * n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 3

4 * Find shortest relationship path using BFS * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: one start; also store parent[neighbour] = person when you first reach v; stop at the target and walk parent back to rebuild the path

QUESTION INFO

QUESTION

People are connected by relationships, given as an adjacency dict that maps each person to the people they are directly related to. shortest_path(adj, s, t) returns the shortest chain of people from s to t (both ends included), or None when t cannot be reached.

Function signature

def shortest_path(adj: dict, s, t) -> list | None

EXAMPLES

QUESTION

Example 1

Input	`adj = {'ann': ['bob', 'cy'], 'bob': ['dee'], 'cy': ['dee', 'eli'], 'dee': ['fay'], 'eli': ['fay'], 'fay': []}`
	`shortest_path(adj, 'ann', 'fay')`
	`shortest_path(adj, 'ann', 'ann')`
	`shortest_path(adj, 'fay', 'ann')`

Output	`shortest_path(adj, 'ann', 'fay') -> ['ann', 'bob', 'dee', 'fay']`
	`shortest_path(adj, 'ann', 'ann') -> ['ann']`
	`shortest_path(adj, 'fay', 'ann') -> null`

Why	**Several chains of 4 people exist; BFS finds the first. A person is trivially related to themselves, and fay has no outgoing relationships.**

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Depth-first search over all simple paths

BRUTE-FORCE IDEA

Try every simple path from the source and keep the shortest that reaches the target. Correct; exponential in the size of the graph.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Depth-first search over all simple paths
Try every simple path from the source
keep the shortest that reaches the target
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(path)

WHY THIS IS WEAKER | The number of simple paths is exponential. Breadth-first search reaches the target along a shortest path first, O(V + E), and bidirectional BFS explores far fewer nodes on big graphs.

HOW TO IMPROVE IT

one start; also store parent[neighbour] = person when you first reach v; stop at the target and walk parent back to rebuild the path

SHORT BEST SOLUTION

Queue the source and mark it visited. Pop a person, queue each unvisited neighbour with parent = this person, and stop when the target is queued. Follow parents back to the source and reverse to get the path.

```python
from collections import deque


# BASE QUESTION (optimal): Breadth-first search with a
# parent map
# Idea: Queue the source and mark it visited. Pop a
# person, queue each unvisited neighbour with parent =
# this person, and stop when the target is queued. Follow
# parents back to the source and reverse to get the path.
def shortest_relationship(connections, source, target):
    if source == target:
        return [source]
    parent, queue = {source: None}, deque([source])
    while queue:
        person = queue.popleft()
        for neighbour in connections.get(person, ()):
            if neighbour not in parent:
                parent[neighbour] = person
                if neighbour == target:
```

```
                path = [neighbour]
                while parent[path[-1]] is not None:
                    path.append(parent[path[-1]])
                return path[::-1]
            queue.append(neighbour)
    return None
```

COMPLEXITY | Time O(V + E) | Space O(V)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 4

5 * Compute distance to nearest taxi in grid * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: starts = every taxi cell; usable = not a wall; the dist table is the answer, cells never reached stay -1

QUESTION INFO

QUESTION

A grid has 'T' for a taxi, '#' for a blocked cell and '.' for an open cell. For every cell return the number of steps (up, down, left or right, never through '#') to the nearest taxi: 0 for a taxi, and -1 for a blocked cell or a cell no taxi can reach.

Function signature

def taxi_distances(grid: list) -> list

EXAMPLES

Example 1	
Input	`['T.#', '...', '#.T']`
Output	`[[0, 1, -1], [1, 2, 1], [-1, 1, 0]]`
Why	The two taxis are in opposite corners; the middle cell is 2 steps from both; the blocked cells show -1.

Example 2	
Input	`['..', '..']`
Output	`[[-1, -1], [-1, -1]]`
Why	No taxi at all: every open cell is unreachable.

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | BFS from every cell to find its nearest taxi

BRUTE-FORCE IDEA

For each open cell run a BFS until the first taxi is met. Correct, but it repeats almost the same search from every cell.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: BFS from every cell to find its nearest taxi
For each open cell run a BFS until the first taxi is met
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O((m * n)^2) | Space O(m * n)

WHY THIS IS WEAKER | A separate BFS per cell, O((m * n)^2). One BFS that starts from all taxis at once reaches every cell at its nearest-taxi distance, O(m * n).

HOW TO IMPROVE IT

starts = every taxi cell; usable = not a wall; the dist table is the answer, cells never reached stay -1

SHORT BEST SOLUTION

Initialise the queue with every taxi at distance 0 and expand in rounds. A cell receives its distance the first time it is reached; blocked cells are never entered; cells that no taxi can reach stay at -1.

```
from collections import deque


# BASE QUESTION (optimal): One multi-source BFS
# Idea: Initialise the queue with every taxi at distance 0
# and expand in rounds. A cell receives its distance the
# first time it is reached; blocked cells are never
# entered; cells that no taxi can reach stay at -1.
def taxi_distances(grid):
    rows, cols = len(grid), len(grid[0])
    distance = [[-1] * cols for _ in range(rows)]
    queue = deque()
    for row in range(rows):
        for col in range(cols):
            if grid[row][col] == "T":
                distance[row][col] = 0
                queue.append((row, col))
    while queue:
        row, col = queue.popleft()
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols and grid[next_row][next_col] != "#" and distance[n
ext_row][next_col] == -1:
                distance[next_row][next_col] = distance[row][col] + 1
                queue.append((next_row, next_col))
    return distance
```

COMPLEXITY | Time O(m * n) | Space O(m * n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 5

6 * Solve Grid and Counting Problems * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the island count again (queue version); the second part of the question is a short counting formula, no search

QUESTION INFO

QUESTION

A software engineering intern phone screen included two coding tasks: 1.
Count islands in a grid Given an `m x n` grid of characters where `'1'` represents land and `'0'` represents water, return the number of islands. An island is formed by connecting adjacent land cells horizontally or vertically. You may assume all four edges of the grid are surrounded by water. 2.
Count ways to form a target word Given an array of uppercase letters, count how many different ways you can choose letters from the array so that the chosen letters can be rearranged to spell **`GOOGLE`**. Each array element can be used at most once, and letters at different indices are considered different choices even if they contain the same character. Design efficient algorithms for both tasks, and discuss the time and space complexity of your approach.

Part 1: Count Islands in a GridGiven a 2D grid of characters where each cell is either '1' (land) or '0' (water), return the number of islands. An island is a group of adjacent land cells connected only horizontally or vertically. Diagonal cells do not count as connected. You may assume the grid edges are surrounded by water.
Constraints
0 <= number of rows <= 3000 <= number of columns <= 300grid[r][c] is either '0' or '1'All rows have the same length

QUESTION

EXAMPLES

```
Input: []
```

Expected Output:
Explanation: Empty grid contains no land, so there are no islands.

```
Input: [['1']]
```

Expected Output:
Explanation: A single land cell forms one island.

Hints

Whenever you find an unvisited land cell, start a DFS or BFS to mark the entire island.Only 4-directional neighbors matter: up, down, left, and right.

Part 2: Count Ways to Form the Word GOOGLEGiven an array of uppercase letters, count how many different ways you can choose letters so that the chosen letters can be rearranged to spell GOOGLE. Each array element can be used at most once. Letters at different indices count as different choices even if they contain the same character. The word GOOGLE requires exactly 2 G's, 2 O's, 1 L, and 1 E.

Constraints

0 <= len(letters) <= 200000Each element is an uppercase English letter from 'A' to 'Z'The answer may be large, but fits in Python's integer type

Examples

Input: []

Expected Output:

Explanation: No letters means it is impossible to form GOOGLE.

Input: ['G','O','O','G','L','E']

Expected Output:

Explanation: There is exactly one way to choose all six letters.

Hints

First count how many G, O, L, and E letters appear in the array.You need to choose 2 of the G positions and 2 of the O positions, then 1 L and 1 E.

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Islands: recursive flood fill

BRUTE-FORCE IDEA

Scan the grid; at each unvisited land cell, count one island and recurse into the four neighbours. Correct and the standard approach; deep recursion can overflow on a large snake-like island.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
seen = empty set; islands = 0
scan every grid cell
when unvisited land is found, increment islands
DFS from it and mark all 4-directional land as seen
return islands
```

BRUTE-FORCE COMPLEXITY | Time O(m * n) | Space O(m * n) recursion

WHY THIS IS WEAKER | Time is O(m * n), but the recursion can be m * n deep on a snake-shaped island and overflow the stack. A queue or an explicit stack avoids that.

BRUTE FORCE 2 | Counting: enumerate every 6-element selection

BRUTE-FORCE IDEA

Try every set of 6 indices and test whether the letters are G, G, O, O, L, E. Exact; infeasible for n beyond a few dozen.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Counting: enumerate every 6-element selection
Try every set of 6 indices
test whether the letters are G
```

BRUTE-FORCE COMPLEXITY | Time O(C(n, 6) * 6) | Space O(1)

WHY THIS IS WEAKER | C(n, 6) selections. The letters needed are fixed, so the answer is C(#G, 2) * C(#O, 2) * #L * #E from one pass of counts.

HOW TO IMPROVE IT

the island count again (queue version); the second part of the question is a short counting formula, no search

SHORT BEST SOLUTION

Part 1: Count each unvisited land cell once, then DFS through its whole component.

Part 2: Count the available letters and multiply the ways to choose the required copies of G, O, L and E.

```python
from collections import Counter
from math import comb


# PART 1: Recursive DFS island count
def numIslands(grid):
    if not grid:
        return 0
    rows, cols = len(grid), len(grid[0])
    seen = set()

    def dfs(row, col):
        if not (0 <= row < rows and 0 <= col < cols):
            return
        if grid[row][col] != "1" or (row, col) in seen:
            return
        seen.add((row, col))
        for row_step, col_step in ((1, 0), (-1, 0), (0, 1), (0, -1)):
            dfs(row + row_step, col + col_step)

    islands = 0
    for row in range(rows):
        for col in range(cols):
            if grid[row][col] == "1" and (row, col) not in seen:
                islands += 1
                dfs(row, col)
    return islands


# PART 2: choose the required copies of each letter
def google_ways(letters):
    count = Counter(letters)
    return comb(count["G"], 2) * comb(count["O"], 2) * count["L"] * count["E"]
```

COMPLEXITY | Part 1: Time O(m * n) | Space O(m * n)
Part 2: Time O(n) | Space O(1) for a fixed alphabet

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 6

7 * Farthest City for a New Data Center * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: same on a graph: starts = the data centres, neighbors = adjacency list; answer = the city with the largest distance

QUESTION INFO

QUESTION

Cities are nodes 0..num_cities-1 of an undirected, unweighted graph; edges[i] = [u, v] is a direct connection. You are given a non-empty list dcs of cities that already host a data center. You want to build a new data center in the city that is as far as possible from the nearest existing data center — that is, the city whose shortest-path distance (in number of edges) to its closest data center is maximized. Return that city's index. Break ties by choosing the smallest index. It is guaranteed every city can reach at least one data center.

Function signature

```
def farthest_city(num_cities: int, edges: list[list[int]], dcs: list[int]) -> int:
    ...
```

Follow-up (as reported): Run a single BFS seeded from all data centers at once (distance 0) instead of a BFS per data center.

EXAMPLES

Example 1	
Input	`num_cities = 5, edges = [[0,1],[1,2],[2,3],[3,4]], dcs = [0]`
Output	`4`
Why	`Distances from city 0 are 0,1,2,3,4; city 4 is farthest.`

Example 2	
Input	`num_cities = 5, edges = [[0,1],[1,2],[2,3],[3,4]], dcs = [0, 4]`
Output	`2`
Why	`Nearest-DC distances are 0->0, 1->1, 2->2, 3->1, 4->0; city 2 is farthest at distance 2.`

CONSTRAINTS

- 1 <= num_cities <= 10**5 , 0 <= len(edges) <= 2*10**5 .
- 1 <= len(dcs) <= num_cities , dcs are distinct valid city indices.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= num_cities <= 10**5 , 0 <= len(edges) <= 2*10**5 .
- 1 <= len(dcs) <= num_cities , dcs are distinct valid city indices.

BRUTE FORCE | A BFS from every data center

BRUTE-FORCE IDEA

Start with: A BFS from every data center. Each data center repeats a full search over the graph.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: A BFS from every data center
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(D * (V + E)) | Space O(V)

WHY THIS IS WEAKER | Each data center repeats a full search over the graph.

HOW TO IMPROVE IT

same on a graph: starts = the data centres, neighbors = adjacency list; answer = the city with the largest distance

SHORT BEST SOLUTION

Seed the queue with every data center at distance 0; the first visit is the nearest one.

```
from collections import deque


# BASE QUESTION (optimal): One multi-source BFS
# Idea: Seed the queue with every data center at distance
# 0; the first visit is the nearest one.
def farthest_city(num_cities, edges, data_centers):
    neighbours = [[] for _ in range(num_cities)]
    for from_city, to_city in edges:
        neighbours[from_city].append(to_city)
        neighbours[to_city].append(from_city)
    distance = [-1] * num_cities
    queue = deque(data_centers)
    for data_center in data_centers:
        distance[data_center] = 0
    while queue:
        from_city = queue.popleft()
        for to_city in neighbours[from_city]:
            if distance[to_city] < 0:
                distance[to_city] = distance[from_city] + 1
                queue.append(to_city)
    return distance.index(max(distance))
```

COMPLEXITY | Time O(V + E) | Space O(V + E)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 7

GROUP 2 OF 43 | 6 QUESTIONS | PART A - PART 1

Sweep over events in time order

PATTERN CONTEXT

Turn every item into events (a start adds one, an end removes one, an arrival joins a queue), sort the events by time, and walk them keeping a small state: a counter, a queue, a heap or a running total. What you record at each event is the only thing that changes.

10 * Meeting Rooms / Interval Scheduling Variants * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the template as written: the largest running count is the number of rooms

QUESTION INFO

QUESTION

LC 253 (Meeting Rooms II) appears in many forms across phone screen and onsite coding rounds — minimum number of rooms, room assignment, cyclic time, daily server jobs, gate entry/exit. Common follow-ups push toward sweep line, min-heap, or distributed processing.

The base task in every variant: given a list of intervals (start, end) — or pickup/return pairs, meeting time pairs, server job runs — answer one or more of:

Min rooms / cars / servers needed to honor all intervals concurrently (the canonical min-meeting-rooms problem).
Assignment: which interval goes to which room/car, so the caller can use the rental record.
Single-interval overlap test between two intervals.

Variant: parking-lot capacity

Arrival and departure intervals represent cars occupying a parking lot. Compute the peak simultaneous occupancy using the same interval-overlap model.

Variant: cyclic time (24h wrap)

Intervals wrap around midnight; if an interval ends after the day boundary, it carries into the next day.
Follow-up: "if N is huge, how do you optimize?" — expect a sweep-line discussion.

Variant: daily-recurring server jobs

Same set of jobs runs every day. If a job can't finish in one day, truncate at midnight and resume next day.
A server can run multiple identical jobs that overlap on the same machine; different jobs need separate servers.
Follow-up: a single job that runs many days needs its own dedicated server.

Variant: on-call rotation consolidation

Input: on-call rotations {name, start, end} using half-open intervals.
Output maximal contiguous segments during which the active on-call set does not change; omit gaps with nobody on call.
Segments are sorted by start, and names in each segment are the active set for that time range.
Clean approach: sort starts, keep a min-heap by end time, and emit a segment whenever the next start or next end changes the active set.

Variant: gate entry / exit with priority rules

Array of (timestamp, action) where action in {enter, exit}; each index is a different person.
Return for each person what time they actually enter/exit, applying tie-break rules:

If the prior tick had an enter, enter wins on a tie.
If the prior tick had an exit, exit wins on a tie.
If the prior tick had no activity, exit wins.
Within the winning class, lower index wins.

QUESTION

Optimization: instead of stepping time tick-by-tick, jump through the sorted input timestamps to reach O(n).

Variant: Google Calendar free-time intersection

Input: requested attendees plus each person's busy calendar intervals.
Return every time range where all requested attendees are free.
Normalize each person's busy blocks into free gaps, then intersect the candidate ranges across attendees.

Variant: car-rental assignment

Input: rental orders containing an order ID, pickup time, and return time.
Assign every order to a car so that no assigned intervals overlap, minimize the number of cars, and return each car's complete order list. Model each car explicitly with its own rental record.
A current output variant instead returns a dictionary from each reservation ID to its assigned car ID; the follow-up asks for the final fleet size.
A return at time t and another pickup at time t may reuse the same car.
Follow-up: if the fleet is capped at K cars and all requests cannot be served, define how the allocator detects and handles the failure.

Variant: exact-concurrency intervals

Compute the maximum number of simultaneously active meetings.
Given a target concurrency X, also return every maximal time range during which exactly X meetings are active.
Define the event ordering when one meeting starts at the same timestamp another ends; this choice controls whether a zero-width overlap is counted.

Variant: maximum-profit job scheduling

Each job has startTime, endTime, and profit. Choose a set of non-overlapping jobs maximizing total profit. The candidate identified it as LC 1235 (Maximum Profit in Job Scheduling).
Asked as the first question of an L4 onsite round, followed by an unrelated range-query problem in the same slot.

On-call employee metadata variant

Each duty record includes employee ID, employee name, start time, and end time. Return the employee count and identifying information for each time segment.

EXAMPLES

```
Canonical min-meeting-rooms example: [[0,30],[5,10],[15,20]] -> 2.
On-call consolidation: A[10,50], B[20,60], C[30,40], D[30,40] -> [10,20]: A, [20,30]: A,B, [30,40]:
    A,B,C,D, [40,50]: A,B, [50,60]: B.
Single-interval overlap: intervalsOverlap([1,5],[4,7]) -> true.
```

Long-running job: a 36-hour daily job is impossible under the truncate rule; reserve a separate server.

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Count the overlap at every start time

BRUTE-FORCE IDEA

For every interval count how many intervals contain its start point and take the maximum. Correct and simple; quadratic.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Count the overlap at every start time
For every interval count how many intervals contain its start point
take the maximum
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1)

WHY THIS IS WEAKER | Every interval is compared with every other, O(n^2). Sorting start and end events and keeping a running count is O(n log n).

HOW TO IMPROVE IT

the template as written: the largest running count is the number of rooms

SHORT BEST SOLUTION

Part 1: Sweep sorted start and end events and keep the largest active count.

Part 2: Process meetings by start time and reuse the room that becomes free first.

Part 3: Sweep change times and emit each interval with its current active people.

```
import heapq
from collections import defaultdict


def min_rooms(intervals):
    events = sorted((time, delta) for start, end in intervals
                    for time, delta in ((start, 1), (end, -1)))
    active = answer = 0
    for _, delta in events:
        active += delta
        answer = max(answer, active)
    return answer


def assign_rooms(intervals):
    active = []
    rooms = [0] * len(intervals)
    room_count = 0
    for start, end, index in sorted((begin, finish, position) for position, (begin, finish) in enumerate(interv
als)):
        if active and active[0][0] <= start:
            _, room = heapq.heappop(active)
        else:
            room = room_count
            room_count += 1
        rooms[index] = room
        heapq.heappush(active, (end, room))
    return rooms


def oncall_segments(rotations):
    changes = defaultdict(list)
    for name, start, end in rotations:
        changes[start].append((name, 1))
        changes[end].append((name, -1))
    active, answer = defaultdict(int), []
    times = sorted(changes)
    for time, next_time in zip(times, times[1:]):
        for name, delta in changes[time]:
            active[name] += delta
        people = frozenset(name for name, count in active.items() if count)
        if not people:
            continue
        if answer and answer[-1][1] == time and answer[-1][2] == people:
            answer[-1] = (answer[-1][0], next_time, people)
        else:
            answer.append((time, next_time, people))
    return answer
```

COMPLEXITY | Part 1: Time O(n log n) | Space O(n)
Part 2: Time O(n log n) | Space O(n)
Part 3: Time O(n log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 10

11 * Train / Flight Schedule Reachability * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: sort legs by departure; a leg is usable if you have already reached its origin before it leaves; keep the earliest arrival per place

QUESTION INFO

QUESTION

Given trains (or flights) as (from, to, departure, arrival), decide whether you can travel from a source to a destination, boarding a later leg only if it departs after you arrive. Graph reachability with time state, solved with BFS / DFS.

Input: a schedule of legs, each with a departure station, an arrival station, a departure time, and an arrival time; plus a source and a destination.
Output: whether the destination is reachable from the source, where a leg can only be taken if it departs after you arrive at its departure station.
The same prompt appears as a flight schedule between airports in new-grad R1.

EXAMPLES

```
trains = [
  ("A", "Y", dep=1, arr=20),
  ("A", "Y", dep=2, arr=5),
  ("Y", "B", dep=6, arr=9)
]
source = "A", destination = "B"

Output: true — the first A->Y train arrives too late for the Y->B connection, but the second one makes it
.
```

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Explore every (station, arrival time) state

BRUTE-FORCE IDEA

Search from the source over states (station, time you arrived), following every leg that departs after the current arrival. Correct and the direct approach, but each state rescans all legs.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Explore every (station, arrival time) state
Search from the source over states (station
time you arrived)
following every leg that departs after the current arrival
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(legs^2) | Space O(legs)

WHY THIS IS WEAKER | Every state rescans all legs and the same station is reached at many times, O(legs^2). Sorting the legs by departure and keeping only the earliest arrival per station answers in O(L log L).

HOW TO IMPROVE IT

sort legs by departure; a leg is usable if you have already reached its origin before it leaves; keep the earliest arrival per place

SHORT BEST SOLUTION

Sort the legs by departure time. Keep earliest[station] = earliest known arrival, with the source at minus infinity. For each leg whose origin has an earliest arrival before (or at) its departure, relax earliest[destination] with its arrival. Sorting guarantees every leg that could feed this one has already been processed, because a leg arrives after it departs.

```python
INFINITY = float("inf")
```

```python
# BASE QUESTION (optimal): Sort by departure and keep
# earliest arrivals
# Idea: Sort the legs by departure time. Keep
# earliest[station] = earliest known arrival, with the
# source at minus infinity. For each leg whose origin has
# an earliest arrival before (or at) its departure, relax
# earliest[destination] with its arrival. Sorting
# guarantees every leg that could feed this one has
# already been processed, because a leg arrives after it
# departs.
def earliest_arrivals(legs, source, strict=True):
    earliest = {source: -INFINITY}
    for origin, destination, departure, arrival in sorted(legs, key=lambda leg: leg[2]):
        earliest_at_origin = earliest.get(origin, INFINITY)
        if earliest_at_origin < departure or (not strict and earliest_at_origin == departure):
            if arrival < earliest.get(destination, INFINITY):
                earliest[destination] = arrival
    return earliest
```

COMPLEXITY | Time O(L log L) | Space O(S)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 11

12 * Solve Rooms and Top-K Streams

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: sort starts and ends separately and walk two pointers: a start before the earliest end needs a new room; the stream part is a size-K heap

QUESTION INFO

QUESTION

Two small tasks. (1) min_rooms(intervals): the minimum number of rooms needed to hold all meetings; a meeting may start exactly when another ends. (2) TopKStream(k): numbers arrive one at a time with add(x); top() returns the k largest numbers seen so far, largest first.

Function signature

def min_rooms(intervals: list) -> int
class TopKStream:
  def __init__(self, k: int)
  def add(self, x: int) -> None
  def top(self) -> list

EXAMPLES

Example 1	
Input	`[(0, 30), (5, 10), (15, 20)]`
Output	`2`
Why	`(5, 10) and (15, 20) both overlap the long meeting (0, 30): 2 rooms.`

Example 2	
Input	`[(7, 10), (2, 4)]`
Output	`1`
Why	`The second meeting ends before the first starts: one room is enough.`

QUESTION

Example 3

Input	`s = TopKStream(3)`
	`s.add(5)`
	`s.add(1)`
	`s.add(9)`
	`s.add(7)`
	`s.top()`

Output	`s.top() -> [9, 7, 5]`
Why	`Of 5, 1, 9, 7 the three largest are 9, 7 and 5.`

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Rooms: count the overlap at every time unit

BRUTE-FORCE IDEA

For each integer time t count the meetings with start <= t < end and take the maximum. Correct (integer times only); it depends on the length of the time range, not just n.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Rooms: count the overlap at every time unit
For each integer time t count the meetings with start <= t < end
take the maximum
```

BRUTE-FORCE COMPLEXITY | Time O(range * n) | Space O(1)

WHY THIS IS WEAKER | Depends on the length of the time range, not only on n: O(range * n). Sorted starts and ends with two pointers (or a heap of end times) are O(n log n).

BRUTE FORCE 2 | Stream: sort everything on every query

BRUTE-FORCE IDEA

Store all values and sort to read the K-th largest after each add. Correct; it re-sorts the whole history each time.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Stream: sort everything on every query
Store all values
sort to read the K-th largest after each add
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n log n) per add | Space O(n)

WHY THIS IS WEAKER | Every add re-sorts the whole history, O(n log n). A min-heap of size K keeps the K-th largest at its root in O(log K) per add.

HOW TO IMPROVE IT

sort starts and ends separately and walk two pointers: a start before the earliest end needs a new room; the stream part is a size-K heap

SHORT BEST SOLUTION

Part 1: Sort starts and ends separately. For each start: if it is at or after the earliest unfinished end, that room is reused (advance the end pointer); otherwise a new room is needed. The count of rooms opened is the maximum overlap.

Part 2: Push each value; if the heap has more than K items pop the smallest. The root is then the K-th largest. Memory is K, regardless of how many values have been seen.

```python
import heapq


# Rooms: sorted starts and ends with two pointers
```

```python
# Idea: Sort starts and ends separately. For each start:
# if it is at or after the earliest unfinished end, that
# room is reused (advance the end pointer); otherwise a
# new room is needed. The count of rooms opened is the
# maximum overlap.
def minMeetingRooms(intervals):
    starts, ends = sorted(start_time for start_time, _ in intervals), sorted(end_time for _, end_time in interv
als)
    rooms = next_end_index = 0
    for start_time in starts:
        if start_time >= ends[next_end_index]:
            next_end_index += 1
        else:
            rooms += 1
    return rooms


# Stream: min-heap of size K
# Idea: Push each value; if the heap has more than K items
# pop the smallest. The root is then the K-th largest.
# Memory is K, regardless of how many values have been
# seen.
class KthLargestStream:
    def __init__(self, top_count, nums=()):
        self.top_count, self.smallest_heap = top_count, []
        for value in nums:
            self.add(value)

    def add(self, value):
        heapq.heappush(self.smallest_heap, value)
        if len(self.smallest_heap) > self.top_count:
            heapq.heappop(self.smallest_heap)
        return self.smallest_heap[0] if len(self.smallest_heap) == self.top_count else None
```

COMPLEXITY | Part 1: Time O(n log n) | Space O(n)
Part 2: Time O(log k) per add | Space O(k)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 12

13 * Gate Entry / Exit Time Resolution with Tie-Break Rules

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: the same two-queue simulation with the tie-break rules from the statement

QUESTION INFO

QUESTION

A gate lets one person through per second. events[i] = (time, action) for person i, where action is 'enter' or 'exit'. When several people want the gate in the same second: if the gate was not used in the previous second, people who exit go first; if it was used, the direction used in the previous second goes first again. Within one direction, people go in order of arrival (person id). Return the second at which each person passes.

Function signature

def gate_pass_times(events: list) -> list

EXAMPLES

QUESTION

Example 1	
Input	`[(0, 'enter'), (0, 'exit'), (1, 'enter'), (5, 'enter')]`
Output	`[1, 0, 2, 5]`
Why	At second 0 the gate was idle, so the exit (person 1) goes first. Person 0 follows at 1, person 2 (arrived at 1) at 2, and person 3 arrives alone at 5.

Example 2	
Input	`[(0, 'enter'), (1, 'enter'), (1, 'exit')]`
Output	`[0, 1, 2]`
Why	The gate was used for entering in the previous second, so the waiting enter keeps priority over the exit.

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Simulate every second

BRUTE-FORCE IDEA

Advance time one second at a time, add arrivals to two queues, and pick a queue by the previous-state rule. Easy to reason about for the event-driven version, but it spends time on every empty second.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Simulate every second
Advance time one second at a time
add arrivals to two queues
pick a queue by the previous-state rule
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(T + n) where T is the last timestamp | Space O(n)

WHY THIS IS WEAKER | It runs through every second between the first and last timestamp, so a timestamp of 10^9 means a billion iterations. Jumping to the next arrival when the queues are empty makes it O(n log n).

HOW TO IMPROVE IT

the same two-queue simulation with the tie-break rules from the statement

SHORT BEST SOLUTION

Sort people by (arrival time, id) and keep one FIFO queue for waiting enters and one for exits. At each second pick from the queues using the previous-second state. When both queues are empty jump straight to the next arrival; if that jump leaves a gap of at least one idle second the previous state resets to 'none', which is the subtle part of the optimization.

```python
from collections import deque


# BASE QUESTION (optimal): Event-driven simulation with
# two queues
# Idea: Sort people by (arrival time, id) and keep one
# FIFO queue for waiting enters and one for exits. At each
# second pick from the queues using the previous-second
# state. When both queues are empty jump straight to the
# next arrival; if that jump leaves a gap of at least one
# idle second the previous state resets to 'none', which
# is the subtle part of the optimization.
def gate_pass_times(events):
    num_people = len(events)
    people_by_arrival = sorted(range(num_people), key=lambda person: (events[person][0], person))
    enter_queue, exit_queue = deque(), deque()
```

```
    pass_time = [0] * num_people
    time, next_arrival, last_direction, last_busy_second = 0, 0, None, -10
    while next_arrival < num_people or enter_queue or exit_queue:
        if not enter_queue and not exit_queue:
            time = max(time, events[people_by_arrival[next_arrival]][0])
            if time > last_busy_second + 1:
                last_direction = None
        while next_arrival < num_people and events[people_by_arrival[next_arrival]][0] <= time:
            arrived_person = people_by_arrival[next_arrival]
            (enter_queue if events[arrived_person][1] == "enter" else exit_queue).append(arrived_person)
            next_arrival += 1
        if enter_queue and exit_queue:
            queue = enter_queue if last_direction == "enter" else exit_queue
        else:
            queue = enter_queue or exit_queue
        person = queue.popleft()
        pass_time[person] = time
        last_direction = events[person][1]
        last_busy_second = time
        time += 1
    return pass_time
```

COMPLEXITY | Time O(n log n) for the sort, O(n) simulation | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 13

14 * Compute servers needed for daily recurring jobs * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: merge each job's own intervals first so one job counts once, then sweep all of them together

QUESTION INFO

QUESTION

You operate a cluster of identical servers that run recurring **daily** jobs. For a single day, you are given a list of job execution intervals. Each interval has:

- **`job_id`** (string or integer),
- **`start_time`** (timestamp within that day),
- **`end_time`** (timestamp within that day), with **`start_time < end_time`**. Interpret each interval as a half-open time range **`[start_time, end_time)`** during which that job must be running on some server **on that day**. Rules and constraints:
- Each physical server can execute at most **one job at a time**.
- However, if multiple intervals in the input correspond to the **same `job_id`**, they may overlap in time **on the same server** (they are considered parts or retries of the same job and do not consume extra capacity).
- If a job's true execution would extend past midnight into the next day, it is conceptually **truncated at the end of the current day** for scheduling purposes; for each day you only see the portion that falls within that day's 24-hour window.

Tasks

1. Given all job intervals for a single day (potentially with repeated **`job_id`**s and overlapping intervals), compute the **minimum number of servers** required to schedule all the jobs under the above rules.
2. Describe the time and space complexity of your algorithm in terms of the number of intervals (m).

Follow-up

QUESTION

Some rare jobs are extremely long-running and conceptually span **multiple days** without restart. For such a multi-day job, you decide to reserve a **dedicated server** that is not shared with any other jobs, but which may still handle different intervals of that same **`job_id`**. Explain how you would extend or adapt your solution to:

- Detect which jobs require a dedicated server based on their multi-day duration or configuration.
- Account for these dedicated servers when computing the total number of servers needed for the system.

Part 1: Minimum Servers for One Day with Overlapping Same-Job IntervalsYou are given all job execution intervals for a single day. Each interval is half-open, meaning [start_time, end_time), and represents a time range during which a job must be running. A server can run at most one job at a time. However, if multiple intervals have the same job_id, they are considered parts or retries of the same job, so overlapping intervals with the same job_id consume only one server total during the overlap. Return the minimum number of servers required to run all intervals for the day.
Constraints
0 <= len(intervals) <= 2000000 <= job_id <= 10^90 <= start_time < end_time <= 1440Intervals are half-open: an interval ending at time t does not overlap an interval starting at time t.Multiple intervals may have the same job_id.

EXAMPLES

```
Input: ([],)
Expected Output: 0
```

Explanation: There are no jobs, so no servers are needed.

```
Input: ([[5, 100, 200], [5, 150, 250], [5, 240, 300]],)
Expected Output: 1
```

Explanation: All intervals belong to the same job_id. Even though they overlap, they can share one server.

Hints

At any moment, one server is needed for each distinct active job_id, not for each active interval.Try merging all intervals belonging to the same job_id first, then run a sweep-line algorithm over the merged intervals.
Part 2: Total Servers with Dedicated Multi-Day JobsYou are given true job execution intervals that may span multiple days. Some jobs require dedicated servers. A job is dedicated if its job_id is listed in configured_dedicated, or if any one of its true intervals has duration at least min_dedicated_duration minutes. A dedicated job uses one reserved server for that job_id and does not share that server with other jobs, though multiple intervals of the same dedicated job_id may use the same dedicated server. For a given target_day, compute the total number of

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | For every time unit count the distinct jobs running

BRUTE-FORCE IDEA

For each integer time t collect the set of job ids with an interval covering t and take the largest set size. Correct; it depends on the length of the day, not only on m.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: For every time unit count the distinct jobs running
For each integer time t collect the set of job ids with an interval covering t
take the largest set size
```

BRUTE-FORCE COMPLEXITY | Time O(T * m) | Space O(m)

WHY THIS IS WEAKER | Depends on the length of the day, O(T * m). Merge each job's own intervals first, then sort start and end events and keep a running count, O(m log m).

HOW TO IMPROVE IT

merge each job's own intervals first so one job counts once, then sweep all of them together

SHORT BEST SOLUTION

Group the intervals by job_id, sort and merge each group's overlapping or touching intervals, then collect all merged intervals. Sort the +1/-1 events (ends before starts at equal times for half-open intervals) and take the maximum running count.

```
from collections import defaultdict


# BASE QUESTION (optimal): Merge each job's intervals,
# then sweep start and end events
# Idea: Group the intervals by job_id, sort and merge each
# group's overlapping or touching intervals, then collect
# all merged intervals. Sort the +1/-1 events (ends before
# starts at equal times for half-open intervals) and take
# the maximum running count.
def servers_needed(intervals):
    by_job = defaultdict(list)
    for job_id, start, end in intervals:
        by_job[job_id].append((start, end))
    events = []
    for job_intervals in by_job.values():
        merged_start, merged_end = None, None
        for start, end in sorted(job_intervals):
            if merged_end is not None and start <= merged_end:
                merged_end = max(merged_end, end)
            else:
                if merged_end is not None:
                    events += [(merged_start, 1), (merged_end, -1)]
                merged_start, merged_end = start, end
        events += [(merged_start, 1), (merged_end, -1)]
    running_count = most_servers = 0
    for _, change in sorted(events):
        running_count += change
        most_servers = max(most_servers, running_count)
    return most_servers
```

COMPLEXITY | Time O(m log m) | Space O(m)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 14

15 * Meeting Rooms III Variant * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: split meetings that cross midnight into two pieces first, then the same sweep

QUESTION INFO

QUESTION

Given an array of meeting time intervals where each interval is [start, end], determine the minimum number of conference rooms required to hold all meetings without overlap. Variant: times are cyclic within a 24-hour period (meetings can wrap around midnight).

EXAMPLES

Example 1	
Input	`intervals = [[0, 30], [5, 10], [15, 20]]`
Output	`2`
Why	`Meeting [0,30] overlaps with [5,10] and [15,20], so we need 2 rooms.`

QUESTION

Example 2	
Input	`intervals = [[7, 10], [2, 4]]`
Output	`1`
Why	`No overlap, 1 room is sufficient.`

Example 3	
Input	`intervals = [[22, 2], [23, 1], [0, 3]]`
Output	`3`
Why	`All three meetings overlap around midnight.`

CONSTRAINTS

- 1 <= intervals.length <= 10^4
- 0 <= start < end <= 24 * 60 (minutes in a day)
- Meetings may wrap around (e.g., [23:00, 01:00])

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= intervals.length <= 10^4
- 0 <= start < end <= 24 * 60 (minutes in a day)
- Meetings may wrap around (e.g., [23:00, 01:00])

BRUTE FORCE | Exact minimum with one room per meeting, by backtracking

BRUTE-FORCE IDEA

Treat each meeting as the set of minute slots it occupies (a wrapped meeting is two ranges of slots) and try to colour the meetings with k rooms for k = 1, 2, and so on; two meetings conflict when their slot sets intersect. It is exact for 'a meeting keeps its room' on small cases, but it is exponential.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Exact minimum with one room per meeting, by backtracking
Treat each meeting as the set of minute slots it occupies (a wrapped meeting is two ranges
of slots)
try to colour the meetings with k rooms for k = 1
so on
two meetings conflict when their slot sets intersect
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(n)

WHY THIS IS WEAKER | Backtracking over room assignments is exponential. Splitting wrapped meetings and sweeping start and end events gives the maximum overlap in O(n log n); that equals the exact answer when meetings do not wrap.

HOW TO IMPROVE IT

split meetings that cross midnight into two pieces first, then the same sweep

SHORT BEST SOLUTION

Turn every meeting into one or two ordinary intervals, create +1 events at starts and -1 events at ends, sort them (ends before starts at the same minute, because intervals are half-open), and take the maximum running count. This equals the minimum number of rooms if wrapped meetings may switch rooms at midnight, and it is always a lower bound on the exact answer.

```python
DAY = 24


def _pieces(meetings, day):
    for start, end in meetings:
```

```
        if end > start:
            yield start, end
        elif end < start:
            yield start, day
            yield 0, end


# BASE QUESTION (optimal): Split wrapped meetings and
# sweep (maximum overlap)
# Idea: Turn every meeting into one or two ordinary
# intervals, create +1 events at starts and -1 events at
# ends, sort them (ends before starts at the same minute,
# because intervals are half-open), and take the maximum
# running count. This equals the minimum number of rooms
# if wrapped meetings may switch rooms at midnight, and it
# is always a lower bound on the exact answer.
def min_rooms_sweep(meetings, day=DAY):
    events = []
    for start, end in _pieces(meetings, day):
        events.append((start, 1))
        events.append((end, -1))
    events.sort()
    running_count = most_rooms = 0
    for _, change in events:
        running_count += change
        most_rooms = max(most_rooms, running_count)
    return most_rooms
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 15

GROUP 3 OF 43 | 4 QUESTIONS | PART A - PART 1

Tree recursion, Tier 1: one function that visits a node, asks its children, and reports upward

PATTERN CONTEXT

Write dfs(node, depth). An empty node returns a base value, a real node calls dfs on both children, updates the best answer so far, and returns one number or flag to its parent. Only three lines change: the base value, the update, and what you return.

22 * Expression Tree Single-Leaf Mutation * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: post-order evaluation (children first, then apply the operator); to test one leaf flip, re-evaluate with that leaf changed

QUESTION INFO

QUESTION

A boolean expression tree has leaves holding True or False and internal nodes AND, OR, XOR (two children) or NOT (one child). For each leaf in turn, imagine flipping only that leaf and report the value of the whole tree after the flip (every flip is independent of the others). Return the leaves and the list of answers, in leaf order.

Function signature

def flip_answers(root) -> (list, list)

(the leaves, the root value

after flipping each one)

EXAMPLES

Example 1	

Input	`t = Node('AND', Node(val=True), Node('OR', Node(val=False), Node(val=True)))`
	`evaluate(t)`
	`[leaf.val for leaf in flip_answers(t)[0]]`
	`flip_answers(t)[1]`

Output	`evaluate(t) -> true`
	`[leaf.val for leaf in flip_answers(t)[0]] -> [true, false, true]`
	`flip_answers(t)[1] -> [false, true, false]`

Why	`The tree is True AND (False OR True) = True. Flipping the first leaf gives False, flipping the False leaf changes nothing (the OR is still True), flipping the last leaf makes the OR False.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Flip, re-evaluate the whole tree, flip back

BRUTE-FORCE IDEA

For each of the L leaves, toggle it, evaluate the entire tree, and toggle it back. It is the correct direct approach for the cached version.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Flip, re-evaluate the whole tree, flip back
For each of the L leaves
toggle it
evaluate the entire tree
toggle it back
```

```
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(h)

WHY THIS IS WEAKER | L leaves times an O(n) evaluation: O(n^2). Caching every node's value and recomputing only the path to the root (stopping when a value does not change) costs O(depth) per leaf.

HOW TO IMPROVE IT

post-order evaluation (children first, then apply the operator); to test one leaf flip, re-evaluate with that leaf changed

SHORT BEST SOLUTION

One traversal stores every node's value and its parent. To flip a leaf, toggle it, then walk up recomputing each ancestor from its children's cached values, stopping as soon as a node's value does not change (then the root is unchanged too). Undo by restoring the saved values on the walked path.

```python
# BASE QUESTION (optimal): Cache each value and parent
# Idea: Build the cache with DFS. For each leaf flip, only
# recompute its path to the root, then restore that path.
def apply(op, left, right=None):
    if op == "AND":
        return left and right
    if op == "OR":
        return left or right
    if op == "XOR":
        return left != right
    return not left


def flip_answers(root):
    parent = {root: None}
    value = {}
    leaves = []

    def build(node):
        if node.op is None:
            value[node] = node.val
            leaves.append(node)
            return
        parent[node.left] = node
        build(node.left)
        right_value = None
        if node.right is not None:
            parent[node.right] = node
            build(node.right)
            right_value = value[node.right]
        value[node] = apply(node.op, value[node.left], right_value)

    build(root)
    answers = []
    for leaf in leaves:
        changed = []
        value[leaf] = not value[leaf]
        changed.append(leaf)
        node = parent[leaf]
        while node is not None:
            right_value = value[node.right] if node.right is not None else None
            new_value = apply(node.op, value[node.left], right_value)
            if new_value == value[node]:
                break
            value[node] = new_value
            changed.append(node)
            node = parent[node]
        answers.append(value[root])
        for node in changed:
            value[node] = not value[node] if node.op is None else apply(
                node.op, value[node.left],
                value[node.right] if node.right is not None else None,
            )
```

```
    return answers
```

COMPLEXITY | Time O(n + L * depth) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 22

23 * Longest Consecutive Path in Binary Tree * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: pass the parent's value and the run length down; extend the run when value == parent + 1, else restart at 1

QUESTION INFO

QUESTION

Given a binary tree, return the length of the longest path that goes from a parent down to a child where every value is exactly 1 more than the previous one (counted in nodes).

Function signature

def longestConsecutive(root: TreeNode) -> int

EXAMPLES

Example 1	
Input	`[1, null, 3, 2, 4, null, null, null, 5]`
Output	`3`
Why	`3 -> 4 -> 5 is the longest increasing-by-one downward path.`

Example 2	
Input	`[2, null, 3, 2, null, 1]`
Output	`2`
Why	`2 -> 3 is the longest; 3 -> 2 and 2 -> 1 go the wrong way.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Start a downward walk from every node

BRUTE-FORCE IDEA

For each node, walk down both children while the next value is exactly one more, tracking the longest run. Correct for the base question; nodes deep in a chain are re-walked from every ancestor.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Start a downward walk from every node
For each node
walk down both children while the next value is exactly one more
tracking the longest run
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n * h) | Space O(h)

WHY THIS IS WEAKER | A node deep in a chain is walked again from every ancestor, O(n * h). Passing the current run length down in one traversal makes it O(n).

HOW TO IMPROVE IT

pass the parent's value and the run length down; extend the run when value == parent + 1, else restart at 1

SHORT BEST SOLUTION

Pass the length of the run ending at the parent. If the child is exactly parent + 1 the run grows, otherwise it restarts at 1. Track the global best. Length is counted in nodes; state that choice out loud.

```python
# BASE QUESTION (optimal): Top-down DFS carrying the
# current run
# Idea: Pass the length of the run ending at the parent.
# If the child is exactly parent + 1 the run grows,
# otherwise it restarts at 1. Track the global best.
# Length is counted in nodes; state that choice out loud.
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def longestConsecutive(root):
    best = 0

    def dfs(node, parent_val, run):
        nonlocal best
        if node is None:
            return
        run = run + 1 if parent_val is not None and node.val == parent_val + 1 else 1
        best = max(best, run)
        dfs(node.left, node.val, run)
        dfs(node.right, node.val, run)

    dfs(root, None, 0)
    return best
```

COMPLEXITY | Time O(n) | Space O(h)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 23

24 * Implement zigzag level-order traversal * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: collect values by depth; reverse every odd level (or fill each level from the other end)

QUESTION INFO

QUESTION

Given the root of a binary tree, return its values level by level, reading the first level from left to right, the next from right to left, and so on, alternating. The result is a list of lists, one per level. Aim for O(n) time.

Function signature

def zigzag_level_order(root: TreeNode) -> list

EXAMPLES

Example 1	
Input	`[3, 9, 20, null, null, 15, 7]`
Output	`[[3], [20, 9], [15, 7]]`
Why	`Level 0 is [3], level 1 is read right to left [20, 9], level 2 left to right [15, 7].`

Example 2	
Input	`[1]`
Output	`[[1]]`
Why	`A single node.`

QUESTION

Example 3

Input	`[]`
Output	`[]`
Why	`An empty tree has no levels.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Level order, then reverse every second level

BRUTE-FORCE IDEA

Do a normal breadth-first traversal collecting each level, and reverse the levels with odd index at the end. Easy to get right, with one extra pass over those levels.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Level order, then reverse every second level
Do a normal breadth-first traversal collecting each level
reverse the levels with odd index at the end
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | Same O(n) time, plus an extra pass copying the reversed levels. Filling each level from the left or right end of a deque as it is built needs no reversal.

HOW TO IMPROVE IT

collect values by depth; reverse every odd level (or fill each level from the other end)

SHORT BEST SOLUTION

Keep a left-to-right flag. For each level build a deque; append on the right when going left to right and on the left when going right to left. Flip the flag after each level.

```python
from collections import deque


# BASE QUESTION (optimal): BFS filling each level from the
# left or the right
# Idea: Keep a left-to-right flag. For each level build a
# deque; append on the right when going left to right and
# on the left when going right to left. Flip the flag
# after each level.
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def zigzagLevelOrder(root):
    levels, queue, left_to_right = [], deque([root] if root else []), True
    while queue:
        level = deque()
        for _ in range(len(queue)):
            node = queue.popleft()
            level.append(node.val) if left_to_right else level.appendleft(node.val)
            queue.extend(child for child in (node.left, node.right) if child)
        levels.append(list(level))
        left_to_right = not left_to_right
    return levels
```

COMPLEXITY | Time O(n) | Space O(width)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 24

25 * Given the root of a binary tree, return the length of the diameter of the tree. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: best = max(best, left + right); return 1 + max(left, right) (the diameter)

QUESTION INFO

QUESTION

Given the root of a binary tree, create a function diameterOfTree that determines and returns the length of the diameter of the tree, where the diameter of a tree is defined as the number of nodes along the longest path between any two nodes in the tree. For example, given the following binary tree: // tree 1 / \ 2 3 / \ 4 5 The diameter is 4, which is the length of the path [4, 2, 1, 3] or [5, 2, 1, 3].Watch Ali, SDE @ Amazon, answer the question: "Given the root of a binary tree, return the length of the diameter of the tree."No answers contributed by the community yet.Interview experiences6 shared‹›Uber · 3 months ago"interview process is very lengthy took around 5 weeks, and expectations is very high in each round, for LLD working code is required, also dont mess your HM round since i felt like it contained more weightage"Software Engineer · Mid Level / L4Meta · 4 months ago"The interview went smoothly overall. 1 round of phone screen followed by the full loop of 4 rounds packed in a day. 2 coding 1 behavior 1 system design."Software Engineer · Senior / L5Google · 9 months ago"I interviewed for an AI Engineer internship, but the process was mostly DSA-heavy. I had an online coding round first, then a technical round with binary tree and DP questions, then another harder DSA round with DP and linked list questions, and finally an HR round. The online round covered arrays and strings, and the later rounds got progressively more difficult. The only non-DSA part was the last

EXAMPLES

Example 1	
Input	`[1, 2, 3, 4, 5]`
Output	`4`
Why	`The longest path 4 - 2 - 1 - 3 has 4 nodes.`

Example 2	
Input	`[1]`
Output	`1`
Why	`A single node.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | At every node compute both heights from scratch

BRUTE-FORCE IDEA

For each node call a height function on its two children and take the maximum of left + right over all nodes. Correct; the same subtrees have their heights recomputed for every ancestor.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: At every node compute both heights from scratch
For each node call a height function on its two children
take the maximum of left + right over all nodes
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) worst case | Space O(h)

WHY THIS IS WEAKER | Heights of the same subtrees are recomputed for every ancestor, O(n^2) on a skewed tree. One post-order pass returns heights and updates the best on the way, O(n).

HOW TO IMPROVE IT

best = max(best, left + right); return 1 + max(left, right) (the diameter)

SHORT BEST SOLUTION

depth(node) returns the height of the node's subtree and, on the way, updates best = max(best, left + right). Every node is visited once, and the answer is best when the traversal ends (edges, not nodes).

```
# BASE QUESTION (optimal): One post-order pass returning
# heights and updating the best
# Idea: depth(node) returns the height of the node's
# subtree and, on the way, updates best = max(best, left +
# right). Every node is visited once, and the answer is
# best when the traversal ends (edges, not nodes).
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def diameterOfBinaryTree(root):
    best = 0

    def depth(node):
        nonlocal best
        if node is None:
            return 0
        left_height, right_height = depth(node.left), depth(node.right)
        best = max(best, left_height + right_height)
        return 1 + max(left_height, right_height)

    depth(root)
    return best
```

COMPLEXITY | Time O(n) | Space O(h)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 25

GROUP 4 OF 43 | 8 QUESTIONS | PART A - PART 1

A heap of what is available: pop the best, put back the changed one

PATTERN CONTEXT

The heap holds the things you can choose from right now: machines and when they are free, rooms, lockers, the biggest remaining count, the two smallest weights. Each step pops the best choice, uses it, and pushes it back changed (or a combined item).

28 * Streaming Median (LC 295) with Follow-ups * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: two heaps: a max-heap for the lower half and a min-heap for the upper half; keep them balanced

QUESTION INFO

QUESTION

Design a structure that receives numbers one at a time and can report the median of everything received so far at any moment. addNum(x) adds a number; findMedian() returns the middle value, or the mean of the two middle values when the count is even.

Function signature

class MedianFinder:
    def addNum(self, x: float) -> None
    def findMedian(self) -> float

EXAMPLES

Example 1

Input	`m = MedianFinder()`
	`m.addNum(5)`
	`m.addNum(2)`
	`m.findMedian()`
	`m.addNum(8)`
	`m.findMedian()`
	`m.addNum(1)`
	`m.findMedian()`

Output	`m.findMedian() -> 3.5`
	`m.findMedian() -> 5`
	`m.findMedian() -> 3.5`

Why	`Sorted so far: [2, 5] -> 3.5; [2, 5, 8] -> 5; [1, 2, 5, 8] -> 3.5.`

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Keep a sorted list and insert with bisect

BRUTE-FORCE IDEA

Insert each value into a sorted list with bisect.insort and read the middle. Very simple; every insert shifts up to n elements, though in practice that is a fast memmove.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Keep a sorted list and insert with bisect
Insert each value into a sorted list with bisect.insort
read the middle
```

BRUTE-FORCE COMPLEXITY | Time O(n) per insert (list shift), O(1) median | Space O(n)

WHY THIS IS WEAKER | Each insert shifts up to n elements, O(n) per add (fast in practice, but linear). Two heaps keep the median available with O(log n) per add and O(1) per query.

HOW TO IMPROVE IT

two heaps: a max-heap for the lower half and a min-heap for the upper half; keep them balanced

SHORT BEST SOLUTION

Part 1: Lower half in a max-heap, upper half in a min-heap, sizes differing by at most one and every lower value <= every upper value. The median is the larger heap's root, or the average of both roots.

Part 2: If values are integers in [0, V), keep a count per value. Adding is O(1); the median scans the counts once (or maintains a moving pointer to make it amortized O(1)).

Part 3: A Fenwick (binary indexed) tree stores counts per value. The k-th smallest is found by descending the tree with binary lifting, so any rank, not just the median, costs O(log V). Compress coordinates if V is huge.

```
import heapq


# BASE QUESTION (optimal): Two heaps
# Idea: Lower half in a max-heap, upper half in a min-
# heap, sizes differing by at most one and every lower
# value <= every upper value. The median is the larger
# heap's root, or the average of both roots.
class MedianFinder:
    def __init__(self):
        self.low = []        # max-heap via negation
        self.high = []       # min-heap

    def addNum(self, number):
        heapq.heappush(self.low, -number)
        heapq.heappush(self.high, -heapq.heappop(self.low))
        if len(self.high) > len(self.low):
            heapq.heappush(self.low, -heapq.heappop(self.high))

    def findMedian(self):
        if len(self.low) > len(self.high):
            return float(-self.low[0])
        return (-self.low[0] + self.high[0]) / 2
```

COMPLEXITY | Part 1: Time O(log n) add, O(1) median | Space O(n)
Part 2: Time O(1) add, O(V) median | Space O(V)
Part 3: Time O(log V) add and query | Space O(V)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 28

29 * Compute minimum number of rooms needed * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: heap of end times; a meeting reuses the room that frees first if it ended, otherwise push a new room

QUESTION INFO

QUESTION

Problem

You are given a list of meetings, each with a start time and end time. A single room can host only one meeting at a time. Two meetings **overlap** if one starts before the other ends (treat meetings as half-open intervals **[start, end)** so a meeting ending at time `t` does not conflict with another starting at time `t`).

Task

Return the **minimum number of rooms** required to host all meetings.

QUESTION

Input

- **`intervals`**: an array of **`n`** pairs **`[start, end]`** where **`0 <= start <= end`**.

Output

- An integer: the minimum number of rooms needed.

Constraints (typical interview bounds)

- **`1 <= n <= 200000`**
- Times are integers in a reasonable range (e.g., **`0 . 10^9`**).

Example

- Input: **`[[0,30],[5,10],[15,20]]`**
- Output: **`2`** (One room can host **`[0,30]`**, the other can host **`[5,10]`** then **`[15,20]`**.)

You are given a list of meetings, where each meeting is represented as [start, end]. A single room can host only one meeting at a time. Meetings are treated as half-open intervals [start, end), so a meeting ending at time t does not conflict with another meeting starting at time t. Return the minimum number of rooms required to host all meetings. If start == end, the meeting occupies no time and does not require a room.
Constraints
0 <= n <= 2000000 <= start <= end <= 10^9

EXAMPLES

```
Input: ([[0,30],[5,10],[15,20]],)
Expected Output: 2
Explanation: [0,30] overlaps with both [5,10] and [15,20], so at most 2 rooms are needed at the same
    time.
Input: ([[7,10],[2,4],[4,7],[10,12]],)
Expected Output: 1
```

Explanation: These meetings only touch at endpoints, so one room can host all of them in sequence.

Hints

Try turning each meeting into two events: one when it starts and one when it ends.Because intervals are half-open, if a meeting ends at the same time another starts, process the ending event first.

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Count the overlap at every time unit

BRUTE-FORCE IDEA

For each integer time count the meetings with start <= t < end and take the maximum. Correct (integer times); it depends on the length of the time range, not only on n.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Count the overlap at every time unit
For each integer time count the meetings with start <= t < end
take the maximum
```

BRUTE-FORCE COMPLEXITY | Time O(range * n) | Space O(1)

WHY THIS IS WEAKER | Depends on the length of the time range, O(range * n). A min-heap of room end times, or sorted starts and ends with two pointers, is O(n log n).

HOW TO IMPROVE IT

heap of end times; a meeting reuses the room that frees first if it ended, otherwise push a new room

SHORT BEST SOLUTION

Part 1: Process meetings by start time. If the earliest-ending room has finished by this meeting's start, reuse it (pop and push the new end); otherwise open a new room (push). The heap size at the end is the number of rooms. Zero-length meetings (start == end) occupy no time under half-open intervals, so they are ignored.

Part 2: Sort the starts and the ends separately. For each start, if it is at or after the earliest unfinished end, a room has been freed (move the end pointer); otherwise a new room is needed. No heap and no pairing of starts with ends is required.

```
import heapq


# BASE QUESTION (optimal): Sort by start with a min-heap
# of end times
# Idea: Process meetings by start time. If the earliest-
# ending room has finished by this meeting's start, reuse
# it (pop and push the new end); otherwise open a new room
# (push). The heap size at the end is the number of rooms.
# Zero-length meetings (start == end) occupy no time under
# half-open intervals, so they are ignored.
def min_meeting_rooms(intervals):
    ends = []
    for start, end in sorted(interval for interval in intervals if interval[0] < interval[1]):          # a zer
o-length meeting [t, t) occupies no time, so it needs no room
        if ends and ends[0] <= start:
            heapq.heapreplace(ends, end)
        else:
            heapq.heappush(ends, end)
    return len(ends)
```

COMPLEXITY | Part 1: Time O(n log n) | Space O(n)
Part 2: Time O(n log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 29

30 * Implement a Reusable Locker Allocation System * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: heap of freed locker numbers; take the smallest freed one before handing out a new number

QUESTION INFO

QUESTION

Problem

A locker bank has lockers numbered from 1 through **`n`**. Process operations in order. **`store(customer)`** assigns the smallest available locker and returns its number, or **`-1`** if that customer already has a locker or none is available. **`pickup(customer)`** releases and returns that customer's locker, or **`-1`** if no locker is assigned. **`last()`** returns the locker number from the most recent successful **`store`**, or **`-1`** if there has not been one. Return one result per operation.

Constraints & Assumptions

- **`1 <= n <= 200,000`**.
- There are at most 200,000 operations.
- Customer IDs are non-empty strings.
- A released locker becomes immediately available for reuse.

Clarifications

- Failed store and pickup operations do not change **`last()`**.
- The allocation rule is always the smallest currently free locker number.
- A customer can store again after a successful pickup.

Examples

QUESTION

```
n = 3 operations = [store("a"), store("b"), pickup("a"), store("c"), last()] output = [1, 2, 1, 1, 1]
```

Hints

```
Track free lockers Use an ordered structure that can remove and reinsert the minimum locker efficiently.
```

```
Track ownership A map from customer ID to locker supports validation and pickup.
```

Process a fresh locker bank with lockers numbered 1 through n. Operations are ["store", customer], ["pickup", customer], or ["last"]. Store assigns and returns the smallest available locker, or -1 if the customer already owns one or none is free. Pickup releases and returns that customer's locker, or -1 if none is assigned. Last reports the locker from the most recent successful store, or -1 before any success. Return one integer per operation.

Constraints

1 <= n <= 200000.operations contains at most 200000 encoded store, pickup, or last lists.Customer IDs are non-empty strings.A released locker becomes immediately available for reuse.Failed store and pickup operations do not change last.A customer can store again after a successful pickup.

EXAMPLES

```
Input: (3, [['store', 'a'], ['store', 'b'], ['pickup', 'a'], ['store', 'c'], ['last']])
Expected Output: [1, 2, 1, 1, 1]
```

Explanation: This is the source example; the released smallest locker is reused.

```
Input: (5, [])
Expected Output: []
```

Explanation: An empty batch produces no per-operation results.

Hints

Keep available locker numbers in a min-oriented structure.Use a customer-to-locker map to validate stores and pickups.

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan the lockers for the first free one

BRUTE-FORCE IDEA

Keep an array of owners and scan from 1 for the first empty slot on store. Correct, and each store can touch every locker.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan the lockers for the first free one
Keep an array of owners
scan from 1 for the first empty slot on store
```

BRUTE-FORCE COMPLEXITY | Time O(n) per operation | Space O(n)

WHY THIS IS WEAKER | Each store scans up to n lockers, O(n). A min-heap of freed numbers plus a counter for the next unused one makes each operation O(log n).

HOW TO IMPROVE IT

heap of freed locker numbers; take the smallest freed one before handing out a new number

SHORT BEST SOLUTION

store: if the customer already has a locker return -1; take the smallest released number from the heap if any, else the next never-used number if it is at most n, else -1. pickup: remove the customer's record and push the locker number on the heap. The heap never holds a number that is currently in use.

```python
import heapq


# BASE QUESTION (optimal): Min-heap of released numbers
# plus a next-unused counter
# Idea: store: if the customer already has a locker return
# -1; take the smallest released number from the heap if
# any, else the next never-used number if it is at most n,
# else -1. pickup: remove the customer's record and push
# the locker number on the heap. The heap never holds a
# number that is currently in use.
class Lockers:
    def __init__(self, locker_count):
        self.locker_count, self.next_new, self.released_lockers, self.locker_of = locker_count, 1, [], {}

    def store(self, customer):
        if customer in self.locker_of:
            return -1
        if self.released_lockers:
            locker_number = heapq.heappop(self.released_lockers)
        elif self.next_new <= self.locker_count:
            locker_number, self.next_new = self.next_new, self.next_new + 1
        else:
            return -1
        self.locker_of[customer] = locker_number
        return locker_number

    def pickup(self, customer):
        locker_number = self.locker_of.pop(customer, None)
        if locker_number is None:
            return -1
        heapq.heappush(self.released_lockers, locker_number)
        return locker_number
```

COMPLEXITY | Time O(log n) per operation | Space O(lockers released and not reused)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 30

31 * Simulate room assignments for scheduled meetings * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: a heap of free room numbers and a heap of (end time, room); a delayed meeting takes the earliest-ending room and shifts later

QUESTION INFO

QUESTION

There are n rooms numbered 0 to n - 1 and a list of meetings (start, end). Meetings are handled in order of start time. A meeting takes the lowest-numbered free room; if every room is busy it waits until the room that frees up first becomes free and then runs for its full original length, so it ends later. Return the room that hosted the most meetings (the lowest room number on a tie).

Function signature

def mostBookedRoom(n: int, meetings: list) -> int

EXAMPLES

Example 1	
Input	`2, [[0, 10], [1, 5], [2, 7], [3, 4]]`
Output	`0`
Why	`Rooms 0 and 1 each host two meetings; the lower number wins the tie.`

QUESTION

Example 2	
Input	`3, [[1, 20], [2, 10], [3, 5], [4, 9], [6, 8]]`
Output	`1`
Why	`With three rooms, room 1 hosts two of the five meetings and no room hosts more.`

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan all rooms for every meeting

BRUTE-FORCE IDEA

Keep the end time of every room in an array. For each meeting in start order, take the lowest-index room that has finished by the start; if none, take the room with the earliest end (lowest index on ties) and delay the meeting. Correct; each meeting scans all N rooms.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan all rooms for every meeting
Keep the end time of every room in an array
process every required item or candidate directly
```

BRUTE-FORCE COMPLEXITY | Time O(M * N) | Space O(N)

WHY THIS IS WEAKER | Every meeting scans all N rooms, O(M * N). Two heaps (free rooms and busy rooms by end time) make each meeting O(log N).

HOW TO IMPROVE IT

a heap of free room numbers and a heap of (end time, room); a delayed meeting takes the earliest-ending room and shifts later

SHORT BEST SOLUTION

Process meetings in start order. Move rooms that finished by the meeting's start into the free heap (lowest index first). If a room is free, use it. Otherwise delay the meeting until the earliest busy room finishes and keep its original length. Count meetings per room and return the busiest (lowest index on ties).

```python
import heapq


# BASE QUESTION (optimal): Two heaps: free rooms and busy
# rooms
# Idea: Process meetings in start order. Move rooms that
# finished by the meeting's start into the free heap
# (lowest index first). If a room is free, use it.
# Otherwise delay the meeting until the earliest busy room
# finishes and keep its original length. Count meetings
# per room and return the busiest (lowest index on ties).
def mostBookedRoom(room_count, meetings):
    free_rooms = list(range(room_count))
    heapq.heapify(free_rooms)
    busy_rooms = []                       # (end time, room)
    meetings_per_room = [0] * room_count
    for start, end in sorted(meetings):
        while busy_rooms and busy_rooms[0][0] <= start:
            heapq.heappush(free_rooms, heapq.heappop(busy_rooms)[1])
        if free_rooms:
            room = heapq.heappop(free_rooms)
            heapq.heappush(busy_rooms, (end, room))
        else:
            finish, room = heapq.heappop(busy_rooms)
            heapq.heappush(busy_rooms, (finish + end - start, room))
```

```
        meetings_per_room[room] += 1
    return meetings_per_room.index(max(meetings_per_room))
```

COMPLEXITY | Time O(M log M + M log N) | Space O(N)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 31

32 * Minimum Makespan on Identical CPUs * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the template as written: heap of the times the CPUs become free

QUESTION INFO

QUESTION

You have C identical CPUs and a list release_times , where release_times[i] is the earliest time job i may start. Every job runs non-preemptively for the same fixed duration S . When a job's release time arrives it may start on any free CPU; if all CPUs are busy it waits and starts on the first CPU that becomes free (never before its release time). Each CPU runs one job at a time. Scheduling optimally, compute the minimum possible makespan: the time at which the last job finishes.

Function signature

def min_makespan(release_times: list[int], num_cpus: int, duration: int) -> int:

...

Follow-up (as reported): (Reported) The naive approach tracked an O(n*C) table of CPU availability; reduce it to a min-heap of the C next-free times, popping the earliest-free CPU per job after sorting jobs by release time.

EXAMPLES

Example 1	
Input	`release_times = [0, 1, 2, 3], num_cpus = 2, duration = 3`
Output	`7`
Why	`CPU-A: job@0 -> [0,3), job@2 starts at 3 -> [3,6). CPU-B: job@1 -> [1,4), job@3 starts at 4 -> [4,7). Last finish = 7.`

Example 2	
Input	`release_times = [0, 0, 0], num_cpus = 1, duration = 2`
Output	`6`
Why	`One CPU runs them back to back: [0,2), [2,4), [4,6).`

Example 3	
Input	`release_times = [5], num_cpus = 3, duration = 4`
Output	`9`
Why	`The single job starts at its release time 5, finishes at 9.`

CONSTRAINTS

- 1 <= C <= 10^5
- 0 <= len(release_times) <= 10^5
- 0 <= release_times[i] <= 10^9
- 1 <= S <= 10^9
- If there are no jobs, the makespan is 0 .
- Target complexity: O(n log C) time, O(C) space.

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= C <= 10^5
- 0 <= len(release_times) <= 10^5
- 0 <= release_times[i] <= 10^9
- 1 <= S <= 10^9

BRUTE FORCE | Try every assignment of jobs to CPUs

BRUTE-FORCE IDEA

It enumerates all C^n ways to distribute the jobs; each CPU runs its jobs in release order.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every assignment of jobs to CPUs
It enumerates all C^n ways to distribute the jobs
each CPU runs its jobs in release order
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(C^n) | Space O(n)

WHY THIS IS WEAKER | It enumerates all C^n ways to distribute the jobs; each CPU runs its jobs in release order.

HOW TO IMPROVE IT

the template as written: heap of the times the CPUs become free

SHORT BEST SOLUTION

Each job takes the CPU that frees first, but never starts before its release.

```python
# BASE QUESTION (optimal): Sorted jobs and a heap of free
# times
# Idea: Each job takes the CPU that frees first, but never
# starts before its release.
import heapq


def min_makespan(release_times, num_cpus, duration):
    cpu_free_at = [0] * num_cpus
    makespan = 0
    for release_time in sorted(release_times):
        start = max(release_time, heapq.heappop(cpu_free_at))
        heapq.heappush(cpu_free_at, start + duration)
        makespan = max(makespan, start + duration)
    return makespan
```

COMPLEXITY | Time O(n log n) | Space O(C)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 32

33 * Design server allocation for multi-type nodes * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: a heap per type of (flexibility, server); skip servers already allocated (lazy deletion)

QUESTION INFO

QUESTION

Design a service that manages a pool of server nodes. Each node supports several instance types. getServer(count, types) hands out up to count servers that together cover all the requested types, preferring the least flexible servers (the ones that support the fewest types) so that versatile servers stay free, and returns None (allocating nothing) if the request cannot be met. releaseServer(ids) returns servers to the pool.

Function signature

class Servers:

def __init__(self, servers: dict)

id -> set of instance types

def getServer(self, count: int, types: list) -> list | None

def releaseServer(self, ids: list) -> None

EXAMPLES

Example 1

Input	`s = Servers({1: {'a'}, 2: {'a', 'b'}, 3: {'b'}, 4: {'a', 'b', 'c'}})`
	`s.getServer(2, ['a', 'b'])`
	`s.getServer(2, ['a', 'b'])`
	`s.getServer(1, ['a'])`
	`s.releaseServer([1])`
	`s.getServer(1, ['a'])`

Output	`s.getServer(2, ['a', 'b']) -> [1, 3]`
	`s.getServer(2, ['a', 'b']) -> [2, 4]`
	`s.getServer(1, ['a']) -> null`
	`s.getServer(1, ['a']) -> [1]`

Why	`The first request takes the two least flexible servers (1 for a, 3 for b); the second takes 2 and 4; nothing is left for the third; after server 1 is released it can serve type a again.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan every free server on each request

BRUTE-FORCE IDEA

For each requested type find the best free server by scanning all servers, then fill the remaining count the same way. Simple; every request touches every server.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan every free server on each request
For each requested type find the best free server by scanning all servers
fill the remaining count the same way
```

BRUTE-FORCE COMPLEXITY | Time O(N * T) per request (N servers) | Space O(N)

WHY THIS IS WEAKER | Every request scans all N servers, O(N * types). Per-type heaps ordered by flexibility with lazy deletion make a request O((count + types) log N).

HOW TO IMPROVE IT

a heap per type of (flexibility, server); skip servers already allocated (lazy deletion)

SHORT BEST SOLUTION

For each type keep a heap of (number of supported types, id) for the servers that support it. A request first covers each requested type (taking the best server not already chosen), then fills the rest from the best heap tops among the requested types. Allocating a server sets its allocated flag, so stale entries in other types' heaps are discarded when they reach the top. If the request cannot be met, every claim is rolled back.

```python
from collections import defaultdict
import heapq


# BASE QUESTION (optimal): Per-type heaps ordered by
# flexibility, with lazy deletion
# Idea: For each type keep a heap of (number of supported
# types, id) for the servers that support it. A request
# first covers each requested type (taking the best server
# not already chosen), then fills the rest from the best
# heap tops among the requested types. Allocating a server
# sets its allocated flag, so stale entries in other
# types' heaps are discarded when they reach the top. If
# the request cannot be met, every claim is rolled back.
class Servers:
    def __init__(self, servers):
        self.types_of_server = {server_id: frozenset(server_types) for server_id, server_types in servers.items
()}
        self.allocated = set()
        self.heaps = defaultdict(list)
        for server_id, supported_types in self.types_of_server.items():
            for server_type in supported_types:
                heapq.heappush(self.heaps[server_type], (len(supported_types), server_id))

    def _top(self, server_type, skipped_ids):
        heap, skipped_entries = self.heaps[server_type], []
        while heap:
            flexibility, server_id = heap[0]
            if server_id in self.allocated:
                heapq.heappop(heap)
            elif server_id in skipped_ids:
                skipped_entries.append(heapq.heappop(heap))
            else:
                break
```

```python
        result = heap[0] if heap else None           # read the best valid entry before putting the skipped on
es back
        for item in skipped_entries:
            heapq.heappush(heap, item)
        return result

    def getServer(self, count, types):
        chosen = []
        for server_type in types:
            if any(server_type in self.types_of_server[chosen_id] for chosen_id in chosen):
                continue
            top = self._top(server_type, set(chosen))
            if top is None:
                return None
            chosen.append(top[1])
        while len(chosen) < count:
            best = None
            for server_type in types:
                top = self._top(server_type, set(chosen))
                if top and (best is None or top < best):
                    best = top
            if best is None:
                return None
            chosen.append(best[1])
        if len(chosen) > count:
            return None
        self.allocated |= set(chosen)
        return chosen

    def releaseServer(self, ids):
        for server_id in ids:
            if server_id in self.allocated:
                self.allocated.discard(server_id)
```

```
            for server_type in self.types_of_server[server_id]:
                heapq.heappush(self.heaps[server_type], (len(self.types_of_server[server_id]), server_id))
```

COMPLEXITY | Time O((count + types) log N) amortized per request | Space O(total type memberships)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 33

34 * Trip Expense Settlement * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: a heap of creditors and a heap of debtors; the biggest of each settles min(amount) and the remainder goes back

QUESTION INFO

QUESTION

A group of friends go on a trip and each friend may pay for others. Every record gives the payer, the amount and the list of people who share it equally (the payer may be one of them). Compute a small set of payments that settles everyone's debts: each payment is {payer, amount, payee}.

Function signature

def settle_greedy(records: list) -> list

EXAMPLES

Example 1	
Input	`trip = [{'payer': 'alice', 'amount': 4000, 'payees': ['bob', 'jess', 'alice', 'sam']}, {'payer': 'jess', 'amount': 2000, 'payees': ['jess', 'alice']}]` `settle_greedy(trip)`
Output	`settle_greedy(trip) -> [{'payer': 'bob', 'amount': 1000, 'payee': 'alice'}, {'payer': 'sam', 'amount': 1000, 'payee': 'alice'}]`
Why	`Alice paid 4000 shared four ways and owes 1000 of Jess's 2000; Bob and Sam each owe 1000, and Jess is even. Two payments settle everything.`

Example 2	
Input	`trip = [{'payer': 'alice', 'amount': 3000, 'payees': ['alice', 'bob', 'cy']}]` `settle_greedy(trip)`
Output	`settle_greedy(trip) -> [{'payer': 'bob', 'amount': 1000, 'payee': 'alice'}, {'payer': 'cy', 'amount': 1000, 'payee': 'alice'}]`
Why	`Bob and Cy each owe Alice their 1000 share.`

CONSTRAINTS

- 1 <= payments.length <= 1000
- 1 <= amount <= 10^6
- 1 <= payees.length <= 100

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= payments.length <= 1000
- 1 <= amount <= 10^6
- 1 <= payees.length <= 100

BRUTE FORCE | Backtracking over who pays whom (exact minimum)

BRUTE-FORCE IDEA

Take the first person with a nonzero balance and try settling them with every later person of the opposite sign, recursing on the rest; keep the fewest payments. This gives the exact minimum, but the search explodes with the number of people.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Backtracking over who pays whom (exact minimum)
Take the first person with a nonzero balance
try settling them with every later person of the opposite sign
recursing on the rest
keep the fewest payments
```

BRUTE-FORCE COMPLEXITY | Time O(n!) worst case | Space O(n)

WHY THIS IS WEAKER | It tries every pairing, up to n! In the worst case. Matching the largest creditor with the largest debtor (heaps) is O(n log n) and uses at most n - 1 payments, though not always the minimum.

HOW TO IMPROVE IT

a heap of creditors and a heap of debtors; the biggest of each settles min(amount) and the remainder goes back

SHORT BEST SOLUTION

Part 1: Push creditors (positive net) in a max-heap and debtors (negative net) in another. Repeatedly pay min(debtors, credit) from the biggest debtor to the biggest creditor, and push back whoever still has a remainder. Each payment settles at least one person, so there are at most n - 1 payments, but it is not always the minimum possible number.

Part 2: A group of k people whose balances sum to zero can settle among themselves with k - 1 payments, so the minimum is (people with a balance) minus (the maximum number of disjoint zero-sum groups). dp[mask] is the largest number of zero-sum groups inside mask: add one person at a time and gain 1 when the mask's total is zero. Feasible for about 20 people.

```python
import heapq


# BASE QUESTION (optimal): Heaps: match the largest debtor
# with the largest creditor
# Idea: Push creditors (positive net) in a max-heap and
# debtors (negative net) in another. Repeatedly pay
# min(debtors, credit) from the biggest debtor to the biggest
# creditor, and push back whoever still has a remainder.
def balances(records):
    # net balance per person: what they paid minus their
    # equal share of every record they are part of
    net = {}
    for record in records:
        payer, amount, people = record["payer"], record["amount"], record["payees"]
        share = amount // len(people)
        net[payer] = net.get(payer, 0) + amount
        for person in people:
            net[person] = net.get(person, 0) - share
    return {person: balance for person, balance in net.items() if balance}


def settle_greedy(records):
    net = balances(records)
    creditors = [(-balance, person) for person, balance in net.items() if balance > 0]
    debtors = [(balance, person) for person, balance in net.items() if balance < 0]
    heapq.heapify(creditors)
    heapq.heapify(debtors)
    payments = []
    while creditors and debtors:
        neg_credit, creditor = heapq.heappop(creditors)
        neg_debt, debtor = heapq.heappop(debtors)
        payment = min(-neg_credit, -neg_debt)
        payments.append({"payer": debtor, "amount": payment, "payee": creditor})
        if -neg_credit > payment:
            heapq.heappush(creditors, (neg_credit + payment, creditor))
        if -neg_debt > payment:
            heapq.heappush(debtors, (neg_debt + payment, debtor))
    return payments
```

COMPLEXITY | Part 1: Time O(n log n) | Space O(n)
Part 2: Time O(2^n * n) | Space O(2^n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 34

35 * Playlist Shuffle No Adjacent Artists * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: pop the artist with the most songs left, hold it for one turn, then push it back with a smaller count

QUESTION INFO

QUESTION

Given the artist of every song in a playlist, return an order of the artists in which as few neighbours as possible are by the same artist: with no repeats next to each other whenever that is possible.

Function signature

def shuffle_no_adjacent(artists: list) -> list

EXAMPLES

Example 1	
Input	`['a', 'a', 'b', 'b', 'c']`
Output	`['a', 'b', 'a', 'b', 'c']`
Why	`Two a's and two b's alternate; c fits wherever it likes.`

Example 2	
Input	`['a', 'a', 'a', 'b']`
Output	`['a', 'b', 'a', 'a']`
Why	`Three songs by 'a' cannot all be separated by a single 'b': the best is to keep two a's together at the end.`

Example 3	
Input	`['a', 'a']`
Output	`['a', 'a']`
Why	`A single artist: nothing can be done.`

CONSTRAINTS

- 1 <= songs.length <= 10^4

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= songs.length <= 10^4

BRUTE FORCE | Try every ordering and keep the fewest adjacent pairs

BRUTE-FORCE IDEA

Enumerate all orderings of the artists' songs (as a multiset) and count neighbours with the same artist. Exact; hopeless beyond about ten songs.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every ordering and keep the fewest adjacent pairs
Enumerate all orderings of the artists' songs (as a multiset)
count neighbours with the same artist
```

BRUTE-FORCE COMPLEXITY | Time O(n! * n) | Space O(n)

WHY THIS IS WEAKER | N! Orderings. A greedy that always plays the artist with the most songs left (not the one that just played), using a heap, is O(n log A).

HOW TO IMPROVE IT

pop the artist with the most songs left, hold it for one turn, then push it back with a smaller count

SHORT BEST SOLUTION

Keep a heap of (remaining songs, artist). Each step pop the artist with the most songs left; play one of their songs, and push them back only after the next artist has played (so they cannot repeat). If only the previous artist is left, play it anyway: that is an unavoidable adjacency. For feasible inputs the result has no adjacent pair; otherwise the number of adjacent pairs equals the minimum, 2m - n - 1 for a dominant artist with m songs.

```
from collections import Counter
import heapq


# BASE QUESTION (optimal): Greedy with a max-heap, holding
# back the last artist
# Idea: Keep a heap of (remaining songs, artist). Each
# step pop the artist with the most songs left; play one
# of their songs, and push them back only after the next
# artist has played (so they cannot repeat). If only the
# previous artist is left, play it anyway: that is an
# unavoidable adjacency. For feasible inputs the result
# has no adjacent pair; otherwise the number of adjacent
# pairs equals the minimum, 2m - n - 1 for a dominant
# artist with m songs.
def shuffle_no_adjacent(artists):
    counts = Counter(artists)
    heap = [(-songs_left, artist) for artist, songs_left in counts.items()]
    heapq.heapify(heap)
    playlist, held = [], None
    while heap or held:
        if not heap:
            songs_left, artist = held
            playlist.append(artist)
            held = (songs_left + 1, artist) if songs_left + 1 else None
            continue
        songs_left, artist = heapq.heappop(heap)
        playlist.append(artist)
        if held:
            heapq.heappush(heap, held)
        held = (songs_left + 1, artist) if songs_left + 1 else None
    return playlist
```

COMPLEXITY | Time O(n log A) | Space O(A)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 35

GROUP 5 OF 43 | 6 QUESTIONS | PART A - PART 1

Recurrence on a position or a range (memo or table)

PATTERN CONTEXT

Define best(i): the answer for everything from position i onward (or best(i, j) for a range, or a table over two sequences). Write what an empty or finished state returns, then combine the answers of the states you can jump to. Cache it, or fill a table in the right order.

37 * Target Expressions from Ordered Digits * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: best(start, end) = every value reachable from digits l..r; split at m and combine the two sides with +, - and *

QUESTION INFO

QUESTION

Given a list of digits in order and a target number, put +, - or * between the digits (never reorder them and never glue two digits into one number) and add parentheses wherever you like. Return every distinct expression that evaluates to the target.

Function signature

def expressions_for_target(digits: list, target: int) -> list

EXAMPLES

Example 1	
Input	`[2, 3, 4], 14`
Output	`['2*(3+4)', '2+3*4']`
Why	`2 * (3 + 4) = 14 and 2 + 3 * 4 = 14 are the only two ways.`

Example 2	
Input	`[1, 2, 3], 6`
Output	`['1*2*3', '1+2+3']`
Why	`Three digits can reach 6 several ways.`

Example 3	
Input	`[1, 2], 100`
Output	`[]`
Why	`No way to reach 100 with these digits.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recursively try every split and operator, without remembering sub-results

BRUTE-FORCE IDEA

Split the digits at every point, combine all value pairs of the two parts with +, - and *, and see whether the target appears. Correct, but the same sub-interval is recomputed many times; memoising by interval is the fix.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recursively try every split and operator, without remembering sub-results
```

```
Split the digits at every point
combine all value pairs of the two parts with +
see whether the target appears
```

BRUTE-FORCE COMPLEXITY | Time exponential (Catalan(n) * 3^(n-1) expressions) | Space O(n) recursion plus the value set

WHY THIS IS WEAKER | The same sub-interval of digits is recomputed from scratch for every split above it, giving Catalan-number growth. Remembering the values per interval (memoisation) makes it polynomial in the number of distinct values.

HOW TO IMPROVE IT

best(start, end) = every value reachable from digits l..r; split at m and combine the two sides with +, - and *

SHORT BEST SOLUTION

Every expression over digits[start:end] is two sub-expressions over a split point joined by +, - or *. Memoize, for each interval, a map from numeric value to the set of expression strings that produce it. Keep expression construction separate from evaluation, and add parentheses only where precedence demands them.

```python
from collections import defaultdict


# BASE QUESTION (optimal): Interval DP: all expressions
# for digits[start:end]
# Idea: Every expression over digits[start:end] is two sub-
# expressions over a split point joined by +, - or *.
# Memoize, for each interval, a map from numeric value to
# the set of expression strings that produce it. Keep
# expression construction separate from evaluation, and
# add parentheses only where precedence demands them.
def expressions_for_target(digits, target):
    memo = {}

    def build(start, end):
        if (start, end) in memo:
            return memo[(start, end)]
        by_value = defaultdict(set)                        # value -> {(text, precedence)}
        if end - start == 1:
            by_value[digits[start]].add((str(digits[start]), 3))
        for split in range(start + 1, end):
            for left_value, left_texts in build(start, split).items():
                for right_value, right_texts in build(split, end).items():
                    for left_text, left_precedence in left_texts:
                        for right_text, right_precedence in right_texts:
                            for op, precedence, value in (("+", 1, left_value + right_value), ("-", 1, left_val
ue - right_value), ("*", 2, left_value * right_value)):
                                left = f"({left_text})" if left_precedence < precedence else left_text
                                right = f"({right_text})" if (right_precedence < precedence or (right_precedenc
e == precedence and op == "-")) else right_text
                                by_value[value].add((f"{left}{op}{right}", precedence))
        memo[(start, end)] = by_value
        return by_value

    return sorted(text for text, _ in build(0, len(digits)).get(target, ()))
```

COMPLEXITY | Time exponential in the number of results; polynomial in the number of distinct values | Space O(distinct values per interval)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 37

38 * Count Word Decompositions into Chemical Element Symbols * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: bottom-up ways[position]: add ways[position+1] if the one-letter symbol exists and ways[position+2] if the two-letter symbol exists

QUESTION INFO

QUESTION

Chemical elements have symbols of one or two letters (H, He, Si, Cs ...). Given a word, count the ways to write it as a sequence of element symbols (ignore upper and lower case). Different splits count as different ways even if they spell the same word.

Function signature

def count_decompositions(word: str) -> int

EXAMPLES

Example 1	
Input	`'Physics'`
Output	`4`
Why	`P-H-Y-Si-C-S, P-H-Y-Si-Cs, P-H-Y-S-I-C-S and P-H-Y-S-I-Cs: 4 ways.`

Example 2	
Input	`'Ho'`
Output	`2`
Why	`'Ho' is one symbol (holmium) and also H + O: 2 ways.`

Example 3	
Input	`'xqz'`
Output	`0`
Why	`No split works (x, y... cannot all be symbols).`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recursion that tries a one-letter and a two-letter symbol at every position

BRUTE-FORCE IDEA

At each position take either the next letter or the next two letters if they are symbols, and add up the ways to finish. Correct; the same suffix is solved many times.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recursion that tries a one-letter and a two-letter symbol at every position
At each position take either the next letter or the next two letters if they are symbols
add up the ways to finish
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential (Fibonacci growth) | Space O(n)

WHY THIS IS WEAKER | The same suffix is reached by many different splits, so the calls grow like the Fibonacci numbers. Computing ways from the end of the word once per position is O(n).

HOW TO IMPROVE IT

bottom-up ways[position]: add ways[position+1] if the one-letter symbol exists and ways[position+2] if the two-letter symbol exists

SHORT BEST SOLUTION

ways[position] is the number of decompositions of word[position:]. A symbol has one or two letters, so ways[position] = ways[position + 1] (if word[position] is a symbol) + ways[position + 2] (if word[position:position + 2] is a symbol). ways[word_length] = 1 for the empty suffix. Matching is case-insensitive.

```
ELEMENTS = set("""h he li be b c n o f ne na mg al si
p s cl ar k ca sc ti v cr mn fe co ni cu zn ga ge as
se br kr rb sr y zr nb mo tc ru rh pd ag cd in sn sb
te i xe cs ba la ce pr nd pm sm eu gd tb dy ho er tm
yb lu hf ta w re os ir pt au hg tl pb bi po at rn fr
ra ac th pa u np pu am cm bk cf es fm md no lr rf db
sg bh hs mt ds rg cn nh fl mc lv ts og""".split())


# BASE QUESTION (optimal): Suffix-count DP
# Idea: ways[position] is the number of decompositions of
# word[position:]. A symbol has one or two letters, so ways[position] =
# ways[position + 1] (if word[position] is a symbol) + ways[position + 2] (if
# word[position:position + 2] is a symbol). ways[word_length] = 1 for the empty
# suffix. Matching is case-insensitive.
def count_element_decompositions(word, symbols=ELEMENTS):
    lowered_word = word.lower()
    word_length = len(lowered_word)
    ways = [0] * (word_length + 2)
    ways[word_length] = 1
    for position in range(word_length - 1, -1, -1):
        if lowered_word[position] in symbols:
            ways[position] += ways[position + 1]
        if position + 2 <= word_length and lowered_word[position:position + 2] in symbols:
            ways[position] += ways[position + 2]
    return ways[0]
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 38

39 * Fence Painting Minimum Operations (Gemini Team) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: range (left, right): paint one stripe at the lowest height, then solve the pieces on both sides; a sparse table finds the lowest point fast

QUESTION INFO

QUESTION

A fence has n planks, plank i is a[i] units tall. One stroke is either horizontal (paints one row of height h across a run of neighbouring planks that are all at least h tall) or vertical (paints one plank completely, bottom to top). Return the smallest number of strokes needed to paint the whole fence.

Function signature

def paint(a: list) -> int

EXAMPLES

Example 1	
Input	`[2, 2, 1, 2, 1]`
Output	`3`
Why	`Row 1 across all five planks, row 2 across the first two planks, and one vertical stroke on plank 4: 3 strokes.`

QUESTION

Example 2	
Input	`[1, 1, 1]`
Output	`1`
Why	`One horizontal stroke of height 1 across all three planks paints the whole fence.`

Example 3	
Input	`[5]`
Output	`1`
Why	`One tall plank: a single vertical stroke beats five horizontal ones.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Divide and conquer, scanning for the minimum each time

BRUTE-FORCE IDEA

For a range with base height b: either paint every plank vertically (cost = number of planks) or paint a horizontal band up to the range minimum (cost = min - b) and recurse on the pieces split at the minimum planks. Finding the minimum by scanning makes a sorted or constant fence quadratic.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Divide and conquer, scanning for the minimum each time
For a range with base height b: either paint every plank vertically (cost = number of
planks) or paint a horizontal band up to the range minimum (cost = min - b)
recurse on the pieces split at the minimum planks
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) worst case | Space O(n) recursion

WHY THIS IS WEAKER | Finding the minimum by scanning makes a sorted or constant fence quadratic, O(n^2). A sparse table for range-minimum gives O(n log n).

HOW TO IMPROVE IT

range (left, right): paint one stripe at the lowest height, then solve the pieces on both sides; a sparse table finds the lowest point fast

SHORT BEST SOLUTION

The recursion is correct; only the minimum lookup is slow. A sparse table answers range-minimum-index queries in O(1), and each plank becomes the split point at most once, so the recursion visits n ranges. Splitting at one minimum plank (instead of all of them) keeps ranges disjoint and makes the count obvious.

```python
# BASE QUESTION (optimal): Same recursion with a sparse
# table for range minimum
# Idea: The recursion is correct; only the minimum lookup
# is slow. A sparse table answers range-minimum-index
# queries in O(1), and each plank becomes the split point
# at most once, so the recursion visits n ranges.
# Splitting at one minimum plank (instead of all of them)
# keeps ranges disjoint and makes the count obvious.
def paint(heights):
    plank_count = len(heights)
    if plank_count == 0:
        return 0
    table = [list(range(plank_count))]
    level = 1
    while (1 << level) <= plank_count:
```

```
        previous_row, next_row = table[-1], []
        for start in range(plank_count - (1 << level) + 1):
            left_candidate, right_candidate = previous_row[start], previous_row[start + (1 << (level - 1))]
            next_row.append(left_candidate if heights[left_candidate] <= heights[right_candidate] else right_ca
ndidate)
        table.append(next_row)
        level += 1

    def argmin(left, right):
        span_level = (right - left + 1).bit_length() - 1
        left_candidate, right_candidate = table[span_level][left], table[span_level][right - (1 << span_level)
+ 1]
        return left_candidate if heights[left_candidate] <= heights[right_candidate] else right_candidate

    def solve(left, right, base):
        if left > right:
            return 0
        lowest_index = argmin(left, right)
        return min(right - left + 1, heights[lowest_index] - base + solve(left, lowest_index - 1, heights[lowes
t_index]) + solve(lowest_index + 1, right, heights[lowest_index]))

    return solve(0, plank_count - 1, 0)
```

COMPLEXITY | Time O(n log n) | Space O(n log n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 39

40 * Maximize 1D deliveries within distance and minimize total distance * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: table over (orders used, drivers used): skip an order, skip a driver, or match them when they are within D (count, then minus distance)

QUESTION INFO

QUESTION

Orders and drivers are points on a line. A driver can deliver an order that is at most D away; every driver takes at most one order and every order is delivered by at most one driver. First maximize the number of deliveries; among all plans with that many deliveries, minimize the total distance driven. Return (number of deliveries, total distance).

Function signature

def deliveries(orders: list, drivers: list, D: int) -> tuple

EXAMPLES

Example 1	
Input	`[1, 5, 9], [2, 6, 20], 2`
Output	`(2, 2)`
Why	`1 is served by the driver at 2, and 5 by the driver at 6: 2 deliveries with total distance 1 + 1 = 2.`

Example 2	
Input	`[1, 5, 9], [2, 6, 20], 11`
Output	`(3, 13)`
Why	`With D = 11 the driver at 20 can also take the order at 9 (distance 11): 1 + 1 + 11 = 13 for 3 deliveries.`

QUESTION

Example 3	
Input	`[3, 4], [], 5`
Output	`(0, 0)`
Why	`No drivers, no deliveries.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every set of matches recursively

BRUTE-FORCE IDEA

For each order either skip it or match it to any unused driver within D; keep the best (most matches, then smallest total distance). It is exact but exponential.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every set of matches recursively
For each order either skip it or match it to any unused driver within D
keep the best (most matches
smallest total distance)
```

BRUTE-FORCE COMPLEXITY | Time O(drivers^orders) | Space O(n)

WHY THIS IS WEAKER | Each order branches over all drivers: exponential. After sorting, a DP over (orders considered, drivers considered) that compares (matches, minus distance) is O(n * m).

HOW TO IMPROVE IT

table over (orders used, drivers used): skip an order, skip a driver, or match them when they are within D (count, then minus distance)

SHORT BEST SOLUTION

Sort both lists. dp[orders_used][drivers_used] is the best (matches, -total distance) using the first i orders and the first j drivers: skip order i, skip driver j, or match them if their distance is at most D. Because an optimal matching can be taken non-crossing, matching in sorted order loses nothing. Compare pairs lexicographically.

```python
# BASE QUESTION (optimal): DP over sorted orders and
# drivers, maximising (matches, -distance)
# Idea: Sort both lists. dp[orders_used][drivers_used] is the best (matches,
# -total distance) using the first i orders and the first
# j drivers: skip order i, skip driver j, or match them if
# their distance is at most D. Because an optimal matching
# can be taken non-crossing, matching in sorted order
# loses nothing. Compare pairs lexicographically.
def deliveries(orders, drivers, max_distance):
    sorted_orders, sorted_drivers = sorted(orders), sorted(drivers)
    dp = [[(0, 0)] * (len(sorted_drivers) + 1) for _ in range(len(sorted_orders) + 1)]
    for orders_used in range(1, len(sorted_orders) + 1):
        for drivers_used in range(1, len(sorted_drivers) + 1):
            best = max(dp[orders_used - 1][drivers_used], dp[orders_used][drivers_used - 1])
            if abs(sorted_orders[orders_used - 1] - sorted_drivers[drivers_used - 1]) <= max_distance:
                matches, negative_distance = dp[orders_used - 1][drivers_used - 1]
                best = max(best, (matches + 1, negative_distance - abs(sorted_orders[orders_used - 1] - sorted_
drivers[drivers_used - 1])))
            dp[orders_used][drivers_used] = best
    matches, negative_distance = dp[-1][-1]
    return (matches, -negative_distance)
```

COMPLEXITY | Time O(n * m) | Space O(n * m)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 40

41 * Find the longest palindromic subsequence using dynamic programming. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: table over (start, end): equal ends add 2 to the inside; otherwise the better of dropping either end

QUESTION INFO

QUESTION

Given a string s, return the length of its longest palindromic subsequence: the longest sequence of letters of s, taken in order but not necessarily next to each other, that reads the same forwards and backwards.

Function signature

def lps_length(s: str) -> int

EXAMPLES

Example 1	
Input	`'bbbab'`
Output	`4`
Why	`'bbbb' is a subsequence of 'bbbab' and a palindrome.`

Example 2	
Input	`'cbbd'`
Output	`2`
Why	`'bb' (the two b's) is the longest.`

Example 3	
Input	`'a'`
Output	`1`
Why	`A single letter.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recursion on the two ends

BRUTE-FORCE IDEA

Lps(i, j): if the ends match add 2 to lps(i + 1, j - 1), otherwise take the max of lps(i + 1, j) and lps(i, j - 1). Exact; the same ranges recur exponentially often.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recursion on the two ends
Lps(i
j): if the ends match add 2 to lps(i + 1
j - 1)
otherwise take the max of lps(i + 1
lps(i
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(2^n) | Space O(n)

WHY THIS IS WEAKER | Two branches per step and the same (i, j) range recurs: O(2^n). A table over all ranges is O(n^2) time, with a rolling row for O(n) memory.

HOW TO IMPROVE IT

table over (start, end): equal ends add 2 to the inside; otherwise the better of dropping either end

SHORT BEST SOLUTION

dp[start][end] is the length of the longest palindromic subsequence of text[start..j]. Fill i from the end to the start and j from start + 1 up: equal ends give dp[start + 1][end - 1] + 2, otherwise max(dp[start + 1][end], dp[start][end - 1]). Only the next row and the current row are needed.

```
# BASE QUESTION (optimal): Interval DP with a rolling row
# Idea: dp[start][end] is the length of the longest palindromic
# subsequence of text[start..j]. Fill i from the end to the start
# and j from start + 1 up: equal ends give dp[start + 1][end - 1] +
# 2, otherwise max(dp[start + 1][end], dp[start][end - 1]). Only the
# next row and the current row are needed.
def longestPalindromeSubseq(text):
    length = len(text)
    if length == 0:
        return 0
    row_below = [0] * length                                    # dp[start + 1][*]
    for start in range(length - 1, -1, -1):
        current_row = [0] * length
        current_row[start] = 1
        for end in range(start + 1, length):
            current_row[end] = row_below[end - 1] + 2 if text[start] == text[end] else max(row_below[end], curr
ent_row[end - 1])
        row_below = current_row
    return row_below[length - 1]
```

COMPLEXITY | Time O(n^2) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 41

42 * Jump Game with Profit

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: best[cell] = max over every j before i that can reach i (earlier_cell + max_jump_from[earlier_cell] >= cell) of best[earlier_cell] + profit[cell]

QUESTION INFO

QUESTION

You stand at index 0 of an array. From index i you may jump to any index from i + 1 up to i + arr[i]. Landing on index i earns profit[i] (the start earns profit[0] too). Return the largest total profit of a path that ends exactly at the last index, or -1 if the last index cannot be reached.

Function signature

def max_profit_fast(arr: list, profit: list) -> int

EXAMPLES

Example 1	
Input	`[2, 3, 1, 1, 4], [1, 5, 2, 3, 4]`
Output	`15`
Why	`Jumping one index at a time visits (and collects) everything: 1 + 5 + 2 + 3 + 4 = 15.`

Example 2	
Input	`[1, 0, 1], [1, 1, 1]`
Output	`-1`
Why	`From index 1 you cannot jump at all (arr[1] = 0), so the last index is unreachable.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Reading A: recursion over every jump length

BRUTE-FORCE IDEA

From each cell try every allowed jump length and add the profit of the landing cell; take the best total that reaches the last index. Exact, and exponential.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Reading A: recursion over every jump length
From each cell try every allowed jump length
add the profit of the landing cell
take the best total that reaches the last index
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(max jump ^ n) | Space O(n)

WHY THIS IS WEAKER | Every cell branches into up to max-jump choices, so the paths multiply (max jump to the power n). A table of the best profit per cell, filled left to right, is O(n * max jump).

BRUTE FORCE 2 | Reading B: recursion over 'jump or skip' choices

BRUTE-FORCE IDEA

At each index either jump exactly arr[i] (collecting profit[i]) or skip to i + 1 for free, and take the larger total. Exact for the exact-jump reading.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Reading B: recursion over 'jump or skip' choices
At each index either jump exactly arr[i] (collecting profit[i]) or skip to i + 1 for free
take the larger total
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(2^n) | Space O(n)

WHY THIS IS WEAKER | Two branches at every cell and the same suffix is solved repeatedly: O(2^n). Computing best(i) once per index from the right is O(n).

HOW TO IMPROVE IT

best[cell] = max over every j before i that can reach i (earlier_cell + max_jump_from[earlier_cell] >= cell) of best[earlier_cell] + profit[cell]

SHORT BEST SOLUTION

Part 1: Let best[cell] be the largest profit achievable when standing on i. Landing on i from any earlier_cell < cell with earlier_cell + max_jump_from[earlier_cell] >= cell gives best[cell] = profit[cell] + max(best[earlier_cell]). Unreachable cells stay at minus infinity. Return best[cell_count - 1], or -1 if the end cannot be reached. Pull-style DP keeps the reachability test explicit.

Part 2: Memoize from the right. dp[cell] is the best profit starting at i, with dp[cell] = 0 for cell >= cell_count (past the end ends the game). dp[cell] = max(skip: dp[cell + 1], jump: profit[cell] + dp[cell + max_jump_from[cell]]). This is the 'exact jump, not range cap' reading the original report warns about.

```python
# BASE QUESTION (optimal): Reading A: max_jump_from[cell] is a maximum
# jump, profit[cell] is gained on landing, maximize total
# profit to the last index
# Idea: Let best[cell] be the largest profit achievable when
# standing on i. Landing on i from any earlier_cell < cell with earlier_cell +
# max_jump_from[earlier_cell] >= cell gives best[cell] = profit[cell] + max(best[earlier_cell]).
# Unreachable cells stay at minus infinity. Return best[cell_count
# - 1], or -1 if the end cannot be reached. Pull-style DP
# keeps the reachability test explicit.
def max_profit_jump(max_jump_from, profit):
    cell_count = len(max_jump_from)
    MINUS_INFINITY = float("-inf")
    best = [MINUS_INFINITY] * cell_count
    best[0] = profit[0]
```

```
    for cell in range(1, cell_count):
        for earlier_cell in range(cell):
            if best[earlier_cell] != MINUS_INFINITY and earlier_cell + max_jump_from[earlier_cell] >= cell:
                best[cell] = max(best[cell], best[earlier_cell] + profit[cell])
    return -1 if best[-1] == MINUS_INFINITY else best[-1]
```

COMPLEXITY | Part 1: Time O(n * maxJump) | Space O(n)
Part 2: Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 42

Best-first search with a heap: Dijkstra and its minimax twin

PATTERN CONTEXT

Keep a heap of (cost so far, node). Pop the cheapest, skip stale entries, and push any neighbour you improved. Dijkstra adds the edge weight to the cost; the minimax questions take the maximum of the cost and the edge instead. Everything else is the same loop.

46 * Path Minimizing the Maximum Cell (Swim in Rising Water) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the same minimax loop with a given start and end cell (Swim in Rising Water)

QUESTION INFO

QUESTION

You get a grid of heights, a start cell and an end cell, and you may move to any of the four neighbouring cells. The cost of a path is the largest height on it (start and end included). Return the smallest cost over all paths from start to end.

Function signature

def min_max_path(g: list, start: tuple, end: tuple) -> int

EXAMPLES

Example 1	
Input	`[[0, 2], [1, 3]], (0, 0), (1, 1)`
Output	`3`
Why	`Both routes pass the cell of height 3 at the end, so the best cost is 3.`

Example 2	
Input	`[[0, 1, 2, 3, 4], [24, 23, 22, 21, 5], [12, 13, 14, 15, 16], [11, 17, 18, 19, 20], [10, 9, 8, 7, 6]], (0, 0), (4, 4)`
Output	`16`
Why	`A 5x5 board: the best route has to climb to 16 somewhere, and no route stays lower.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every threshold and test reachability

BRUTE-FORCE IDEA

For each candidate value t in increasing order run a BFS using only cells <= t and stop at the first t that connects start to end. Correct, but it repeats a full search per value.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every threshold and test reachability
For each candidate value t in increasing order run a BFS using only cells <= t
stop at the first t that connects start to end
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(V^2) | Space O(V)

WHY THIS IS WEAKER | One full BFS per distinct value, O(V^2). Dijkstra on the running maximum finds the answer in one pass, O(V log V).

HOW TO IMPROVE IT

the same minimax loop with a given start and end cell (Swim in Rising Water)

SHORT BEST SOLUTION

Heap entries are (cost so far, cell). Popping a cell fixes its cost; each neighbour gets max(cost, neighbour value). Stop when the end cell is popped. Skip stale heap entries. The answer is the cost of the end cell.

```
import heapq


# BASE QUESTION (optimal): Dijkstra on the running maximum
# Idea: Heap entries are (cost so far, cell). Popping a
# cell fixes its cost; each neighbour gets max(cost,
# neighbour value). Stop when the end cell is popped. Skip
# stale heap entries. The answer is the cost of the end
# cell.
def min_max_path(grid, start, end):
    rows, cols = len(grid), len(grid[0])
    best = {start: grid[start[0]][start[1]]}
    heap = [(best[start], start)]
    while heap:
        cost, (row, col) = heapq.heappop(heap)
        if cost > best[(row, col)]:
            continue
        if (row, col) == end:
            return cost
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols:
                new_cost = max(cost, grid[next_row][next_col])
                if new_cost < best.get((next_row, next_col), float("inf")):
                    best[(next_row, next_col)] = new_cost
                    heapq.heappush(heap, (new_cost, (next_row, next_col)))
    return None
```

COMPLEXITY | Time O(V log V) | Space O(V)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 46

47 * Nearest Favorite City by Travel Time * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: combine = cost + minutes; stop when a favourite is popped (keep popping equal distances to break ties on the smaller index)

QUESTION INFO

QUESTION

You are given a network of num_cities cities (0..num_cities-1) connected by weighted, undirected roads roads[i] = [u, v, w] where w > 0 is the travel time. Given a start city and a non-empty list favorites of favorite cities, return the favorite city reachable from start in the shortest total travel time. Break ties by smallest city index. If none of the favorite cities is reachable, return -1 .

Function signature

```
def nearest_favorite(num_cities: int, roads: list[list[int]],
                     start: int, favorites: list[int]) -> int:
    ...
```

EXAMPLES

QUESTION

Example 1	
Input	`num_cities = 5, roads = [[0,1,2],[0,2,5],[1,3,1],[2,4,1]], start = 0, favorites = [3, 4]`
Output	`3`
Why	`Shortest times from 0: to 3 = 2+1 = 3, to 4 = 5+1 = 6. City 3 is nearer.`

Example 2	
Input	`num_cities = 3, roads = [[0,1,4]], start = 0, favorites = [2]`
Output	`-1`
Why	`City 2 is disconnected from start, so no favorite is reachable.`

CONSTRAINTS

- 1 <= num_cities <= 10**5 , 0 <= len(roads) <= 2*10**5 , 1 <= w <= 10**9 .
- The interviewer required the optimal solution (Dijkstra with a priority queue).

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= num_cities <= 10**5 , 0 <= len(roads) <= 2*10**5 , 1 <= w <= 10**9 .
- The interviewer required the optimal solution (Dijkstra with a priority queue).

BRUTE FORCE | Relax every road V - 1 times

BRUTE-FORCE IDEA

Start with: Relax every road V - 1 times. Bellman-Ford repeats the full relaxation even when most distances are final.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Relax every road V - 1 times
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(V * E) | Space O(V)

WHY THIS IS WEAKER | Bellman-Ford repeats the full relaxation even when most distances are final.

HOW TO IMPROVE IT

combine = cost + minutes; stop when a favourite is popped (keep popping equal distances to break ties on the smaller index)

SHORT BEST SOLUTION

Cities are finalised in order of distance, so the first favourite reached is the nearest.

```python
# BASE QUESTION (optimal): Dijkstra from the start
# Idea: Cities are finalised in order of distance, so the
# first favourite reached is the nearest.
import heapq


def nearest_favorite(num_cities, roads, start, favorites):
    neighbours = [[] for _ in range(num_cities)]
    for city, next_city, minutes in roads:
        neighbours[city].append((next_city, minutes))
        neighbours[next_city].append((city, minutes))
    best_time = [float("inf")] * num_cities
    best_time[start] = 0
    heap, wanted = [(0, start)], set(favorites)
    nearest_time, nearest_city = None, -1
    while heap:
```

```
        time, city = heapq.heappop(heap)
        if time > best_time[city]:
            continue
        if nearest_time is not None and time > nearest_time:
            break
        if city in wanted and (nearest_time is None or city < nearest_city):
            nearest_time, nearest_city = time, city
        for next_city, minutes in neighbours[city]:
            if time + minutes < best_time[next_city]:
                best_time[next_city] = time + minutes
                heapq.heappush(heap, (best_time[next_city], next_city))
    return nearest_city
```

COMPLEXITY | Time O((V + E) log V) | Space O(V + E)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 47

48 * Given a train schedule with departure city, arrival city, start time, end time, and duration, and given a source and destination, find the sequence of trains that gets you there at the earliest possible time. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: cost = arrival time; neighbors only offers trains that leave at or after the current time; combine = the train's arrival time

QUESTION INFO

QUESTION

A schedule lists trains as (from, to, departure time, arrival time). Starting in city src at time start, find the sequence of trains that gets you to dst at the earliest possible time. You can only board a train that leaves at or after the time you arrive in its departure city. Return (arrival time, the trains you take), or None if dst cannot be reached.

Function signature

def earliest_arrival(trains: list, src, dst, start=0) -> tuple | None

EXAMPLES

Example 1	
Input	`[('A', 'B', 1, 3), ('B', 'C', 4, 6), ('A', 'C', 2, 9), ('B', 'C', 2, 5)], 'A', 'C'`
Output	`(6, [('A', 'B', 1, 3), ('B', 'C', 4, 6)])`
Why	`A -> B arrives at 3; the train B -> C that leaves at 2 is already gone, so take the one leaving at 4 and arrive at 6 (the direct A -> C train arrives later, at 9).`

Example 2	
Input	`[('A', 'B', 1, 3), ('B', 'C', 4, 6)], 'C', 'A'`
Output	`null`
Why	`No train leaves C, so A cannot be reached from C.`

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Depth-first search over every sequence of trains

BRUTE-FORCE IDEA

From the source, try every train that can still be boarded, recursively, and keep the earliest arrival at the destination. Exact; the number of sequences can explode.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
DFS from (source city, starting time)
try every train that departs after the current arrival time
when the destination is reached, compare arrival times
backtrack and try the other train sequences
return the sequence with the earliest arrival
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(trains)

WHY THIS IS WEAKER | The number of train sequences can be exponential. Because waiting is free, an earlier arrival is never worse, so Dijkstra on earliest arrival times is O((T + C) log C).

HOW TO IMPROVE IT

cost = arrival time; neighbors only offers trains that leave at or after the current time; combine = the train's arrival time

SHORT BEST SOLUTION

earliest_at[city] is the earliest time you can be there. Pop the city with the smallest time, and for every train from it that departs at or after that time, improve earliest_at[next_city] to the train's arrival time and remember the train. Because waiting is free, a later arrival never beats an earlier one, so the first pop of the destination is optimal.

```python
from collections import defaultdict
import heapq


# BASE QUESTION (optimal): Dijkstra on earliest arrival
# times, with the train sequence
# Idea: earliest_at[city] is the earliest time you can be there.
# Pop the city with the smallest time, and for every train
# from it that departs at or after that time, improve
# earliest_at[next_city] to the train's arrival time and remember the
# train. Because waiting is free, a later arrival never
# beats an earlier one, so the first pop of the
# destination is optimal.
def earliest_arrival(trains, source, destination, start=0):
    by_city = defaultdict(list)
    for train in trains:
        by_city[train[0]].append(train)
    earliest_at, came_by, heap = {source: start}, {}, [(start, source)]
    while heap:
        time_now, city = heapq.heappop(heap)
        if time_now > earliest_at[city]:
            continue
        if city == destination:
            route = []
            while city != source:
                train, previous_city = came_by[city]
                route.append(train)
                city = previous_city
            return time_now, route[::-1]
        for train in by_city[city]:
            origin, next_city, departure, arrival = train
            if departure >= time_now and arrival < earliest_at.get(next_city, float("inf")):
                earliest_at[next_city], came_by[next_city] = arrival, (train, city)
                heapq.heappush(heap, (arrival, next_city))
    return None
```

COMPLEXITY | Time O((T + C) log C) with trains sorted by departure per city | Space O(T + C)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 48

49 * Find the shortest path in a graph. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: combine = cost + weight; also store parent[neighbour]; stop at the target and rebuild the path

QUESTION INFO

QUESTION

A directed graph is given as a list of weighted edges (u, v, w) with non-negative weights. Return the total length and the nodes of a shortest path from s to t, or None if t cannot be reached.

Function signature

def dijkstra_path(edges: list, s, t) -> tuple | None

(distance, path)

EXAMPLES

Example 1	
Input	`[('A', 'B', 1), ('A', 'C', 4), ('B', 'C', 2), ('C', 'D', 1)], 'A', 'D'`
Output	`(4, ['A', 'B', 'C', 'D'])`
Why	`A -> B -> C -> D costs 1 + 2 + 1 = 4; the direct edge A -> C costs 4, which is longer overall.`

Example 2	
Input	`[('A', 'B', 1), ('B', 'C', 2), ('C', 'D', 1)], 'D', 'A'`
Output	`null`
Why	`There is no edge leading from D back to A.`

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Bellman-Ford: relax every edge V - 1 times

BRUTE-FORCE IDEA

Start with distance 0 at the source and repeatedly try to improve every edge's endpoint. It works with negative weights as well and is; it simply repeats full passes over all edges.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Bellman-Ford: relax every edge V - 1 times
Start with distance 0 at the source
repeatedly try to improve every edge's endpoint
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(V * E) | Space O(V)

WHY THIS IS WEAKER | Each round relaxes every edge: O(V * E). It is the right tool for negative weights; for non-negative weights Dijkstra's heap is O((V + E) log V), and for unweighted graphs BFS is O(V + E).

HOW TO IMPROVE IT

combine = cost + weight; also store parent[neighbour]; stop at the target and rebuild the path

SHORT BEST SOLUTION

Keep the best known distance to each node and a min-heap of (distance, node). Pop the closest unsettled node, relax its edges, and remember the parent of every improvement. The first time the target is popped its distance is final; rebuild the path by following parents. It needs non-negative weights.

```python
from collections import defaultdict
import heapq


# BASE QUESTION (optimal): Dijkstra with a heap (non-
# negative weights), returning the path
# Idea: Keep the best known distance to each node and a
# min-heap of (distance, node). Pop the closest unsettled
# node, relax its edges, and remember the parent of every
# improvement. The first time the target is popped its
# distance is final; rebuild the path by following
# parents. It needs non-negative weights.
def shortest_path(node_count, edges, source, target):
    graph = defaultdict(list)
    for node, neighbour, weight in edges:
        graph[node].append((neighbour, weight))
    shortest, parent, heap = {source: 0}, {source: None}, [(0, source)]
    while heap:
        distance_so_far, node = heapq.heappop(heap)
        if distance_so_far > shortest[node]:
            continue
        if node == target:
            path = []
            while node is not None:
                path.append(node)
                node = parent[node]
            return distance_so_far, path[::-1]
        for neighbour, weight in graph[node]:
            if distance_so_far + weight < shortest.get(neighbour, float("inf")):
                shortest[neighbour], parent[neighbour] = distance_so_far + weight, node
                heapq.heappush(heap, (distance_so_far + weight, neighbour))
    return None
```

COMPLEXITY | Time O((V + E) log V) | Space O(V)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 49

50 * Given a matrix with a rat, bread, cat, and obstacles, find a path from the rat to the bread that maximizes the minimum distance from cat. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: maximise the minimum distance to the cat along the path: push negated values, combine = min(value, cat distance of the cell)

QUESTION INFO

QUESTION

A grid contains a rat R, a piece of bread B, a cat C, walls

and open cells '.'. Move

the rat from R to B through open cells (up, down, left, right). The safety of a path is the smallest distance to the cat of any cell on it (distances count steps around the walls). Return the largest safety you can achieve, or -1 if the bread cannot be reached.

Function signature

def safest_path(grid: list) -> int

EXAMPLES

QUESTION

Example 1	
Input	`['R..', '...', 'C.B']`
Output	`2`
Why	The path along the top and right side keeps at least 2 steps from the cat at the bottom left.

Example 2	
Input	`['R#B']`
Output	`-1`
Why	The wall blocks the only route to the bread.

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every simple path from the rat to the bread

BRUTE-FORCE IDEA

Depth-first search over all simple paths, computing the minimum cat distance along each, and keep the best. Exact; the number of paths explodes on larger grids.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every simple path from the rat to the bread
Depth-first search over all simple paths
computing the minimum cat distance along each
keep the best
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(path length)

WHY THIS IS WEAKER | The number of simple paths grows exponentially with the grid. A max-heap search that extends the path with the best bottleneck so far (or binary search on the threshold) is O(m * n log(m * n)).

HOW TO IMPROVE IT

maximise the minimum distance to the cat along the path: push negated values, combine = min(value, cat distance of the cell)

SHORT BEST SOLUTION

BFS from every cat gives the distance of each cell. Then run Dijkstra from the rat where the 'cost' of a path is the minimum cat distance along it and we maximise it: pop the cell with the best bottleneck so far, and extend to a neighbour with min(bottleneck, neighbour distance). The first time the bread is popped its bottleneck is optimal.

```python
import heapq
from collections import deque


def _cat_dist(grid):
    rows, cols = len(grid), len(grid[0])
    cat_distance = {(row, col): 0 for row in range(rows) for col in range(cols) if grid[row][col] == "C"}
    queue = deque(cat_distance)
    while queue:
        row, col = queue.popleft()
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols and grid[next_row][next_col] != "#" and (next_row,
 next_col) not in cat_distance:
                cat_distance[(next_row, next_col)] = cat_distance[(row, col)] + 1
                queue.append((next_row, next_col))
    return cat_distance
```

```
def _find(grid, symbol):
    return next((row, col) for row in range(len(grid)) for col in range(len(grid[0])) if grid[row][col] == symb
ol)


# BASE QUESTION (optimal): Multi-source BFS from the cat,
# then a max-heap bottleneck search
# Idea: BFS from every cat gives the distance of each
# cell. Then run Dijkstra from the rat where the 'cost' of
# a path is the minimum cat distance along it and we
# maximise it: pop the cell with the best bottleneck so
# far, and extend to a neighbour with min(bottleneck,
# neighbour distance). The first time the bread is popped
# its bottleneck is optimal.
def safest_path(grid):
    rows, cols = len(grid), len(grid[0])
    cat = _cat_dist(grid)
    start, goal = _find(grid, "R"), _find(grid, "B")
    best = {start: cat.get(start, 10**9)}
    heap = [(-best[start], start)]
    while heap:
        negative_bottleneck, position = heapq.heappop(heap)
        if -negative_bottleneck < best[position]:
            continue
        if position == goal:
            return -negative_bottleneck
        row, col = position
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols and grid[next_row][next_col] != "#":
                value = min(-negative_bottleneck, cat.get((next_row, next_col), 10**9))
                if value > best.get((next_row, next_col), -1):
                    best[(next_row, next_col)] = value
                    heapq.heappush(heap, (-value, (next_row, next_col)))
    return -1
```

COMPLEXITY | Time O(m * n log(m * n)) | Space O(m * n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 50

GROUP 7 OF 43 | 8 QUESTIONS | PART A - PART 1

Backtracking: choose, recurse, undo

PATTERN CONTEXT

Build the answer one choice at a time in a list. When the list is complete, record it. Otherwise loop over the allowed choices: append one, recurse, and then remove exactly what you added so the next choice starts from the same state. Pruning is just a check that skips a choice.

55 * Mahjong Winning Hand Detection (14 Tiles) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: take the smallest tile; try it as a triple, then as the start of a run; undo the counts after each try

QUESTION INFO

QUESTION

Mountain View onsite coding round 1: given a hand of 14 tiles, decide whether it is a winning mahjong hand. The candidate must clarify the rule set (typically: 4 melds of three identical tiles or three consecutive ranks + 1 pair).

Input: array of 14 integers representing tile values (clarify suit / range during the interview).
A winning hand consists of:

One pair (two identical tiles), and
Four melds, each being either:

A triplet (3 identical tiles), or
A run (3 consecutive ranks in the same suit).

Minority variant: A newer version instead specifies one pair plus two melds while retaining a 14-tile input. Because that accounts for only eight tiles, clarify the intended meld count and how the remaining six tiles should be handled before coding.

Output: boolean.

EXAMPLES

Example 1	
Input	`[1, 1, 1, 2, 3, 4, 11, 12, 13, 21, 21, 21, 31, 31]`
Output	`true`
Why	`Tiles are numbers: 1-9 one suit, 11-19 a second suit, 21-29 a third, 31 and up honours. Pair 31-31, triplets 1-1-1 and 21-21-21, and sequences 2-3-4 and 11-12-13.`

Example 2	
Input	`[1, 1, 1, 2, 3, 4, 11, 12, 13, 21, 21, 21, 31, 32]`
Output	`false`
Why	`31 and 32 are different honours, so there is no pair to finish the hand.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Choose any three tiles for every meld

BRUTE-FORCE IDEA

For each pair, try every set of three remaining tiles that forms a meld and recurse. Correct; it ignores that the smallest tile must be used first.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Choose any three tiles for every meld
For each pair
try every set of three remaining tiles that forms a meld
recurse
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time roughly C(12,3) * C(9,3) * C(6,3) per pair | Space O(1)

WHY THIS IS WEAKER | Tries every set of three tiles at every step. The smallest remaining tile can only be in a triplet of itself or a run starting at it, so at most two choices remain.

HOW TO IMPROVE IT

take the smallest tile; try it as a triple, then as the start of a run; undo the counts after each try

SHORT BEST SOLUTION

Tiles are encoded as suit * 10 + rank for the numbered suits (ranks 1 to 9, suits 0 to 2) and 30 and above for honours, which never form runs. Count the tiles, remove each possible pair, then take the smallest tile left and try a triplet of it or a run starting at it, backtracking if neither works.

```
from collections import Counter


# BASE QUESTION (optimal): Try each pair, then greedy-
# smallest with two options
# Idea: Tiles are encoded as suit * 10 + rank for the
# numbered suits (ranks 1 to 9, suits 0 to 2) and 30 and
# above for honours, which never form runs. Count the
# tiles, remove each possible pair, then take the smallest
# tile left and try a triplet of it or a run starting at
# it, backtracking if neither works.
def is_winning_hand(hand):
    def melds(tile_counts):
        smallest_tile = next((tile for tile in sorted(tile_counts) if tile_counts[tile] > 0), None)
        if smallest_tile is None:
            return True
        if tile_counts[smallest_tile] >= 3:
            tile_counts[smallest_tile] -= 3
            rest_works = melds(tile_counts)
            tile_counts[smallest_tile] += 3
            if rest_works:
                return True
        if smallest_tile < 30 and smallest_tile % 10 <= 7 and tile_counts.get(smallest_tile + 1, 0) > 0 and til
e_counts.get(smallest_tile + 2, 0) > 0:
            for run_tile in (smallest_tile, smallest_tile + 1, smallest_tile + 2):
                tile_counts[run_tile] -= 1
            rest_works = melds(tile_counts)
            for run_tile in (smallest_tile, smallest_tile + 1, smallest_tile + 2):
                tile_counts[run_tile] += 1
            if rest_works:
                return True
        return False
    tile_counts = Counter(hand)
    for pair_tile in list(tile_counts):
        if tile_counts[pair_tile] >= 2:
            tile_counts[pair_tile] -= 2
            rest_works = melds(tile_counts)
            tile_counts[pair_tile] += 2
            if rest_works:
                return True
    return False
```

COMPLEXITY | Time O(1) for 14 tiles (about 2^4 branches per pair) | Space O(distinct tiles)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 55

56 * Count Prefix-Matching Simple Paths in a Labeled Graph * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: extend the path along edges whose label matches the next query letter, with a bitmask of visited nodes; count per depth

QUESTION INFO

QUESTION

A directed graph has labelled vertices (labels[i] is the label of vertex i) and edges (u, v). A simple path never visits a vertex twice. For a query word, count for every prefix length L (1 up to the length of the word) the number of simple paths with L vertices whose labels spell the first L letters of the word. Return the list of counts.

Function signature

def prefix_paths(labels: list, edges: list, query: str) -> list

EXAMPLES

Example 1	
Input	`['a', 'b', 'c', 'a'], [(0, 1), (1, 2), (3, 1)], 'abc'`
Output	`[2, 2, 2]`
Why	`Two paths spell 'a' (vertices 0 and 3), and both continue to b (vertex 1) and then c (vertex 2).`

Example 2	
Input	`['a', 'a'], [(0, 1), (1, 0)], 'aaa'`
Output	`[2, 2, 0]`
Why	`Two paths spell 'a' and two spell 'aa' (0 -> 1 and 1 -> 0); none spells 'aaa' because that would reuse a vertex.`

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Enumerate all simple paths, then test the labels

BRUTE-FORCE IDEA

List every simple path of every length and compare its labels with each prefix of the query. Correct; it explores paths whose first label already disagrees.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Enumerate all simple paths, then test the labels
List every simple path of every length
compare its labels with each prefix of the query
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(path)

WHY THIS IS WEAKER | It explores paths whose first label already disagrees with the query. Extending a path only when the next label matches the next query letter prunes those branches.

HOW TO IMPROVE IT

extend the path along edges whose label matches the next query letter, with a bitmask of visited nodes; count per depth

SHORT BEST SOLUTION

Start from every vertex whose label equals the first query letter. Extend to an unvisited neighbour only if its label equals the next query letter, adding one to the count for the current length at each step. Branches that stop matching are cut at once.

```python
from collections import defaultdict


# BASE QUESTION (optimal): DFS with label pruning and a
# visited bitmask
# Idea: Start from every vertex whose label equals the
# first query letter. Extend to an unvisited neighbour
# only if its label equals the next query letter, adding
# one to the count for the current length at each step.
# Branches that stop matching are cut at once.
def prefix_paths(labels, edges, query):
    neighbours = defaultdict(list)
    for node, neighbour in edges:
        neighbours[node].append(neighbour)
    counts = [0] * len(query)

    def go(node, depth, visited_mask):
        counts[depth] += 1
        if depth + 1 < len(query):
            for next_node in neighbours[node]:
                if not visited_mask >> next_node & 1 and labels[next_node] == query[depth + 1]:
                    go(next_node, depth + 1, visited_mask | 1 << next_node)

    for start_vertex in range(len(labels)):
        if query and labels[start_vertex] == query[0]:
            go(start_vertex, 0, 1 << start_vertex)
    return counts
```

COMPLEXITY | Time O(number of matching paths * degree) | Space O(path)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 56

57 * Determine if a 14-tile hand is winning

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: the same try-triple-then-run recursion over a count array, after choosing the pair

QUESTION INFO

QUESTION

A hand is 14 tiles with values 1 to 9 of one suit (you may reorder them). It wins if it splits into one pair (two identical tiles) and four melds of three tiles, where a meld is either a triplet (three identical tiles) or a sequence (three consecutive values). Return whether the hand is winning.

Function signature

def is_winning_hand(tiles: list) -> bool

EXAMPLES

Example 1	
Input	`[1, 1, 1, 2, 3, 4, 5, 5, 5, 6, 7, 8, 9, 9]`
Output	`true`
Why	`Pair 9-9, triplets 1-1-1 and 5-5-5, sequences 2-3-4 and 6-7-8.`

Example 2	
Input	`[1, 2, 3, 4, 5, 6, 7, 8, 9, 1, 1, 2, 4, 6]`
Output	`false`
Why	`These tiles cannot be split into a pair and four melds.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Remove any triplet or sequence from anywhere, exhaustively

BRUTE-FORCE IDEA

For each pair candidate, repeatedly try removing any triplet or any sequence available among the counts (in any order) and see whether everything can be removed. Correct; it tries the same groupings in many orders, unlike the smallest-first search.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Remove any triplet or sequence from anywhere, exhaustively
For each pair candidate
repeatedly try removing any triplet or any sequence available among the counts (in any
order)
see whether everything can be removed
```

BRUTE-FORCE COMPLEXITY | Time exponential but small for 14 tiles | Space O(depth)

WHY THIS IS WEAKER | It tries the same groupings in many different orders (still tiny for 14 tiles). Always grouping the smallest remaining tile tries each grouping once.

HOW TO IMPROVE IT

the same try-triple-then-run recursion over a count array, after choosing the pair

SHORT BEST SOLUTION

Keep counts of 1..9. For each value with at least two tiles, take it as the pair. Then repeatedly look at the smallest value still present: it must start a triplet (count at least 3) or a sequence (rank, rank + 1, rank + 2 all present); try both. Because the smallest value has to be used by one of those melds, no grouping is missed and no order is repeated.

```python
# BASE QUESTION (optimal): Counts, try each pair, then
# always group the smallest remaining tile
# Idea: Keep counts of 1..9. For each value with at least
# two tiles, take it as the pair. Then repeatedly look at
# the smallest value still present: it must start a
# triplet (count at least 3) or a sequence (rank, rank + 1, rank +
# 2 all present); try both. Because the smallest value has
# to be used by one of those melds, no grouping is missed
# and no order is repeated.
def is_winning_hand(tiles):
    count_of_rank = [0] * 12

    for tile in tiles:
        count_of_rank[tile] += 1

    def melds(rank):
        while rank <= 9 and count_of_rank[rank] == 0:
            rank += 1
        if rank > 9:
            return True
        if count_of_rank[rank] >= 3:
            count_of_rank[rank] -= 3
            rest_works = melds(rank)
            count_of_rank[rank] += 3
            if rest_works:
                return True
        if rank <= 7 and count_of_rank[rank + 1] and count_of_rank[rank + 2]:
            count_of_rank[rank] -= 1; count_of_rank[rank + 1] -= 1; count_of_rank[rank + 2] -= 1
            rest_works = melds(rank)
            count_of_rank[rank] += 1; count_of_rank[rank + 1] += 1; count_of_rank[rank + 2] += 1
            if rest_works:
                return True
        return False

    for pair in range(1, 10):
```

```
        if count_of_rank[pair] >= 2:
            count_of_rank[pair] -= 2
            rest_works = melds(1)
            count_of_rank[pair] += 2
            if rest_works:
                return True
    return False
```

COMPLEXITY | Time O(9 * small branching) = constant for 14 tiles | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 57

58 * American Football Score Combinations * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: choices are the scoring plays from index i onward (the same play may repeat); complete when the remaining score is 0

QUESTION INFO

QUESTION

Bryan's DP question was: enumerate all the ways a football team can accumulate a given total number of points. In American football a single scoring play is worth one of: 2 (safety), 3 (field goal), 6 (touchdown), 7 (touchdown + extra point), or 8 (touchdown + 2-point conversion). Given a target total , return every distinct combination of scoring plays whose values sum to total . Two combinations are the same if they use the same multiset of play values (order does not matter). Each combination should be returned as a list of play values sorted in non-decreasing order. The overall list of combinations may be in any order. A play value may be used any number of times.

Function signature

def football_combinations(total: int) -> list[list[int]]:
    """Return all distinct multisets of {2,3,6,7,8} that sum to total, each sorted ascending."""
    ...

Follow-up (as reported): If you only need the count of combinations (not the lists), do it with an O(total) DP table (unbounded coin-change counting).

EXAMPLES

Example 1	
Input	`total = 6`
Output	`[[2, 2, 2], [3, 3], [6]]`
Why	`2+2+2 = 3+3 = 6 = 6.`

Example 2	
Input	`total = 7`
Output	`[[2, 2, 3], [7]]`
Why	`2+2+3 = 7 and the single play 7.`

Example 3	
Input	`total = 0`
Output	`[[]]`
Why	`The empty combination is the only way to score 0.`

CONSTRAINTS

- 0 <= total <= 60

QUESTION

- Allowed play values are exactly [2, 3, 6, 7, 8] .
- If total == 0 , the only combination is the empty one, so return [[]] .

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= total <= 60
- Allowed play values are exactly [2, 3, 6, 7, 8] .
- If total == 0 , the only combination is the empty one, so return [[]] .

BRUTE FORCE | Ordered sequences, then de-duplicate

BRUTE-FORCE IDEA

Start with: Ordered sequences, then de-duplicate. It builds every ordering of the plays and only then collapses permutations.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Ordered sequences, then de-duplicate
Start with: Ordered sequences
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space exponential

WHY THIS IS WEAKER | It builds every ordering of the plays and only then collapses permutations.

HOW TO IMPROVE IT

choices are the scoring plays from index i onward (the same play may repeat); complete when the remaining score is 0

SHORT BEST SOLUTION

Choosing plays in non-decreasing order lists every multiset exactly once.

```python
# BASE QUESTION (optimal): Backtracking in increasing
# order
# Idea: Choosing plays in non-decreasing order lists every
# multiset exactly once.
def football_combinations(total):
    plays = (2, 3, 6, 7, 8)
    combinations = []

    def dfs(start, remaining, path):
        if remaining == 0:
            combinations.append(path[:])
            return
        for play_index in range(start, len(plays)):
            if plays[play_index] <= remaining:
                path.append(plays[play_index])
                # the same play may be used again, so i stays
                dfs(play_index, remaining - plays[play_index], path)
                path.pop()

    dfs(0, total, [])
    return combinations
```

COMPLEXITY | Time O(number of combinations * length) | Space O(total) recursion

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 58

59 * Count Passcode Patterns on a Keypad

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: the choice is the next adjacent unused key; complete at k keys; count instead of storing

QUESTION INFO

QUESTION

A keypad is a rows x cols grid of keys (the classic phone dial is 3x3):
1 2 3 4 5 6 7 8 9 A passcode pattern is a path across keys where each move goes only to an orthogonally adjacent key (horizontal or vertical — no diagonals), and no key is visited more than once. The order matters: 1 -> 2 -> 3 -> 6 and 6 -> 3 -> 2 -> 1 are different patterns. Given the grid dimensions and an integer k , return the number of distinct passcode patterns that use exactly k keys.

Function signature

```
def count_patterns(rows: int, cols: int, k: int) -> int:
    """Return the number of distinct passcode patterns using exactly k keys,
    moving only to orthogonally adjacent, unvisited keys on a rows x cols grid."""
    ...
```

Follow-up (as reported): Count patterns of length at least k instead of exactly k . Hint: sum the DFS counts over all lengths from k to rows*cols .

EXAMPLES

Example 1	
Input	`rows = 3, cols = 3, k = 1`
Output	`9`
Why	`Each of the 9 keys alone is one pattern.`

Example 2	
Input	`rows = 3, cols = 3, k = 2`
Output	`24`
Why	`There are 12 adjacent key pairs; each can be traversed in 2 directions, giving 24 patterns.`

Example 3	
Input	`rows = 3, cols = 3, k = 3`
Output	`44`
Why	`Summing over each possible middle key b, deg(b)*(deg(b)-1) directed 3-key paths: corners give 4*2=8, edge-mids 4*6=24, center 12; total 44.`

CONSTRAINTS

- 1 <= rows, cols <= 4 .
- 1 <= k <= rows * cols .
- A single key (k = 1) is a valid pattern of length 1.
- Count directed patterns (a path and its reverse are counted separately).

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= rows, cols <= 4 .

- 1 <= k <= rows * cols .
- A single key (k = 1) is a valid pattern of length 1.
- Count directed patterns (a path and its reverse are counted separately).

BRUTE FORCE | Every ordered selection of k keys

BRUTE-FORCE IDEA

Start with: Every ordered selection of k keys. It builds selections that are not even connected before testing them.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Every ordered selection of k keys
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(P(rc, k) * k) | Space O(k)

WHY THIS IS WEAKER | It builds selections that are not even connected before testing them.

HOW TO IMPROVE IT

the choice is the next adjacent unused key; complete at k keys; count instead of storing

SHORT BEST SOLUTION

Extend the path one adjacent unused key at a time and undo the choice afterwards.

```python
# BASE QUESTION (optimal): DFS with backtracking from
# every key
# Idea: Extend the path one adjacent unused key at a time
# and undo the choice afterwards.
def count_patterns(rows, cols, pattern_length):
    def dfs(row, col, keys_used, seen):
        if keys_used == pattern_length:
            return 1
        total = 0
        for row_step, col_step in ((1, 0), (-1, 0), (0, 1), (0, -1)):
            next_row, next_col = row + row_step, col + col_step
            if 0 <= next_row < rows and 0 <= next_col < cols and (next_row, next_col) not in seen:
                seen.add((next_row, next_col))
                total += dfs(next_row, next_col, keys_used + 1, seen)
                seen.remove((next_row, next_col))
        return total

    return sum(dfs(row, col, 1, {(row, col)})
               for row in range(rows) for col in range(cols))
```

COMPLEXITY | Time O(number of simple paths) | Space O(k)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 59

60 * Match people to questions using tags and priorities

* MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: try each question of a person; if it is taken, recursively ask its owner to move (augmenting path); undo by not recording on failure

QUESTION INFO

QUESTION

A set of people and a set of interview questions: adj[p] lists the questions person p is allowed to answer (their tags match). Each person is given at most one question and each question goes to at most one person. Return the largest number of people that can be matched.

Function signature

def max_matching(adj: list) -> int

EXAMPLES

QUESTION

Example 1	
Input	`[[0, 1], [0], [1, 2]]`
Output	`3`
Why	`Person 0 takes question 0, person 1 has only question 0 so person 0 moves to question 1, person 2 takes question 2: all three are matched.`

Example 2	
Input	`[[0], [0], [0]]`
Output	`1`
Why	`Three people can only answer the same single question: just one match.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every assignment recursively

BRUTE-FORCE IDEA

For each person either skip them or give them any unused question they share a tag with, and keep the largest number of matches. Exact, so it checks the fast algorithm, but exponential.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
process people one at a time
try each compatible unused question, or leave the person unmatched
recurse to the next person, then undo the choice
keep the assignment with the most matches and best priority
```

BRUTE-FORCE COMPLEXITY | Time O((Q + 1)^P) | Space O(P)

WHY THIS IS WEAKER | Each person branches over all questions: exponential. Augmenting paths (move an earlier person to another question when needed) solve it in O(people * edges).

HOW TO IMPROVE IT

try each question of a person; if it is taken, recursively ask its owner to move (augmenting path); undo by not recording on failure

SHORT BEST SOLUTION

For each person run a DFS that tries each acceptable question: if it is free take it, otherwise try to move its current owner to another question (recursively), using a visited set so no question is tried twice in one attempt. Each successful DFS increases the matching by one. Hopcroft-Karp improves this to O(E * sqrt(V)).

```python
# BASE QUESTION (optimal): Augmenting paths (Kuhn's
# algorithm)
# Idea: For each person run a DFS that tries each
# acceptable question: if it is free take it, otherwise
# try to move its current owner to another question
# (recursively), using a visited set so no question is
# tried twice in one attempt. Each successful DFS
# increases the matching by one. Hopcroft-Karp improves
# this to O(E * sqrt(V)).
def max_matching(acceptable):
    owner_of = {}

    def try_assign(person, seen):
        for question in acceptable[person]:
            if question in seen:
                continue
            seen.add(question)
            if question not in owner_of or try_assign(owner_of[question], seen):
```

```
                owner_of[question] = person
                return True
        return False

    return sum(try_assign(person, set()) for person in range(len(acceptable)))
```

COMPLEXITY | Time O(P * E) | Space O(P + Q)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 60

61 * Generate Parentheses * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: choices are '(' and ')'; allowed when opened < pair_count, and ')' only when closed < opened; complete at length 2n

QUESTION INFO

QUESTION

Generate all combinations of well-formed parentheses. A parenthesis combination is well-formed if each opening parenthesis "(" is closed by a matching closing parenthesis ")" in the correct order. For example, "()", "()()", and "((()))" are all combinations of well-formed parentheses, while ")(", "((" and "(()" are not. Examples n = 1 output: ["()"] n = 2 output: ["(())", "()()"] n = 3 output: ["((()))", "(()())", "(())()", "()(())", "()()()"] n = 4 output should contain: ["(((())))", "((()()))", "((())())", "((()))()", "(()(()))", "(()()())", "(())(())", "(())()()", "()((()))", "()(()())", "()(())()", "()()(())", "()()()()"]The problem is to generate all combinations of well-formed parentheses given n pairs of parentheses. A combination is well-formed if every opening parenthesis has a corresponding closing parenthesis and they are properly nested. Recursive Approach The most common and intuitive way to solve this problem is by using recursion with backtracking. Here's a step-by-step breakdown of the approach: Define the Problem: We need to build strings of length 2n using n opening (and n closing) parentheses. At any point in the string, the number of closing parentheses should not exceed the number of opening parentheses to ensure the sequence is valid. Recursive Function: We use a recursive helper function to build the strings character by character. The function tracks the remaining number of opening and closing parentheses that can still be added. It also

EXAMPLES

Example 1	
Input	`3`
Output	`['((()))', '(()())', '(())()', '()(())', '()()()']`
Why	`Five well-formed combinations of three pairs.`

Example 2	
Input	`1`
Output	`['()']`
Why	`One pair.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Generate all 2^(2n) strings and keep the balanced ones

BRUTE-FORCE IDEA

Enumerate every sequence of n opens and n closes and test balance. Correct; it builds exponentially many invalid strings that backtracking never creates.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Generate all 2^(2n) strings and keep the balanced ones
Enumerate every sequence of n opens
n closes
test balance
```

BRUTE-FORCE COMPLEXITY | Time O(2^(2n) * n) | Space O(2^(2n))

WHY THIS IS WEAKER | Builds exponentially many invalid strings. Backtracking only adds '(' while fewer than n are used and ')' while fewer than the opens, so it builds only valid prefixes (the Catalan number of results).

HOW TO IMPROVE IT

choices are '(' and ')'; allowed when opened < pair_count, and ')' only when closed < opened; complete at length 2n

SHORT BEST SOLUTION

Track how many '(' and ')' have been placed. Add '(' if open < pair_count; add ')' if close < open. When the string reaches length 2n it is valid. Only valid prefixes are extended, so no work is wasted; the number of results is the Catalan number C(pair_count) = (2n choose pair_count) / (pair_count + 1).

```python
# BASE QUESTION (optimal): Backtracking with open and
# close counters
# Idea: Track how many '(' and ')' have been placed. Add
# '(' if open < pair_count; add ')' if close < open. When the
# string reaches length 2n it is valid. Only valid
# prefixes are extended, so no work is wasted; the number
# of results is the Catalan number C(pair_count) = (2n choose pair_count) /
# (pair_count + 1).
def generateParenthesis(pair_count):
    results, current = [], []

    def go(opened, closed):
        if len(current) == 2 * pair_count:
            results.append("".join(current))
            return
        if opened < pair_count:
            current.append("(")
            go(opened + 1, closed)
            current.pop()
        if closed < opened:
            current.append(")")
            go(opened, closed + 1)
            current.pop()

    go(0, 0)
    return results
```

COMPLEXITY | Time O(4^n / sqrt(n)) | Space O(n) beyond the output

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 61

62 * Permutations * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the template as written

QUESTION INFO

QUESTION

Given an array of distinct integers, return all possible permutations of the array (every rearrangement of its elements), in any order.

Function signature

def permute(nums: list) -> list

EXAMPLES

Example 1	
Input	`[1, 2, 3]`
Output	`[[1, 2, 3], [1, 3, 2], [2, 1, 3], [2, 3, 1], [3, 1, 2], [3, 2, 1]]`
Why	`Six orders of three numbers.`

Example 2	
Input	`[0, 1]`
Output	`[[0, 1], [1, 0]]`
Why	`Two numbers have two orders.`

Example 3	
Input	`[1]`
Output	`[[1]]`
Why	`One number has a single permutation.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Generate all n^n sequences and keep the valid ones

BRUTE-FORCE IDEA

Enumerate every sequence of length n over the values and filter those with all values distinct. It shows the idea but wastes almost all of its work (n^n versus n!).

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Generate all n^n sequences and keep the valid ones
Enumerate every sequence of length n over the values
filter those with all values distinct
```

BRUTE-FORCE COMPLEXITY | Time O(n^n * n) | Space O(n!)

WHY THIS IS WEAKER | N^n sequences when only n! Are valid, so almost all the work is wasted. Backtracking with a used marker only builds valid prefixes, O(n * n!).

HOW TO IMPROVE IT

the template as written

SHORT BEST SOLUTION

Fill positions left to right. At each position try every unused value, mark it, recurse, and unmark. Only valid prefixes are ever extended, so every step leads to a permutation.

```python
# BASE QUESTION (optimal): Backtracking with a used array
# Idea: Fill positions left to right. At each position try
# every unused value, mark it, recurse, and unmark. Only
# valid prefixes are ever extended, so every step leads to
# a permutation.
def permute(nums):
```

```
    results, current, used = [], [], [False] * len(nums)

    def go():
        if len(current) == len(nums):
            results.append(current[:])
            return
        for index, value in enumerate(nums):
            if not used[index]:
                used[index] = True
                current.append(value)
                go()
                current.pop()
                used[index] = False

    go()
    return results
```

COMPLEXITY | Time O(n * n!) | Space O(n) beyond the output

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 62

Runs and groups: extend while it fits, restart or close when it does not

PATTERN CONTEXT

Walk the sequence once with a counter. If the next item fits the rule, the run grows; if not, the run restarts (or the group closes). Longest run, first break, number of groups, and sums over runs are all small changes to what you record.

66 * Longest Non-Decreasing Subarray with One Replacement * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: left[index] and right[index] run lengths; the best answer joins the run before and after one changed element

QUESTION INFO

QUESTION

Find the maximum length of a contiguous non-decreasing subarray if at most one element may be changed to any value. Some rounds start from the plain longest-increasing-subarray question and add the one-replacement follow-up; those versions ask for an increasing run, so clarify strict vs non-strict.

Input: an integer array nums.
You may change at most one element to any value.
Return the maximum length of a contiguous subarray that is non-decreasing (nums[i] <= nums[i + 1] for every adjacent pair inside it) after the change.

Variant: other rounds ask for a strictly increasing run and open with the no-replacement version (longest increasing subarray) before adding the one-replacement follow-up — clarify strict vs non-strict before coding.

Expect to discuss and compare the complexity of several approaches; a brute-force first pass is usually followed by a request to optimize.

EXAMPLES

```
nums = [1, 0, 3, 4, 5, 2, 3]
Output: 5
Change nums[1] from 0 to 1 -> [1, 1, 3, 4, 5, 2, 3]
[1, 1, 3, 4, 5] is a non-decreasing subarray of length 5.
```

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every change and re-scan

BRUTE-FORCE IDEA

For every position and every candidate value, rewrite that element and find the longest non-decreasing run. This takes cubic time.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every change and re-scan
For every position
every candidate value
rewrite that element
find the longest non-decreasing run
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^3) | Space O(n)

WHY THIS IS WEAKER | O(n^3). The run lengths on both sides of each position, computed once, answer every change in O(1).

HOW TO IMPROVE IT

left[index] and right[index] run lengths; the best answer joins the run before and after one changed element

SHORT BEST SOLUTION

left[index] and right[index] are the non-decreasing run lengths ending and starting at i. Changing values[index] gives left[index-1] + 1 + right[index+1] if values[index-1] <= values[index+1], otherwise max(left[index-1], right[index+1]) + 1. The ends need only one side. Because changing a value to itself is allowed, this also covers the no-change answer.

```
# BASE QUESTION (optimal): Left and right run lengths
# Idea: left[index] and right[index] are the non-decreasing run
# lengths ending and starting at i. Changing values[index] gives
# left[index-1] + 1 + right[index+1] if values[index-1] <= values[index+1],
# otherwise max(left[index-1], right[index+1]) + 1. The ends need
# only one side. Because changing a value to itself is
# allowed, this also covers the no-change answer.
def longest_with_one_change(values):
    length = len(values)
    if length <= 2:
        return length
    left, right = [1] * length, [1] * length
    for index in range(1, length):
        if values[index - 1] <= values[index]:
            left[index] = left[index - 1] + 1
    for index in range(length - 2, -1, -1):
        if values[index] <= values[index + 1]:
            right[index] = right[index + 1] + 1
    best = 1
    for index in range(length):
        run_before = left[index - 1] if index else 0
        run_after = right[index + 1] if index + 1 < length else 0
        best = max(best, run_before + 1 + run_after if (not run_before or not run_after or values[index - 1] <=
 values[index + 1]) else max(run_before, run_after) + 1)
    return best
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 66

67 * Piano Hand Movement Grouping * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: start a new group when adding the note would make max - min exceed k

QUESTION INFO

QUESTION

A pianist plays a sequence of notes in order (a bigger number is a higher key). The hand can stay in one position while every note it plays lies within a span of k (highest minus lowest at most k). Starting a new hand position costs a move. Return the smallest number of hand positions needed to play all the notes in order.

Function signature

def groups(notes: list, k: int) -> int

EXAMPLES

Example 1	
Input	`[1, 3, 5, 7, 9, 11], 4`
Output	`2`
Why	`[1, 3, 5] fits in a span of 4, then [7, 9, 11]: 2 positions.`

QUESTION

Example 2	
Input	`[1, 2, 3, 4, 5], 4`
Output	`1`
Why	`All five notes fit in a span of 4: one position.`

Example 3	
Input	`[5, 9, 1], 4`
Output	`2`
Why	`5 and 9 fit together, then 1 is too far from them: 2 positions.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | DP over every way to cut the notes into consecutive groups

BRUTE-FORCE IDEA

Best[j] is the fewest groups for the first j notes: try every start i of the last group and require max - min of notes[i:j] to be at most k. Quadratic but easy to trust.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: DP over every way to cut the notes into consecutive groups
Best[j] is the fewest groups for the first j notes: try every start i of the last group
require max - min of notes[i:j] to be at most k
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Every end tries every start and re-checks the range, O(n^2). A greedy pass that starts a new group only when the next note no longer fits is O(n).

HOW TO IMPROVE IT

start a new group when adding the note would make max - min exceed k

SHORT BEST SOLUTION

Keep lo and hi of the current group. Extend while highest - lowest stays within k, otherwise close the group and start a new one at this note. Cutting as late as possible cannot be worse because any later cut leaves a suffix that is a subset of what an earlier cut would have to cover.

```python
# BASE QUESTION (optimal): Greedy scan, tracking the
# group's smallest and largest note
# Idea: Keep lo and hi of the current group. Extend while
# highest - lowest stays within k, otherwise close the group and
# start a new one at this note. Cutting as late as
# possible cannot be worse because any later cut leaves a
# suffix that is a subset of what an earlier cut would
# have to cover.
def groups(notes, max_spread):
    if not notes:
        return 0
    count, lowest, highest = 1, notes[0], notes[0]
    for note in notes[1:]:
        new_lowest, new_highest = min(lowest, note), max(highest, note)
        if new_highest - new_lowest <= max_spread:
            lowest, highest = new_lowest, new_highest
        else:
            count += 1
            lowest = highest = note
```

```
    return count
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 67

68 * Find longest increasing contiguous subarray * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: fits = values[index] > values[index-1]; also keep a left run and a right run per index so one change can join two runs

QUESTION INFO

QUESTION

Given an array you may change at most one element to any value (an integer when integers=True, any real number otherwise). Return the length of the longest strictly increasing contiguous subarray you can obtain.

Function signature

def longest_one_change(a: list, integers: bool = True) -> int

EXAMPLES

Example 1	
Input	`[1, 2, 3, 2, 5, 6]`
Output	`6`
Why	`Change the second 2 to 4 and the whole array 1, 2, 3, 4, 5, 6 increases.`

Example 2	
Input	`[1, 3, 2, 4]`
Output	`3`
Why	`Between 1 and 2 there is no integer, so the array 1, 3, 2, 4 only improves to a run of 3.`

Example 3	
Input	`[1, 3, 2, 4], integers = false`
Output	`4`
Why	`With real numbers, 3 can become 1.5 and the whole array increases.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Test every subarray for 'increasing after at most one change'

BRUTE-FORCE IDEA

For each subarray count the places where an element is not larger than its predecessor, and check whether one change can fix it (one such spot and the neighbours leave room, or none). Exact and slow.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Test every subarray for 'increasing after at most one change'
For each subarray count the places where an element is not larger than its predecessor
check whether one change can fix it (one such spot
the neighbours leave room
or none)
```

BRUTE-FORCE COMPLEXITY | Time O(n^3) | Space O(1)

WHY THIS IS WEAKER | O(n^2) subarrays, each tested in O(n): O(n^3). Run lengths ending and starting at each index, then trying each index as the changed element, is O(n).

HOW TO IMPROVE IT

fits = values[index] > values[index-1]; also keep a left run and a right run per index so one change can join two runs

SHORT BEST SOLUTION

left[index] is the length of the strictly increasing run ending at i and right[index] the length of the one starting at i. If you change values[index], you can attach the run before it and the run after it when values[index + 1] - values[index - 1] >= 2 (there is an integer strictly between), giving left[index - 1] + 1 + right[index + 1]. Otherwise you can only extend one side: left[index - 1] + 1 or right[index + 1] + 1. Also consider the unchanged runs.

```
# BASE QUESTION (optimal): Left and right run lengths, try
# every position as the changed element
# Idea: left[index] is the length of the strictly increasing
# run ending at i and right[index] the length of the one
# starting at i. If you change values[index], you can attach the
# run before it and the run after it when values[index + 1] - values[index -
# 1] >= 2 (there is an integer strictly between), giving
# left[index - 1] + 1 + right[index + 1]. Otherwise you can only
# extend one side: left[index - 1] + 1 or right[index + 1] + 1.
# Also consider the unchanged runs.
def longest_one_change(values):
    length = len(values)
    if length == 0:
        return 0
    left, right = [1] * length, [1] * length
    for index in range(1, length):
        if values[index] > values[index - 1]:
            left[index] = left[index - 1] + 1
    for index in range(length - 2, -1, -1):
        if values[index] < values[index + 1]:
            right[index] = right[index + 1] + 1
    best = max(left)
    for index in range(length):
        if index > 0:
            best = max(best, left[index - 1] + 1)
        if index < length - 1:
            best = max(best, right[index + 1] + 1)
        if 0 < index < length - 1 and values[index + 1] - values[index - 1] >= 2:
            best = max(best, left[index - 1] + 1 + right[index + 1])
    return min(best, length)
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 68

69 * Sum of Good Subarrays (Adjacent Diff ±1) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: keep the running length and running sum of the current up-run and down-run; add both to the total at every index

QUESTION INFO

QUESTION

MLE L5 coding round: sum over all contiguous subarrays whose elements form a strictly monotonic ±1 step sequence (every adjacent pair differs by exactly +1, OR every adjacent pair differs by exactly -1). Required time complexity is O(N).

QUESTION

Input: integer array arr of length N.
A subarray is good if its consecutive differences are either all +1 (monotone increasing by 1) or all -1 (monotone decreasing by 1).
Singleton subarrays are always good.
Output: the sum of element-sums across all good subarrays (each good subarray contributes the sum of its elements once).
Required: O(N) time.

EXAMPLES

```
arr = [3, 5, 6, 7, 6]
```

Good subarrays

```
  singletons:           [3], [5], [6], [7], [6]
  monotone +1 runs:     [5, 6], [6, 7], [5, 6, 7]
  monotone -1 runs:     [7, 6]
Not good: [3, 5] (diff 2), [6, 7, 6] (mixes +1 then -1).
```

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Enumerate every subarray

BRUTE-FORCE IDEA

Check every subarray's differences and add up its elements. Correct; quadratic or worse, but the problem asks for O(n).

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Enumerate every subarray
Check every subarray's differences
add up its elements
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) or O(n^3) | Space O(1)

WHY THIS IS WEAKER | O(n^2) subarrays (O(n^3) with the sums). Run totals per direction update in O(1) per element, so one pass is O(n).

HOW TO IMPROVE IT

keep the running length and running sum of the current up-run and down-run; add both to the total at every index

SHORT BEST SOLUTION

For an increasing run of length L ending at i let S be the sum of all suffix sums of the run. When the next element x extends it, S becomes S + (L + 1) * value and L becomes L + 1; a break restarts at S = value, L = 1. Do the same for decreasing runs. Add S_up + S_down - value at each index, because the singleton [value] appears in both totals.

```python
# BASE QUESTION (optimal): Two running run totals
# Idea: For an increasing run of length L ending at i let
# S be the sum of all suffix sums of the run. When the
# next element x extends it, S becomes S + (L + 1) * value and
# L becomes L + 1; a break restarts at S = value, L = 1. Do
# the same for decreasing runs. Add S_up + S_down - value at
# each index, because the singleton [value] appears in both
# totals.
def good_subarray_sum(values):
    total = 0
    up_total = up_length = down_total = down_length = 0
    for index, value in enumerate(values):
        step = value - values[index - 1] if index else None
        up_total, up_length = (up_total + (up_length + 1) * value, up_length + 1) if step == 1 else (value, 1)
        down_total, down_length = (down_total + (down_length + 1) * value, down_length + 1) if step == -1 else
(value, 1)
```

```
        total += up_total + down_total - value
    return total
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 69

70 * Sustained-Error Time Window * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: from the given second extend left and right while errors >= baseline; compare the time span with the interval

QUESTION INFO

QUESTION

Reconstructing the "two arrays - Error and Time" question. You are given two parallel arrays: times (timestamps in seconds, strictly increasing) and errors (the error reading recorded at each timestamp), where errors[i] was recorded at times[i] . You are also given three integers: second , interval , and baseline . Determine whether there exists a contiguous run of readings that:

1. Includes the reading taken at timestamp second (guaranteed to exist exactly once in times), 2. Has every reading in the run satisfy error >= baseline , and 3. Spans a duration of at least interval seconds — that is, times[right] - times[left] >= interval , where left and right are the endpoints of the run. Return True if such a run exists, otherwise False . Note that if the reading at second is itself below baseline , the answer is immediately False .

Function signature

```
def has_sustained_error_window(
    times: list[int],
    errors: list[int],
    second: int,
    interval: int,
    baseline: int,
) -> bool:
    """Return True if a run of readings all >= baseline includes second and spans >=
interval seconds."""
    ...
```

EXAMPLES

Example 1	
Input	`times = [1, 2, 3, 4, 5], errors = [10, 20, 5, 30, 40], second = 1, interval = 1, baseline = 8`
Output	`True`
Why	`Starting at time 1 (err 10 >= 8) the run extends to time 2 (err 20 >= 8) but stops at time 3 (err 5 < 8). The run covers times 1..2, a span of 1 >= interval 1.`

Example 2	
Input	`times = [1, 2, 3, 4, 5], errors = [10, 20, 5, 30, 40], second = 4, interval = 2, baseline = 8`
Output	`False`
Why	`Time 4 (err 30 >= 8) can only extend right to time 5 (err 40 >= 8); time 3 (err 5) blocks going left. The run covers times 4..5, span 1 < interval 2.`

QUESTION

Example 3	
Input	`times = [1, 2, 3], errors = [10, 4, 10], second = 2, interval = 0, baseline = 8`
Output	`False`
Why	`The reading at second=2 is 4 < baseline 8, so no valid run includes it.`

CONSTRAINTS

- 1 <= len(times) == len(errors) <= 10^5 times is strictly increasing; 0 <= times[i] <= 10^9
- -10^9 <= errors[i] <= 10^9 , baseline in the same range second is one of the values in times ; 0 <= interval <= 10^9

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(times) == len(errors) <= 10^5 times is strictly increasing; 0 <= times[i] <= 10^9
- -10^9 <= errors[i] <= 10^9 , baseline in the same range second is one of the values in times ; 0 <= interval <= 10^9

BRUTE FORCE | Try every pair of endpoints

BRUTE-FORCE IDEA

Start with: Try every pair of endpoints. Every candidate run is re-checked reading by reading.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every pair of endpoints
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^3) | Space O(1)

WHY THIS IS WEAKER | Every candidate run is re-checked reading by reading.

HOW TO IMPROVE IT

from the given second extend left and right while errors >= baseline; compare the time span with the interval

SHORT BEST SOLUTION

The longest valid run around the index is the best one, so only its span matters.

```python
# BASE QUESTION (optimal): Expand around the reading
# Idea: The longest valid run around the index is the best
# one, so only its span matters.
def has_sustained_error_window(times, errors, second, interval, baseline):
    index = times.index(second)
    if errors[index] < baseline:
        return False
    left = right = index
    while left > 0 and errors[left - 1] >= baseline:
        left -= 1
    while right + 1 < len(times) and errors[right + 1] >= baseline:
        right += 1
    return times[right] - times[left] >= interval
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 70

71 * Longest Non-Decreasing Subarray From Two Arrays * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: two run lengths per index, one for each array you may take the value from

QUESTION INFO

QUESTION

You are given two integer arrays nums1 and nums2 , both of length n . You must construct a third array nums3 of length n where, for each index i , nums3[i] is chosen to be either nums1[i] or nums2[i] (chosen independently per index). Return the length of the longest non-decreasing subarray (a contiguous run) achievable in nums3 over all valid choices.

Function signature

def longest_non_decreasing_subarray(nums1: list[int], nums2: list[int]) -> int:

...

Follow-up (as reported): (Reported) The interviewer first asked for the longest contiguous non-decreasing subarray of a single array, then generalized to picking per index from two arrays; also asked for the time complexity and to write test cases.

EXAMPLES

Example 1	
Input	`nums1 = [2, 3, 1], nums2 = [1, 2, 1]`
Output	`2`
Why	`Choosing nums3 = [2, 3, ...] gives the run [2, 3] of length 2. No choice yields a non-decreasing run of length 3.`

Example 2	
Input	`nums1 = [1, 3, 2, 1], nums2 = [2, 2, 3, 4]`
Output	`4`
Why	`nums3 = [1, 2, 3, 4] (pick nums1[0], nums2[1], nums2[2], nums2[3]) is non-decreasing over all 4 positions.`

Example 3	
Input	`nums1 = [1, 1], nums2 = [2, 2]`
Output	`2`
Why	`Any choice, e.g. [1, 1], is non-decreasing across both.`

CONSTRAINTS

- 1 <= n <= 10^5
- nums1 and nums2 have the same length n .
- 1 <= nums1[i], nums2[i] <= 10^9
- Target complexity: O(n) time, O(1) extra space.

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= n <= 10^5
- nums1 and nums2 have the same length n .
- 1 <= nums1[i], nums2[i] <= 10^9
- Target complexity: O(n) time, O(1) extra space.

BRUTE FORCE | Try every choice of arrays

BRUTE-FORCE IDEA

Start with: Try every choice of arrays. Start with: Try every choice of arrays. It builds all 2^n possible nums3 arrays.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every choice of arrays
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(2^n * n) | Space O(n)

WHY THIS IS WEAKER | It builds all 2^n possible nums3 arrays.

HOW TO IMPROVE IT

two run lengths per index, one for each array you may take the value from

SHORT BEST SOLUTION

dp1 and dp2 are the best run ending at i using nums1[index] or nums2[index].

```python
# BASE QUESTION (optimal): Two-state DP
# Idea: dp1 and dp2 are the best run ending at i using
# nums1[index] or nums2[index].
def longest_non_decreasing_subarray(nums1, nums2):
    dp1 = dp2 = best = 1
    for index in range(1, len(nums1)):
        new_dp1 = new_dp2 = 1
        for previous_value, previous_length in ((nums1[index - 1], dp1), (nums2[index - 1], dp2)):
            if previous_value <= nums1[index]:
                new_dp1 = max(new_dp1, previous_length + 1)
            if previous_value <= nums2[index]:
                new_dp2 = max(new_dp2, previous_length + 1)
        dp1, dp2 = new_dp1, new_dp2
        best = max(best, dp1, dp2)
    return best
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 71

72 * Count Super Streaks * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: sort by time; the same type within T continues the streak; when it breaks, count it if size >= min_events and span >= min_duration

QUESTION INFO

QUESTION

A player produces a list of events, each (event_id, event_type, event_time) . Sort events by event_time . A streak is a maximal run of events, consecutive in time order, in which every event has the same event_type AND the gap between each event and the previous one in the run is at most T seconds. A streak ends when the next event has a different event_type or its gap from the previous event exceeds T . A streak is a Super Streak if it contains at least N events AND spans at least X seconds (span = last event time minus first event time within the streak). Return the number of Super Streaks.

Function signature

```
def count_super_streaks(events: list[tuple[int, str, int]],
                        N: int, T: int, X: int) -> int:
    ...
```

QUESTION

Follow-up (as reported): Follow-up 1: events are tagged with user_id ; return a per-user count of Super Streaks. Follow-up 2: after processing, support count_streaks(user_id, time_A, time_B) returning how many of that user's Super Streaks completed entirely within [time_A, time_B] . Store each user's Super-Streak end-times sorted and answer with two binary searches in O(log n) per query.

EXAMPLES

Example 1	
Input	`events = [(1,"A",0),(2,"A",10),(3,"A",20),(4,"A",30),(5,"A",40)], N=4, T=10, X=30`
Output	`1`
Why	`One streak of 5 A-events, gaps all 10 (<=T), span 40 (>=30), size 5 (>=4). Super.`

Example 2	
Input	`events = [(1,"A",0),(2,"A",10),(3,"A",20),(4,"A",30),(5,"A",40), (6,"A",55),(7,"A",56),(8,"A",57)], N=4, T=10, X=30`
Output	`1`
Why	`The jump 40->55 (gap 15 > T) breaks the run. The first streak [0..40] is Super; the second [55,56,57] has only 3 events and span 2, so not Super.`

Example 3	
Input	`events = [(1,"A",0),(2,"B",5),(3,"A",10)], N=2, T=10, X=1`
Output	`0`
Why	`Type changes each step, so every streak has length 1 — none reaches N=2.`

CONSTRAINTS

- 0 <= len(events) <= 10^5 .
- N >= 1 , T >= 0 , X >= 0 ; times are non-negative integers, possibly with duplicate timestamps.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(events) <= 10^5 .
- N >= 1 , T >= 0 , X >= 0 ; times are non-negative integers, possibly with duplicate timestamps.

BRUTE FORCE | Build the streak lists, then test them

BRUTE-FORCE IDEA

It stores every streak as a list although only the size and the first and last time matter.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Build the streak lists, then test them
It stores every streak as a list although only the size
the first
last time matter
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n log n) | Space O(n)

WHY THIS IS WEAKER | It stores every streak as a list although only the size and the first and last time matter.

HOW TO IMPROVE IT

sort by time; the same type within T continues the streak; when it breaks, count it if size >= min_events and span >= min_duration

SHORT BEST SOLUTION

Keep only the open streak's start, last time, size and type.

```
# BASE QUESTION (optimal): One pass over the sorted events
# Idea: Keep only the open streak's start, last time, size
# and type.
def count_super_streaks(events, min_events, max_gap, min_duration):
    count = 0
    start = last_time = streak_type = None
    size = 0
    for _, event_type, event_time in sorted(events, key=lambda event: event[2]):
        if streak_type == event_type and event_time - last_time <= max_gap:
            size += 1
        else:
            if size >= min_events and last_time - start >= min_duration:
                count += 1
            start, streak_type, size = event_time, event_type, 1
        last_time = event_time
    if size >= min_events and last_time - start >= min_duration:
        count += 1
    return count
```

COMPLEXITY | Time O(n log n) | Space O(1) extra

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 72

Union-find: merge the pairs, then ask who is together

PATTERN CONTEXT

Keep a parent array. find climbs to the root (with path compression), union joins two roots. The only thing that changes between questions is where the pairs to merge come from and what you read at the end: same group, group size, number of groups, or the moment everything became one group.

73 * Count Islands in a Grid and Track the Count as Land Cells Are Added * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: each new land cell: groups += 1, then union it with every land neighbour (groups -= 1 per real merge); report groups after each addition

QUESTION INFO

QUESTION

A map is a grid with **`m`** rows and **`n`** columns in which every cell is land (**`1`**) or water (**`0`**). An island is a maximal group of land cells connected horizontally or vertically. Cells that touch only at a corner are not connected. You are given the initial map and a sequence of additions, each of which turns one cell into land. Report the number of islands before any addition, and again after each addition.

Function Signature

```
def island_counts(grid: list[list[int]], additions: list[tuple[int, int]]) -> list[int]:
```

Rules

- **`grid[r][c]`** is **`1`** for land and **`0`** for water, where **`r`** is the row index and **`c`** is the column index.
- An addition **`(r, c)`** turns cell **`(r, c)`** into land. If that cell is already land, either initially or because of an earlier addition, the map does not change.
- Additions are applied in the given order. Land never turns back into water.
- Return a list of length **`len(additions) + 1`**. Element **`0`** is the number of islands in the initial grid. For **`i >= 1`**, element **`i`** is the number of islands after the first **`i`** additions have been applied.

Constraints

- **`1 <= m <= 1000`**, **`1 <= n <= 1000`** and **`m * n <= 200000`**
- Every row of **`grid`** has exactly **`n`** entries, each **`0`** or **`1`**.
- **`0 <= len(additions) <= 200000`**
- Every addition **`(r, c)`** satisfies **`0 <= r < m`** and **`0 <= c < n`**. The same cell may appear more than once.
- Every count is at most **`m * n`**, so all values fit in a 32-bit signed integer.

Examples

Example 1

```
Input: grid = [[1, 1, 0, 0], [0, 0, 0, 1], [1, 0, 0, 1]], additions = []
Output: [3]
```

The islands are {(0, 0), (0, 1)}, {(1, 3), (2, 3)} and {(2, 0)}. With no additions, the only element is the initial count.

Example 2

```
Input: grid = [[0, 0, 0], [0, 0, 0], [0, 0, 0]], additions = [(0, 0), (0, 1), (1, 2), (2, 1), (1, 1), (1, 1)]
Output: [0, 1, 1, 2, 3, 1, 1]
```

QUESTION

The grid starts with no land. (0, 0) creates an island and (0, 1) joins it. (1, 2) and (2, 1) each touch no land, so each creates a new island. (1, 1) touches all three islands and merges them into one. The second (1, 1) is already land, so nothing changes.

Example 3

```
Input: grid = [[1, 1, 0, 0], [0, 0, 0, 1], [1, 0, 0, 1]], additions = [(0, 2), (0, 3), (1, 0)]
Output: [3, 3, 2, 1]
```

(0, 2) joins the top-left island. It touches (1, 3) only at a corner, so the count stays 3. (0, 3) connects the top-left island to the island at (1, 3) and (2, 3), giving 2. (1, 0) connects the merged island to (2, 0), giving 1.

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recount all islands after every addition

BRUTE-FORCE IDEA

Apply the addition and run a full flood-fill count again. Correct; every addition costs a pass over the entire grid.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recount all islands after every addition
Apply the addition
run a full flood-fill count again
```

BRUTE-FORCE COMPLEXITY | Time O(A * m * n) | Space O(m * n)

WHY THIS IS WEAKER | Each addition costs a pass over the whole grid, O(A * m * n). Union-Find merges the new cell with its land neighbours and updates the count in near-constant time.

HOW TO IMPROVE IT

each new land cell: groups += 1, then union it with every land neighbour (groups -= 1 per real merge); report groups after each addition

SHORT BEST SOLUTION

Build the initial islands with unions (count them), then for each addition: ignore it if the cell is already land; otherwise make it a new set (count + 1) and union it with each land neighbour, subtracting one from the count every time two different sets merge. Path compression and union by size make each step near-constant.

```python
# BASE QUESTION (optimal): Union-Find updated cell by cell
# Idea: Build the initial islands with unions (count
# them), then for each addition: ignore it if the cell is
# already land; otherwise make it a new set (count + 1)
# and union it with each land neighbour, subtracting one
# from the count every time two different sets merge. Path
# compression and union by size make each step near-
# constant.
def island_counts(grid, additions):
    rows, cols = len(grid), len(grid[0])
    parent = {}

    def find(cell):
        while parent[cell] != cell:
            parent[cell] = parent[parent[cell]]
            cell = parent[cell]
        return cell

    count = 0

    def add(row, col):
        nonlocal count
```

```
        if (row, col) in parent:
            return
        parent[(row, col)] = (row, col)
        count += 1
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if (next_row, next_col) in parent:
                root_a, root_b = find((row, col)), find((next_row, next_col))
                if root_a != root_b:
                    parent[root_a] = root_b
                    count -= 1

    for row in range(rows):
        for col in range(cols):
            if grid[row][col] == 1:
                add(row, col)
    counts = [count]
    for row, col in additions:
        add(row, col)
        counts.append(count)
    return counts
```

COMPLEXITY | Time O(m * n + A * alpha(m * n)) | Space O(m * n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 73

74 * 2D Points Connected Components by Radius * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: pairs = points within distance r, found by bucketing points into cells of size r and checking only the 9 nearby cells; answer = groups

QUESTION INFO

QUESTION

Given 2D points and radius **r**, connect two points when their distance is at most **r**; connectivity is transitive, and the output is the final number of clusters. The interviewer accepted or preferred a brute-force neighbor scan after discussing optimization ideas.

Input: a list of 2D points and a radius r.
Two points are connected if their Euclidean distance is <= r.
Connectivity is transitive: if A connects to B and B connects to C, all three belong to the same cluster.
Return the number of final clusters.

EXAMPLES

Example 1	
Input	`[(0, 0), (1, 0), (10, 10), (10, 11), (20, 20)], 2`
Output	`3`
Why	`Points within distance 2 of each other form a cluster: two near the origin, two near (10, 10), one alone.`

Example 2	
Input	`[(0, 0), (1, 0), (2, 0), (3, 0)], 1`
Output	`1`
Why	`A chain is one cluster even though its ends are far apart.`

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Compare every pair and run a DFS

BRUTE-FORCE IDEA

Compare every pair of points, build the neighbour lists and count components with a DFS. This is the direct approach the interviewer accepted, and it is fine for a few thousand points.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compare every pair and run a DFS
Compare every pair of points
build the neighbour lists
count components with a DFS
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Every pair is compared, O(n^2). Putting points in cells of side r means each point is compared only with the 3 x 3 cells around it.

HOW TO IMPROVE IT

pairs = points within distance r, found by bucketing points into cells of size r and checking only the 9 nearby cells; answer = groups

SHORT BEST SOLUTION

Put each point in the cell (point_x // radius, point_y // radius). Two points within distance r are in the same or adjacent cells, so only those 9 cells are compared. Union the pairs whose squared distance is <= radius^2 and count the roots. With radius = 0 only identical points are linked, which the same code handles by using cell size 1.

```python
from collections import defaultdict


# BASE QUESTION (optimal): Union-find with a spatial grid
# Idea: Put each point in the cell (point_x // radius, point_y // radius). Two
# points within distance r are in the same or adjacent
# cells, so only those 9 cells are compared. Union the
# pairs whose squared distance is <= radius^2 and count the
# roots. With radius = 0 only identical points are linked,
# which the same code handles by using cell size 1.
def count_components(points, radius):
    cell_size = radius if radius > 0 else 1
    parent = list(range(len(points)))
    def find(node):
        while parent[node] != node:
            parent[node] = parent[parent[node]]
            node = parent[node]
        return node
    cells = defaultdict(list)
    for point_index, (point_x, point_y) in enumerate(points):
        cells[(point_x // cell_size, point_y // cell_size)].append(point_index)
    for point_index, (point_x, point_y) in enumerate(points):
        cell_x, cell_y = point_x // cell_size, point_y // cell_size
        for cell_dx in (-1, 0, 1):
            for cell_dy in (-1, 0, 1):
                for other_index in cells.get((cell_x + cell_dx, cell_y + cell_dy), ()):
                    if other_index > point_index and (point_x - points[other_index][0]) ** 2 + (point_y - point
s[other_index][1]) ** 2 <= radius * radius:
                        parent[find(point_index)] = find(other_index)
    return len({find(point_index) for point_index in range(len(points))})
```

COMPLEXITY | Time about O(n) when points are spread out (O(n^2) if they all share a cell) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 74

75 * Count Clusters of 2D Points Within a Radius * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: identical to the line above: cell buckets give the candidate pairs, answer = groups

QUESTION INFO

QUESTION

Given 2D points and a radius r, two points are connected when their distance is at most r. A cluster is a group of points connected directly or through chains of other points. Return the number of clusters.

Function signature

def count_clusters(points: list, r: float) -> int

EXAMPLES

Example 1	
Input	`[(0, 0), (1, 0), (10, 10), (10, 11), (20, 20)], 2`
Output	`3`
Why	`Three groups: the two points near the origin, the two near (10, 10), and the lone point at (20, 20).`

Example 2	
Input	`[(0, 0), (1, 0), (2, 0)], 1`
Output	`1`
Why	`A chain: (0,0)-(1,0)-(2,0) are linked neighbour to neighbour, so it is one cluster.`

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Union-Find over all pairs of points

BRUTE-FORCE IDEA

Compare every pair, union the ones within r, and count the roots. Correct and simple; the quadratic pair loop is the cost.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Union-Find over all pairs of points
Compare every pair
union the ones within r
count the roots
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | N^2 / 2 distance checks. Hashing points into grid cells of side r means each point is compared only with points in its own and neighbouring cells.

HOW TO IMPROVE IT

identical to the line above: cell buckets give the candidate pairs, answer = groups

SHORT BEST SOLUTION

Put each point in the cell (point_x // radius, point_y // radius). A point within distance r of another can only be in the same cell or one of the 8 neighbouring cells, so compare only with those. Union the pairs within r and count the distinct roots.

```python
from collections import defaultdict


def _find_factory(point_count):
    parent = list(range(point_count))
```

```
    def find(node):
        while parent[node] != node:
            parent[node] = parent[parent[node]]
            node = parent[node]
        return node

    return parent, find


# BASE QUESTION (optimal): Grid hashing with cells of side
# r, then Union-Find
# Idea: Put each point in the cell (point_x // radius, point_y // radius). A
# point within distance r of another can only be in the
# same cell or one of the 8 neighbouring cells, so compare
# only with those. Union the pairs within r and count the
# distinct roots.
def count_clusters(points, radius):
    cells = defaultdict(list)
    for point_index, (point_x, point_y) in enumerate(points):
        cells[(point_x // radius, point_y // radius)].append(point_index)
    parent, find = _find_factory(len(points))
    for (cell_x, cell_y), members in cells.items():
        for cell_dx in (-1, 0, 1):
            for cell_dy in (-1, 0, 1):
                for neighbour_index in cells.get((cell_x + cell_dx, cell_y + cell_dy), ()):
                    for member_index in members:
                        if member_index < neighbour_index and (points[member_index][0] - points[neighbour_index
][0]) ** 2 + (points[member_index][1] - points[neighbour_index][1]) ** 2 <= radius * radius:
                            parent[find(member_index)] = find(neighbour_index)
    return len({find(point_index) for point_index in range(len(points))})
```

COMPLEXITY | Time O(n) expected for evenly spread points | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 75

76 * Find A Threshold-Limited Path With Minimum Required Safety * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: pairs = edges sorted from the highest safety down; union until find(source) == find(target); the answer is the safety of that last edge

QUESTION INFO

QUESTION

A graph has n nodes and edges (u, v, safety). The safety of a path is the smallest safety among its edges. Using only edges whose safety is at least threshold (no limit when threshold is None), return the best (largest) safety you can reach going from s to t, or -1 if no such path exists. A path from a node to itself has infinite safety.

Function signature

def best_path_safety(n: int, edges: list, s: int, t: int, threshold=None) -> int

EXAMPLES

Example 1	
Input	`3, [(0, 1, 5), (1, 2, 3), (0, 2, 1)], 0, 2`
Output	`3`
Why	`Direct edge 0-2 has safety 1; the route 0-1-2 has safety min(5, 3) = 3, which is better.`

QUESTION

Example 2	
Input	`3, [(0, 1, 5), (1, 2, 3), (0, 2, 1)], 0, 2, threshold = 4`
Output	`-1`
Why	`Only edges with safety at least 4 may be used: 0-1 is allowed but nothing leads to node 2.`

Example 3	
Input	`3, [(0, 1, 5)], 1, 1`
Output	`infinity`
Why	`Source and target are the same node.`

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every simple path

BRUTE-FORCE IDEA

Depth-first search over simple paths from the source, tracking the minimum edge safety, and keep the best at the target. It is exact but exponential.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every simple path
Depth-first search over simple paths from the source
tracking the minimum edge safety
keep the best at the target
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(path)

WHY THIS IS WEAKER | Exponential. Adding edges from strongest to weakest in a Union-Find until the source and target connect is O(E log E).

HOW TO IMPROVE IT

pairs = edges sorted from the highest safety down; union until find(source) == find(target); the answer is the safety of that last edge

SHORT BEST SOLUTION

Sort the edges by safety, strongest first, and union their endpoints one by one. The moment the source and the target become connected, the safety of the edge just added is the best possible minimum along a path. If they never connect return -1.

```python
# BASE QUESTION (optimal): Union-Find adding edges from
# strongest to weakest
# Idea: Sort the edges by safety, strongest first, and
# union their endpoints one by one. The moment the source
# and the target become connected, the safety of the edge
# just added is the best possible minimum along a path. If
# they never connect return -1.
def best_path_safety(node_count, edges, source, target):
    if source == target:
        return float("inf")
    parent = list(range(node_count))

    def find(node):
        while parent[node] != node:
            parent[node] = parent[parent[node]]
            node = parent[node]
        return node

    for from_node, to_node, safety in sorted(edges, key=lambda edge: -edge[2]):
```

```
        parent[find(from_node)] = find(to_node)
        if find(source) == find(target):
            return safety
    return -1
```

COMPLEXITY | Time O(E log E) | Space O(V)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 76

77 * Determine Reporting Relationships * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: pairs = peer links; keep the boss at the root and carry it over when two groups merge

QUESTION INFO

QUESTION

Build an in-memory organization model that supports three operations on employee IDs:

1. **`add_manager(a, b)`**: employee **a** is the direct manager of employee **b**.
2. **`add_peer(a, b)`**: employees **a** and **b** share the same direct manager.
3. **`is_manager(a, b)`**: return whether employee **a** is a manager of employee **b**. The answer should account for relationships that can be inferred from prior operations. For example, if **`add_manager(m, x)`** and **`add_peer(x, y)`** were previously recorded, then **`is_manager(m, y)`** should return **`true`**. You may assume the input is logically consistent and does not create invalid cycles in the org chart. Explain how you would design the data structures and implement these operations efficiently.

Implement an in-memory organization model that processes operations on employee IDs in order. Each operation is a tuple **`(op, a, b)`** where:

- **`("add_manager", a, b)`**: employee **a** becomes the direct manager of employee **b**.
- **`("add_peer", a, b)`**: employees **a** and **b** share the same direct manager.
- **`("is_manager", a, b)`**: ask whether employee **a** is a direct or indirect manager of employee **b**. Relationships can be inferred from earlier operations. For example, after **`add_manager(m, x)`** and **`add_peer(x, y)`**, **`is_manager(m, y)`** should be **`True`**. Process the operations sequentially and return a list of answers for all **`is_manager`** queries, in order. Notes:
- The input is logically consistent.
- No operation creates a cycle in the org chart.
- Repeating the same valid relationship is allowed.
- An employee is not considered their own manager.

Constraints
0 <= len(operations) <= 20000Employee IDs are integers in the range [-10^9, 10^9]At most 10000 distinct employee IDs appearThe sequence of operations is logically consistent and does not create cycles

EXAMPLES

```
Input: ([("add_manager", 1, 2), ("add_peer", 2, 3), ("is_manager", 1, 3)])
Expected Output: [True]
```

Explanation: Employee 2 reports to 1. Since 2 and 3 are peers, 3 also reports to 1.

```
Input: ([("add_manager", 2, 4), ("is_manager", 1, 4), ("add_manager", 1, 2), ("is_manager", 1, 4),
    ("is_manager", 2, 4), ("is_manager", 4, 2)])
Expected Output: [False, True, True, False]
```

QUESTION

Explanation: Before 2 gets a manager, 1 is not a manager of 4. After adding 1 as manager of 2, 1 becomes an indirect manager of 4. Hints**`add_peer(a, b)`** means **`a`** and **`b`** share one direct manager, but they are not interchangeable as managers. Group such employees with Union-Find.Once a peer group learns its direct manager, every current member of that group gets the same parent pointer. To answer **`is_manager(a, b)`**, walk upward from **`b`** through parent pointers.

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recompute all inferences from the stored facts on every query

BRUTE-FORCE IDEA

Keep the list of facts. For a query, repeatedly merge peers into groups and attach managers until nothing changes, then walk up from b. Correct; every query redoes the whole inference.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recompute all inferences from the stored facts on every query
Keep the list of facts
process every required item or candidate directly
```

BRUTE-FORCE COMPLEXITY | Time O(F * employees) per query | Space O(F)

WHY THIS IS WEAKER | Each query redoes the whole inference over all facts. Union-Find for peer groups with a manager stored on each group keeps every update near-constant and a query O(depth).

HOW TO IMPROVE IT

pairs = peer links; keep the boss at the root and carry it over when two groups merge

SHORT BEST SOLUTION

Each employee starts in its own group. add_peer unions the two groups (and keeps the manager, if either has one). add_manager sets the manager of b's group to a. is_manager(a, b) repeatedly takes the manager of the current employee's group and moves up; it is true if a appears in that chain.

```python
# BASE QUESTION (optimal): Union-Find for peer groups with
# a manager stored on each group
# Idea: Each employee starts in its own group. add_peer
# unions the two groups (and keeps the manager, if either
# has one). add_manager sets the manager of b's group to
# a. is_manager(a, b) repeatedly takes the manager of the
# current employee's group and moves up; it is true if a
# appears in that chain.
class Org:
    def __init__(self):
        self.parent, self.manager_of = {}, {}

    def _find(self, person):
        self.parent.setdefault(person, person)
        while self.parent[person] != person:
            self.parent[person] = self.parent[self.parent[person]]
            person = self.parent[person]
        return person

    def add_peer(self, peer_a, peer_b):
        root_a, root_b = self._find(peer_a), self._find(peer_b)
        if root_a != root_b:
            self.parent[root_b] = root_a
            if root_b in self.manager_of and root_a not in self.manager_of:
                self.manager_of[root_a] = self.manager_of[root_b]

    def add_manager(self, manager, employee):
        self._find(manager)
```

```
        self.manager_of[self._find(employee)] = manager

    def is_manager(self, manager, employee):
        seen, current = set(), employee
        while current not in seen:
            seen.add(current)
            root = self._find(current)
            if root not in self.manager_of:
                return False
            current = self.manager_of[root]
            if current == manager:
                return True
        return False
```

COMPLEXITY | Time O(alpha) per update, O(depth) per query | Space O(employees)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 77

78 * Determine whether two nodes are related * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: pairs = the given edges; the query is find(person_a) == find(person_b)

QUESTION INFO

QUESTION

Nodes 0 to n - 1 are linked by edges, and two nodes are related when a chain of edges joins them (links work in both directions; the graph may be disconnected and may contain cycles). Support union(a, b) to add a link and related(a, b) to answer the query.

Function signature

class Relations:
  def __init__(self, n: int, edges=())
  def union(self, a: int, b: int) -> None
  def related(self, a: int, b: int) -> bool

EXAMPLES

Example 1	

Input	`r = Relations(5, [(0, 1), (1, 2)])`
	`r.related(0, 2)`
	`r.related(0, 3)`
	`r.union(3, 4)`
	`r.related(3, 4)`
	`r.related(2, 4)`

Output	`r.related(0, 2) -> true`
	`r.related(0, 3) -> false`
	`r.related(3, 4) -> true`
	`r.related(2, 4) -> false`

Why	`0, 1, 2 form one group and 3, 4 another until they are joined; the two groups are never linked to each other.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Search from the first node for every query

BRUTE-FORCE IDEA

Build adjacency lists and run a DFS from a looking for b each time. Correct, and each query repeats the same traversal work.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Search from the first node for every query
Build adjacency lists
run a DFS from a looking for b each time
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(V + E) per query | Space O(V)

WHY THIS IS WEAKER | Every query repeats the same traversal, O(V + E) each. A Union-Find built once answers each query in near-constant time.

HOW TO IMPROVE IT

pairs = the given edges; the query is find(person_a) == find(person_b)

SHORT BEST SOLUTION

Union the endpoints of every relationship; two nodes are related exactly when they have the same root. Path compression and union by size keep every operation near constant time, and the structure also handles cycles and disconnected components without any visited set.

```python
# BASE QUESTION (optimal): Union-Find built once,
# constant-time queries
# Idea: Union the endpoints of every relationship; two
# nodes are related exactly when they have the same root.
# Path compression and union by size keep every operation
# near constant time, and the structure also handles
# cycles and disconnected components without any visited
# set.
class Relations:
    def __init__(self, person_count, edges):
        self.parent, self.size = list(range(person_count)), [1] * person_count
        for person_a, person_b in edges:
            self.union(person_a, person_b)

    def find(self, person):
        while self.parent[person] != person:
            self.parent[person] = self.parent[self.parent[person]]
            person = self.parent[person]
        return person

    def union(self, person_a, person_b):
        root_a, root_b = self.find(person_a), self.find(person_b)
        if root_a != root_b:
            if self.size[root_a] < self.size[root_b]:
                root_a, root_b = root_b, root_a
            self.parent[root_b] = root_a
            self.size[root_a] += self.size[root_b]

    def related(self, person_a, person_b):
        return self.find(person_a) == self.find(person_b)
```

COMPLEXITY | Time O(E alpha(V)) to build, O(alpha(V)) per query | Space O(V)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 78

79 * Movie Recommendations via Transitive Similarity * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: pairs = similar movies (use a dict instead of a list); group = every movie with the same root as the target, sorted by rating

QUESTION INFO

QUESTION

You are given a target movie name, a dictionary ratings mapping each movie to its numeric rating, an integer n , and a list similarities of [movieX, movieY] pairs meaning the two movies are directly similar. Similarity is transitive: if A is similar to B and B is similar to C, then A, B, and C all belong to the same similarity group. Return up to n recommended movies from the target's similarity group, excluding the target itself, ordered by rating descending (break ties alphabetically by name). If n == 1 , this returns the single highest- rated similar movie. If the group has fewer than n other movies, return all of them.

Function signature

```
def recommend(target: str, ratings: dict[str, float], n: int,
              similarities: list[list[str]]) -> list[str]:
    ...
```

EXAMPLES

Example 1	
Input	`target = "A", ratings = {"A": 8, "B": 5, "C": 9, "D": 7, "E": 6}, n = 1, similarities = [["A", "B"], ["B", "C"]]`
Output	`["C"]`
Why	`A, B, C form one group. Excluding A, the highest rated is C (9).`

Example 2	
Input	`target = "A", ratings = {"A": 8, "B": 5, "C": 9, "D": 7, "E": 6}, n = 2, similarities = [["A", "B"], ["B", "C"]]`
Output	`["C", "B"]`
Why	`Similar to A are {B (5), C (9)}; sorted by rating descending -> [C, B].`

Example 3	
Input	`target = "D", ratings = {"A": 8, "B": 5, "C": 9, "D": 7, "E": 6}, n = 3, similarities = [["A", "B"], ["B", "C"]]`
Output	`[]`
Why	`D is in no similarity pair, so its group is just {D}; excluding D leaves nothing.`

CONSTRAINTS

- Every movie mentioned appears as a key in ratings .
- 1 <= n ; ratings are numbers, names are distinct strings.

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- Every movie mentioned appears as a key in ratings .
- 1 <= n ; ratings are numbers, names are distinct strings.

BRUTE FORCE | Grow the group until nothing changes

BRUTE-FORCE IDEA

Start with: Grow the group until nothing changes. Every round rescans all pairs until the group stops growing.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Grow the group until nothing changes
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(P * rounds) | Space O(P)

WHY THIS IS WEAKER | Every round rescans all pairs until the group stops growing.

HOW TO IMPROVE IT

pairs = similar movies (use a dict instead of a list); group = every movie with the same root as the target, sorted by rating

SHORT BEST SOLUTION

The target's component is its group; rank it by rating.

```python
# BASE QUESTION (optimal): Union-find and a sort
# Idea: The target's component is its group; rank it by
# rating.
def recommend(target, ratings, how_many, similarities):
    parent = {}

    def find(movie):
        parent.setdefault(movie, movie)
        while parent[movie] != movie:
            parent[movie] = parent[parent[movie]]
            movie = parent[movie]
        return movie

    for movie_a, movie_b in similarities:
        parent[find(movie_a)] = find(movie_b)
    root = find(target)
    group = [movie for movie in list(parent) if movie != target and find(movie) == root]
    group.sort(key=lambda movie: (-ratings[movie], movie))
    return group[:how_many]
```

COMPLEXITY | Time O(P alpha + g log g) | Space O(P)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 79

GROUP 10 OF 43 | 7 QUESTIONS | PART A - PART 1

A dict of per-key state that you update, expire, or rank

PATTERN CONTEXT

Keep a dict from a key (message, user, word, activity) to the little bit of state you need: the last time seen, a Counter, a set, a list of timestamps. Every event updates one key's state; the rule that decides what to accept, expire or report is the only part that changes.

82 * Active Communication Top Users * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: a set of seen pairs, user_a Counter of partners per user, and buckets by count to read the top users in O(1)

QUESTION INFO

QUESTION

A stream of messages like 'A-B' means users A and B talked. Each distinct pair counts once (A-B and B-A are the same pair) and a pair of a user with themselves is ignored. A user's count is how many different partners they have. Support add(pair), most_active() (the set of users with the highest count) and top_k(k) (k users by count, ties broken by name).

Function signature

class Active:
    def add(self, pair: str) -> None
    def most_active(self) -> set
    def top_k(self, k: int) -> list

EXAMPLES

Example 1

Input	`a = Active()`
	`a.add('A-B')`
	`a.add('B-C')`
	`a.add('B-A')`
	`a.most_active()`
	`a.top_k(2)`

Output	`a.most_active() -> {'B'}` `a.top_k(2) -> ['B', 'A']`
Why	`B talked to A and C (2 partners). The repeated pair B-A adds nothing. A and C have 1 each, so the top 2 are B, then A (ties by name).`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recompute counts from all edges on every query

BRUTE-FORCE IDEA

Keep every edge and, for each query, rebuild the distinct-partner counts and sort the users. It works, but the work is repeated on each call.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recompute counts from all edges on every query
Keep every edge
for each query
rebuild the distinct-partner counts
```

sort the users

BRUTE-FORCE COMPLEXITY | Time O(E + U log U) per query | Space O(E)

WHY THIS IS WEAKER | Each query rebuilds all counts and sorts, O(E + U log U). Maintaining counts incrementally with count buckets gives O(1) updates and an O(1) most-active lookup.

HOW TO IMPROVE IT

a set of seen pairs, user_a Counter of partners per user, and buckets by count to read the top users in O(1)

SHORT BEST SOLUTION

A new edge increments the count of both users. Buckets map each count to the set of users with it, so a user moves from bucket c to bucket old_count + 1. Because counts never decrease, the highest non-empty bucket index only grows, which makes the most active users available in O(1). Top-K uses a heap over the count dict.

```
from collections import Counter
from collections import defaultdict
import heapq


def _norm(pair):
    user_a, user_b = pair.split("-")
    return (user_a, user_b) if user_a <= user_b else (user_b, user_a)


# BASE QUESTION (optimal): Edge set, counts per user,
# count buckets for the maximum
# Idea: A new edge increments the count of both users.
# Buckets map each count to the set of users with it, so a
# user moves from bucket c to bucket old_count + 1. Because counts
# never decrease, the highest non-empty bucket index only
# grows, which makes the most active users available in
# O(1). Top-K uses a heap over the count dict.
class Active:
    def __init__(self, count_self=False):
        self.edges, self.connection_count, self.buckets, self.highest_count, self.count_self = set(), Counter()
, defaultdict(set), 0, count_self

    def _bump(self, user):
        old_count = self.connection_count[user]
        if old_count:
            self.buckets[old_count].discard(user)
        self.connection_count[user] = old_count + 1
        self.buckets[old_count + 1].add(user)
        self.highest_count = max(self.highest_count, old_count + 1)

    def add(self, pair):
        user_a, user_b = _norm(pair)
        if (user_a == user_b and not self.count_self) or (user_a, user_b) in self.edges:
            return
        self.edges.add((user_a, user_b))
        self._bump(user_a)
        if user_a != user_b:
            self._bump(user_b)

    def top_k(self, how_many):
        return heapq.nsmallest(how_many, self.connection_count, key=lambda user: (-self.connection_count[user],
 user))

    def most_active(self):
        return set(self.buckets[self.highest_count]) if self.highest_count else set()
```

COMPLEXITY | Time O(1) per pair, O(U log k) for top-K, O(1) for the most active user | Space O(E)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 82

83 * Filter Repeated Robot Status Messages * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the same rule applied to a stream of (timestamp, message) pairs; keep what is accepted

QUESTION INFO

QUESTION

A robot status stream is a list of (timestamp, message). Show a message when it has never been shown, or when strictly more than gap seconds (default 10) have passed since it was last shown. Messages are compared exactly (case-sensitive), a hidden repeat does not restart the interval, and the shown messages keep their input order. Return the shown (timestamp, message) pairs.

Function signature

def filter_status(stream: list, gap: int = 10) -> list

EXAMPLES

Example 1	
Input	`[(0, 'ok'), (5, 'ok'), (10, 'ok'), (11, 'ok'), (11, 'OK')]`
Output	`[(0, 'ok'), (11, 'ok'), (11, 'OK')]`
Why	`'ok' at 5 and 10 are hidden (not more than 10 seconds after 0); at 11 it is 11 seconds later, so it shows; 'OK' is a different message.`

Example 2	
Input	`[(0, 'low'), (3, 'low'), (7, 'low')], gap = 2`
Output	`[(0, 'low'), (3, 'low'), (7, 'low')]`
Why	`A smaller gap lets repeats through sooner.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan the already shown messages for each new one

BRUTE-FORCE IDEA

Keep the list of shown (time, message) pairs and, for each new message, look backward for its last shown time. Correct; each message searches the whole history.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan the already shown messages for each new one
Keep the list of shown (time
message) pairs
for each new message
look backward for its last shown time
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Each message searches the whole history, O(n^2). A dictionary from message to its last displayed time makes each decision O(1).

HOW TO IMPROVE IT

the same rule applied to a stream of (timestamp, message) pairs; keep what is accepted

SHORT BEST SOLUTION

For each message in order: if it was never shown or now - last_shown > 10, emit it and set last_shown = now; otherwise drop it and leave last unchanged. One dictionary lookup per message.

```python
# BASE QUESTION (optimal): Dictionary from message to the
# time it was last shown
# Idea: For each message in order: if it was never shown
# or now - last_shown > 10, emit it and set last_shown = now;
# otherwise drop it and leave last unchanged. One
# dictionary lookup per message.
def filter_status(stream):
    last_shown, shown = {}, []
    for time, message in stream:
        if message not in last_shown or time - last_shown[message] > 10:
            last_shown[message] = time
            shown.append((time, message))
    return shown
```

COMPLEXITY | Time O(n) | Space O(distinct messages)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 83

84 * Next-Word Predictor * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: defaultdict(Counter): counts[word][following_word] += 1 for neighbouring words; remember the best next word per word

QUESTION INFO

QUESTION

Train on sentences, then predict the word that most often follows a given word. train(sentence) learns every pair of neighbouring words; predict(word) returns the word that followed it most often (ties go to the alphabetically smaller word) or None if nothing ever followed it. predict must run in O(1).

Function signature

class Predictor:
    def train(self, sentence: str) -> None
    def predict(self, word: str) -> str | None

EXAMPLES

Example 1

Input	`p = Predictor()`
	`p.train('I love apples')`
	`p.train('I love pears')`
	`p.train('you love pears too')`
	`p.predict('love')`
	`p.predict('I')`
	`p.predict('too')`

Output	`p.predict('love') -> 'pears'`
	`p.predict('I') -> 'love'`
	`p.predict('too') -> null`

Why	`After 'love' came 'apples' once and 'pears' twice, so 'pears' wins. Nothing ever followed 'too'.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan the whole corpus on every prediction

BRUTE-FORCE IDEA

Keep the sentences and count successors of the query word from scratch on each call. Simple and correct, but every query is as slow as one full pass over the data.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan the whole corpus on every prediction
Keep the sentences
count successors of the query word from scratch on each call
```

BRUTE-FORCE COMPLEXITY | Time O(corpus size) per prediction | Space O(corpus size)

WHY THIS IS WEAKER | Each prediction is a full pass over the corpus. Counting pairs while training and keeping the best successor per word makes prediction O(1).

HOW TO IMPROVE IT

defaultdict(Counter): counts[word][following_word] += 1 for neighbouring words; remember the best next word per word

SHORT BEST SOLUTION

counts[word][next] is incremented for every adjacent pair. After the increment compare that one count with the current best for the word (higher count wins, then the smaller word). Because only one count changed, no other successor can overtake the best, so one comparison keeps it correct.

```python
from collections import Counter
from collections import defaultdict


# BASE QUESTION (optimal): Counts plus a maintained best
# successor
# Idea: counts[word][next] is incremented for every
# adjacent pair. After the increment compare that one
# count with the current best for the word (higher count
# wins, then the smaller word). Because only one count
# changed, no other successor can overtake the best, so
# one comparison keeps it correct.
class Predictor:
    def __init__(self):
        self.counts = defaultdict(Counter)
        self.best = {}

    def train(self, sentence):
        words = sentence.split()
        for word, following_word in zip(words, words[1:]):
            self.counts[word][following_word] += 1
            current_best = self.best.get(word)
            if current_best is None or self.counts[word][following_word] > self.counts[word][current_best] or (
self.counts[word][following_word] == self.counts[word][current_best] and following_word < current_best):
                self.best[word] = following_word

    def predict(self, word):
        return self.best.get(word)
```

COMPLEXITY | Time O(1) per pair when training and per prediction | Space O(number of distinct pairs)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 84

85 * Batch and Streaming Log Deduplication * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: dict message -> first (or latest) timestamp depending on the policy; sort the result at the end

QUESTION INFO

QUESTION

Given log entries (timestamp, message), keep one entry per message and return the kept entries sorted by timestamp. With policy 'first' keep the entry seen first for each message; with policy 'latest' keep the one with the latest timestamp.

Function signature

def dedupe(logs: list, policy: str = 'first') -> list

EXAMPLES

Example 1	
Input	`[(5, 'a'), (1, 'b'), (9, 'a'), (3, 'c'), (2, 'b')]`
Output	`[(1, 'b'), (3, 'c'), (5, 'a')]`
Why	`Messages 'a' and 'b' each appear twice; the first one seen is kept.`

Example 2	
Input	`[(5, 'a'), (1, 'b'), (9, 'a'), (3, 'c'), (2, 'b')], policy = 'latest'`
Output	`[(2, 'b'), (3, 'c'), (9, 'a')]`
Why	`With 'latest', the later timestamp of each repeated message wins.`

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Quadratic scan for earlier or later duplicates

BRUTE-FORCE IDEA

For each log line, scan the rest (or the earlier lines) to see if it is the one to keep. Clear logic, quadratic cost.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Quadratic scan for earlier or later duplicates
For each log line
scan the rest (or the earlier lines) to see if it is the one to keep
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Each line scans the others: O(n^2). A dictionary from message to the kept entry is one pass, and a heap with lazy deletion serves the streaming version.

HOW TO IMPROVE IT

dict message -> first (or latest) timestamp depending on the policy; sort the result at the end

SHORT BEST SOLUTION

Part 1: One pass with a dict from message to the retained timestamp (first seen: keep the first; latest: replace when the new timestamp is larger), then sort the retained entries for the output.

Part 2: current[msg] holds the retained timestamp. add pushes (timestamp, msg) on the heap only when it becomes the retained value (new message, or a larger timestamp under the 'latest' policy), leaving the older heap entry behind. get_next pops entries until one matches current[msg] (the others are stale), returns it and removes the message.

```python
import heapq
```

```python
# PART 1: Keep one batch entry per message, then sort by time.
def dedupe(logs, policy="first"):
    kept = {}
    for timestamp, message in logs:
        if message not in kept or (policy == "latest" and timestamp > kept[message]):
            kept[message] = timestamp
    return sorted((timestamp, message) for message, timestamp in kept.items())


# PART 2: Stream entries in time order with lazy heap deletion.
class LogStream:
    def __init__(self, policy="latest"):
        self.policy = policy
        self.current = {}
        self.heap = []

    def add(self, timestamp, message):
        if (message not in self.current
                or self.policy == "latest" and timestamp > self.current[message]):
            self.current[message] = timestamp
            heapq.heappush(self.heap, (timestamp, message))

    def get_next(self):
        while self.heap:
            timestamp, message = heapq.heappop(self.heap)
            if self.current.get(message) == timestamp:
                del self.current[message]
                return timestamp, message
        return None
```

COMPLEXITY | Part 1: Time O(n log n) | Space O(n)
Part 2: Time O(log n) amortized per add and get_next | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 85

86 * Design a logger that returns top-K messages

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: dict message -> list of timestamps; top_k counts each list inside the time range with bisect, then takes the k largest

QUESTION INFO

QUESTION

Design a logger. log(ts, msg) records a message at time ts (timestamps arrive in non-decreasing order). top_k(k, start, end) returns the k most frequent messages whose timestamps fall in [start, end]; leave start and end out to use the whole history. Ties are broken alphabetically.

Function signature

class Logger:
    def log(self, ts: int, msg: str) -> None
    def top_k(self, k: int, start=None, end=None) -> list

EXAMPLES

QUESTION

Example 1

Input	`lg = Logger()`
	`lg.log(1, 'disk full')`
	`lg.log(2, 'timeout')`
	`lg.log(3, 'timeout')`
	`lg.log(4, 'disk full')`
	`lg.log(5, 'timeout')`
	`lg.top_k(1)`
	`lg.top_k(2, 4, 5)`
	`lg.top_k(2, 1, 1)`

Output	`lg.top_k(1) -> ['timeout']`
	`lg.top_k(2, 4, 5) -> ['disk full', 'timeout']`
	`lg.top_k(2, 1, 1) -> ['disk full']`

Why	`Overall 'timeout' occurred 3 times. In the window [4, 5] 'disk full' and 'timeout' occur once each (tie, alphabetical). The window [1, 1] only holds 'disk full'.`

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Keep every log entry and scan it for each query

BRUTE-FORCE IDEA

Store all (timestamp, message) pairs; for a query count the messages inside the window and sort. Correct; every query reads the whole log.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Keep every log entry and scan it for each query
Store all (timestamp
message) pairs
for a query count the messages inside the window
sort
```

BRUTE-FORCE COMPLEXITY | Time O(n + d log d) per query | Space O(n)

WHY THIS IS WEAKER | Every query reads the whole log, O(n + d log d). Per-message sorted timestamps with binary search answer a window count in O(log n), and a heap picks the top k.

HOW TO IMPROVE IT

dict message -> list of timestamps; top_k counts each list inside the time range with bisect, then takes the k largest

SHORT BEST SOLUTION

log appends the timestamp to the message's list (timestamps arrive in order, so lists stay sorted). A window count for one message is bisect_right(end) - bisect_left(start). top_k computes the counts for the d distinct messages and takes the K best with a heap, ordering by (-count, message).

```python
from bisect import bisect_left
from bisect import bisect_right
from collections import defaultdict
import heapq


# BASE QUESTION (optimal): Per-message sorted timestamps
# plus a heap for the top K
# Idea: log appends the timestamp to the message's list
# (timestamps arrive in order, so lists stay sorted). A
# window count for one message is bisect_right(end) -
# bisect_left(start). top_k computes the counts for the d
# distinct messages and takes the K best with a heap,
```

```
# ordering by (-count, message).
class Logger:
    def __init__(self):
        self.times = defaultdict(list)

    def log(self, timestamp, message):
        self.times[message].append(timestamp)

    def top_k(self, how_many, start=None, end=None):
        counts = {}
        for message, message_times in self.times.items():
            first_in_window = 0 if start is None else bisect_left(message_times, start)
            after_window = len(message_times) if end is None else bisect_right(message_times, end)
            if after_window > first_in_window:
                counts[message] = after_window - first_in_window
        return heapq.nsmallest(how_many, counts, key=lambda message: (-counts[message], message))
```

COMPLEXITY | Time O(1) per log, O(d log n + d log k) per query | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 86

87 * Website Activity Analytics: Unique Users + Session Time * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: dict activity -> set of users for unique counts; dict user -> sorted events to add up login-to-logout session time

QUESTION INFO

QUESTION

Each record has a user_id, a timestamp ('YYYY-MM-DD HH:MM:SS') and an activity. Return (1) the number of distinct users per activity and (2) the average number of seconds a user spends on the site. A user's time is the sum of the gaps from each 'login' to the next 'logout'; a login without a logout counts until the user's last event.

Function signature

def activity_stats(records: list) -> (dict, float)

EXAMPLES

Example 1	

Input	`rec = lambda u, t, a: {'user_id': u, 'timestamp': '2026-01-01 ' + t, 'activity': a}`
	`records = [rec(1, '10:00:00', 'login'), rec(1, '10:10:00', 'click'), rec(1, '10:30:00', 'logout'), rec(2, '10:05:00', 'login'), rec(2, '10:20:00', 'logout')]`
	`activity_stats(records)`

Output	`activity_stats(records) -> ({'click': 1, 'login': 2, 'logout': 2}, 1350)`
Why	`User 1 spent 30 minutes (1800 s), user 2 spent 15 minutes (900 s): the average is 1350 s. Both users logged in and out; only user 1 clicked.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Rescan all records for every user and every activity

BRUTE-FORCE IDEA

For each activity loop over all records to collect users, and for each user loop over all records again to compute the session time. Correct, but it rescans the data many times.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Rescan all records for every user and every activity
For each activity loop over all records to collect users
for each user loop over all records again to compute the session time
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(U * N + A * N) | Space O(U + A)

WHY THIS IS WEAKER | Rescans the whole log for every activity and every user. One pass that groups by activity and by user, then sorts each user's events once, is O(N log N).

HOW TO IMPROVE IT

dict activity -> set of users for unique counts; dict user -> sorted events to add up login-to-logout session time

SHORT BEST SOLUTION

Parse each record once. unique[activity] collects user ids in a set. Each user's events are collected and sorted by time; a login opens a session and the next logout closes it, and an open session ends at the user's last event. The average is the total seconds divided by the number of users (state the denominator you chose).

```python
from collections import defaultdict


# BASE QUESTION (optimal): One pass to group, then
# sessions per user
# Idea: Parse each record once. unique[activity] collects
# user ids in a set. Each user's events are collected and
# sorted by time; a login opens a session and the next
# logout closes it, and an open session ends at the user's
# last event. The average is the total seconds divided by
# the number of users (state the denominator you chose).
def activity_stats(records):
    from datetime import datetime
    unique, events = defaultdict(set), defaultdict(list)
    for record in records:
        unique[record["activity"]].add(record["user_id"])
        events[record["user_id"]].append((datetime.strptime(record["timestamp"], "%Y-%m-%d %H:%M:%S"), record["
activity"]))
    total = 0
    for user_events in events.values():
        user_events.sort()
        start = None
        for event_time, activity in user_events:
            if activity == "login" and start is None:
                start = event_time
            elif activity == "logout" and start is not None:
                total += (event_time - start).total_seconds()
                start = None
        if start is not None:
            total += (user_events[-1][0] - start).total_seconds()
    return {activity: len(users) for activity, users in sorted(unique.items())}, (total / len(events) if events
 else 0.0)
```

COMPLEXITY | Time O(N log N) | Space O(N)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 87

88 * Hit Counter * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: one deque of hit times; drop the ones older than 300 seconds; the answer is the length

QUESTION INFO

QUESTION

Design a hit counter. hit(timestamp) records a hit at that second, and get_hits(timestamp) returns the number of hits in the past 300 seconds (timestamps never go backwards).

Function signature

class QueueCounter:
    def hit(self, ts: int) -> None
    def get_hits(self, ts: int) -> int

EXAMPLES

Example 1

Input	`h = QueueCounter()`
	`h.hit(1)`
	`h.hit(2)`
	`h.hit(3)`
	`h.get_hits(4)`
	`h.hit(300)`
	`h.get_hits(300)`
	`h.get_hits(301)`

Output	`h.get_hits(4) -> 3`
	`h.get_hits(300) -> 4`
	`h.get_hits(301) -> 3`

Why	`At time 301 the hit at time 1 is exactly 300 seconds old and no longer counts.`

CONSTRAINTS

- 1 <= timestamp <= 2 * 10^9
- Timestamps are monotonically increasing

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= timestamp <= 2 * 10^9
- Timestamps are monotonically increasing

BRUTE FORCE | Store every hit and count in the window

BRUTE-FORCE IDEA

Append every timestamp to a list and count those with timestamp > now - 300. Correct, but memory grows forever and each query scans everything.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Store every hit and count in the window
Append every timestamp to a list
count those with timestamp > now - 300
```

BRUTE-FORCE COMPLEXITY | Time O(n) per getHits | Space O(n)

WHY THIS IS WEAKER | Memory grows with every hit forever, and each query scans them all, O(n). A queue that drops timestamps older than 300 seconds (or 300 per-second buckets) keeps both bounded.

HOW TO IMPROVE IT

one deque of hit times; drop the ones older than 300 seconds; the answer is the length

SHORT BEST SOLUTION

Keep a deque of timestamps in arrival order. getHits pops from the front while the oldest is outside the window, and the answer is the deque length. Each timestamp is pushed and popped once.

```python
from collections import deque


# BASE QUESTION (optimal): Queue of timestamps with
# eviction
# Idea: Keep a deque of timestamps in arrival order.
# getHits pops from the front while the oldest is outside
# the window, and the answer is the deque length. Each
# timestamp is pushed and popped once.
class HitCounter:
    def __init__(self):
        self.timestamps = deque()

    def hit(self, time):
        self.timestamps.append(time)

    def getHits(self, time):
        while self.timestamps and self.timestamps[0] <= time - 300:
            self.timestamps.popleft()
        return len(self.timestamps)
```

COMPLEXITY | Time O(1) amortized | Space O(hits in the window)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 88

Binary search on a yes/no condition

PATTERN CONTEXT

If answers go no, no, no, yes, yes, yes as the value grows, binary search finds the first yes. Write ok(x), set lo and hi, and halve. The questions only differ in what x is (a length, a speed, a time, an index, a version) and how ok(x) is checked.

90 * Cutting Ribbons (LC 1891) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: x = the piece length; ok(x) = the ribbons give at least k pieces; you want the LARGEST ok, so search the first x that fails and subtract 1

QUESTION INFO

QUESTION

You have ribbons of given lengths and an integer k. Each ribbon may be cut into pieces of one common integer length L (leftover bits are thrown away). Return the largest L that gives at least k pieces in total, or 0 if even length 1 cannot give k pieces.

Function signature

def max_length(ribbons: list, k: int) -> int

EXAMPLES

Example 1	
Input	`[9, 7, 5], 3`
Output	`5`
Why	`Length 5 gives 1 + 1 + 1 = 3 pieces, length 6 only 2.`

Example 2	
Input	`[5, 7], 12`
Output	`1`
Why	`Length 1 already gives 12 pieces.`

Example 3	
Input	`[1, 2], 10`
Output	`0`
Why	`Total length is 3 but 10 pieces are needed: impossible, so 0.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every length from the longest ribbon down

BRUTE-FORCE IDEA

For each length L from the maximum ribbon down to 1 count the pieces and return the first L that gives at least K. Correct, but with ribbons up to 10^9 it would try a billion lengths.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every length from the longest ribbon down
For each length L from the maximum ribbon down to 1 count the pieces
```

return the first L that gives at least K

BRUTE-FORCE COMPLEXITY | Time O(n * max) | Space O(1)

WHY THIS IS WEAKER | Tries up to max(ribbon) lengths, each costing O(n); with ribbons up to 10^9 that is a billion probes. The check is monotone, so binary search needs only log2(max) probes.

HOW TO IMPROVE IT

x = the piece length; ok(x) = the ribbons give at least k pieces; you want the LARGEST ok, so search the first x that fails and subtract 1

SHORT BEST SOLUTION

Part 1: If you can cut k pieces of length L you can cut k pieces of any shorter length, so feasibility is monotone. Binary search the largest L in [1, max(ribbons)] with sum(r // L) >= pieces_needed, and return 0 if even L = 1 fails.

Part 2: With real-valued lengths there is no 'next integer', so search the real line and stop after a fixed number of halvings (each halves the error) instead of looping on an epsilon comparison, which can fail to terminate near rounding noise. Feasibility uses floor(r / L) as before.

```
# BASE QUESTION (optimal): Binary search the answer
# (integer lengths)
# Idea: If you can cut k pieces of length L you can cut k
# pieces of any shorter length, so feasibility is
# monotone. Binary search the largest L in [1,
# max(ribbons)] with sum(r // L) >= pieces_needed, and return 0 if
# even L = 1 fails.
def maxLength(ribbons, pieces_needed):
    low_length, high_length, best_length = 1, max(ribbons), 0
    while low_length <= high_length:
        trial_length = (low_length + high_length) // 2
        if sum(ribbon // trial_length for ribbon in ribbons) >= pieces_needed:
            best_length, low_length = trial_length, trial_length + 1
        else:
            high_length = trial_length - 1
    return best_length
```

COMPLEXITY | Part 1: Time O(n log(max length)) | Space O(1)
Part 2: Time O(n * iterations) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 90

91 * First Bad Version with Parallel Search Follow-up * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: x = the version number; ok(x) = is_bad(x)

QUESTION INFO

QUESTION

Versions are numbered 1 to n and from some version on every version is bad. The function is_bad(v) tells whether version v is bad. Find the first bad version with as few calls to is_bad as possible.

Function signature

def first_bad_version(n: int, is_bad) -> int

EXAMPLES

Example 1	
Input	`bad_from = 4` `first_bad_version(10, lambda v: v >= bad_from)`
Output	`first_bad_version(10, lambda v: v >= bad_from) -> 4`
Why	`Versions 4 to 10 are bad, so 4 is the first; binary search needs about log2(10) calls.`

QUESTION

Example 2	
Input	`bad_from = 1` `first_bad_version(1, lambda v: v >= bad_from)`
Output	`first_bad_version(1, lambda v: v >= bad_from) -> 1`
Why	`With a single version that is bad, the answer is 1.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Check versions one by one

BRUTE-FORCE IDEA

Call isBadVersion for 1, 2, 3, and so on and stop at the first bad one. Correct; with a billion versions and a slow check it is far too many calls.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Check versions one by one
Call isBadVersion for 1
so on
stop at the first bad one
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) calls | Space O(1)

WHY THIS IS WEAKER | One call per version, O(n) calls; with a billion versions and a slow check that is far too many. The check is monotone, so binary search needs only log2(n) calls, and probing several points in parallel needs fewer rounds.

HOW TO IMPROVE IT

x = the version number; ok(x) = is_bad(x)

SHORT BEST SOLUTION

Part 1: isBad is monotone (all good, then all bad), so keep the invariant 'the first bad version is in [low, high]' and halve the interval. Use low + (high - low) // 2 in languages with fixed-width integers.

Part 2: Each round fires up to `workers` probes concurrently at evenly spaced points. Because the predicate is monotone, exactly one gap between consecutive probes contains the flip, so the interval shrinks by a factor of workers + 1 per round. This trades more total calls for fewer sequential rounds, which is the right trade when each probe is slow (a network call).

```python
# BASE QUESTION (optimal): Base problem: binary search
# Idea: isBad is monotone (all good, then all bad), so
# keep the invariant 'the first bad version is in [low,
# high]' and halve the interval. Use low + (high - low) // 2 in
# languages with fixed-width integers.
def first_bad_version(version_count, is_bad):
    low, high = 1, version_count
    while low < high:
        middle = low + (high - low) // 2
        if is_bad(middle):
            high = middle
        else:
            low = middle + 1
    return low
```

COMPLEXITY | Part 1: Time O(log n) calls | Space O(1)
Part 2: Time O(log_{k+1} n) rounds | Space O(k)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 91

92 * Solve Banana Speed and Interval Merge * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: x = the eating speed; ok(x) = total hours (ceil of pile / x summed) <= hours_allowed; first_true over 1..max pile

QUESTION INFO

QUESTION

During two technical phone screens for a new graduate software engineering role, the candidate was asked to solve two coding problems: minimum eating speed and merging overlapping intervals.

Constraints & Assumptions

- Provide clear algorithms, correctness reasoning, complexity, and edge cases.
- For minimum eating speed, each hour can be spent on only one pile.
- For interval merging, intervals overlap when the next start is less than or equal to the current merged end.
- Use sorted or binary-search approaches where appropriate.

Clarifying Questions

to Ask - Are **`piles`** and **h** always positive?

- Is **h** at least the number of piles?
- Are interval endpoints inclusive?
- Can intervals be empty or unsorted?

Part 1

- Minimum Eating Speed Given **`piles`** and **h**, return the minimum integer speed **k** such that all piles can be eaten within **h** hours.

What This Part Should Cover

- Binary search on answer from 1 to **`max(piles)`**.
- Feasibility function using sum of **`ceil(pile / k)`**.
- Why feasibility is monotonic.
- Complexity and edge cases.

Part 2

- Merge Overlapping Intervals Given intervals **`[start, end]`**, merge all overlapping intervals and return non-overlapping intervals sorted by start.

What This Part Should Cover

- Sort by start.
- Maintain current merged interval.
- Extend on overlap, otherwise append and start a new interval.
- Complexity and edge cases.

What a Strong

Answer Covers - Recognizes monotonic binary search for eating speed.

- Uses integer-safe ceiling division.
- Correctly defines interval overlap.
- Gives clean code and complexity for both problems.

Follow-up Questions

- How would you avoid floating-point division in the banana problem?
- What if **h** is smaller than the number of piles?
- How do you merge touching intervals like **`[1,3]`** and **`[3,5]`**?

QUESTION

- How would you handle streaming intervals?Minimum Eating Speed (Koko Eating Bananas)Koko loves to eat bananas. There are **n** piles of bananas; the i-th pile has **`piles[i]`** bananas. The guards will return in **h** hours. Koko can decide her constant bananas-per-hour eating speed **k**. Each hour she chooses one pile and eats **k** bananas from it. If that pile has fewer than **k** bananas, she eats all of them and does not eat more during that hour. An hour is spent on only one pile. Return the minimum integer speed **k** such that she can finish all the bananas within **h** hours. Example:
- **`piles = [3, 6, 7, 11]`**, **`h = 8`** -> **`4`**
- **`piles = [30, 11, 23, 4, 20]`**, **`h = 5`** -> **`30`**
- **`piles = [30, 11, 23, 4, 20]`**, **`h = 6`** -> **`23`** Constraints: **`1 <= piles.length <= h`**, **`1 <= piles[i] <= 10^9`**.

Constraints

1 <= piles.length <= h1 <= piles[i] <= 10^9h can be as large as 10^9

EXAMPLES

```
Input: ([3, 6, 7, 11], 8)
Expected Output: 4
Explanation: At k=4: ceil(3/4)+ceil(6/4)+ceil(7/4)+ceil(11/4) = 1+2+2+3 = 8 hours. k=3 needs 1+2+3+4 =
    10 > 8, so 4 is minimal.
Input: ([30, 11, 23, 4, 20], 5)
Expected Output: 30
```

Explanation: With exactly 5 piles and 5 hours, each pile must be eaten in one hour, so k must be at least max(piles) = 30.

Hints

If speed k finishes in time, any larger speed also finishes in time. This monotonicity lets you binary search on the answer.The answer lies in [1, max(piles)]: speed 1 is the slowest, and max(piles) finishes every pile in at most one hour.Hours needed at speed k is sum of ceil(pile / k). Use integer-safe ceiling: (pile + k - 1) // k to avoid floating point.Merge Overlapping IntervalsGiven an array of intervals where **`intervals[i] = [start_i, end_i]`**, merge all overlapping intervals and return an array of the non-overlapping intervals that cover all the intervals in the input, sorted by start. Two intervals overlap when the next start is less than or equal to the current merged end

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Bananas: try every speed upward

BRUTE-FORCE IDEA

For speed 1, 2, 3, and so on compute the hours needed and return the first speed that fits in h. Correct; the speed can be 10^9.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Bananas: try every speed upward
For speed 1
so on compute the hours needed
return the first speed that fits in h
```

BRUTE-FORCE COMPLEXITY | Time O(max * n) | Space O(1)

WHY THIS IS WEAKER | Up to max(piles) speeds, each costing O(n); piles can be 10^9. Hours are non-increasing in the speed, so binary search over speeds is O(n log max).

BRUTE FORCE 2 | Intervals: merge any overlapping pair until none is left

BRUTE-FORCE IDEA

Repeatedly find two overlapping intervals and replace them with their union. Correct; each pass is quadratic and several passes may be needed.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Intervals: merge any overlapping pair until none is left
Repeatedly find two overlapping intervals
replace them with their union
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Each pass compares all pairs, O(n^2), and several passes may be needed. Sorting by start and sweeping once is O(n log n).

HOW TO IMPROVE IT

x = the eating speed; ok(x) = total hours (ceil of pile / x summed) <= hours_allowed; first_true over 1..max pile

SHORT BEST SOLUTION

Part 1: Hours needed are non-increasing in the speed, so keep low_speed = 1 and high_speed = max(piles) and halve: if speed mid finishes within h, try slower, otherwise faster. Use ceiling division (an hour finishes at most one pile).

Part 2: Sort by start. If an interval starts at or before the end of the last merged one, extend that end with max; otherwise start a new interval.

```python
# Bananas: binary search the speed
# Idea: Hours needed are non-increasing in the speed, so
# keep low_speed = 1 and high_speed = max(piles) and halve: if speed mid
# finishes within h, try slower, otherwise faster. Use
# ceiling division (an hour finishes at most one pile).
def minEatingSpeed(piles, hours_allowed):
    low_speed, high_speed = 1, max(piles)
    while low_speed < high_speed:
        speed = (low_speed + high_speed) // 2
        if sum(-(-pile // speed) for pile in piles) <= hours_allowed:
            high_speed = speed
        else:
            low_speed = speed + 1
    return low_speed


# Intervals: sort and sweep
# Idea: Sort by start. If an interval starts at or before
# the end of the last merged one, extend that end with
# max; otherwise start a new interval.
def merge_intervals(intervals):
    merged = []
    for start, end in sorted(intervals):
        if merged and start <= merged[-1][1]:
            merged[-1][1] = max(merged[-1][1], end)
        else:
            merged.append([start, end])
    return merged
```

COMPLEXITY | Part 1: Time O(n log max) | Space O(1)
Part 2: Time O(n log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 92

93 * Given a list with response time for each server, calculate how long time it will take to process n requests. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: x = the elapsed time; ok(x) = sum(x // time_i) >= request_count requests

QUESTION INFO

QUESTION

Servers work in parallel and server i needs times[i] time units to process one request (optionally server i only starts at starts[i]). Return the earliest time at which n requests have been processed in total.

Function signature

def time_for(times: list, n: int, starts=None) -> int

EXAMPLES

Example 1	
Input	`[1, 2], 3`
Output	`2`
Why	`By time 2 the first server has done 2 requests and the second one: 3 in total.`

Example 2	
Input	`[3], 4`
Output	`12`
Why	`One server needs 3 units per request: 4 requests take 12.`

Example 3	
Input	`[2, 1], 6, starts = [0, 5]`
Output	`8`
Why	`The second server only starts at time 5, so the first one does most of the work.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Simulate: give each request to the server that frees up first

BRUTE-FORCE IDEA

Keep a heap of the time each server becomes free; for each request take the earliest and add its response time back. The answer is the largest finishing time. Correct, but it handles requests one at a time, so n = 10^9 is too slow.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Simulate: give each request to the server that frees up first
Keep a heap of the time each server becomes free
for each request take the earliest
add its response time back
```

BRUTE-FORCE COMPLEXITY | Time O(n log S) | Space O(S)

WHY THIS IS WEAKER | One heap operation per request, O(n log S); with n up to 10^9 requests that is too slow. The finish time is a monotone threshold, so binary search on T with sum(T // response time) >= n is O(S log answer).

HOW TO IMPROVE IT

x = the elapsed time; ok(x) = sum(x // time_i) >= request_count requests

SHORT BEST SOLUTION

In time T server i completes T // r_i requests, so the servers finish at least n requests exactly when sum(T // r_i) >= request_count, which is monotone in T. Search T between 0 and min(response_times) * request_count (one fastest server alone is enough).

```
# BASE QUESTION (optimal): Binary search on the total time
# Idea: In time T server i completes T // r_i requests, so
# the servers finish at least n requests exactly when
# sum(T // r_i) >= request_count, which is monotone in T. Search T
```

```python
# between 0 and min(response_times) * request_count (one fastest server alone
# is enough).
def time_for_requests(response_times, request_count):
    if request_count == 0:
        return 0
    low, high = 1, min(response_times) * request_count
    while low < high:
        time_limit = (low + high) // 2
        if sum(time_limit // response_time for response_time in response_times) >= request_count:
            high = time_limit
        else:
            low = time_limit + 1
    return low
```

COMPLEXITY | Time O(S log(min(times) * n)) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 93

GROUP 12 OF 43 | 5 QUESTIONS | PART A - PART 1

Greedy: scan once, carry the best choice so far, act only when forced

PATTERN CONTEXT

Decide locally and never look back: sort or scan, carry one value (how far you can reach, how far you are covered, the largest seen so far), and take an action only at the moment the current choice stops being enough.

100 * Visible People in a Line (Last Person, Then Everyone) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: scan from the end carrying the tallest person seen between; count a person when no taller one blocks the view

QUESTION INFO

QUESTION

Given heights of people in a line, A can see B if nobody between them is taller than both. Part 1: count how many people the last person can see. Part 2: return that count for every person. Written in a plain editor with no compilation.

Input: an array of heights representing a line of people.
Visibility rule: person A can see person B if no one standing between them is taller than both A and B.
Part 1: return how many people the last person in line can see.
Part 2: for the same array, return, for every person in the line, how many people they can see.

EXAMPLES

```
[1, 10, 6, 7, 9, 8, 2, 4, 3, 5] -> 6 (the last person, height 5, can see 3, 4, 2, 8, 9, 10).
```

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Test every pair

BRUTE-FORCE IDEA

For a pair (i, j) look at everyone between them and check that none is taller than both. Counting over all pairs is O(n^3). Useful for hand-tracing the example.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Test every pair
For a pair (i
j) look at everyone between them
check that none is taller than both
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^3) for everyone, O(n^2) for the last person | Space O(1)

WHY THIS IS WEAKER | Each pair rescans the people between it, O(n^3) for everyone and O(n^2) for the last person. One pass with a running maximum answers the last person in O(n), and a monotonic stack answers everyone in O(n).

HOW TO IMPROVE IT

scan from the end carrying the tallest person seen between; count a person when no taller one blocks the view

SHORT BEST SOLUTION

Part 1: Walk from the person next to the last one towards the front, keeping the largest height seen strictly between the last person and the current one. The current person is visible when that maximum is at most the larger of the two heights. Then fold the current height into the maximum.

Part 2: Count what each person sees to the right, then run the same code on the reversed line for the left. To the right of i: everyone up to and including the first strictly taller person t is visible, and after t only the people at least as tall as everyone from t

onward. next_at_least_as_tall[k] is the next person at least as tall as k, and chain[k] counts how far that chain runs, so the right count is (t - index) + chain[t], or the whole rest of the line when nobody is taller.

```python
# PART 1: Count the people visible to the last person.
def last_sees(heights):
    if not heights:
        return 0
    last = heights[-1]
    tallest_between = float("-inf")
    visible = 0
    for index in range(len(heights) - 2, -1, -1):
        if tallest_between <= max(last, heights[index]):
            visible += 1
        tallest_between = max(tallest_between, heights[index])
    return visible


# PART 2: Monotonic stacks handle every person, including equal heights.
def visible_to_right(heights):
    people_count = len(heights)
    next_at_least_as_tall = [None] * people_count
    next_taller = [None] * people_count
    stack = []
    for index in range(people_count - 1, -1, -1):
        while stack and heights[stack[-1]] < heights[index]:
            stack.pop()
        next_at_least_as_tall[index] = stack[-1] if stack else None
        stack.append(index)
    stack = []
    for index in range(people_count - 1, -1, -1):
        while stack and heights[stack[-1]] <= heights[index]:
            stack.pop()
        next_taller[index] = stack[-1] if stack else None
        stack.append(index)
    chain = [0] * people_count
    for index in range(people_count - 1, -1, -1):
        if next_at_least_as_tall[index] is not None:
            chain[index] = 1 + chain[next_at_least_as_tall[index]]
    answer = []
    for index in range(people_count):
        taller = next_taller[index]
        answer.append(people_count - 1 - index if taller is None else taller - index + chain[taller])
    return answer
```

COMPLEXITY | Part 1: Time O(n) | Space O(1)
Part 2: Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 100

101 * Bounded Knapsack Variant

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: sort items by value, highest first; take min(available count, remaining capacity) of each

QUESTION INFO

QUESTION

Each item type has a value and an available count; every item weighs 1. Given capacity W, maximize total value. DP or DFS + memo works at O(n·W), greedy by value after sorting is O(n log n), and the best answer uses quickselect for O(n).

Input: for each item type, its value and the number of copies available. Every item has weight 1.
Input: a maximum total weight W.
Output: the maximum total value achievable without exceeding W.

EXAMPLES

QUESTION

Example 1	
Input	`[(10, 2), (6, 3), (3, 5)], 4`
Output	`32`
Why	`Items are (value, available count), each weighs 1. Take both 10s and two 6s: 20 + 12 = 32.`

Example 2	
Input	`[(10, 2), (6, 3), (3, 5)], 7`
Output	`44`
Why	`A bigger capacity takes more items.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | DP over the capacity

BRUTE-FORCE IDEA

Treat every copy as its own 0/1 item and run the classic 1D knapsack table. It is correct for any weights, but it is far too slow when the counts or W are large.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: DP over the capacity
Treat every copy as its own 0/1 item
run the classic 1D knapsack table
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(total copies * W) | Space O(W)

WHY THIS IS WEAKER | Every copy is its own pass over the table, O(total copies * W). With unit weights the most valuable items are always the right ones, so sorting by value and taking greedily is O(n log n).

HOW TO IMPROVE IT

sort items by value, highest first; take min(available count, remaining capacity) of each

SHORT BEST SOLUTION

Part 1: Sort the item types by value, highest first. For each type take min(copies, slots left) and add value times that. Stop when the bag is full. Greedy is correct because all items weigh 1: any optimal bag can swap an item for a more valuable one without breaking the limit.

Part 2: Pick a random pivot value and split the types into higher, equal and lower. If the higher types alone fill the bag, recurse into them; otherwise take all of them, take as many equal ones as fit, and recurse into the lower ones with the slots that remain. Each round discards a constant fraction in expectation.

```python
# BASE QUESTION (optimal): Sort by value and take greedily
# Idea: Sort the item types by value, highest first. For
# each type take min(copies, slots left) and add value
# times that. Stop when the bag is full. Greedy is correct
# because all items weigh 1: any optimal bag can swap an
# item for a more valuable one without breaking the limit.
def max_value_sorted(items, capacity):
    total = 0
    for value, copies in sorted(items, reverse=True):
        take = min(copies, capacity)
        total += value * take
        capacity -= take
        if capacity == 0:
```

```
            break
    return total
```

COMPLEXITY | Part 1: Time O(n log n) | Space O(1) extra
Part 2: Time O(n) expected | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 101

102 * Find the Largest Monotone Increasing Number * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: scan the digits right to left; when a digit is smaller than the one before it, lower the earlier digit and fill 9s after it

QUESTION INFO

QUESTION

Given an integer n, return the largest number not greater than n whose digits never decrease from left to right (for example 1223 qualifies, 1232 does not).

Function signature

def largest_monotone(n: int) -> int

EXAMPLES

Example 1	
Input	**10**
Output	**9**
Why	**10 itself has a drop (1 then 0); the largest valid number below it is 9.**

Example 2	
Input	**1234**
Output	**1234**
Why	**1234 is already monotone.**

Example 3	
Input	**332**
Output	**299**
Why	**332: lower the first 3 to 2 and fill the rest with 9s: 299.**

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Decrement n until its digits are monotone

BRUTE-FORCE IDEA

Starting at n, subtract one until every digit is at least the one before it. Correct, but n can be 10^9, so it can take a billion steps.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Decrement n until its digits are monotone
Starting at n
subtract one until every digit is at least the one before it
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n * d) | Space O(1)

WHY THIS IS WEAKER | N can be 10^9, so it may take a billion steps. A right-to-left scan that lowers a digit where order breaks and fills the rest with 9s is O(digits).

HOW TO IMPROVE IT

scan the digits right to left; when a digit is smaller than the one before it, lower the earlier digit and fill 9s after it

SHORT BEST SOLUTION

Walk i from the last digit to the first. When digit[position - 1] > digit[position], decrease digit[position - 1] by one and remember i as the start of the nines. After the scan set every digit from the marked position to the end to 9. The scan goes right to left so a decrement that breaks the pair on its left is repaired on the next step.

```
# BASE QUESTION (optimal): Right-to-left scan, then fill
# with nines
# Idea: Walk i from the last digit to the first. When
# digit[position - 1] > digit[position], decrease digit[position - 1] by one
# and remember i as the start of the nines. After the scan
# set every digit from the marked position to the end to
# 9. The scan goes right to left so a decrement that
# breaks the pair on its left is repaired on the next
# step.
def monotoneIncreasingDigits(number):
    digits = list(str(number))
    nines_start = len(digits)
    for position in range(len(digits) - 1, 0, -1):
        if digits[position - 1] > digits[position]:
            digits[position - 1] = str(int(digits[position - 1]) - 1)
            nines_start = position
    for position in range(nines_start, len(digits)):
        digits[position] = "9"
    return int("".join(digits))
```

COMPLEXITY | Time O(d) | Space O(d)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 102

103 * Min boards to cover roof holes * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the template as written

QUESTION INFO

QUESTION

Holes in a roof are at integer positions along a line (the same position may be listed twice). A board of the given length covers that many consecutive positions. Return the fewest boards that cover every hole.

Function signature

def min_boards(holes: list, length: int) -> int

EXAMPLES

Example 1	
Input	[1, 2, 4, 7, 8], 3
Output	3
Why	Boards cover [1..3], [4..6] and [7..9].

Example 2	
Input	[1, 10], 5
Output	2
Why	The holes are far apart: one board each.

QUESTION

Example 3	
Input	`[5, 5, 5], 1`
Output	`1`
Why	`Repeated positions need only one board.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | DP over the number of holes covered

BRUTE-FORCE IDEA

Best[i] is the fewest boards that cover the first i holes: try every start j of the last board (it must cover holes j..i-1 within length L). It is exact but quadratic.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: DP over the number of holes covered
Best[i] is the fewest boards that cover the first i holes: try every start j of the last
board (it must cover holes j..i-1 within length L)
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Every i tries every j, O(n^2). Starting a board at the leftmost uncovered hole and skipping everything it covers is optimal and O(n log n) including the sort.

HOW TO IMPROVE IT

the template as written

SHORT BEST SOLUTION

Sort the holes. Place a board starting at the first uncovered hole, which covers every hole below start + board_length; skip those and repeat. Each board is placed as far right as it can be while still covering the leftmost uncovered hole, so no better placement exists (exchange argument).

```python
# BASE QUESTION (optimal): Greedy: start a board at the
# leftmost uncovered hole
# Idea: Sort the holes. Place a board starting at the
# first uncovered hole, which covers every hole below
# start + board_length; skip those and repeat. Each board is placed
# as far right as it can be while still covering the
# leftmost uncovered hole, so no better placement exists
# (exchange argument).
def min_boards(holes, board_length):
    boards, covered_until = 0, float("-inf")
    for hole in sorted(set(holes)):
        if hole > covered_until:
            boards += 1
            covered_until = hole + board_length - 1
    return boards
```

COMPLEXITY | Time O(n log n) | Space O(1) extra

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 103

104 * Jump Game II (Minimum Jumps) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: scan the indices carrying the farthest reach; when you hit the end of the current jump, you must jump again: jumps += 1, end = farthest

QUESTION INFO

QUESTION

You are given a 0-indexed array of non-negative integers nums . You start at index 0. From any index i , you can jump to any index in the range [i + 1, i + nums[i]] (i.e., nums[i] is the maximum forward jump length from i). Return the minimum number of jumps required to reach the last index len(nums)

- 1 . The test cases guarantee that the last index is always reachable.

Function signature

```
def jump(nums: list[int]) -> int:
    """Return the minimum number of jumps to reach the last index."""
    ...
```

Follow-up (as reported): Solve it in O(n) time and O(1) extra space with a greedy "current reach vs. farthest reach" scan rather than O(n) DP.

EXAMPLES

Example 1	
Input	`nums = [2, 3, 1, 1, 4]`
Output	`2`
Why	`Jump 1 step from index 0 to 1, then 3 steps to the last index.`

Example 2	
Input	`nums = [2, 3, 0, 1, 4]`
Output	`2`
Why	`0 -> 1 -> 4.`

Example 3	
Input	`nums = [0]`
Output	`0`
Why	`Already at the last index.`

CONSTRAINTS

- 1 <= len(nums) <= 10^4
- 0 <= nums[i] <= 1000
- If the array has length 1 you are already at the last index, so the answer is 0.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(nums) <= 10^4
- 0 <= nums[i] <= 1000
- If the array has length 1 you are already at the last index, so the answer is 0.

BRUTE FORCE | DP over the minimum jumps to each index

BRUTE-FORCE IDEA

Start with: DP over the minimum jumps to each index. Every index looks back at every earlier index.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: DP over the minimum jumps to each index
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Every index looks back at every earlier index.

HOW TO IMPROVE IT

scan the indices carrying the farthest reach; when you hit the end of the current jump, you must jump again: jumps += 1, end = farthest

SHORT BEST SOLUTION

Each jump extends the covered range to the farthest index seen so far.

```python
# BASE QUESTION (optimal): Greedy by ranges
# Idea: Each jump extends the covered range to the
# farthest index seen so far.
def jump(nums):
    jumps, current_end, farthest = 0, 0, 0
    for position in range(len(nums) - 1):
        farthest = max(farthest, position + nums[position])
        if position == current_end:
            # the range of the current jump is used up: take another
            jumps += 1
            current_end = farthest
    return jumps
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 104

GROUP 13 OF 43 | 3 QUESTIONS | PART A - PART 1

Sets and counters: remember the first collection, check the second against it

PATTERN CONTEXT

Build a set (or a Counter) from one collection, then walk the other and ask 'is this in it?'. Membership in a set is O(1), which is the whole trick. Duplicates, intersections, filters, and missing values are the same loop with a different question.

109 * Reconstruct original array from doubled shuffle * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: Counter of the values; go from the smallest, pairing each x with 2x and using both counts up

QUESTION INFO

QUESTION

An array of numbers was doubled: for every element x of the original, the value 2x was added, and then all the numbers were shuffled. Given the shuffled array changed, return an original array (in any order), or an empty list if changed cannot be a doubled array. Zero and negative numbers are possible.

Function signature

def find_original(changed: list) -> list

EXAMPLES

Example 1	
Input	`[1, 3, 4, 2, 6, 8]`
Output	`[1, 3, 4]`
Why	`Original [1, 3, 4] gives doubles [2, 6, 8].`

Example 2	
Input	`[1, 3, 2]`
Output	`[]`
Why	`An odd number of values cannot be doubled.`

Example 3	
Input	`[0, 0, 0, 0]`
Output	`[0, 0]`
Why	`Zeros pair with zeros.`

Example 4	
Input	`[-2, -1, -4, -2]`
Output	`[-1, -2]`
Why	`Negatives work too: -2 is the double of -1.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try to remove each smallest element together with its double from a list

BRUTE-FORCE IDEA

Sort by absolute value and, for each element still in the list, remove it and one copy of its double (list.remove is a linear search). Correct and short; the repeated linear searches make it quadratic.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try to remove each smallest element together with its double from a list
Sort by absolute value
for each element still in the list
remove it
one copy of its double (list.remove is a linear search)
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | List.remove is a linear search, so each step is O(n) and the loop is O(n^2). A Counter processed in order of absolute value is O(n log n).

HOW TO IMPROVE IT

Counter of the values; go from the smallest, pairing each x with 2x and using both counts up

SHORT BEST SOLUTION

Count the values. For each value x in order of absolute value with a remaining count, it needs count[value] copies of 2x: if count[2x] is smaller, fail; otherwise consume them and append x that many times. Zero is handled by the same rule because 2 * 0 = 0 (the check becomes 'the count is even' via consuming pairs).

```python
from collections import Counter


# BASE QUESTION (optimal): Counter processed in order of
# absolute value
# Idea: Count the values. For each value x in order of
# absolute value with a remaining count, it needs count[value]
# copies of 2x: if count[2x] is smaller, fail; otherwise
# consume them and append x that many times. Zero is
# handled by the same rule because 2 * 0 = 0 (the check
# becomes 'the count is even' via consuming pairs).
def findOriginalArray(changed):
    if len(changed) % 2:
        return []
    count, original = Counter(changed), []
    for value in sorted(count, key=lambda value: (abs(value), value)):
        if count[value] == 0:
            continue
        if value == 0:
            if count[0] % 2:
                return []
            original += [0] * (count[0] // 2)
            count[0] = 0
            continue
        if count[2 * value] < count[value]:
            return []
        original += [value] * count[value]
        count[2 * value] -= count[value]
        count[value] = 0
    return original
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 109

110 * Min deletions to avoid overlap in first k

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: Counter of the left part and of the right part; for each value in both, delete min(left count, right count)

QUESTION INFO

QUESTION

Split an array into its first k elements and the rest. Return the minimum number of elements to delete so that no value appears on both sides (for every shared value delete all copies from the side that has fewer).

Function signature

def min_deletions(a: list, k: int) -> int

EXAMPLES

Example 1	
Input	`[1, 2, 2, 3, 2, 1], 3`
Output	`2`
Why	`Left [1, 2, 2], right [3, 2, 1]: value 1 appears once on each side (delete 1), value 2 appears twice on the left and once on the right (delete 1): 2 deletions.`

Example 2	
Input	`[1, 2, 3, 4], 2`
Output	`0`
Why	`No value is shared between [1, 2] and [3, 4].`

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every set of deletions

BRUTE-FORCE IDEA

Enumerate subsets of elements to delete in increasing size, and return the first size that leaves the two parts disjoint. Exact, but exponential.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every set of deletions
Enumerate subsets of elements to delete in increasing size
return the first size that leaves the two parts disjoint
```

BRUTE-FORCE COMPLEXITY | Time O(2^n * n) | Space O(n)

WHY THIS IS WEAKER | 2^n sets to try: exponential. Each shared value must disappear from one side, so the cost is the sum over shared values of the smaller of its two counts, in O(n).

HOW TO IMPROVE IT

Counter of the left part and of the right part; for each value in both, delete min(left count, right count)

SHORT BEST SOLUTION

Count each value in the first k elements and in the rest. For every value with a positive count on both sides, one side has to lose all of its copies, so add the smaller count. Different values never affect each other.

```python
from collections import Counter


# BASE QUESTION (optimal): Sum of min(count in the first
# k, count in the rest) over shared values
```

```python
# Idea: Count each value in the first k elements and in
# the rest. For every value with a positive count on both
# sides, one side has to lose all of its copies, so add
# the smaller count. Different values never affect each
# other.
def min_deletions(values, prefix_length):
    left, right = Counter(values[:prefix_length]), Counter(values[prefix_length:])
    return sum(min(left[value], right[value]) for value in left.keys() & right.keys())
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 110

111 * Filter Common Elements Per Window (Two Lists) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: build a set per window of the first list; keep the matching window of the second list minus that set

QUESTION INFO

QUESTION

You are given two lists a and b of equal length n and a window size k that divides n . Split both lists into consecutive, non-overlapping windows of size k (window 0 is indices 0..k-1 , window 1 is k..2k-1 , and so on). For each window, remove from b 's window every value that also appears in the corresponding window of a . Concatenate the surviving elements of b , preserving their original order, and return that list.

Function signature

```
def filter_windows(a: list[int], b: list[int], k: int) -> list[int]:
    ...
```

Follow-up (as reported): The interviewer asked to do the removal in place / O(1) extra space. Naive in-place removal becomes O(n*k); discuss compaction with a write pointer to keep it O(n) time.

EXAMPLES

Example 1	
Input	`a = [1, 2, 3, 4, 5, 6], b = [3, 7, 8, 6, 9, 5], k = 3`
Output	`[7, 8, 9]`
Why	`Window 0: a={1,2,3}, b=[3,7,8] -> remove 3 -> [7,8]. Window 1: a={4,5,6}, b=[6,9,5] -> remove 6 and 5 -> [9]. Concatenate -> [7,8,9].`

Example 2	
Input	`a = [1, 1], b = [1, 2], k = 2`
Output	`[2]`
Why	`Single window: a={1}, b=[1,2] -> remove 1 -> [2].`

CONSTRAINTS

- 0 <= n <= 10**5 , n % k == 0 , 1 <= k <= n (for n > 0).
- Values are integers.

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= n <= 10**5 , n % k == 0 , 1 <= k <= n (for n > 0).

- Values are integers.

BRUTE FORCE | Scan a's window for every element

BRUTE-FORCE IDEA

Start with: Scan a's window for every element. Every element of b searches a whole window of a linearly.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan a's window for every element
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n * k) | Space O(n)

WHY THIS IS WEAKER | Every element of b searches a whole window of a linearly.

HOW TO IMPROVE IT

build a set per window of the first list; keep the matching window of the second list minus that set

SHORT BEST SOLUTION

Membership in the window of a is a set lookup.

```python
# BASE QUESTION (optimal): A set per window
# Idea: Membership in the window of a is a set lookup.
def filter_windows(first, second, window_size):
    kept = []
    for window_start in range(0, len(first), window_size):
        seen = set(first[window_start:window_start + window_size])
        kept.extend(value for value in second[window_start:window_start + window_size] if value not in seen)
    return kept
```

COMPLEXITY | Time O(n) | Space O(k)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 111

bisect on a sorted list: find a position, count a range, keep it sorted

PATTERN CONTEXT

bisect_left and bisect_right give the position of a value in a sorted list in O(log n). Subtract two positions to count a range, slice between them to list a range, insort to keep inserting in sorted order, and replace an element at a position for the 'longest increasing' trick.

114 * Sliding-Window Mean After Removing the Largest K Values * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: keep the window in a sorted list; insort the new value, remove the old one with bisect_left, and track the sum of the top-K region

QUESTION INFO

QUESTION

For every window of a fixed size over the array, remove the k largest values of the window (repeated values count separately) and compute the mean of the rest. Return the means in window order as reduced fractions written 'numerator/denominator'. Update the windows incrementally instead of sorting each one again.

Function signature

def trimmed_window_means(nums: list, window: int, k: int) -> list

EXAMPLES

Example 1	
Input	`[1, 2, 3, 4, 5], 3, 1`
Output	`['3/2', '5/2', '7/2']`
Why	`Window [1,2,3] without its largest value 3 averages (1+2)/2 = 3/2; then 5/2 and 7/2.`

Example 2	
Input	`[2, 4, 6, 8], 2, 0`
Output	`['3/1', '5/1', '7/1']`
Why	`With k = 0 nothing is removed: plain window means.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Sort every window and sum the kept values

BRUTE-FORCE IDEA

For each window sort a copy, drop the largest k values, and compute the exact fraction. Correct; every window is sorted from scratch.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Sort every window and sum the kept values
For each window sort a copy
drop the largest k values
compute the exact fraction
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n * w log w) | Space O(w)

WHY THIS IS WEAKER | Each window is sorted from scratch, O(n * w log w). A sorted list updated by one insertion and one removal, with the sum of the top k adjusted at the boundary, avoids it.

HOW TO IMPROVE IT

keep the window in a sorted list; insort the new value, remove the old one with bisect_left, and track the sum of the top-K region

SHORT BEST SOLUTION

Keep the window values in a sorted list, the total window sum, and top_sum = the sum of its K largest elements. Inserting x at position p: if p falls into the top K region, add x and subtract the element that was pushed out of it; removing is the mirror image. The mean of the kept values is (window sum - top_sum) / (window - remove_count), reduced with gcd.

```
from bisect import bisect_left
from bisect import bisect_right
import math


def _frac(numerator, denominator):
    divisor = math.gcd(abs(numerator), denominator)
    return f"{numerator // divisor}/{denominator // divisor}"


# BASE QUESTION (optimal): Sorted window with an
# incrementally maintained top-K sum
# Idea: Keep the window values in a sorted list, the total
# window sum, and top_sum = the sum of its K largest elements.
# Inserting x at position p: if p falls into the top K
# region, add x and subtract the element that was pushed
# out of it; removing is the mirror image. The mean of the
# kept values is (window sum - top_sum) / (window - remove_count),
# reduced with gcd.
def trimmed_window_means(nums, window, remove_count):
    window_sorted, window_sum, top_sum, means = [], 0, 0, []

    def insert(value):
        nonlocal window_sum, top_sum
        old_size = len(window_sorted)
        position = bisect_right(window_sorted, value)
        window_sorted.insert(position, value)
        window_sum += value
        if remove_count:
            if old_size < remove_count:
                top_sum += value
            elif position >= old_size + 1 - remove_count:
                top_sum += value - window_sorted[old_size - remove_count]                  # window_sorted[old
_size - remove_count] is the element pushed out of the top-K region

    def remove(value):
        nonlocal window_sum, top_sum
        old_size = len(window_sorted)
        position = bisect_left(window_sorted, value)
        window_sum -= value
        if remove_count:
            if old_size <= remove_count:
                top_sum -= value
            elif position >= old_size - remove_count:
                top_sum += window_sorted[old_size - remove_count - 1] - value               # the element just
 below the region joins it
        del window_sorted[position]

    for index, value in enumerate(nums):
        insert(value)
        if index >= window:
            remove(nums[index - window])
        if index >= window - 1:
            means.append(_frac(window_sum - top_sum, window - remove_count))
    return means
```

COMPLEXITY | Time O(n * (log w + w) in memmove, O(n log w) with a balanced tree) | Space O(w)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 114

115 * Streaming Insert + Find K-th Largest * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: insort every new value; the rank-th largest is sorted_list[-1 - rank]

QUESTION INFO

QUESTION

Design a structure with insert(x), which adds a number, and findLargest(k), which returns the k-th largest number inserted so far (k = 0 is the largest), counting repeated values separately, or None when fewer than k + 1 numbers exist.

Function signature

```
class StreamKth:
    def insert(self, x: float) -> None
    def findLargest(self, k: int) -> float | None
```

EXAMPLES

Example 1

Input	`s = StreamKth()`
	`s.insert(3)`
	`s.insert(3)`
	`s.insert(2)`
	`s.findLargest(0)`
	`s.findLargest(1)`
	`s.findLargest(2)`
	`s.findLargest(3)`

Output	`s.findLargest(0) -> 3`
	`s.findLargest(1) -> 3`
	`s.findLargest(2) -> 2`
	`s.findLargest(3) -> null`

Why	`Sorted descending the numbers are 3, 3, 2, so the 0th, 1st and 2nd largest are 3, 3 and 2, and there is no 3rd.`

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Store everything and sort for every query

BRUTE-FORCE IDEA

Append values to a list; on a query sort it in descending order and read index k. Correct; each query re-sorts everything.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Store everything and sort for every query
Append values to a list
on a query sort it in descending order
read index k
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n log n) per query | Space O(n)

WHY THIS IS WEAKER | Every query re-sorts the whole history, O(n log n). A sorted list with binary search makes each query O(1) after an O(log n) search plus shift; for a fixed k a size-k heap is O(log k).

HOW TO IMPROVE IT

insort every new value; the rank-th largest is sorted_list[-1 - rank]

SHORT BEST SOLUTION

Part 1: Keep all values in an ascending list with insort. The rank-th largest (0-indexed as in the examples) is a[-1 - rank]. Python's list shift is a fast memmove, which is why this beats a hand-written tree for realistic n; mention a balanced BST or Fenwick tree when n is huge.

Part 2: If k never changes, only the k largest values can ever matter. A min-heap of size k keeps them and the root is the rank-th largest. Pop only when the heap would exceed k.

```
from bisect import insort


# BASE QUESTION (optimal): Sorted list with bisect (rank
# chosen per query)
# Idea: Keep all values in an ascending list with insort.
# The rank-th largest (0-indexed as in the examples) is a[-1
# - rank]. Python's list shift is a fast memmove, which is
# why this beats a hand-written tree for realistic n;
# mention a balanced BST or Fenwick tree when n is huge.
class StreamKth:
    def __init__(self):
        self.sorted_values = []

    def insert(self, value):
        insort(self.sorted_values, value)

    def findLargest(self, rank):
        return self.sorted_values[-1 - rank] if 0 <= rank < len(self.sorted_values) else None
```

COMPLEXITY | Part 1: Time O(log n) search + O(n) shift per insert, O(1) query | Space O(n)
Part 2: Time O(log k) per insert, O(1) query | Space O(k)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 115

116 * Count Sorted Strings With a Given Prefix * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: first_match = bisect_left(words, prefix); after_last_match = bisect_left(words, prefix + a very large character); the answer is after_last_match - first_match

QUESTION INFO

QUESTION

Given a lexicographically sorted array of strings words and a query string prefix , return the number of strings in words that start with prefix . Because words is sorted, all matching strings occupy a contiguous range; find that range with binary search in O(log n * L) time (where L is the prefix length), not by scanning.

Function signature

def count_with_prefix(words: list[str], prefix: str) -> int:

...

Follow-up (as reported): Many repeated queries: cache counts in a hash map keyed by prefix. All queries unique: build a trie of all words storing subtree sizes, so each query is O(L) .

EXAMPLES

Example 1	
Input	`words = ["app", "apple", "application", "banana", "bat"], prefix = "app"`
Output	**3**
Why	`"app", "apple", "application" all start with "app".`

QUESTION

Example 2	
Input	`words = ["app", "apple", "application", "banana", "bat"], prefix = "ba"`
Output	`2`
Why	`"banana" and "bat".`

Example 3	
Input	`words = ["app", "apple", "application", "banana", "bat"], prefix = "cat"`
Output	`0`
Why	`No string starts with "cat".`

CONSTRAINTS

- 0 <= len(words) <= 10**6 , sorted in non-decreasing lexicographic order.
- 0 <= len(prefix) <= 100 . An empty prefix matches every string.

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(words) <= 10**6 , sorted in non-decreasing lexicographic order.
- 0 <= len(prefix) <= 100 . An empty prefix matches every string.

BRUTE FORCE | Scan and count

BRUTE-FORCE IDEA

Start with: Scan and count. Start with: Scan and count. It ignores that the list is sorted and tests every word.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan and count
Start with: Scan
process every required item or candidate directly
```

BRUTE-FORCE COMPLEXITY | Time O(n * L) | Space O(1)

WHY THIS IS WEAKER | It ignores that the list is sorted and tests every word.

HOW TO IMPROVE IT

first_match = bisect_left(words, prefix); after_last_match = bisect_left(words, prefix + a very large character); the answer is after_last_match - first_match

SHORT BEST SOLUTION

The matching words form one contiguous block of the sorted list; find both ends.

```python
# BASE QUESTION (optimal): Two binary searches
# Idea: The matching words form one contiguous block of
# the sorted list; find both ends.
from bisect import bisect_left


def count_with_prefix(words, prefix):
    first_match = bisect_left(words, prefix)
    after_last_match = bisect_left(words, prefix + "\uffff")
    return after_last_match - first_match
```

COMPLEXITY | Time O(L log n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 116

117 * Russian Doll Envelopes * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: sort by width ascending and height descending, then the longest increasing run of heights using a tails list and bisect_left

QUESTION INFO

QUESTION

LeetCode 354. Maximum chain of envelopes that strictly fit inside each other. Sort + LIS reduction.

Input: list of (width, height) envelopes.
Output: longest chain where each envelope strictly fits in the next.
Trick: sort by width ascending, height descending (when widths tie); run LIS (Longest Increasing Subsequence) on heights.
LIS with patience-sort gives O(n log n).

EXAMPLES

```
[[5,4],[6,4],[6,7],[2,3]] -> 3.
```

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | O(n^2) DP over sorted envelopes

BRUTE-FORCE IDEA

Sort by width and for each envelope take the best chain of an earlier envelope that is strictly smaller in both dimensions. Correct; quadratic is too slow for 10^5 envelopes.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: O(n^2) DP over sorted envelopes
Sort by width
for each envelope take the best chain of an earlier envelope that is strictly smaller in
both dimensions
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Each envelope is compared with all earlier ones, O(n^2). After sorting, a longest increasing subsequence of the heights with binary search is O(n log n).

HOW TO IMPROVE IT

sort by width ascending and height descending, then the longest increasing run of heights using a tails list and bisect_left

SHORT BEST SOLUTION

Sort by (width ascending, height descending). Keep a list `tails` where tails[k] is the smallest height ending a chain of length k + 1; for each height bisect_left finds where it belongs (replace) or extends the list. The final length is the answer.

```python
from bisect import bisect_left


# BASE QUESTION (optimal): Sort, then LIS with binary
# search
# Idea: Sort by (width ascending, height descending). Keep
# a list `tails` where tails[k] is the smallest height
# ending a chain of length k + 1; for each height
# bisect_left finds where it belongs (replace) or extends
# the list. The final length is the answer.
def max_envelopes(envelopes):
    tails = []
    for _, height in sorted(envelopes, key=lambda envelope: (envelope[0], -envelope[1])):
        position = bisect_left(tails, height)
```

```
        if position == len(tails):
            tails.append(height)
        else:
            tails[position] = height
    return len(tails)
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 117

118 * Given positions of N kids and M ice cream sellers on a 1D line, find the smallest distance E such that every kid lies within the range [P - E, P + E] of at least one seller. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: for each kid bisect into the sorted sellers and take the nearer of the two neighbours; the answer is the largest of those distances

QUESTION INFO

QUESTION

Kids and ice cream sellers stand at positions on a line. Find the smallest distance E such that every kid is within E of at least one seller (the kid at position K is served by a seller at P when P - E <= K <= P + E).

Function signature

def smallest_e(kids: list, sellers: list) -> int

EXAMPLES

Example 1	
Input	`[1, 5, 9], [2, 8]`
Output	`3`
Why	`The kid at 5 is 3 away from both sellers (at 2 and 8); the others are 1 away.`

Example 2	
Input	`[4], [4]`
Output	`0`
Why	`A kid standing on a seller needs distance 0.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | For each kid check every seller

BRUTE-FORCE IDEA

For each kid compute the distance to every seller and take the minimum, then take the maximum over kids. Correct; every kid scans all sellers.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: For each kid check every seller
For each kid compute the distance to every seller
take the minimum
take the maximum over kids
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(N * M) | Space O(1)

WHY THIS IS WEAKER | N * M distance checks. With the sellers sorted, binary search finds each kid's nearest seller among two neighbours, O((N + M) log M).

HOW TO IMPROVE IT

for each kid bisect into the sorted sellers and take the nearer of the two neighbours; the answer is the largest of those distances

SHORT BEST SOLUTION

With the sellers sorted, bisect_left gives the first seller at or after the kid; the nearest seller is that one or the one before it. E is the largest of these nearest distances.

```
from bisect import bisect_left


# BASE QUESTION (optimal): Sort the sellers, binary search
# each kid
# Idea: With the sellers sorted, bisect_left gives the
# first seller at or after the kid; the nearest seller is
# that one or the one before it. E is the largest of these
# nearest distances.
def smallest_radius(kids, sellers):
    sellers = sorted(sellers)
    worst = 0
    for kid in kids:
        position = bisect_left(sellers, kid)
        nearest_distance = min(abs(kid - sellers[seller_index]) for seller_index in (position - 1, position) if
 0 <= seller_index < len(sellers))
        worst = max(worst, nearest_distance)
    return worst
```

COMPLEXITY | Time O((N + M) log M) | Space O(1) extra

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 118

Prefix sums, Tier 1: precompute once, answer ranges and balance points in O(1)

PATTERN CONTEXT

pre[i] is the total of the first i items. Any range sum is pre[r + 1] - pre[l], and a balance point is an index where the left total equals the right total. Count something other than values (equal pairs, ones, repeats) by accumulating that instead.

124 * Range Duplicate Check with Constant-Time Queries * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: next[position] = next occurrence of the same value; reach[left] = smallest next over l..end; the range l..r is all distinct when reach[left] > right

QUESTION INFO

QUESTION

Preprocess an array once so that each query unique(l, r) answers in O(1) whether all elements from index l to index r (inclusive) are different.

Function signature

class RangeUnique:
  def __init__(self, a: list)
  def unique(self, l: int, r: int) -> bool

EXAMPLES

Example 1	

Input	`ru = RangeUnique([1, 2, 3, 1, 4])`
	`ru.unique(0, 2)`
	`ru.unique(0, 3)`
	`ru.unique(1, 4)`

Output	`ru.unique(0, 2) -> true`
	`ru.unique(0, 3) -> false`
	`ru.unique(1, 4) -> true`

Why	`Indexes 0..3 hold 1, 2, 3, 1: the value 1 repeats. Ranges 0..2 and 1..4 have no repeats.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Put the range in a set

BRUTE-FORCE IDEA

Insert the elements of the range into a set and compare sizes. Correct; every query costs the length of the range.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Put the range in a set
Insert the elements of the range into a set
compare sizes
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(r - l) per query | Space O(r - l)

WHY THIS IS WEAKER | Each query costs the length of the range. A next-occurrence array with suffix minimums answers each query with one comparison.

HOW TO IMPROVE IT

next[position] = next occurrence of the same value; reach[left] = smallest next over l..end; the range l..r is all distinct when reach[left] > right

SHORT BEST SOLUTION

Scan right to left with a map from value to its latest index to fill next[position] (infinity if none). Then reach[position] = min(next[position], reach[position + 1]). Query(left, right) is reach[left] > right. If the array changes, rebuild in O(length) (fine when reads dominate), or keep a sorted structure per value when updates are frequent.

```
# BASE QUESTION (optimal): Next-occurrence array with
# suffix minimum
# Idea: Scan right to left with a map from value to its
# latest index to fill next[position] (infinity if none). Then
# reach[position] = min(next[position], reach[position + 1]). Query(left, right) is
# reach[left] > right. If the array changes, rebuild in O(length)
# (fine when reads dominate), or keep a sorted structure
# per value when updates are frequent.
class RangeUnique:
    def __init__(self, values):
        length, INFINITY = len(values), float("inf")
        last_index_of, next_same_index = {}, [INFINITY] * length
        for position in range(length - 1, -1, -1):
            next_same_index[position] = last_index_of.get(values[position], INFINITY)
            last_index_of[values[position]] = position
        self.reach = [INFINITY] * (length + 1)
        for position in range(length - 1, -1, -1):
            self.reach[position] = min(next_same_index[position], self.reach[position + 1])

    def unique(self, left, right):
        return self.reach[left] > right
```

COMPLEXITY | Time O(n) preprocessing, O(1) per query | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 124

125 * Find a Horizontal Cut That Bisects Rectangle Area * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: events by height with a running width; accumulate area strip by strip until half the total, then solve inside that strip

QUESTION INFO

QUESTION

Given many disjoint axis-aligned rectangles (x1, y1, x2, y2), return the y of the horizontal line that splits their combined area in half, so the area below the line equals the area above it. Return an exact fraction (no floating-point error).

Function signature

def bisecting_cut(rects: list) -> Fraction

EXAMPLES

Example 1	
Input	`[(0, 0, 2, 2)]`
Output	`1`
Why	`One 2x2 square: the middle line is y = 1.`

QUESTION

Example 2	
Input	`[(0, 0, 1, 1), (0, 2, 1, 4)]`
Output	`5/2`
Why	`Areas 1 and 2, total 3, so half is 1.5: all of the first rectangle (area 1) plus 0.5 of the second, which starts at y = 2: y = 5/2.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Binary search on the cut height

BRUTE-FORCE IDEA

The area below y is non-decreasing in y, so bisect y between the lowest and highest edge until the area below is half the total. It is simple and general, but it works in floating point (limited precision) and costs a full pass over all rectangles per step.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Binary search on the cut height
The area below y is non-decreasing in y
so bisect y between the lowest
highest edge until the area below is half the total
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(R * iterations) | Space O(1)

WHY THIS IS WEAKER | Each step sums over all rectangles and the answer is only as precise as floating point allows. A sweep over sorted edges with exact fractions finds the slab and the cut directly in O(R log R).

HOW TO IMPROVE IT

events by height with a running width; accumulate area strip by strip until half the total, then solve inside that strip

SHORT BEST SOLUTION

Make events (y, +width) at each rectangle's bottom and (y, -width) at its top and sort them. Between consecutive event heights the slope is the running sum of widths. Accumulate area slab by slab; in the slab where the accumulated area would reach half the total, the cut is y0 + (half - accumulated) / slope. Fractions keep it exact; slabs with slope zero are skipped, which returns the lowest valid cut.

```python
from fractions import Fraction
from collections import defaultdict


# BASE QUESTION (optimal): Sweep the y events and
# interpolate inside one slab
# Idea: Make events (y, +width) at each rectangle's bottom
# and (y, -width) at its top and sort them. Between
# consecutive event heights the slope is the running sum
# of widths. Accumulate area slab by slab; in the slab
# where the accumulated area would reach half the total,
# the cut is y0 + (half - accumulated) / slope. Fractions
# keep it exact; slabs with slope zero are skipped, which
# returns the lowest valid cut.
def bisecting_cut(rectangles):
    total = sum((x2 - x1) * (y2 - y1) for x1, y1, x2, y2 in rectangles)
    half = Fraction(total, 2)
    events = defaultdict(int)
    for x1, y1, x2, y2 in rectangles:
        events[y1] += x2 - x1
        events[y2] -= x2 - x1
    event_heights = sorted(events)
    total_width, area_so_far = 0, Fraction(0)
```

```
    for y0, y1 in zip(event_heights, event_heights[1:]):
        total_width += events[y0]
        gain = total_width * (y1 - y0)
        if total_width and area_so_far + gain >= half:
            return y0 + (half - area_so_far) / total_width
        area_so_far += gain
    return Fraction(event_heights[-1])
```

COMPLEXITY | Time O(R log R) | Space O(R)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 125

126 * Check pivot after removing one element * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: prefix sums plus first/last position maps: look up the value whose removal would balance both sides

QUESTION INFO

QUESTION

A pivot index of an array is an index whose left-side sum equals its right-side sum (the pivot element itself is excluded). Remove exactly one element from the array, then ask whether the remaining array has a pivot index. Return one pair (removed index, pivot index), both as indexes in the original array, or None if no removal creates a pivot. Negative numbers and duplicates are allowed.

Function signature

def pivot_after_removal(a: list) -> tuple | None

EXAMPLES

Example 1	
Input	`[2, 1, -1, 3]`
Output	`(0, 3)`
Why	`Removing the 2 at index 0 leaves [1, -1, 3]; the 3 (original index 3) has left sum 1 + (-1) = 0 and nothing on its right.`

Example 2	
Input	`[1, 2, 3]`
Output	`null`
Why	`Whichever element is removed, no pivot appears.`

Example 3	
Input	`[0, 0]`
Output	`(0, 1)`
Why	`Remove either 0: the other one is a pivot of an array with one element.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every removal and every pivot

BRUTE-FORCE IDEA

Start with: Try every removal and every pivot. For each removed index build the shorter array and test every pivot with sums.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every removal and every pivot
Start with: Try every removal
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^3), O(n^2) with prefix sums | Space O(n)

WHY THIS IS WEAKER | N removals times n pivots times a sum: O(n^3) (O(n^2) with prefix sums). For each pivot the value to remove is determined by prefix sums, so a dictionary lookup makes it O(n).

HOW TO IMPROVE IT

prefix sums plus first/last position maps: look up the value whose removal would balance both sides

SHORT BEST SOLUTION

Let P be the prefix sum before i and S the total. If the removed element r lies to the right of the pivot i, the sides are equal exactly when values[r] = total - 2P - values[pivot]; if r lies to the left, when values[r] = 2P + values[pivot] - total. Keep the last index of each value (needs an index greater than pivot) and the first index (needs an index smaller than pivot).

```python
# BASE QUESTION (optimal): Closed-form value to remove,
# found in a dictionary of positions
# Idea: Let P be the prefix sum before i and S the total.
# If the removed element r lies to the right of the pivot
# i, the sides are equal exactly when values[r] = total - 2P -
# values[pivot]; if r lies to the left, when values[r] = 2P + values[pivot] - total.
# Keep the last index of each value (needs an index
# greater than pivot) and the first index (needs an index
# smaller than pivot).
def pivot_after_removal(values):
    length, total = len(values), sum(values)
    first, last = {}, {}
    for index, value in enumerate(values):
        first.setdefault(value, index)
        last[value] = index
    prefix_sum = 0
    for pivot in range(length):
        need_right = total - 2 * prefix_sum - values[pivot]
        if need_right in last and last[need_right] > pivot:
            return True
        need_left = 2 * prefix_sum + values[pivot] - total
        if need_left in first and first[need_left] < pivot:
            return True
        prefix_sum += values[pivot]
    return False
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 126

127 * Gold Chain Split into Two Equal Halves * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: prefix sums with first/last position maps: find two cut points that leave equal halves around one removed link

QUESTION INFO

QUESTION

A gold chain is an array of segment weights. Remove exactly one segment, reconnect the rest, then cut the chain once so that the two parts have equal total weight. Return whether this is possible for some choice of the removed segment and the cut.

Function signature

def gold_chain(a: list) -> bool

QUESTION

EXAMPLES

Example 1

Input	`[1, 2, 3, 4]`
Output	`true`
Why	`Remove the 4: [1, 2, 3] splits as [1, 2] and [3], both 3.`

Example 2

Input	`[1, 1, 5]`
Output	`true`
Why	`Remove the 5: [1, 1] splits into 1 and 1.`

Example 3

Input	`[1, 2, 4]`
Output	`false`
Why	`Removing 1 leaves 6 (halves of 3 need [2] | [4]: no), removing 2 leaves 5, removing 4 leaves 3: nothing works.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | For each removal rebuild the chain and try every cut

BRUTE-FORCE IDEA

Remove each segment, then test every cut position with a running sum. Correct, with a quadratic loop that is the starting point to improve.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: For each removal rebuild the chain and try every cut
Remove each segment
test every cut position with a running sum
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | N removals, each rebuilt and scanned: O(n^2). Prefix sums with a lookup of the needed prefix value answer each removal in O(1).

HOW TO IMPROVE IT

prefix sums with first/last position maps: find two cut points that leave equal halves around one removed link

SHORT BEST SOLUTION

Let P[cut_index] be the sum of the first j segments. For removed index r with half = (total - segments[removed_index]) / 2 (skip if odd): a cut before r exists if half equals some P[cut_index] with 1 <= cut_index <= removed_index; a cut after r exists if half + segments[removed_index] equals some P[k] with removed_index < k <= segment_count - 1. Keep the first index and the last index of every prefix value, so each check is one lookup.

```
# BASE QUESTION (optimal): Prefix sums with first and last
# occurrence maps
# Idea: Let P[cut_index] be the sum of the first j segments. For
# removed index r with half = (total - segments[removed_index]) / 2 (skip if
# odd): a cut before r exists if half equals some P[cut_index]
# with 1 <= cut_index <= removed_index; a cut after r exists if half + segments[removed_index]
# equals some P[k] with removed_index < k <= segment_count - 1. Keep the first
# index and the last index of every prefix value, so each
```

```python
# check is one lookup.
def gold_chain(segments):
    segment_count = len(segments)
    if segment_count < 3:
        return False
    prefix_sums = [0]
    for segment in segments:
        prefix_sums.append(prefix_sums[-1] + segment)
    total = prefix_sums[-1]
    first, last = {}, {}
    for cut_index in range(1, segment_count):
        first.setdefault(prefix_sums[cut_index], cut_index)
        last[prefix_sums[cut_index]] = cut_index
    for removed_index in range(segment_count):
        rest = total - segments[removed_index]
        if rest % 2:
            continue
        half = rest // 2
        if half in first and first[half] <= min(removed_index, segment_count - 2):
            return True
        if (half + segments[removed_index]) in last and last[half + segments[removed_index]] > removed_index:
            return True
    return False
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 127

128 * Count the number of ways to split a string. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: positions of the ones; the answer multiplies the gap sizes around the ones_per_part-th and 2k-th one (zero ones: choose two cut points)

QUESTION INFO

QUESTION

A binary string s is split into three non-empty parts by two cuts. Count the ways in which the three parts contain the same number of 1s, and return the count modulo 1,000,000,007.

Function signature

def numWays(s: str) -> int

EXAMPLES

Example 1	
Input	`'10101'`
Output	`4`
Why	`Four ways to split '10101' so each part has one 1 (for example 1|010|1 and 10|10|1).`

Example 2	
Input	`'1001'`
Output	`0`
Why	`Two 1s cannot be divided into three equal groups.`

QUESTION

Example 3	
Input	`'0000'`
Output	`3`
Why	`No 1s: any two cuts work: choose 2 of the 3 gaps = 3 ways.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every pair of cut positions

BRUTE-FORCE IDEA

With prefix counts of ones, check all pairs (i, j) of cut points and count those giving three equal parts. Correct and fine for short strings.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every pair of cut positions
With prefix counts of ones
check all pairs (i
j) of cut points
count those giving three equal parts
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | O(n^2) pairs. The cuts only depend on where the ones are, so the answer is a product of two gap sizes, computed in O(n).

HOW TO IMPROVE IT

positions of the ones; the answer multiplies the gap sizes around the ones_per_part-th and 2k-th one (zero ones: choose two cut points)

SHORT BEST SOLUTION

Let ones be the total number of ones. If it is not divisible by 3 the answer is 0; if it is 0, any two of the length - 1 cut positions work, C(length - 1, 2). Otherwise each part holds ones_per_part = ones / 3 ones; the first cut can fall anywhere in the run of zeros between the ones_per_part-th and (ones_per_part + 1)-th one, and the second between the 2k-th and (2k + 1)-th one, so the answer is (gap1 + 1) * (gap2 + 1).

```python
MOD = 10 ** 9 + 7


# BASE QUESTION (optimal): Count the ones, multiply the
# gap sizes
# Idea: Let ones be the total number of ones. If it is not
# divisible by 3 the answer is 0; if it is 0, any two of
# the length - 1 cut positions work, C(length - 1, 2). Otherwise
# each part holds ones_per_part = ones / 3 ones; the first cut can
# fall anywhere in the run of zeros between the ones_per_part-th and
# (ones_per_part + 1)-th one, and the second between the 2k-th and (2k
# + 1)-th one, so the answer is (gap1 + 1) * (gap2 + 1).
def numWays(bits):
    ones = [index for index, bit in enumerate(bits) if bit == "1"]
    length, ones_count = len(bits), len(ones)
    if ones_count % 3:
        return 0
    if ones_count == 0:
        return ((length - 1) * (length - 2) // 2) % MOD if length >= 3 else 0
    ones_per_part = ones_count // 3
    return ((ones[ones_per_part] - ones[ones_per_part - 1]) * (ones[2 * ones_per_part] - ones[2 * ones_per_part
 - 1])) % MOD
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 128

GROUP 16 OF 43 | 5 QUESTIONS | PART A - PART 1

Sorted intervals: merge while they touch, then read the gaps or search

PATTERN CONTEXT

Sort by start. Walk once: if the next interval starts before the last one ends, extend the last; otherwise start a new one. The merged list is sorted and disjoint, so gaps are the spaces between neighbours and any point can be found with bisect.

130 * Find Containing Range * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: merge once, keep the starts in a list, and bisect to find the one interval that could contain x

QUESTION INFO

QUESTION

Given a collection of inclusive ranges [start, end] (overlapping or touching ranges are merged), find(x) returns the merged range that contains the number x, or None when no range contains it.

Function signature

class RangeIndex:
    def __init__(self, ranges: list)
    def find(self, x: int) -> tuple | None

EXAMPLES

Example 1	

Input	`r = RangeIndex([(1, 3), (2, 6), (8, 10)])`
	`r.find(5)`
	`r.find(7)`
	`r.find(10)`

Output	`r.find(5) -> (1, 6)`
	`r.find(7) -> null`
	`r.find(10) -> (8, 10)`

Why	`[1, 3] and [2, 6] overlap and merge into [1, 6]; 7 falls in the gap between the merged ranges.`

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan all ranges for each query

BRUTE-FORCE IDEA

Merge the ranges (or not) and for each query test every range for containment. Correct; each query walks the whole list.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan all ranges for each query
Merge the ranges (or not)
for each query test every range for containment
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) per query | Space O(n)

WHY THIS IS WEAKER | Every query walks all n ranges. Merging once (O(n log n)) and binary searching the sorted starts makes each query O(log n).

HOW TO IMPROVE IT

merge once, keep the starts in a list, and bisect to find the one interval that could contain x

SHORT BEST SOLUTION

After sorting and merging, the starts are increasing, so bisect_right(starts, point) - 1 is the only candidate; it contains x if its end is at least x. Preprocessing is paid once.

```
from bisect import bisect_right


# BASE QUESTION (optimal): Merge once, then binary search
# the starts
# Idea: After sorting and merging, the starts are
# increasing, so bisect_right(starts, point) - 1 is the only
# candidate; it contains x if its end is at least x.
# Preprocessing is paid once.
class RangeIndex:
    def __init__(self, ranges):
        merged = []
        for start, end in sorted(ranges):
            if merged and start <= merged[-1][1]:
                merged[-1][1] = max(merged[-1][1], end)
            else:
                merged.append([start, end])
        self.starts = [start for start, _ in merged]
        self.merged = merged

    def find(self, point):
        position = bisect_right(self.starts, point) - 1
        if position >= 0 and self.merged[position][1] >= point:
            return tuple(self.merged[position])
        return None
```

COMPLEXITY | Time O(n log n) to build, O(log n) per query | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 130

131 * Union of Sorted Interval Lists * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: walk both sorted lists with two indices, always taking the smaller start next, then the same merge test

QUESTION INFO

QUESTION

Two lists of intervals are each sorted and non-overlapping on their own. Return the union of both lists as sorted, non-overlapping intervals. Intervals that touch, such as (1, 3) and (3, 5), are merged unless touching=False.

Function signature

def union_two(a: list, b: list, touching=True) -> list

EXAMPLES

Example 1	
Input	[(1, 3), (6, 9)], [(2, 5), (10, 12)]
Output	[(1, 5), (6, 9), (10, 12)]
Why	(1, 3) and (2, 5) overlap and become (1, 5); (6, 9) and (10, 12) stay.

Example 2	
Input	[(1, 3)], [(3, 5)]
Output	[(1, 5)]
Why	(1, 3) and (3, 5) touch at 3, so they merge.

QUESTION

Example 3	
Input	`[(1, 3)], [(3, 5)], touching = false`
Output	`[(1, 3), (3, 5)]`
Why	`With touching=False they only merge when they truly overlap.`

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Concatenate, sort and merge

BRUTE-FORCE IDEA

Put all intervals in one list, sort by start, and fold overlapping ones. Correct and short; the sort ignores that both inputs were already sorted.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Concatenate, sort and merge
Put all intervals in one list
sort by start
fold overlapping ones
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O((m + n) log (m + n)) | Space O(m + n)

WHY THIS IS WEAKER | Sorting ignores that both lists are already sorted: O((m + n) log (m + n)) instead of O(m + n) for a two-pointer merge.

HOW TO IMPROVE IT

walk both sorted lists with two indices, always taking the smaller start next, then the same merge test

SHORT BEST SOLUTION

Part 1: Always take the interval with the smaller start next. If it touches or overlaps the last output interval (start <= last end, closed intervals) extend that one, otherwise append it. Invariant: everything before the pointers is already folded into a canonical output.

Part 2: heapq.merge keeps one pointer per list in a heap and yields intervals in start order without materializing the inputs, so it works for lists too large for memory. The same fold step builds the union lazily.

```python
# BASE QUESTION (optimal): Two-pointer merge
# Idea: Always take the interval with the smaller start
# next. If it touches or overlaps the last output interval
# (start <= last end, closed intervals) extend that one,
# otherwise append it. Invariant: everything before the
# pointers is already folded into a canonical output.
def union_two(first, second):
    merged, first_index, second_index = [], 0, 0
    while first_index < len(first) or second_index < len(second):
        if second_index == len(second) or (first_index < len(first) and first[first_index][0] <= second[second_
index][0]):
            next_interval = first[first_index]; first_index += 1
        else:
            next_interval = second[second_index]; second_index += 1
        if merged and next_interval[0] <= merged[-1][1]:
            merged[-1][1] = max(merged[-1][1], next_interval[1])
        else:
            merged.append([next_interval[0], next_interval[1]])
    return merged
```

COMPLEXITY | Part 1: Time O(m + n) | Space O(m + n) for the output
Part 2: Time O(N log K) | Space O(K) beyond the output

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 131

132 * Range Module (Insert and Point Query) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: keep sorted lefts and rights; adding bisects to the first interval that touches and the first that starts after, and replaces that block by one merged interval

QUESTION INFO

QUESTION

Design a data structure that tracks a set of half-open ranges [left, right) over the number line. add_range(left, right) adds the interval; overlapping or touching intervals are merged so that only disjoint, non-adjacent intervals remain. query_point(x) returns True when x lies inside a tracked interval; because the intervals are half-open, a point equal to a right endpoint is not inside.

Function signature

class RangeModule:
    def add_range(self, left: float, right: float) -> None
    def query_point(self, x: float) -> bool

EXAMPLES

Example 1

Input	`rm = RangeModule()`
	`rm.add_range(10, 20)`
	`rm.query_point(15)`
	`rm.query_point(20)`
	`rm.query_point(10)`

Output	`rm.query_point(15) -> true`
	`rm.query_point(20) -> false`
	`rm.query_point(10) -> true`

Why	`[10, 20) contains 10 and 15 but not 20.`

Example 2

Input	`rm = RangeModule()`
	`rm.add_range(10, 20)`
	`rm.add_range(20, 30)`
	`rm.query_point(20)`
	`rm.query_point(25)`
	`rm.query_point(30)`

Output	`rm.query_point(20) -> true`
	`rm.query_point(25) -> true`
	`rm.query_point(30) -> false`

Why	`The touching ranges merge into [10, 30), so 20 is now covered; 30 still is not.`

Example 3

Input	`rm = RangeModule()` `rm.query_point(5)`
Output	`rm.query_point(5) -> false`
Why	`Nothing has been added yet.`

CONSTRAINTS

- Endpoints and query points are integers or floats with -10^9 <= value <= 10^9 .
- For every add_range call, left < right .
- Up to 10^5 total operations.

QUESTION

- Touching intervals merge: add_range(1, 5) followed by add_range(5, 10) yields one interval [1,
- 10) .

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- Endpoints and query points are integers or floats with -10^9 <= value <= 10^9 .
- For every add_range call, left < right .
- Up to 10^5 total operations.
- Touching intervals merge: add_range(1, 5) followed by add_range(5, 10) yields one interval [1,

BRUTE FORCE | A plain list scanned linearly

BRUTE-FORCE IDEA

Start with: A plain list scanned linearly. Start with: A plain list scanned linearly. Every query walks every added interval.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: A plain list scanned linearly
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n) per query | Space O(n)

WHY THIS IS WEAKER | Every query walks every added interval.

HOW TO IMPROVE IT

keep sorted lefts and rights; adding bisects to the first interval that touches and the first that starts after, and replaces that block by one merged interval

SHORT BEST SOLUTION

Disjoint sorted intervals mean one bisect locates the only candidate for a point.

```python
# BASE QUESTION (optimal): Merged intervals with bisect
# Idea: Disjoint sorted intervals mean one bisect locates
# the only candidate for a point.
from bisect import bisect_left, bisect_right


class RangeModule:
    def __init__(self):
        self.lefts = []
        self.rights = []

    def add_range(self, left, right):
        first_overlap = bisect_left(self.rights, left)
        after_last_overlap = bisect_right(self.lefts, right)
        if first_overlap < after_last_overlap:
            left = min(left, self.lefts[first_overlap])
            right = max(right, self.rights[after_last_overlap - 1])
        self.lefts[first_overlap:after_last_overlap] = [left]
        self.rights[first_overlap:after_last_overlap] = [right]

    def query_point(self, point):
        first_overlap = bisect_right(self.lefts, point) - 1
        return first_overlap >= 0 and point < self.rights[first_overlap]
```

COMPLEXITY | Time O(log n) query | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 132

133 * Meetings Minus a Do-Not-Schedule Block * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: merge first, then for each merged meeting keep the part before the blocked window and the part after it

QUESTION INFO

QUESTION

You are given a list of meetings , each a (start, end) interval with start < end , and one "do-not- schedule" (DNS) block (dns_start, dns_end) . Consider the set of all time points covered by at least one meeting. Return the smallest list of non-overlapping, sorted intervals that covers exactly those points except any point lying inside the DNS block (treat the DNS block as removing the open coverage on (dns_start, dns_end) ; equivalently split coverage at those bounds). Meetings may overlap and are given in any order. Return the result sorted by start time.

Function signature

```
def subtract_dns(meetings: list[tuple[int, int]],
                 dns: tuple[int, int]) -> list[tuple[int, int]]:
    ...
```

Follow-up (as reported): Generalize to a list of non-overlapping DNS blocks: merge the meetings first, then walk both sorted lists together subtracting each block.

EXAMPLES

Example 1	
Input	`meetings = [(1,7),(5,10),(12,30),(22,40),(50,65)], dns = (20,35)`
Output	`[(1,10),(12,20),(35,40),(50,65)]`
Why	`Merged coverage is [1,10],[12,40],[50,65]. Removing (20,35) splits [12,40] into [12,20] and [35,40].`

Example 2	
Input	`meetings = [(0,10)], dns = (0, 10)`
Output	`[]`
Why	`The DNS block removes the entire covered range.`

Example 3	
Input	`meetings = [(0,5),(6,9)], dns = (100, 200)`
Output	`[(0,5),(6,9)]`
Why	`The DNS block covers nothing that was scheduled.`

CONSTRAINTS

- 1 <= len(meetings) <= 10^5 .
- Coordinates are integers; 0 <= start < end <= 10^9 .

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(meetings) <= 10^5 .
- Coordinates are integers; 0 <= start < end <= 10^9 .

BRUTE FORCE | Mark every unit of time

BRUTE-FORCE IDEA

Start with: Mark every unit of time. It touches every unit of time covered, which is huge when times are large.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Mark every unit of time
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(total length) | Space O(total length)

WHY THIS IS WEAKER | It touches every unit of time covered, which is huge when times are large.

HOW TO IMPROVE IT

merge first, then for each merged meeting keep the part before the blocked window and the part after it

SHORT BEST SOLUTION

After merging, every interval keeps at most a left and a right piece around the block.

```python
# BASE QUESTION (optimal): Merge, then cut
# Idea: After merging, every interval keeps at most a left
# and a right piece around the block.
def merge_intervals(meetings):
    merged = []
    for start, end in sorted(meetings):
        if merged and start <= merged[-1][1]:
            merged[-1] = (merged[-1][0], max(merged[-1][1], end))
        else:
            merged.append((start, end))
    return merged


def subtract_dns(meetings, block):
    block_start, block_end = block
    remaining = []
    for start, end in merge_intervals(meetings):
        if start < block_start:
            remaining.append((start, min(end, block_start)))
        if end > block_end:
            remaining.append((max(start, block_end), end))
    return remaining
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 133

134 * Merge Intervals * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the template as written

QUESTION INFO

QUESTION

Merge Intervals. An interval is a representation of a range of values along a number line. For example, [1,3] represents all numbers between 1 and 3, inclusive. Given a collection of intervals, the task is to merge all overlapping intervals. For example, consider the intervals [1,3],[2,6],[8,10],[15,18]. The function should return merged intervals where overlapping intervals are combined. In this case, intervals [1,3] and [2,6] overlap, so they should be merged into [1,6]. Examples intervals = [[1,3], [2,6], [8,10], [15,18]] output: [[1,6], [8,10], [15,18]] intervals = [[1,4], [4,5]] output: [[1,5]] intervals = [[1,4], [2,3]] output: [[1,4]] intervals = [[1,2], [3,4], [5,6], [7,8]] output: [[1,2], [3,4], [5,6], [7,8]]The mergeIntervals function takes an array of intervals and merges all overlapping intervals into a single interval. The process involves sorting the intervals by their start times and then iterating through them to merge where necessary. Here is a step-by-step breakdown: Initial Check: If the input array is empty, the function immediately returns an empty array, as there are no intervals to merge. Sorting: The intervals are sorted by their start times. This ensures that intervals can be processed in a linear sweep, merging adjacent intervals when they overlap. Merging Intervals: Initialize a result list with the first interval. Then, iterate through the sorted intervals starting from the second interval. For each interval, compare it with the last interval in th

QUESTION

EXAMPLES

Example 1	
Input	`[[1, 3], [2, 6], [8, 10], [15, 18]]`
Output	`[[1, 6], [8, 10], [15, 18]]`
Why	`[1, 3] and [2, 6] overlap and merge into [1, 6].`

Example 2	
Input	`[[1, 4], [4, 5]]`
Output	`[[1, 5]]`
Why	`Intervals that touch at an endpoint are merged as well.`

Example 3	
Input	`[[8, 10], [1, 3], [2, 4]]`
Output	`[[1, 4], [8, 10]]`
Why	`Unsorted input is fine.`

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Repeatedly merge any overlapping pair

BRUTE-FORCE IDEA

Look for two intervals that overlap, replace them with their union, and repeat until none overlap. Correct; many passes over all pairs.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Repeatedly merge any overlapping pair
Look for two intervals that overlap
replace them with their union
repeat until none overlap
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) or worse | Space O(n)

WHY THIS IS WEAKER | Each pass compares all pairs, O(n^2), and several passes may be needed. Sorting by start and sweeping once is O(n log n).

HOW TO IMPROVE IT

the template as written

SHORT BEST SOLUTION

Sort by start. If an interval starts at or before the end of the last output interval, extend that end with max; otherwise append it. The sort guarantees nothing earlier can overlap the new interval.

```python
# BASE QUESTION (optimal): Sort by start and sweep
# Idea: Sort by start. If an interval starts at or before
# the end of the last output interval, extend that end
# with max; otherwise append it. The sort guarantees
# nothing earlier can overlap the new interval.
def merge_intervals(intervals):
    merged = []
    for start, end in sorted(intervals):
        if merged and start <= merged[-1][1]:
            merged[-1][1] = max(merged[-1][1], end)
        else:
```

```
            merged.append([start, end])
    return merged
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 134

Trie: nested dicts, one character per level

PATTERN CONTEXT

Store words as nested dicts, one character per level, with an end marker that holds whatever you want to remember. Insert walks and creates; search walks and fails when a character is missing; prefix queries stop in the middle; longest-match keeps the last end marker you passed.

137 * URL Request Handler Router * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: one trie level per URL segment; a node has literal children and one wildcard child; try the literal first

QUESTION INFO

QUESTION

Design a request router. add(pattern) registers a URL pattern whose segments are either literal text or a placeholder written {name}; match(url) returns the registered pattern that handles the URL, or None. When a URL matches both a literal segment and a placeholder, the literal wins.

Function signature

class Router:
    def add(self, pattern: str) -> None
    def match(self, url: str) -> str | None

EXAMPLES

Example 1

Input	`r = Router()`
	`r.add('/users/{userId}/photo/{photoId}')`
	`r.add('/users/me/photo/{photoId}')`
	`r.match('/users/123/photo/456')`
	`r.match('/users/me/photo/9')`
	`r.match('/orders/1')`

Output	`r.match('/users/123/photo/456') -> '/users/{userId}/photo/{photoId}'`
	`r.match('/users/me/photo/9') -> '/users/me/photo/{photoId}'`
	`r.match('/orders/1') -> null`

Why	`The placeholder pattern handles any user id, but for /users/me/... the literal 'me' pattern wins; an unknown path matches nothing.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan every registered pattern

BRUTE-FORCE IDEA

Compare the URL's segments with each pattern's segments (same count, each one equal or a placeholder), collect the matches and keep the best by specificity. It is the direct approach.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan every registered pattern
Compare the URL's segments with each pattern's segments (same count
each one equal or a placeholder)
collect the matches
keep the best by specificity
```

BRUTE-FORCE COMPLEXITY | Time O(H * L) per lookup | Space O(1) extra

WHY THIS IS WEAKER | Each lookup compares against every registered pattern, O(patterns * segments). A trie of segments with a literal child and a wildcard child follows only the compatible branches.

HOW TO IMPROVE IT

one trie level per URL segment; a node has literal children and one wildcard child; try the literal first

SHORT BEST SOLUTION

Each trie node has literal children and one wildcard child; the pattern is stored at its last node. Lookup tries the literal child first, and if that whole branch fails it backs up and tries the wildcard child. A match must consume every segment and end on a node that stores a pattern.

```
def _segments(path):
    return [segment for segment in path.strip("/").split("/") if segment != ""]


# BASE QUESTION (optimal): Segment trie with literal-first
# fallback
# Idea: Each trie node has literal children and one
# wildcard child; the pattern is stored at its last node.
# Lookup tries the literal child first, and if that whole
# branch fails it backs up and tries the wildcard child. A
# match must consume every segment and end on a node that
# stores a pattern.
class Router:
    def __init__(self, patterns=()):
        self.root = {"lit": {}, "wild": None, "handler": None}
        for pattern in patterns:
            self.add(pattern)

    def add(self, pattern):
        node = self.root
        for segment in _segments(pattern):
            if segment.startswith("{") and segment.endswith("}"):
                if node["wild"] is None:
                    node["wild"] = {"lit": {}, "wild": None, "handler": None}
                node = node["wild"]
            else:
                node = node["lit"].setdefault(segment, {"lit": {}, "wild": None, "handler": None})
        if node["handler"] is None:
            node["handler"] = pattern

    def match(self, url):
        url_segments = _segments(url)

        def walk(node, position):
            if position == len(url_segments):
                return node["handler"]
            child = node["lit"].get(url_segments[position])
            if child is not None:
                found = walk(child, position + 1)
                if found is not None:
                    return found
            if node["wild"] is not None:
                return walk(node["wild"], position + 1)
            return None

        return walk(self.root, 0)
```

COMPLEXITY | Time O(L * branching) per lookup, O(total pattern length) to build | Space O(total pattern length)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 137

138 * Trie Prefix Search / Autocomplete * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: walk to the prefix node; return the words under it (autocomplete)

QUESTION INFO

QUESTION

Build a trie from a list of words. search_prefix(prefix) returns every word that starts with the prefix, sorted alphabetically (empty list when none).

Function signature

class Trie:
    def insert(self, word: str) -> None
    def search_prefix(self, prefix: str) -> list

EXAMPLES

Example 1	

Input	`t = Trie(['car', 'card', 'care', 'cat', 'dog'])`
	`t.search_prefix('car')`
	`t.search_prefix('ca')`
	`t.search_prefix('x')`

Output	`t.search_prefix('car') -> ['car', 'card', 'care']`
	`t.search_prefix('ca') -> ['car', 'card', 'care', 'cat']`
	`t.search_prefix('x') -> []`

Why	`'car' is itself a word that starts with 'car'; the prefix 'x' matches nothing.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Filter the whole word list

BRUTE-FORCE IDEA

Scan every word, keep those starting with the prefix and sort them. Correct; every query pays for the whole dictionary.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Filter the whole word list
Scan every word
keep those starting with the prefix
sort them
```

BRUTE-FORCE COMPLEXITY | Time O(n * L + n log n) per query | Space O(output)

WHY THIS IS WEAKER | Every query reads the whole dictionary. A trie reaches the prefix in O(length of the prefix) and reads only the completions.

HOW TO IMPROVE IT

walk to the prefix node; return the words under it (autocomplete)

SHORT BEST SOLUTION

Insert each word, marking the last node. For a query walk the prefix; if a step is missing return an empty list. From the prefix node run a DFS that visits children in alphabetical order and records the word at every end node, which yields sorted words.

```
# BASE QUESTION (optimal): Trie with a DFS over sorted
# children
# Idea: Insert each word, marking the last node. For a
# query walk the prefix; if a step is missing return an
# empty list. From the prefix node run a DFS that visits
# children in alphabetical order and records the word at
```

```python
# every end node, which yields sorted words.
class Trie:
    def __init__(self, words=()):
        self.root = {}
        for word in words:
            self.insert(word)

    def insert(self, word):
        node = self.root
        for character in word:
            node = node.setdefault(character, {})
        node["$"] = True

    def search_prefix(self, prefix):
        node = self.root
        for character in prefix:
            if character not in node:
                return []
            node = node[character]
        words = []
        def dfs(node, path):
            if "$" in node:
                words.append("".join(path))
            for character in sorted(key for key in node if key != "$"):
                path.append(character)
                dfs(node[character], path)
                path.pop()
        dfs(node, list(prefix))
        return words
```

COMPLEXITY | Time O(L + output) per query after O(total characters) to build | Space O(total characters)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 138

139 * Build Prefix Lookup with a Trie

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: build_trie, then walk to the prefix node and collect every word below it

QUESTION INFO

QUESTION

Build a trie from a list of words. completions(prefix) returns every inserted word that starts with the prefix (including the prefix itself when it is a word), sorted alphabetically, or an empty list when none does.

Function signature

class PrefixTrie:
    def insert(self, word: str) -> None
    def completions(self, prefix: str) -> list

EXAMPLES

QUESTION

Example 1

Input	`t = PrefixTrie(['apple', 'app', 'apply', 'apt', 'bat'])`
	`t.completions('app')`
	`t.completions('ap')`
	`t.completions('z')`

Output	`t.completions('app') -> ['app', 'apple', 'apply']`
	`t.completions('ap') -> ['app', 'apple', 'apply', 'apt']`
	`t.completions('z') -> []`

Why	`'app' is itself a word and a prefix of two more; the prefix 'z' matches nothing.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan every word for the prefix

BRUTE-FORCE IDEA

For each query test word.startswith(prefix) on every dictionary word. Correct, but each query pays for the whole dictionary.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan every word for the prefix
For each query test word.startswith(prefix) on every dictionary word
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(W * L) per query | Space O(1) extra

WHY THIS IS WEAKER | Every query touches every word, O(W * L). A trie walks only the prefix and then the subtree below it.

HOW TO IMPROVE IT

build_trie, then walk to the prefix node and collect every word below it

SHORT BEST SOLUTION

Insert every word into nested dicts, marking word ends. To answer a prefix, walk down one character at a time (no match means no completions), then DFS the subtree collecting words.

```python
# BASE QUESTION (optimal): Trie with a DFS below the
# prefix node
# Idea: Insert every word into nested dicts, marking word
# ends. To answer a prefix, walk down one character at a
# time (no match means no completions), then DFS the
# subtree collecting words.
class PrefixTrie:
    END = "$"

    def __init__(self, words=()):
        self.root = {}
        for word in words:
            self.insert(word)

    def insert(self, word):
        node = self.root
        for character in word:
            node = node.setdefault(character, {})
        node[self.END] = True

    def completions(self, prefix):
        node = self.root
        for character in prefix:
            if character not in node:
```

```
            return []
        node = node[character]
    words, stack = [], [(node, prefix)]
    while stack:
        current_node, text = stack.pop()
        if self.END in current_node:
            words.append(text)
        for character, child in current_node.items():
            if character != self.END:
                stack.append((child, text + character))
    return sorted(words)
```

COMPLEXITY | Time O(L) to walk the prefix + O(size of the answer) | Space O(total characters)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 139

140 * Greedy Longest-Match Dictionary Tokenizer * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the same longest-match walk, with a priority rule to choose between equal-length matches

QUESTION INFO

QUESTION

Phone-screen coding round (R1). Tokenize a text against a dictionary of **`key:id`** entries using greedy longest-match; emit the matched id when a key matches, otherwise output the unmatched character as a literal. One follow-up on changing the tie-breaking / priority rule for which token wins. A second follow-up appears when the dictionary is very large: discuss how to keep the matcher memory- and lookup-efficient at scale.

Input: a text string and a dictionary mapping key (string) -> id.
Sweep the text left-to-right. At each position, find the longest dictionary key that matches a prefix of the remaining text; emit its id. If no key matches at the current position, emit the single character as a literal and advance by one.
Output the concatenation (or list) of ids and literal characters in order.
Follow-up: the interviewer changes the tie-breaking / priority rule for which key wins when multiple match (e.g., not longest but highest-priority by dictionary order). Adapt the data structure so the rule swap is cheap.
Follow-up: the dictionary becomes very large. Discuss how to keep memory and per-position lookup efficient — a compressed / radix trie (or sharding the key space) instead of holding every key in a flat map.

EXAMPLES

Example 1	

Input	`t = Tokenizer({'ab': 1, 'abc': 2, 'c': 3})`
	`t.tokenize('abcab')`
	`t.tokenize('abx')`

Output	`t.tokenize('abcab') -> [2, 1]` `t.tokenize('abx') -> [1, 'x']`
Why	`At each position the longest dictionary key wins: 'abc' (not 'ab' followed by 'c'), then 'ab'. A character that starts no key is emitted as it is.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Hash map lookup for every prefix length

BRUTE-FORCE IDEA

At every position try the prefix lengths from longest to shortest in a dictionary lookup. Correct and easy; each lookup hashes a substring, so it costs O(L^2) per position.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Hash map lookup for every prefix length
At every position try the prefix lengths from longest to shortest in a dictionary lookup
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n * L) lookups, each O(L) | Space O(dictionary)

WHY THIS IS WEAKER | Every lookup hashes a substring, O(L^2) per position. Walking a trie of the keys one character at a time costs O(L) per position.

HOW TO IMPROVE IT

the same longest-match walk, with a priority rule to choose between equal-length matches

SHORT BEST SOLUTION

Insert every key into a trie with its id at the last node. From each text position walk down while the next character exists, remembering the last terminal passed. Emit its id and jump past its length, or emit the single character if no terminal was found. For a priority rule keep the terminal with the best priority instead of the last one.

```python
# BASE QUESTION (optimal): Trie walk keeping the deepest
# (or best-priority) terminal
# Idea: Insert every key into a trie with its id at the
# last node. From each text position walk down while the
# next character exists, remembering the last terminal
# passed. Emit its id and jump past its length, or emit
# the single character if no terminal was found. For a
# priority rule keep the terminal with the best priority
# instead of the last one.
def tokenize_trie(text, table, priority=None):
    root = {}
    for key, token_id in table.items():
        node = root
        for character in key:
            node = node.setdefault(character, {})
        node["$"] = (key, token_id)
    tokens, start = [], 0
    while start < len(text):
        node, best_match, end = root, None, start
        while end < len(text) and text[end] in node:
            node = node[text[end]]
            end += 1
            if "$" in node and (best_match is None or priority is None or priority(node["$"][0]) < priority(bes
t_match[0])):
                best_match = node["$"]
        if best_match is None:
            tokens.append(text[start])
            start += 1
        else:
            tokens.append(best_match[1])
            start += len(best_match[0])
    return tokens
```

COMPLEXITY | Time O(n * L) | Space O(total key length)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 140

141 * Check if all substrings are dictionary words * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: for every start walk the trie; a substring of length 3 or more that leaves the trie or lacks END fails the check

QUESTION INFO

QUESTION

You are given a string **`s`** (letters only) and access to an English dictionary **`dict`** (a set of valid words). Return **`true`** if **every contiguous substring** of **`s`** with length **`>= 3`** is a valid dictionary word (i.e., for all **`i, j`** with **`0 <= i <= j < n`** and **`j - i + 1 >= 3`**, **`s[i..j] in dict`**). Otherwise return **`false`**. Clarifications:

- Substring means **contiguous**.
- You may assume dictionary lookups are available (e.g., **`dict.contains(word)`**), or you can preprocess the dictionary. Example:
- If **`s = "cats"`**, you must check **`"cat"`**, **`"ats"`**, and **`"cats"`** (and any other substrings of length >= 3).

EXAMPLES

Example 1	
Input	`'abcd', {'abc', 'abcd', 'bcd'}`
Output	`true`
Why	`Every substring of length 3 or more (abc, bcd, abcd) is in the dictionary.`

Example 2	
Input	`'abcd', {'abc', 'abcd'}`
Output	`false`
Why	`'bcd' is missing from the dictionary.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Enumerate each substring and look it up in a set

BRUTE-FORCE IDEA

Slice s[i:j] for every i, j with j - i >= 3 and test membership. Correct and simple; every lookup re-hashes the whole substring.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Enumerate each substring and look it up in a set
Slice s[i:j] for every i
j with j - i >= 3
test membership
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) lookups, O(n^3) characters hashed | Space O(1) extra

WHY THIS IS WEAKER | About n^2 / 2 lookups and each hashes a whole substring, O(n^3) characters. Walking a trie from each start, one character at a time, is O(n^2) with no hashing of long strings.

HOW TO IMPROVE IT

for every start walk the trie; a substring of length 3 or more that leaves the trie or lacks END fails the check

SHORT BEST SOLUTION

For each start i descend the trie along text[start:]. At depth 1 and 2 only walk; from depth 3 on the current node must be marked as a word end. If the walk leaves the trie (and at least three characters remain from this start) or hits a non-word node at depth >= 3, return False immediately. No substring is built or rehashed.

```python
# BASE QUESTION (optimal): Walk a trie from each start
# position
# Idea: For each start i descend the trie along text[start:]. At
# depth 1 and 2 only walk; from depth 3 on the current
# node must be marked as a word end. If the walk leaves
# the trie (and at least three characters remain from this
# start) or hits a non-word node at depth >= 3, return
# False immediately. No substring is built or rehashed.
def all_substrings_words(text, words):
    root = {}
    for word in words:
        node = root
        for character in word:
            node = node.setdefault(character, {})
        node["$"] = True
    for start in range(len(text)):
        node = root
        for depth, character in enumerate(text[start:], 1):
            node = node.get(character)
            if node is None:
                return len(text) - start < 3                # a start with 3 or more characters left has a subs
tring that is not a word
            if depth >= 3 and "$" not in node:
                return False
    return True
```

COMPLEXITY | Time O(n^2) | Space O(dictionary size)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 141

142 * Return dictionary words matching a prefix * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: no trie needed: sort the words and bisect to the first match, then read forward while startswith(prefix)

QUESTION INFO

QUESTION

You are given a dictionary of words. Return all words that start with a given prefix. Matching is case-sensitive, duplicates in the dictionary are returned as often as they occur, and the result is an empty list when nothing matches. The index returns the matches in alphabetical order; limit (optional) stops after that many words.

Function signature

class PrefixIndex:
  def __init__(self, words: list)
  def matches(self, prefix: str, limit=None) -> list

EXAMPLES

Example 1	

Input	`idx = PrefixIndex(['apple', 'app', 'Apple', 'apply', 'apple', 'bat'])`
	`idx.matches('app')`
	`idx.matches('app', limit=2)`
	`idx.matches('cat')`

Output	`idx.matches('app') -> ['app', 'apple', 'apple', 'apply']`
	`idx.matches('app', limit=2) -> ['app', 'apple']`
	`idx.matches('cat') -> []`

Why	`'Apple' starts with a capital letter so it does not match 'app'; 'apple' is listed twice because it is in the dictionary twice.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Filter the whole list per query

BRUTE-FORCE IDEA

Test startswith on every word. Correct and fine for one query; each query pays for the whole list.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Filter the whole list per query
Test startswith on every word
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(W * L) per query | Space O(m)

WHY THIS IS WEAKER | Each query scans the whole dictionary, O(W * L). With the words sorted, binary search finds the block of matches directly.

HOW TO IMPROVE IT

no trie needed: sort the words and bisect to the first match, then read forward while startswith(prefix)

SHORT BEST SOLUTION

With the words sorted, the matches are the words from bisect_left(prefix) until the first word that no longer starts with the prefix. The upper edge can be found by a second bisect with prefix + a character larger than any other, or by walking forward while startswith holds.

```python
from bisect import bisect_left


# BASE QUESTION (optimal): Sort once, binary search the
# range
# Idea: With the words sorted, the matches are the words
# from bisect_left(prefix) until the first word that no
# longer starts with the prefix. The upper edge can be
# found by a second bisect with prefix + a character
# larger than any other, or by walking forward while
# startswith holds.
class PrefixIndex:
    def __init__(self, words):
        self.words = sorted(words)

    def matches(self, prefix, limit=None):
        position = bisect_left(self.words, prefix)
        matches = []
        while position < len(self.words) and self.words[position].startswith(prefix):
            matches.append(self.words[position])
            if limit is not None and len(matches) == limit:
                break
            position += 1
        return matches
```

COMPLEXITY | Time O(log W + m) per query after O(W log W) sorting | Space O(W)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 142

143 * Implement Trie Insert and Exact-Word Search * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: insert is build_trie; search is walk and then END in node

QUESTION INFO

QUESTION

Implement a trie with insert(word), search(word) and startsWith(prefix). search is true only when the complete word was inserted earlier (a longer stored word that merely starts with it does not count); startsWith is true when some inserted word begins with the prefix.

Function signature

class Trie:
    def insert(self, word: str) -> None
    def search(self, word: str) -> bool
    def startsWith(self, prefix: str) -> bool

EXAMPLES

Example 1

Input	`t = Trie()`
	`t.insert('apple')`
	`t.search('apple')`
	`t.search('app')`
	`t.startsWith('app')`
	`t.insert('app')`
	`t.search('app')`

Output	`t.search('apple') -> true`
	`t.search('app') -> false`
	`t.startsWith('app') -> true`
	`t.search('app') -> true`

Why	`'app' is only a prefix of 'apple' until it is inserted on its own.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Keep a list of words and scan

BRUTE-FORCE IDEA

Store the words in a list; search compares equality with each, startsWith tests the prefix on each. Correct, with every call linear in the dictionary size.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Keep a list of words and scan
Store the words in a list
search compares equality with each
startsWith tests the prefix on each
```

BRUTE-FORCE COMPLEXITY | Time O(W * L) per operation | Space O(total characters)

WHY THIS IS WEAKER | Each call scans every stored word, O(W * L). A trie with an end-of-word flag makes every call O(L).

HOW TO IMPROVE IT

insert is build_trie; search is walk and then END in node

SHORT BEST SOLUTION

Each node has children by character and a flag. insert creates missing nodes and marks the last one. search walks the word and requires the flag at the end; startsWith only requires that the walk succeeds.

```python
# BASE QUESTION (optimal): Trie of nodes with an end-of-
# word flag
# Idea: Each node has children by character and a flag.
# insert creates missing nodes and marks the last one.
# search walks the word and requires the flag at the end;
# startsWith only requires that the walk succeeds.
class Trie:
    def __init__(self):
        self.root = {"children": {}, "end": False}

    def _walk(self, text):
        node = self.root
        for character in text:
            node = node["children"].get(character)
            if node is None:
                return None
        return node

    def insert(self, word):
        node = self.root
        for character in word:
            node = node["children"].setdefault(character, {"children": {}, "end": False})
        node["end"] = True

    def search(self, word):
        node = self._walk(word)
        return node is not None and node["end"]

    def startsWith(self, prefix):
        node = self._walk(prefix)
        return node is not None and (node["end"] or bool(node["children"]))     # a node counts only if some wo
rd passes through it
```

COMPLEXITY | Time O(L) per operation | Space O(total characters)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 143

A dict for lookup plus an ordered container for who is next (lazy deletion)

PATTERN CONTEXT

Keep a dict for O(1) lookup and a deque, list or heap for the order. When something is removed, delete it from the dict only and skip stale entries when you read from the order. LRU caches, waitlists, undo stacks and schedulers all fit.

146 * Design a restaurant waitlist system * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: a segment tree of party sizes gives the first entry that fits in O(log n); the dict maps ids to slots

QUESTION INFO

QUESTION

You are implementing the waitlist system for a restaurant. Parties arrive over time, can cancel, and get seated when a table becomes available.

Rules

- Each party has a unique **`partyId`**, a **`name`**, and a **`size`** (number of people).
- Parties are seated in **arrival order**, but only if they **fit** the available table.
- When a table of capacity **c** becomes available, you must seat the **earliest-arrived** party whose **`size <= c`**.
- If no party fits, the table remains unused for that event.

Operations to support Design a

data structure / class that supports the following operations efficiently:

1. **`addParty(name, size) -> partyId`**
- Adds a new party to the end of the waitlist and returns its id.
2. **`cancelParty(partyId) -> bool`**
- Removes the party from the waitlist if present.
- Returns **`true`** if removed, otherwise **`false`**.
3. **`seatTable(capacity) -> partyId or null`**
- Finds and removes the earliest party with **`size <= capacity`**.
- Returns that party's id, or **`null`** if nobody fits.
4. **`getPosition(partyId) -> int`**
- Returns how many parties are ahead of this party in the current waitlist order (0-based).
- If the party is not in the waitlist, return **`-1`**.

Constraints

- Up to **`2 * 10^5`** operations.
- Party size is a small positive integer (e.g., 1–20).
- Aim for better-than-linear time per operation (especially for **`seatTable`** and **`getPosition`**).

Example

(one possible interaction) - addParty("A", 4) -> id1 - addParty("B", 2) -> id2 - seatTable(2) -> id2 (B fits and is earliest among those that fit) - seatTable(4) -> id1

EXAMPLES

QUESTION

Example 1

Input	`w = Waitlist()`
	`w.add(1, 'ann', 4)`
	`w.add(2, 'bob', 2)`
	`w.add(3, 'cy', 3)`
	`w.seat(3)`
	`w.cancel(3)`
	`w.seat(3)`
	`w.seat(4)`

Output	`w.seat(3) -> 2`
	`w.cancel(3) -> true`
	`w.seat(3) -> null`
	`w.seat(4) -> 1`

Why	`A table for 3 takes the earliest party that fits: bob (2), not ann (4). After cy cancels, nobody left fits a table for 3; a table for 4 seats ann.`

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan the waiting list in arrival order

BRUTE-FORCE IDEA

Keep parties in arrival order in a list. Seat scans from the front for the first party whose size fits; cancel removes by id. Correct; every operation walks the list.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan the waiting list in arrival order
Keep parties in arrival order in a list
process every required item or candidate directly
```

BRUTE-FORCE COMPLEXITY | Time O(n) per seat and per cancel | Space O(n)

WHY THIS IS WEAKER | Both operations walk the list, O(n). A segment tree of the smallest size per range finds the leftmost party that fits in O(log n).

HOW TO IMPROVE IT

a segment tree of party sizes gives the first entry that fits in O(log n); the dict maps ids to slots

SHORT BEST SOLUTION

Each arrival gets the next position. The tree stores, for each range, the smallest waiting party size (infinity for empty or seated slots). seat(c) descends from the root: go left if the left range's minimum is at most c, else right, ending at the leftmost fitting party. Cancel and seat set a leaf to infinity and recompute the path up. The tree has fixed capacity here; double and rebuild to grow.

```python
# BASE QUESTION (optimal): Segment tree of minimum size
# over arrival order
# Idea: Each arrival gets the next position. The tree
# stores, for each range, the smallest waiting party size
# (infinity for empty or seated slots). seat(c) descends
# from the root: go left if the left range's minimum is at
# most c, else right, ending at the leftmost fitting
# party. Cancel and seat set a leaf to infinity and
# recompute the path up. The tree has fixed capacity here;
# double and rebuild to grow.
class Waitlist:
    INF = float("inf")
```

```
    def __init__(self, capacity=1024):
        self.leaf_count = capacity
        self.tree = [self.INF] * (2 * capacity)
        self.arrivals, self.slot_of, self.party_at = 0, {}, {}

    def _set(self, node_index, value):
        node_index += self.leaf_count
        self.tree[node_index] = value
        while node_index > 1:
            node_index //= 2
            self.tree[node_index] = min(self.tree[2 * node_index], self.tree[2 * node_index + 1])

    def add(self, party_id, name, size):
        self.slot_of[party_id] = self.arrivals
        self.party_at[self.arrivals] = party_id
        self._set(self.arrivals, size)
        self.arrivals += 1

    def cancel(self, party_id):
        if party_id not in self.slot_of or self.tree[self.slot_of[party_id] + self.leaf_count] == self.INF:
            return False
        self._set(self.slot_of[party_id], self.INF)
        return True

    def seat(self, capacity):
        if self.tree[1] > capacity:
            return None
        node_index = 1
        while node_index < self.leaf_count:
            node_index = 2 * node_index if self.tree[2 * node_index] <= capacity else 2 * node_index + 1
        self._set(node_index - self.leaf_count, self.INF)
        return self.party_at[node_index - self.leaf_count]
```

COMPLEXITY | Time O(log n) per operation | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 146

147 * Implement Python's itertools.tee

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: one deque per consumer; whoever runs out pulls the next value from the source and appends it to every queue

QUESTION INFO

QUESTION

Implement itertools.tee: given an iterable and n, return n independent iterators that each produce the same values from the one source. The source must be read only once, and a value must be kept only until every iterator has taken it.

Function signature

def tee_queues(iterable, n=2) -> tuple

EXAMPLES

QUESTION

Example 1

Input	`a, b = tee_queues([1, 2, 3])`
	`next(a)`
	`next(a)`
	`next(b)`
	`list(b)`
	`list(a)`

Output	`next(a) -> 1`
	`next(a) -> 2`
	`next(b) -> 1`
	`list(b) -> [2, 3]`
	`list(a) -> [3]`

Why	`Iterator a runs ahead by two values; b still sees every value from the start, and neither disturbs the other.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Build the whole source into a list

BRUTE-FORCE IDEA

Read everything into a list and give each consumer an index. Correct for finite data, but it consumes the source eagerly, never works on an infinite iterator and holds all items in memory.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Build the whole source into a list
Read everything into a list
give each consumer an index
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) up front | Space O(n)

WHY THIS IS WEAKER | It reads the source completely before anything is consumed, never works on an infinite or lazy source, and keeps every item in memory. One buffer per consumer (or a shared linked list) holds only what the slowest consumer has not read.

HOW TO IMPROVE IT

one deque per consumer; whoever runs out pulls the next value from the source and appends it to every queue

SHORT BEST SOLUTION

When a consumer's deque is empty it pulls the next item from the source and appends it to every consumer's deque, then pops its own. Each item is read from the source exactly once and is dropped by a consumer after being read, so memory is only the gap between the leader and the slowest follower.

```python
from collections import deque


# BASE QUESTION (optimal): One deque per consumer, filled
# lazily
# Idea: When a consumer's deque is empty it pulls the next
# item from the source and appends it to every consumer's
# deque, then pops its own. Each item is read from the
# source exactly once and is dropped by a consumer after
# being read, so memory is only the gap between the leader
# and the slowest follower.
def tee(iterable, copies=2):
    source_iterator = iter(iterable)
    queues = [deque() for _ in range(copies)]
```

```
    def consumer(own_queue):
        while True:
            if not own_queue:
                try:
                    value = next(source_iterator)
                except StopIteration:
                    return
                for queue in queues:
                    queue.append(value)
            yield own_queue.popleft()

    return tuple(consumer(queue) for queue in queues)
```

COMPLEXITY | Time O(n) total, O(1) amortized per item per consumer | Space O(lag between fastest and slowest consumer)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 147

148 * Restaurant Waitlist: join / delete / find_first_match — * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: a deque per party size with tickets; delete only removes the name from the dict; find_first_match skips dead tickets

QUESTION INFO

QUESTION

Design a restaurant waitlist. join(name, size) adds a party with its size at the back. delete(name) removes a party (it may be anywhere in the queue). find_first_match(table) returns the name of the party that joined earliest among those whose size fits the table (size at most table), or None.

Function signature

class Waitlist:
  def join(self, name: str, size: int) -> None
  def delete(self, name: str) -> None
  def find_first_match(self, table: int) -> str | None

EXAMPLES

Example 1

Input	`w = Waitlist()`
	`w.join('ann', 4)`
	`w.join('bob', 2)`
	`w.join('cy', 3)`
	`w.find_first_match(3)`
	`w.delete('bob')`
	`w.find_first_match(3)`
	`w.find_first_match(1)`

Output	`w.find_first_match(3) -> 'bob'`
	`w.find_first_match(3) -> 'cy'`
	`w.find_first_match(1) -> null`

Why	`A table for 3 skips ann (4 is too big) and takes bob, who joined before cy. After bob leaves it is cy. No party fits a table for 1.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Keep one list in join order and scan it

BRUTE-FORCE IDEA

Append on join, remove by value on delete and scan from the front on find. Correct; delete and find are linear.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Keep one list in join order and scan it
Append on join
remove by value on delete
scan from the front on find
```

BRUTE-FORCE COMPLEXITY | Time O(n) per find and per delete | Space O(n)

WHY THIS IS WEAKER | Delete and find are linear in the list. One queue per party size with lazy deletion makes join and delete O(1) and find touch only the queue fronts.

HOW TO IMPROVE IT

a deque per party size with tickets; delete only removes the name from the dict; find_first_match skips dead tickets

SHORT BEST SOLUTION

join stores (sequence number, name) in the queue for its size and in a map from name to its record. delete just removes the record from the map. find_first_match, for each size up to the table size, pops dead entries from the front of its queue, then picks the live front with the smallest sequence number. No one is removed by find.

```python
from collections import defaultdict
from collections import deque


# BASE QUESTION (optimal): Queue per party size with lazy
# deletion
# Idea: join stores (sequence number, name) in the queue
# for its size and in a map from name to its record.
# delete just removes the record from the map.
# find_first_match, for each size up to the table size,
# pops dead entries from the front of its queue, then
# picks the live front with the smallest sequence number.
# No one is removed by find.
class Waitlist:
    def __init__(self):
        self.last_number, self.live_number, self.queues = 0, {}, defaultdict(deque)

    def join(self, name, size):
        self.last_number += 1
        self.live_number[name] = self.last_number
        self.queues[size].append((self.last_number, name))

    def delete(self, name):
        self.live_number.pop(name, None)

    def find_first_match(self, table_size):
        oldest = None
        for size, queue_of_size in self.queues.items():
            if size > table_size:
                continue
            while queue_of_size and self.live_number.get(queue_of_size[0][1]) != queue_of_size[0][0]:
                queue_of_size.popleft()
            if queue_of_size and (oldest is None or queue_of_size[0][0] < oldest[0]):
                oldest = queue_of_size[0]
        return oldest[1] if oldest else None
```

COMPLEXITY | Time O(1) join and delete, O(S) find (S distinct sizes up to the table size) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 148

149 * Store With Cheapest-Item Removal * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the order is a heap keyed by (price, timestamp); remove_cheapest pops it

QUESTION INFO

QUESTION

Design a Store that holds items. Each item has a string value , an integer price in cents, and an integer timestamp . Support: add(value, price, timestamp) : insert an item. remove_cheapest() : remove and return the value of the item with the smallest price. If several items share the smallest price, remove the one with the smallest timestamp . Return None if the store is empty. A heap keyed by (price, timestamp) gives O(log n) per operation.

Function signature

```
class Store:
    def __init__(self) -> None:
        ...
    def add(self, value: str, price: int, timestamp: int) -> None:
        ...
    def remove_cheapest(self) -> str | None:
        """Remove and return the value of the lowest-priced item (ties broken by
earliest timestamp)."""
        ...
```

Follow-up (as reported): Improve remove_cheapest from O(log n) to O(1) amortized using the price bound: keep a length-10001 array indexed by price (each slot a queue/count of items) and track the smallest non-empty price index seen so far.

EXAMPLES

Example 1	
Input	`add("a",5,1); add("b",2,2); add("c",8,3); remove_cheapest()`
Output	`"b"`
Why	`"b" has the smallest price (2 cents).`

Example 2	
Input	`add("x",7,10); add("y",7,4); remove_cheapest()`
Output	`"y"`
Why	`Prices tie at 7; "y" wins on the earlier timestamp (4 < 10).`

Example 3	
Input	`remove_cheapest() # on an empty store`
Output	`None`

CONSTRAINTS

- Prices are integers with 1 <= price <= 10000 (i.e. 1 cent up to $100.00).
- Up to 10^6 operations.

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- Prices are integers with 1 <= price <= 10000 (i.e. 1 cent up to $100.00).
- Up to 10^6 operations.

BRUTE FORCE | A list scanned on every removal

BRUTE-FORCE IDEA

Start with: A list scanned on every removal. Start with: A list scanned on every removal. Remove_cheapest looks at every item.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: A list scanned on every removal
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n) per removal | Space O(n)

WHY THIS IS WEAKER | Remove_cheapest looks at every item.

HOW TO IMPROVE IT

the order is a heap keyed by (price, timestamp); remove_cheapest pops it

SHORT BEST SOLUTION

The smallest (price, timestamp) is always at the top of the heap.

```python
# BASE QUESTION (optimal): A heap keyed by price and
# timestamp
# Idea: The smallest (price, timestamp) is always at the
# top of the heap.
import heapq


class Store:
    def __init__(self):
        self.heap = []

    def add(self, value, price, timestamp):
        heapq.heappush(self.heap, (price, timestamp, value))

    def remove_cheapest(self):
        if not self.heap:
            return None
        return heapq.heappop(self.heap)[2]
```

COMPLEXITY | Time O(log n) per operation | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 149

150 * Round-Robin Job Scheduler * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: a deque of [job, time left]; pop, run for min(quantum, left); push back at the end if unfinished, else record the finish time

QUESTION INFO

QUESTION

The third question was "not really leetcode style" but was a job-scheduling problem solved with a round- robin approach. Reconstruct it as follows. You are given a list of jobs, each with a unique string id and a positive integer burst (the number of time units of CPU it needs). All jobs are available at time 0 and are placed in a ready queue in the given order. The CPU runs jobs round-robin with a fixed time quantum : it takes the job at the front of the queue, runs it for min(quantum, remaining) time units, and then — if the job still has work left — puts it at the back of the queue; otherwise the job completes at that moment. Return the list of (id, completion_time) pairs in the order the jobs complete.

QUESTION

Function signature

def round_robin(jobs: list[tuple[str, int]], quantum: int) -> list[tuple[str, int]]:
    """Simulate round-robin scheduling; return (id, completion_time) pairs in completion order."""
    ...

Follow-up (as reported): Report note ("a constraint was changed"): add per-job arrival times so a job only enters the queue once the clock reaches its arrival. Hint: keep a pointer over arrival-sorted jobs and admit them each time the clock advances.

EXAMPLES

Example 1	
Input	`jobs = [("A", 4), ("B", 3), ("C", 2)], quantum = 2`
Output	`[("C", 6), ("A", 8), ("B", 9)]`
Why	`t=0-2 A (rem 2), t=2-4 B (rem 1), t=4-6 C completes at 6, t=6-8 A completes at 8, t=8-9 B completes at 9.`

Example 2	
Input	`jobs = [("X", 5)], quantum = 2`
Output	`[("X", 5)]`
Why	`One job, run in slices 2+2+1, finishing at time 5.`

Example 3	
Input	`jobs = [("P", 1), ("Q", 1)], quantum = 3`
Output	`[("P", 1), ("Q", 2)]`
Why	`Each finishes within a single quantum: P at 1, Q at 2.`

CONSTRAINTS

- 1 <= len(jobs) <= 10^4
- 1 <= burst <= 10^6 , 1 <= quantum <= 10^6
- Job ids are unique, non-empty strings.

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(jobs) <= 10^4
- 1 <= burst <= 10^6 , 1 <= quantum <= 10^6
- Job ids are unique, non-empty strings.

BRUTE FORCE | Cycle over the job list until all are done

BRUTE-FORCE IDEA

Start with: Cycle over the job list until all are done. It rescans the whole list every round, including jobs that already finished.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Cycle over the job list until all are done
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n * rounds) | Space O(n)

WHY THIS IS WEAKER | It rescans the whole list every round, including jobs that already finished.

HOW TO IMPROVE IT

a deque of [job, time left]; pop, run for min(quantum, left); push back at the end if unfinished, else record the finish time

SHORT BEST SOLUTION

Run the front job for one slice and send it to the back if it is not finished.

```python
# BASE QUESTION (optimal): A queue of jobs
# Idea: Run the front job for one slice and send it to the
# back if it is not finished.
from collections import deque


def round_robin(jobs, quantum):
    queue = deque([job_id, remaining_time] for job_id, remaining_time in jobs)
    clock, done = 0, []
    while queue:
        job = queue.popleft()
        time_slice = min(quantum, job[1])
        clock += time_slice
        job[1] -= time_slice
        if job[1] > 0:
            # not finished: back of the line
            queue.append(job)
        else:
            done.append((job[0], clock))
    return done
```

COMPLEXITY | Time O(total burst / quantum + n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 150

151 * Aggregate FizzBuzz Offer Matches * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: a deque of offers per type; a new offer matches against the front of the opposite queue, partial quantities stay at the front

QUESTION INFO

QUESTION

You are given a list of offers in priority order (earliest = highest priority). Each offer is a dict with keys type (either "Fizz" or "Buzz"), name (a string), and quantity (a positive integer). Fizz offers match against Buzz offers on a first-come, first-served basis, and offers may be partially fulfilled. Process the offers in order, maintaining a FIFO queue of unmatched Fizz offers and a FIFO queue of unmatched Buzz offers. When a new offer arrives, match it against the front of the opposite queue repeatedly, each time consuming min(remaining quantities) and recording a match (fizz_name, buzz_name, matched_quantity) ; drop an offer from the queue once its quantity hits 0. If the arriving offer still has quantity left when the opposite queue empties, enqueue it in its own queue. Aggregate matches so the same (fizz_name, buzz_name) pair is reported once with its summed quantity. Return the aggregated matches, in the order each pair was first produced, formatted as: "FizzBuzz! <quantity> <FizzName>:<BuzzName>" .

Function signature

def fizzbuzz_matches(offers: list[dict]) -> list[str]:

...

EXAMPLES

QUESTION

Example 1

Input	`offers = [ {"type": "Fizz", "name": "A", "quantity": 5}, {"type": "Buzz", "name": "X", "quantity": 3}, {"type": "Buzz", "name": "Y", "quantity": 4}, ]`
Output	`["FizzBuzz! 3 A:X", "FizzBuzz! 2 A:Y"]`
Why	`A(5) waits. X(3) matches A -> (A,X,3), A now 2. Y(4) matches A's remaining 2 -> (A,Y,2); Y's leftover 2 waits with no Fizz to match.`

Example 2

Input	`offers = [ {"type": "Fizz", "name": "A", "quantity": 3}, {"type": "Buzz", "name": "X", "quantity": 1}, {"type": "Buzz", "name": "X", "quantity": 2}, ]`
Output	`["FizzBuzz! 3 A:X"]`
Why	`A matches the first X for 1 and the second X for 2. Both are the pair (A, X), so they aggregate to a single "FizzBuzz! 3 A:X".`

CONSTRAINTS

- 0 <= len(offers) <= 10**4 ; quantities are positive integers.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(offers) <= 10**4 ; quantities are positive integers.

BRUTE FORCE | Lists with pop(0)

BRUTE-FORCE IDEA

Start with: Lists with pop(0). Removing from the front of a list shifts every other element.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Lists with pop(0)
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n * matches) | Space O(n)

WHY THIS IS WEAKER | Removing from the front of a list shifts every other element.

HOW TO IMPROVE IT

a deque of offers per type; a new offer matches against the front of the opposite queue, partial quantities stay at the front

SHORT BEST SOLUTION

Match against the opposite queue's front until the offer or the queue is empty.

```python
# BASE QUESTION (optimal): Two FIFO queues and an ordered
# total
# Idea: Match against the opposite queue's front until the
# offer or the queue is empty.
from collections import deque


def fizzbuzz_matches(offers):
    queues = {"Fizz": deque(), "Buzz": deque()}
    totals = {}
    for offer in offers:
        mine = queues[offer["type"]]
        other = queues["Buzz" if offer["type"] == "Fizz" else "Fizz"]
        quantity_left = offer["quantity"]
        while quantity_left and other:
```

```
            front = other[0]
            matched = min(quantity_left, front[1])
            fizz_name = offer["name"] if offer["type"] == "Fizz" else front[0]
            buzz_name = front[0] if offer["type"] == "Fizz" else offer["name"]
            totals[(fizz_name, buzz_name)] = totals.get((fizz_name, buzz_name), 0) + matched
            quantity_left -= matched
            front[1] -= matched
            if front[1] == 0:
                other.popleft()
        if quantity_left:
            mine.append([offer["name"], quantity_left])
    return [f"FizzBuzz! {quantity} {fizz}:{buzz}" for (fizz, buzz), quantity in totals.items()]
```

COMPLEXITY | Time O(offers + matches) | Space O(offers)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 151

GROUP 19 OF 43 | 4 QUESTIONS | PART A - PART 1

Rolling state: the next state comes only from the previous one

PATTERN CONTEXT

You never need the whole table, only the previous state: one number, a pair, or one row. Write the state, then one line that builds the new state from the old one, and loop.

155 * Maximize sum without choosing adjacent elements * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: exactly the house robber update on the array

QUESTION INFO

QUESTION

Given a list of non-negative numbers, choose some of them so that no two chosen numbers are neighbours in the list, and return the largest possible sum of the chosen numbers.

Function signature

def rob_line(nums: list) -> int

EXAMPLES

Example 1	
Input	`[2, 7, 9, 3, 1]`
Output	`12`
Why	`Choose 2, 9 and 1 (positions 0, 2, 4): 12.`

Example 2	
Input	`[5]`
Output	`5`
Why	`One number: take it.`

Example 3	
Input	`[]`
Output	`0`
Why	`Nothing to choose: 0.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recursion over take or skip

BRUTE-FORCE IDEA

At index i either skip it (go to i + 1) or take it and jump to i + 2, and keep the larger total. Exact; the same suffixes are solved over and over.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recursion over take or skip
At index i either skip it (go to i + 1) or take it
jump to i + 2
keep the larger total
```

BRUTE-FORCE COMPLEXITY | Time O(2^n) | Space O(n)

WHY THIS IS WEAKER | Two calls per index and the same suffix is solved repeatedly: O(2^n). Keeping only the best total with and without the previous element is O(n) time and O(1) space.

HOW TO IMPROVE IT

exactly the house robber update on the array

SHORT BEST SOLUTION

take is the best total that includes the current element, skip the best that excludes it. For the next element: new take = skip + house_value (the previous element must have been skipped), new skip = max(take, skip). The answer is the larger of the two at the end.

```
# BASE QUESTION (optimal): Two running values (take, skip)
# Idea: take is the best total that includes the current
# element, skip the best that excludes it. For the next
# element: new take = skip + house_value (the previous element must
# have been skipped), new skip = max(take, skip). The
# answer is the larger of the two at the end.
def _house_dp(house_money):
    take = skip = 0
    for house_value in house_money:
        take, skip = skip + house_value, max(take, skip)
    return max(take, skip)
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 155

156 * Paint House II * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: state = a whole row of costs, one per colour; new[j] = cost[j] + the smallest previous value that is not colour j (keep the two smallest)

QUESTION INFO

QUESTION

There are n houses in a row, each to be painted one of k colors. costs[i][j] is the cost to paint house i with color j . No two adjacent houses may share the same color. Return the minimum total cost to paint all houses.

Function signature

def min_cost_paint_houses(costs: list[list[int]]) -> int:

    ...

Follow-up (as reported): Reduce the per-row transition from O(k) per cell to O(1) amortized by tracking the two smallest values of the previous row.

EXAMPLES

Example 1	
Input	costs = [[1, 5, 3], [2, 9, 4]]
Output	5
Why	Paint house 0 color 0 (1) and house 1 color 2 (4) -> 5; or house 0 color 2 (3) and house 1 color 0 (2) -> 5.

Example 2	
Input	costs = [[1, 3], [2, 4]]
Output	5
Why	1 (house0 color0) + 4 (house1 color1) = 5, the cheapest valid pairing.

QUESTION

Example 3	
Input	`costs = [[7]]`
Output	`7`
Why	`A single house, one color.`

CONSTRAINTS

- costs is an n x k grid, 0 <= n , k >= 1 (and k >= 2 whenever n >= 2 , so a valid painting exists).
- Costs are non-negative integers. Return 0 for n == 0 .
- Aim for O(n*k) time.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- costs is an n x k grid, 0 <= n , k >= 1 (and k >= 2 whenever n >= 2 , so a valid painting exists).
- Costs are non-negative integers. Return 0 for n == 0 .
- Aim for O(n*k) time.

BRUTE FORCE | Try every colouring

BRUTE-FORCE IDEA

Start with: Try every colouring. Start with: Try every colouring. It enumerates all k^n ways to colour the row.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every colouring
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(k^n * n) | Space O(n)

WHY THIS IS WEAKER | It enumerates all k^n ways to colour the row.

HOW TO IMPROVE IT

state = a whole row of costs, one per colour; new[j] = cost[j] + the smallest previous value that is not colour j (keep the two smallest)

SHORT BEST SOLUTION

A cell only needs the cheapest previous colour that differs from its own.

```python
# BASE QUESTION (optimal): Two smallest values of the
# previous row
# Idea: A cell only needs the cheapest previous colour
# that differs from its own.
def min_cost_paint_houses(costs):
    previous_total = [0] * len(costs[0])
    for house_costs in costs:
        cheapest_two = sorted(range(len(previous_total)), key=lambda colour: previous_total[colour])[:2]
        first = cheapest_two[0]
        second = cheapest_two[1] if len(cheapest_two) > 1 else cheapest_two[0]
        previous_total = [house_costs[colour] + (previous_total[second] if colour == first else previous_total[
first])
                for colour in range(len(house_costs))]
    return min(previous_total)
```

COMPLEXITY | Time O(n k) | Space O(1) extra

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 156

157 * Given n houses in a line with money in each, find the maximum amount a robber can steal, without stealing from two adjacent houses. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: state = (best if you take this house, best if you skip it): take, skip = skip + house_value, max(take, skip)

QUESTION INFO

QUESTION

Houses stand in a line and house i holds nums[i] money. A robber cannot rob two houses that are next to each other. Return the largest total amount he can steal.

Function signature

def rob_line(nums: list) -> int

EXAMPLES

Example 1	
Input	`[1, 2, 3, 1]`
Output	`4`
Why	`Rob houses 0 and 2: 1 + 3 = 4.`

Example 2	
Input	`[2, 7, 9, 3, 1]`
Output	`12`
Why	`Rob houses 0, 2 and 4: 2 + 9 + 1 = 12.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recursion over take or skip

BRUTE-FORCE IDEA

At index i either skip it (go to i + 1) or take it and jump to i + 2, and keep the larger total. Exact; the same suffixes are solved over and over.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recursion over take or skip
At index i either skip it (go to i + 1) or take it
jump to i + 2
keep the larger total
```

BRUTE-FORCE COMPLEXITY | Time O(2^n) | Space O(n)

WHY THIS IS WEAKER | Two calls per index and the same suffix is solved repeatedly: O(2^n). Keeping only the best total with and without the previous element is O(n) time and O(1) space.

HOW TO IMPROVE IT

state = (best if you take this house, best if you skip it): take, skip = skip + house_value, max(take, skip)

SHORT BEST SOLUTION

take is the best total that includes the current element, skip the best that excludes it. For the next element: new take = skip + house_value (the previous element must have been skipped), new skip = max(take, skip). The answer is the larger of the two at the end.

```
# BASE QUESTION (optimal): Two running values (take, skip)
# Idea: take is the best total that includes the current
```

```
# element, skip the best that excludes it. For the next
# element: new take = skip + house_value (the previous element must
# have been skipped), new skip = max(take, skip). The
# answer is the larger of the two at the end.
def _house_dp(house_money):
    take = skip = 0
    for house_value in house_money:
        take, skip = skip + house_value, max(take, skip)
    return max(take, skip)
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 157

158 * Write SQL code to publish the Fibonacci series. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: WITH RECURSIVE keeps (n, a, b); every new row is (n + 1, b, a + b): the same rolling pair, written as a query

QUESTION INFO

QUESTION

Write a SQL query that publishes the first N Fibonacci numbers (0, 1, 1, 2, 3, 5, ...), one per row and in order, where N is given as the parameter :limit. Use a recursive common table expression.

Function signature

-- result: one column named fibonacci, N rows

EXAMPLES

Example 1	
Input	`:limit = 8`
Output	`0, 1, 1, 2, 3, 5, 8, 13`
Why	`Eight rows: each number is the sum of the two before it.`

Example 2	
Input	`:limit = 1`
Output	`0`
Why	`A single row.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Procedural loop that issues one statement per term

BRUTE-FORCE IDEA

Compute each term in application code (or a stored procedure) and INSERT it row by row. It works anywhere, but it needs a loop outside the query language and one round trip per term.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Procedural loop that issues one statement per term
Compute each term in application code (or a stored procedure)
INSERT it row by row
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) statements | Space O(n)

WHY THIS IS WEAKER | One statement and one round trip per term, and a loop outside the query language. A recursive CTE produces all rows in one set-based statement.

HOW TO IMPROVE IT

WITH RECURSIVE keeps (n, a, b); every new row is (n + 1, b, a + b): the same rolling pair, written as a query

SHORT BEST SOLUTION

The anchor row is (1, 0, 1): the first term is 0 and the next pair is (0, 1). Each recursive step produces (n + 1, b, a + b) while n is below the limit, so the term in column a is the Fibonacci number. Without the WHERE n < limit the recursion never stops. Integer overflow appears around the 93rd term for 64-bit integers.

```
-- BASE QUESTION (optimal): Recursive CTE carrying (n, a, b)
-- Idea: The anchor row is (1, 0, 1): the first term is 0 and
-- the next pair is (0, 1). Each recursive step produces
-- (n + 1, b, a + b) while n is below the limit, so the
-- term in column a is the Fibonacci number. Without the
-- WHERE n < limit the recursion never stops. Integer
-- overflow appears around the 93rd term for 64-bit
-- integers.
WITH RECURSIVE fib(n, a, b) AS (
    SELECT 1, 0, 1
    UNION ALL
    SELECT n + 1, b, a + b FROM fib WHERE n < :limit
)
SELECT a AS fibonacci FROM fib ORDER BY n;
```

COMPLEXITY | Time O(n) | Space O(1) working set (O(n) result)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 158

GROUP 20 OF 43 | 3 QUESTIONS | PART A - PART 1

Two pointers: move i and j toward each other (or along two lists)

PATTERN CONTEXT

Put one pointer at each end (or one at the start of each of two sorted lists). Look at the two items, decide which pointer moves, and stop when they meet. Most of these are O(n) because each pointer only moves one way.

164 * Find median of merged RLE arrays * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: walk both sorted run lists, always taking the smaller value, adding its count until you pass the middle position

QUESTION INFO

QUESTION

Two sorted arrays are stored run-length encoded as (value, count) pairs in increasing order of value. Return the median of all their elements together, without expanding the runs (a count can be huge).

Function signature

def median_rle(a: list, b: list) -> float

EXAMPLES

Example 1	
Input	`[(1, 2), (5, 1)], [(3, 3)]`
Output	`3`
Why	`Combined sequence 1, 1, 3, 3, 3, 5: the two middle elements are 3 and 3.`

Example 2	
Input	`[(1, 1)], [(2, 1), (9, 1)]`
Output	`2`
Why	`Combined sequence 1, 2, 9: the middle one is 2.`

Example 3	
Input	`[(1, 1000000)], [(5, 1000000)]`
Output	`3`
Why	`Counts may be enormous: a million 1s and a million 5s have median 3.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Expand both arrays, merge and take the middle

BRUTE-FORCE IDEA

Decompress each run into repeated values, merge, and read the middle. Correct; counts like 10^9 make the expansion impossible.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Expand both arrays, merge and take the middle
Decompress each run into repeated values
read the middle
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(total count) | Space O(total count)

WHY THIS IS WEAKER | Needs time and memory proportional to the total count, which can be 10^9 or more. Walking the two run lists in value order and counting up to the middle rank costs only the number of runs.

HOW TO IMPROVE IT

walk both sorted run lists, always taking the smaller value, adding its count until you pass the middle position

SHORT BEST SOLUTION

Let N be the sum of all counts. Walk the two run lists in value order (the merge step of merge sort), adding each run's count to a running total; the value of the run that makes the total pass a rank is the element at that rank. Ask for ranks (total_count - 1) // 2 and total_count // 2 and average them.

```
# BASE QUESTION (optimal): Merge the runs and walk to the
# two middle ranks
# Idea: Let N be the sum of all counts. Walk the two run
# lists in value order (the merge step of merge sort),
# adding each run's count to a running total; the value of
# the run that makes the total pass a rank is the element
# at that rank. Ask for ranks (total_count - 1) // 2 and total_count // 2 and
# average them.
def median_rle(first_runs, second_runs):
    total_count = sum(run_count for _, run_count in first_runs) + sum(run_count for _, run_count in second_runs
)

    def kth(rank):                                          # zero-based rank in the merged order
        first_index = second_index = elements_passed = 0
        while True:
            take_a = second_index >= len(second_runs) or (first_index < len(first_runs) and first_runs[first_in
dex][0] <= second_runs[second_index][0])
            value, run_count = first_runs[first_index] if take_a else second_runs[second_index]
            elements_passed += run_count
            if elements_passed > rank:
                return value
            first_index, second_index = (first_index + 1, second_index) if take_a else (first_index, second_ind
ex + 1)

    return kth(total_count // 2) if total_count % 2 else (kth(total_count // 2 - 1) + kth(total_count // 2)) /
2
```

COMPLEXITY | Time O(r1 + r2) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 164

165 * Run-Length Vector Storage + Dot Product * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: one pointer per run list; multiply the values and take min(remaining counts) of the two current runs

QUESTION INFO

QUESTION

Two vectors of the same length are stored run-length encoded as (value, count) pairs. Compute their dot product without expanding them to full arrays.

Function signature

def rle_dot(a: list, b: list) -> int

EXAMPLES

QUESTION

Example 1	
Input	`[(3, 3), (0, 3), (4, 2)], [(1, 4), (2, 4)]`
Output	`25`
Why	`A = [3,3,3,0,0,0,4,4], B = [1,1,1,1,2,2,2,2]: 3*1*3 + 0 + 4*2*2 = 9 + 16 = 25.`

Example 2	
Input	`[(2, 1)], [(5, 1)]`
Output	`10`
Why	`One element each: 2 * 5.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Expand both vectors and multiply

BRUTE-FORCE IDEA

Decompress each run into repeated values and add up the element-wise products. Correct; with logical lengths like 10^9 it cannot even be stored.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Expand both vectors and multiply
Decompress each run into repeated values
add up the element-wise products
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(logical length) | Space O(logical length)

WHY THIS IS WEAKER | It needs memory and time proportional to the logical length, which can be 10^9. Walking the two run lists together with two pointers costs only the number of runs.

HOW TO IMPROVE IT

one pointer per run list; multiply the values and take min(remaining counts) of the two current runs

SHORT BEST SOLUTION

Each run is (value, count). Compare the remaining lengths of the two current runs, multiply the values by the overlap, then advance whichever run was exhausted (or both). Never expand the vectors: the work depends on the number of runs, not the logical length.

```python
# BASE QUESTION (optimal): Two-pointer over runs
# Idea: Each run is (value, count). Compare the remaining
# lengths of the two current runs, multiply the values by
# the overlap, then advance whichever run was exhausted
# (or both). Never expand the vectors: the work depends on
# the number of runs, not the logical length.
def rle_dot(first_runs, second_runs):
    first_index = second_index = 0
    first_left = first_runs[0][1] if first_runs else 0
    second_left = second_runs[0][1] if second_runs else 0
    total = 0
    while first_index < len(first_runs) and second_index < len(second_runs):
        overlap = min(first_left, second_left)
        total += first_runs[first_index][0] * second_runs[second_index][0] * overlap
        first_left -= overlap
        second_left -= overlap
        if first_left == 0:
            first_index += 1
            first_left = first_runs[first_index][1] if first_index < len(first_runs) else 0
```

```
        if second_left == 0:
            second_index += 1
            second_left = second_runs[second_index][1] if second_index < len(second_runs) else 0
    return total
```

COMPLEXITY | Time O(r1 + r2) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 165

166 * Count Unique Triplets That Sum to Zero * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: sort, fix the first number, then run the pair template on the rest; skip equal neighbours so each triple counts once

QUESTION INFO

QUESTION

Given an integer array nums , count how many unique triplets (nums[i], nums[j], nums[k]) with three distinct indices i < j < k satisfy nums[i] + nums[j] + nums[k] == 0 . Two triplets are considered the same if they consist of the same multiset of values, regardless of the indices they came from. Return the number of distinct value-triplets (not the number of index combinations).

Function signature

def count_zero_sum_triplets(nums: list[int]) -> int:

    ...

Follow-up (as reported): Return the actual list of unique triplets instead of the count. Hint: sort the array, fix one element, and use two pointers while skipping duplicate values.

EXAMPLES

Example 1	
Input	nums = [-1, 0, 1, 2, -1, -4]
Output	2
Why	The distinct value-triplets are [-1, -1, 2] and [-1, 0, 1]. Even though [-1, 0, 1] can be formed with different index sets, it is counted once.

Example 2	
Input	nums = [0, 0, 0, 0]
Output	1
Why	The only triplet is [0, 0, 0].

Example 3	
Input	nums = [1, 2, 3]
Output	0
Why	No three elements sum to 0.

CONSTRAINTS

- 0 <= len(nums) <= 3000
- -10^5 <= nums[i] <= 10^5
- The array may contain duplicates.
- If len(nums) < 3 , the answer is 0 .

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(nums) <= 3000
- -10^5 <= nums[i] <= 10^5
- The array may contain duplicates.
- If len(nums) < 3 , the answer is 0 .

BRUTE FORCE | Three nested loops into a set

BRUTE-FORCE IDEA

Start with: Three nested loops into a set. Tries all index triples and stores the sorted value triple to remove repeats.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Three nested loops into a set
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^3) | Space O(triplets)

WHY THIS IS WEAKER | Tries all index triples and stores the sorted value triple to remove repeats.

HOW TO IMPROVE IT

sort, fix the first number, then run the pair template on the rest; skip equal neighbours so each triple counts once

SHORT BEST SOLUTION

Fix the smallest value and sweep the other two inward, skipping duplicates.

```python
# BASE QUESTION (optimal): Sort and two pointers
# Idea: Fix the smallest value and sweep the other two
# inward, skipping duplicates.
def count_zero_sum_triplets(nums):
    nums = sorted(nums)
    count = 0
    for fixed_index in range(len(nums) - 2):
        if fixed_index and nums[fixed_index] == nums[fixed_index - 1]:
            continue
        left, right = fixed_index + 1, len(nums) - 1
        while left < right:
            total = nums[fixed_index] + nums[left] + nums[right]
            if total < 0:
                left += 1
            elif total > 0:
                right -= 1
            else:
                count += 1
                left += 1
                right -= 1
                while left < right and nums[left] == nums[left - 1]:
                    left += 1
    return count
```

COMPLEXITY | Time O(n^2) | Space O(1) extra

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 166

Walk two strings at once with two indices

PATTERN CONTEXT

Keep i on one string and j on the other. Skip characters that do not matter, compare the rest, and decide what to do at the first difference. Substring search, ordering, similarity and move-based transforms are all this loop.

170 * Can board states be transformed? * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: skip the X squares in both strings; the remaining letters must match in order, and L may only move left, R only right

QUESTION INFO

QUESTION

A board is a string of 'L', 'R', '#' and 'X'. 'X' is an empty square. An 'L' piece may move left into an empty square, an 'R' piece may move right into an empty square, pieces never jump over each other, and '#' (a wall piece) never moves. Return whether start can be turned into end by any number of moves.

Function signature

def can_transform(start: str, end: str) -> bool

EXAMPLES

Example 1	
Input	`'RXXLRXRXL', 'XRLXXRRLX'`
Output	`true`
Why	`L moves left and R moves right while keeping their order: this classic board is reachable.`

Example 2	
Input	`'XL', 'LX'`
Output	`true`
Why	`L can move left.`

Example 3	
Input	`'LX', 'XL'`
Output	`false`
Why	`L cannot move to the right.`

Example 4	
Input	`'#X', 'X#'`
Output	`false`
Why	`A wall never moves.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | BFS over every reachable board

BRUTE-FORCE IDEA

From the start string, repeatedly apply the legal moves (XL becomes LX, RX becomes XR) and see whether the target appears. Correct, for the invariant-based check, but the number of boards explodes.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: BFS over every reachable board
From the start string
repeatedly apply the legal moves (XL becomes LX
RX becomes XR)
see whether the target appears
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential in the length | Space exponential

WHY THIS IS WEAKER | The number of reachable boards grows exponentially. Pieces never cross, so compare the sequence of L and R pieces and check that no L moved right and no R moved left, in O(n).

HOW TO IMPROVE IT

skip the X squares in both strings; the remaining letters must match in order, and L may only move left, R only right

SHORT BEST SOLUTION

Skip blanks in both strings and compare the next piece in each: they must be the same letter. An 'L' must not have moved right (its start index is at least its target index) and an 'R' must not have moved left (start index at most target index). If the strings end together the transformation is possible.

```python
# BASE QUESTION (optimal): Two pointers over the pieces
# Idea: Skip blanks in both strings and compare the next
# piece in each: they must be the same letter. An 'L' must
# not have moved right (its start index is at least its
# target index) and an 'R' must not have moved left (start
# index at most target index). If the strings end together
# the transformation is possible.
def canTransform(start, end):
    if len(start) != len(end):
        return False
    start_index = end_index = 0
    length = len(start)
    while start_index < length or end_index < length:
        while start_index < length and start[start_index] == "X":
            start_index += 1
        while end_index < length and end[end_index] == "X":
            end_index += 1
        if start_index == length or end_index == length:
            return start_index == end_index
        if start[start_index] != end[end_index]:
            return False
        if start[start_index] == "L" and start_index < end_index:
            return False
        if start[start_index] == "R" and start_index > end_index:
            return False
        start_index, end_index = start_index + 1, end_index + 1
    return True
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 170

171 * Decide string transform with directional tokens and walls

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: the same walk with dots as blanks and the movement rule from the statement (walls block)

QUESTION INFO

QUESTION

Two equal-length strings start and target use the characters 'L', 'R', '.' (an empty slot) and 'X' (an immovable wall). An 'L' token can move left into empty slots, an 'R' token can move right into empty slots, tokens can never pass through each other or through a wall, and 'X' never moves. Return whether start can be turned into target by any number of moves.

Function signature

def can_transform(start: str, end: str) -> bool

EXAMPLES

Example 1	
Input	`'R..L', '.RL.'`
Output	`true`
Why	`R moves one step right and L one step left: both stay in order.`

Example 2	
Input	`'L.', '.L'`
Output	`false`
Why	`L can only move left, never right.`

Example 3	
Input	`'X.', '.X'`
Output	`false`
Why	`A wall cannot move.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | BFS over boards with walls

BRUTE-FORCE IDEA

Apply the legal single-step moves ('.L' becomes 'L.', 'R.' becomes '.R'), walls never move, and search for the target. Exact, but it explodes with the length.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: BFS over boards with walls
Apply the legal single-step moves ('.L' becomes 'L.'
'R.' becomes '.R')
walls never move
search for the target
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space exponential

WHY THIS IS WEAKER | The number of reachable boards grows exponentially with the length. Comparing the non-dot sequence (walls and pieces in order) with position rules (L never moves right, R never moves left, walls stay) is O(n).

HOW TO IMPROVE IT

the same walk with dots as blanks and the movement rule from the statement (walls block)

SHORT BEST SOLUTION

Walk both strings skipping dots. The next symbol must be equal; a wall must be at the same index; an L must satisfy start index >= target index; an R must satisfy start index <= target index. Since the order of symbols is the same, a piece cannot cross a wall.

```
# BASE QUESTION (optimal): Compare the non-dot sequences
# with position rules
# Idea: Walk both strings skipping dots. The next symbol
# must be equal; a wall must be at the same index; an L
# must satisfy start index >= target index; an R must
# satisfy start index <= target index. Since the order of
# symbols is the same, a piece cannot cross a wall.
def can_transform_walls(start, end):
    if len(start) != len(end):
        return False
    start_index = end_index = 0
    length = len(start)
    while True:
        while start_index < length and start[start_index] == ".":
            start_index += 1
        while end_index < length and end[end_index] == ".":
            end_index += 1
        if start_index == length or end_index == length:
            return start_index == end_index
        if start[start_index] != end[end_index]:
            return False
        if start[start_index] == "X" and start_index != end_index:
            return False
        if start[start_index] == "L" and start_index < end_index:
            return False
        if start[start_index] == "R" and start_index > end_index:
            return False
        start_index, end_index = start_index + 1, end_index + 1
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 171

172 * Determine if two sentences are similar. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: zip the two sentences; each pair of words must be equal or listed as similar (store pairs in both directions)

QUESTION INFO

QUESTION

Determine if Two Sentences Are Similar. Two sentences are similar if they have the same length and each pair of corresponding words in the two sentences is similar. The similarity between words is defined by the provided list of similar word pairs. A word is always similar to itself. For example, if we have the list of similar word pairs as [("great", "good"), ("acting","drama"), ("skills","talent")], then the sentences "You have great acting skills" and "You have good drama talent" are similar. Examples sentence1 = ["Let's", "code", "in", "Python"] sentence2 = ["Let's", "program", "in", "Python"] similarPairs = [("code", "program"),] output: true sentence1 = ["I", "love", "to", "play", "football"] sentence2 = ["I", "love", "playing", "soccer"] similarPairs = [("play", "playing"), ("football", "soccer")] output: false, different sentence lengths sentence1 = ["Do", "you", "like", "coffee"] sentence2 = ["Do", "you", "love", "coffee"] similarPairs = [("like", "enjoy"), ("coffee", "tea"),] output: false

"like" is not similar to

QUESTION

"love" based on the given pairs. sentence1 = ["I", "really", "love", "leetcode", "and", "apples"] sentence2 = ["I", "so", "like", "codesignal", "and", "oranges"] similarPairs = [("very", "so"), ("love", "adore"), ("really", "very"), ("leetcode", "codesignal"), ("apples", "oranges"), ("like", "adore"),] output: trueThe provided solution checks if two given sentences are similar by using a graph-based approach. It constructs a graph where words are c

EXAMPLES

Example 1	
Input	`['great', 'acting', 'skills'], ['fine', 'drama', 'talent'], [['great', 'fine'], ['acting', 'drama'], ['skills', 'talent']]`
Output	`true`
Why	`Each word is paired with a similar word.`

Example 2	
Input	`['a'], ['a', 'b'], []`
Output	`false`
Why	`Sentences of different lengths are never similar.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan the pair list for each word pair

BRUTE-FORCE IDEA

For each position, check equality or scan the whole list of similar pairs in both orders. Correct, but every word pair walks the entire pair list.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan the pair list for each word pair
For each position
check equality or scan the whole list of similar pairs in both orders
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n * P) | Space O(1)

WHY THIS IS WEAKER | Each word pair scans the whole pair list, O(n * P). Putting the pairs into a set makes each check O(1).

HOW TO IMPROVE IT

zip the two sentences; each pair of words must be equal or listed as similar (store pairs in both directions)

SHORT BEST SOLUTION

Put each pair into a set in both directions, so a word pair is similar if the words are equal or the pair is in the set. Sentences of different length are never similar.

```python
# BASE QUESTION (optimal): Set of pairs (not transitive)
# Idea: Put each pair into a set in both directions, so a
# word pair is similar if the words are equal or the pair
# is in the set. Sentences of different length are never
# similar.
def areSentencesSimilar(sentence_one, sentence_two, pairs):
    if len(sentence_one) != len(sentence_two):
        return False
    known = set()
    for word_a, word_b in pairs:
        known.add((word_a, word_b))
        known.add((word_b, word_a))
    return all(word_one == word_two or (word_one, word_two) in known for word_one, word_two in zip(sentence_one
```

```
, sentence_two))
```

COMPLEXITY | Time O(n + P) | Space O(P)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 172

Simulate a machine: a class with state and one branch per kind of event

PATTERN CONTEXT

Put the state in __init__, give it one method that takes an event, and write one branch per event kind. Board games keep a grid or counters and check for a win after each move; game-of-life style updates compute every cell from the old state before changing it.

175 * Tic-Tac-Toe System with K Players and 3-in-a-Row Win

* MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: store the board; after each move count the run through the new cell in four directions and compare with the target length

QUESTION INFO

QUESTION

Design Tic-Tac-Toe for k players on an n x n board. Players move in turn (1, 2, ..., k, then 1 again). move(player, r, c) places a mark and returns 'player p wins' as soon as that player has 3 marks in a straight line (row, column or diagonal), 'draw' when the board is full, and 'continue' otherwise. A move out of turn, off the board or onto a taken cell raises an error.

Function signature

```
class TicTacToe:
    def __init__(self, n: int, k: int, target: int = 3)
    def move(self, player: int, r: int, c: int) -> str
```

EXAMPLES

Example 1

Input	`g = TicTacToe(3, 2)`
	`g.move(1, 0, 0)`
	`g.move(2, 1, 0)`
	`g.move(1, 0, 1)`
	`g.move(2, 1, 1)`
	`g.move(1, 0, 2)`

Output	`g.move(1, 0, 0) -> 'continue'`
	`g.move(2, 1, 0) -> 'continue'`
	`g.move(1, 0, 1) -> 'continue'`
	`g.move(2, 1, 1) -> 'continue'`
	`g.move(1, 0, 2) -> 'player 1 wins'`

Why	Player 1 takes the whole top row while player 2 has two in the middle row: player 1 wins.

Example 2

Input	`g = TicTacToe(3, 3)`
	`g.move(1, 0, 0)`
	`g.move(2, 0, 1)`
	`g.move(3, 0, 2)`
	`g.move(1, 1, 1)`

Output	`g.move(1, 0, 0) -> 'continue'`
	`g.move(2, 0, 1) -> 'continue'`
	`g.move(3, 0, 2) -> 'continue'`
	`g.move(1, 1, 1) -> 'continue'`

Why	With three players the turns go 1, 2, 3, 1; nobody has three in a line yet.

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Rescan the whole board after every move

BRUTE-FORCE IDEA

After each move scan every cell and every direction for a run of 3. Correct; wasteful, because only lines through the new mark can have changed.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Rescan the whole board after every move
After each move scan every cell
every direction for a run of 3
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) per move | Space O(n^2)

WHY THIS IS WEAKER | Scans O(n^2) cells per move. Only lines through the cell just played can have changed, so checking four directions around it is O(1).

HOW TO IMPROVE IT

store the board; after each move count the run through the new cell in four directions and compare with the target length

SHORT BEST SOLUTION

Place the mark, then for each of the four directions walk outward both ways while the cells hold the same player and add the two counts plus one. A total of at least 3 wins immediately. Validate the turn order and empty cell first; a full board with no win is a draw.

```python
# BASE QUESTION (optimal): Check the four lines through
# the new mark
# Idea: Place the mark, then for each of the four
# directions walk outward both ways while the cells hold
# the same player and add the two counts plus one. A total
# of at least 3 wins immediately. Validate the turn order
# and empty cell first; a full board with no win is a
# draw.
class TicTacToe:
    def __init__(self, size, player_count, target=3):
        self.size, self.player_count, self.target = size, player_count, target
        self.board = [[0] * size for _ in range(size)]
        self.turn, self.moves, self.over = 0, 0, False

    def move(self, player, row, col):
        if self.over or player != self.turn % self.player_count + 1 or not (0 <= row < self.size and 0 <= col <
 self.size) or self.board[row][col]:
            raise ValueError("illegal move")
        self.board[row][col] = player
        self.turn += 1; self.moves += 1
        for row_step, col_step in ((0, 1), (1, 0), (1, 1), (1, -1)):
            line_length = 1
            for sign in (1, -1):
                scan_row, scan_col = row + sign * row_step, col + sign * col_step
                while 0 <= scan_row < self.size and 0 <= scan_col < self.size and self.board[scan_row][scan_col
] == player:
                    line_length += 1
                    scan_row, scan_col = scan_row + sign * row_step, scan_col + sign * col_step
            if line_length >= self.target:
                self.over = True
                return f"player {player} wins"
        if self.moves == self.size * self.size:
            self.over = True
            return "draw"
        return "continue"
```

COMPLEXITY | Time O(1) per move (O(target) when generalised) | Space O(n^2)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 175

176 * Design Tic-Tac-Toe With K Players

* MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: per player Counters for rows, columns and both diagonals; a move wins when a counter reaches n

QUESTION INFO

QUESTION

Design and implement a Tic-Tac-Toe game engine. You are given:

- **k** players, each with a unique player ID or mark.
- An **`n x n`** board, initially empty.
- Players take turns placing their own mark on an empty cell. Unlike classic Tic-Tac-Toe, a player wins as soon as they form **3 consecutive marks** in a straight line, regardless of how large **n** is. A straight line may be horizontal, vertical, diagonal down-right, or diagonal down-left. Implement a class or set of functions that supports making moves and returns the game status after each move, such as:
- **`"Game Over"`** if a move is attempted after the game has ended.
- **`"<player> won"`** when the latest move causes that player to have 3 consecutive marks.
- A non-terminal status such as **`"Continue"`** if the game should keep going. Your implementation should handle invalid moves, such as placing a mark outside the board or on an occupied cell. Discuss the time and space complexity of your approach, especially for large **n**.

Design a Tic-Tac-Toe game engine for k players on an n x n board. The board starts empty, and players must play in cyclic order: player 1, player 2, ..., player k, then back to player 1. A player wins immediately when their latest valid move creates 3 consecutive marks in a straight line. A straight line may be horizontal, vertical, diagonal down-right, or diagonal down-left. For this problem, implement a function that processes a list of move attempts and returns the status after each attempt. Each move is given as [player, row, col] using 0-based indexing. Return one status string per move:

- "Game Over" if a move is attempted after someone has already won.
- "Invalid move" if the move is not legal. A move is invalid if: - the player ID is not in [1, k] - it is not that player's turn - the cell is outside the board - the cell is already occupied - "<player> won" if the move is valid and causes that player to have 3 consecutive marks - "Continue" if the move is valid and the game should continue Important: an invalid move does not change the board and does not advance the turn. You do not need to detect draws. If no one wins, valid moves simply return "Continue".

Constraints
1 <= n <= 10^91 <= k <= 10^50 <= len(moves) <= 2 * 10^5Each move is a list of three integers: [player, row, col]Row and column in the input may be outside the board and should then be treated as an invalid moveYour approach should be efficient for large n, so avoid scanning the whole board after every move

EXAMPLES

```
Input: (3, 2, [])
Expected Output: []
```

Explanation: No moves are made, so there are no statuses to report.

```
Input: (5, 2, [[1, 0, 0], [2, 1, 0], [1, 0, 1], [2, 1, 1], [1, 0, 2], [2, 2, 2]])
Expected Output: ['Continue', 'Continue', 'Continue', 'Continue', '1 won', 'Game Over']
```

Explanation: Player 1 forms three consecutive marks horizontally at (0,0), (0,1), and (0,2). Any later move returns 'Game Over'.

QUESTION

Hints

Only the latest valid move can create a new 3-in-a-row, so you never need to rescan the entire board.For large boards, store only occupied cells in a hash map keyed by (row, col), then check the 4 line directions around the new move.

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan the lines through the last move

BRUTE-FORCE IDEA

Store the board and, after each move, check the move's row, column and (if on one) the diagonals for n equal marks. Simple and fast enough for small n, but every move rescans up to four lines.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan the lines through the last move
Store the board
after each move
check the move's row
column
(if on one) the diagonals for n equal marks
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) per move | Space O(n^2)

WHY THIS IS WEAKER | Each move rescans up to four lines, O(n). Keeping a counter per player per row, column and diagonal makes each move O(1).

HOW TO IMPROVE IT

per player Counters for rows, columns and both diagonals; a move wins when a counter reaches n

SHORT BEST SOLUTION

rows[p][row], cols[p][col], diag[p] and anti[p] count each player's marks on those lines. A move increments the counters for its row, column and, if applicable, the diagonals; the player wins when one reaches n. A separate set of occupied cells rejects repeated moves.

```python
from collections import Counter
from collections import defaultdict


# BASE QUESTION (optimal): Counters per player and per
# line
# Idea: rows[p][row], cols[p][col], diag[p] and anti[p] count
# each player's marks on those lines. A move increments
# the counters for its row, column and, if applicable, the
# diagonals; the player wins when one reaches n. A
# separate set of occupied cells rejects repeated moves.
class TicTacToe:
    def __init__(self, size):
        self.size = size
        self.rows, self.cols = defaultdict(Counter), defaultdict(Counter)
        self.diagonal, self.anti_diagonal, self.taken = Counter(), Counter(), set()

    def move(self, row, col, player):
        if (row, col) in self.taken:
            return -1
        self.taken.add((row, col))
        self.rows[player][row] += 1
        self.cols[player][col] += 1
        won = self.rows[player][row] == self.size or self.cols[player][col] == self.size
        if row == col:
            self.diagonal[player] += 1
```

```
            won = won or self.diagonal[player] == self.size
        if row + col == self.size - 1:
            self.anti_diagonal[player] += 1
            won = won or self.anti_diagonal[player] == self.size
        return player if won else 0
```

COMPLEXITY | Time O(1) per move | Space O(k * n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 176

177 * Implement iterator merging lists with filtering * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: keep a position per list and a turn counter; advance() looks for the next unblocked, unseen value and stores it for next()/has_next()

QUESTION INFO

QUESTION

Design an iterator over several lists that takes one element from each list in turn (a zigzag), skipping any value in the blocked set and any value that was already returned. hasNext() says whether another value is available and next() returns it.

Function signature

class FilteredZigzag:
  def __init__(self, lists: list, blocked=())
  def hasNext(self) -> bool
  def next(self) -> int

EXAMPLES

Example 1	

Input	`it = FilteredZigzag([[1, 2, 3], [4, 5, 6, 7], [8, 9]], blocked=[2, 5])`
	`out = []`
	`while it.hasNext():`
	`out.append(it.next())`
	`out`

Output	`out -> [1, 4, 8, 9, 3, 6, 7]`
Why	`Order of visit: 1, 4, 8, then 2 (blocked), 5 (blocked), 9, then 3, 6, then 7. The blocked values never appear.`

Example 2	

Input	`it = FilteredZigzag([[1, 1], [1, 2]])`
	`out = []`
	`while it.hasNext():`
	`out.append(it.next())`
	`out`

Output	`out -> [1, 2]`
Why	`Repeated values are returned only once.`

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Build the full merged, filtered list first

BRUTE-FORCE IDEA

Interleave the lists, drop blocked and repeated ids, and iterate over the resulting list. Correct, but it is not lazy: it reads everything up front and holds it all in memory.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Build the full merged, filtered list first
Interleave the lists
drop blocked
repeated ids
iterate over the resulting list
```

BRUTE-FORCE COMPLEXITY | Time O(total) | Space O(total)

WHY THIS IS WEAKER | It reads everything up front and keeps it all in memory, so it is not lazy and never works on a long or unbounded source. A lookahead value per iterator computes only the next valid item.

HOW TO IMPROVE IT

keep a position per list and a turn counter; advance() looks for the next unblocked, unseen value and stores it for next()/has_next()

SHORT BEST SOLUTION

Keep a pointer per list and a rotation index. _advance() scans round robin for the next value that is not blocked and not already yielded and stores it as the lookahead (or None when everything is exhausted). hasNext() is 'lookahead is not None'; next() returns the lookahead and calls _advance().

```python
# BASE QUESTION (optimal): Lazy iterator with one pointer
# per list and a lookahead
# Idea: Keep a pointer per list and a rotation index.
# _advance() scans round robin for the next value that is
# not blocked and not already yielded and stores it as the
# lookahead (or None when everything is exhausted).
# hasNext() is 'lookahead is not None'; next() returns the
# lookahead and calls _advance().
class FilteredZigzag:
    def __init__(self, lists, blocked=()):
        self.lists, self.blocked = lists, set(blocked)
        self.positions, self.turn, self.seen, self.lookahead = [0] * len(lists), 0, set(), None
        self._advance()

    def _advance(self):
        self.lookahead = None
        misses = 0
        while misses < len(self.lists):
            list_index = self.turn
            self.turn = (self.turn + 1) % len(self.lists)
            if self.positions[list_index] < len(self.lists[list_index]):
                value = self.lists[list_index][self.positions[list_index]]
                self.positions[list_index] += 1
                misses = 0
                if value not in self.blocked and value not in self.seen:
                    self.lookahead = value
                    return
            else:
                misses += 1

    def hasNext(self):
        return self.lookahead is not None

    def next(self):
        value = self.lookahead
        self.seen.add(value)
        self._advance()
        return value
```

COMPLEXITY | Time O(1) amortized per value | Space O(number of lists + distinct ids yielded)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 177

178 * Infinite Deck Card Duel

* MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: an infinite deck is just a random draw; play rounds, compare the highest cards, and decide the champion by the lead in wins

QUESTION INFO

QUESTION

Design an InfiniteDeck that can be drawn from forever: each call to draw() returns a uniformly random integer card value in [1, 13] (cards are never exhausted). Make the deck seedable so games are reproducible. Then simulate a duel between two players. In each round, each player draws hand_size cards. The round winner is the player holding the single highest card that round; if both players' highest cards are equal, the round is a draw (nobody scores). Play num_rounds rounds and tally round wins per player. Finally, declare a champion: the player is champion only if they lead the other in round wins by at least 2; otherwise there is no champion. Implement: InfiniteDeck.draw() returning the next card. round_winner(hand1, hand2) -> 1 , 2 , or 0 (0 meaning draw), based on the highest card in each hand. decide_champion(wins1, wins2) -> 1 , 2 , or 0 (0 meaning no champion), applying the "lead by at least 2" rule. play_game(...) tying it together (uses the deck for randomness).

Function signature

```
import random
class InfiniteDeck:
    def __init__(self, seed: int | None = None) -> None:
        """Create a deck backed by a (optionally seeded) RNG."""
        ...
    def draw(self) -> int:
        """Return a uniformly random card value in [1, 13]."""
        ...
def round_winner(hand1: list[int], hand2: list[int]) -> int:
    """Return 1 if player 1's max card is higher, 2 if player 2's is higher, 0 on a
tie."""
    ...
def decide_champion(wins1: int, wins2: int) -> int:
    """Return the champion (1 or 2) only if they lead by >= 2 round wins, else 0."""
    ...
def play_game(deck: InfiniteDeck, hand_size: int = 5, num_rounds: int = 100) -> int:
    """Play the duel and return the champion via decide_champion, or 0 if none."""
    ...
```

Follow-up (as reported): Extend to n players: replace the two win counters with a dictionary keyed by player id and pick the highest card across all hands each round. Introduce multiple independent InfiniteDeck instances (or one shared deck with per-card metadata) to support probabilistic house rules.

EXAMPLES

Example 1	
Input	`round_winner([3, 7, 2], [7, 1, 9])`
Output	`2`
Why	`Player 2's highest card (9) beats player 1's highest (7).`

Example 2	
Input	`round_winner([5, 5], [5, 4])`
Output	`0`
Why	`Both players' highest card is 5, so the round is a draw.`

QUESTION

CONSTRAINTS

- 1 <= hand_size <= 20 , 1 <= num_rounds <= 100000 , card values in [1, 13] .

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= hand_size <= 20 , 1 <= num_rounds <= 100000 , card values in [1, 13] .

BRUTE FORCE | Inline simulation with one Random

BRUTE-FORCE IDEA

It repeats the same logic without the classes; it is only here to check that the class version gives the same game.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Inline simulation with one Random
It repeats the same logic without the classes
it is only here to check that the class version gives the same game
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(rounds * hand) | Space O(1)

WHY THIS IS WEAKER | It repeats the same logic without the classes; it is only here to check that the class version gives the same game.

HOW TO IMPROVE IT

an infinite deck is just a random draw; play rounds, compare the highest cards, and decide the champion by the lead in wins

SHORT BEST SOLUTION

The deck owns the Random; the rules are plain functions of hands and win counts.

```python
import random


# BASE QUESTION (optimal): A seedable deck and pure game
# rules
# Idea: The deck owns the Random; the rules are plain
# functions of hands and win counts.
class InfiniteDeck:
    def __init__(self, seed=None):
        self.rng = random.Random(seed)

    def draw(self):
        return self.rng.randint(1, 13)


def round_winner(hand1, hand2):
    if max(hand1) == max(hand2):
        return 0
    return 1 if max(hand1) > max(hand2) else 2


def decide_champion(wins1, wins2):
    if wins1 - wins2 >= 2:
        return 1
    return 2 if wins2 - wins1 >= 2 else 0


def play_game(deck, hand_size=5, num_rounds=100):
    wins = {1: 0, 2: 0}
    for _ in range(num_rounds):
        first_hand = [deck.draw() for _ in range(hand_size)]
        second_hand = [deck.draw() for _ in range(hand_size)]
        winner = round_winner(first_hand, second_hand)
```

```
        if winner:
            wins[winner] += 1
    return decide_champion(wins[1], wins[2])
```

COMPLEXITY | Time O(rounds * hand) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 178

179 * Design an elevator control system * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: each car keeps floor, direction and stops; a request goes to the car with the lowest cost; step() moves every car one floor

QUESTION INFO

QUESTION

Design the control logic for a multi-elevator building. A Request(floor, direction) is a call from a floor (direction +1 for up, -1 for down). Each Elevator keeps its current floor, direction and a set of stops. cost(request) estimates how expensive it is for this car to take the request (the distance, plus a detour penalty when the car is heading away); the dispatcher gives the request to the cheapest car. step() moves the car one floor towards its stops and returns the floor it served, if any.

Function signature

class Elevator:
  def __init__(self, car_id, floor=0)
  def cost(self, req: Request) -> float
  def step(self) -> int | None

EXAMPLES

Example 1

Input	`e = Elevator(1, floor=0)`
	`e.stops.update({3, 5})`
	`[e.step() for _ in range(6)]`

Output	`[e.step() for _ in range(6)] -> [null, null, null, 3, null, 5]`
Why	The car climbs one floor per step and stops at floors 3 and 5 (served when it is on the floor at the start of a step).

Example 2

Input	`idle = Elevator(2, floor=7)`
	`busy = Elevator(3, floor=2)`
	`busy.direction = 1`
	`busy.stops.update({9})`
	`idle.cost(Request(4, -1))`
	`busy.cost(Request(4, 1))`
	`busy.cost(Request(4, -1))`

Output	`idle.cost(Request(4, -1)) -> 3`
	`busy.cost(Request(4, 1)) -> 2`
	`busy.cost(Request(4, -1)) -> 14`

Why	An idle car costs its distance (3). A car already going up that will pass floor 4 going the same way costs its distance (2); going the other way it must finish its run first, so it costs much more.

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the

largest input size so I target the expected complexity.

BRUTE FORCE | First-come-first-served on the nearest idle car

BRUTE-FORCE IDEA

Serve requests in arrival order, each by whichever car is closest at that moment, visiting stops in the order they were requested. Easy to explain and fair, but cars zig-zag across the building.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: First-come-first-served on the nearest idle car
Serve requests in arrival order
each by whichever car is closest at that moment
visiting stops in the order they were requested
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(requests * cars) | Space O(requests)

WHY THIS IS WEAKER | Cars zig-zag across the building and requests wait behind far-away ones, so total travel and waiting are high. Serving stops in sweep order (SCAN) and assigning by cost cuts both.

HOW TO IMPROVE IT

each car keeps floor, direction and stops; a request goes to the car with the lowest cost; step() moves every car one floor

SHORT BEST SOLUTION

Each car keeps a set of stops and a direction. step() moves one floor in its direction and removes a stop it reaches; when no stops remain ahead it reverses, and when none remain at all it goes idle. The controller assigns a hall call to the car with the lowest cost: distance, plus a penalty if the car is moving away or is out of service. Cars that fail drop out and their stops are reassigned.

```python
# BASE QUESTION (optimal): Elevator (SCAN) per car and
# cost-based assignment of calls
# Idea: Each car keeps a set of stops and a direction.
# step() moves one floor in its direction and removes a
# stop it reaches; when no stops remain ahead it reverses,
# and when none remain at all it goes idle. The controller
# assigns a hall call to the car with the lowest cost:
# distance, plus a penalty if the car is moving away or is
# out of service. Cars that fail drop out and their stops
# are reassigned.
class Request:
    def __init__(self, floor, direction):
        self.floor, self.direction = floor, direction              # direction: +1 up, -1 down

class Elevator:
    def __init__(self, car_id, floor=0):
        self.car_id, self.floor, self.direction, self.stops, self.in_service = car_id, floor, 0, set(), True

    def cost(self, call):
        if not self.in_service:
            return float("inf")
        distance = abs(self.floor - call.floor)
        if self.direction == 0:
            return distance
        ahead = (call.floor - self.floor) * self.direction >= 0
        return distance if ahead and call.direction == self.direction else distance + 2 * (max(self.stops) - mi
n(self.stops) + 1 if self.stops else 0) + 10

    def step(self):
        served = None
        if self.floor in self.stops:
            self.stops.discard(self.floor)
            served = self.floor
        if not self.stops:
            self.direction = 0
            return served
        up = [stop for stop in self.stops if stop > self.floor]
        down = [stop for stop in self.stops if stop < self.floor]
        if self.direction >= 0 and up or (self.direction < 0 and not down):
```

```
            self.direction = 1
        else:
            self.direction = -1
        self.floor += self.direction
        return served
```

COMPLEXITY | Time O(cars) per call, O(1) per car per step | Space O(stops)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 179

GROUP 23 OF 43 | 6 QUESTIONS | PART A - PART 1

Top-K by frequency: Counter plus nsmallest

PATTERN CONTEXT

Count with a Counter, then ask heapq.nsmallest for the k best using a key of (negative count, tie-break). That is the whole solution for almost every top-K question; only what you count and the tie-break change.

181 * Top-K Frequent in Large Logs * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: count the first word of every line (the speaker); return the names of the k most frequent

QUESTION INFO

QUESTION

A chat log is a list of lines, each starting with the speaker's name ('A says hi'). Return the k speakers who appear most often; ties are broken alphabetically by name.

Function signature

def top_k_speakers(chat_log: list, k: int) -> list

EXAMPLES

Example 1	
Input	`['A says hi', 'B says hi', 'A says bye', 'C says hi', 'A says yo'], 2`
Output	`['A', 'B']`
Why	`A spoke 3 times, B and C once each; B comes before C alphabetically.`

Example 2	
Input	`['A says hi', 'B says hi', 'A says bye', 'C says hi', 'A says yo'], 3`
Output	`['A', 'B', 'C']`
Why	`Asking for 3 returns everyone, most frequent first.`

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Count everything, then sort all users

BRUTE-FORCE IDEA

Count the speakers and sort every distinct user by (count, name). Correct; sorting all d users does more work than keeping the best K.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Count everything, then sort all users
Count the speakers
sort every distinct user by (count
name)
```

BRUTE-FORCE COMPLEXITY | Time O(n + d log d) | Space O(d)

WHY THIS IS WEAKER | Sorts all d distinct speakers, O(d log d), when only K are wanted; a size-K heap is O(d log k), and a stream needs a counter sketch instead of exact counts.

HOW TO IMPROVE IT

count the first word of every line (the speaker); return the names of the k most frequent

SHORT BEST SOLUTION

Part 1: Parse the speaker from each line and count in a hash map. nsmallest keeps a heap of size k, so it never sorts all d distinct users. Ties break by user name so the output is deterministic.

Part 2: Partition the (count, user) pairs around a random pivot until exactly the k largest sit on the left. Random pivots make the bad case unlikely; mention median-of-medians if a guaranteed O(d) bound is asked for.

```
from collections import Counter
import heapq


# BASE QUESTION (optimal): Count, then keep a size-K heap
# Idea: Parse the speaker from each line and count in a
# hash map. nsmallest keeps a heap of size k, so it never
# sorts all d distinct users. Ties break by user name so
# the output is deterministic.
def top_k_speakers(chat_log, how_many):
    counts = Counter(line.split(" ", 1)[0] for line in chat_log)
    return [user for user, _ in heapq.nsmallest(how_many, counts.items(), key=lambda entry: (-entry[1], entry[0
]))]
```

COMPLEXITY | Part 1: Time O(n + d log k) | Space O(d)
Part 2: Time O(n + d) average, O(d^2) worst case | Space O(d)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 181

182 * Rearrange Tasks With Cooldown * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: only the biggest count matters: answer = max(len(tasks), (biggest - 1) * (cooldown + 1) + how many tasks share the biggest count)

QUESTION INFO

QUESTION

You are given a list of tasks (letters; the same letter means the same kind of task). Every task takes one unit of time, and two tasks of the same kind must have at least n other tasks or idle units between them (the cooldown). The CPU may stay idle. Return the least number of time units needed to finish all the tasks.

Function signature

def least_interval(tasks: list, n: int) -> int

EXAMPLES

Example 1	
Input	['A', 'A', 'A', 'B', 'B', 'B'], 2
Output	8
Why	A B idle A B idle A B: three A's and three B's with a cooldown of 2 take 8 units.

Example 2	
Input	['A', 'A', 'A', 'B', 'B', 'B'], 0
Output	6
Why	No cooldown: just run the 6 tasks back to back.

Example 3	
Input	['A', 'B', 'C'], 5
Output	3
Why	All tasks differ, so no cooldown is ever needed.

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Simulate each time slot, always running the available task with the most copies left

BRUTE-FORCE IDEA

At every time unit pick, among tasks not cooling down, the one with the most remaining copies; if none is available the slot is idle. A heap for ready tasks and a queue of tasks in cooldown make it work. Correct, but it simulates every slot, including long idle stretches.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Simulate each time slot, always running the available task with the most copies...
At every time unit pick
among tasks not cooling down
the one with the most remaining copies
if none is available the slot is idle
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(T log k) where T is the answer | Space O(k)

WHY THIS IS WEAKER | It simulates every slot, including long idle stretches, O(T log k). The answer follows from the most frequent task: max(len(tasks), (maxCount - 1) * (n + 1) + number of tasks with that count).

HOW TO IMPROVE IT

only the biggest count matters: answer = max(len(tasks), (biggest - 1) * (cooldown + 1) + how many tasks share the biggest count)

SHORT BEST SOLUTION

Let highest_frequency be the highest frequency and tasks_at_highest the number of tasks that have that frequency. The skeleton needs (highest_frequency - 1) * (cooldown + 1) + tasks_at_highest slots. If there are more tasks than that, no idle time is needed and the answer is len(tasks). So the answer is max(len(tasks), (highest_frequency - 1) * (cooldown + 1) + tasks_at_highest).

```python
from collections import Counter


# BASE QUESTION (optimal): Closed form from the most
# frequent task
# Idea: Let highest_frequency be the highest frequency and tasks_at_highest the number of
# tasks that have that frequency. The skeleton needs (highest_frequency -
# 1) * (cooldown + 1) + tasks_at_highest slots. If there are more tasks than
# that, no idle time is needed and the answer is
# len(tasks). So the answer is max(len(tasks), (highest_frequency - 1) *
# (cooldown + 1) + tasks_at_highest).
def leastInterval(tasks, cooldown):
    frequencies = Counter(tasks).values()
    highest_frequency = max(frequencies, default=0)
    tasks_at_highest = sum(1 for frequency in frequencies if frequency == highest_frequency)
    return max(len(tasks), (highest_frequency - 1) * (cooldown + 1) + tasks_at_highest) if tasks else 0
```

COMPLEXITY | Time O(len(tasks)) | Space O(1) for a fixed alphabet

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 182

183 * Implement frequency + distance top-K queries * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: count the point tuples; the tie-break is distance from the origin, then x, then y

QUESTION INFO

QUESTION

You will implement solutions for two coding interview questions.

Question

1: Return the top-K points by frequency (with distance tie-break) You are given an array **`points`** of 2D integer points, where each point is represented as **`[x, y]`**. The array may contain duplicates (the same **`[x, y]`** can appear multiple times). Define:

- **`freq(p)`** = number of times point **`p`** appears in the array.
- **`dist(p)`** = squared Euclidean distance to the origin, **`x^2 + y^2`**. Return **`k` distinct points** sorted by the following priority:

1. Higher frequency first (**`freq`** descending)
2. If frequencies tie, smaller distance first (**`dist`** ascending)
3. If still tied, sort by **`x`** ascending, then **`y`** ascending

Input

- **`points`**: list of length **`n`**, each element **`[x, y]`**
- **`k`**: integer, **`1 <= k <= (# of distinct points)`**

Output

- A list of **`k`** distinct points following the ranking above.

Constraints (typical interview assumptions)

- **`1 <= n <= 2 * 10^5`**
- **`-10^4 <= x, y <= 10^4`** ---

Question

2: Time-based key-value store with “closest timestamp” lookup Design a data structure supporting the following operations:

- **`set(key, value, timestamp)`**: store the string **`value`** for string **`key`** at integer **`timestamp`**.
- Assume timestamps for the same **`key`** are inserted in **non-decreasing** order.
- **`getClosest(key, timestamp) -> value or ""`**:
- If **`key`** has no stored values, return **`""`**.
- Otherwise, return the **`value`** whose stored timestamp is **closest** to the query **`timestamp`**.
- If there is a tie (equal distance to two timestamps), return the value with the **smaller** timestamp.

Example

If for key **`"a"`** you stored timestamps **`[2, 10, 15]`** with values **`["x", "y", "z"]`**:

- **`getClosest("a", 11)`** returns **`"y"`** (10 is closest)
- **`getClosest("a", 14)`** returns **`"z"`** (15 is closest)
- **`getClosest("a", 12)`** returns **`"y"`** (10 and 15 are equally far; choose smaller timestamp)

What to discuss

- Time and space complexity of your approach - How you would improve a “naive” solution (e.g., linear scan) to a more optimal one

Constraints (typical interview assumptions)

- Up to **`2 * 10^5`** total operations - Keys and values are strings; timestamps are integers within typical 32-bit range

QUESTION

Part 1: Top-K Points by Frequency with Distance Tie-BreakGiven a list of 2D integer points, where duplicate points may appear, return the top **k** distinct points ranked by these rules: (1) higher frequency first, (2) if frequencies tie, smaller squared distance to the origin first, and (3) if still tied, smaller **x** first, then smaller **y**. The result must contain exactly **k** distinct points in ranked order.

Constraints

1 <= len(points) <= 2 * 10^5-10^4 <= x, y <= 10^41 <= k <= number of distinct points

EXAMPLES

```
Input: ([[1,1],[2,2],[2,2],[1,1],[2,2],[3,3]], 2)
Expected Output: [[2, 2], [1, 1]]
Explanation: Point [2,2] appears 3 times and [1,1] appears 2 times, so they are the top 2 distinct
    points.
Input: ([[1,2],[0,2],[1,2],[2,1],[0,2],[2,1],[3,0]], 3)
Expected Output: [[0, 2], [1, 2], [2, 1]]
Explanation: The points [0,2], [1,2], and [2,1] each appear twice. Their squared distances are 4, 5, and
    5, so [0,2] comes first. Between [1,2] and [2,1], x is smaller for [1,2].
```

Hints

First count how many times each point appears. After that, each distinct point only needs to be ranked once.Python can sort tuples directly, so think about building a sort key like **`(-frequency, distance, x, y)`**.

Part 2: Time-Based Key-Value Store with Closest Timestamp QueriesProcess a sequence of operations for a time-based key-value store. Each operation is either **`['set', key, value, timestamp]`** or **`['getClosest', key, timestamp]`**. For every **`getClosest`** operation, return the stored value whose timestamp is closest to the query tim

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Q1: sort all distinct points by the full key

BRUTE-FORCE IDEA

Count the points, sort the distinct ones by (-frequency, distance, x, y), take K. Correct; sorting all d distinct points does more work than a heap of size K.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Q1: sort all distinct points by the full key
Count the points
sort the distinct ones by (-frequency
distance
take K
```

BRUTE-FORCE COMPLEXITY | Time O(d log d) | Space O(d)

WHY THIS IS WEAKER | Sorts all d distinct points, O(d log d); a size-K heap with the same key is O(d log k).

BRUTE FORCE 2 | Q2: scan every timestamp of the key

BRUTE-FORCE IDEA

Walk the key's (timestamp, value) list and keep the closest, preferring the smaller timestamp on a tie. Correct; linear per query.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Q2: scan every timestamp of the key
Walk the key's (timestamp
value) list
keep the closest
preferring the smaller timestamp on a tie
```

BRUTE-FORCE COMPLEXITY | Time O(m) per query | Space O(total entries)

WHY THIS IS WEAKER | Linear in the number of stored timestamps per query. Timestamps arrive in order, so binary search on the sorted list finds the neighbours in O(log m).

HOW TO IMPROVE IT

count the point tuples; the tie-break is distance from the origin, then x, then y

SHORT BEST SOLUTION

Part 1: Count equal points and keep the best k by frequency, distance, then coordinates.

Part 2: Keep each key's records sorted and use binary search around the requested time.

```
import heapq
from bisect import bisect_left, bisect_right, insort
from collections import Counter, defaultdict


def top_points(points, how_many):
    counts = Counter(points)
    best = heapq.nsmallest(
        how_many, counts,
        key=lambda point: (-counts[point], point[0] ** 2 + point[1] ** 2, point),
    )
    return [list(point) for point in best]


class TimeStore:
    def __init__(self):
        self.data = defaultdict(list)

    def set(self, key, value, timestamp):
        insort(self.data[key], (timestamp, value), key=lambda item: item[0])

    def get_closest(self, key, timestamp):
        items = self.data.get(key, [])
        if not items:
            return ""
        position = bisect_left(items, timestamp, key=lambda item: item[0])
        choices = items[max(0, position - 1):position + 1]
        return min(choices, key=lambda item: (abs(item[0] - timestamp), item[0]))[1]

    def get_at_or_before(self, key, timestamp):
        items = self.data.get(key, [])
        position = bisect_right(items, timestamp, key=lambda item: item[0]) - 1
        return items[position][1] if position >= 0 else ""
```

COMPLEXITY | Part 1: Time O(n + u log k) | Space O(u)
Part 2: Time set O(n), get O(log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 183

184 * Find top/bottom-k words in list or stream * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the same Counter; run nsmallest once with -count and once with +count for the bottom-how_many

QUESTION INFO

QUESTION

You are given words (strings) either as a finite list or as an unbounded stream. 1)
List version: Given an array **`words`** and an integer **k**, return: - the **k** most frequent distinct words, and - the **k** least frequent distinct words. Specify how you break ties (e.g., lexicographically, or any order). 2)
Stream version (online): Design a data structure that supports:

- **`ingest(word)`**: process the next word in the stream - **`topK(k)`**: return the **k** most frequent distinct words seen so far - **`bottomK(k)`**: return the **k** least frequent distinct words seen so far Discuss time/space complexity and how you would implement these operations efficiently. 3)

QUESTION

Variant (discuss/design): Extend the stream design to support returning the **first k** and/or **last k** words that are currently **non-repeating** (i.e., words with frequency exactly 1) according to stream arrival order.

Part 1: Top and Bottom K Frequent Words in a ListGiven an array of strings words and a non-negative integer k, return two lists: the k most frequent distinct words and the k least frequent distinct words. Use deterministic tie-breaking: if two words have the same frequency, the lexicographically smaller word comes first. If k is larger than the number of distinct words, return all distinct words.

Constraints

0 <= len(words) <= 2 * 10^50 <= k <= 2 * 10^5Each word is a non-empty stringComparison is case-sensitive unless otherwise stated

EXAMPLES

```
Input: (['i', 'love', 'leetcode', 'i', 'love', 'coding'], 2)
Expected Output: [['i', 'love'], ['coding', 'leetcode']]
Explanation: `i` and `love` both appear twice, so lexicographical order gives `['i', 'love']`. `coding`
    and `leetcode` both appear once, so lexicographical order gives `['coding', 'leetcode']`.
Input: (['apple', 'banana', 'apple', 'cherry', 'banana', 'date'], 3)
Expected Output: [['apple', 'banana', 'cherry'], ['cherry', 'date', 'apple']]
```

Explanation: Frequencies are: apple=2, banana=2, cherry=1, date=1. Top 3 are ordered by frequency descending, then lexicographically: **`apple`**, **`banana`**, **`cherry`**. Bottom 3 are ordered by frequency ascending, then lexicographically: **`cherry`**, **`date`**, **`apple`**.

Hints

Count how many times each distinct word appears before deciding the answer.You can sort the distinct words twice: once by (-frequency, word) and once by (frequency, word).

Part 2: Online Top/Bottom K Words from a StreamYou are given a sequence of stream operations. Each operation is a 2-element list of strings: ['ingest', word], ['topK', str(k)], or ['bottomK', str(k)]. Process the operations in order. For each query, return the k most frequent or k least frequent distinct words seen so far. Ties are broken lexicographically ascending. If k is larger than the number of distinct words seen so far, return all distinct words.

Constraints

1 <= len(operations) <= 2 * 10^5Each operation is one of ['ingest', word], ['topK', str(k)], or ['bottomK', str(k)]0 <= k <= 2 * 10^5Queries may appear before any word is ingested

Examples

Input: [['ingest', 'apple'], ['ingest', 'banana'], ['ingest', 'apple'], ['topK', '2'], ['bottomK', '2']]

Expected Output: [['apple', 'banana'], ['banana', 'apple']]

Explanation: After the ingests, counts are apple=2 and banana=1. topK(2) returns the most frequent first, while bottomK(2) returns the least frequent first.

Input: [['topK', '3'], ['ingest', 'dog'], ['ingest', 'cat'], ['ingest', 'cat'], ['topK', '2'], ['bottomK', '2'], ['topK', '5']]

Expected Output: [[], ['cat', 'dog'], ['dog', 'cat'], ['cat', 'dog']]

Explanation: The first query happens before any words are seen, so it returns []. Later, cat has frequency 2 and dog has frequency 1.

Hints

Keep an exact frequency map for all words seen so far.A max-heap and a min-heap with lazy deletion let you support online updates without fully resorting all words

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Sort all distinct words by (count, word)

BRUTE-FORCE IDEA

Count, sort the distinct words by count and then alphabetically, and take the first or last K.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Sort all distinct words by (count, word)
sort the distinct words by count
alphabetically
take the first or last K
```

BRUTE-FORCE COMPLEXITY | Time O(d log d) | Space O(d)

WHY THIS IS WEAKER | Sorts all d distinct words, O(d log d). Two heaps of size K are O(d log k); a stream only needs the counts, and a Space-Saving sketch bounds memory for the top K.

HOW TO IMPROVE IT

the same Counter; run nsmallest once with -count and once with +count for the bottom-how_many

SHORT BEST SOLUTION

One pass counts the words. heapq.nsmallest with key (-count, word) gives the K most frequent and with key (count, word) the K least frequent, each in O(d log how_many) without a full sort.

```python
from collections import Counter
import heapq


# BASE QUESTION (optimal): Counter plus two heaps of size
# K
# Idea: One pass counts the words. heapq.nsmallest with
# key (-count, word) gives the K most frequent and with
# key (count, word) the K least frequent, each in O(d log
# how_many) without a full sort.
def top_bottom(words, how_many):
    counts = Counter(words)
    return (heapq.nsmallest(how_many, counts, key=lambda word: (-counts[word], word)), heapq.nsmallest(how_many
, counts, key=lambda word: (counts[word], word)))
```

COMPLEXITY | Time O(n + d log k) | Space O(d)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 184

185 * Solve chat, grid paths, and car rentals * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the top-K part is the template as written; the other parts are a rolling-column path count and a greedy car assignment

QUESTION INFO

QUESTION

You are given three independent programming tasks. For each, describe and implement an efficient algorithm, and be prepared to analyze its time and space complexity. ---

Q1.

Find top-K most active chat users You are given a log of chat messages from a messaging application.

Input:

- An array **`messages`** of length **n**, where each element **`messages[i]`** is the **`user_id`** (string or integer) of the sender of the i-th message.
- An integer **k** (1 <= k <= number of distinct users).

Output:

- Return a list of **k** user IDs corresponding to the users who have sent the most messages.

Details & requirements:

- The "activity" of a user is defined as the number of messages they have sent (i.e., their frequency in **`messages`**).

QUESTION

- If there are fewer than **k** distinct users, return all distinct users.
- If multiple users have the same frequency, you may break ties arbitrarily, or (for determinism) sort tied users by **user_id** in ascending order.
- Aim for an algorithm that is more efficient than sorting all users by count when **k** is much smaller than the number of distinct users. Design and implement a function, for example:
- **List<UserId> topKActiveUsers(List<UserId> messages, int k)** Also explain:
- At least two different approaches (e.g., based on sorting, heaps, or selection algorithms).
- The time and space complexity of each approach. ---

Q2.

Count paths in a 2D grid with constrained moves You have a 2D grid with **m** rows and **n** columns. Index rows from top to bottom as **0..m-1** and columns from left to right as **0..n-1**.

- The **start** cell is the bottom-left corner: **(m-1, 0)**.
- The **end** cell is the bottom-right corner: **(m-1, n-1)**. From any cell **(r, c)**, you may move **only** to the next column to the right, in one of up to three ways (if within bounds):
- Right-up: **(r-1, c+1)**
- Right: **(r, c+1)**
- Right-down: **(r+1, c+1)** You cannot move left or stay in the same column. All cells are passable (no obstacles).

Task:

- Given integers **m** and **n**, compute the total number of distinct paths from the start cell **(m-1, 0)** to the end cell **(m-1, n-1)** using only the allowed moves.

Requirements:

- Implement a function such as **long countPaths(int m, int n)**.
- The result may be large; assume it fits in a 64-bit integer, or, if you prefer, you may return the count modulo a large prime (e.g., 1,000,000,007) — specify which convention you are using.
- First design a straightforward dynamic programming solution using O(m·n) time and O(m·n) space.
- Then describe how to optimize the space usage (e.g., to O(m)) while keeping the same time complexity, and explain why the optimization is correct. ---

Q3.

Minimum cars needed for rentals and assignment You work at a car rental company. You are given a list of rental bookings, each with a pickup and return time.

Input:

- An array of **R** rental records. Each record has:
- **rental_id**: a unique identifier (e.g., integer or string).
- **pickup_time**: an integer start time.
- **return_time**: an integer end time, with **pickup_time < return_time**. Assume times are on a single timeline (e.g., minutes from some epoch). A car can be reassigned immediately after it is returned, i.e., if one rental ends at time **t** and another starts at time **t**, the same car can serve both.

Tasks: 1.

Capacity: Compute the **minimum number of cars** required so that all rentals can be fulfilled without time conflicts on any single car. 2.

Assignment: Produce one valid assignment of rentals to cars that uses exactly that minimum number of cars.

Output:

- An integer **C** = minimal number of cars needed.
- A mapping from each **rental_id** to a **car_id** in the range **1..C**, such that no two rentals assigned to the same **car_id** overlap in time.

QUESTION

Requirements & clarifications:

- Overlap definition: two rentals with intervals `[s1, e1)` and `[s2, e2)` overlap if their time intervals intersect with non-empty duration. Using half-open intervals, if `e1 == s2` they do **not** over

EXAMPLES

Example 1	
Input	`['ann', 'bob', 'ann', 'cy', 'bob', 'ann'], 2`
Output	`['ann', 'bob']`
Why	`Task 1: ann sent 3 messages, bob 2, cy 1; the top 2 are ann and bob.`

Example 2	
Input	`3, 3`
Output	`2`
Why	`Task 2: paths over 3 columns of a 3-row grid that start and end in the bottom row, moving up-right, right or down-right.`

Example 3	
Input	`[(1, 0, 5), (2, 1, 4), (3, 5, 8)]`
Output	`(2, {1: 0, 2: 1, 3: 1})`
Why	`Task 3: rentals are (id, pickup, return). Rental 3 starts at 5, after rental 2 returned its car at 4, so it reuses that car: 2 cars in total.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Q1: sort all users by (count, user id)

BRUTE-FORCE IDEA

Count the messages and sort all distinct users, then take the first K. Correct; sorting all d users does more work than a heap of size K.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Q1: sort all users by (count, user id)
Count the messages
sort all distinct users
take the first K
```

BRUTE-FORCE COMPLEXITY | Time O(d log d) | Space O(d)

WHY THIS IS WEAKER | Sorting all d distinct users costs O(d log d) even though only K are needed; a heap of size K costs O(d log k).

BRUTE FORCE 2 | Q2: plain recursion over the moves

BRUTE-FORCE IDEA

From the start cell try each of the three moves recursively and count the ways to reach the end. Exact; the same cells are recomputed exponentially often.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Q2: plain recursion over the moves
From the start cell try each of the three moves recursively
count the ways to reach the end
```

BRUTE-FORCE COMPLEXITY | Time O(3^n) | Space O(n)

WHY THIS IS WEAKER | Three calls per cell and the same cell is reached by many paths, so the call tree has about 3^n nodes. One column at a time with a rolling array is O(m * n).

BRUTE FORCE 3 | Q3: count the overlap at every time and assign greedily without a heap

BRUTE-FORCE IDEA

For each rental, look for any car whose last return is at or before its pickup by scanning all cars; open a new one if none is free. Correct for the number of cars; each rental scans every car.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Q3: count the overlap at every time and assign greedily without a heap
For each rental
look for any car whose last return is at or before its pickup by scanning all cars
open a new one if none is free
```

BRUTE-FORCE COMPLEXITY | Time O(R^2) | Space O(R)

WHY THIS IS WEAKER | Each rental scans every car, so the cost is O(R * cars), up to O(R^2). A heap of free times finds the earliest-free car in O(log R).

HOW TO IMPROVE IT

the top-K part is the template as written; the other parts are a rolling-column path count and a greedy car assignment

SHORT BEST SOLUTION

Part 1: Count messages and keep the best k users.

Part 2: Carry one column of path counts into the next column.

Part 3: Keep current return times in a min-heap and reuse a returned car.

```python
import heapq
from collections import Counter


def top_k_users(messages, how_many):
    counts = Counter(messages)
    return heapq.nsmallest(how_many, counts, key=lambda user: (-counts[user], user))


def count_paths(rows, cols, blocked=(), mod=None):
    blocked = set(blocked)
    ways = [0] * rows
    ways[-1] = 1
    for col in range(1, cols):
        following = [0] * rows
        for row in range(rows):
            if (row, col) not in blocked:
                following[row] = ways[row]
                if row:
                    following[row] += ways[row - 1]
                if row + 1 < rows:
                    following[row] += ways[row + 1]
                if mod:
                    following[row] %= mod
        ways = following
    return ways[-1]


def min_cars(rentals):
    returns = []
    for pickup, back in sorted(rentals):
        if returns and returns[0] <= pickup:
            heapq.heapreplace(returns, back)
        else:
            heapq.heappush(returns, back)
    return len(returns)
```

COMPLEXITY | Part 1: Time O(n + u log k) | Space O(u)
Part 2: Time O(m * n) | Space O(m)
Part 3: Time O(n log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 185

186 * Top-K Chatters by Word Count

* MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: count len(body.split()) per speaker instead of lines; ties by name

QUESTION INFO

QUESTION

You are given a chat transcript as a list of message lines. Each line has the form "Name: message text" , where everything before the first ": " is the author's name and everything after it is the message body. Count the number of words each person typed across the entire transcript (a word is a maximal run of non- whitespace characters in the body). A person may appear on many lines; sum their words. Return the k people who typed the most words, as a list of (name, word_count) tuples sorted by word_count descending. Break ties by name in ascending (lexicographic) order.

Function signature

```
def top_k_chatters(lines: list[str], k: int) -> list[tuple[str, int]]:
    ...
```

Follow-up (as reported): The interviewer's real ask was to implement the parsing inside a class alongside stubbed API methods. Wrap this in a ChatAnalyzer class that ingests lines incrementally via add_line(line) and answers top_k(k) on demand.

EXAMPLES

Example 1	
Input	`lines = ["Alice: hello there friend", "Bob: hi", "Alice: how are you"], k = 1`
Output	`[("Alice", 6)]`
Why	`Alice typed 3 + 3 = 6 words, Bob typed 1. Top 1 is Alice.`

Example 2	
Input	`lines = ["Ann: a b", "Bob: c d", "Cara: e"], k = 2`
Output	`[("Ann", 2), ("Bob", 2)]`
Why	`Ann and Bob both have 2 words; tie broken alphabetically, so Ann precedes Bob. Cara (1 word) is excluded.`

Example 3	
Input	`lines = [], k = 3`
Output	`[]`

CONSTRAINTS

- 0 <= len(lines) <= 10^5
- Every line contains exactly one ": " separating name from body; the name is non-empty and contains no ": " .
- The body may be empty (0 words).
- 1 <= k ; if k exceeds the number of distinct people, return all of them.

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(lines) <= 10^5
- Every line contains exactly one ": " separating name from body; the name is non-empty and contains no ": " .

- The body may be empty (0 words).
- 1 <= k ; if k exceeds the number of distinct people, return all of them.

BRUTE FORCE | Count, sort everyone, slice

BRUTE-FORCE IDEA

Start with: Count, sort everyone, slice. It sorts all p authors although only the top k are needed.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Count, sort everyone, slice
Start with: Count
process every required item or candidate directly
```

BRUTE-FORCE COMPLEXITY | Time O(p log p) | Space O(p)

WHY THIS IS WEAKER | It sorts all p authors although only the top k are needed.

HOW TO IMPROVE IT

count len(body.split()) per speaker instead of lines; ties by name

SHORT BEST SOLUTION

Sum the words per author, then pick the k best with a heap keyed by (-count, name).

```python
# BASE QUESTION (optimal): Counter and a bounded selection
# Idea: Sum the words per author, then pick the k best
# with a heap keyed by (-count, name).
import heapq
from collections import Counter


def top_k_chatters(lines, how_many):
    counts = Counter()
    for line in lines:
        name, body = line.split(": ", 1)
        counts[name] += len(body.split())
    return heapq.nsmallest(how_many, counts.items(), key=lambda entry: (-entry[1], entry[0]))
```

COMPLEXITY | Time O(n + p log k) | Space O(p)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 186

Digits and characters: peel one at a time and carry

PATTERN CONTEXT

Work one digit or character at a time: n % 10 peels the last digit, n // 10 removes it, a carry moves overflow to the next place, and (ord(c) - 97 + k) % 26 shifts a letter. Build the answer in a list and reverse or join at the end.

188 * Solve several coding interview problems * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the big-number addition on strings, with int(ch) per digit; the other parts are list-meeting and overlap sweeps

QUESTION INFO

QUESTION

Three small tasks. (1) add_strings(a, b): add two non-negative integers given as digit strings and return the sum as a string, without converting the whole numbers. (2) intersection(head_a, head_b): two singly linked lists may share a common tail; return the first shared node, or None. (3) overlap_depth(rects): rectangles (x1, y1, x2, y2) include their left and bottom edges but not their right and top edges; return the largest number of rectangles that cover a single point.

Function signature

def add_strings(a: str, b: str) -> str
def intersection(head_a, head_b)
def overlap_depth(rects: list) -> int

EXAMPLES

Example 1

Input	`'999', '1'`
Output	`'1000'`
Why	`A carry runs through every digit: 999 + 1 = 1000.`

Example 2

Input	`'456', '77'`
Output	`'533'`
Why	`456 + 77 = 533.`

Example 3

Input	`class N:`
	`def __init__(self, val, next=None):`
	`self.val, self.next = val, next`
	`shared = N(8, N(9))`
	`a = N(1, N(2, shared))`
	`b = N(5, shared)`
	`intersection(a, b).val`

Output	`intersection(a, b).val -> 8`
Why	`List a is 1, 2, 8, 9 and list b is 5, 8, 9: they join at the node holding 8.`

Example 4

Input	`[(0, 0, 4, 4), (2, 2, 6, 6), (3, 3, 5, 5)]`
Output	`3`
Why	`The point (3, 3) lies inside all three rectangles.`

QUESTION

Example 5	
Input	`[(0, 0, 2, 2), (2, 0, 4, 2)]`
Output	`1`
Why	`Rectangles that only share an edge do not overlap.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Big numbers: convert to integers and add

BRUTE-FORCE IDEA

Int(a) + int(b), then str. Valid in Python, but it hides the algorithm and relies on arbitrary-precision integers.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Big numbers: convert to integers and add
Int(a) + int(b)
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) for the conversions | Space O(n)

WHY THIS IS WEAKER | Relies on arbitrary-precision integers (not in every language) and the string-to-integer conversions are quadratic for very long inputs. Adding digit by digit from the right with a carry is O(max length).

BRUTE FORCE 2 | Chains that meet: store one chain in a set

BRUTE-FORCE IDEA

Put every node of the first chain in a set and walk the second until a node is in the set. Correct, using memory for the set.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Chains that meet: store one chain in a set
Put every node of the first chain in a set
walk the second until a node is in the set
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(m + n) | Space O(m)

WHY THIS IS WEAKER | O(m + n) time but O(m) extra memory. Two pointers that switch to the other chain's head at the end meet at the shared node with O(1) memory.

BRUTE FORCE 3 | Overlap: test every time point

BRUTE-FORCE IDEA

For each integer time count the intervals containing it (half-open) and take the maximum. Correct; it depends on the time range.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Overlap: test every time point
For each integer time count the intervals containing it (half-open)
take the maximum
```

BRUTE-FORCE COMPLEXITY | Time O(range * n) | Space O(1)

WHY THIS IS WEAKER | Depends on the length of the time range, O(range * n). Sorting start and end events and keeping a running count is O(n log n).

HOW TO IMPROVE IT

the big-number addition on strings, with int(ch) per digit; the other parts are list-meeting and overlap sweeps

SHORT BEST SOLUTION

Part 1: Walk both strings from the right, add the two digits and the carry, append total % 10, carry total // 10; after the loop add a final carry digit if needed and reverse. Never convert the whole string to an integer.

Part 2: Walk p along a and q along b; when a pointer reaches the end, restart it at the other chain's head. Both pointers then travel m + n steps in total, so they meet at the first shared node (or both become None together).

Part 3: Create +1 at each start and -1 at each end, sort the events (ends before starts at the same time for half-open intervals), and track the running count; its maximum is the answer.

```
# Big numbers: digit by digit from the right with a carry
# Idea: Walk both strings from the right, add the two
# digits and the carry, append total % 10, carry total //
# 10; after the loop add a final carry digit if needed and
# reverse. Never convert the whole string to an integer.
def addStrings(first_number, second_number):
    first_index, second_index, carry, digits = len(first_number) - 1, len(second_number) - 1, 0, []
    while first_index >= 0 or second_index >= 0 or carry:
        total = carry + (int(first_number[first_index]) if first_index >= 0 else 0) + (int(second_number[second
_index]) if second_index >= 0 else 0)
        digits.append(str(total % 10))
        carry = total // 10
        first_index, second_index = first_index - 1, second_index - 1
    return "".join(digits[::-1])


# Chains that meet: two pointers that switch heads at the
# end
# Idea: Walk p along a and q along b; when a pointer
# reaches the end, restart it at the other chain's head.
# Both pointers then travel m + n steps in total, so they
# meet at the first shared node (or both become None
# together).
class LNode:
    def __init__(self, val, nxt=None):
        self.val, self.next = val, nxt

def getIntersectionNode(first_head, second_head):
    walker_a, walker_b = first_head, second_head
    while walker_a is not walker_b:
        walker_a = walker_a.next if walker_a else second_head
        walker_b = walker_b.next if walker_b else first_head
    return walker_a


# Overlap: sweep over sorted start and end events
# Idea: Create +1 at each start and -1 at each end, sort
# the events (ends before starts at the same time for
# half-open intervals), and track the running count; its
# maximum is the answer.
def max_overlap(intervals):
    events = sorted([(start, 1) for start, _ in intervals] + [(end, -1) for _, end in intervals])
    running_count = best = 0
    for _, change in events:
        running_count += change
        best = max(best, running_count)
    return best
```

COMPLEXITY | Part 1: Time O(max(m, n)) | Space O(max(m, n))
Part 2: Time O(m + n) | Space O(1)
Part 3: Time O(n log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 188

189 * Check Whether Digits Are Monotone Increasing

* MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: peel digits from the right; each digit must not exceed the previous one you saw

QUESTION INFO

QUESTION

Return true when the decimal digits of a non-negative integer never decrease from left to right (repeated digits are allowed). Zero and single-digit numbers qualify.

Function signature

def is_monotone_digits(n: int) -> bool

EXAMPLES

Example 1	
Input	1234
Output	true
Why	1, 2, 3, 4 only goes up.

Example 2	
Input	1223
Output	true
Why	Equal neighbours are allowed.

Example 3	
Input	332
Output	false
Why	3, 3, 2 goes down at the end.

Example 4	
Input	0
Output	true
Why	Zero qualifies.

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | String comparison with its sorted version

BRUTE-FORCE IDEA

Convert to a string and compare it with the sorted string: a number has non-decreasing digits exactly when its digits are already in sorted order. Clear, with an unneeded sort.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: String comparison with its sorted version
Convert to a string
compare it with the sorted string: a number has non-decreasing digits exactly when its
digits are already in sorted order
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(d log d) | Space O(d)

WHY THIS IS WEAKER | Sorting costs O(d log d) and copies the digits. Comparing each digit with its neighbour while peeling digits with % 10 is O(d) with O(1) space.

HOW TO IMPROVE IT

peel digits from the right; each digit must not exceed the previous one you saw

SHORT BEST SOLUTION

Take the last digit with number % 10, then the next one, and so on. Because we move right to left, the digit just read must not exceed the one read before it. A single digit and zero are monotone.

```
# BASE QUESTION (optimal): Peel digits from the right and
# compare with the previous one
# Idea: Take the last digit with number % 10, then the next
# one, and so on. Because we move right to left, the digit
# just read must not exceed the one read before it. A
# single digit and zero are monotone.
def is_monotone_digits(number):
    previous_digit = 9
    while number:
        digit = number % 10
        if digit > previous_digit:
            return False
        previous_digit = digit
        number //= 10
    return True
```

COMPLEXITY | Time O(d) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 189

190 * Add two big integers from digit lists * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the template as written

QUESTION INFO

QUESTION

You are given two non-empty singly linked lists (or arrays) representing two non-negative integers. The digits are stored in **reverse order**, and each node contains a single digit from 0 to 9. Your task is to add the two numbers and return the sum as a new linked list, also in reverse order. You may assume:

- Each list has at least one node.
- The integers do not contain leading zeros, except the number 0 itself.

Example

Input:

- List 1: 2 -> 4 -> 3 (represents the integer 342)
- List 2: 5 -> 6 -> 4 (represents the integer 465)

Output:

- 7 -> 0 -> 8 (represents the integer 807, since 342 + 465 = 807)

Another example

Input:

- List 1: 9 -> 9 -> 9 (represents 999)
- List 2: 1 (represents 1)

Output:

- 0 -> 0 -> 0 -> 1 (represents 1000)

QUESTION

Constraints

- Let n be the length of the first list and m be the length of the second list.
- 1 <= n, m <= 10^5.
- Each node value is an integer in the range [0, 9].

Task: Describe an algorithm to perform this addition and construct the resulting linked list in O(n + m) time and O(1) additional space (excluding the space for the output list).

You are given two non-empty arrays of digits representing two non-negative integers. The digits are stored in **reverse order** (least significant digit first), and each element is a single digit from 0 to 9. Add the two numbers and return the sum as a new array of digits, also in reverse order.

Assumptions

- Each array has at least one element.
- The integers do not contain leading zeros, except the number 0 itself.

Example 1

```
Input: l1 = [2, 4, 3], l2 = [5, 6, 4] (342 + 465)
Output: [7, 0, 8] (807)
```

Example 2

```
Input: l1 = [9, 9, 9], l2 = [1] (999 + 1)
Output: [0, 0, 0, 1] (1000)
```

Implement **`add_two_numbers(l1, l2)`** (camelCase **`addTwoNumbers`** in Java/C++/JS) returning the digit array of the sum in reverse order. Aim for O(n + m) time and O(1) extra space beyond the output.

Constraints

1 <= n, m <= 10^5 where n = len(l1), m = len(l2)Each element is an integer in the range [0, 9]Digits are stored in reverse order (least significant digit first)No leading zeros except for the number 0 itself

EXAMPLES

```
Input: ([2, 4, 3], [5, 6, 4])
Expected Output: [7, 0, 8]
Explanation: 342 + 465 = 807, stored reversed as [7, 0, 8].
Input: ([9, 9, 9], [1])
Expected Output: [0, 0, 0, 1]
```

Explanation: 999 + 1 = 1000; carry propagates through all three digits and creates a new leading digit.

Hints

Walk both arrays from index 0 (the least significant digit) toward the end, just like grade-school column addition.Maintain a single carry variable. At each position, sum the available digits plus the carry; append (sum % 10) and set carry = sum // 10.When one array runs out, keep going with just the carry and the remaining longer array. Don't forget a final carry can add one extra digit (e.g. 999 + 1 -> [0,0,0,1]).

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Convert both lists to integers and add

BRUTE-FORCE IDEA

Join the digits, int() them, add, and split the result into digits. It is a valid shortcut in Python (big integers), but it hides the algorithm, depends on the language having big integers, and the conversions are quadratic for very long inputs.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Convert both lists to integers and add
Join the digits
int() them
split the result into digits
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) for the string-to-integer conversions | Space O(n)

WHY THIS IS WEAKER | It relies on arbitrary-size integers and the conversions are quadratic for very long inputs. Adding digit by digit from the right with a carry is O(max length).

HOW TO IMPROVE IT

the template as written

SHORT BEST SOLUTION

Walk both lists from the right with indices first_index and second_index; at each step add first_digits[first_index] (or 0 once exhausted), second_digits[second_index] (or 0) and the carry, append sum % 10, and carry = sum // 10. After the loop add a final 1 if the carry remains, and reverse the collected digits.

```python
# BASE QUESTION (optimal): Digit by digit with a carry
# Idea: Walk both lists from the right with indices first_index and
# second_index; at each step add first_digits[first_index] (or 0 once exhausted), second_digits[second_index] (
or
# 0) and the carry, append sum % 10, and carry = sum //
# 10. After the loop add a final 1 if the carry remains,
# and reverse the collected digits.
def add_digits(first_digits, second_digits):
    first_index, second_index, carry, result = len(first_digits) - 1, len(second_digits) - 1, 0, []
    while first_index >= 0 or second_index >= 0 or carry:
        total = carry + (first_digits[first_index] if first_index >= 0 else 0) + (second_digits[second_index] i
f second_index >= 0 else 0)
        result.append(total % 10)
        carry = total // 10
        first_index, second_index = first_index - 1, second_index - 1
    return result[::-1] or [0]
```

COMPLEXITY | Time O(max(m, n)) | Space O(max(m, n))

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 190

GROUP 25 OF 43 | 4 QUESTIONS | PART A - PART 1

Look at the input once and keep only a small summary (and count your calls)

PATTERN CONTEXT

When the input is too big to hold, read it one item at a time and keep a tiny summary: a maximum, a set from the smaller file, a sorted window. When asking a question is expensive (an oracle), use binary search on how many items you include so the number of calls stays small.

193 * Streaming Points: Emit K Within Distance

* MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: keep the points in a sorted list; each new point is inserted with bisect and only windows around it are re-checked

QUESTION INFO

QUESTION

Positions (floats) arrive one at a time. Whenever k stored points fit inside a span of d (largest minus smallest is at most d), output those k points and remove them. add(x) returns the groups emitted because of this arrival (a list of lists).

Function signature

class PointStream:
  def __init__(self, k: int, d: float)
  def add(self, x: float) -> list

EXAMPLES

Example 1

Input	`s = PointStream(3, 2.0)`
	`s.add(1.0)`
	`s.add(5.0)`
	`s.add(2.0)`
	`s.add(1.5)`

Output	`s.add(1.0) -> []`
	`s.add(5.0) -> []`
	`s.add(2.0) -> []`
	`s.add(1.5) -> [[1, 1.5, 2]]`

Why	`After 1, 5, 2 no three points fit in a span of 2 (1 and 5 are too far apart); adding 1.5 gives 1, 1.5, 2 with span 1, which is emitted and removed.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Re-sort and scan every window after each point

BRUTE-FORCE IDEA

Sort all points, slide a window of K, and emit the lowest group that fits, repeating until none does. Correct; the whole list is re-examined for every new point.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Re-sort and scan every window after each point
Sort all points
slide a window of K
emit the lowest group that fits
```

```
repeating until none does
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n log n) per point | Space O(n)

WHY THIS IS WEAKER | Every arrival re-sorts and rescans the whole list, O(n log n). A sorted list with binary search checks only the windows around the new point.

HOW TO IMPROVE IT

keep the points in a sorted list; each new point is inserted with bisect and only windows around it are re-checked

SHORT BEST SOLUTION

Insert with bisect. Because no window existed before, a new window must contain the new point, so test the K windows that include its position. After emitting a group, the points on both sides of the gap become neighbours, so test the windows around that seam too. A balanced tree (sorted container) makes the insert and delete O(log n).

```python
from bisect import bisect_left
from bisect import insort


# BASE QUESTION (optimal): Sorted list, check only windows
# touching the change
# Idea: Insert with bisect. Because no window existed
# before, a new window must contain the new point, so test
# the K windows that include its position. After emitting
# a group, the points on both sides of the gap become
# neighbours, so test the windows around that seam too. A
# balanced tree (sorted container) makes the insert and
# delete O(log n).
class PointStream:
    def __init__(self, group_size, max_spread):
        self.group_size, self.max_spread, self.points = group_size, max_spread, []

    def _first_window(self, position):
        first_start = max(0, position - self.group_size + 1)
        last_start = min(position, len(self.points) - self.group_size)
        for start in range(first_start, last_start + 1):
            if self.points[start + self.group_size - 1] - self.points[start] <= self.max_spread:
                return start
        return None

    def add(self, point):
        insort(self.points, point)
        emitted, position = [], bisect_left(self.points, point)
        while True:
            start = self._first_window(position)
            if start is None:
                return emitted
            emitted.append(self.points[start:start + self.group_size])
            del self.points[start:start + self.group_size]
            position = start
```

COMPLEXITY | Time O(log n + K) per check, plus list shifting | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 193

194 * Two-Day Log Intersection under Memory Limit * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: dict object -> set of event codes for each day; keep the objects that appear on both days with at least two distinct codes

QUESTION INFO

QUESTION

PhD intern coding round 1: two log files with **`(timestamp, obj_id, client_id)`** entries. Find **`obj_id`**s that (a) appear in both files and (b) have at least two distinct **`client_id`**s. Follow-up imposes a memory limit, expecting external sort + two-pointer.

Input: two log files; each line is (timestamp, obj_id, client_id).
Find every obj_id such that:

(a) appears in both files, and
(b) is associated with at least 2 distinct client_ids (aggregated across both files).

Follow-up 1: complexity of set-intersection on the file contents.
Follow-up 2: memory cannot hold either file fully — design a streaming solution.

EXAMPLES

```
File A:  (t1, X, c1), (t2, X, c2), (t3, Y, c1)
File B:  (t4, X, c1), (t5, Z, c2)
-> Only X is in both files; X has 2 distinct clients {c1, c2}.
Output: [X]
```

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | For every object scan the other file

BRUTE-FORCE IDEA

For each object in the first file scan the second file and the first file again to count clients. Correct, quadratic, and it needs both files in memory.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: For every object scan the other file
For each object in the first file scan the second file
the first file again to count clients
```

BRUTE-FORCE COMPLEXITY | Time O(n * m) | Space O(n + m)

WHY THIS IS WEAKER | O(n * m) time and both files must fit in memory. Sorting both files by object and merging them with two pointers is O(n log n) and holds one object's clients at a time.

HOW TO IMPROVE IT

dict object -> set of event codes for each day; keep the objects that appear on both days with at least two distinct codes

SHORT BEST SOLUTION

Read externally sorted logs together and hold only the client sets for the current object from each day.

```python
# BASE QUESTION (optimal): Merge two object-sorted streams
# Each row is (timestamp, object_id, client_id). The inputs
# are sorted externally by object id, so memory holds only
# one object's client set from each day at a time.
def log_objects_sorted(day1, day2):
    day_one_rows = iter(day1)
    day_two_rows = iter(day2)

    def next_group(rows, first_row):
```

```
        if first_row is None:
            return None, None
        object_id = first_row[1]
        clients = {first_row[2]}
        for row in rows:
            if row[1] != object_id:
                return (object_id, clients), row
            clients.add(row[2])
        return (object_id, clients), None

    group_a, next_a = next_group(day_one_rows, next(day_one_rows, None))
    group_b, next_b = next_group(day_two_rows, next(day_two_rows, None))
    answer = []
    while group_a and group_b:
        if group_a[0] == group_b[0]:
            if len(group_a[1] | group_b[1]) >= 2:
                answer.append(group_a[0])
            group_a, next_a = next_group(day_one_rows, next_a)
            group_b, next_b = next_group(day_two_rows, next_b)
        elif group_a[0] < group_b[0]:
            group_a, next_a = next_group(day_one_rows, next_a)
        else:
            group_b, next_b = next_group(day_two_rows, next_b)
    return answer
```

COMPLEXITY | Time O(n + m) | Space O(clients for one object)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 194

195 * Find shared objects across two log files * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: load only the smaller file's object ids into a set, then stream the big file and keep ids found in the set

QUESTION INFO

QUESTION

You are given two large log files representing activity on two different days. Each line of each log has three fields:

- **`timestamp`**: a time value (you may treat it as an opaque string or integer)
- **`obj_id`**: identifier of an object (string or integer)
- **`client_id`**: identifier of a client (string or integer) The logs are not necessarily sorted by any field. Define an object as **interesting** if:

1. Its **`obj_id`** appears at least once in day 1's log **and** at least once in day 2's log; and 2. Across all of its appearances in both days combined, it is associated with at least **two distinct `client_id`** values.

Tasks

1. Design and implement a function that, given file handles (or streaming iterators) for the two log files, returns the set of all such interesting **`obj_id`**s.
2. For your initial, straightforward in-memory solution (you may assume both logs fit in memory), state the time and space complexity in terms of the total number of log records.
3. Follow-up (memory constrained): Now assume the logs are very large and **cannot** both fit into memory at once.

- You may still sequentially scan each file and you may perform external sorting.
- Describe how you would modify your approach to work under this memory constraint.
- Analyze the time and space complexity of this memory-efficient solution.

QUESTION

Return sorted obj_ids that appear in both day logs and have at least two distinct client_ids across both days combined.
Constraints
Rows are [timestamp, obj_id, client_id] or dicts with those fields

EXAMPLES

```
Input: ([['t1', 'o1', 'c1'], ['t2', 'o2', 'c1']], [['t3', 'o1', 'c2'], ['t4', 'o2', 'c1']])
Expected Output: ['o1']
Input: ([], [['t', 'o', 'c']])
Expected Output: []
```

Hints

Track which days each object appears in and its set of clients.

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Compare every pair of lines

BRUTE-FORCE IDEA

For each line in A scan all of B for the same object id. Correct, but quadratic in the file sizes.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compare every pair of lines
For each line in A scan all of B for the same object id
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n * m) | Space O(1) extra

WHY THIS IS WEAKER | O(n * m) comparisons. Loading the smaller file's ids into a hash set and streaming the other file is O(n + m); if neither fits, partition both by a hash of the id.

HOW TO IMPROVE IT

load only the smaller file's object ids into a set, then stream the big file and keep ids found in the set

SHORT BEST SOLUTION

Load the object ids of the smaller file into a set, then stream the other file and keep ids that are in the set. Each line is read once; memory depends only on the smaller file.

```python
def _obj_ids(lines):
    return {line.split()[1] for line in lines if line.strip()}


# BASE QUESTION (optimal): Hash set of the smaller file's
# ids
# Idea: Load the object ids of the smaller file into a
# set, then stream the other file and keep ids that are in
# the set. Each line is read once; memory depends only on
# the smaller file.
def shared_objects(small_lines, big_lines):
    small = _obj_ids(small_lines)
    return {line.split()[1] for line in big_lines if line.strip() and line.split()[1] in small}
```

COMPLEXITY | Time O(n + m) | Space O(distinct ids of the smaller file)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 195

196 * Minimize calls to find all bad test pairs * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the same group testing: put a whole group in one call and split only the groups that fail

QUESTION INFO

QUESTION

You have (n) atomic tests (t_1, t_2, \dots, t_n). You are given access to a black-box function:

```
bool runTest(set<Test> S)
```

which behaves as follows:

- **`runTest(S) == True`** if the subset (S) contains **no** incompatible (bad) pair of tests.
- **`runTest(S) == False`** if the subset (S) contains **at least one** incompatible (bad) pair of tests. There is some unknown set of unordered **bad pairs** ((t_i, t_j)), where (i\ne j). You do **not** know in advance which pairs are bad or how many bad pairs there are. Your goal is to discover **all** bad pairs while making as few calls to **`runTest`** as possible. You may adaptively choose each subset (S) based on the results of previous calls.

Tasks

1. Design an algorithm that, using only calls to **`runTest`**, finds **all** bad pairs ((t_i, t_j)).
2. Analyze the number of calls to **`runTest`** your algorithm requires, as a function of:

- (n): the total number of tests, and - (k): the total number of bad pairs.

3. Argue why your algorithm is correct (i.e., why it will eventually find every bad pair and not report any good pair as bad). You do not need to provide full production code, but describe the algorithm clearly enough that it could be implemented.

You have n atomic tests numbered 0 to n - 1. Some unknown unordered pairs of tests are incompatible, called bad pairs. A black-box oracle runTest(S) returns True if the subset S contains no bad pair, and False if S contains at least one bad pair. In this non-interactive version, you are given bad_pairs only so your code can simulate the oracle. Implement the following deterministic oracle-discovery strategy and return both the bad pairs it discovers and the number of oracle calls it makes. Strategy:

1. Process vertices from right to left: v = n - 1, n - 2, ..., 0.
2. Maintain all bad pairs already discovered among vertices greater than the current v.
3. For the suffix vertices v + 1 through n - 1, greedily partition them into color classes such that no discovered bad pair appears inside one color class. Since all suffix-suffix bad pairs have already been discovered, every color class is truly compatible internally.
4. For each color class C, find all bad partners of v inside C using recursive group testing:

- Call runTest({v} union C).
- If the oracle returns True, v has no bad pair with any test in C.
- If C has size 1 and the oracle returns False, that single pair is bad.
- Otherwise split C into two halves and recurse on both halves. Return all discovered bad pairs sorted lexicographically, where each pair is represented as [i, j] with i < j, along with the exact number of runTest calls made by this strategy.

Constraints
0 <= n <= 5000 <= len(bad_pairs) <= n * (n - 1) / 2Each bad pair contains two distinct test indices in the range [0, n - 1]bad_pairs may be given in any endpoint order

EXAMPLES

```
Input: (0, [])
Expected Output: ([], 0)
```

Explanation: There are no tests and therefore no oracle calls are needed.

QUESTION

```
Input: (4, [])
Expected Output: ([], 3)
```

Explanation: For v = 2, 1, and 0, the algorithm queries the compatible suffix once and finds no bad pairs. v = 3 has an empty suffix.

Hints

If a candidate set C is internally compatible, then runTest({v} union C) tells you whether v conflicts with at least one element of C.Process tests from right to left so that when handling v, all bad pairs inside the suffix v + 1 ... n - 1 are already known and can be used to build compatible groups.

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Call runTest on every pair

BRUTE-FORCE IDEA

Try each pair {a, b} on its own. Always correct, but the number of calls grows quadratically even if there are no bad pairs at all.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Call runTest on every pair
Try each pair {a
b} on its own
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time n(n - 1) / 2 calls | Space O(1)

WHY THIS IS WEAKER | N(n - 1) / 2 calls even when nothing is bad. Keeping groups of mutually compatible tests and binary searching only when a group fails needs about n + p log n calls for p bad pairs.

HOW TO IMPROVE IT

the same group testing: put a whole group in one call and split only the groups that fail

SHORT BEST SOLUTION

Process tests one by one. Keep groups of tests that pass runTest. For a new test test and a group G: if runTest(G + {test}) passes there is no partner in G; otherwise binary search for a partner: the smallest prefix of G that fails with test identifies it, remove it from consideration and repeat until the rest passes. Put test into a group with no partner (or a new group). Pairs inside a group cannot exist, and pairs across groups are found when the later test arrives.

```python
# BASE QUESTION (optimal): Independent groups plus binary
# search for partners
# Idea: Process tests one by one. Keep groups of tests
# that pass runTest. For a new test test and a group G: if
# runTest(G + {test}) passes there is no partner in G;
# otherwise binary search for a partner: the smallest
# prefix of G that fails with test identifies it, remove it
# from consideration and repeat until the rest passes. Put
# test into a group with no partner (or a new group). Pairs
# inside a group cannot exist, and pairs across groups are
# found when the later test arrives.
def find_bad_pairs(tests, runTest):
    groups, pairs = [], set()

    def partners(group, test):
        found, rest = [], list(group)
        while rest and not runTest(set(rest) | {test}):
            low, high = 1, len(rest)
            while low < high:
                middle = (low + high) // 2
                if runTest(set(rest[:middle]) | {test}):
```

```
                low = middle + 1
            else:
                high = middle
        found.append(rest[low - 1])
        rest = rest[:low - 1] + rest[low:]
    return found

for test in tests:
    placed = False
    for group in groups:
        found_partners = partners(group, test)
        for partner in found_partners:
            pairs.add((min(partner, test), max(partner, test)))
        if not found_partners and not placed:
            group.append(test)
            placed = True
    if not placed:
        groups.append([test])
return pairs
```

COMPLEXITY | Time O(n * g + p log n) calls (g = number of groups, p = number of bad pairs) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 196

Grid DP: each cell from its neighbours above and to the left

PATTERN CONTEXT

Walk the grid row by row and keep only the previous row. A cell's value depends on the cell above, the cell to the left, and the diagonal. Change the condition for a cell to count and the formula that combines the three neighbours.

199 * Grid Path Count with Diagonal Moves and Checkpoints * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: advance column by column keeping the number of ways in each row; new[row] = old[row-1] + old[row] + old[row+1]; zero out rows that miss a checkpoint

QUESTION INFO

QUESTION

Count the paths in an n x m grid from the bottom-left cell (n - 1, 0) to the bottom-right cell (n - 1, m - 1). From (r, c) you may move up-right (r - 1, c + 1), right (r, c + 1) or down-right (r + 1, c + 1) and must stay inside the grid. Optionally you are given checkpoint cells that every path has to pass through; with ordered=True they must be visited in the order given.

Function signature

def count_paths(n: int, m: int, checkpoints=(), ordered=False) -> int

EXAMPLES

Example 1	
Input	`3, 3`
Output	`2`
Why	`Two paths: through (1, 1) and then down-right, or through (2, 1) and then right.`

Example 2	
Input	`3, 3, checkpoints = [(1, 1)]`
Output	`1`
Why	`The path must also pass through (1, 1), the middle cell.`

Example 3	
Input	`3, 4, checkpoints = [(1, 1), (2, 2)], ordered = true`
Output	`1`
Why	`Two checkpoints in different columns, visited in column order.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Enumerate every path

BRUTE-FORCE IDEA

Depth-first search over all move sequences, counting those that end at the bottom-right with the checkpoints visited. Exponential in the number of columns.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Enumerate every path
Depth-first search over all move sequences
```

counting those that end at the bottom-right with the checkpoints visited

BRUTE-FORCE COMPLEXITY | Time O(3^m) | Space O(m)

WHY THIS IS WEAKER | Up to 3^m paths. Every move advances one column, so a dp over (column, row) with a rolling array counts them in O(n * m).

HOW TO IMPROVE IT

advance column by column keeping the number of ways in each row; new[row] = old[row-1] + old[row] + old[row+1]; zero out rows that miss a checkpoint

SHORT BEST SOLUTION

ways[row] is the number of paths that are in row r at the current column. The next column is ways'[row] = ways[row - 1] + ways[row] + ways[row + 1]. At a checkpoint column keep only the checkpoint's row. Two checkpoints in one column make the answer 0. For the ordered version the checkpoints must be given by strictly increasing column, because one move always advances one column.

```
# BASE QUESTION (optimal): Column DP with a rolling array
# and checkpoint filters
# Idea: ways[row] is the number of paths that are in row r
# at the current column. The next column is ways'[row] =
# ways[row - 1] + ways[row] + ways[row + 1]. At a checkpoint
# column keep only the checkpoint's row. Two checkpoints
# in one column make the answer 0. For the ordered version
# the checkpoints must be given by strictly increasing
# column, because one move always advances one column.
def count_paths(rows, cols, checkpoints=(), ordered=False):
    checkpoint_cells = [tuple(col) for col in checkpoints]
    by_col = {}
    for row, col in checkpoint_cells:
        if by_col.setdefault(col, row) != row or len({checkpoint_col for _, checkpoint_col in checkpoint_cells}
) != len(checkpoint_cells):
            return 0
    if ordered and [col for _, col in checkpoint_cells] != sorted(col for _, col in checkpoint_cells):
        return 0
    ways = [0] * rows
    ways[rows - 1] = 1
    def keep(column_ways, col):
        if col in by_col:
            column_ways = [ways_value if row == by_col[col] else 0 for row, ways_value in enumerate(column_ways
)]
        return column_ways
    ways = keep(ways, 0)
    for col in range(1, cols):
        ways = keep([(ways[row - 1] if row else 0) + ways[row] + (ways[row + 1] if row + 1 < rows else 0) for r
ow in range(rows)], col)
    return ways[rows - 1]
```

COMPLEXITY | Time O(n * m) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 199

200 * Longest Same-Color Grid Sequence * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: keep four run lengths per cell (horizontal, vertical, two diagonals) from the previous row and the left cell; the answer is the largest

QUESTION INFO

QUESTION

Given a grid of colours (letters such as R, G, B), return the length of the longest run of one colour along a straight line: horizontal, vertical, or diagonal (in either diagonal direction).

Function signature

def longest_same_color_line(g: list) -> int

EXAMPLES

Example 1	
Input	`['RRRG', 'RRBB', 'GRRR', 'BGBR']`
Output	`4`
Why	`The main diagonal is R, R, R, R from corner to corner: 4. No other line is longer.`

Example 2	
Input	`['RGB', 'BGR', 'RGB']`
Output	`3`
Why	`The longest line is the vertical run of three G in the middle column.`

Example 3	
Input	`['RR', 'RR']`
Output	`2`
Why	`A single colour fills the grid: the longest line is a full row or column.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Extend a line from every cell in every direction

BRUTE-FORCE IDEA

Start at each cell and walk in each of the four directions while the colour matches. Correct; it re-walks the same cells many times.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Extend a line from every cell in every direction
Start at each cell
walk in each of the four directions while the colour matches
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(m * n * max(m, n)) | Space O(1)

WHY THIS IS WEAKER | Re-walks the same cells in every line, O(m * n * max(m, n)). Keeping the run length ending at each cell in each direction is O(m * n).

HOW TO IMPROVE IT

keep four run lengths per cell (horizontal, vertical, two diagonals) from the previous row and the left cell; the answer is the largest

SHORT BEST SOLUTION

Keep, for the previous row, the run length ending at each column in the vertical and both diagonal directions, and a running value for the horizontal run in the current row. A direction extends only when the predecessor has the same colour. The best value seen is the answer.

```
# BASE QUESTION (optimal): Four run lengths per cell
# Idea: Keep, for the previous row, the run length ending
# at each column in the vertical and both diagonal
# directions, and a running value for the horizontal run
```

```python
# in the current row. A direction extends only when the
# predecessor has the same colour. The best value seen is
# the answer.
def longest_same_color_line(colors):
    if not colors or not colors[0]:
        return 0
    width, best = len(colors[0]), 0
    up_run, down_left_run, down_right_run = [0] * width, [0] * width, [0] * width          # vertical, down-lef
t diagonal, down-right diagonal runs of the previous row
    for row_index, row in enumerate(colors):
        new_up_run, new_down_left_run, new_down_right_run = [1] * width, [1] * width, [1] * width
        left_run = 0
        for col_index, color in enumerate(row):
            left_run = left_run + 1 if col_index and row[col_index - 1] == color else 1
            if row_index:
                if colors[row_index - 1][col_index] == color:
                    new_up_run[col_index] = up_run[col_index] + 1
                if col_index and colors[row_index - 1][col_index - 1] == color:
                    new_down_right_run[col_index] = down_right_run[col_index - 1] + 1
                if col_index + 1 < width and colors[row_index - 1][col_index + 1] == color:
                    new_down_left_run[col_index] = down_left_run[col_index + 1] + 1
            best = max(best, left_run, new_up_run[col_index], new_down_left_run[col_index], new_down_right_run[
col_index])
        up_run, down_left_run, down_right_run = new_up_run, new_down_left_run, new_down_right_run
    return best
```

COMPLEXITY | Time O(m * n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 200

201 * Count Same-Color Squares in Matrix

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: the template as written: a square ends here if the three neighbours match; its size is 1 + min of their sizes

QUESTION INFO

QUESTION

LeetCode 221 variant — count maximal same-color square sub-grids. DP on each cell as the bottom-right corner.

Input: m × n grid of colored cells.
Output: count of square sub-grids whose cells are all the same color (or maximum side length, depending on prompt).
DP: dp[i][j] = 1 + min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]) if grid[i][j] == grid[i-1][j] == grid[i][j-1] == grid[i-1][j-1] else 1.
Sum dp[i][j] across the grid for the count variant; max for the side variant.

EXAMPLES

The plain "count all-1 squares" DP generalizes to same-color directly.

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Check every square directly

BRUTE-FORCE IDEA

For every corner and every side length look at all cells in the square. Correct; far too slow for a large grid.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Check every square directly
For every corner
```

```
every side length look at all cells in the square
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(m * n * min(m, n)^3) | Space O(1)

WHY THIS IS WEAKER | Re-checks the same cells again and again, O(m * n * min(m, n)^3). A dp on the bottom-right corner reuses the neighbours' results in O(m * n).

HOW TO IMPROVE IT

the template as written: a square ends here if the three neighbours match; its size is 1 + min of their sizes

SHORT BEST SOLUTION

dp[col_index] holds the best square side ending in the current row at column j. A cell extends the squares of its up, left and up-left neighbours only when all three have the same colour as the cell; otherwise its square has side 1. Add dp[col_index] to the count (use max instead of the sum for the 'largest side' version). A rolling row keeps the space at O(width).

```python
# BASE QUESTION (optimal): dp on the bottom-right corner
# Idea: dp[col_index] holds the best square side ending in the
# current row at column j. A cell extends the squares of
# its up, left and up-left neighbours only when all three
# have the same colour as the cell; otherwise its square
# has side 1. Add dp[col_index] to the count (use max instead of
# the sum for the 'largest side' version). A rolling row
# keeps the space at O(width).
def count_same_color_squares(colors):
    if not colors:
        return 0
    width, total = len(colors[0]), 0
    previous_row = [0] * width
    for row_index, row in enumerate(colors):
        current_row = [0] * width
        for col_index, color in enumerate(row):
            if row_index and col_index and colors[row_index - 1][col_index] == color and colors[row_index][col_
index - 1] == color and colors[row_index - 1][col_index - 1] == color:
                current_row[col_index] = 1 + min(previous_row[col_index], current_row[col_index - 1], previous_
row[col_index - 1])
            else:
                current_row[col_index] = 1
            total += current_row[col_index]
        previous_row = current_row
    return total
```

COMPLEXITY | Time O(m * n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 201

GROUP 27 OF 43 | 5 QUESTIONS | PART A - PART 1

Nested structures: a stack of contexts, or recursion with a cache

PATTERN CONTEXT

When something opens (a bracket, a parenthesis, a placeholder), save the current context on a stack and start a fresh one. When it closes, pop the saved context and fold the result back in. Recursive expansion does the same with a function call and a cache.

204 * Decode String * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the template as written

QUESTION INFO

QUESTION

An encoded string uses k[text] to mean that text is repeated k times; the brackets can be nested. Return the decoded string. (k is a positive integer, possibly with several digits; the input is always valid.)

Function signature

def decode(s: str) -> str

EXAMPLES

Example 1	
Input	'3[a]2[bc]'
Output	'aaabcbc'
Why	3 copies of a, then 2 copies of bc.

Example 2	
Input	'2[abc]3[cd]ef'
Output	'abcabccdcdcdef'
Why	Plain letters after the brackets are kept as they are.

Example 3	
Input	'3[a2[c]]'
Output	'accaccacc'
Why	Nested: the inner 2[c] first (cc), then 3 copies of acc.

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recursive descent over the string

BRUTE-FORCE IDEA

A function that reads letters until it meets a digit (then reads the number, the bracketed substring recursively, and repeats it) or a closing bracket. It mirrors the grammar; the recursion depth equals the nesting depth.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
read plain letters until a digit or closing bracket
after a digit, parse its bracketed substring recursively
repeat that decoded substring by the parsed count
```

`return when the matching closing bracket is reached`

BRUTE-FORCE COMPLEXITY | Time O(output length) | Space O(depth)

WHY THIS IS WEAKER | Time is proportional to the output, which is optimal; the cost is the recursion depth, equal to the nesting depth. A stack of (text so far, repeat count) avoids recursion.

HOW TO IMPROVE IT

the template as written

SHORT BEST SOLUTION

Part 1: On '[' push the text built so far and the repeat count, then start fresh. On ']' pop and append the current text repeated count times to the saved prefix. Digits accumulate into a multi-digit number (the usual bug is reading only one digit). No recursion is needed.

Part 2: Run-length encode the string into count[char] groups. This is the simple flat inverse; producing the shortest nested encoding (LeetCode 471) needs interval DP and is a hard follow-up.

```python
# BASE QUESTION (optimal): Stack of (prefix, repeat)
# frames
# Idea: On '[' push the text built so far and the repeat
# count, then start fresh. On ']' pop and append the
# current text repeated count times to the saved prefix.
# Digits accumulate into a multi-digit number (the usual
# bug is reading only one digit). No recursion is needed.
def decodeString(encoded):
    stack, current, pending_count = [], [], 0
    for char in encoded:
        if char.isdigit():
            pending_count = pending_count * 10 + int(char)
        elif char == "[":
            stack.append((current, pending_count))
            current, pending_count = [], 0
        elif char == "]":
            text_before, repeat_count = stack.pop()
            text_before.append("".join(current) * repeat_count)
            current = text_before
        else:
            current.append(char)
    return "".join(current)
```

COMPLEXITY | Part 1: Time O(output length) | Space O(output + nesting depth)
Part 2: Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 204

205 * Template Variable Expansion with Follow-Ups * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: resolve(name) with a cache and a 'being resolved' set; meeting a name that is being resolved is a cycle

QUESTION INFO

QUESTION

Replace **%KEY%** placeholders using a key-value map whose values may contain other placeholders. Clarify malformed delimiters and literal-percent escaping, flatten nested values, and detect cyclic dependencies before expansion.

QUESTION

Input: a key-value map and a string containing placeholders delimited by %, such as %USER%. Replace every placeholder with the corresponding value.
Map values can themselves contain placeholders, so resolve or flatten dependencies before expanding the final string. Define behavior for malformed delimiters such as unmatched percent signs, for a literal % in output, and for nested-looking forms before coding; the prompt intentionally leaves these contracts open.
Follow-up: detect cyclic dependencies among map values. Model keys as a dependency graph and use indegrees plus a queue to determine whether a complete topological resolution is possible.

EXAMPLES

```
Given {USER: "admin", HOME: "/%USER%/home"}:
```

Input: I am %USER% My home is %HOME% Output: I am admin My home is /admin/home

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recursive expansion without memo or cycle check, capped by depth

BRUTE-FORCE IDEA

Recursively substitute placeholders and stop with an error after a fixed depth. Correct for acyclic maps; repeats shared work and detects cycles only by running out of depth.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recursive expansion without memo or cycle check, capped by depth
Recursively substitute placeholders
stop with an error after a fixed depth
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential in the worst case | Space O(depth)

WHY THIS IS WEAKER | Re-expands shared values every time and finds cycles only by running out of depth. Memoising each value and keeping a visiting set expands each key once and reports a cycle at once.

HOW TO IMPROVE IT

resolve(name) with a cache and a 'being resolved' set; meeting a name that is being resolved is a cycle

SHORT BEST SOLUTION

expand(text) scans the text once: %% becomes %, %NAME% becomes the resolved value of NAME. resolve(name) memoises each value and keeps a visiting set; meeting a name that is already being resolved means a cycle and raises an error.

```python
# BASE QUESTION (optimal): Depth-first expansion with memo
# and a visiting set
# Idea: expand(text) scans the text once: %% becomes %,
# %NAME% becomes the resolved value of NAME. resolve(name)
# memoises each value and keeps a visiting set; meeting a
# name that is already being resolved means a cycle and
# raises an error.
def expand_template(text, mapping):
    done, visiting = {}, set()
    def resolve(name):
        if name in done:
            return done[name]
        if name in visiting:
            raise ValueError("cycle: " + name)
        if name not in mapping:
            raise KeyError(name)
        visiting.add(name)
        done[name] = scan(mapping[name])
        visiting.discard(name)
```

```
        return done[name]
    def scan(raw):
        pieces, position = [], 0
        while position < len(raw):
            if raw[position] != "%":
                pieces.append(raw[position]); position += 1
            elif raw[position:position + 2] == "%%":
                pieces.append("%"); position += 2
            else:
                closing_percent = raw.find("%", position + 1)
                if closing_percent < 0:
                    raise ValueError("unmatched %")
                pieces.append(resolve(raw[position + 1:closing_percent]))
                position = closing_percent + 1
        return "".join(pieces)
    return scan(text)
```

COMPLEXITY | Time O(total length of the expanded text + keys) | Space O(keys)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 205

206 * Expand Recursive Key-Value Placeholders in a Template

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: regex-find {key}, expand each key recursively with a cache and a visiting set to stop cycles

QUESTION INFO

QUESTION

Expand %KEY% placeholders in a template. A placeholder is replaced by its value from the mapping, and placeholders inside the values are expanded too, recursively. An unknown key is left exactly as written. A cycle (A needs B needs A) raises an error.

Function signature

def expand_template(template: str, mapping: dict) -> str

EXAMPLES

Example 1	
Input	`'%GREETING%, welcome!', {'NAME': 'Ada', 'GREETING': 'Hello %NAME%'}`
Output	`'Hello Ada, welcome!'`
Why	`%GREETING% expands to 'Hello %NAME%', and %NAME% is expanded in turn.`

Example 2	
Input	`'Hi %WHO%', {}`
Output	`'Hi %WHO%'`
Why	`An unknown key is kept as written.`

Example 3	
Input	`'%A%', {'A': '%B%', 'B': '%A%'}`
Output	`raises ValueError: cycle at A`
Why	`A needs B and B needs A: a cycle is reported instead of looping forever.`

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Keep substituting until the text stops changing

BRUTE-FORCE IDEA

Replace every %KEY% found in the text by its value and repeat until nothing changes. Simple, but it never stops on a cycle (so it needs an iteration cap) and re-expands the same key every time it appears.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Keep substituting until the text stops changing
Replace every %KEY% found in the text by its value
repeat until nothing changes
```

BRUTE-FORCE COMPLEXITY | Time O(depth * length) | Space O(length)

WHY THIS IS WEAKER | It re-expands the same key every time it appears and needs an arbitrary cap to stop on cycles. A memoised recursive expansion with a visiting set expands each key once and detects cycles exactly.

HOW TO IMPROVE IT

regex-find {key}, expand each key recursively with a cache and a visiting set to stop cycles

SHORT BEST SOLUTION

expand(key) substitutes every placeholder in the key's value by expand(inner), caches the result, and uses a visiting set to raise on a cycle. Unknown keys stay as written. The template itself is expanded the same way.

```python
import re


_PH = re.compile(r"%([A-Za-z_][A-Za-z0-9_]*)%")


# BASE QUESTION (optimal): Memoised recursive expansion
# with a visiting set
# Idea: expand(key) substitutes every placeholder in the
# key's value by expand(inner), caches the result, and
# uses a visiting set to raise on a cycle. Unknown keys
# stay as written. The template itself is expanded the
# same way.
def expand_template(template, mapping):
    cache, visiting = {}, set()

    def expand_text(text):
        return _PH.sub(lambda match: expand_key(match.group(1)) if match.group(1) in mapping else match.group(0
), text)

    def expand_key(key):
        if key in cache:
            return cache[key]
        if key in visiting:
            raise ValueError(f"cycle at {key}")
        visiting.add(key)
        cache[key] = expand_text(mapping[key])
        visiting.discard(key)
        return cache[key]

    return expand_text(template)
```

COMPLEXITY | Time O(total size of all values + template) | Space O(number of keys)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 206

207 * Nested List Weight Sum (LC 339) with Follow-Ups

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: depth counter: '[' adds 1, ']' subtracts 1; each number adds value * depth (read digits and a minus sign as you go)

QUESTION INFO

QUESTION

A nested list holds integers and further lists. The weight of an integer is its depth: 1 at the top level, 2 inside one inner list, 3 inside two, and so on. Return the sum of every integer times its depth. A second version reads the nested list from a string like '[1,[4,[6]]]'.

Function signature

def weight_sum(nested: list) -> int

def weight_sum_string(s: str) -> int

EXAMPLES

Example 1	
Input	`[3, 8, [2, 14], [2, [91]]]`
Output	`320`
Why	`3 + 8 at depth 1, 2 + 14 and 2 at depth 2, and 91 at depth 3: 11 + 32 + 4 + 273 = 320.`

Example 2	
Input	`'[1,[4,[6]]]'`
Output	`27`
Why	`The string version of [1, [4, [6]]]: 1*1 + 4*2 + 6*3 = 27.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Parse the string into lists, then recurse with the depth

BRUTE-FORCE IDEA

Json.loads turns the text into nested lists; a recursive function adds each integer times its depth. Clear, but it builds the whole structure and recurses as deep as the nesting.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Parse the string into lists, then recurse with the depth
Json.loads turns the text into nested lists
a recursive function adds each integer times its depth
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(depth)

WHY THIS IS WEAKER | It builds the whole nested structure and recurses as deep as the nesting, so memory and the call stack grow with depth. A single left-to-right scan with a bracket counter needs O(1) extra space.

HOW TO IMPROVE IT

depth counter: '[' adds 1, ']' subtracts 1; each number adds value * depth (read digits and a minus sign as you go)

SHORT BEST SOLUTION

Part 1: Every integer is weighted by the number of '[' currently open, so no tree needs to be built. Accumulate digits in a buffer and flush the buffer when a non-digit arrives or the string ends. Forgetting that last flush is the classic bug.

Part 2: Once the string is parsed into Python lists, process one depth level at a time with a queue. BFS never recurses, so a pathologically deep input cannot overflow the call stack.

```python
# BASE QUESTION (optimal): Single pass over the string
# with a depth counter
# Idea: Every integer is weighted by the number of '['
# currently open, so no tree needs to be built. Accumulate
# digits in a buffer and flush the buffer when a non-digit
# arrives or the string ends. Forgetting that last flush
# is the classic bug.
def nestedWeightSumFromString(text):
    depth = total = 0
    number, sign, has_digits = 0, 1, False
    for char in text + "]":                    # a sentinel flushes a trailing number
        if char.isdigit():
            number, has_digits = number * 10 + int(char), True
            continue
        if has_digits:
            total += sign * number * depth
            number, sign, has_digits = 0, 1, False
        if char == "[":
            depth += 1
        elif char == "]":
            depth -= 1
        elif char == "-":
            sign = -1
    return total
```

COMPLEXITY | Part 1: Time O(n) | Space O(1)
Part 2: Time O(n) | Space O(width)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 207

208 * Nested Function Expression Evaluator * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: tokenise, then a recursive parse(position): a number returns itself; a name reads its arguments until ')' and applies the function

QUESTION INFO

QUESTION

Evaluate an expression made of integers and nested function calls: add and mul (any number of arguments), sub and pow (two), div (integer division truncating toward zero), min and max (any number). Return the integer result.

Function signature

def evaluate(expr: str) -> int

EXAMPLES

Example 1	
Input	`'add(1, sub(1, 0))'`
Output	`2`
Why	`sub(1, 0) is 1, so add(1, 1) = 2.`

Example 2	
Input	`'add(2, mul(3, pow(4, 5)))'`
Output	`3074`
Why	`pow(4, 5) = 1024, times 3 = 3072, plus 2 = 3074.`

QUESTION

Example 3	
Input	`'max(3, div(-7, 2), min(5, 9))'`
Output	`5`
Why	`div(-7, 2) truncates toward zero to -3; max of 3, -3 and 5 is 5.`

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Repeatedly replace the innermost call with its value

BRUTE-FORCE IDEA

Find a call whose arguments contain no parentheses (a regular expression can do this), compute it, replace the text with the result, and repeat until only a number is left. It needs no parser, and it is quadratic because every replacement rescans and rebuilds the string.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Repeatedly replace the innermost call with its value
Find a call whose arguments contain no parentheses (a regular expression can do this)
compute it
replace the text with the result
repeat until only a number is left
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Every replacement rescans and rebuilds the string, O(n) per call and O(n^2) in total, and it cannot report errors well. A tokenizer plus recursive descent reads the text once, O(n).

HOW TO IMPROVE IT

tokenise, then a recursive parse(position): a number returns itself; a name reads its arguments until ')' and applies the function

SHORT BEST SOLUTION

Tokenize into identifiers, integers, commas and parentheses. parse(position) returns (value, next index): a number returns itself; an identifier must be followed by '(', then arguments are collected until ')' so any number of arguments works. Apply the function table to the collected values. Keeping the parser and the function table separate makes new functions a one-line change.

```python
import math
import re


# BASE QUESTION (optimal): Tokenizer + recursive descent
# over name(arg, arg, ...)
# Idea: Tokenize into identifiers, integers, commas and
# parentheses. parse(position) returns (value, next index): a
# number returns itself; an identifier must be followed by
# '(', then arguments are collected until ')' so any
# number of arguments works. Apply the function table to
# the collected values. Keeping the parser and the
# function table separate makes new functions a one-line
# change.
def evaluate(expr):
    tokens = re.findall(r"-?\d+|[A-Za-z_]\w*|[(),]", expr)
    ops = {
        "add": lambda *a: sum(a),
        "sub": lambda a, b: a - b,
        "mul": lambda *a: math.prod(a),
        "pow": lambda a, b: a ** b,
        "div": lambda a, b: int(a / b),            # truncates toward zero; ask the interviewer about negatives
        "min": min,
        "max": max,
```

```
    }

    def parse(position):
        token = tokens[position]
        if re.fullmatch(r"-?\d+", token):
            return int(token), position + 1
        if token not in ops:
            raise ValueError(f"unknown function {token!r}")
        if tokens[position + 1] != "(":
            raise ValueError("expected '('")
        position += 2
        args = []
        while tokens[position] != ")":
            value, position = parse(position)
            args.append(value)
            if tokens[position] == ",":
                position += 1
        return ops[token](*args), position + 1

    value, end = parse(0)
    if end != len(tokens):
        raise ValueError("trailing tokens")
    return value
```

COMPLEXITY | Time O(n) | Space O(depth)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 208

GROUP 28 OF 43 | 5 QUESTIONS | PART A - PART 1

Frontier search (BFS or DFS), Tier 2: free steps, early exit, neighbours worked out on the fly

PATTERN CONTEXT

The same frontier loop as Tier 1, with three small upgrades: a step can cost 0 or 1 (free steps go to the front of the queue), you can return the moment you reach the goal, and the neighbours of a node can be computed on the fly instead of stored.

209 * Shortest Path Nodes and Escaping a Pursuer * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: run it from a, from t and from b; a node is on a shortest alice-to-target path when da + dt == da[target]; compare with the pursuer's distance db

QUESTION INFO

QUESTION

In an unweighted, undirected graph (an adjacency dict) Alice is at node a and wants to reach the target t. Return the length (number of edges) of the shortest path from a to t found with BFS. The follow-ups ask for every node that lies on some shortest path, and whether Alice can reach t before a pursuer Bob.

Function signature

def shortest_length(adj: dict, a, t) -> int

EXAMPLES

Example 1	
Input	`{0: [1, 2], 1: [0, 3], 2: [0, 3], 3: [1, 2]}, 0, 3`
Output	`2`
Why	`0 -> 1 -> 3 and 0 -> 2 -> 3 both take 2 edges.`

Example 2	
Input	`{0: []}, 0, 0`
Output	`0`
Why	`A start that is the target needs 0 edges.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Enumerate every shortest path

BRUTE-FORCE IDEA

Depth-first search all simple paths of the shortest length from Alice to the target and check each node against Bob's arrival time. Correct; the number of shortest paths can be huge.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Enumerate every shortest path
Depth-first search all simple paths of the shortest length from Alice to the target
check each node against Bob's arrival time
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential in the worst case | Space O(paths)

WHY THIS IS WEAKER | The number of shortest paths can be exponential. Three BFS runs (from Alice, the target and Bob) decide every node in O(V + E).

HOW TO IMPROVE IT

run it from a, from t and from b; a node is on a shortest alice-to-target path when da + dt == da[target]; compare with the pursuer's distance db

SHORT BEST SOLUTION

dA, dT and dB come from BFS. The nodes on some shortest path are those with dA[neighbour] + dT[neighbour] == dA[target]. For the escape question walk only those nodes and only while dB[neighbour] > dA[neighbour], layer by layer from Alice; if the target is reached Alice has a safe shortest path.

```
from collections import deque


# BASE QUESTION (optimal): Three BFS runs and a safe
# shortest-path DAG
# Idea: dA, dT and dB come from BFS. The nodes on some
# shortest path are those with dA[neighbour] + dT[neighbour] == dA[target]. For
# the escape question walk only those nodes and only while
# dB[neighbour] > dA[neighbour], layer by layer from Alice; if the target
# is reached Alice has a safe shortest path.
def shortest_path_info(neighbours, alice, target, bob):
    def bfs(start):
        distance = {start: 0}
        queue = deque([start])
        while queue:
            node = queue.popleft()
            for neighbour in neighbours[node]:
                if neighbour not in distance:
                    distance[neighbour] = distance[node] + 1
                    queue.append(neighbour)
        return distance
    da, dt, db = bfs(alice), bfs(target), bfs(bob)
    if target not in da:
        return None, set(), False
    length = da[target]
    nodes = {node for node in da if node in dt and da[node] + dt[node] == length}
    safe_nodes, frontier = set(), [alice] if db.get(alice, 10 ** 9) > 0 else []
    seen = set(frontier)
    while frontier:
        next_frontier = []
        for node in frontier:
            for neighbour in neighbours[node]:
                if neighbour in nodes and da[neighbour] == da[node] + 1 and neighbour not in seen and db.get(ne
ighbour, 10 ** 9) > da[neighbour]:
                    seen.add(neighbour); next_frontier.append(neighbour)
        frontier = next_frontier
    return length, nodes, target in seen
```

COMPLEXITY | Time O(V + E) | Space O(V)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 209

210 * Radio Broadcast Reachability

* MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: neighbors(current) = every radio within the radius, worked out with squared distances; goal = the target radio, answer = whether it was reached

QUESTION INFO

QUESTION

You are given a list of radios placed on a 2D plane, each described by its (x, y) coordinates. Every radio can broadcast to any other radio within a fixed Euclidean distance radius (inclusive). A message can travel from one radio to another either directly (within radius) or through a chain of intermediate radios that each relay it to a next radio within radius . Given the radios, the broadcast radius , a source index, and a target index, determine whether a message originating at source can reach target through zero or more hops. Return True or False .

Function signature

```
from typing import List, Tuple
def can_reach(radios: List[Tuple[float, float]],
              radius: float,
              source: int,
              target: int) -> bool:
    ...
```

Follow-up (as reported): Model the radios as nodes and connect any two within radius , then run BFS (or DFS / union-find) from source . To avoid floating-point issues, compare squared distances against radius * radius .

EXAMPLES

Example 1	
Input	`radios = [(0, 0), (1, 0), (2, 0), (5, 0)], radius = 1.5, source = 0, target = 2`
Output	`True`
Why	`0->1 (dist 1) and 1->2 (dist 1) are each within 1.5, so 2 is reachable.`

Example 2	
Input	`radios = [(0, 0), (1, 0), (2, 0), (5, 0)], radius = 1.5, source = 0, target = 3`
Output	`False`
Why	`Radio 3 is 3 units from radio 2 (> 1.5) and farther from the rest; no chain of hops reaches it.`

Example 3	
Input	`radios = [(0, 0), (3, 4)], radius = 5.0, source = 0, target = 1`
Output	`True`
Why	`The Euclidean distance is exactly 5.0, which is within radius.`

CONSTRAINTS

- 1 <= len(radios) <= 2000
- Coordinates are integers or floats with -10^4 <= x, y <= 10^4 . radius >= 0 .
- 0 <= source, target < len(radios) .
- Two radios a and b are directly connected when sqrt((ax-bx)^2 + (ay-by)^2) <= radius .
- If source == target , the answer is True .

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I

target the expected complexity.

Known bounds to confirm:
- 1 <= len(radios) <= 2000
- Coordinates are integers or floats with -10^4 <= x, y <= 10^4 . radius >= 0 .
- 0 <= source, target < len(radios) .
- Two radios a and b are directly connected when sqrt((ax-bx)^2 + (ay-by)^2) <= radius .

BRUTE FORCE | Grow the reached set until it stops changing

BRUTE-FORCE IDEA

Start with: Grow the reached set until it stops changing. It re-scans all pairs after every round although a queue visits each radio once.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Grow the reached set until it stops changing
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^3) | Space O(n)

WHY THIS IS WEAKER | It re-scans all pairs after every round although a queue visits each radio once.

HOW TO IMPROVE IT

neighbors(current) = every radio within the radius, worked out with squared distances; goal = the target radio, answer = whether it was reached

SHORT BEST SOLUTION

Compare squared distances; every radio enters the queue once.

```python
# BASE QUESTION (optimal): BFS over the implicit graph
# Idea: Compare squared distances; every radio enters the
# queue once.
from collections import deque


def can_reach(radios, radius, source, target):
    limit = radius * radius
    seen = [False] * len(radios)
    seen[source] = True
    queue = deque([source])
    while queue:
        current = queue.popleft()
        if current == target:
            return True
        for other, (other_x, other_y) in enumerate(radios):
            x_gap, y_gap = radios[current][0] - other_x, radios[current][1] - other_y
            if not seen[other] and x_gap * x_gap + y_gap * y_gap <= limit:
                seen[other] = True
                queue.append(other)
    return False
```

COMPLEXITY | Time O(n^2) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 210

211 * Minimum Direction Violations in a Directed Graph * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: an edge in its own direction costs cost = 0, the same edge walked backwards costs cost = 1; free steps use appendleft

QUESTION INFO

QUESTION

A directed graph has n nodes. Walking along an edge in its own direction is free; walking against it costs 1 (a direction violation). Return the minimum total cost of a walk from start to end, or -1 if end cannot be reached at all.

Function signature

def min_violations(n: int, edges: list, start: int, end: int) -> int

EXAMPLES

Example 1	
Input	`5, [[0, 1], [2, 1], [2, 3], [4, 3]], 0, 4`
Output	`2`
Why	`Path 0 -> 1 (free), 1 -> 2 against edge 2 -> 1 (cost 1), 2 -> 3 (free), 3 -> 4 against edge 4 -> 3 (cost 1): total 2.`

Example 2	
Input	`3, [[0, 1]], 0, 2`
Output	`-1`
Why	`Node 2 has no edge at all, so it cannot be reached.`

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Bellman-Ford on the doubled graph

BRUTE-FORCE IDEA

Give every edge a forward copy with cost 0 and a backward copy with cost 1, and relax all edges V - 1 times. Correct and simple for the 0-1 BFS; it is slower because it repeats full passes over the edges.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Bellman-Ford on the doubled graph
Give every edge a forward copy with cost 0
a backward copy with cost 1
relax all edges V - 1 times
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(V * E) | Space O(V)

WHY THIS IS WEAKER | Every round scans all edges, V - 1 rounds: O(V * E). Costs of only 0 and 1 allow a deque search (0-1 BFS) in O(V + E).

HOW TO IMPROVE IT

an edge in its own direction costs cost = 0, the same edge walked backwards costs cost = 1; free steps use appendleft

SHORT BEST SOLUTION

Add every edge twice: forward with weight 0, backward with weight 1. Shortest path with weights in {0, 1} is a deque BFS: push a zero-weight neighbour to the front and a one-weight neighbour to the back. Plain BFS minimizes edge count, not violations, which is the common mistake.

```
from collections import deque
```

```python
# BASE QUESTION (optimal): 0-1 BFS on the doubled graph
# Idea: Add every edge twice: forward with weight 0,
# backward with weight 1. Shortest path with weights in
# {0, 1} is a deque BFS: push a zero-weight neighbour to
# the front and a one-weight neighbour to the back. Plain
# BFS minimizes edge count, not violations, which is the
# common mistake.
def minDirectionViolations(node_count, edges, start, end):
    graph = [[] for _ in range(node_count)]
    for node, neighbour in edges:
        graph[node].append((neighbour, 0))
        graph[neighbour].append((node, 1))
    violations = [float("inf")] * node_count
    violations[start] = 0
    queue = deque([start])
    while queue:
        node = queue.popleft()
        for neighbour, cost in graph[node]:
            if violations[node] + cost < violations[neighbour]:
                violations[neighbour] = violations[node] + cost
                queue.appendleft(neighbour) if cost == 0 else queue.append(neighbour)
    return -1 if violations[end] == float("inf") else violations[end]
```

COMPLEXITY | Time O(n + m) | Space O(n + m)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 211

212 * Compute shortest paths with blocked nodes

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: neighbors skip every blocked node; if the source itself is blocked the answer is all -1; unreachable nodes stay -1

QUESTION INFO

QUESTION

Given a graph with nodes and edges and a designated source node s, compute the shortest distance from s to every other node. Some nodes are inaccessible (blocked) and cannot be traversed or reached; treat them as removed from the graph. Return a distance for each node where unreachable nodes (including blocked ones) are -1. Describe your algorithm, data structures, and time/space complexity, and note any differences for directed vs. undirected graphs.

Given n nodes labeled 0..n-1, edges, a source, and blocked nodes, return shortest unweighted distances from the source. Blocked and unreachable nodes must be -1. Set directed=True to treat edges as directed.
Constraints
Nodes are labeled 0 through n-1Edges are unweighted

EXAMPLES

```
Input: (5, [[0, 1], [1, 2], [0, 3], [3, 4]], 0, [3], False)
Expected Output: [0, 1, 2, -1, -1]
Input: (4, [[0, 1], [1, 2], [2, 3]], 0, [], True)
Expected Output: [0, 1, 2, 3]
```

Hints

Treat blocked nodes as removed before BFS.For directed graphs, only add the given edge direction.

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Floyd-Warshall on the graph without the blocked nodes

BRUTE-FORCE IDEA

Compute all-pairs distances ignoring blocked nodes and read the row of the source. Correct; cubic in the number of nodes.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Floyd-Warshall on the graph without the blocked nodes
Compute all-pairs distances ignoring blocked nodes
read the row of the source
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(V^3) | Space O(V^2)

WHY THIS IS WEAKER | All-pairs distances cost O(V^3) and O(V^2) memory when one source is needed. A BFS from the source that never enters a blocked node is O(V + E).

HOW TO IMPROVE IT

neighbors skip every blocked node; if the source itself is blocked the answer is all -1; unreachable nodes stay -1

SHORT BEST SOLUTION

Build adjacency lists (one direction only if directed). If the source is blocked return all -1. Run BFS from the source, skipping blocked neighbours; the distance array starts at -1 and is set when a node is first reached.

```python
from collections import defaultdict
from collections import deque


# BASE QUESTION (optimal): BFS that never enters a blocked
# node
# Idea: Build adjacency lists (one direction only if
# directed). If the source is blocked return all -1. Run
# BFS from the source, skipping blocked neighbours; the
# distance array starts at -1 and is set when a node is
# first reached.
def blocked_distances(node_count, edges, source, blocked, directed=False):
    blocked_set = set(blocked)
    if source in blocked_set:
        return [-1] * node_count
    neighbours = defaultdict(list)
    for node, neighbour in edges:
        neighbours[node].append(neighbour)
        if not directed:
            neighbours[neighbour].append(node)
    distance = [-1] * node_count
    distance[source] = 0
    queue = deque([source])
    while queue:
        node = queue.popleft()
        for neighbour in neighbours[node]:
            if neighbour not in blocked_set and distance[neighbour] == -1:
                distance[neighbour] = distance[node] + 1
                queue.append(neighbour)
    return distance
```

COMPLEXITY | Time O(V + E) | Space O(V + E)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 212

213 * Shortest Directed Cycle Through a Node

* MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: start = start_node; the goal is an edge that leads back to v, so return distance_from_start[node] + 1 the moment a neighbour equals v

QUESTION INFO

QUESTION

Given a directed graph with n nodes and a list of edges (a, b), return the length (number of edges) of the shortest cycle that passes through node v, or -1 if v is on no cycle.

Function signature

def shortest_cycle_through(n: int, edges: list, v: int) -> int

EXAMPLES

Example 1	
Input	`3, [(0, 1), (1, 2), (2, 0)], 0`
Output	`3`
Why	`0 -> 1 -> 2 -> 0 is a cycle of 3 edges.`

Example 2	
Input	`3, [(0, 1), (1, 2)], 0`
Output	`-1`
Why	`No edge leads back to 0.`

Example 3	
Input	`3, [(0, 1), (1, 0), (1, 2), (2, 0)], 0`
Output	`2`
Why	`A shorter cycle 0 -> 1 -> 0 exists next to the longer one.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Enumerate every simple cycle through v with DFS

BRUTE-FORCE IDEA

Explore all simple paths from v and record the length whenever an edge returns to v. Exact, but the number of simple paths can be exponential.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Enumerate every simple cycle through v with DFS
Explore all simple paths from v
record the length whenever an edge returns to v
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(V)

WHY THIS IS WEAKER | The number of simple paths can be exponential in the graph size. A breadth-first search from v reaches the shortest cycle in O(V + E) because it visits nodes in order of distance.

HOW TO IMPROVE IT

start = start_node; the goal is an edge that leads back to v, so return distance_from_start[node] + 1 the moment a neighbour equals v

SHORT BEST SOLUTION

A directed cycle through v is a path v to u followed by an edge u to v. BFS visits nodes in nondecreasing distance, so the first time we see an edge back to v its length is distance_from_start[node] + 1 and is the shortest. A self-loop gives 1. Return -1 if BFS ends without closing.

```
from collections import deque


# BASE QUESTION (optimal): BFS from v, close the cycle on
# the first edge back to v
# Idea: A directed cycle through v is a path v to u
# followed by an edge u to v. BFS visits nodes in
# nondecreasing distance, so the first time we see an edge
# back to v its length is distance_from_start[node] + 1 and is the shortest.
# A self-loop gives 1. Return -1 if BFS ends without
# closing.
def shortest_cycle_through(node_count, edges, start_node):
    neighbours = [[] for _ in range(node_count)]
    for from_node, to_node in edges:
        neighbours[from_node].append(to_node)
    distance_from_start = {start_node: 0}
    queue = deque([start_node])
    while queue:
        node = queue.popleft()
        for next_node in neighbours[node]:
            if next_node == start_node:
                return distance_from_start[node] + 1
            if next_node not in distance_from_start:
                distance_from_start[next_node] = distance_from_start[node] + 1
                queue.append(next_node)
    return -1
```

COMPLEXITY | Time O(V + E) | Space O(V)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 213

Topological sort (Kahn) and cycle detection

PATTERN CONTEXT

Count the incoming edges of every node, start from the nodes with none, and peel them off one by one. If every node got peeled there is a valid order; if some never did, there is a cycle. The questions only differ in what the nodes are called and whether you return the order or a yes/no.

214 * Course Scheduling and Dependency Variants * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: build the vertex set from the edges too; return the order or None

QUESTION INFO

QUESTION

Topological-ordering family. Variants: course ordering with validity checks and a nested-list extension; merging two arrays whose internal order defines precedence (return false on a conflict cycle); and a damaged-node dependency-path variant that minimizes broken nodes on a route.

Course-ordering variant

Work through the course-ordering problem, including validity checks, complexity analysis, and test cases.
Extend the result to a nested-list format. The grouping rule is unspecified; clarify what each inner list represents before implementing the extension.

Damaged-node and path variant

Remove damaged nodes and their downstream effects, then find a valid path.
A follow-up minimizes broken nodes on a required route. Clarify the graph direction, path endpoints, and meaning of broken nodes before choosing an algorithm.

Merge-two-ordered-sequences variant

Input: two integer arrays a and b. The order of elements inside each array is an ordering constraint — a = [1, 3, 5] means 1 must come before 3, and 3 before 5.
Determine whether the two arrays can be merged into one sequence that preserves the relative order of both. Return false if the constraints conflict; otherwise return true (or any valid merged sequence).

EXAMPLES

Merge-two-ordered-sequences

```
a = [1, 2, 3]
b = [2, 4, 5]
-> true   (one valid merged order: [1, 2, 3, 4, 5])

a = [1, 6, 4]
b = [4, 1]
-> false  (a requires 1 -> 6 -> 4, b requires 4 -> 1: contradiction)
```

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every permutation of the vertices

BRUTE-FORCE IDEA

Generate permutations and check every constraint. Correct; factorial time is unusable beyond about ten vertices.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every permutation of the vertices
Generate permutations
check every constraint
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(V! * E) | Space O(V)

WHY THIS IS WEAKER | V! Orderings. Kahn's algorithm repeatedly takes a vertex with no unmet prerequisites and finishes in O(V + E).

HOW TO IMPROVE IT

build the vertex set from the edges too; return the order or None

SHORT BEST SOLUTION

Build the graph (prerequisite -> dependent, including isolated vertices), count indegrees and keep a queue of vertices with indegree 0. Pop one, append it to the order and lower the indegree of its dependents. If fewer than V vertices were emitted there is a cycle. For the merge question, build the edges from adjacent pairs of both arrays and answer True exactly when a full order exists (a repeated value inside one array is a self-loop, so it fails).

```python
from collections import defaultdict
from collections import deque


# BASE QUESTION (optimal): Kahn's algorithm (course order,
# or can two sequences merge)
# Idea: Build the graph (prerequisite -> dependent,
# including isolated vertices), count indegrees and keep a
# queue of vertices with indegree 0. Pop one, append it to
# the order and lower the indegree of its dependents. If
# fewer than V vertices were emitted there is a cycle. For
# the merge question, build the edges from adjacent pairs
# of both arrays and answer True exactly when a full order
# exists (a repeated value inside one array is a self-
# loop, so it fails).
def topological_order(vertices, edges):
    dependents, indegree = defaultdict(list), {vertex: 0 for vertex in vertices}
    for prerequisite, dependent in edges:
        dependents[prerequisite].append(dependent)
        indegree[dependent] = indegree.get(dependent, 0) + 1
        indegree.setdefault(prerequisite, indegree.get(prerequisite, 0))
    queue = deque(vertex for vertex, incoming in indegree.items() if incoming == 0)
    order = []
    while queue:
        prerequisite = queue.popleft()
        order.append(prerequisite)
        for dependent in dependents[prerequisite]:
            indegree[dependent] -= 1
            if indegree[dependent] == 0:
                queue.append(dependent)
    return order if len(order) == len(indegree) else None
```

COMPLEXITY | Time O(V + E) | Space O(V + E)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 214

215 * Validate Race Results (Cycle Detection) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: edge = winner -> loser; the results are consistent exactly when there is no cycle

QUESTION INFO

QUESTION

You are given a list of pairwise race results as (winner, loser) string pairs, meaning winner beat loser . The results are consistent if they can all be true simultaneously under a strict ranking — equivalently, if the directed graph (edge winner -> loser) contains no cycle. Return True if the results are valid (acyclic) and False otherwise.

Function signature

def results_are_valid(results: list[tuple[str, str]]) -> bool:

...

Follow-up (as reported): Recursive DFS can hit Python's recursion limit on long chains; convert to an iterative DFS with an explicit stack, or use Kahn's topological-sort algorithm.

EXAMPLES

Example 1	
Input	`results = [("A","B"), ("B","C")]`
Output	`True`
Why	`A beat B beat C — a consistent ranking A > B > C, no cycle.`

Example 2	
Input	`results = [("A","B"), ("B","C"), ("C","A")]`
Output	`False`
Why	`A > B > C > A is a contradiction (a cycle).`

Example 3	
Input	`results = []`
Output	`True`
Why	`No results impose no contradiction.`

CONSTRAINTS

- 0 <= len(results) <= 10^5 .
- Player names are non-empty strings; duplicate result pairs may appear.
- An empty result list is valid (True).

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(results) <= 10^5 .
- Player names are non-empty strings; duplicate result pairs may appear.
- An empty result list is valid (True).

BRUTE FORCE | Recursive DFS with colours

BRUTE-FORCE IDEA

Start with: Recursive DFS with colours. Correct, but a chain of 100,000 results exceeds Python's recursion limit.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recursive DFS with colours
```

```
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(V + E) | Space O(V) recursion

WHY THIS IS WEAKER | Correct, but a chain of 100,000 results exceeds Python's recursion limit.

HOW TO IMPROVE IT

edge = winner -> loser; the results are consistent exactly when there is no cycle

SHORT BEST SOLUTION

Peel off nodes with no incoming edge; everything peels off exactly when there is no cycle.

```python
# BASE QUESTION (optimal): Kahn's algorithm
# Idea: Peel off nodes with no incoming edge; everything
# peels off exactly when there is no cycle.
from collections import deque


def results_are_valid(results):
    graph, indegree = {}, {}
    for winner, loser in set(results):
        graph.setdefault(winner, []).append(loser)
        graph.setdefault(loser, [])
        indegree[loser] = indegree.get(loser, 0) + 1
        indegree.setdefault(winner, 0)
    queue = deque(racer for racer in graph if indegree[racer] == 0)
    removed = 0
    while queue:
        racer = queue.popleft()
        removed += 1
        for slower in graph[racer]:
            indegree[slower] -= 1
            if indegree[slower] == 0:
                queue.append(slower)
    return removed == len(graph)
```

COMPLEXITY | Time O(V + E) | Space O(V + E)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 215

216 * Solve tree, graph, sliding-window problems * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the course-schedule part is can-finish: len(order) == course_count; the zigzag and window parts are separate small loops

QUESTION INFO

QUESTION

Question

LeetCode 103. Binary Tree Zigzag Level Order Traversal – given a binary tree root, return its zigzag level-order traversal. LeetCode 207. Course Schedule – given course prerequisites, determine if all courses can be finished by detecting cycles with topological sort. LeetCode 340. Longest Substring with At Most K Distinct Characters – given string s and integer k, return the length of the longest substring containing at most k distinct characters.
https://leetcode.com/problems/binary-tree-zigzag-level-order-traversal/description/
https://leetcode.com/problems/course-schedule/description/
https://leetcode.com/problems/longest-substring-with-at-most-k-distinct-characters/description/

QUESTION

Part 1: Binary Tree Zigzag Level Order TraversalYou are given a binary tree in level-order array form, where each element is either an integer or None for a missing child. Return the zigzag level-order traversal of the tree's node values. In a zigzag traversal, the first level is read from left to right, the second from right to left, the third from left to right, and so on.

Constraints

0 <= len(root) <= 2000Each non-None node value is an integer in the range [-10^4, 10^4]The input list represents a valid binary tree in level-order form

EXAMPLES

```
Input: ([3, 9, 20, None, None, 15, 7],)
Expected Output: [[3], [20, 9], [15, 7]]
```

Explanation: Level 1 is read left-to-right, level 2 right-to-left, and level 3 left-to-right.

```
Input: ([1],)
Expected Output: [[1]]
```

Explanation: A single-node tree has only one level.

Hints

A breadth-first search naturally processes the tree one level at a time.You can collect each level left-to-right, then reverse every other level before appending it to the answer.

Part 2: Course ScheduleThere are num_courses courses labeled from 0 to num_courses - 1. You are given a list of prerequisite pairs [a, b], meaning you must take course b before course a. Determine whether it is possible to finish all courses. This is equivalent to checking whether the prerequisite graph contains a cycle.

Constraints

1 <= num_courses <= 10^50 <= len(prerequisites) <= 2 * 10^5Each prerequisite pair has length 20 <= course, prerequisite < num_courses

Examples

Input: (2, [[1, 0]])

Expected Output: True

Explanation: Course 0 can be taken first, then course 1.

Input: (2, [[1, 0], [0, 1]])

Expected Output: False

Explanation: The two courses depend on each other, forming a cycle.

Hints

Think of courses as nodes in a directed graph and prerequisites as directed edges.If you repeatedly take courses with indegree 0 and cannot process all nodes, a cycle exists.

Part 3: Longest Substring with At Most K Distinct CharactersGiven a string s and an integer k, return the length of the longest substring that contains at most k distinct characters.

Constraints

0 <= len(s) <= 2 * 10^50 <= k <= len(s)s may contain any standard characters

QUESTION

Examples
Input: ("eceba", 2)
Expected Output: 3
Explanation: The longest valid substring is 'ece', which has length 3.
Input: ("aa", 1)
Expected Output: 2
Explanation: The whole string contains only one distinct character.
Hints
A sliding window can maintain a valid substring while expanding to the right.Keep character counts in the current window, and shrink from the left whenever the number of distinct characters exceeds k.

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Course schedule: DFS from every course looking for a cycle

BRUTE-FORCE IDEA

From each course follow prerequisites; if the walk returns to the start there is a cycle. Correct; the same parts of the graph are walked again and again.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Course schedule: DFS from every course looking for a cycle
From each course follow prerequisites
if the walk returns to the start there is a cycle
```

BRUTE-FORCE COMPLEXITY | Time O(V * (V + E)) | Space O(V)

WHY THIS IS WEAKER | Each search can walk the whole graph again, O(V * (V + E)). Kahn's algorithm (remove courses with no unmet prerequisites) is O(V + E).

BRUTE FORCE 2 | Longest window: check every substring

BRUTE-FORCE IDEA

For each start extend while the number of distinct characters stays within K and record the longest. This takes quadratic time.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Longest window: check every substring
For each start extend while the number of distinct characters stays within K
record the longest
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(k)

WHY THIS IS WEAKER | O(n^2) in the worst case. A sliding window with a count map that shrinks from the left when it exceeds K distinct characters is O(n).

BRUTE FORCE 3 | Zigzag: level order, then reverse the odd levels

BRUTE-FORCE IDEA

Collect each level with a queue and reverse every second level at the end. Simple; each reversed level is copied once more.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Zigzag: level order, then reverse the odd levels
Collect each level with a queue
reverse every second level at the end
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | Same O(n) time with one extra copy of every second level; filling each level from alternating ends of a deque avoids it.

HOW TO IMPROVE IT

the course-schedule part is can-finish: len(order) == course_count; the zigzag and window parts are separate small loops

SHORT BEST SOLUTION

Part 1: Count prerequisites per course and start from courses with none. Remove a course, decrease the count of the courses that depend on it, and queue those that reach zero. All courses can be taken exactly when every course is removed; a leftover means a cycle.

Part 2: Add text[right] to the counts. While more than K distinct characters are present, remove text[left] (delete the key when its count hits 0) and advance left. The window is always valid after the shrink, so record its length. Each index enters and leaves once.

Part 3: Keep a direction flag; for each level build a deque, appending on the right when going left to right and on the left otherwise. Flip the flag after each level; no reversal is needed.

```
from collections import defaultdict
from collections import deque


# Course schedule: Kahn's topological sort
# Idea: Count prerequisites per course and start from
# courses with none. Remove a course, decrease the count
# of the courses that depend on it, and queue those that
# reach zero. All courses can be taken exactly when every
# course is removed; a leftover means a cycle.
def canFinish(course_count, prereqs):
    dependents, indegree = defaultdict(list), [0] * course_count
    for course, prerequisite in prereqs:
        dependents[prerequisite].append(course)
        indegree[course] += 1
    queue = deque(course for course in range(course_count) if indegree[course] == 0)
    taken_count = 0
    while queue:
        ready_course = queue.popleft()
        taken_count += 1
        for nxt in dependents[ready_course]:
            indegree[nxt] -= 1
            if indegree[nxt] == 0:
                queue.append(nxt)
    return taken_count == course_count


# Longest window: sliding window with a count map
# Idea: Add text[right] to the counts. While more than K
# distinct characters are present, remove text[left] (delete
# the key when its count hits 0) and advance left. The
# window is always valid after the shrink, so record its
# length. Each index enters and leaves once.
```

```
def lengthOfLongestSubstringKDistinct(text, max_distinct):
    count, left, best = {}, 0, 0
    for right, char in enumerate(text):
        count[char] = count.get(char, 0) + 1
        while len(count) > max_distinct:
            count[text[left]] -= 1
            if count[text[left]] == 0:
                del count[text[left]]
            left += 1
        best = max(best, right - left + 1)
    return best


# Zigzag: fill each level from the correct end
# Idea: Keep a direction flag; for each level build a
# deque, appending on the right when going left to right
# and on the left otherwise. Flip the flag after each
# level; no reversal is needed.
```

```python
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def zigzagLevelOrder(root):
    levels, queue, left_to_right = [], deque([root] if root else []), True
    while queue:
        level = deque()
        for _ in range(len(queue)):
            node = queue.popleft()
            level.append(node.val) if left_to_right else level.appendleft(node.val)
            queue.extend(child for child in (node.left, node.right) if child)
        levels.append(list(level))
        left_to_right = not left_to_right
    return levels
```

COMPLEXITY | Part 1: Time O(V + E) | Space O(V + E)
Part 2: Time O(n) | Space O(k)
Part 3: Time O(n) | Space O(width)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 216

217 * Validate course catalog dependencies * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: first reject prerequisites naming courses that do not exist; then the same peel; report the missing names or the cycle

QUESTION INFO

QUESTION

You are given a set of course ids and prerequisite pairs (u, v), meaning course v requires course u. The catalog is valid when no prerequisite mentions a course that does not exist and there is no cycle. Return (True, [], order) with a possible study order when valid; (False, missing, None) listing the unknown course ids; or (False, [], cycle) with one cycle that was found.

Function signature

def validate_catalog(courses: set, prereqs: list) -> tuple

EXAMPLES

Example 1	
Input	{1, 2, 3, 4}, [(1, 2), (2, 3), (1, 3)]
Output	(true, [], [1, 4, 2, 3])
Why	Course 1 comes first, 4 is unrelated, then 2 and 3.

Example 2	
Input	{1, 2}, [(1, 9)]
Output	(false, [9], null)
Why	Course 9 is not in the catalog.

Example 3	
Input	{1, 2}, [(1, 2), (2, 1)]
Output	(false, [], [1, 2, 1])
Why	1 requires 2 and 2 requires 1: a cycle.

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | DFS from every course looking for a cycle

BRUTE-FORCE IDEA

First collect references to unknown courses; then, for each course, search for a path back to itself. Correct; the same parts of the graph are searched repeatedly.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: DFS from every course looking for a cycle
First collect references to unknown courses
for each course
search for a path back to itself
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(V * (V + E)) | Space O(V)

WHY THIS IS WEAKER | Every course can trigger a walk of the whole graph, O(V * (V + E)). Kahn's algorithm is O(V + E) and its leftovers also reveal a cycle to report.

HOW TO IMPROVE IT

first reject prerequisites naming courses that do not exist; then the same peel; report the missing names or the cycle

SHORT BEST SOLUTION

Report unknown courses (any referenced id not in the set). If none, run Kahn's algorithm: repeatedly remove courses with no remaining prerequisites, building an order. If every course is removed return (True, order). Otherwise the courses left include a cycle: every leftover course still has an unmet prerequisite inside the leftover set, so walk backwards along prerequisites until a course repeats and return that cycle in forward order.

```python
from collections import defaultdict
from collections import deque


# BASE QUESTION (optimal): Missing references, then Kahn's
# algorithm, then a cycle from the leftovers
# Idea: Report unknown courses (any referenced id not in
# the set). If none, run Kahn's algorithm: repeatedly
# remove courses with no remaining prerequisites, building
# an order. If every course is removed return (True,
# order). Otherwise the courses left include a cycle:
# every leftover course still has an unmet prerequisite
# inside the leftover set, so walk backwards along
# prerequisites until a course repeats and return that
# cycle in forward order.
def validate_catalog(courses, prereqs):
    course_set = set(courses)
    missing = sorted({course_id for prerequisite, course in prereqs for course_id in (prerequisite, course) if
course_id not in course_set}, key=str)
    if missing:
        return False, missing, None
    dependents, indegree = defaultdict(list), {course: 0 for course in course_set}
    for prerequisite, course in prereqs:
        dependents[prerequisite].append(course)
        indegree[course] += 1
    queue = deque(sorted((course for course in course_set if indegree[course] == 0), key=str))
    order = []
    while queue:
        ready_course = queue.popleft()
        order.append(ready_course)
        for dependent in dependents[ready_course]:
            indegree[dependent] -= 1
            if indegree[dependent] == 0:
                queue.append(dependent)
    if len(order) == len(course_set):
        return True, [], order
```

```
    left = {course for course in course_set if indegree[course] > 0}
    prerequisites_of = defaultdict(list)
    for prerequisite, course in prereqs:
        if prerequisite in left and course in left:
            prerequisites_of[course].append(prerequisite)
    start = min(left, key=str)
    walk, walk_position, course_id = [], {}, start
    while course_id not in walk_position:                                  # every leftover course has a leftov
er prerequisite, so this walk never stops early
        walk_position[course_id] = len(walk)
        walk.append(course_id)
        course_id = prerequisites_of[course_id][0]
    cycle = walk[walk_position[course_id]:] + [course_id]
    return False, [], cycle[::-1]                          # walked against the edges, so reverse to list prerequ
isites first
```

COMPLEXITY | Time O(V + E) | Space O(V + E)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 217

218 * Course Schedule * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: edge = prerequisite -> course; return the order, or an empty list when there is a cycle

QUESTION INFO

QUESTION

You are given numCourses and a list of prerequisites, where each element in prerequisites[i] = [a_i, b_i] means you must complete course b_i before taking course a_i. Write a function canFinish that returns True if it's possible to finish all courses, and False if it's not. You need to complete all the courses for the result to be True. Assume there are no duplicate prerequisites, and numCourses is a non-negative number. Examples Input: numCourses = 3, prerequisites = [[2, 1], [1, 0]]
Output: True Explanation: You can complete course 0 first, then course 1, and finally course 2.
Input: numCourses = 4, prerequisites = [[3, 2], [2, 1], [1, 0], [0, 3]]
Output: False Explanation: The prerequisites form a cycle, making it impossible to finish all courses.
Input: numCourses = 5, prerequisites = [[4, 2], [3, 1], [2, 0]]
Output: True Explanation: You can complete the courses in order from course 0 to course 4.Hint 1Try modeling the courses as a graph, with edges representing prerequisites. Consider how you can detect cycles to determine if all courses can be completed.

EXAMPLES

Example 1	
Input	`2, [[1, 0]]`
Output	`true`
Why	`Course 1 requires course 0: finishing both is possible.`

Example 2	
Input	`2, [[1, 0], [0, 1]]`
Output	`false`
Why	`Course 1 requires 0 and course 0 requires 1: a cycle.`

QUESTION

Example 3	
Input	`2, [[1, 0]]`
Output	`[0, 1]`
Why	`A valid order: take 0 before 1.`

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | DFS from every course looking for a path back to itself

BRUTE-FORCE IDEA

For each course follow dependencies; reaching the start again is a cycle. Correct; the same parts of the graph are walked repeatedly.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: DFS from every course looking for a path back to itself
For each course follow dependencies
reaching the start again is a cycle
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(V * (V + E)) | Space O(V)

WHY THIS IS WEAKER | The same parts of the graph are walked again for many starting courses, O(V * (V + E)). Kahn's algorithm (take courses with no unmet prerequisites) is O(V + E).

HOW TO IMPROVE IT

edge = prerequisite -> course; return the order, or an empty list when there is a cycle

SHORT BEST SOLUTION

Count each course's unmet prerequisites and queue those with none. Take a course, reduce the counts of the courses that need it, and queue any that reach zero. All courses can be finished exactly when every course was taken; the order taken is a valid schedule.

```python
from collections import defaultdict
from collections import deque


# BASE QUESTION (optimal): Kahn's algorithm (topological
# sort)
# Idea: Count each course's unmet prerequisites and queue
# those with none. Take a course, reduce the counts of the
# courses that need it, and queue any that reach zero. All
# courses can be finished exactly when every course was
# taken; the order taken is a valid schedule.
def find_order(course_count, prereqs):
    dependents, indegree = defaultdict(list), [0] * course_count
    for course, prerequisite in prereqs:
        dependents[prerequisite].append(course)
        indegree[course] += 1
    queue, order = deque(course for course in range(course_count) if indegree[course] == 0), []
    while queue:
        ready_course = queue.popleft()
        order.append(ready_course)
        for dependent in dependents[ready_course]:
            indegree[dependent] -= 1
            if indegree[dependent] == 0:
                queue.append(dependent)
    return order if len(order) == course_count else []
```

COMPLEXITY | Time O(V + E) | Space O(V + E)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 218

Counting pairs by looking up each partner in a Counter or a bucket

PATTERN CONTEXT

For each value, compute the partner it would need (v + k, 2v, the other corner of a square) and look it up in a Counter or a set. When the partner depends on two things, enumerate the pairs once and bucket them by a shared key.

219 * Progressive Array Partitioning * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: Counter of the values; stage 1 asks whether all the counts are different: len(set(counts.values())) == len(counts)

QUESTION INFO

QUESTION

The problem is given in stages; stage 1 is: group the equal values of an array together. Return true when every group has a different size (no two values occur the same number of times). The function takes the occurrence count of each value as a dict, so counting stays separate from the rule and later stages can change only the rule.

Function signature

def stage1(counts: dict) -> bool

counts maps value -> how

many times it occurs

EXAMPLES

Example 1	

Input	`from collections import Counter`
	`nums = [1, 2, 3, 4, 2, 3, 3, 4, 4, 4]`
	`stage1(Counter(nums))`

Output	`stage1(Counter(nums)) -> true`
Why	`The groups have sizes 1, 2, 3, 4: all different, so true.`

Example 2	

Input	`from collections import Counter`
	`nums = [1, 2, 3]`
	`stage1(Counter(nums))`

Output	`stage1(Counter(nums)) -> false`
Why	`Every value occurs once, so all three groups have size 1: false.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Compare every pair of counts

BRUTE-FORCE IDEA

Count each value, then check every pair of counts for equality. Fine for small d and a clear statement of the rule.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compare every pair of counts
Count each value
check every pair of counts for equality
```

BRUTE-FORCE COMPLEXITY | Time O(d^2) where d is the number of distinct values | Space O(d)

WHY THIS IS WEAKER | O(d^2) pair checks for d distinct values. Comparing the number of distinct counts with the number of distinct values (a set) is O(n).

HOW TO IMPROVE IT

Counter of the values; stage 1 asks whether all the counts are different: len(set(counts.values())) == len(counts)

SHORT BEST SOLUTION

All counts are different exactly when the set of counts has as many elements as there are distinct values. One pass to count, one pass over the counts.

```
from collections import Counter


# BASE QUESTION (optimal): Counter, then compare the set
# of counts with the number of values
# Idea: All counts are different exactly when the set of
# counts has as many elements as there are distinct
# values. One pass to count, one pass over the counts.
def distinct_counts(nums):
    counts = Counter(nums)
    return len(set(counts.values())) == len(counts)
```

COMPLEXITY | Time O(n) | Space O(d)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 219

220 * K-diff Pairs in an Array

DIFFICULTY | * MEDIUM *

Better approach: the template as written: partner = v + difference; for difference == 0 count values that appear twice

QUESTION INFO

QUESTION

Given an integer array nums and an integer k , return the number of unique k-diff pairs in the array. A k-diff pair is a pair of integers (nums[i], nums[j]) where: i != j , and abs(nums[i] - nums[j]) == k . Two k-diff pairs (a, b) and (c, d) are considered the same (counted once) if they consist of the same pair of values. Count each distinct value pair only once.

Function signature

def count_k_diff_pairs(nums: list[int], k: int) -> int:
    """Return the number of unique value-pairs with absolute difference k."""
    ...

EXAMPLES

Example 1	
Input	`nums = [3, 1, 4, 1, 5], k = 2`
Output	`2`
Why	`The unique 2-diff value pairs are (1, 3) and (3, 5).`

Example 2	
Input	`nums = [1, 2, 3, 4, 5], k = 1`
Output	`4`
Why	`(1,2), (2,3), (3,4), (4,5).`

QUESTION

Example 3	
Input	`nums = [1, 3, 1, 5, 4], k = 0`
Output	`1`
Why	`With k = 0 count values that appear at least twice; only 1 does.`

CONSTRAINTS

- 1 <= len(nums) <= 10^4
- -10^7 <= nums[i] <= 10^7
- 0 <= k <= 10^7

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(nums) <= 10^4
- -10^7 <= nums[i] <= 10^7
- 0 <= k <= 10^7

BRUTE FORCE | Every pair of indexes into a set

BRUTE-FORCE IDEA

Start with: Every pair of indexes into a set. Compares all index pairs and stores the value pairs to remove repeats.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Every pair of indexes into a set
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(pairs)

WHY THIS IS WEAKER | Compares all index pairs and stores the value pairs to remove repeats.

HOW TO IMPROVE IT

the template as written: partner = v + difference; for difference == 0 count values that appear twice

SHORT BEST SOLUTION

A value v starts a pair when v + difference is present; for difference = 0 it must appear twice.

```python
# BASE QUESTION (optimal): Count the values
# Idea: A value v starts a pair when v + difference is present; for
# difference = 0 it must appear twice.
from collections import Counter


def count_k_diff_pairs(nums, difference):
    if difference < 0:
        return 0
    counts = Counter(nums)
    if difference == 0:
        return sum(1 for copies in counts.values() if copies >= 2)
    return sum(1 for value in counts if value + difference in counts)
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 220

221 * Count Axis-Aligned Squares Among Grid Points

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: a set of points; two opposite corners fix the other two, so look both up; every square is found twice, divide by 2

QUESTION INFO

QUESTION

The interviewer described a diagram of squares and rectangles and asked which data structure to use and how to count the squares. Framed concretely: you are given a list of distinct integer lattice points in the plane. Count the number of axis-aligned squares whose four corners all appear in points . A square is axis-aligned when its sides are parallel to the x- and y-axes. Return the count of such squares.

Function signature

```
from typing import List, Tuple
def count_axis_aligned_squares(points: List[Tuple[int, int]]) -> int:
    """Return the number of axis-aligned squares whose 4 corners are all in points."""
    ...
```

Follow-up (as reported): A cleaner O(n^2) approach: put all points in a set, then for every pair that could be two opposite corners of a square (equal absolute x- and y-difference), check the other two corners; each square is found via both its diagonals, so divide the total by 2.

EXAMPLES

Example 1

Input	`points = [(0, 0), (0, 1), (1, 0), (1, 1)]`
Output	`1`
Why	`The four points form exactly one unit square.`

Example 2

Input	`points = [(0,0),(0,1),(1,0),(1,1),(0,2),(1,2),(2,0),(2,1),(2,2)]`
Output	`5`
Why	`A 3x3 grid of points has four 1x1 squares and one 2x2 square.`

Example 3

Input	`points = [(0, 0), (2, 0), (0, 1), (2, 1)]`
Output	`0`
Why	`These form a 2x1 rectangle, which is not a square.`

CONSTRAINTS

- 0 <= len(points) <= 2000 ; all points are distinct.
- Each point is (x, y) with -10^9 <= x, y <= 10^9 .
- A square must have positive side length (four distinct corners); degenerate zero-size "squares" do not count.
- Only axis-aligned squares count (tilted squares are ignored).

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(points) <= 2000 ; all points are distinct.
- Each point is (x, y) with -10^9 <= x, y <= 10^9 .
- A square must have positive side length (four distinct corners); degenerate zero-size "squares" do not count.

- Only axis-aligned squares count (tilted squares are ignored).

BRUTE FORCE | Test every four points

BRUTE-FORCE IDEA

It looks at every group of four points although two opposite corners already fix the other two.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Test every four points
It looks at every group of four points although two opposite corners already fix the other
two
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^4) | Space O(1)

WHY THIS IS WEAKER | It looks at every group of four points although two opposite corners already fix the other two.

HOW TO IMPROVE IT

a set of points; two opposite corners fix the other two, so look both up; every square is found twice, divide by 2

SHORT BEST SOLUTION

Two opposite corners decide the other two, so one set lookup per pair is enough.

```python
# BASE QUESTION (optimal): Opposite corners and a set
# Idea: Two opposite corners decide the other two, so one
# set lookup per pair is enough.
def count_axis_aligned_squares(points):
    seen = set(points)
    total = 0
    for index, (x_first, y_first) in enumerate(points):
        for x_second, y_second in points[index + 1:]:
            if abs(x_first - x_second) == abs(y_first - y_second) != 0:
                if (x_first, y_second) in seen and (x_second, y_first) in seen:
                    total += 1
    return total // 2
```

COMPLEXITY | Time O(n^2) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 221

222 * Sum of Two Squares in Two Ways * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: enumerate every pair (a, b) with a <= b once, add a*a + b*b to a Counter, answer = sums counted exactly twice

QUESTION INFO

QUESTION

Given an integer n, find all integers s < n that can be written as a sum of two positive squares in exactly two different ways (the order of the two squares does not matter). For example 50 = 1^2 + 7^2 = 5^2 + 5^2. Return them in increasing order.

Function signature

def two_squares_two_ways(n: int) -> list

EXAMPLES

Example 1	
Input	`100`
Output	`[50, 65, 85]`
Why	`50 = 1 + 49 = 25 + 25 and 65 = 1 + 64 = 16 + 49 both have exactly two ways.`

QUESTION

Example 2	
Input	`50`
Output	`[]`
Why	`Nothing below 50 qualifies.`

CONSTRAINTS

- 1 <= n <= 2^31 - 1

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= n <= 2^31 - 1

BRUTE FORCE | For each s below n, try every a and b

BRUTE-FORCE IDEA

For every s count the pairs (a, b) with a <= b and a^2 + b^2 = s. Correct, but it re-searches pairs for every s.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: For each s below n, try every a and b
For every s count the pairs (a
b) with a <= b
a^2 + b^2 = s
```

BRUTE-FORCE COMPLEXITY | Time O(n * sqrt(n)) | Space O(1)

WHY THIS IS WEAKER | Every s searches all pairs again, O(n * sqrt(n)). Generating each pair once and counting sums in a hash map is O(n).

HOW TO IMPROVE IT

enumerate every pair (a, b) with a <= b once, add a*a + b*b to a Counter, answer = sums counted exactly twice

SHORT BEST SOLUTION

Loop a from 1 while 2a^2 < limit and b from a while a^2 + b^2 < limit, adding each sum to a Counter. There are about pi * limit / 8 pairs. The answer is the sorted list of sums whose count is exactly 2.

```python
from collections import Counter


# BASE QUESTION (optimal): Count every pair once in a hash
# map
# Idea: Loop a from 1 while 2a^2 < limit and b from a while
# a^2 + b^2 < limit, adding each sum to a Counter. There are
# about pi * limit / 8 pairs. The answer is the sorted list of
# sums whose count is exactly 2.
def two_squares_two_ways(limit):
    sum_counts = Counter()
    a = 1
    while 2 * a * a < limit:
        b = a
        while a * a + b * b < limit:
            sum_counts[a * a + b * b] += 1
            b += 1
        a += 1
    return sorted(square_sum for square_sum, ways_count in sum_counts.items() if ways_count == 2)
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 222

GROUP 31 OF 43 | 3 QUESTIONS | PART A - PART 1

Tree DP: build a parent order once, then add up from the leaves

PATTERN CONTEXT

Get an order where every parent comes before its children (a BFS order from the root), then sweep it backwards so every child is finished before its parent. Subtree sizes and sums fall out of that one backwards sweep; the re-rooting questions add a second forward sweep.

224 * Compute File System Total Size * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: add each file's size while walking down from the root with a stack; a shared directory is counted once (seen set)

QUESTION INFO

QUESTION

A file system is a tree: File(size) is a file and Dir(*children) is a directory holding files and directories. Return the total size of all files. A directory object can be shared (it appears under more than one parent); its files are counted once.

Function signature

def total_size(root) -> int

EXAMPLES

Example 1	
Input	`root = Dir(Dir(File(10), File(20)), File(5), Dir(File(30)))` `total_size(root)`
Output	`total_size(root) -> 65`
Why	`10 + 20 + 5 + 30 = 65.`

Example 2	

Input	`shared = Dir(File(7))`
	`root = Dir(shared, shared, File(1))`
	`total_size(root)`

Output	`total_size(root) -> 8`
Why	`The same directory appears twice but its 7 is counted once: 7 + 1 = 8.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Plain recursion

BRUTE-FORCE IDEA

Sum a file's size, or the sum over a directory's children. Correct for ordinary trees and the answer most interviews want first. It overflows on very deep trees and loops forever on a cycle.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
size(node):
    if node is a file, return its size
    total = 0
    for each child: total += size(child)
    return total
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(depth) call stack

WHY THIS IS WEAKER | Linear time, which is optimal; the weaknesses are the recursion depth on very deep trees and an endless loop if a link creates a cycle. An explicit stack with a visited set fixes both.

HOW TO IMPROVE IT

add each file's size while walking down from the root with a stack; a shared directory is counted once (seen set)

SHORT BEST SOLUTION

An explicit stack removes the recursion limit, and a set of visited node ids makes each node count once, so shared directories are not double counted and cycles terminate.

```
# BASE QUESTION (optimal): Iterative traversal with a
# visited set
# Idea: An explicit stack removes the recursion limit, and
# a set of visited node ids makes each node count once, so
# shared directories are not double counted and cycles
# terminate.
class File:
    def __init__(self, size):
        self.size = size

class Dir:
    def __init__(self, *children):
        self.children = list(children)

def total_size(root):
    total, seen, stack = 0, set(), [root]
    while stack:
        node = stack.pop()
        if id(node) in seen:
            continue
        seen.add(id(node))
        if isinstance(node, File):
            total += node.size
        else:
            stack.extend(node.children)
    return total
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 224

225 * Minimum-Cost Tree Disconnection * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: backwards sweep with two numbers per node: the best cost when the node is cut off, and the cost if it is free

QUESTION INFO

QUESTION

A rooted tree is given as directed edges (parent, child, cost) with node 0 as the root. Cut edges so that no path from the root to any original leaf remains. The price of a cut set is the sum of the costs of the edges cut (costs may be negative). Return the minimum total price.

Function signature

def disconnect_cost(n: int, edges: list) -> int

EXAMPLES

Example 1	
Input	`3, [(0, 1, 3), (0, 2, 2)]`
Output	`5`
Why	`Two leaves hang off the root: both edges must be cut, 3 + 2 = 5.`

QUESTION

Example 2	
Input	`3, [(0, 1, 4), (1, 2, 1)]`
Output	`1`
Why	`A chain 0 - 1 - 2: cutting the cheaper edge 1 -> 2 (cost 1) already separates the leaf.`

Example 3	
Input	`3, [(0, 1, -1), (1, 2, -1)]`
Output	`-2`
Why	`Negative costs: cutting both edges costs -2, which beats cutting only one.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every subset of edges

BRUTE-FORCE IDEA

Enumerate each subset of edges to cut, check that the root can reach no leaf, and keep the cheapest. Exponential, but exact, so it checks the DP.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every subset of edges
Enumerate each subset of edges to cut
check that the root can reach no leaf
keep the cheapest
```

BRUTE-FORCE COMPLEXITY | Time O(2^E * n) | Space O(n)

WHY THIS IS WEAKER | 2^E subsets, each checked with a search: O(2^E * n). A tree DP (cut the edge above a child, or require the child to be disconnected from its leaves) is O(n).

HOW TO IMPROVE IT

backwards sweep with two numbers per node: the best cost when the node is cut off, and the cost if it is free

SHORT BEST SOLUTION

For a child c with edge weight w there are two options: cut the edge, paying w plus whatever extra negative-weight edges you can still cut below c (free_total(child) = sum of min(0, weight) over its subtree), or keep the edge and require best(child), which is impossible when c is a leaf. best(v) is the sum of the cheaper option over its children. With non-negative weights free_total(child) is 0 and this is the classic recurrence.

```python
from collections import defaultdict


# BASE QUESTION (optimal): Tree DP: best(v) = cheapest
# cuts inside v's subtree that disconnect v from its
# leaves
# Idea: For a child c with edge weight w there are two
# options: cut the edge, paying w plus whatever extra
# negative-weight edges you can still cut below c (free_total(child)
# = sum of min(0, weight) over its subtree), or keep the
# edge and require best(child), which is impossible when c is
# a leaf. best(v) is the sum of the cheaper option over
# its children. With non-negative weights free_total(child) is 0 and
# this is the classic recurrence.
def disconnect_cost(node_count, edges):
    kids = defaultdict(list)
```

```
    for parent, child, weight in edges:
        kids[parent].append((child, weight))
    INFINITY = float("inf")

    def solve(node):
        free_total = 0
        best = 0
        for child, weight in kids[node]:
            child_best, child_free = solve(child)
            free_total += min(0, weight) + child_free
            keep_cost = INFINITY if not kids[child] else child_best
            best += min(weight + child_free, keep_cost)
        return best, free_total

    return solve(0)[0] if kids[0] or node_count > 1 else 0
```

COMPLEXITY | Time O(n) | Space O(h)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 225

226 * Given a large n-ary tree where the root node can pass information to one child per iteration, and in subsequent iterations, nodes can pass information to one child while the root continues with others, find the minimum number of iterations to reach all nodes. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: rounds[node] = max over children (rank among children sorted by time, longest first, + rounds[child]); read t at the root

QUESTION INFO

QUESTION

Information starts at the root of a tree. In every iteration a node that already has the information can pass it to one of its children (the tree maps each node to its children as (child, cost) pairs; in the basic problem every cost is 1 iteration). Return the minimum number of iterations until every node has the information.

Function signature

def min_rounds(tree: dict, root) -> int

EXAMPLES

Example 1	
Input	{'r': [('a', 1), ('b', 1), ('c', 1)], 'a': [('d', 1)]}, 'r'
Output	3
Why	The root informs a, b and c in iterations 1, 2, 3, while a informs d in iteration 2: everyone knows after 3 iterations.

Example 2	
Input	{'r': [('a', 1)], 'a': [('b', 1)]}, 'r'
Output	2
Why	A path r - a - b needs one iteration per edge.

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every calling order at every node

BRUTE-FORCE IDEA

For each node try every permutation of its children and take the order that minimises the finish time. Exact for the sorted-order rule; factorial in the number of children.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every calling order at every node
For each node try every permutation of its children
take the order that minimises the finish time
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(product of d! over nodes) | Space O(n)

WHY THIS IS WEAKER | A node with d children has d! Orderings, so the cost is the product of d! Over all nodes. Sorting the children by decreasing T gives the same answer in O(d log d).

HOW TO IMPROVE IT

rounds[node] = max over children (rank among children sorted by time, longest first, + rounds[child]); read t at the root

SHORT BEST SOLUTION

Let T(node) be the rounds needed to inform the subtree of v once v knows. v informs its children one per round. If children are served in decreasing order of T, the child served in round i finishes at i + T(child), so T(node) = max over i of (i + T(child_i)) with i starting at 1. Computed iteratively so a very deep tree cannot overflow the recursion stack.

```python
# BASE QUESTION (optimal): Post-order DP: serve the
# slowest subtree first
# Idea: Let T(node) be the rounds needed to inform the
# subtree of v once v knows. v informs its children one
# per round. If children are served in decreasing order of
# T, the child served in round i finishes at i + T(child),
# so T(node) = max over i of (i + T(child_i)) with i starting
# at 1. Computed iteratively so a very deep tree cannot
# overflow the recursion stack.
def minRoundsToInformAll(node_count, children, root=0):
    order, stack = [], [root]
    while stack:
        node = stack.pop()
        order.append(node)
        stack.extend(children.get(node, []))
    rounds = {}
    for node in reversed(order):
        children_rounds = sorted((rounds[child] for child in children.get(node, [])), reverse=True)
        rounds[node] = max((turn + 1 + child_time for turn, child_time in enumerate(children_rounds)), default=
0)
    return rounds[root]
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 226

Counting or reaching an amount: the knapsack loop

PATTERN CONTEXT

Make an array indexed by amount. For each item, sweep the amounts and let ways[t] depend on ways[t - item]. Sweep upwards when an item can be reused, downwards when each item can be used once. Swap += for min(...) + 1 (fewest items) or or (reachable).

229 * Maximize Shelf Space (Unbounded Knapsack) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: reachable[t] = any(reachable[t - size]); the answer is the largest reachable total up to the capacity

QUESTION INFO

QUESTION

A shelf has a fixed integer capacity capacity . You have boxes of various integer sizes given in sizes , and you may use any number of each box, including duplicates (each size is available infinitely). Choose boxes so their total size does not exceed capacity , maximizing the total size used. Return that maximum total.

Function signature

```
def max_shelf_space(capacity: int, sizes: list[int]) -> int:
    ...
```

EXAMPLES

Example 1	
Input	`capacity = 10, sizes = [3, 5, 7]`
Output	`10`
Why	`3 + 7 = 10 exactly fills the shelf.`

Example 2	
Input	`capacity = 11, sizes = [5, 3]`
Output	`11`
Why	`5 + 3 + 3 = 11.`

Example 3	
Input	`capacity = 7, sizes = [4, 5]`
Output	`5`
Why	`Options within 7 are 4 or 5 (two boxes would be >= 8). The best is 5.`

CONSTRAINTS

- 0 <= capacity <= 10**4 .
- 1 <= len(sizes) <= 100 , each size is a positive integer.

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= capacity <= 10**4 .
- 1 <= len(sizes) <= 100 , each size is a positive integer.

BRUTE FORCE | Recurse over every choice

BRUTE-FORCE IDEA

Start with: Recurse over every choice. The same remaining capacity is solved again and again.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recurse over every choice
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(capacity)

WHY THIS IS WEAKER | The same remaining capacity is solved again and again.

HOW TO IMPROVE IT

reachable[t] = any(reachable[t - size]); the answer is the largest reachable total up to the capacity

SHORT BEST SOLUTION

A total is reachable if removing one box leaves a reachable total.

```python
# BASE QUESTION (optimal): Reachable totals DP
# Idea: A total is reachable if removing one box leaves a
# reachable total.
def max_shelf_space(capacity, sizes):
    reachable = [True] + [False] * capacity
    for total in range(1, capacity + 1):
        reachable[total] = any(box_size <= total and reachable[total - box_size] for box_size in sizes)
    return max(total for total in range(capacity + 1) if reachable[total])
```

COMPLEXITY | Time O(capacity * boxes) | Space O(capacity)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 229

230 * Coin Change * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: fewest coins: start with infinity, best[t] = min(best[t], best[t - coin] + 1); -1 when it stays infinity

QUESTION INFO

QUESTION

You are given an integer array coins representing different coin denominations and an integer amount representing a total amount of money. Write a function coinChange that returns the fewest number of coins needed to make up that amount. If that amount cannot be made up by any combination of the coins, return -1. You may assume that you have an infinite number of each kind of coin. Examples PythonInput: coins = [1, 2, 5], amount = 11 Output: 3 Explanation: The minimum number of coins needed is 3 (11 = 5 + 5 + 1).
Input: coins = [4, 5], amount = 7 Output: -1 Explanation: It is not possible to make up the amount 7 with coins of denominations 4 and 5.Solution 1: Brute-Force approach The naive approach to solving the coin change problem involves a recursive strategy to explore all possible combinations of coins to find the minimum number needed to make up the target amount. It starts by recursively trying each coin denomination to reduce the target amount. For each coin, the algorithm subtracts the coin's value from the current amount and solves the reduced problem. If the amount becomes zero, it indicates that no more coins are needed, and the function returns zero. If the amount becomes negative, it signals that this path is not feasible, so the function returns infinity or a large number. The algorithm combines the results from each recursive call with the current coin to determine the minimum number of coins required. Finally, after exploring all possibilities, it returns the

EXAMPLES

QUESTION

Example 1	
Input	`[1, 2, 5], 11`
Output	`3`
Why	`5 + 5 + 1 = 11: 3 coins.`

Example 2	
Input	`[2], 3`
Output	`-1`
Why	`A coin of 2 can never make 3.`

Example 3	
Input	`[1], 0`
Output	`0`
Why	`Zero needs no coins.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recursion over the last coin

BRUTE-FORCE IDEA

Min_coins(a) = 1 + min over coins c <= a of min_coins(a - c). Exact; the same amounts are recomputed exponentially often.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
fewest(amount):
    if amount == 0, return 0
    try every coin not larger than amount
    answer = min(1 + fewest(amount - coin))
    return answer, or impossible if no coin works
```

BRUTE-FORCE COMPLEXITY | Time O(coins^amount) | Space O(amount)

WHY THIS IS WEAKER | The same amount is recomputed through many coin orders, exponential. Filling fewest[a] for a from 0 upward is O(amount * coins).

HOW TO IMPROVE IT

fewest coins: start with infinity, best[t] = min(best[t], best[t - coin] + 1); -1 when it stays infinity

SHORT BEST SOLUTION

best[0] = 0, and for sub_amount from 1 to amount, best[sub_amount] = 1 + min(best[sub_amount - coin]) over coins coin <= sub_amount (infinity if none works). The answer is best[amount] or -1. Each amount is solved once.

```python
# BASE QUESTION (optimal): Bottom-up DP over amounts
# Idea: best[0] = 0, and for sub_amount from 1 to amount, best[sub_amount] =
# 1 + min(best[sub_amount - coin]) over coins coin <= sub_amount (infinity if none
# works). The answer is best[amount] or -1. Each amount is
# solved once.
def coinChange(coins, amount):
    UNREACHABLE = amount + 1
    best = [0] + [UNREACHABLE] * amount
    for sub_amount in range(1, amount + 1):
        for coin in coins:
            if coin <= sub_amount and best[sub_amount - coin] + 1 < best[sub_amount]:
                best[sub_amount] = best[sub_amount - coin] + 1
    return best[amount] if best[amount] != UNREACHABLE else -1
```

COMPLEXITY | Time O(amount * coins) | Space O(amount)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 230

231 * Partition an array into two sub-arrays with equal sum. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: subset sum to total / 2, written with a bitset: reachable_sums |= reachable_sums << number

QUESTION INFO

QUESTION

Given an array of integers, determine if it can be partitioned into two subsets such that the sum of the elements in each subset is equal. A partition is considered valid if it divides the array into two subsets where both subsets have the same sum. Examples Input: [2, 4, 11, 5]
Output: True Explanation: The array can be partitioned into two subsets [2, 4, 5] and [11] with equal sum.
Input: [2, 7, 3, 1]
Output: False Explanation: The array cannot be partitioned into two subsets with equal sums.

EXAMPLES

Example 1	
Input	`[1, 5, 11, 5]`
Output	`true`
Why	`[1, 5, 5] and [11] both add up to 11.`

Example 2	
Input	`[1, 2, 3, 5]`
Output	`false`
Why	`The total is 11, which is odd.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every subset

BRUTE-FORCE IDEA

Enumerate all subsets and check whether any sums to half the total. It is exact but exponential.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every subset
Enumerate all subsets
check whether any sums to half the total
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(2^n * n) | Space O(n)

WHY THIS IS WEAKER | 2^n subsets. Subset-sum as a reachability table (or a bitset shifted by each number) is O(n * sum).

HOW TO IMPROVE IT

subset sum to total / 2, written with a bitset: reachable_sums |= reachable_sums << number

SHORT BEST SOLUTION

reachable_sums is an integer whose bit s is set when some subset sums to s; start with bit 0. For each number x do reachable_sums |= reachable_sums << number. The answer is whether bit total / 2 is set. A boolean array updated from high to low sums is the same idea in the table form.

```python
# BASE QUESTION (optimal): Subset-sum as a bitset
# Idea: reachable_sums is an integer whose bit s is set when some
# subset sums to s; start with bit 0. For each number x do
# reachable_sums |= reachable_sums << number. The answer is whether bit total / 2
# is set. A boolean array updated from high to low sums is
# the same idea in the table form.
def canPartition(nums):
    total = sum(nums)
    if total % 2:
        return False
    reachable_sums = 1
    for number in nums:
        reachable_sums |= reachable_sums << number
    return bool(reachable_sums >> (total // 2) & 1)
```

COMPLEXITY | Time O(n * sum / 64) with big-integer shifts (O(n * sum) as a boolean table) | Space O(sum)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 231

232 * Explain how to find a target sum in an array. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: turn +/- signs into a subset sum of (total + target) / 2, items used once (sweep downwards)

QUESTION INFO

QUESTION

Determine the Number of Ways to Assign Symbols to Reach a Target Sum. You are given a list of non-negative integers, a1, a2, ..., an, and a target sum S. Your task is to assign each integer in the list a symbol + or - and calculate how many different ways there are to arrange these symbols such that the sum of the integers equals S after applying the assigned symbols. For example, if you have the list [1, 1, 1, 1, 1] and the target sum 3, one way to reach the target sum is to assign + to three 1s and - to two 1s, like so: +1 +1 +1 +1 -1. Another way could be +1 -1 +1 +1 +1. You need to return the total number of unique arrangements that meet the target sum S. Examples nums = [1, 1, 1, 1, 1] targetSum = 3 output: 5 nums = [1, 2, 1] targetSum = 2 output: 2 nums = [1, 2, 7, 1, 5, 3] targetSum = 8 output: -1 (Explanation: Since the total sum of the numbers plus the target is odd, it's not possible to reach the target sum.)The original problem asks for the number of ways to assign symbols (+ or -) to a list of numbers such that their sum equals a target sum S. This problem can be converted into a problem of finding subsets of the given numbers whose sum equals (totalSum + S) / 2. This is because if we can find subsets whose sum equals (totalSum + S) / 2, the remaining numbers will naturally form the other subset that sums to (totalSum - S) / 2, which effectively gives us the desired configuration of symbols. NoteWhy (totalSum + S) / 2?Given a list of numbers nums=[a1,a2,...,an]\te

EXAMPLES

Example 1	
Input	`[1, 1, 1, 1, 1], 3`
Output	`5`
Why	`Put + or - in front of every number; five ways to reach 3 with five 1s.`

Example 2	
Input	`[1], 1`
Output	`1`
Why	`+1 is the only way.`

QUESTION

Example 3	
Input	`[1, 2], 5`
Output	`-1`
Why	`No sign choice gives 5.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every combination of signs

BRUTE-FORCE IDEA

Recursively choose + or - for each number and count the ways to land on S. It is exact but exponential.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
search(index, total):
    if index == n, count this choice when total == target
    recurse after adding nums[index]
    recurse after subtracting nums[index]
return the total number of valid choices
```

BRUTE-FORCE COMPLEXITY | Time O(2^n) | Space O(n)

WHY THIS IS WEAKER | Two branches per number: O(2^n). The numbers with a + sign form a subset with sum (total + S) / 2, so a subset-sum counting table is O(n * sum).

HOW TO IMPROVE IT

turn +/- signs into a subset sum of (total + target) / 2, items used once (sweep downwards)

SHORT BEST SOLUTION

Let wanted_sum = (total + target) / 2 (return 0 if it is not a non-negative integer). ways[subset_sum] counts the subsets with sum subset_sum; for each number x update ways from high to low: ways[subset_sum] += ways[subset_sum - number]. The answer is ways[wanted_sum]. Zeros double the count, which this loop handles naturally.

```python
# BASE QUESTION (optimal): Subset-sum counting: choose P
# with sum (total + target) / 2
# Idea: Let wanted_sum = (total + target) / 2 (return 0 if it is not
# a non-negative integer). ways[subset_sum] counts the subsets with
# sum subset_sum; for each number x update ways from high to low:
# ways[subset_sum] += ways[subset_sum - number]. The answer is ways[wanted_sum]. Zeros
# double the count, which this loop handles naturally.
def findTargetSumWays(nums, target):
    total = sum(nums)
    if (total + target) % 2 or total + target < 0 or abs(target) > total:
        return 0
    wanted_sum = (total + target) // 2
    ways = [1] + [0] * wanted_sum
    for number in nums:
        for subset_sum in range(wanted_sum, number - 1, -1):
            ways[subset_sum] += ways[subset_sum - number]
    return ways[wanted_sum]
```

COMPLEXITY | Time O(n * sum) | Space O(sum)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 232

233 * Reverse Coin Change * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: given the ways table, find the coin values: the first amount whose count is wrong must be a coin; add it and rebuild

QUESTION INFO

QUESTION

In the coin change counting problem, dp[i] is the number of ways to make amount i from a set of coins (dp[0] = 1). In the reverse problem you are given the dp array and must find a set of coins that produces it, or None if no coin set does.

Function signature

def coins_from_dp(dp: list) -> list | None

EXAMPLES

Example 1	
Input	`[1, 1, 2, 2, 3, 3]`
Output	`[1, 2]`
Why	`Coins 1 and 2 make 1, 1, 2, 2, 3, 3 ways for the amounts 0 to 5.`

Example 2	
Input	`[1, 0, 1, 0, 1]`
Output	`[2]`
Why	`Only the coin 2 can make 2 and 4 but not 1 and 3.`

Example 3	
Input	`[1, 2, 3]`
Output	`null`
Why	`No coin set can produce these counts.`

CONSTRAINTS

- 1 <= dp.length <= 1000
- dp[0] = 1

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= dp.length <= 1000
- dp[0] = 1

BRUTE FORCE | Try every subset of coin values 1..n

BRUTE-FORCE IDEA

Enumerate each subset of {1..n} as a candidate coin set, build its ways table and compare with the target. It gives the exact answer.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every subset of coin values 1..n
Enumerate each subset of {1..n} as a candidate coin set
build its ways table
compare with the target
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(2^n * n * coins) | Space O(n)

WHY THIS IS WEAKER | 2^n candidate sets, each with an O(n * coins) table. Scanning amounts upward and adding a coin whenever the target has one more way than the table is O(n^2).

HOW TO IMPROVE IT

given the ways table, find the coin values: the first amount whose count is wrong must be a coin; add it and rebuild

SHORT BEST SOLUTION

Keep the ways table for the coins found so far. For each amount a from 1 up: if the target equals the current table, no coin; if the target is exactly one larger, amount is a coin (the single coin a adds exactly one way at amount amount), so add it and update the table; otherwise the input is impossible. dp[0] must be 1. At the end the table must equal the target.

```
# BASE QUESTION (optimal): Scan upward and add a coin
# whenever the table falls short
# Idea: Keep the ways table for the coins found so far.
# For each amount a from 1 up: if the target equals the
# current table, no coin; if the target is exactly one
# larger, amount is a coin (the single coin a adds exactly one
# way at amount amount), so add it and update the table;
# otherwise the input is impossible. dp[0] must be 1. At
# the end the table must equal the target.
def ways_table(coins, max_amount):
    dp = [1] + [0] * max_amount
    for coin in coins:
        for amount in range(coin, max_amount + 1):
            dp[amount] += dp[amount - coin]
    return dp

def coins_from_dp(dp):
    max_amount = len(dp) - 1
    if not dp or dp[0] != 1:
        return None
    current_table, coins = [1] + [0] * max_amount, []
    for amount in range(1, max_amount + 1):
        if dp[amount] == current_table[amount]:
            continue
        if dp[amount] != current_table[amount] + 1:
            return None
        coins.append(amount)
        for updated_amount in range(amount, max_amount + 1):
            current_table[updated_amount] += current_table[updated_amount - amount]
    return coins if current_table == list(dp) else None
```

COMPLEXITY | Time O(n * coins) = O(n^2) worst case | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 233

Prefix sums, Tier 2: running total plus a dict of earlier totals

PATTERN CONTEXT

Keep a running total. At each step ask the dict about an earlier total: total - k (subarray sums to k), total % k (sum divisible by k), or the smallest earlier total (best subarray ending here). Store the first index, a count, or the minimum.

234 * Solve array-sum and city-community problems * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: count subarrays summing to k with the template as written; the second task is union-find over building cells

QUESTION INFO

QUESTION

Two tasks. (1) count_subarrays(nums, k): count the contiguous subarrays whose elements add up to k. (2) communities(grid): a grid has 1 for a building and 0 for empty land; buildings that touch up, down, left or right belong to the same community. Return the number of communities.

Function signature

def count_subarrays(nums: list, k: int) -> int
def communities(grid: list) -> int

EXAMPLES

Example 1	
Input	`[1, 1, 1], 2`
Output	`2`
Why	`[1, 1] appears twice (positions 0-1 and 1-2).`

Example 2	
Input	`[1, 2, 3], 3`
Output	`2`
Why	`[1, 2] and [3] both add up to 3.`

Example 3	
Input	`[[1, 1, 0], [0, 0, 1], [1, 0, 1]]`
Output	`3`
Why	`Top-left pair, the single building at the right of the middle row joined to the one below it, and the bottom-left: 3 communities.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Task 1: try every subarray

BRUTE-FORCE IDEA

Extend each start with a running sum and count the ends where it equals K. This takes quadratic time.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
count = 0
```

```
for each start index:
    running_sum = 0
    extend the end index and add each value
    increment count whenever running_sum == K
return count
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1)

WHY THIS IS WEAKER | O(n^2). A dictionary of prefix sums (how many earlier prefixes equal current - K) is O(n) and also works on a stream and with negative numbers.

BRUTE FORCE 2 | Task 2: relabel cells until the labels stop changing

BRUTE-FORCE IDEA

Give every building its own label and repeatedly set each cell's label to the smallest label among itself and its neighbours until nothing changes; the number of distinct labels is the number of communities. Correct and easy to trust, and slow on long snakes.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Task 2: relabel cells until the labels stop changing
Give every building its own label
repeatedly set each cell's label to the smallest label among itself
its neighbours until nothing changes
the number of distinct labels is the number of communities
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O((m * n)^2) worst case | Space O(m * n)

WHY THIS IS WEAKER | Labels spread one step per round, so a long snake needs about m * n rounds of O(m * n): O((m * n)^2). Union-Find or one flood fill is near-linear.

HOW TO IMPROVE IT

count subarrays summing to k with the template as written; the second task is union-find over building cells

SHORT BEST SOLUTION

Part 1: Walk the array keeping the prefix sum. The number of subarrays ending at the current index with sum K equals how many earlier prefixes were equal to prefix - K, so look that up and then record the current prefix. Seed the dictionary with {0: 1} for subarrays that start at index 0. Works with negative numbers and online.

Part 2: Make each building its own set and union it with the building to its right and below. The number of communities is the number of distinct roots. Union-Find also supports adding buildings later.

```python
# Task 1: prefix sums in a dictionary
# Idea: Walk the array keeping the prefix sum. The number
# of subarrays ending at the current index with sum K
# equals how many earlier prefixes were equal to prefix -
# K, so look that up and then record the current prefix.
# Seed the dictionary with {0: 1} for subarrays that start
# at index 0. Works with negative numbers and online.
def subarraySum(nums, target_sum):
    prefix_counts, prefix, count = {0: 1}, 0, 0
    for number in nums:
        prefix += number
        count += prefix_counts.get(prefix - target_sum, 0)
        prefix_counts[prefix] = prefix_counts.get(prefix, 0) + 1
    return count


# Task 2: Union-Find over building cells
# Idea: Make each building its own set and union it with
# the building to its right and below. The number of
# communities is the number of distinct roots. Union-Find
# also supports adding buildings later.
def communities(grid):
    cells = [(row, col) for row in range(len(grid)) for col in range(len(grid[0])) if grid[row][col] == "1"]
    parent = {cell: cell for cell in cells}

    def find(cell):
        while parent[cell] != cell:
            parent[cell] = parent[parent[cell]]
            cell = parent[cell]
        return cell
```

```
    for row, col in cells:
        for neighbour in ((row + 1, col), (row, col + 1)):
            if neighbour in parent:
                parent[find((row, col))] = find(neighbour)
    return len({find(cell) for cell in cells})
```

COMPLEXITY | Part 1: Time O(n) | Space O(n)
Part 2: Time O(m * n * alpha(m * n)) | Space O(m * n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 234

235 * Maximum-Sum Subarray With Equal Endpoints * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: per value keep the earliest start with the smallest total before it; candidate = total now - that earlier total

QUESTION INFO

QUESTION

Given an integer array nums , choose a (contiguous) subarray whose first and last elements are equal and whose sum is as large as possible. A single element is a valid subarray (its first and last elements coincide). Return the chosen subarray itself (as a list). If several subarrays tie for the maximum sum, return the one with the smallest starting index; if they still tie, return the shortest.

Function signature

def max_sum_equal_endpoints(nums: list[int]) -> list[int]:
    ...

Follow-up (as reported): Achieve O(n): for each value, you only ever care about pairing the earliest occurrence (with the best prefix sum) against later occurrences.

EXAMPLES

Example 1	
Input	`nums = [1, 3, -2, 4, 1]`
Output	`[1, 3, -2, 4, 1]`
Why	`The value 1 appears at indices 0 and 4; that whole window sums to 7, beating any single element.`

Example 2	
Input	`nums = [5, -10, 5]`
Output	`[5]`
Why	`The window from the first 5 to the last 5 sums to 0, which is worse than taking a single 5 (sum 5).`

Example 3	
Input	`nums = [-3, -1, -2]`
Output	`[-1]`
Why	`All values are distinct, so only single elements qualify; -1 is the largest.`

CONSTRAINTS

- 1 <= len(nums) <= 10^5 .
- -10^4 <= nums[i] <= 10^4 .

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(nums) <= 10^5 .
- -10^4 <= nums[i] <= 10^4 .

BRUTE FORCE | Every pair of equal endpoints

BRUTE-FORCE IDEA

Start with: Every pair of equal endpoints. Each end index looks back at every earlier index with the same value.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Every pair of equal endpoints
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1)

WHY THIS IS WEAKER | Each end index looks back at every earlier index with the same value.

HOW TO IMPROVE IT

per value keep the earliest start with the smallest total before it; candidate = total now - that earlier total

SHORT BEST SOLUTION

For each end the best start is the same value with the smallest prefix sum so far.

```python
# BASE QUESTION (optimal): Best earlier prefix per value
# Idea: For each end the best start is the same value with
# the smallest prefix sum so far.
def max_sum_equal_endpoints(nums):
    prefix = [0]
    for value in nums:
        prefix.append(prefix[-1] + value)
    best_start, best_key = {}, None
    for end_index, value in enumerate(nums):
        if value not in best_start or prefix[end_index] < prefix[best_start[value]]:
            best_start[value] = end_index
        start_index = best_start[value]
        candidate_key = (-(prefix[end_index + 1] - prefix[start_index]), start_index, end_index - start_index)
        if best_key is None or candidate_key < best_key:
            best_key = candidate_key
    return nums[best_key[1]:best_key[1] + best_key[2] + 1] if best_key else []
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 235

236 * Maximum Subarray Sum With a Minimum Length Constraint * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the smallest earlier total is only allowed from min_len positions back, so update it one step late

QUESTION INFO

QUESTION

This is the classic "maximum subarray" problem with the extra constraint the report mentioned ("Max sub array question with a weird constraint"). Given an integer array nums and an integer min_len , find the contiguous subarray (containing at least min_len elements) that has the largest sum, and return that sum.

QUESTION

Function signature

def max_subarray_min_len(nums: list[int], min_len: int) -> int:

"""Return the maximum sum over all contiguous subarrays of length >= min_len."""

...

Follow-up (as reported): The report said each question had a follow-up where a constraint was changed. Now bound the subarray length to the range [min_len, max_len] . Hint: maintain a sliding window of prefix-sum minima with a monotonic deque.

EXAMPLES

Example 1	
Input	`nums = [1, -2, 3, 4, -1], min_len = 2`
Output	`7`
Why	`The subarray [3, 4] (length 2) has sum 7, the largest among all subarrays of length >= 2.`

Example 2	
Input	`nums = [-1, -2, -3], min_len = 2`
Output	`-3`
Why	`Every element is negative. The best length->=2 subarray is [-1, -2] with sum -3.`

Example 3	
Input	`nums = [5, 4, -1, 7, 8], min_len = 1`
Output	`23`
Why	`With min_len = 1 this reduces to ordinary Kadane; the whole array sums to 23.`

CONSTRAINTS

- 1 <= min_len <= len(nums) <= 10^5
- -10^4 <= nums[i] <= 10^4
- The array is guaranteed to be non-empty and min_len never exceeds len(nums) , so a valid subarray always exists.
- Values may be negative; the answer can therefore be negative.

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= min_len <= len(nums) <= 10^5
- -10^4 <= nums[i] <= 10^4
- The array is guaranteed to be non-empty and min_len never exceeds len(nums) , so a valid subarray always exists.
- Values may be negative; the answer can therefore be negative.

BRUTE FORCE | Try every start and end

BRUTE-FORCE IDEA

Start with: Try every start and end. Every start index extends to every end index at least min_len long, O(n^2) sums.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every start and end
Start with: Try every start
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1)

WHY THIS IS WEAKER | Every start index extends to every end index at least min_len long, O(n^2) sums.

HOW TO IMPROVE IT

the smallest earlier total is only allowed from min_len positions back, so update it one step late

SHORT BEST SOLUTION

Sweep the end of the subarray; the best start is the smallest prefix sum that leaves at least min_len elements.

```
# BASE QUESTION (optimal): Prefix sums and a running
# minimum
# Idea: Sweep the end of the subarray; the best start is
# the smallest prefix sum that leaves at least min_len
# elements.
def max_subarray_min_len(nums, min_len):
    prefix = [0]
    for number in nums:
        prefix.append(prefix[-1] + number)
    best, smallest = None, float("inf")
    for end in range(min_len, len(nums) + 1):
        # the prefix that is allowed to start a valid subarray
        smallest = min(smallest, prefix[end - min_len])
        candidate = prefix[end] - smallest
        best = candidate if best is None else max(best, candidate)
    return best
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 236

237 * Find the maximum subarray sum. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the dict is just one number: the smallest earlier total; the best_sum subarray ends here with total - smallest (Kadane is the same idea)

QUESTION INFO

QUESTION

Given an array of integers nums, write a function maxSubarraySum to find the maximum sum of a contiguous subarray within the array and return that maximum sum. The subarray must be contiguous, meaning that the elements must appear consecutively in the original array. Examples Input: nums = [2, 3, -2, 4]
Output: 7 Explanation: Maximum sum is 2 + 3 + (-2) + 4 = 7.
Input: nums = [1, -1, -5, -4]
Output: 1 Explanation: The maximum sum is 1, which is the single element with the highest value.Challenge 1Can you come up with a solution with a O(n) time complexity?Challenge 2Can you come up with a solution with a O(1) space complexity?

EXAMPLES

Example 1	
Input	[-2, 1, -3, 4, -1, 2, 1, -5, 4]
Output	6
Why	[4, -1, 2, 1] has the largest sum, 6.

Example 2	
Input	[1]
Output	1
Why	A single element.

QUESTION

Example 3	
Input	`[5, 4, -1, 7, 8]`
Output	`23`
Why	`All numbers are positive: take them all.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every subarray

BRUTE-FORCE IDEA

For each start extend the end with a running sum and keep the maximum. This takes quadratic time.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
best = negative infinity
for each start index:
    running_sum = 0
    extend the end index and add each value
    update best with running_sum
return best
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1)

WHY THIS IS WEAKER | O(n^2) subarrays. Kadane's algorithm keeps the best sum ending at each position (extend or restart) in O(n) time and O(1) space.

HOW TO IMPROVE IT

the dict is just one number: the smallest earlier total; the best_sum subarray ends here with total - smallest (Kadane is the same idea)

SHORT BEST SOLUTION

current_sum is the best_sum sum of a subarray ending at the current element: either the element alone or the element added to the previous current_sum. best_sum is the maximum current_sum seen. A negative running sum is never worth carrying forward.

```python
# BASE QUESTION (optimal): Kadane's algorithm
# Idea: current_sum is the best_sum sum of a subarray ending at the
# current element: either the element alone or the element
# added to the previous current_sum. best_sum is the maximum current_sum seen.
# A negative running sum is never worth carrying forward.
def maxSubArray(nums):
    current_sum = best_sum = nums[0]
    for number in nums[1:]:
        current_sum = max(number, current_sum + number)
        best_sum = max(best_sum, current_sum)
    return best_sum
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 237

Group by a canonical key

PATTERN CONTEXT

Find a key that is the same for everything that should be grouped (sorted letters, letter counts, shifts between letters, the keyboard row) and put the items into a dict under that key. Equal key means same group.

240 * Check if all substrings are anagrams of words * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: key = the tuple of letter counts; keep the keys of the allowed words in a set and test every substring of length 3 or more against it

QUESTION INFO

QUESTION

You are given a string **`s`** (letters only) and access to an English dictionary **`dict`** (a set of valid words). Return **`true`** if **every contiguous substring** of **`s`** with length **`>= 3`** can be **rearranged (anagrammed)** into a valid dictionary word. Formally, for every substring **`t = s[i..j]`** with **`|t| >= 3`**, there exists some word **`w in dict`** such that **`w`** is a permutation of **`t`** (same multiset of characters). Otherwise return **`false`**. Clarifications:

- Substring means **contiguous**. - “Can be rearranged into a word” means the substring and the dictionary word have exactly the same character counts. Example:
- If **`t = "act"`**, it passes if the dictionary contains any of **`{ "act", "cat", "tac" }`**.

EXAMPLES

Example 1	
Input	`'abca', ['abc', 'cab', 'bca', 'abca']`
Output	`true`
Why	`Every substring of length 3 or more of 'abca' is an anagram of a dictionary word.`

Example 2	
Input	`'abca', ['abc']`
Output	`false`
Why	`'bca' is an anagram of 'abc' but 'abca' has no match.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Compare each substring with every word's sorted letters

BRUTE-FORCE IDEA

For each substring, sort its letters and compare with the sorted letters of every dictionary word. Correct, and the dictionary is rescanned for each substring.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compare each substring with every word's sorted letters
For each substring
sort its letters
compare with the sorted letters of every dictionary word
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2 * W * L log L) | Space O(1) extra

WHY THIS IS WEAKER | Each substring is compared with every word and sorted again: O(n^2 * W * L log L). A set of word signatures, with the substring's count vector updated incrementally, makes each check O(alphabet).

HOW TO IMPROVE IT

key = the tuple of letter counts; keep the keys of the allowed words in a set and test every substring of length 3 or more against it

SHORT BEST SOLUTION

Precompute the set of count-vector signatures of the dictionary words. For each start extend one letter at a time, update the count vector in O(1), and look up its tuple (once the length is at least 3). Return False at the first missing signature.

```
# BASE QUESTION (optimal): Set of signatures, incremental
# count vectors
# Idea: Precompute the set of count-vector signatures of
# the dictionary words. For each start extend one letter
# at a time, update the count vector in O(1), and look up
# its tuple (once the length is at least 3). Return False
# at the first missing signature.
def all_anagram_words(text, words, alphabet="abcdefghijklmnopqrstuvwxyz"):
    letter_index = {letter: position for position, letter in enumerate(alphabet)}
    signatures = set()
    for word in words:
        counts = [0] * len(alphabet)
        for letter in word:
            counts[letter_index[letter]] += 1
        signatures.add(tuple(counts))
    for start in range(len(text)):
        counts = [0] * len(alphabet)
        for end in range(start, len(text)):
            counts[letter_index[text[end]]] += 1
            if end - start + 1 >= 3 and tuple(counts) not in signatures:
                return False
    return True
```

COMPLEXITY | Time O(n^2 * A) for alphabet size A | Space O(dictionary size)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 240

241 * Group anagrams * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: key = the sorted letters (or a tuple of 26 letter counts)

QUESTION INFO

QUESTION

Given a list of words, group together the words that are anagrams of each other (the same letters in a different order). Return the groups in any order.

Function signature

def group_by_sorting(words: list) -> list

EXAMPLES

Example 1	
Input	`['eat', 'tea', 'tan', 'ate', 'nat', 'bat']`
Output	`[['eat', 'tea', 'ate'], ['tan', 'nat'], ['bat']]`
Why	`eat, tea and ate use the same letters; tan and nat; bat is alone.`

QUESTION

Example 2	
Input	`['']`
Output	`[['']]`
Why	`An empty string is its own group.`

Example 3	
Input	`['a']`
Output	`[['a']]`
Why	`One word.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Compare each word with a representative of every group

BRUTE-FORCE IDEA

For each word compare its sorted letters with those of the first word of every existing group; start a new group when none match. Correct; each word is compared with many groups.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compare each word with a representative of every group
For each word compare its sorted letters with those of the first word of every existing
group
start a new group when none match
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n * g * L log L) | Space O(n)

WHY THIS IS WEAKER | Each word is compared with many groups, O(n * groups * L log L). Using the sorted letters (or a letter-count tuple) as a dictionary key groups all words in one pass.

HOW TO IMPROVE IT

key = the sorted letters (or a tuple of 26 letter counts)

SHORT BEST SOLUTION

Sort the letters of each word and use the resulting string as the key; words with the same key are anagrams. One pass, one dictionary.

```python
from collections import defaultdict


# BASE QUESTION (optimal): Dictionary keyed by the sorted
# letters
# Idea: Sort the letters of each word and use the
# resulting string as the key; words with the same key are
# anagrams. One pass, one dictionary.
def groupAnagrams(words):
    groups = defaultdict(list)
    for word in words:
        groups["".join(sorted(word))].append(word)
    return list(groups.values())
```

COMPLEXITY | Time O(n * L log L) | Space O(n * L)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 241

242 * Given an array of strings, return groups of "buddy strings," where the strings have the same length and the distance between corresponding characters is the same. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: key = the differences between neighbouring letters, modulo 26 (a shifted word has the same differences)

QUESTION INFO

QUESTION

Two strings are buddy strings when they have the same length and the distance between each pair of neighbouring letters is the same (the alphabet wraps around after z), so one is the other shifted by a constant. Group the strings into buddy groups and return the groups in any order.

Function signature

def group_buddies(words: list) -> list

EXAMPLES

Example 1	
Input	`['abc', 'bcd', 'acf', 'xyz', 'az', 'ba', 'a', 'z']`
Output	`[['abc', 'bcd', 'xyz'], ['acf'], ['az', 'ba'], ['a', 'z']]`
Why	`'abc', 'bcd' and 'xyz' all step +1, +1; 'az' and 'ba' both step +25 (a to z wraps); single letters 'a' and 'z' form a group.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Compare each string with one representative of every existing group

BRUTE-FORCE IDEA

For each string look through the existing groups and test it against the group's first member (same length and a constant shift); start a new group if none fits. Correct; each string is compared with many groups.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compare each string with one representative of every existing group
For each string look through the existing groups
test it against the group's first member (same length
a constant shift)
start a new group if none fits
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n * groups * L) | Space O(n)

WHY THIS IS WEAKER | Each string is compared with the representative of every group, O(n * groups * L). Using the sequence of gaps between neighbouring letters as a dictionary key groups everything in one pass.

HOW TO IMPROVE IT

key = the differences between neighbouring letters, modulo 26 (a shifted word has the same differences)

SHORT BEST SOLUTION

Two strings are buddies when every letter is shifted by the same amount, which means the gaps between neighbouring letters (mod 26) are identical. Use those gaps as the hash key.

```python
from collections import defaultdict


# BASE QUESTION (optimal): Group by the tuple of
# successive differences
```

```python
# Idea: Two strings are buddies when every letter is
# shifted by the same amount, which means the gaps between
# neighbouring letters (mod 26) are identical. Use those
# gaps as the hash key.
def groupBuddyStrings(strings):
    groups = defaultdict(list)
    for word in strings:
        key = tuple((ord(word[position]) - ord(word[position - 1])) % 26 for position in range(1, len(word)))
        groups[key].append(word)
    return list(groups.values())
```

COMPLEXITY | Time O(total characters) | Space O(total characters)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 242

K-way merge: a heap of the next item from each source

PATTERN CONTEXT

Put the first item of every list (with its list number and position) on a heap. Pop the smallest, record it, and push the next item from the same list. After k pops you have the k best without sorting everything.

245 * K Largest Elements Across Sorted Lists * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: seed with the LAST item of each list (negated); the successor is the item before it; reverse the result

QUESTION INFO

QUESTION

You are given a list of inner lists, where each inner list is sorted in ascending order. Return the k largest elements taken from the union of all the inner lists, and return them sorted in ascending order. Duplicate values are kept (an element counts once per position it occupies). The interviewer first accepted a simple solution, then asked for an optimized one.

Function signature

```
from typing import List
def k_largest_sorted(lists: List[List[int]], k: int) -> List[int]:
    """Return the k largest elements across all (individually ascending) inner
    lists, in ascending sorted order."""
    ...
```

Follow-up (as reported): Optimize beyond concatenate-and-sort. Hint: because each list is already sorted, seed a max-heap with the last element of every non-empty list and pop k times, pushing the predecessor within a list after each pop; collect the popped values and reverse them.

EXAMPLES

Example 1	
Input	`lists = [[1, 4, 7], [2, 5, 8], [3, 6, 9]], k = 4`
Output	`[6, 7, 8, 9]`
Why	`The four largest values overall are 9, 8, 7, 6, returned ascending.`

Example 2	
Input	`lists = [[10, 20, 30]], k = 2`
Output	`[20, 30]`

Example 3	
Input	`lists = [[5], [1, 3]], k = 5`
Output	`[1, 3, 5]`
Why	`k exceeds the total count, so all elements are returned sorted.`

Example 4	
Input	`lists = [[1, 2], [3, 4]], k = 0`
Output	`[]`

CONSTRAINTS

- Let m be the number of inner lists and T the total number of elements across all of them.
- 0 <= m <= 10^4 , 0 <= T <= 10^6 .

QUESTION

- Each element is an integer in [-10^9, 10^9] .
- 0 <= k . If k == 0 return [] . If k >= T , return every element sorted ascending.
- Some inner lists may be empty.

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- Let m be the number of inner lists and T the total number of elements across all of them.
- 0 <= m <= 10^4 , 0 <= T <= 10^6 .
- Each element is an integer in [-10^9, 10^9] .
- 0 <= k . If k == 0 return [] . If k >= T , return every element sorted ascending.

BRUTE FORCE | Concatenate and sort

BRUTE-FORCE IDEA

Start with: Concatenate and sort. Start with: Concatenate and sort. It sorts all N elements although only k are needed.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Concatenate and sort
Start with: Concatenate
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(N log N) | Space O(N)

WHY THIS IS WEAKER | It sorts all N elements although only k are needed.

HOW TO IMPROVE IT

seed with the LAST item of each list (negated); the successor is the item before it; reverse the result

SHORT BEST SOLUTION

The next largest element is always at the end of one of the lists.

```python
# BASE QUESTION (optimal): A max-heap over the ends of the
# lists
# Idea: The next largest element is always at the end of
# one of the lists.
import heapq


def k_largest_sorted(lists, how_many):
    heap = [(-numbers[-1], list_index, len(numbers) - 1) for list_index, numbers in enumerate(lists) if numbers
]
    heapq.heapify(heap)
    largest = []
    while heap and len(largest) < how_many:
        negated, list_index, position = heapq.heappop(heap)
        largest.append(-negated)
        if position:
            heapq.heappush(heap, (-lists[list_index][position - 1], list_index, position - 1))
    return largest[::-1]
```

COMPLEXITY | Time O(m + k log m) | Space O(m)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 245

246 * Sort a nearly sorted (or K sorted) array. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: keep a heap of the first max_distance + 1 items; pop the smallest and push the next input item (heapreplace)

QUESTION INFO

QUESTION

In a k-sorted array every element is at most k positions away from where it would be in the sorted array. Sort such an array in O(n log k) time (better than a full sort when k is small).

Function signature

def sort_k_sorted(arr: list, k: int) -> list

EXAMPLES

Example 1	
Input	`[6, 5, 3, 2, 8, 10, 9], 3`
Output	`[2, 3, 5, 6, 8, 9, 10]`
Why	`Every number is within 3 positions of its place in the sorted order.`

Example 2	
Input	`[2, 1, 4, 3], 1`
Output	`[1, 2, 3, 4]`
Why	`k = 1: neighbours may be swapped.`

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Sort the whole array

BRUTE-FORCE IDEA

The library sort is correct and the right answer when K is large; it ignores the structure, so it does more work than needed when K is small.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Sort the whole array
The library sort is correct
the right answer when K is large
it ignores the structure
so it does more work than needed when K is small
```

BRUTE-FORCE COMPLEXITY | Time O(n log n) | Space O(n)

WHY THIS IS WEAKER | O(n log n) regardless of K. Each element is at most K places from its sorted position, so a min-heap holding a window of K + 1 elements emits the sorted order in O(n log K).

HOW TO IMPROVE IT

keep a heap of the first max_distance + 1 items; pop the smallest and push the next input item (heapreplace)

SHORT BEST SOLUTION

Push the first K + 1 elements. Then for each remaining element, pop the smallest into the output and push the new element. At the end pop the heap empty. The smallest of any K + 1 consecutive pending elements is the next element of the sorted order.

```python
import heapq


# BASE QUESTION (optimal): Min-heap of size K + 1
```

```python
# Idea: Push the first K + 1 elements. Then for each
# remaining element, pop the smallest into the output and
# push the new element. At the end pop the heap empty. The
# smallest of any K + 1 consecutive pending elements is
# the next element of the sorted order.
def sort_nearly(values, max_distance):
    heap = list(values[:max_distance + 1])
    heapq.heapify(heap)
    sorted_values = []
    for value in values[max_distance + 1:]:
        sorted_values.append(heapq.heapreplace(heap, value))
    while heap:
        sorted_values.append(heapq.heappop(heap))
    return sorted_values
```

COMPLEXITY | Time O(n log K) | Space O(K)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 246

Tree recursion, Tier 2: pass a rule down, return a node or a yes/no up

PATTERN CONTEXT

Same recursive walk as Tier 1, but now the information also travels down: lower and upper bounds for a binary search tree, or the target nodes you are looking for. The BST rule (left < node < right) lets you go left or right instead of visiting everything.

250 * Find mode in a trinary search tree * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: in-order walk (left, middle, right); equal values are neighbours, so one running count finds the mode(s)

QUESTION INFO

QUESTION

You are given the root of a **trinary search tree** where each node has up to three children:

```
class Node { int val; Node left; // all values < val Node middle; // all values = val Node right; // all v
alues > val }
```

Assume the tree is intended to satisfy the ordering rules above.

Task

Return the **mode(s)** of the tree: the value(s) that occur most frequently among all nodes.

- If there is one mode, return a single-element list.
- If multiple values tie for highest frequency, return **all** such values (in any order).

Constraints (typical)

- Number of nodes **n** can be large (e.g., up to **`10^5`**).
- Values can be any 32-bit signed integers.

Follow-up

How would your approach change if the tree is **broken** (i.e., it may not respect the **<**, **=**, **>** constraints), but you still must return the correct mode(s)?

Part 1: Find Mode(s) in a Valid Trinary Search TreeYou are given a valid trinary search tree. Each node has an integer value and up to three children: left, middle, and right. For every node, all values in its left subtree are strictly less than the node's value, all values in its middle subtree are equal to the node's value, and all values in its right subtree are strictly greater than the node's value. The tree is encoded using parallel arrays instead of Node objects. Return the value or values that occur most frequently in the tree. To make the output deterministic, return the modes in increasing order. If the tree is empty, return an empty list.
Constraints
0 <= n <= 100000, where n == len(values)len(left) == len(middle) == len(right) == nEach child index is either -1 or an integer in [0, n - 1]root == -1 if n == 0; otherwise 0 <= root < nThe nodes reachable from root form a valid tree with no cyclesThe tree satisfies the trinary search ordering rules recursivelyEach value is a 32-bit signed integer

EXAMPLES

```
Input: ([], [], [], [], -1)
Expected Output: []
```

Explanation: The tree is empty, so there are no modes.

```
Input: ([42], [-1], [-1], [-1], 0)
Expected Output: [42]
```

Explanation: A single-node tree has that node's value as the only mode.

QUESTION

Hints

In a valid trinary search tree, an inorder traversal using left, node, middle, right visits values in nondecreasing order.Because equal values appear consecutively during that traversal, you can track only the current run length and the best run length so far.

Part 2: Find Mode(s) in a Broken Trinary TreeYou are given a trinary tree with integer values. Each node has up to three children: left, middle, and right. Unlike a valid trinary search tree, this tree may be broken: the left, middle, and right children do not necessarily follow any ordering rules. Return the value or values that occur most frequently among all reachable nodes. To make the output deterministic, return the modes in increasing order. If the tree is empty, return an empty list.

Constraints

0 <= n <= 100000, where n == len(values)len(left) == len(middle) == len(right) == nEach child index is either -1 or an integer in [0, n - 1]root == -1 if n == 0; otherwise 0 <= root < nThe nodes reachable from root form a tree with no cyclesNo ordering rule is guaranteed between a node and its childrenEach value is a 32-bit signed integer

Examples

Input: ([], [], [], [], -1)

Expected Output: []

Explanation: The tree is empty, so there are no modes.

Input: ([-2147483648], [-1], [-1], [-1], 0)

Expected Output: [-2147483648]

Explanation: A single-node tree has that node's value as the only mode.

Hints

Without the ordering property, equal values are not guaranteed to be adjacent in any traversal.Use a frequency table while traversing the entire tree.

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Collect every value in a Counter

BRUTE-FORCE IDEA

Traverse the whole tree, count each value and return the ones with the highest count. Correct for any tree (even a broken one); it uses memory proportional to the number of distinct values.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Collect every value in a Counter
Traverse the whole tree
count each value
return the ones with the highest count
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | O(n) time but memory for every distinct value. In a valid tree an in-order walk (left, middle, right) gives equal values consecutively, so a running counter needs only O(height) space.

HOW TO IMPROVE IT

in-order walk (left, middle, right); equal values are neighbours, so one running count finds the mode(s)

SHORT BEST SOLUTION

Left, node, middle copies and right produces sorted values, so equal values form one run and no frequency dictionary is needed.

```python
# BASE QUESTION (optimal): Recursive in-order traversal
# Idea: Left, node, middle copies, right produces sorted
# values, so count equal-value runs without a dictionary.
def find_modes(root):
```

```
    best = run = 0
    current = None
    modes = []

    def add(value):
        nonlocal best, run, current, modes
        run = run + 1 if value == current else 1
        current = value
        if run > best:
            best, modes = run, [value]
        elif run == best:
            modes.append(value)

    def inorder(node):
        if node is None:
            return
        inorder(node.left)
        add(node.val)
        middle = node.middle
        while middle:
            add(middle.val)
            middle = middle.middle
        inorder(node.right)

    inorder(root)
    return modes
```

COMPLEXITY | Time O(n) | Space O(h) recursion

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 250

251 * Next Upcoming Birthday in a Family Tree * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: walk the ancestors with a stack and a seen set, keep the best birthday by the circular distance from today

QUESTION INFO

QUESTION

You are given a family tree rooted at "me". Each Person node has a name , a birthday given as (month, day) , and pointers to parent1 and parent2 (either may be None). Following parent pointers reaches the whole ancestry; treat the reachable set of Person nodes as the tree to search (each distinct person counted once, even if reachable by two paths). Given today's date (today_month, today_day) , implement alarm_next_birthday(me, today_month, today_day) that scans every person and returns the (month, day) of the soonest upcoming birthday, where a birthday landing exactly on today counts as upcoming (distance 0) and the calendar wraps around the year. If the tree is empty (me is None), return the empty string "" . Use this ordering key to compare dates: key(m, d) = (m - 1) * 31 + d , and cyclic distance (key(person) - key(today)) mod 372 . Return the person with the smallest such distance; if several people share that soonest date, return that (month, day) .

QUESTION

Function signature

```
from typing import Optional, Union
class Person:
    def __init__(self, name: str, month: int, day: int,
                 parent1: "Optional[Person]" = None,
                 parent2: "Optional[Person]" = None) -> None:
        self.name = name
        self.month = month
        self.day = day
        self.parent1 = parent1
        self.parent2 = parent2
def alarm_next_birthday(me: Optional[Person],
                        today_month: int,
                        today_day: int) -> Union[tuple[int, int], str]:
    """Return (month, day) of the nearest upcoming birthday, or "" if there is
nobody."""
    ...
```

Follow-up (as reported): Support repeated alarms: after reporting one birthday, the next call returns the following-soonest, and once all have been reported, return "" .

EXAMPLES

Example 1	
Input	`me = Me(12,1) with parent1 = Alice(9,25), parent2 = Bob(1,10); today = (9, 12)`
Output	`(9, 25)`
Why	`keys — today=260, Alice=273 (dist 13), Me=342 (dist 82), Bob=10 (dist (10- 260) mod 372 = 122). Alice is soonest.`

Example 2	
Input	`me = None; today = (5, 5)`
Output	`""`
Why	`No people to check.`

Example 3	
Input	`me = Me(3,3) alone; today = (3, 3)`
Output	`(3, 3)`
Why	`The only birthday is today, distance 0.`

CONSTRAINTS

- 1 <= month <= 12 , 1 <= day <= 31 .
- Up to 10^5 people; parent pointers form a DAG (no cycles).

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= month <= 12 , 1 <= day <= 31 .
- Up to 10^5 people; parent pointers form a DAG (no cycles).

BRUTE FORCE | Collect everyone and sort by distance

BRUTE-FORCE IDEA

Start with: Collect everyone and sort by distance. It sorts all p people although only the nearest is needed.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Collect everyone and sort by distance
Start with: Collect everyone
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(p log p) | Space O(p)

WHY THIS IS WEAKER | It sorts all p people although only the nearest is needed.

HOW TO IMPROVE IT

walk the ancestors with a stack and a seen set, keep the best birthday by the circular distance from today

SHORT BEST SOLUTION

DFS through both parents, visit each person once and keep the birthday with the smallest wrapped distance from today.

```python
# BASE QUESTION (optimal): Recursive family-tree walk
# Idea: DFS through both parents, visit each person once,
# and keep the birthday with the smallest wrapped distance.
def next_birthday(me, today_month, today_day):
    today = (today_month - 1) * 31 + today_day
    seen = set()
    best = None

    def dfs(person):
        nonlocal best
        if person is None or person in seen:
            return
        seen.add(person)
        day = (person.month - 1) * 31 + person.day
        candidate = ((day - today) % 372, person.month, person.day)
        if best is None or candidate < best:
            best = candidate
        dfs(person.parent1)
        dfs(person.parent2)

    dfs(me)
    return "" if best is None else (best[1], best[2])
```

COMPLEXITY | Time O(n) | Space O(n) for seen and recursion

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 251

252 * Lowest Common Ancestor * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: dfs returns the node if it is p or q; a node that gets an answer from both sides is the ancestor

QUESTION INFO

QUESTION

Given the root of a binary tree and two of its nodes p and q, find their lowest common ancestor: the lowest node that has both p and q as descendants (a node can be a descendant of itself).

Function signature

def lowestCommonAncestor_tree(root: BNode, p: BNode, q: BNode) -> BNode

EXAMPLES

QUESTION

Example 1

Input	`root = [3, [5, [6], [2]], [1]] (value, left, right)`
	`first query: p = 5, q = 1`
	`second query: p = 5, q = 6`

Output	`lowestCommonAncestor_tree(root, p, q).val -> 3` `lowestCommonAncestor_tree(root, p, p.left).val -> 5`
Why	`5 and 1 are in different subtrees of 3, so 3 is their lowest common ancestor. 5 is an ancestor of 6, and a node may be its own ancestor, so the answer is 5.`

CONSTRAINTS

- Number of nodes in tree: [2, 10^5]
- -10^9 <= Node.val <= 10^9
- All node values are unique
- p != q
- p and q exist in the tree

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- Number of nodes in tree: [2, 10^5]
- -10^9 <= Node.val <= 10^9
- All node values are unique
- p != q

BRUTE FORCE | Find the root-to-node path for each node and compare

BRUTE-FORCE IDEA

Compute the path from the root to p and to q, then the last common node on the two paths is the LCA. Correct and easy to explain; it stores two full paths.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Find the root-to-node path for each node and compare
Compute the path from the root to p
to q
the last common node on the two paths is the LCA
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | Two full paths stored, O(n) space, and two traversals. One recursive pass that reports whether each subtree contains p or q finds the answer in O(n) time and O(h) space.

HOW TO IMPROVE IT

dfs returns the node if it is p or q; a node that gets an answer from both sides is the ancestor

SHORT BEST SOLUTION

dfs(node) returns node if node is p or q, otherwise combines its children: if both children return something, node is the LCA; if only one does, pass that up. The first node that sees both p and q below it is returned all the way to the top.

```python
# BASE QUESTION (optimal): One recursive pass
# Idea: dfs(node) returns node if node is p or q,
# otherwise combines its children: if both children return
# something, node is the LCA; if only one does, pass that
# up. The first node that sees both p and q below it is
# returned all the way to the top.
class BNode:
    def __init__(self, val, left=None, right=None):
```

```
        self.val, self.left, self.right = val, left, right

def lowestCommonAncestor_tree(root, first_target, second_target):
    def dfs(node):
        if node is None or node is first_target or node is second_target:
            return node
        left, right = dfs(node.left), dfs(node.right)
        return node if left and right else left or right

    return dfs(root)
```

COMPLEXITY | Time O(n) | Space O(h)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 252

253 * Determine if a given binary tree is a binary search tree (BST). * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: check(): pass lo and hi bounds down; fail when a value leaves its allowed window

QUESTION INFO

QUESTION

Given the root of a binary tree, determine whether it is a valid binary search tree: for every node, all values in its left subtree are smaller and all values in its right subtree are larger (strictly).

Function signature

def isValidBST(root: TreeNode) -> bool

EXAMPLES

Example 1	
Input	`[2, 1, 3]`
Output	`true`
Why	`2 with 1 on the left and 3 on the right.`

Example 2	
Input	`[5, 1, 4, null, null, 3, 6]`
Output	`false`
Why	`The 3 sits in the right subtree of 5 but is smaller than 5.`

Example 3	
Input	`[2, 2, 2]`
Output	`false`
Why	`A node equal to its parent breaks the strict rule.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | In-order list must be strictly increasing

BRUTE-FORCE IDEA

Collect the in-order traversal into a list and check it is strictly increasing. Correct; it stores every value.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: In-order list must be strictly increasing
Collect the in-order traversal into a list
check it is strictly increasing
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | O(n) time but stores every value, O(n) space. Passing the allowed (low, high) range down the tree checks each node once with O(height) space and stops at the first violation.

HOW TO IMPROVE IT

check(): pass lo and hi bounds down; fail when a value leaves its allowed window

SHORT BEST SOLUTION

Every node must be strictly between the bounds set by its ancestors; each child tightens one bound.

```python
# BASE QUESTION (optimal): Pass valid bounds down recursively
# Idea: Every node must be strictly between the bounds set
# by its ancestors; each child tightens one bound.
def isValidBST(root):
    def valid(node, low, high):
        if node is None:
            return True
        if not low < node.val < high:
            return False
        return (valid(node.left, low, node.val)
                and valid(node.right, node.val, high))

    return valid(root, float("-inf"), float("inf"))
```

COMPLEXITY | Time O(n) | Space O(h) recursion

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 253

Random pick without repeats: swap the chosen item with the last one and pop

PATTERN CONTEXT

To pick a random item and never pick it again in O(1), keep the items in an array. Choose a random index, swap that item with the last one, pop the last, and fix its position in the index dict. The same trick gives O(1) delete and O(1) random access.

254 * Design set with O(1) random access

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: add appends and records the position; delete swaps with the last item and pops; get_random picks a random index

QUESTION INFO

QUESTION

Design a data structure ("FancySet") that stores unique integers and supports the following operations, each in average O(1) time:

- **`add(x)`**: Insert value **`x`** if it is not already present. Return whether the insert happened.
- **`delete(x)`**: Remove value **`x`** if it exists. Return whether a deletion happened.
- **`getRandom()`**: Return a uniformly random element currently in the set. Requirements / follow-ups:
- All operations should be average O(1).
- Handle corner cases such as calling **`getRandom()`** on an empty set (define and justify behavior).
- Discuss how you would make this data structure thread-safe. Compare using a single lock for the whole set vs finer-grained locking (e.g., element-level locks), and explain trade-offs.

Design a data structure called FancySet that stores unique integers and supports these operations in average O(1) time: add(x), delete(x), and getRandom(). For this coding problem, implement a function **`solution(operations, values, seed)`** that simulates a newly created empty FancySet and returns the result of every operation. Use the standard dynamic-array + hash-map behavior:

- Store current elements in an array-like list.
- Store each element's current index in a hash map.
- When deleting an element that is not at the end of the list, move the last element into the removed slot, then pop the last slot. To make test cases deterministic, **`getRandom()`** does not use the language's built-in RNG. Instead, keep an internal 32-bit state initialized to **`seed`**. When **`getRandom()`** is called on a non-empty set, update the state with: **`state = (1664525 * state + 1013904223) mod 2^32`** and return the element at index **`state % current_size`** from the current internal list. If **`getRandom()`** is called on an empty set, return **`None`** and do not advance the state. In a real interview, picking a random index from the array gives uniform random access; the fixed generator above is only for reproducible grading. Follow-up discussion (not graded by the code runner): explain how you would make FancySet thread-safe, and compare using one lock for the whole structure versus finer-grained locking.

Constraints

1 <= len(operations) <= 2 * 10^5len(values) == len(operations)Each operation is one of 'add', 'delete', or 'getRandom'For 'add' and 'delete', values[i] contains exactly one integer; for 'getRandom', values[i] is []-10^9 <= x <= 10^9All operations should run in O(1) average time

EXAMPLES

```
Input: (['add','add','getRandom','delete','getRandom','add','getRandom'], [[1],[2],[],[1],[],[2],[]], 7)
Expected Output: [True, True, 1, True, 2, False, 2]
```

QUESTION

Explanation: After inserting 1 and 2, the first deterministic random choice picks index 0, so it returns 1. Deleting 1 leaves only 2. Adding 2 again fails because it is already present.

```
Input: (['getRandom','add','add','delete','delete','getRandom'], [[],[5],[5],[7],[5],[]], 1)
Expected Output: [None, True, False, False, True, None]
```

Explanation: Calling getRandom on an empty set returns None. Adding 5 twice inserts it only once. Deleting 7 fails because it is not present. After deleting 5, the set becomes empty again.

Hints

You need both fast membership/index lookup and fast access by numeric position. What two data structures combine to give that?Deleting from the middle of an array is expensive unless you replace the removed element with the last element and update that last element's stored index.

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | A plain list

BRUTE-FORCE IDEA

Store values in a list; add checks for membership with a scan and delete removes by search. Correct; membership and deletion are linear.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: A plain list
Store values in a list
add checks for membership with a scan
delete removes by search
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) for add and delete, O(1) getRandom | Space O(n)

WHY THIS IS WEAKER | Membership checks and deletion are O(n). An array plus a dictionary from value to index, deleting by swapping with the last element, makes all three operations O(1).

HOW TO IMPROVE IT

add appends and records the position; delete swaps with the last item and pops; get_random picks a random index

SHORT BEST SOLUTION

add appends to the array and records the index. delete finds the index from the dict, moves the last element into that slot, updates its index in the dict, pops the array and removes the key. getRandom picks a uniform index with random.randrange; every element is equally likely.

```python
import random


# BASE QUESTION (optimal): Array of values plus a dict
# from value to index
# Idea: add appends to the array and records the index.
# delete finds the index from the dict, moves the last
# element into that slot, updates its index in the dict,
# pops the array and removes the key. getRandom picks a
# uniform index with random.randrange; every element is
# equally likely.
class FancySet:
    def __init__(self):
        self.values, self.index_of = [], {}

    def add(self, value):
        if value in self.index_of:
```

```
            return False
        self.index_of[value] = len(self.values)
        self.values.append(value)
        return True

    def delete(self, value):
        if value not in self.index_of:
            return False
        slot, last_value = self.index_of[value], self.values[-1]
        self.values[slot] = last_value
        self.index_of[last_value] = slot
        self.values.pop()
        del self.index_of[value]
        return True

    def getRandom(self):
        if not self.values:
            raise IndexError("empty set")
        return self.values[random.randrange(len(self.values))]
```

COMPLEXITY | Time O(1) average per operation | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 254

255 * Random Generator Without Repeats

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: the template as written: values 0..n-1, every call returns a new one; raise when empty

QUESTION INFO

QUESTION

Design a class RandomGenerator initialized with a positive integer n . Each call to generate() returns one integer chosen uniformly at random from the values in [0, n-1] that have not yet been returned. After n calls every value in [0, n-1] has been returned exactly once; a further call must raise IndexError (the pool is empty). Each generate() call must run in O(1) time. The intended approach maintains an array of the remaining values and, on each call, swaps a randomly chosen remaining value to the end of the active region and shrinks that region (partial Fisher-Yates).

Function signature

```
import random
class RandomGenerator:
    def __init__(self, n: int) -> None:
        """Prepare to yield a random permutation of [0, n-1], one value per
generate()."""
        ...
    def generate(self) -> int:
        """Return a uniformly random not-yet-returned value in O(1).
        Raise IndexError if all values have already been generated."""
        ...
```

Follow-up (as reported): Add a reset() that makes all n values available again in O(1) amortized.

EXAMPLES

QUESTION

Example 1	
Input	`g = RandomGenerator(3); [g.generate() for _ in range(3)]`
Output	`e.g. [2, 0, 1]`
Why	`The three calls return a permutation of {0, 1, 2} with no repeats; any of the 6 permutations is a valid output. A 4th call would raise IndexError.`

Example 2	
Input	`g = RandomGenerator(1); g.generate()`
Output	`0`
Why	`Only one value exists.`

CONSTRAINTS

- 1 <= n <= 10**6 .

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= n <= 10**6 .

BRUTE FORCE | Shuffle once and pop

BRUTE-FORCE IDEA

Start with: Shuffle once and pop. A reset needs a new O(n) shuffle, and the whole permutation is fixed in advance.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Shuffle once and pop
Start with: Shuffle once
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) setup, O(1) per call | Space O(n)

WHY THIS IS WEAKER | A reset needs a new O(n) shuffle, and the whole permutation is fixed in advance.

HOW TO IMPROVE IT

the template as written: values 0..n-1, every call returns a new one; raise when empty

SHORT BEST SOLUTION

Swap the drawn value to the end of the active region and shrink it.

```python
import random


# BASE QUESTION (optimal): Partial Fisher-Yates with an
# active region
# Idea: Swap the drawn value to the end of the active
# region and shrink it.
class RandomGenerator:
    def __init__(self, count, seed=None):
        self.count = count
        self.values = list(range(count))
        self.size = count
        self.rng = random.Random(seed)

    def generate(self):
        if self.size == 0:
            raise IndexError("all values have been generated")
```

```
        pick = self.rng.randrange(self.size)
        self.size -= 1
        self.values[pick], self.values[self.size] = self.values[self.size], self.values[pick]
        return self.values[self.size]
```

COMPLEXITY | Time O(1) per call | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 255

256 * Random Song Shuffler With a No-Repeat Window * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the same pool, plus a deque of the last k songs; the played song leaves the pool and returns to it when it drops off the deque

QUESTION INFO

QUESTION

Design a shuffler that plays songs at random from a fixed playlist, subject to one rule: the last k songs played may not be replayed. In other words, a song becomes eligible again only after k other (distinct) plays have occurred since it was last played. Implement the class so that each call to play() runs in O(1) time (no scanning of the playlist). The intended technique is a valid_songs list of currently eligible songs plus a song_to_index dictionary, using swap-and-pop to move a just-played song out of the eligible region and to re-admit the song that is leaving the window.

Function signature

```
import random
class SongShuffler:
    def __init__(self, songs: list[str], k: int) -> None:
        """Initialize with the playlist and the no-repeat window size k."""
        ...
    def play(self) -> str:
        """Return a uniformly random eligible song in O(1); it must not be
        one of the last k songs returned. Update state so the window slides."""
        ...
```

Follow-up (as reported): The interviewer wanted the naive deque solution replaced by O(1). Show why swap-and-pop with an index map achieves constant time per play() .

EXAMPLES

Example 1	
Input	`SongShuffler(["A", "B", "C"], k=1); then play(), play(), play(), play()`
Output	`e.g. "B", "A", "C", "B"`
Why	`k=1 forbids repeating the single most recent song. "B" then "A" is fine (A != B). Then "C" (C != A). Then "B" is allowed again because the only forbidden song is the immediately previous one ("C"). Any sequence with no two adjacent equal songs is valid.`

Example 2	
Input	`SongShuffler(["X", "Y"], k=0); play(), play()`
Output	`e.g. "Y", "Y"`
Why	`With k=0 there is no restriction, so repeats are allowed.`

CONSTRAINTS

- 1 <= len(songs) <= 10**5 , song titles are distinct strings.

QUESTION

- 0 <= k < len(songs) (so at least one song is always eligible). play() returns any eligible song chosen uniformly at random; every returned song must not be among the previous k returned songs.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(songs) <= 10**5 , song titles are distinct strings.
- 0 <= k < len(songs) (so at least one song is always eligible). play() returns any eligible song chosen uniformly at random; every returned song must not be among the previous k returned songs.

BRUTE FORCE | Rebuild the eligible list on every play

BRUTE-FORCE IDEA

Start with: Rebuild the eligible list on every play. Every play scans the whole playlist to find the songs not in the window.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Rebuild the eligible list on every play
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n) per play | Space O(n)

WHY THIS IS WEAKER | Every play scans the whole playlist to find the songs not in the window.

HOW TO IMPROVE IT

the same pool, plus a deque of the last k songs; the played song leaves the pool and returns to it when it drops off the deque

SHORT BEST SOLUTION

Remove the played song from the eligible array by swapping it with the last element; the song leaving the window is appended back.

```python
from collections import deque
import random


# BASE QUESTION (optimal): Eligible array, index map and
# swap-and-pop
# Idea: Remove the played song from the eligible array by
# swapping it with the last element; the song leaving the
# window is appended back.
class SongShuffler:
    def __init__(self, songs, window_size, seed=None):
        self.eligible = list(songs)
        self.index_of = {song: position for position, song in enumerate(self.eligible)}
        self.recent = deque()
        self.window_size = window_size
        self.rng = random.Random(seed)

    def play(self):
        pick = self.rng.randrange(len(self.eligible))
        song = self.eligible[pick]
        last = self.eligible[-1]
        # swap-and-pop: the last song takes the hole
        self.eligible[pick] = last
        self.index_of[last] = pick
        self.eligible.pop()
        del self.index_of[song]
        self.recent.append(song)
        if len(self.recent) > self.window_size:
            returning_song = self.recent.popleft()
            self.index_of[returning_song] = len(self.eligible)
```

```
            self.eligible.append(returning_song)
        return song
```

COMPLEXITY | Time O(1) per play | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 256

257 * Unique Random Number Generator * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the lazy version for huge n: a dict 'moved' stores only the swapped slots, so memory stays O(calls)

QUESTION INFO

QUESTION

Design a class UniqueRandom initialized with an integer n . Each call to generate() returns a number drawn uniformly at random from [0, n-1] that has not been returned before, so across successive calls no value ever repeats. After all n distinct values have been produced, further calls return None . Every unused value must remain equally likely on each draw (no bias), and generate() must run in O(1) time per call.

Function signature

```
import random
class UniqueRandom:
    def __init__(self, n: int, seed: int | None = None) -> None:
        ...
    def generate(self) -> int | None:
        """Return a not-yet-returned number in [0, n-1], or None once all n are
exhausted."""
        ...
```

Follow-up (as reported): Use a partial Fisher-Yates shuffle: keep a virtual array 0..n-1 , and on the i -th call swap a random unused slot into position i using a sparse dict of moved entries, giving O(1) time and O(number-of- draws) space.

EXAMPLES

Example 1	
Input	`g = UniqueRandom(3); [g.generate(), g.generate(), g.generate()]`
Output	`a permutation of [0, 1, 2] (e.g. [2, 0, 1])`
Why	`The three calls return all of 0,1,2 in some random order with no repeats.`

Example 2	
Input	`(continuing above) g.generate()`
Output	`None`
Why	`All 3 values already returned, so the pool is exhausted.`

Example 3	
Input	`g = UniqueRandom(1); [g.generate(), g.generate()]`
Output	`[0, None]`
Why	`The only value 0 is returned once, then the generator is exhausted.`

CONSTRAINTS

- 1 <= n <= 10^7 .
- Support an optional seed for reproducible tests.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= n <= 10^7 .
- Support an optional seed for reproducible tests.

BRUTE FORCE | Shuffle everything up front

BRUTE-FORCE IDEA

Start with: Shuffle everything up front. It stores all n values even if only a few are ever drawn.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Shuffle everything up front
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n) setup, O(1) per call | Space O(n)

WHY THIS IS WEAKER | It stores all n values even if only a few are ever drawn.

HOW TO IMPROVE IT

the lazy version for huge n: a dict 'moved' stores only the swapped slots, so memory stays O(calls)

SHORT BEST SOLUTION

Swap a random unused slot into position i, remembering only the moved slots.

```python
import random


# BASE QUESTION (optimal): Lazy Fisher-Yates with a
# dictionary
# Idea: Swap a random unused slot into position i,
# remembering only the moved slots.
class UniqueRandom:
    def __init__(self, count, seed=None):
        self.count = count
        self.drawn = 0
        self.swapped = {}
        self.rng = random.Random(seed)

    def generate(self):
        if self.drawn == self.count:
            return None
        slot = self.drawn
        pick = self.rng.randrange(slot, self.count)
        value = self.swapped.get(pick, pick)
        # the value that sat at slot i takes the place of the one just drawn
        self.swapped[pick] = self.swapped.get(slot, slot)
        self.drawn += 1
        return value
```

COMPLEXITY | Time O(1) per call | Space O(draws)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 257

Build and serialize: write a structure out, read it back with the same walk

PATTERN CONTEXT

Encoding is one walk that writes tokens; decoding is the same walk reading tokens back (an iterator or an index). Put a length or a child count before the data so the reader always knows how much to take.

259 * Print a Directory Tree * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: build nested dicts from the path parts, then print them with a recursive walk that indents by depth

QUESTION INFO

QUESTION

You are given a list of file/directory paths , each a '/' -separated string (for example "dir/subdir/file.txt"). Reproduce what a Unix directory listing tool prints: build the directory hierarchy implied by the paths and return it as a single newline-separated string. Rules: Each path component is a node. A component that appears under the same parent in several paths must be printed only once. Indent each node with 2 * depth spaces, where the top-level components have depth 0 . Children are printed in the order they are first encountered across the input list (preserve first-seen order; do not sort). The output must not have a trailing newline.

Function signature

```
def print_directory_tree(paths: list[str]) -> str:
    ...
```

Follow-up (as reported): Add box-drawing connectors (, ,) like the real tree command. Hint: track, per level, whether the current node is the last child of its parent.

EXAMPLES

Example 1	
Input	`paths = ["a/b/c", "a/b/d", "a/e"]`
Output	`"a\n b\n c\n d\n e"`
Why	`Rendered: a b c d e`

Example 2	
Input	`paths = ["src/main.py", "src/util.py", "README.md"]`
Output	`"src\n main.py\n util.py\nREADME.md"`
Why	`"src" and its two files come first (first-seen order), then the top-level "README.md".`

Example 3	
Input	`paths = []`
Output	`""`

CONSTRAINTS

- 0 <= len(paths) <= 10^4
- Each path is non-empty, has no leading/trailing '/' , and no empty components.
- If paths is empty, return the empty string "" .

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the

largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(paths) <= 10^4
- Each path is non-empty, has no leading/trailing '/' , and no empty components.
- If paths is empty, return the empty string "" .

BRUTE FORCE | Nested lists with a linear search

BRUTE-FORCE IDEA

Start with: Nested lists with a linear search. Every component searches the list of children of its parent one by one.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Nested lists with a linear search
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(total length * width) | Space O(total length)

WHY THIS IS WEAKER | Every component searches the list of children of its parent one by one.

HOW TO IMPROVE IT

build nested dicts from the path parts, then print them with a recursive walk that indents by depth

SHORT BEST SOLUTION

Insert every path into a dict of dicts, then print depth first with two spaces per level.

```python
# BASE QUESTION (optimal): Nested dictionaries and a DFS
# Idea: Insert every path into a dict of dicts, then print
# depth first with two spaces per level.
def print_directory_tree(paths):
    root = {}
    for path in paths:
        node = root
        for part in path.split("/"):
            node = node.setdefault(part, {})
    lines = []

    def walk(node, depth):
        for name, kids in node.items():
            lines.append("  " * depth + name)
            walk(kids, depth + 1)

    walk(root, 0)
    return "\n".join(lines)
```

COMPLEXITY | Time O(total length) | Space O(total length)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 259

260 * Construct a binary tree given its preorder and inorder traversals. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the first preorder value is the root; its position in a value-to-index map of the inorder list splits left from right

QUESTION INFO

QUESTION

Given two integer arrays, preorder and inorder, which represent the pre-order and in-order traversal of a binary tree, construct and return the binary tree. Pre-order traversal visits nodes in this order: root -> left subtree -> right subtree In-order traversal visits nodes in this order: left subtree -> root -> right subtree You need to construct the binary tree and return its root node. Examples preorder = [3, 9, 20, 15, 7], inorder = [9, 3, 15, 20, 7] output: [3, 9, 20, null, null, 15, 7] Explanation: Given the two arrays, the binary tree constructed is: 3 / \ 9 20 / \ 15 7 preorder = [1, 2], inorder = [2, 1] output: [1, 2]
Explanation: Given the two arrays, the binary tree constructed is: 1 / 2 Constraints: The number of nodes in the tree is in the range [1, 3000]. preorder and inorder consist of unique integers. The elements of inorder will always be the same as the elements of preorder but in a different order.

EXAMPLES

Example 1	
Input	**preorder = [3, 9, 20, 15, 7] inorder = [9, 3, 15, 20, 7]**
Output	**values(build_pre_in([3, 9, 20, 15, 7], [9, 3, 15, 20, 7])) -> [3, 9, 20, null, null, 15, 7]**
Why	**The first preorder value 3 is the root; in the inorder list 9 is left of it and 15, 20, 7 are right. The tree is shown level by level.**

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Find the root with list.index and slice the lists

BRUTE-FORCE IDEA

Take preorder[0] as the root, find it in inorder with a linear search, and recurse on slices. Correct; the search and the copying make it quadratic on a skewed tree.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Find the root with list.index and slice the lists
Take preorder[0] as the root
find it in inorder with a linear search
recurse on slices
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n^2) for the slices

WHY THIS IS WEAKER | The linear search and the list copies make a skewed tree O(n^2) in time and space. A dictionary of inorder positions with index bounds is O(n).

HOW TO IMPROVE IT

the first preorder value is the root; its position in a value-to-index map of the inorder list splits left from right

SHORT BEST SOLUTION

Map each value to its inorder index. Walk preorder with one shared pointer; for a subtree covering inorder[low:high] take the next preorder value as the root, split at its inorder index, and build the left subtree first (preorder visits it before the right). No slicing, no searching.

```
# BASE QUESTION (optimal): Hash map of inorder positions
# and index bounds
# Idea: Map each value to its inorder index. Walk preorder
# with one shared pointer; for a subtree covering
# inorder[low:high] take the next preorder value as the root,
# split at its inorder index, and build the left subtree
# first (preorder visits it before the right). No slicing,
```

```python
# no searching.
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def buildTree(preorder, inorder):
    inorder_index = {value: position for position, value in enumerate(inorder)}
    next_preorder = 0

    def build(low, high):
        nonlocal next_preorder
        if low >= high:
            return None
        root = TreeNode(preorder[next_preorder])
        next_preorder += 1
        root_position = inorder_index[root.val]
        root.left = build(low, root_position)
        root.right = build(root_position + 1, high)
        return root

    return build(0, len(inorder))
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 260

261 * Write functions to serialize and deserialize a list of strings. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: write len + '#' + text for each string; read the length, then take exactly that many characters

QUESTION INFO

QUESTION

Serialize and Deserialize a List of Strings. You are tasked with writing two functions: serialize and deserialize. The serialize function will take a list of strings as input and return a single string. The deserialize function will take this single string and reconstruct the original list of strings. Details The serialize function should concatenate all strings in the list into a single string, with each original string's length encoded as a single character prefix. The deserialize function should be able to take the single string and accurately split it back into the original list of strings, using the encoded length characters as guides. For example, consider a list ["hello", "world"]. The serialization of this list might be "\x05hello\x05world", where \x05 indicates that the following string has a length of 5 characters. Constraints The maximum length of any string is 65535 (0xFFFF). If a string's length exceeds the maximum, the serialize function should raise an exception. The deserialize function should accurately handle any edge cases, such as empty strings or single-character strings. Assume that all strings consist of ASCII characters only. Examples lst = ['hello', 'world'] serialized = serialize(lst)

serialized output

: "\x05hello\x05world" deserialized = deserialize(serialized)

deserialized output should be

: ['hello', 'world'] lst = [''] serialized = serialize(lst)

serialized output

: "\x00" deserialized = deserialize(serialized)

deserialized output should be

EXAMPLES

QUESTION

Example 1

Input	`items = ['hello', 'wor#ld', '']`
	`serialize(items)`
	`deserialize(serialize(items)) == items`

Output	`serialize(items) -> '5#hello6#wor#ld0#'` `deserialize(serialize(items)) == items -> true`
Why	`Each string is written as its length, a #, and its text, so a # inside a string cannot confuse the reader.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Join with a delimiter

BRUTE-FORCE IDEA

Join with a separator like ',' and split on decode. It works only while no string contains the separator, and it cannot tell an empty list from a list with one empty string.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Join with a delimiter
Join with a separator like ','
split on decode
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(total length) | Space O(total length)

WHY THIS IS WEAKER | It is fast (O(total length)) but wrong: a string that contains the delimiter breaks it, and an empty list cannot be told from a list of one empty string. Writing each string's length before it fixes both.

HOW TO IMPROVE IT

write len + '#' + text for each string; read the length, then take exactly that many characters

SHORT BEST SOLUTION

Write length#text for each string. The decoder reads the length first, so every character is safe inside the text.

```python
# BASE QUESTION (optimal): Length-prefix each string
# Idea: Write length#text for every string. While reading,
# the length tells us exactly where that string ends.
def serialize(strings):
    return "".join(f"{len(text)}#{text}" for text in strings)


def deserialize(data):
    strings = []
    position = 0
    while position < len(data):
        separator = data.index("#", position)
        length = int(data[position:separator])
        start = separator + 1
        strings.append(data[start:start + length])
        position = start + length
    return strings
```

COMPLEXITY | Time O(total characters) | Space O(total characters) for the output

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 261

Games: win or lose with a cache, and guessing with feedback

PATTERN CONTEXT

For two-player games, a state is winning if some move leads to a state where the opponent loses; cache every state. For guessing games, pick the guess that shrinks the worst remaining group of candidates the most, then keep only the candidates that agree with the feedback.

264 * Choose optimal guesses for green-only Wordle

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: bitmask per (position, letter) of the words that have it; the best guess splits the words into the most even groups

QUESTION INFO

QUESTION

You are guessing a hidden word from a candidate list. After each guess you only learn which positions are green, that is, where your letter equals the answer's letter. best_guess(words) picks the candidate whose answer pattern splits the remaining words most evenly: the one with the smallest expected size of the group left (sum of squared group sizes divided by the number of words); ties go to the alphabetically smaller word.

Function signature

def green(guess: str, answer: str) -> tuple
def best_guess(words: list) -> str

EXAMPLES

Example 1	
Input	`'cat', 'car'`
Output	`(true, true, false)`
Why	`c and a match, the last letters differ.`

Example 2	
Input	`['cat', 'car', 'cap', 'dog']`
Output	`'cap'`
Why	`'cap', 'car' and 'cat' split the other words equally well and 'cap' wins the alphabetical tie-break; 'dog' shares no letter positions with the others, so it tells less.`

Example 3	
Input	`['tea']`
Output	`'tea'`
Why	`With a single candidate the answer is that candidate.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Compute the feedback of every guess against every answer

BRUTE-FORCE IDEA

For each guess compute its green pattern against each candidate answer, count the sizes of the pattern groups, and evaluate sum(s^2) / W. Correct; it compares every pair of words.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compute the feedback of every guess against every answer
For each guess compute its green pattern against each candidate answer
count the sizes of the pattern groups
evaluate sum(s^2) / W
```

BRUTE-FORCE COMPLEXITY | Time O(W^2 * L) | Space O(W)

WHY THIS IS WEAKER | Compares every pair of words, O(W^2 * L). Bitsets per (position, letter) split the candidates into pattern groups with a few AND operations.

HOW TO IMPROVE IT

bitmask per (position, letter) of the words that have it; the best guess splits the words into the most even groups

SHORT BEST SOLUTION

Index the words: bits[pos][letter] is an integer whose bit i is set when word i has that letter at pos. For a guess, start with one group containing all words and split it position by position into 'green here' (AND with bits[pos][guess letter]) and 'not green here' (the remainder). Group sizes are popcounts; the score is sum of s^2 over word_count. Choose the minimum score, ties by the smaller word.

```python
from fractions import Fraction
from collections import defaultdict


# BASE QUESTION (optimal): Bitsets per (position, letter),
# splitting groups with AND
# Idea: Index the words: bits[pos][letter] is an integer
# whose bit i is set when word i has that letter at pos.
# For a guess, start with one group containing all words
# and split it position by position into 'green here' (AND
# with bits[pos][guess letter]) and 'not green here' (the
# remainder). Group sizes are popcounts; the score is sum
# of s^2 over word_count. Choose the minimum score, ties by the
# smaller word.
def best_guess(words):
    word_count = len(words)
    bits = [defaultdict(int) for _ in range(len(words[0]))]
    for word_index, word in enumerate(words):
        for pos, letter in enumerate(word):
            bits[pos][letter] |= 1 << word_index
    best = None
    for guess in words:
        groups = [(1 << word_count) - 1]
        for pos, letter in enumerate(guess):
            mask = bits[pos][letter]
            next_groups = []
            for group in groups:
                green_part, other_part = group & mask, group & ~mask
                if green_part:
                    next_groups.append(green_part)
                if other_part:
                    next_groups.append(other_part)
            groups = next_groups
        score = Fraction(sum(group.bit_count() ** 2 for group in groups), word_count)
        if best is None or (score, guess) < best:
            best = (score, guess)
    return best[1]
```

COMPLEXITY | Time O(W * L * groups) with bit-parallel work per group | Space O(W * L)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 264

Sliding window: grow on the right, shrink on the left while it is too bad

PATTERN CONTEXT

Move the right end one step at a time and add that item to your state. While the window breaks the rule, remove items from the left end. After each step the window is valid, so record its length (or the shortest valid one).

269 * Temperature Monitoring Data Structure * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: a deque of the last n readings with a running total; a second deque of decreasing values gives the maximum

QUESTION INFO

QUESTION

Design a monitor over the last n temperature readings. insert(x) adds a reading (the oldest one drops out once more than n are stored), average() returns the mean of the stored readings, and maximum() returns the largest of them in O(1).

Function signature

```
class Monitor:
    def __init__(self, n: int)
    def insert(self, x: float) -> None
    def average(self) -> float
    def maximum(self) -> float
```

EXAMPLES

Example 1

Input	`m = Monitor(3)`
	`m.insert(20)`
	`m.insert(22)`
	`m.insert(19)`
	`m.average()`
	`m.insert(25)`
	`m.average()`
	`m.maximum()`
	`m.insert(10)`
	`m.maximum()`

Output	`m.average() -> 20.333333`
	`m.average() -> 22`
	`m.maximum() -> 25`
	`m.maximum() -> 25`

Why	`With n = 3 the window slides: after 25 arrives the window is 22, 19, 25 (average 22, maximum 25); after 10 it is 19, 25, 10, so the maximum is still 25.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Keep a list and scan the window on every query

BRUTE-FORCE IDEA

Store every reading and compute sum and max over the last N each time. Correct, but a full window scan for each query.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Keep a list and scan the window on every query
Store every reading
compute sum
max over the last N each time
```

BRUTE-FORCE COMPLEXITY | Time O(N) per average or max | Space O(N)

WHY THIS IS WEAKER | Every query scans the window, O(N). A running sum gives the average in O(1), and a deque of decreasing values gives the maximum in O(1) amortised.

HOW TO IMPROVE IT

a deque of the last n readings with a running total; a second deque of decreasing values gives the maximum

SHORT BEST SOLUTION

Keep the window in a deque and a running sum; on insert add the value and drop the oldest when the window is full. For the maximum keep a second deque of (index, value) with decreasing values: pop smaller ones from the back before pushing, and pop the front when its index leaves the window. The front is the maximum.

```python
from collections import deque


# BASE QUESTION (optimal): Running sum plus a monotonic
# deque
# Idea: Keep the window in a deque and a running sum; on
# insert add the value and drop the oldest when the window
# is full. For the maximum keep a second deque of (index,
# value) with decreasing values: pop smaller ones from the
# back before pushing, and pop the front when its index
# leaves the window. The front is the maximum.
class Monitor:
    def __init__(self, window_size):
        self.window_size, self.window, self.total = window_size, deque(), 0
        self.max_queue, self.readings_seen = deque(), 0

    def insert(self, reading):
        self.window.append(reading)
        self.total += reading
        if len(self.window) > self.window_size:
            self.total -= self.window.popleft()
        while self.max_queue and self.max_queue[-1][1] <= reading:
            self.max_queue.pop()
        self.max_queue.append((self.readings_seen, reading))
        self.readings_seen += 1
        while self.max_queue[0][0] <= self.readings_seen - 1 - self.window_size:
            self.max_queue.popleft()

    def average(self):
        return self.total / len(self.window)

    def maximum(self):
        return self.max_queue[0][1]
```

COMPLEXITY | Time O(1) amortized per operation | Space O(N)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 269

270 * Footpath Beauty After Repainting

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: bad = a block that is not the wanted colour; allowed = the repaint budget m; the longest window is the answer

QUESTION INFO

QUESTION

A footpath is built from n blocks laid in a line, painted with one of k colors. block_colors[i] is the color of block i (colors are integers in 1..k). The beauty value for a given favorite color c is the length of the longest run of consecutive blocks that are all color c . You are allowed to repaint at most m blocks (each to any color you choose). Return the maximum possible beauty value for favorite color c after repainting at most m blocks.

Function signature

```
def max_beauty_after_repaint(block_colors: list[int], c: int, m: int) -> int:
    ...
```

Follow-up (as reported): (Reported) Answer q queries, each with a different favorite color, and return the beauty value (with m fixed) for each. Hint: a sliding window per query, or precompute runs. (Reported) The base version with m = 0 was asked first; the repaint version is the twist.

EXAMPLES

Example 1	
Input	`block_colors = [1, 2, 1, 1, 2, 1], c = 1, m = 1`
Output	`4`
Why	`Repaint one of the 2's. The best window has at most one non-c block, e.g. indices 2..5 [1,1,2,1] -> repaint the 2 -> 4.`

Example 2	
Input	`block_colors = [3, 3, 3], c = 3, m = 0`
Output	`3`
Why	`All blocks are already color 3; no repaint needed.`

Example 3	
Input	`block_colors = [2, 2, 2], c = 1, m = 2`
Output	`2`
Why	`Only 2 repaints allowed, so at most 2 consecutive c blocks.`

CONSTRAINTS

- 1 <= n <= 10^5 , block_colors has length n
- 1 <= k , 1 <= c <= k , 1 <= block_colors[i] <= k
- 0 <= m <= n
- With m = 0 , the answer is simply the longest existing run of color c .

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= n <= 10^5 , block_colors has length n
- 1 <= k , 1 <= c <= k , 1 <= block_colors[i] <= k
- 0 <= m <= n
- With m = 0 , the answer is simply the longest existing run of color c .

BRUTE FORCE | Count the other colours in every window

BRUTE-FORCE IDEA

Start with: Count the other colours in every window. Every start and end pair recounts its blocks.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Count the other colours in every window
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1)

WHY THIS IS WEAKER | Every start and end pair recounts its blocks.

HOW TO IMPROVE IT

bad = a block that is not the wanted colour; allowed = the repaint budget m; the longest window is the answer

SHORT BEST SOLUTION

The longest window holding at most m other-colour blocks is the answer.

```python
# BASE QUESTION (optimal): Sliding window with at most m
# repaints
# Idea: The longest window holding at most m other-colour
# blocks is the answer.
def max_beauty_after_repaint(block_colors, target_color, max_repaints):
    left, other_colored, best = 0, 0, 0
    for right, color in enumerate(block_colors):
        other_colored += color != target_color
        while other_colored > max_repaints:
            other_colored -= block_colors[left] != target_color
            left += 1
        best = max(best, right - left + 1)
    return best
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 270

271 * Shortest Subarray with K Distinct Integers

* MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: state = a count dict; while it holds at least k distinct values, record the window and shrink from the left

QUESTION INFO

QUESTION

Given an array nums and an integer k, return the length of the shortest contiguous subarray that contains at least k different integers, or -1 if there is no such subarray.

Function signature

def shortest_k_distinct(nums: list, k: int) -> int

EXAMPLES

Example 1	
Input	`[1, 2, 1, 3, 2], 3`
Output	`3`
Why	`[2, 1, 3] is the shortest window holding three different numbers.`

QUESTION

Example 2	
Input	`[1, 1, 1], 2`
Output	`-1`
Why	`Only one distinct number exists.`

Example 3	
Input	`[1, 2], 2`
Output	`2`
Why	`The whole array is needed.`

CONSTRAINTS

- 1 <= nums.length <= 10^5
- 1 <= nums[i] <= 10^9
- 1 <= k <= nums.length

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= nums.length <= 10^5
- 1 <= nums[i] <= 10^9
- 1 <= k <= nums.length

BRUTE FORCE | Check every subarray

BRUTE-FORCE IDEA

For every start, extend the end while counting distinct values with a set, and record the shortest that reaches k. Correct, and quadratic.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Check every subarray
For every start
extend the end while counting distinct values with a set
record the shortest that reaches k
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | O(n^2) in the worst case. A sliding window with a count map expands on the right and shrinks on the left, so each element is handled twice: O(n).

HOW TO IMPROVE IT

state = a count dict; while it holds at least k distinct values, record the window and shrink from the left

SHORT BEST SOLUTION

Expand the right end, counting each value. While the window has at least k distinct values, record its length and shrink from the left (deleting a value from the map when its count reaches 0). Each index is added once and removed once.

```python
# BASE QUESTION (optimal): Sliding window with a frequency
# map
# Idea: Expand the right end, counting each value. While
# the window has at least k distinct values, record its
# length and shrink from the left (deleting a value from
# the map when its count reaches 0). Each index is added
# once and removed once.
def shortest_k_distinct(nums, distinct_needed):
    if distinct_needed <= 0:
        return 0
```

```
    count, best, left = {}, None, 0
    for right, value in enumerate(nums):
        count[value] = count.get(value, 0) + 1
        while len(count) >= distinct_needed:
            best = right - left + 1 if best is None else min(best, right - left + 1)
            leaving_value = nums[left]
            count[leaving_value] -= 1
            if count[leaving_value] == 0:
                del count[leaving_value]
            left += 1
    return -1 if best is None else best
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 271

272 * Find the minimum window substring. * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: state = how many of each needed letter are still missing; while nothing is missing, record the window and shrink

QUESTION INFO

QUESTION

Given strings s and t, return the shortest substring of s that contains every character of t (a character that appears twice in t must appear at least twice in the substring). Return the empty string if there is no such substring.

Function signature

def min_window(s: str, t: str) -> str

EXAMPLES

Example 1	
Input	'ADOBECODEBANC', 'ABC'
Output	'BANC'
Why	BANC is the shortest window that holds an A, a B and a C.

Example 2	
Input	'a', 'a'
Output	'a'
Why	The whole string is the only window.

Example 3	
Input	'a', 'aa'
Output	''
Why	t needs two a's but s has only one.

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Check every substring

BRUTE-FORCE IDEA

For every substring test whether its letter counts cover t, and keep the shortest. Correct; cubic-ish in practice.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Check every substring
For every substring test whether its letter counts cover t
keep the shortest
```

BRUTE-FORCE COMPLEXITY | Time O(n^2 * m) | Space O(m)

WHY THIS IS WEAKER | O(n^2) substrings, each checked in O(m) or more. A sliding window that grows until valid and then shrinks from the left is O(n + m).

HOW TO IMPROVE IT

state = how many of each needed letter are still missing; while nothing is missing, record the window and shrink

SHORT BEST SOLUTION

need[c] is how many more of c the window requires, and missing counts the distinct requirements still unmet. Move right adding characters; when missing reaches 0 shrink from the left while the window stays valid, recording the shortest. Removing a needed character that drops below its requirement makes missing go up again.

```python
from collections import Counter


# BASE QUESTION (optimal): Sliding window with a 'still
# needed' counter
# Idea: need[c] is how many more of c the window requires,
# and missing counts the distinct requirements still
# unmet. Move right adding characters; when missing
# reaches 0 shrink from the left while the window stays
# valid, recording the shortest. Removing a needed
# character that drops below its requirement makes missing
# go up again.
def minWindow(text, required):
    if not required:
        return ""
    need, missing = Counter(required), len(set(required))
    best, left = "", 0
    for right, char in enumerate(text):
        if char in need:
            need[char] -= 1
            if need[char] == 0:
                missing -= 1
        while missing == 0:
            if best == "" or right - left + 1 < len(best):
                best = text[left:right + 1]
            leaving_char = text[left]
            if leaving_char in need:
                need[leaving_char] += 1
                if need[leaving_char] > 0:
                    missing += 1
            left += 1
    return best
```

COMPLEXITY | Time O(n + m) | Space O(m)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 272

Parent pointers: store who is above each node and climb

PATTERN CONTEXT

Keep parent[x] (or manager[x]). Climbing is a while loop that follows parent until it runs out. Questions about common ancestors, cycles and who reports to whom are all small variations of that climb.

275 * Employee Hierarchy Add / Delete * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: manager[employee] and a set of reports per manager; delete re-attaches the reports to the deleted person's manager

QUESTION INFO

QUESTION

Maintain a company hierarchy. Org(ceo) starts with the CEO. add(emp, mgr) adds a new employee under an existing manager (False if emp already exists or mgr does not). delete(emp) removes an employee and their direct reports move up to the deleted person's manager. The CEO can only be deleted when exactly one person reports to them, who then becomes the CEO.

Function signature

class Org:
  def __init__(self, ceo)
  def add(self, emp, mgr) -> bool
  def delete(self, emp) -> bool

EXAMPLES

Example 1

Input	`o = Org('ceo')`
	`o.add('vp', 'ceo')`
	`o.add('dev', 'vp')`
	`o.add('qa', 'vp')`
	`o.add('dev', 'ceo')`
	`o.delete('vp')`
	`o.manager['dev']`
	`o.delete('ceo')`

Output	`o.add('vp', 'ceo') -> true`
	`o.add('dev', 'vp') -> true`
	`o.add('qa', 'vp') -> true`
	`o.add('dev', 'ceo') -> false`
	`o.delete('vp') -> true`
	`o.manager['dev'] -> 'ceo'`
	`o.delete('ceo') -> false`

Why	`'dev' already exists so the second add fails. Deleting 'vp' moves dev and qa up to the ceo. The ceo now has two reports, so the ceo cannot be deleted.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Store only parent pointers and search for children

BRUTE-FORCE IDEA

Keep a dict from employee to manager. To delete, scan all entries to find the reports and reassign them. Correct, but every delete is a full scan.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Store only parent pointers and search for children
Keep a dict from employee to manager
process every required item or candidate directly
```

BRUTE-FORCE COMPLEXITY | Time O(n) per delete | Space O(n)

WHY THIS IS WEAKER | Every delete scans all employees, O(n). Keeping a children set per employee makes it O(number of reports).

HOW TO IMPROVE IT

manager[employee] and a set of reports per manager; delete re-attaches the reports to the deleted person's manager

SHORT BEST SOLUTION

Keep manager[employee] and reports[employee] (a set). Add inserts into both. Delete reattaches each report to the deleted node's manager and removes the node from its manager's set. Cost depends only on the number of direct reports.

```python
# BASE QUESTION (optimal): Parent map plus children sets
# Idea: Keep manager[employee] and reports[employee] (a set). Add
# inserts into both. Delete reattaches each report to the
# deleted node's manager and removes the node from its
# manager's set. Cost depends only on the number of direct
# reports.
class Org:
    def __init__(self, ceo):
        self.manager = {ceo: None}
        self.reports = {ceo: set()}

    def add(self, employee, boss):
        if employee in self.manager or boss not in self.manager:
            return False
        self.manager[employee] = boss
        self.reports[employee] = set()
        self.reports[boss].add(employee)
        return True

    def delete(self, employee):
        if employee not in self.manager:
            return False
        boss, direct_reports = self.manager[employee], self.reports[employee]
        if boss is None and len(direct_reports) != 1:
            return False
        for report in direct_reports:
            self.manager[report] = boss
            if boss is not None:
                self.reports[boss].add(report)
        if boss is not None:
            self.reports[boss].discard(employee)
        del self.manager[employee], self.reports[employee]
        return True
```

COMPLEXITY | Time O(r) per delete where r is the number of reports | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 275

276 * Validate parent array forms a tree * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: climb from every node with three states (unseen, on this walk, verified); meeting your own walk is a cycle; there must be exactly one root

QUESTION INFO

QUESTION

Problem

You are given an integer array **`parent`** of length **`n`** describing a directed parent pointer for each node **`i`** (nodes are labeled **`0..n-1`**).

- **`parent[i]`** is the parent of node **`i`**.
- Exactly one node should be the root, indicated by **`parent[root] = -1`**. Determine whether these **`n`** nodes form a **valid rooted tree**. A valid rooted tree must satisfy all of the following: 1.

Exactly one root (exactly one index **`i`** with **`parent[i] = -1`**). 2.
No cycles (following parent pointers from any node must eventually reach the root). 3.
Connectivity (every node is reachable from the root; equivalently, every node has exactly one simple path to the root).

Input

- **`parent`**: integer array of length **`n`** where each value is either **`-1`** or in **`[0, n-1]`**.

Output

- Return **`true`** if **`parent`** represents a valid rooted tree; otherwise return **`false`**.

Notes

/ Edge Cases - **`n`** can be 1 (then the only valid tree is **`parent[0] = -1`**).

- Self-parenting like **`parent[i] = i`** is invalid.
- Multiple **`-1`** entries (multiple roots) is invalid.
- A cycle among non-root nodes is invalid.
- A disconnected component (some nodes not reachable from the root) is invalid.

You are given an integer array **`parent`** of length **`n`** describing a parent pointer for each node **`i`** where nodes are labeled **`0`** to **`n-1`**.

- **`parent[i]`** is the parent of node **`i`**.
- Exactly one node should be the root, indicated by **`parent[root] = -1`**. Determine whether these nodes form a valid rooted tree. A valid rooted tree must satisfy all of the following:

1. Exactly one root.
2. No cycles.
3. Every node belongs to the same connected structure and has a path to the root. Return **`True`** if **`parent`** represents a valid rooted tree; otherwise return **`False`**.

Constraints
1 <= n <= 2 * 10^5Each **`parent[i]`** is either **`-1`** or an integer in **`[0, n-1]parent[i] = i`** is invalid

EXAMPLES

```
Input: [-1, 0, 0, 1, 1]
Expected Output: True
```

Explanation: Node 0 is the only root. All other nodes are connected under it, and there are no cycles.

```
Input: [2, -1, 1, 2]
Expected Output: True
Explanation: Node 1 is the root. The paths are 0->2->1, 2->1, and 3->2->1, so every node reaches the
    same root.
```

QUESTION

Hints

First count how many roots there are. A valid tree must have exactly one **`-1`** entry.Build children lists from the parent array, then traverse from the root. If you do not visit every node, the structure is disconnected or contains a cycle away from the root.

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Follow the parent chain from every node, up to n steps

BRUTE-FORCE IDEA

From each node follow parents for at most n steps; if you do not reach the root in n steps there is a cycle. Correct; each node walks its whole chain separately.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Follow the parent chain from every node, up to n steps
From each node follow parents for at most n steps
if you do not reach the root in n steps there is a cycle
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1)

WHY THIS IS WEAKER | Every node walks its whole chain, O(n^2). Marking nodes as 'on this walk' or 'verified' lets every node be processed once, O(n).

HOW TO IMPROVE IT

climb from every node with three states (unseen, on this walk, verified); meeting your own walk is a cycle; there must be exactly one root

SHORT BEST SOLUTION

Check there is exactly one -1 and all other values are in range. Then walk up from each unvisited node marking nodes as 'on this walk'. Meeting a node that is on the current walk is a cycle; meeting a node already verified (or the root) ends the walk, and all nodes on it are marked verified. Each node is visited a constant number of times.

```python
# BASE QUESTION (optimal): One root, then colour each node
# once while walking up
# Idea: Check there is exactly one -1 and all other values
# are in range. Then walk up from each unvisited node
# marking nodes as 'on this walk'. Meeting a node that is
# on the current walk is a cycle; meeting a node already
# verified (or the root) ends the walk, and all nodes on
# it are marked verified. Each node is visited a constant
# number of times.
def validate_parent_array(parent):
    node_count = len(parent)
    if any(parent_id != -1 and not (0 <= parent_id < node_count) for parent_id in parent) or parent.count(-1) !
= 1:
        return False
    state = [0] * node_count                                    # 0 unseen, 1 on the current walk, 2 verified
    for start in range(node_count):
        path, current = [], start
        while current != -1 and state[current] == 0:
            state[current] = 1
            path.append(current)
            current = parent[current]
        if current != -1 and state[current] == 1:
            return False
        for node in path:
            state[node] = 2
    return True
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 276

277 * Root a Tree So It Becomes a Binary Tree * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the root has no parent so it needs degree <= 2; everyone else spends one edge on its parent; answer = the first node of degree <= 2

QUESTION INFO

QUESTION

You are given a connected, undirected, acyclic graph (i.e. a tree) with n nodes labeled 0 .. n-1 , described by an edge list. It is guaranteed that every node has at most 3 neighbors. Choose a node to be the root. When rooted at node r , one neighbor of each non-root node becomes its parent and the rest become its children; the root has no parent, so all of its neighbors become children. The rooted tree is a binary tree if every node has at most 2 children. Return the smallest label of a node that can serve as such a root. If no node works, return -1 . Key observation (the trick): a non-root node with degree d has d - 1 children, so it needs d <= 3 ; the root has degree children, so the root needs degree <= 2 . Since all nodes already have degree <= 3 , any node with degree <= 2 is a valid root — and leaves (degree 1) always exist in a nonempty tree, so a valid root always exists for n >= 1 .

Function signature

```
from typing import List, Tuple
def find_binary_tree_root(n: int, edges: List[Tuple[int, int]]) -> int:
    """Return the smallest node label that can root the tree as a binary tree
    (every node has at most 2 children), or -1 if none exists."""
    ...
```

Follow-up (as reported): Now nodes must be 2-colored (black/white) so that every parent differs in color from its children and all nodes at the same depth share a color. Determine whether a valid root plus coloring exists and return one. Hint: coloring each node by depth % 2 from any chosen root always satisfies both rules in a tree, so a valid coloring exists whenever a valid root does.

EXAMPLES

Example 1	
Input	`n = 4, edges = [(0, 1), (0, 2), (0, 3)]`
Output	`1`
Why	`Node 0 has degree 3, so as root it would have 3 children (invalid). Nodes 1, 2, 3 have degree 1; the smallest valid root is 1.`

Example 2	
Input	`n = 3, edges = [(0, 1), (1, 2)]`
Output	`0`
Why	`A path; every node has degree <= 2, so the smallest label 0 works.`

Example 3	
Input	`n = 1, edges = []`
Output	`0`
Why	`A single node is trivially a binary tree.`

CONSTRAINTS

- 1 <= n <= 10^5 ; the input is a valid tree with n - 1 edges (for n == 1 , no edges).
- Every node has at most 3 neighbors.

QUESTION

- Return the smallest valid root label.

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= n <= 10^5 ; the input is a valid tree with n - 1 edges (for n == 1 , no edges).
- Every node has at most 3 neighbors.
- Return the smallest valid root label.

BRUTE FORCE | Root at every node and count children

BRUTE-FORCE IDEA

Start with: Root at every node and count children. It re-roots the whole tree for each candidate even though only the root's degree matters.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Root at every node and count children
Start with: Root at every node
process every required item or candidate directly
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | It re-roots the whole tree for each candidate even though only the root's degree matters.

HOW TO IMPROVE IT

the root has no parent so it needs degree <= 2; everyone else spends one edge on its parent; answer = the first node of degree <= 2

SHORT BEST SOLUTION

Only the root has as many children as its degree; every other node has one fewer.

```python
# BASE QUESTION (optimal): First node of degree at most 2
# Idea: Only the root has as many children as its degree;
# every other node has one fewer.
def find_binary_tree_root(node_count, edges):
    degree = [0] * node_count
    for node_a, node_b in edges:
        degree[node_a] += 1
        degree[node_b] += 1
    for node in range(node_count):
        if degree[node] <= 2:
            return node
    return -1
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 277

278 * Lowest Common Ancestor of a Binary Tree III

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: two pointers climb; when one falls off the top it restarts at the other node, so both meet at the common ancestor

QUESTION INFO

QUESTION

Given two nodes p and q of a binary tree, return their lowest common ancestor: the deepest node that has both of them as descendants (a node counts as a descendant of itself). Every node has a parent pointer, but you are not given the root.

Function signature

def lca_swap(p: PNode, q: PNode) -> PNode

EXAMPLES

Example 1

Input	`class PNode:`
	`def __init__(self, val):`
	`self.val, self.parent = val, None`
	`root, a, b, c = PNode(3), PNode(5), PNode(1), PNode(6)`
	`a.parent = b.parent = root`
	`c.parent = a`
	`lca_swap(c, b).val`
	`lca_swap(c, a).val`
	`lca_swap(a, b).val`

Output	`lca_swap(c, b).val -> 3`
	`lca_swap(c, a).val -> 5`
	`lca_swap(a, b).val -> 3`

Why	`Node 6 hangs under 5, which hangs under the root 3 next to node 1. The ancestor of 6 and 1 is the root 3; the ancestor of 6 and its own parent 5 is 5 itself.`

CONSTRAINTS

- Number of nodes: [2, 10^5]
- -10^9 <= Node.val <= 10^9
- All node values are unique
- p != q
- p and q exist in the tree

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- Number of nodes: [2, 10^5]
- -10^9 <= Node.val <= 10^9
- All node values are unique
- p != q

BRUTE FORCE | Collect the ancestors of one node in a set

BRUTE-FORCE IDEA

Walk from p to the root storing every ancestor (including p), then walk from q upward until you hit a stored node. Simple and correct; it uses memory for the ancestor set.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Collect the ancestors of one node in a set
Walk from p to the root storing every ancestor (including p)
walk from q upward until you hit a stored node
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(h) | Space O(h)

WHY THIS IS WEAKER | Uses O(h) extra space for the set. Two pointers that switch to the other node's start when they fall off the root meet at the answer with O(1) space.

HOW TO IMPROVE IT

two pointers climb; when one falls off the top it restarts at the other node, so both meet at the common ancestor

SHORT BEST SOLUTION

Move a from p and b from q one step up per round. When a pointer falls off the root, restart it at the other node. After at most two passes both pointers have walked the same total length, so they are equal at the lowest common ancestor (or both None if the nodes are in different trees).

```
# BASE QUESTION (optimal): Two pointers that switch to the
# other node at the end
# Idea: Move a from p and b from q one step up per round.
# When a pointer falls off the root, restart it at the
# other node. After at most two passes both pointers have
# walked the same total length, so they are equal at the
# lowest common ancestor (or both None if the nodes are in
# different trees).
class PNode:
    def __init__(self, val):
        self.val, self.left, self.right, self.parent = val, None, None, None

def lowestCommonAncestor(first_node, second_node):
    pointer_a, pointer_b = first_node, second_node
    while pointer_a is not pointer_b:
        pointer_a = pointer_a.parent if pointer_a else second_node
        pointer_b = pointer_b.parent if pointer_b else first_node
    return pointer_a
```

COMPLEXITY | Time O(h) | Space O(1)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 278

Build or transform a grid with index arithmetic

PATTERN CONTEXT

Decide where each output cell comes from with a formula in (r, c), then build the whole grid with one comprehension. Rotation, diamonds, block summaries and prefix tables are all 'cell(r, c) = something computed from r and c'.

280 * Connected Crop Arrangement on a Grid

*** MEDIUM ***

DIFFICULTY | * MEDIUM *

Better approach: fill the cells in snake order (left to right, then right to left) so equal crops stay in one connected strip

QUESTION INFO

QUESTION

A garden is an n x m grid. There are k crop types numbered 1..k, and crop i must fill exactly counts[i - 1] cells; the counts add up to n * m so every cell is used. Fill the grid so that the cells of each crop form one connected region (cells joined up, down, left or right). Return any valid grid.

Function signature

def arrange_crops(n: int, m: int, counts: list) -> list

EXAMPLES

Example 1	
Input	`2, 3, [2, 3, 1]`
Output	`[[1, 1, 2], [3, 2, 2]]`
Why	`One valid answer: crop 3 is the single cell at the bottom left; crops 1 and 2 are each connected.`

Example 2	
Input	`1, 4, [1, 3]`
Output	`[[1, 2, 2, 2]]`
Why	`A single row: the crops simply sit side by side.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Backtrack over every assignment

BRUTE-FORCE IDEA

Try every way to assign the required crop counts to the grid, then check whether each crop's cells form one connected region. This is exact, but the number of assignments grows exponentially.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
assign cells one at a time
try each crop whose required count is not full
when the grid is filled, test connectivity for every crop
keep the first valid arrangement
backtrack after each failed choice
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(n * m)

WHY THIS IS WEAKER | Exponential in the number of cells. Filling the crops along a snake path through the grid is O(n * m) and every crop stays connected by construction.

HOW TO IMPROVE IT

fill the cells in snake order (left to right, then right to left) so equal crops stay in one connected strip

SHORT BEST SOLUTION

Generate the snake order of the cells and hand out count[0] cells to crop 1, the next count[1] to crop 2, and so on. Consecutive cells of the snake share an edge, so each crop is connected.

```
def snake_cells(rows, cols):
    # the cells of an n x m grid in "snake" order: row 0 left to right,
    # row 1 right to left, row 2 left to right, ... Consecutive cells in
    # this order always share an edge.
    path = []
    for row in range(rows):
        for col in range(cols):
            if row % 2 == 0:
                path.append((row, col))
            else:
                path.append((row, cols - 1 - col))
    return path


# BASE QUESTION (optimal): Fill crops along a snake path
# Idea: Generate the snake order of the cells and hand out
# count[0] cells to crop 1, the next count[1] to crop 2,
# and so on. Consecutive cells of the snake share an edge,
# so each crop is connected.
def arrange_crops(rows, cols, counts):
    assert sum(counts) == rows * cols
    grid = [[0] * cols for _ in range(rows)]
    path = snake_cells(rows, cols)
    path_position = 0
    for crop, cells_for_crop in enumerate(counts, 1):
        for _ in range(cells_for_crop):
            row, col = path[path_position]
            grid[row][col] = crop
            path_position += 1
    return grid
```

COMPLEXITY | Time O(n * m) | Space O(n * m)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 280

281 * Solve matrix groups and recipe inventory * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: block (r // block_size, c // block_size) holds the minimum of every cell that maps to it; the recipe part is a topological sort

QUESTION INFO

QUESTION

Two tasks. (1) block_mins(matrix, k): cut the matrix into k x k blocks (blocks at the right and bottom edge may be smaller) and return the minimum of every block as a smaller matrix. (2) possible_recipes(recipes, supplies): recipes maps a recipe name to the ingredients it needs, and an ingredient may itself be another recipe; return the recipes that can be made from the supplies.

Function signature

def block_mins(matrix: list, k: int) -> list

def possible_recipes(recipes: dict, supplies: list) -> list

EXAMPLES

QUESTION

Example 1	
Input	`[[3, 1, 4, 1], [5, 9, 2, 6], [5, 3, 5, 8]], 2`
Output	`[[1, 1], [3, 5]]`
Why	`Top-left block (3, 1, 5, 9) has minimum 1, top-right (4, 1, 2, 6) has 1; the bottom row forms smaller blocks with minimum 3 and 5.`

Example 2	
Input	`{'bread': ['flour', 'yeast'], 'sandwich': ['bread', 'ham'], 'cake': ['flour', 'egg', 'sugar']}, ['flour', 'yeast', 'ham', 'egg']`
Output	`['bread', 'sandwich']`
Why	`bread needs flour and yeast (available); sandwich needs bread and ham; cake needs sugar, which is missing.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Blocks: collect each block's values and call min

BRUTE-FORCE IDEA

For every block gather its cells into a list and take the minimum. Correct; it copies every block.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Blocks: collect each block's values and call min
For every block gather its cells into a list
take the minimum
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(m * n) | Space O(m * n)

WHY THIS IS WEAKER | Copies every block into a list. One pass that updates the minimum of cell (r // k, c // k) needs no copies.

BRUTE FORCE 2 | Recipes: keep re-scanning until no recipe changes

BRUTE-FORCE IDEA

Repeatedly look for a recipe whose ingredients are all available and add it to the available set, until a full scan adds nothing. Correct; many passes over all recipes.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recipes: keep re-scanning until no recipe changes
Repeatedly look for a recipe whose ingredients are all available
add it to the available set
until a full scan adds nothing
```

BRUTE-FORCE COMPLEXITY | Time O(R^2 * I) | Space O(R)

WHY THIS IS WEAKER | Many passes over all recipes, O(R^2 * I). Counting each recipe's missing ingredients and processing items as they become available (Kahn's algorithm) is O(R + I).

HOW TO IMPROVE IT

block (r // block_size, c // block_size) holds the minimum of every cell that maps to it; the recipe part is a topological sort

SHORT BEST SOLUTION

Part 1: Each cell updates the minimum of block (r // block_size, c // block_size). No block list is built; the last blocks may be smaller when k does not divide the size.

Part 2: For each recipe count the ingredients it still lacks (those not in supplies) and record, for each ingredient, the recipes that need it. Start from the supplies; when an item becomes available decrease the counts of the recipes waiting for it, and queue a recipe when its count hits zero. Recipes on a cycle never reach zero.

```python
from collections import defaultdict
from collections import deque


# Blocks: one pass updating a minimum per block
# Idea: Each cell updates the minimum of block (r // block_size, c
# // block_size). No block list is built; the last blocks may be
# smaller when k does not divide the size.
def block_mins(matrix, block_size):
    rows, cols = len(matrix), len(matrix[0])
    out = [[float("inf")] * ((cols + block_size - 1) // block_size) for _ in range((rows + block_size - 1) // b
lock_size)]
    for row in range(rows):
        for col in range(cols):
            out[row // block_size][col // block_size] = min(out[row // block_size][col // block_size], matrix[r
ow][col])
    return out


# Recipes: Kahn's algorithm on the ingredient graph
# Idea: For each recipe count the ingredients it still
# lacks (those not in supplies) and record, for each
# ingredient, the recipes that need it. Start from the
# supplies; when an item becomes available decrease the
# counts of the recipes waiting for it, and queue a recipe
# when its count hits zero. Recipes on a cycle never reach
# zero.
def findAllRecipes(recipes, ingredients, supplies):
    need, waiting = {}, defaultdict(list)
    for recipe, ingredient_list in zip(recipes, ingredients):
        need[recipe] = len(set(ingredient_list))
        for ingredient in set(ingredient_list):
            waiting[ingredient].append(recipe)
    queue, made = deque(supplies), []
    while queue:
        item = queue.popleft()
        for recipe in waiting[item]:
            need[recipe] -= 1
            if need[recipe] == 0:
                made.append(recipe)
                queue.append(recipe)
    return sorted(made)
```

COMPLEXITY | Part 1: Time O(m * n) | Space O(blocks)
Part 2: Time O(R + I) | Space O(R + I)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 281

282 * Range Sum Query 2D (Immutable Matrix) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: prefix table prefix[row + 1][col + 1] = cell + prefix[row][col + 1] + prefix[row + 1][col] - prefix[row][col]; a query is four lookups

QUESTION INFO

QUESTION

Given a 2D integer matrix, handle multiple queries of the following type: return the sum of the elements inside the rectangle defined by its upper-left corner (row1, col1) and lower-right corner (row2, col2) , inclusive. Implement a class NumMatrix that is initialized once with the matrix and can then answer many sum_region queries efficiently (each query in O(1) after preprocessing).

QUESTION

Function signature

```
class NumMatrix:
    def __init__(self, matrix: list[list[int]]) -> None:
        """Preprocess the matrix (e.g., build a 2D prefix-sum table)."""
        ...
    def sum_region(self, row1: int, col1: int, row2: int, col2: int) -> int:
        """Return the sum of the submatrix with corners (row1, col1) and (row2, col2),
inclusive."""
        ...
```

Follow-up (as reported): Follow-up constraint change from the report: what if the matrix is mutable (values can be updated between queries)? Hint: switch to a 2D Binary Indexed Tree (Fenwick) for O(log rows * log cols) updates and queries.

EXAMPLES

Example 1	
Input	`matrix = [[3, 0, 1, 4, 2], [5, 6, 3, 2, 1], [1, 2, 0, 1, 5], [4, 1, 0, 1, 7], [1, 0, 3, 0, 5]] sum_region(2, 1, 4, 3)`
Output	`8`
Why	`The 3x3 block rows 2..4, cols 1..3 is [[2,0,1],[1,0,1],[0,3,0]] which sums to 8.`

Example 2	
Input	`same matrix; sum_region(1, 1, 2, 2)`
Output	`11`
Why	`Block [[6,3],[2,0]] sums to 11.`

Example 3	
Input	`same matrix; sum_region(1, 2, 2, 4)`
Output	`12`
Why	`Block [[3,2,1],[0,1,5]] sums to 12.`

CONSTRAINTS

- 1 <= rows, cols <= 200
- -10^5 <= matrix[i][j] <= 10^5
- 0 <= row1 <= row2 < rows , 0 <= col1 <= col2 < cols
- There may be up to 10^4 calls to sum_region .

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= rows, cols <= 200
- -10^5 <= matrix[i][j] <= 10^5
- 0 <= row1 <= row2 < rows , 0 <= col1 <= col2 < cols
- There may be up to 10^4 calls to sum_region .

BRUTE FORCE | Add up the rectangle for every query

BRUTE-FORCE IDEA

Start with: Add up the rectangle for every query. Each query walks every cell of its rectangle.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Add up the rectangle for every query
```

```
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(R * C) per query | Space O(1)

WHY THIS IS WEAKER | Each query walks every cell of its rectangle.

HOW TO IMPROVE IT

prefix table prefix[row + 1][col + 1] = cell + prefix[row][col + 1] + prefix[row + 1][col] - prefix[row][col]; a query is four lookups

SHORT BEST SOLUTION

P holds the sum of the rectangle from the top-left corner; a query combines four entries.

```python
# BASE QUESTION (optimal): 2D prefix sums and inclusion-
# exclusion
# Idea: P holds the sum of the rectangle from the top-left
# corner; a query combines four entries.
class NumMatrix:
    def __init__(self, matrix):
        rows, cols = len(matrix), len(matrix[0]) if matrix else 0
        self.prefix = [[0] * (cols + 1) for _ in range(rows + 1)]
        for row in range(rows):
            for col in range(cols):
                # the cell plus the block above and the block to the left,
                # minus the corner block that was counted twice
                self.prefix[row + 1][col + 1] = (matrix[row][col] + self.prefix[row][col + 1]
                                    + self.prefix[row + 1][col] - self.prefix[row][col])

    def sum_region(self, top_row, left_col, bottom_row, right_col):
        prefix = self.prefix
        return (prefix[bottom_row + 1][right_col + 1] - prefix[top_row][right_col + 1]
                - prefix[bottom_row + 1][left_col] + prefix[top_row][left_col])
```

COMPLEXITY | Time O(R * C) build, O(1) per query | Space O(R * C)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 282

283 * Rotate Any m×n 2D Matrix * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: clockwise: new[r][c] = old[rows - 1 - c][r] on a cols-by-rows result (or zip(*matrix[::-1]))

QUESTION INFO

QUESTION

Rotate an m x n matrix 90 degrees clockwise and return the resulting n x m matrix.

Function signature

def rotate_cw(matrix: list) -> list

EXAMPLES

Example 1	
Input	`[[1, 2, 3], [4, 5, 6]]`
Output	`[[4, 1], [5, 2], [6, 3]]`
Why	The last row becomes the first column: [4, 1] is the old first column read from the bottom.

Example 2	
Input	`[[1, 2], [3, 4]]`
Output	`[[3, 1], [4, 2]]`
Why	A square matrix.

QUESTION

Example 3	
Input	`[[1, 2, 3]]`
Output	`[[1], [2], [3]]`
Why	`A single row becomes a single column.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Allocate the new matrix and apply the index formula

BRUTE-FORCE IDEA

Create an n x m matrix and copy each cell old[i][j] to new[j][m - 1 - i]. Correct and the clearest statement of the rotation; the zip trick below is the same thing in one line.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Allocate the new matrix and apply the index formula
Create an n x m matrix
copy each cell old[i][j] to new[j][m - 1 - i]
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(m * n) | Space O(m * n)

WHY THIS IS WEAKER | This is already optimal in time, O(m * n); its cost is the second matrix, O(m * n) extra space. A rectangle cannot be rotated in place (its shape changes), so it is the expected answer; the zip(reversed rows) one-liner is the same idea.

HOW TO IMPROVE IT

clockwise: new[r][c] = old[rows - 1 - c][r] on a cols-by-rows result (or zip(*matrix[::-1]))

SHORT BEST SOLUTION

Part 1: Rotating clockwise puts the last row first, so reverse the rows and transpose (zip with *). A non-square matrix changes shape from m x n to n x m, which is why it cannot be done in place the way LeetCode 48 is.

Part 2: Counter-clockwise: transpose then reverse the rows (or reverse each row, then transpose). 180 degrees: reverse the rows and each row. Square matrices can be rotated in place by cycling four cells at a time, layer by layer.

```python
# BASE QUESTION (optimal): Rotate 90 degrees clockwise by
# reverse + transpose
# Idea: Rotating clockwise puts the last row first, so
# reverse the rows and transpose (zip with *). A non-
# square matrix changes shape from m x n to n x m, which
# is why it cannot be done in place the way LeetCode 48
# is.
def rotate_cw(matrix):
    return [list(row) for row in zip(*matrix[::-1])]
```

COMPLEXITY | Part 1: Time O(mn) | Space O(mn)
Part 2: Time O(mn) | Space O(1) for the in-place square version

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 283

GROUP 43 OF 43 | 2 QUESTIONS | PART A - PART 1

Split into pieces, fix each piece, join (and pack words into lines)

PATTERN CONTEXT

Parsing questions split the text on a separator, transform every piece with one small rule, and join again. Layout questions go word by word and decide whether the next word still fits on the current line.

285 * Snake Case to Camel Case with Repository-Wide Refactor * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: the same function applied to every identifier found in the files

QUESTION INFO

QUESTION

Convert a valid snake_case identifier to camelCase, then design a safe repository-wide migration for variables and function names without changing third-party symbols, public interfaces, or the tests' ability to detect regressions.

Implement a conversion from a valid snake_case identifier to camelCase.
The base input is guaranteed valid: no leading underscore, trailing underscore, or doubled separator.
Example: foo_bar -> fooBar.
Extend the discussion to a repository too large for manual review: replace snake-case variables and functions across the codebase.
Explain how to verify that the repository still works after the automated rewrite, including when test-package identifiers are rewritten too.
Preserve third-party functions and public interfaces; explain how the migration distinguishes symbols that may change from contracts that must remain stable.

EXAMPLES

```
foo_bar -> fooBar
```

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Plain text replace across the files

BRUTE-FORCE IDEA

Regex-replace every snake_case word in every file. Fast and wrong: it rewrites strings, comments, third-party calls and public interface names.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Plain text replace across the files
Regex-replace every snake_case word in every file
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(size of the repository) | Space O(1)

WHY THIS IS WEAKER | Also rewrites strings, comments, third-party calls and public interface names. Renaming only the identifiers the code defines, found with the syntax tree and tokens, keeps those intact.

HOW TO IMPROVE IT

the same function applied to every identifier found in the files

SHORT BEST SOLUTION

snake_to_camel capitalises each part after the first. owned_names collects the functions, arguments and variables the code defines (using the syntax tree), minus an allow-list of names that must stay. rename_identifiers rewrites only NAME tokens that are owned and not after a dot, so strings, comments, attribute accesses and third-party calls are untouched.

```
# BASE QUESTION (optimal): Convert names, rename only
# defined identifiers via tokens
# Idea: snake_to_camel capitalises each part after the
# first. owned_names collects the functions, arguments and
# variables the code defines (using the syntax tree),
# minus an allow-list of names that must stay.
# rename_identifiers rewrites only NAME tokens that are
# owned and not after a dot, so strings, comments,
# attribute accesses and third-party calls are untouched.
def snake_to_camel(name):
    head, *rest = name.split("_")
    return head + "".join(part[:1].upper() + part[1:] for part in rest)
```

COMPLEXITY | Time O(size of the source) | Space O(names)

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 285

286 * Google Doc Line Wrap (Word Wrap with Whitespace and Newlines) * MEDIUM *

DIFFICULTY | * MEDIUM *

Better approach: count lines: go paragraph by paragraph, add a word if it fits (current + 1 + length <= width), else start a new line

QUESTION INFO

QUESTION

EXAMPLES

```
text  = "Hello world\nFoo bar baz qux"
width = 10
Line 1: "Hello"             (5 chars; "world" + space would be 11 -> break)
Line 2: "world"
Line 3: "Foo bar"            (7 chars; "baz" -> "Foo bar baz" = 11 -> break)
Line 4: "baz qux"
Output: 4
```

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Build each line as a string and measure it

BRUTE-FORCE IDEA

Try appending each word to the current line string and check its length, starting a new line when it exceeds the width. Correct, but joining strings repeatedly costs more than the length counter in the optimal version.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Build each line as a string and measure it
Try appending each word to the current line string
check its length
starting a new line when it exceeds the width
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(total characters * words per line) | Space O(width)

WHY THIS IS WEAKER | Re-measuring a joined string for every word costs extra copying. Keeping a running length makes the whole thing O(total characters).

HOW TO IMPROVE IT

count lines: go paragraph by paragraph, add a word if it fits (current + 1 + length <= width), else start a new line

SHORT BEST SOLUTION

Split on newlines first (each paragraph ends its line), then for each word compare cur_len + 1 + len(word) with the width. Only the length is tracked, never the joined line. The empty paragraph produces one empty line, and a word longer than the width sits alone on a line.

```python
# BASE QUESTION (optimal): Greedy with a running line
# length
# Idea: Split on newlines first (each paragraph ends its
# line), then for each word compare cur_len + 1 +
# len(word) with the width. Only the length is tracked,
# never the joined line. The empty paragraph produces one
# empty line, and a word longer than the width sits alone
# on a line.
def wrap_count(text, width):
    total = 0
    for paragraph in text.split("\n"):
        current_length = 0
        lines = 1
        for word in paragraph.split():
            if current_length == 0:
                current_length = len(word)
            elif current_length + 1 + len(word) <= width:
                current_length += 1 + len(word)
            else:
                lines += 1
                current_length = len(word)
        total += lines
    return total
```

COMPLEXITY | Time O(n) | Space O(1) extra beyond the output

GO TO PART A - PART 2: FOLLOW-UPS FOR QUESTION 286

PART A - PART 2 FOLLOW-UPS

PART A - Part 2: follow-up information and code

These are all 507 follow-ups for the 197 Medium questions in PART A. When implementation is useful, the code is included. The blue button returns to the matching PART A - Part 1 solution.

Frontier search (BFS or DFS), Tier 1: spread, count and measure distance

2 * Number of Islands * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | What if land can also be removed?

Union-Find cannot split sets. Either rebuild the affected component with a local BFS from each former neighbour, or process the whole timeline offline with a segment tree over time and union-find with rollback (the offline dynamic connectivity technique).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | How do you handle a grid too large for memory?

Process it in horizontal strips: label islands within the strip, keep only the boundary row's labels, and merge labels across strips with a union-find over label ids. Memory is O(width + number of labels on the boundary).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Find the area of the largest island.

Count the cells during each search and keep the maximum. Same traversal, one counter.

FOLLOW-UP CODE

```
def largest_island(grid):
    if not grid:
        return 0
    rows, cols = len(grid), len(grid[0])
    seen = set()

    def dfs(row, col):
        if not (0 <= row < rows and 0 <= col < cols):
            return 0
        if grid[row][col] != 1 or (row, col) in seen:
            return 0
        seen.add((row, col))
        return 1 + sum(dfs(row + row_step, col + col_step)
                       for row_step, col_step in ((1, 0), (-1, 0), (0, 1), (0, -1)))

    best = 0
    for row in range(rows):
        for col in range(cols):
            best = max(best, dfs(row, col))
    return best
```

FOLLOW-UP 4 | The grid is huge and does not fit in memory.

Process it in strips of rows, label islands in each strip, and merge labels across strip boundaries with a Union-Find over the labels on the boundary row. Cost: O(m * n) time overall and O(width + labels on the boundary row) memory per strip.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 2

3 * Rotten Oranges / Multi-Source BFS (taxis) * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Sources move or new taxis appear. How do you keep the distances fresh?

Adding a source is cheap: push it at distance 0 and run a BFS that only continues into cells where it beats the stored distance (a decrease-only update, amortized small). Removing or moving a source is hard because distances can increase; either rebuild from scratch with a multi-source BFS, or rebuild periodically and accept bounded staleness in between.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Why is multi-source BFS the same cost as single-source BFS?

A virtual super-source connected to all sources by zero-length edges turns the problem into one BFS. Each cell enters the queue once, so it is O(mn) no matter how many sources exist.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Signal variant: a signal spreads from sources, how long until every target gets it?

Run the BFS and take the maximum distance over the target cells. If any target is unreachable the answer is infeasible (-1); with no targets it is 0. To allow destroying at most one blocker, compute distances from the sources and also from the targets and test each blocker cell as a bridge.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | A signal spreads from sources to targets: how long until every target is reached?

One multi-source BFS from the sources, then the answer is the largest distance over the targets (-1 if any target is unreachable, 0 if there are no targets). Per-source BFS would cost O(mn * sources).

FOLLOW-UP CODE

```
def signal_time(grid):
    """'S' sources, 'T' targets, '#' walls, '.' open: time until every target is reached; -1 if one is
unreachable; 0 with no targets."""
    rows, cols = len(grid), len(grid[0])
    time_reached = {(row, col): 0 for row in range(rows) for col in range(cols) if grid[row][col] == "S"}
    queue = deque(time_reached)
    while queue:
        row, col = queue.popleft()
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols and grid[next_row][next_col] != "#" and (
next_row, next_col) not in time_reached:
                time_reached[(next_row, next_col)] = time_reached[(row, col)] + 1
                queue.append((next_row, next_col))
    targets = [(row, col) for row in range(rows) for col in range(cols) if grid[row][col] == "T"]
    if not targets:
        return 0
    return -1 if any(target not in time_reached for target in targets) else max(time_reached[target] for
target in targets)
```

BACK TO PART A - PART 1: QUESTION 3

4 * Find shortest relationship path using BFS * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The graph is huge (a social network).

Bidirectional BFS searches from both ends and stops when the frontiers meet; with branching factor b and distance d it explores about 2 * b^(d/2) nodes instead of b^d.

FOLLOW-UP CODE

```
def shortest_relationship_bidirectional(connections, source, target):
    if source == target:
        return [source]
    parent_forward, parent_backward = {source: None}, {target: None}
    forward_queue, backward_queue = deque([source]), deque([target])
    meeting_point = None
    while forward_queue and backward_queue and meeting_point is None:
        if len(forward_queue) <= len(backward_queue):
            for _ in range(len(forward_queue)):
                person = forward_queue.popleft()
                for neighbour in connections.get(person, ()):
                    if neighbour not in parent_forward:
                        parent_forward[neighbour] = person
                        forward_queue.append(neighbour)
                        if neighbour in parent_backward:
                            meeting_point = neighbour
                            break
                if meeting_point is not None:
                    break
        else:
            for _ in range(len(backward_queue)):
                person = backward_queue.popleft()
                for neighbour in connections.get(person, ()):
                    if neighbour not in parent_backward:
```

```
                    parent_backward[neighbour] = person
                    backward_queue.append(neighbour)
                    if neighbour in parent_forward:
                        meeting_point = neighbour
                        break
            if meeting_point is not None:
                break
    if meeting_point is None:
        return None
    source_half, node = [], meeting_point
    while node is not None:
        source_half.append(node)
        node = parent_forward[node]
    target_half, node = [], parent_backward[meeting_point]
    while node is not None:
        target_half.append(node)
        node = parent_backward[node]
    return source_half[::-1] + target_half
```

FOLLOW-UP 2 | Return every shortest path, or count them.

Keep, for each node, its distance and a list of all parents at that distance; counting is a DP over the BFS layers, and listing the paths is a DFS over the parent lists.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Relationships have different strengths (weights).

BFS is no longer right: use Dijkstra with a heap for non-negative weights. Keep the same parent map to rebuild the path.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 4

5 * Compute distance to nearest taxi in grid * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | A new taxi appears.

Start a BFS from the new taxi but only continue into cells where you improve the stored distance. The distances can only decrease, so the work is bounded by the cells whose answer actually changes.

FOLLOW-UP CODE

```
class TaxiMap:
    """A taxi appears: only cells that become closer change, so relax outward from the new taxi."""
    def __init__(self, grid):
        self.grid, self.dist = [list(grid_row) for grid_row in grid], taxi_distances(grid)

    def add_taxi(self, row, col):
        self.grid[row][col] = "T"
        self.dist[row][col] = 0
        queue = deque([(row, col)])
        while queue:
            cell_row, cell_col = queue.popleft()
            for next_row, next_col in ((cell_row + 1, cell_col), (cell_row - 1, cell_col), (cell_row,
cell_col + 1), (cell_row, cell_col - 1)):
                if 0 <= next_row < len(self.grid) and 0 <= next_col < len(self.grid[0]) and self.grid[
next_row][next_col] != "#":
                    if self.dist[next_row][next_col] == -1 or self.dist[next_row][next_col] > self.dist[
cell_row][cell_col] + 1:
                        self.dist[next_row][next_col] = self.dist[cell_row][cell_col] + 1
                        queue.append((next_row, next_col))
```

FOLLOW-UP 2 | Taxis move or disappear.

Distances can increase, which the decrease-only update cannot handle. Rebuild with one multi-source BFS (O(mn)), perhaps lazily every few updates and accepting bounded staleness between rebuilds.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Return which taxi is nearest, not only the distance.

Store a source id with each cell in the BFS: a cell copies the id of the cell that first reached it. Ties go to whichever reached it first, so say how you want ties broken.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 5

6 * Solve Grid and Counting Problems * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Any target word, not only GOOGLE.

Count the needed letters of the word; the answer is the product over distinct letters of C(available, needed).

FOLLOW-UP CODE

```
def ways_to_form_word(letters, word):
    needed, available = Counter(word), Counter(letters)
    result = 1
    for letter, needed_count in needed.items():
        result *= math.comb(available[letter], needed_count)
    return result
```

FOLLOW-UP 2 | The letters must appear in order (a subsequence of the array spells the word).

Then it is the 'distinct subsequences' DP: ways[j] counts the ways to spell the first j letters; scan the array and update j from high to low. O(n * len(word)).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Islands arrive one cell at a time.

Use Union-Find: each new land cell makes the count go up by one and each union with a neighbouring different island makes it go down by one (LeetCode 305).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 6

7 * Farthest City for a New Data Center * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | The roads have lengths. How does it change?

Replace the BFS queue with a min-heap (Dijkstra) seeded with every data center at distance 0. The first time a city is popped its distance is final. O((V + E) log V).

FOLLOW-UP CODE

```
import heapq


def farthest_city_weighted(num_cities, roads, data_centers):
    neighbours = [[] for _ in range(num_cities)]
    for from_city, to_city, road_length in roads:
        neighbours[from_city].append((to_city, road_length))
        neighbours[to_city].append((from_city, road_length))
    distance = [float("inf")] * num_cities
    heap = [(0, data_center) for data_center in data_centers]
    for d in data_centers:
        distance[d] = 0
    while heap:
        d, from_city = heapq.heappop(heap)
        if d > distance[from_city]:
            continue
        for to_city, road_length in neighbours[from_city]:
            if d + road_length < distance[to_city]:
                distance[to_city] = d + road_length
                heapq.heappush(heap, (distance[to_city], to_city))
    return distance.index(max(distance))
```

BACK TO PART A - PART 1: QUESTION 7

Sweep over events in time order

10 * Meeting Rooms / Interval Scheduling Variants * MEDIUM *

DIFFICULTY | * MEDIUM * | 2 FOLLOW-UPS

FOLLOW-UP 1 | Meetings wrap around midnight (cyclic time).

Split every wrapped meeting [start, end) with start >= end into [start, 24h) and [0, end) and run the same sweep: the overlap at any clock time is unchanged. A meeting that is exactly 24 hours long covers the whole day.

FOLLOW-UP CODE

```
def min_rooms(intervals):
    events = sorted([(start, 1) for start, end in intervals] + [(end, -1) for start, end in intervals],
key=lambda event: (event[0], event[1]))
    running_count = most_rooms = 0
    for _, change in events:
        running_count += change
        most_rooms = max(most_rooms, running_count)
    return most_rooms

def min_rooms_cyclic(intervals, day=24 * 60):
    """Meetings may wrap past midnight: start > end means 'until tomorrow'. Split a wrapped meeting at the
day boundary."""
    parts = []
    for start, end in intervals:
        if start < end:
            parts.append((start, end))
        else:
            parts += [(start, day), (0, end)]
    return min_rooms(parts)
```

FOLLOW-UP 2 | The input is huge and spread over many machines.

Each shard sorts its events and produces a compressed timeline of (time, delta) changes; merge the sorted timelines with a k-way merge and take the running maximum, or add the per-shard counts at fixed time buckets. The answer is a max over time of a sum over shards.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 10

11 * Train / Flight Schedule Reachability * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the earliest arrival time at the destination.

The table already holds it: earliest_arrivals(legs, source)[dest], or infinity when the station is unreachable. That is the same one pass, so it costs nothing extra.

FOLLOW-UP CODE

```
def earliest_arrivals(legs, source, strict=True):
    earliest = {source: -INF}
    for origin, destination, departure, arrival in sorted(legs, key=lambda leg: leg[2]):
        earliest_at_origin = earliest.get(origin, INF)
        if earliest_at_origin < departure or (not strict and earliest_at_origin == departure):
            if arrival < earliest.get(destination, INF):
                earliest[destination] = arrival
    return earliest
```

FOLLOW-UP 2 | Print the route, not just yes or no.

Remember the leg that last improved each station's earliest arrival, then walk back from the destination to the source and reverse. Because only a strictly better arrival replaces the stored leg, the stored legs form a valid chain in time.

FOLLOW-UP CODE

```
def itinerary(legs, source, dest, strict=True):
    """The legs of a route that reaches dest as early as possible, or None."""
    earliest, leg_into = {source: -INF}, {}
    for leg in sorted(legs, key=lambda leg: leg[2]):
        origin, next_station, departure, arrival = leg
        earliest_at_origin = earliest.get(origin, INF)
        if (earliest_at_origin < departure or (not strict and earliest_at_origin == departure)) and arrival <
earliest.get(next_station, INF):
            earliest[next_station], leg_into[next_station] = arrival, leg
    if dest not in earliest:
        return None
    route = []
    while dest != source:
        leg = leg_into[dest]
        route.append(leg)
        dest = leg[0]
    return route[::-1]
```

FOLLOW-UP 3 | Many queries on the same schedule: what changes in the graph representation?

Group the legs by origin station and sort each list by departure time. For a query the earliest arrival is relaxed with a heap ordered by time (Dijkstra on time), and the first leg you may board at a station is found with bisect on its departure list, so each query costs O(L log S) instead of re-sorting every time. If the source is fixed and the queries only differ by destination, keep the earliest-arrival table and answer each in O(1).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 11

12 * Solve Rooms and Top-K Streams * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Rooms: return which room each meeting uses.

Sort by start and keep a min-heap of (end time, room). If the earliest-ending room is free at the next start, reuse it, otherwise open a new room. The heap size never exceeds the answer.

FOLLOW-UP CODE

```
def meeting_room_assignment(intervals):
    """Which room each meeting uses: min-heap of (end time, room id)."""
    free, room_of, room_count = [], {}, 0
    for index in sorted(range(len(intervals)), key=lambda index: intervals[index][0]):
        start, end = intervals[index]
        if free and free[0][0] <= start:
            _, room = heapq.heappop(free)
        else:
            room, room_count = room_count, room_count + 1
        room_of[index] = room
        heapq.heappush(free, (end, room))
    return room_count, room_of
```

FOLLOW-UP 2 | Stream: K changes over time, or you need the median.

A fixed-size heap cannot answer a different K. Use a sorted structure with order statistics (balanced tree or Fenwick tree over values), or two heaps for the median. Cost: a balanced tree or Fenwick tree gives O(log n) per insert and per k-th query; two heaps give O(log n) per insert and O(1) per median.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Rooms: meetings on several floors with capacities.

Group meetings by required capacity, schedule each group separately with the sweep, and assign rooms from the smallest sufficient capacity (best fit) when capacities differ.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 12

13 * Gate Entry / Exit Time Resolution with Tie-Break Rules * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Why does the optimization need care across an empty gap?

If the door sits idle for even one second the previous state becomes 'none', which favours exits again. Jumping time forward without resetting the state would wrongly keep 'enter' priority from before the gap. Compare t with last_used + 1 to know whether the gap really was idle.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What is the complexity, and what if events come online?

Offline it is O(n log n) for the sort, then O(n) for the simulation, independent of how large the timestamps are. Online, keep the two queues and the previous-state variable and advance the clock on every arrival or tick; the work per person stays O(1).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Timestamps go up to 10^9.

That is why the event-driven loop jumps to the next arrival: its running time is O(n log n) and does not depend on the size of the timestamps. Remember to reset the previous state to idle across a gap.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | The tie-break when the gate was idle differs in this version of the question.

Keep the simulation and make the idle winner a parameter: only the line that picks the queue changes. With 'exit first' it equals the standard rule; with 'enter first' an enter and an exit arriving together swap places.

FOLLOW-UP CODE

```
def turnstile_times_priority(events, idle_first="exit"):
    """Same simulation with a configurable idle priority (some versions let entering go first when the door
was idle)."""
    num_people = len(events)
    people_by_arrival = sorted(range(num_people), key=lambda person: (events[person][0], person))
    enter_queue, exit_queue, pass_time = deque(), deque(), [0] * num_people
    time, next_arrival, last_direction, last_busy_second = 0, 0, None, -10
    while next_arrival < num_people or enter_queue or exit_queue:
        if not enter_queue and not exit_queue:
            time = max(time, events[people_by_arrival[next_arrival]][0])
            if time > last_busy_second + 1:
                last_direction = None
        while next_arrival < num_people and events[people_by_arrival[next_arrival]][0] <= time:
            (enter_queue if events[people_by_arrival[next_arrival]][1] == "enter" else exit_queue).append(
people_by_arrival[next_arrival])
            next_arrival += 1
        if enter_queue and exit_queue:
            direction = last_direction if last_direction is not None else idle_first
            queue = enter_queue if direction == "enter" else exit_queue
        else:
            queue = enter_queue or exit_queue
        person = queue.popleft()
        pass_time[person], last_direction, last_busy_second = time, events[person][1], time
        time += 1
    return pass_time

def gate_pass_times_priority(events, idle_first="exit"):
    return turnstile_times_priority(events, idle_first)
```

BACK TO PART A - PART 1: QUESTION 13

14 * Compute servers needed for daily recurring jobs * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Some rare jobs run for several days without restart and get a dedicated server.

Remove those jobs from the daily intervals and add one server per such job: servers = (number of dedicated jobs) + (maximum overlap of the remaining jobs).

FOLLOW-UP CODE

```
def servers_needed(intervals):
    by_job = defaultdict(list)
    for job_id, start, end in intervals:
        by_job[job_id].append((start, end))
    events = []
    for job_intervals in by_job.values():
        merged_start, merged_end = None, None
        for start, end in sorted(job_intervals):
            if merged_end is not None and start <= merged_end:
                merged_end = max(merged_end, end)
            else:
                if merged_end is not None:
                    events += [(merged_start, 1), (merged_end, -1)]
                merged_start, merged_end = start, end
        events += [(merged_start, 1), (merged_end, -1)]
    running_count = most_servers = 0
    for _, change in sorted(events):
        running_count += change
        most_servers = max(most_servers, running_count)
    return most_servers

def servers_with_dedicated(intervals, long_jobs):
    """Follow-up: jobs listed in long_jobs reserve a dedicated server for the whole period; the others share
servers."""
    shared = [(job_id, start, end) for job_id, start, end in intervals if job_id not in long_jobs]
    return len({job_id for job_id, _, _ in intervals if job_id in long_jobs}) + servers_needed(shared)
```

FOLLOW-UP 2 | Jobs that cross midnight.

Truncate at the end of the day for each day's schedule (as the statement says). For a recurring daily job that wraps, split it into two pieces (the end of the day and the start of the day); the sweep then handles both on the same circular day.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Return which server each job runs on.

Sort merged intervals by start and keep a min-heap of (end time, server id): reuse the server that frees up first if it is free, otherwise open a new one. The number of servers opened equals the maximum overlap.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 14

15 * Meeting Rooms III Variant * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | A wrapped meeting must stay in a single room.

Then it is circular-arc colouring, NP-hard in general. A common practical method: cut the circle at the minute with the fewest active meetings; the meetings crossing the cut use fixed rooms, and the rest are scheduled greedily with a heap. It can use more rooms than the maximum overlap, and the five-arc example shows why.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Return which room each meeting uses.

Sort by start and keep a min-heap of (end time, room id); reuse the earliest-ending room if it is free, otherwise open a new one. For wrapped meetings schedule them first so their rooms are fixed.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Days can repeat (a wrapped meeting recurs every day).

A recurring meeting occupies its slots every day, so the circle model above is exactly right, and the no-wrap shortcut of ordering by start time is not.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Return which room each meeting uses.

Sort by start and keep a heap of (end time, room): reuse the earliest-ending room if it is free, otherwise open a new one. For meetings that do not wrap this uses exactly the minimum number of rooms (verified against the sweep); wrapped meetings need the cut-the-circle method.

FOLLOW-UP CODE

```
def assign_rooms(meetings):
    free, out, rooms = [], {}, 0
    for i in sorted(range(len(meetings)), key=lambda i: meetings[i][0]):
        s, e = meetings[i]
        if free and free[0][0] <= s:
            _, room = heapq.heappop(free)
        else:
            room, rooms = rooms, rooms + 1
        out[i] = room
        heapq.heappush(free, (e, room))
    return rooms, out
```

BACK TO PART A - PART 1: QUESTION 15

Tree recursion, Tier 1: one function that visits a node, asks its children, and reports upward

22 * Expression Tree Single-Leaf Mutation * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | The node interface has no parent pointer.

Build the path while collecting leaves: carry the list of ancestors with each stack entry, and use that list for the walk up. It costs O(depth) memory per leaf; the parent dict does the same job with O(n) total.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Apply several flips at once, or flips that stay.

The same cached values work: toggle, walk up with early stop, and keep the changes. For many flips between queries, recompute each affected path in turn.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Which leaves change the answer at all?

Compute top-down 'sensitivity': a leaf matters if every ancestor's other input lets it through (AND with true, OR with false, any XOR or NOT). One pass gives all of them in O(n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Which leaves can change the answer at all?

Propagate a 'matters' flag downward: through AND the other side must be true, through OR the other side must be false, through XOR and NOT it always passes. A leaf matters exactly when the flag reaches it (checked against flipping every leaf). Subtree values are cached in practice, giving O(n).

FOLLOW-UP CODE

```
def sensitive_leaves(root):
    def evaluate(node):
        if node.op is None:
            return node.val
        left = evaluate(node.left)
        right = evaluate(node.right) if node.right is not None else None
        if node.op == "AND": return left and right
        if node.op == "OR": return left or right
        if node.op == "XOR": return left != right
        return not left

    answer = []

    def dfs(node, matters):
        if node.op is None:
            if matters:
                answer.append(node)
            return
        if node.op == "NOT":
            dfs(node.left, matters)
            return
        left, right = evaluate(node.left), evaluate(node.right)
        if node.op == "AND":
            left_matters, right_matters = matters and right, matters and left
        elif node.op == "OR":
            left_matters, right_matters = matters and not right, matters and not left
        else:
            left_matters = right_matters = matters
        dfs(node.left, left_matters)
        dfs(node.right, right_matters)

    dfs(root, True)
    return answer
```

BACK TO PART A - PART 1: QUESTION 22

23 * Longest Consecutive Path in Binary Tree * MEDIUM *

DIFFICULTY | * MEDIUM * | 5 FOLLOW-UPS

FOLLOW-UP 1 | What does the any-direction version return for a single node, or for duplicate values?

A single node is a path of length 1 (0 edges), and an empty tree is 0. Equal values never extend a chain because the step must be exactly +1 or -1, so duplicates reset both counters to 1 at that node.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Can you avoid recursion?

Yes: do an iterative post-order or a BFS that stores (node, run) pairs for the top-down version. The bending version needs child results before the parent, so use an explicit stack with a post-order visit or process nodes in reverse BFS order using a dict from node to (inc, dec). It matters because a chain-shaped tree makes the recursive depth n.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Return the path itself, not only its length.

Carry the current run as a list (or its start node and length) in the traversal and copy it into the best answer when it becomes longer.

FOLLOW-UP CODE

```
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def longest_consecutive_path_values(root):
    best = []

    def dfs(node, parent_val, path):
        if node is None:
            return
        path = path + [node.val] if parent_val is not None and node.val == parent_val + 1 else [node.val]
        if len(path) > len(best):
            best[:] = path
        dfs(node.left, node.val, path)
        dfs(node.right, node.val, path)

    dfs(root, None, [])
    return best
```

FOLLOW-UP 4 | The path may go in any direction (LeetCode 549).

Return the pair (longest increasing chain down, longest decreasing chain down) from each node, and update the global best with inc + dec - 1 at the node. Verified against exhaustive path enumeration for small trees.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 5 | The values must increase by 1 or decrease by 1 along the path but not mix.

That is exactly the any-direction version: the increasing and decreasing parts meet at the bend; a path cannot zig-zag between increasing and decreasing steps.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 23

24 * Implement zigzag level-order traversal * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Bottom-up level order (leaf level first).

Run the normal traversal and reverse the list of levels at the end.

FOLLOW-UP CODE

```
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def level_order_bottom_up(root):
    levels, queue = [], deque([root] if root else [])
    while queue:
        levels.append([node.val for node in queue])
        for _ in range(len(queue)):
            node = queue.popleft()
            queue.extend(child for child in (node.left, node.right) if child)
    return levels[::-1]
```

FOLLOW-UP 2 | Do it with DFS.

Pass the depth; append the value to levels[depth], at the end when the depth is even and at the front (insert at 0 or a deque) when odd. Recursion depth is the tree height.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | N-ary tree.

Replace the two children by the list of children; the direction flag works the same way.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 24

25 * Given the root of a binary tree, return the length of the diameter of the tree. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | An n-ary tree.

Take the two largest child heights at each node (not just left and right): best = max(best, top1 + top2). Sorting the children's heights costs O(d log d) per node, or track the top two in O(d).

FOLLOW-UP CODE

```
class NaryNode:
    def __init__(self, val, children=None):
        self.val, self.children = val, children or []

def nary_diameter(root):
    best = 0

    def depth(node):
        nonlocal best
        if node is None:
            return 0
        tops = sorted((depth(child) for child in node.children), reverse=True)[:2]
        best = max(best, sum(tops))
        return 1 + (tops[0] if tops else 0)

    depth(root)
    return best
```

FOLLOW-UP 2 | Edges have weights, or you want the path itself.

Add the edge weight to each child's height before comparing; for the path, also store which child gave each of the two best heights and rebuild from the node that produced the best sum.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Count the diameter in nodes instead of edges.

Add one to the answer (a path of k edges has k + 1 nodes). State which convention the interviewer wants before you code.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 25

A heap of what is available: pop the best, put back the changed one

28 * Streaming Median (LC 295) with Follow-ups * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Almost every value is in a small range (say 0 to 100). Can you do better than two heaps?

Yes: a counting array gives O(1) inserts, and the median needs one pass over at most V counts, so O(V) per query with V fixed. Keep a moving pointer to the median bucket to make queries amortized O(1).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Now support an arbitrary k-th smallest, not only the median.

Two heaps only work for one fixed rank. Use an order-statistics structure: a Fenwick tree over (compressed) values, or a balanced BST augmented with subtree sizes. Both give O(log n) add and O(log n) k-th, and the Fenwick tree is far shorter to write in an interview.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | What if the data stream is too large to store?

Exact medians need memory proportional to the data, so use a quantile sketch (t-digest, GK, KLL). They answer any quantile within a chosen error bound in O(1/epsilon) memory.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | The median of only the last K values.

Keep the window in a deque (to know what expires) and a sorted list (to read the middle); insert and remove with bisect. O(K) per update because of list shifting; two heaps with lazy deletion give O(log K).

FOLLOW-UP CODE

```
class SlidingWindowMedian:
    """Median of the last k values, with a sorted window (bisect for insert and remove)."""
    def __init__(self, window_size):
        self.window_size, self.queue, self.sorted_values = window_size, deque(), []

    def add(self, value):
        self.queue.append(value)
        bisect.insort(self.sorted_values, value)
        if len(self.queue) > self.window_size:
            oldest = self.queue.popleft()
            del self.sorted_values[bisect.bisect_left(self.sorted_values, oldest)]
        count = len(self.sorted_values)
        return float(self.sorted_values[count // 2]) if count % 2 else (self.sorted_values[count // 2 - 1] +
self.sorted_values[count // 2]) / 2
```

BACK TO PART A - PART 1: QUESTION 28

29 * Compute minimum number of rooms needed * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Can one person attend all meetings (LeetCode 252)?

Sort by start and check that each meeting ends no later than the next one starts. O(n log n), O(1) extra.

FOLLOW-UP CODE

```
def can_attend_all(intervals):
    ordered = sorted(intervals)
    return all(earlier[1] <= later[0] for earlier, later in zip(ordered, ordered[1:]))
```

FOLLOW-UP 2 | Return which room each meeting uses.

Keep a heap of (end time, room id): reuse the room that frees first, otherwise open the next id, and record the id per meeting.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Meetings arrive online.

Keep the end-time heap between calls: pop every end at or before the new start (those rooms are free), then push the new end. The heap size is the number of rooms in use right now; the peak is the answer so far.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 29

30 * Implement a Reusable Locker Allocation System * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | There are 10^9 lockers.

The counter and the heap already make the structure independent of n: nothing is allocated up front, and memory depends only on how many lockers have been released. Cost: O(log R) per operation and O(R) memory, where R is the number of lockers released and not yet reused; independent of n.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Many threads call store and pickup.

Protect the heap, the counter and the customer map with one lock so a locker number is never handed out twice (or fine-grained: a lock around the (heap, counter) pair).

FOLLOW-UP CODE

```
class Lockers:
    def __init__(self, locker_count):
        self.locker_count, self.next_new, self.released_lockers, self.locker_of = locker_count, 1, [], {}

    def store(self, customer):
        if customer in self.locker_of:
            return -1
        if self.released_lockers:
            locker_number = heapq.heappop(self.released_lockers)
        elif self.next_new <= self.locker_count:
            locker_number, self.next_new = self.next_new, self.next_new + 1
        else:
            return -1
        self.locker_of[customer] = locker_number
        return locker_number

    def pickup(self, customer):
        locker_number = self.locker_of.pop(customer, None)
        if locker_number is None:
            return -1
        heapq.heappush(self.released_lockers, locker_number)
        return locker_number

class LockersThreadSafe(Lockers):
    def __init__(self, locker_count):
        super().__init__(locker_count)
        self.lock = threading.Lock()

    def store(self, customer):
        with self.lock:
            return super().store(customer)

    def pickup(self, customer):
        with self.lock:
            return super().pickup(customer)
```

FOLLOW-UP 3 | Lockers of different sizes.

Keep one heap and counter per size class, and let a customer choose the smallest class that fits. The structure per class is the same.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 30

31 * Simulate room assignments for scheduled meetings * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | What if meetings arrive online and cannot be reordered?

The same two heaps work if you process meetings in arrival order and treat the arrival time as the earliest allowed start, but you can no longer rely on sorted starts, so you must first move every room whose end time is <= the arrival into the free heap. Each call stays O(log N).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | How do you report the busiest room by total busy time instead of meeting count?

Add the meeting's actual duration to used[room] instead of 1 when you assign it. Rooms are chosen the same way; only the tally changes, and ties still break toward the lowest index.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Rank rooms by total busy time instead of the number of meetings.

Add each meeting's duration (its original length, not the delay) to the room's tally; the choice of room is unchanged.

FOLLOW-UP CODE

```
def most_booked_by_time(room_count, meetings):
    free_rooms = list(range(room_count))
    heapq.heapify(free_rooms)
    busy_rooms, busy_time = [], [0] * room_count
    for start, end in sorted(meetings):
        while busy_rooms and busy_rooms[0][0] <= start:
            heapq.heappush(free_rooms, heapq.heappop(busy_rooms)[1])
        if free_rooms:
            room, finish = heapq.heappop(free_rooms), end
        else:
            finish, room = heapq.heappop(busy_rooms)
            finish += end - start
        heapq.heappush(busy_rooms, (finish, room))
        busy_time[room] += end - start
    return busy_time.index(max(busy_time))
```

BACK TO PART A - PART 1: QUESTION 31

32 * Minimum Makespan on Identical CPUs * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | The first idea tracked every CPU's availability in a list. Why is the heap better?

With a list you scan all C CPUs for the earliest free one on every job: O(n * C). A min-heap returns the earliest free time in O(log C) and updates in O(log C), so the total is O(n log C) after the sort.

FOLLOW-UP CODE

```
def min_makespan_scan(release_times, num_cpus, duration):
    cpu_free_at = [0] * num_cpus
    for release_time in sorted(release_times):
        cpu = cpu_free_at.index(min(cpu_free_at))
        cpu_free_at[cpu] = max(release_time, cpu_free_at[cpu]) + duration
    return max(cpu_free_at) if release_times else 0
```

BACK TO PART A - PART 1: QUESTION 32

33 * Design server allocation for multi-type nodes * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Many clients call at the same time.

Make allocation atomic: one lock around the allocate step, or shard the pool by type or by id range and route requests to shards. The roll-back-on-failure design means a failed request leaves no trace.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Clients keep asking for rare types and starve.

Use a FIFO queue per request class and hold back versatile servers for waiting rare-type requests, or age requests so an old one outranks new ones. Fragmentation control (least flexible first) already reduces the pressure.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Servers fail while allocated.

Track a lease or heartbeat per allocation; when a server stops responding mark it unavailable, remove it from the heaps (lazy deletion handles stale entries), and tell the client so it can request a replacement.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | A client dies without releasing its servers.

Give every allocation a lease (expiry time). A periodic sweep returns servers whose lease has expired to the free pool (the lazy-deletion indexes make a release cheap), and clients renew the lease while they are alive.

FOLLOW-UP CODE

```
class Servers:
    def __init__(self, servers):
        self.types_of_server = {server_id: frozenset(server_types) for server_id, server_types in servers.
items()}
        self.allocated = set()
        self.heaps = defaultdict(list)
        for server_id, supported_types in self.types_of_server.items():
            for server_type in supported_types:
                heapq.heappush(self.heaps[server_type], (len(supported_types), server_id))

    def _top(self, server_type, skipped_ids):
        heap, skipped_entries = self.heaps[server_type], []
        while heap:
            flexibility, server_id = heap[0]
            if server_id in self.allocated:
                heapq.heappop(heap)
            elif server_id in skipped_ids:
                skipped_entries.append(heapq.heappop(heap))
            else:
                break
        result = heap[0] if heap else None            # read the best valid entry before putting the skipped
ones back
        for item in skipped_entries:
            heapq.heappush(heap, item)
        return result

    def getServer(self, count, types):
        chosen = []
        for server_type in types:
            if any(server_type in self.types_of_server[chosen_id] for chosen_id in chosen):
                continue
            top = self._top(server_type, set(chosen))
            if top is None:
                return None
            chosen.append(top[1])
        while len(chosen) < count:
            best = None
            for server_type in types:
                top = self._top(server_type, set(chosen))
                if top and (best is None or top < best):
                    best = top
            if best is None:
                return None
            chosen.append(best[1])
        if len(chosen) > count:
            return None
```

```
        self.allocated |= set(chosen)
        return chosen

    def releaseServer(self, ids):
        for server_id in ids:
            if server_id in self.allocated:
                self.allocated.discard(server_id)
                for server_type in self.types_of_server[server_id]:
                    heapq.heappush(self.heaps[server_type], (len(self.types_of_server[server_id]), server_id))

class ServersLeased(Servers):
    def __init__(self, servers):
        super().__init__(servers)
        self.lease = {}

    def getServer(self, count, types, now=0, ttl=10):
        got = super().getServer(count, types)
        if got:
            for server_id in got:
                self.lease[server_id] = now + ttl
        return got

    def expire(self, now):
        gone = [server_id for server_id, expires_at in self.lease.items() if expires_at <= now]
        for server_id in gone:
            del self.lease[server_id]
        self.releaseServer(gone)
        return sorted(gone)
```

BACK TO PART A - PART 1: QUESTION 33

34 * Trip Expense Settlement * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Do you need the true minimum number of payments?

That is NP-hard in general (it is the zero-sum partition problem). The greedy answer is at most person_count - 1 payments and is usually accepted; for exact answers use the bitmask DP above (about 20 people) or backtracking with pruning.

FOLLOW-UP CODE

```
def settle_min_dp(records):
    balance_list = list(balances(records).values())
    person_count = len(balance_list)
    subset_total = [0] * (1 << person_count)
    max_zero_groups = [0] * (1 << person_count)
    for mask in range(1, 1 << person_count):
        low = (mask & -mask).bit_length() - 1
        subset_total[mask] = subset_total[mask & (mask - 1)] + balance_list[low]
        best_groups = 0
        for person in range(person_count):
            if mask >> person & 1:
                best_groups = max(best_groups, max_zero_groups[mask ^ (1 << person)])
        max_zero_groups[mask] = best_groups + (1 if subset_total[mask] == 0 else 0)
    return person_count - max_zero_groups[(1 << person_count) - 1]
```

FOLLOW-UP 2 | Amounts that do not divide evenly (cents).

Work in integer cents and decide where the remainder goes (to the payer, or round-robin among payees). Never use floats for money, and make the rule deterministic.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Different shares per person instead of equal split.

Replace the equal share with an explicit amount per payee in each record; the net balance per person is computed the same way.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 34

35 * Playlist Shuffle No Adjacent Artists * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The shuffle should look random, not always the same arrangement.

Break ties between equally frequent artists randomly and pick the song within the artist at random, as in the code. The result is valid but not uniform among all valid shuffles; a uniform sampler needs counting arrangements.

FOLLOW-UP CODE

```
def shuffle_random_no_adjacent(artists, rng):
    """Random order among valid choices: break ties between equally frequent artists at random (valid, but
not uniform over all valid shuffles)."""
    counts = Counter(artists)
    out, previous_artist = [], None
    for _ in range(len(artists)):
        options = [artist for artist in counts if counts[artist] and artist != previous_artist] or [artist
for artist in counts if counts[artist]]
        highest_count = max(counts[artist] for artist in options)
        chosen_artist = rng.choice([artist for artist in options if counts[artist] == highest_count])
        out.append(chosen_artist)
        counts[chosen_artist] -= 1
        previous_artist = chosen_artist
    return out
```

FOLLOW-UP 2 | Songs by the same artist must be at least K apart.

Same idea with a waiting queue of length K: after playing an artist, hold them for K - 1 more songs before they are eligible again (the LeetCode 358 construction).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | What is the minimum number of unavoidable repeats?

With n songs and a dominant artist with m songs, the other n - m songs separate at most n - m + 1 groups, so at least m - (n - m + 1) = 2m - n - 1 neighbouring pairs of that artist are forced (if positive).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 35

Recurrence on a position or a range (memo or table)

37 * Target Expressions from Ordered Digits * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | How do you handle N digits where listing every expression is too much?

Agree on the contract: if you only need to know whether a target is reachable, store just the set of values per interval (a boolean DP over (l, r, value)), which is far smaller than the set of strings. If you need one witness, store one expression string per value. Values are bounded by the digits, so the table stays practical for moderate N.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if concatenating digits (23) or division is allowed?

Concatenation adds base cases: for every l and r the number int(digits[l:r]) is also an atom of the interval. Division adds a fourth operator with a zero check and, if exactness matters, Fraction values instead of floats.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Only a yes/no answer is needed for many digits.

Store just the set of reachable values per interval (as target_reachable does) instead of the expression strings; the strings blow up long before the value sets do. Build the expression only for the winning target by remembering one split per value.

FOLLOW-UP CODE

```
def target_reachable(digits, target):
    memo = {}

    def values(start, end):
        if (start, end) in memo:
            return memo[(start, end)]
        if end - start == 1:
            values_found = {digits[start]}
        else:
            values_found = set()
            for split in range(start + 1, end):
                for left_value in values(start, split):
                    for right_value in values(split, end):
                        values_found |= {left_value + right_value, left_value - right_value, left_value *
right_value}
        memo[(start, end)] = values_found
        return values_found

    return target in values(0, len(digits))
```

BACK TO PART A - PART 1: QUESTION 37

38 * Count Word Decompositions into Chemical Element Symbols * MEDIUM *

DIFFICULTY | * MEDIUM * | 5 FOLLOW-UPS

FOLLOW-UP 1 | How would you list the decompositions, not just count them?

Keep the same suffix recursion but return the list of splits instead of a number, memoized by start index. The count version is O(n); the list version is output-sensitive because there can be exponentially many splits, so agree on whether the interviewer wants a count, one witness, or all of them.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if the dictionary is large and symbols can be long?

Put the dictionary in a trie (or a set plus the maximum word length L) and only look at substrings up to length L from each index. That makes the DP O(n * L) instead of O(n^2), which is the standard Word Break optimization.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | List the decompositions instead of counting them.

Same recursion, but carry the parts chosen so far and record them at the end. The number of results can be exponential, so ask whether a count or a list is wanted.

FOLLOW-UP CODE

```
def list_decompositions(word, symbols=ELEMENTS):
    lowered_word = word.lower()
    decompositions = []

    def go(position, parts):
        if position == len(lowered_word):
            decompositions.append("-".join(symbol.capitalize() for symbol in parts))
            return
        for symbol_length in (1, 2):
            if position + symbol_length <= len(lowered_word) and lowered_word[position:position +
symbol_length] in symbols:
                go(position + symbol_length, parts + [lowered_word[position:position + symbol_length]])

    go(0, [])
    return decompositions
```

FOLLOW-UP 4 | A larger dictionary with long entries.

Store the dictionary in a set plus the maximum word length L (only look at substrings up to L), or in a trie, so each position costs O(L) lookups instead of O(n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 5 | Count modulo a prime, or only test whether any decomposition exists.

Replace addition by addition modulo the prime, or by logical or for a yes/no answer. The DP structure is the same.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 38

39 * Fence Painting Minimum Operations (Gemini Team) * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The horizontal brush is only W planks wide.

A band over a range needs ceil(width / W) strokes per row instead of one, so the band cost becomes (lowest - base) * ceil(len / W). The vertical option and the splitting are unchanged.

FOLLOW-UP CODE

```
def paint_brush_width(heights, brush_width):
    """Follow-up: a horizontal brush paints exactly at most w consecutive planks per stroke."""
    def solve(left, right, base):
        if left > right:
            return 0
        lowest = min(heights[left:right + 1])
        cost = 0
        run_start = left
        while run_start <= right:
            if heights[run_start] == lowest:
                run_start += 1
                continue
            run_end = run_start
            while run_end <= right and heights[run_end] != lowest:
                run_end += 1
            cost += solve(run_start, run_end - 1, lowest)
            run_start = run_end
        # a band of height (lowest - base) over right - left + 1 planks needs ceil(width / brush_width)
strokes per row
        band = (lowest - base) * (-(-(right - left + 1) // brush_width))
        return min(right - left + 1, band + cost)

    return solve(0, len(heights) - 1, 0)
```

FOLLOW-UP 2 | Return the strokes, not just the count.

Record which option won in each range and rebuild the list by walking the recursion again: horizontal bands as (row, l, r), vertical strokes as (plank).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Is it the same as largest rectangle in a histogram?

It shares the idea of splitting at the minimum (a Cartesian tree on the heights), but the objective differs: here you minimise strokes, there you maximise area. Cost: the fence recursion is O(n log n) with a sparse table (O(n^2) worst case with a plain scan); the histogram problem has an O(n) stack solution.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 39

40 * Maximize 1D deliveries within distance and minimize total distance * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Only the number of deliveries matters.

Two pointers on the sorted lists: match when within D, otherwise advance whichever is further left. O(n log n). It maximises the count (checked against exhaustive search) but not the total distance.

FOLLOW-UP CODE

```
def max_deliveries_greedy(orders, drivers, max_distance):
    """Maximum number of matches only: two pointers (does not minimise distance)."""
    sorted_orders, sorted_drivers, order_index, driver_index, count = sorted(orders), sorted(drivers), 0, 0, 0
    while order_index < len(sorted_orders) and driver_index < len(sorted_drivers):
        if abs(sorted_orders[order_index] - sorted_drivers[driver_index]) <= max_distance:
            count, order_index, driver_index = count + 1, order_index + 1, driver_index + 1
        elif sorted_drivers[driver_index] < sorted_orders[order_index]:
            driver_index += 1
        else:
            order_index += 1
    return count
```

FOLLOW-UP 2 | A driver can take several orders (capacity c).

Replicate each driver c times (as separate sorted points) and use the same DP, or treat it as min-cost flow with capacities.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Minimise the largest single distance instead of the total.

Binary search the largest allowed distance D and test whether all orders can be served with the two-pointer matching; the check is monotone in D. Cost: O(n log n + (n + m) log(range)): sorting plus a binary search whose check is a two-pointer pass.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 40

41 * Find the longest palindromic subsequence using dynamic programming. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the subsequence itself.

Keep the full table and walk from (0, length - 1): equal ends are taken (both go into the answer), otherwise move toward the side with the larger value. O(length^2) memory is needed for the walk back.

FOLLOW-UP CODE

```
def lps_string(text):
    length = len(text)
    dp = [[0] * length for _ in range(length)]
    for start in range(length - 1, -1, -1):
        dp[start][start] = 1
        for end in range(start + 1, length):
            dp[start][end] = dp[start + 1][end - 1] + 2 if text[start] == text[end] else max(dp[start + 1][
end], dp[start][end - 1])
    left, right, start, end = [], [], 0, length - 1
```

```
    while start < end:
        if text[start] == text[end]:
            left.append(text[start]); right.append(text[end]); start, end = start + 1, end - 1
        elif dp[start + 1][end] >= dp[start][end - 1]:
            start += 1
        else:
            end -= 1
    middle_letter = [text[start]] if start == end else []
    return "".join(left + middle_letter + right[::-1])
```

FOLLOW-UP 2 | Minimum number of insertions (or deletions) to make the string a palindrome.

It is length - LPS(text): keep the longest palindromic subsequence and add (or remove) the other characters.

FOLLOW-UP CODE

```
def longestPalindromeSubseq(text):
    length = len(text)
    if length == 0:
        return 0
    row_below = [0] * length                                    # dp[start + 1][*]
    for start in range(length - 1, -1, -1):
        current_row = [0] * length
        current_row[start] = 1
        for end in range(start + 1, length):
            current_row[end] = row_below[end - 1] + 2 if text[start] == text[end] else max(row_below[end],
current_row[end - 1])
        row_below = current_row
    return row_below[length - 1]

def min_insertions_to_palindrome(text):
    return len(text) - longestPalindromeSubseq(text)
```

FOLLOW-UP 3 | Longest palindromic substring instead (contiguous).

Expand around each of the 2n - 1 centres in O(n^2), or use Manacher's algorithm in O(n). A subsequence DP is not needed.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 41

42 * Jump Game with Profit

*** MEDIUM ***

DIFFICULTY | * MEDIUM * | 5 FOLLOW-UPS

FOLLOW-UP 1 | Speed up reading A to near-linear time.

The inner loop asks for the max of best[j] over all j that can still reach i. Maintain a max-heap of (best[j], j + arr[j]) and pop entries whose reach is below i (lazy deletion): O(n log n). If the jump limit is the same for all cells, a monotonic deque over a sliding window makes it O(n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if you want to know whether the end is reachable at all (plain Jump Game)?

Greedy: scan left to right keeping the farthest reachable index; if i exceeds it you are stuck. O(n) time, O(1) space. Use the DP when profits matter, and the greedy when only reachability matters.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Only reachability, no profit (LeetCode 55).

Greedy: keep the farthest index reachable so far; if the current index is beyond it you are stuck. O(n), O(1).

FOLLOW-UP CODE

```
def can_reach_end(max_jumps):
    farthest = 0
    for position, jump_length in enumerate(max_jumps):
        if position > farthest:
            return False
        farthest = max(farthest, position + jump_length)
    return True
```

FOLLOW-UP 4 | Make the profit DP faster.

best[i] needs the maximum of best[j] over j that can reach i. A max-heap with lazy deletion of positions that can no longer reach i makes it O(n log n); with a constant maximum jump a monotonic deque gives O(n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 5 | Fewest jumps (LeetCode 45).

BFS by levels or a greedy that tracks the end of the current jump range and the farthest reach, O(n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 42

Best-first search with a heap: Dijkstra and its minimax twin

46 * Path Minimizing the Maximum Cell (Swim in Rising Water) * MEDIUM *

DIFFICULTY | * MEDIUM * | 2 FOLLOW-UPS

FOLLOW-UP 1 | Maximise the minimum distance to a cat instead.

Same bottleneck shape with the objective flipped: score each cell by its Manhattan distance to the cat, run Dijkstra with a max-heap on the running minimum, and skip water. The same monotonicity argument holds (extending a path can only lower its minimum).

FOLLOW-UP CODE

```
def safest_path_score(grid, start, target, cat):
    """Largest possible minimum Manhattan distance to the cat over a path of land cells ('#' is water); None
if no path."""
    rows, cols = len(grid), len(grid[0])
    cat_distance = lambda row, col: abs(row - cat[0]) + abs(col - cat[1])
    best = {start: cat_distance(*start)}
    heap = [(-best[start], start)]
    while heap:
        negative_score, (row, col) = heapq.heappop(heap)
        if -negative_score < best[(row, col)]:
            continue
        if (row, col) == target:
            return -negative_score
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols and grid[next_row][next_col] != "#":
                score = min(-negative_score, cat_distance(next_row, next_col))
                if score > best.get((next_row, next_col), -1):
                    best[(next_row, next_col)] = score
                    heapq.heappush(heap, (-score, (next_row, next_col)))
    return None
```

FOLLOW-UP 2 | Why is Dijkstra correct for a max-along-path cost?

The cost max(path) never decreases when a path is extended, so a vertex's cheapest cost cannot be improved by going through a vertex that is popped later. That is the only property Dijkstra needs from its cost function (monotone, non-negative extension).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 46

47 * Nearest Favorite City by Travel Time * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Answer it for every possible start city in one run.

Reverse the search: run one Dijkstra seeded with every favourite at distance 0, storing for each city the pair (distance, favourite). Because roads are undirected the distance from a city to its nearest favourite is the same, and comparing the pairs lexicographically resolves ties to the smaller favourite. O((V + E) log V) total.

FOLLOW-UP CODE

```
def nearest_favorite_for_all(num_cities, roads, favorites):
    neighbours = [[] for _ in range(num_cities)]
    for from_city, to_city, road_length in roads:
        neighbours[from_city].append((to_city, road_length))
        neighbours[to_city].append((from_city, road_length))
    best = [(float("inf"), -1)] * num_cities
    heap = []
    for favorite in favorites:
        best[favorite] = (0, favorite)
        heapq.heappush(heap, (0, favorite, favorite))
    while heap:
        distance, favorite_city, from_city = heapq.heappop(heap)
        if (distance, favorite_city) > best[from_city]:
            continue
```

```
        for to_city, road_length in neighbours[from_city]:
            candidate = (distance + road_length, favorite_city)
            if candidate < best[to_city]:
                best[to_city] = candidate
                heapq.heappush(heap, (candidate[0], favorite_city, to_city))
    return [favorite_city if distance < float("inf") else -1 for distance, favorite_city in best]
```

BACK TO PART A - PART 1: QUESTION 47

48 * Given a train schedule with departure city, arrival city, start time, end time, and duration, and given a source and destination, find the sequence of trains that gets you there at the earliest possible time. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | A minimum layover is required when changing trains.

Replace the boarding test departure >= time_now with departure >= time_now + layover for every city except the origin. The earliest-arrival property still holds (a later arrival never helps because the layover is the same), so Dijkstra remains valid.

FOLLOW-UP CODE

```
def earliest_arrival_with_layover(trains, source, destination, start=0, layover=0):
    by_city = defaultdict(list)
    for train in trains:
        by_city[train[0]].append(train)
    earliest_at, heap = {source: start}, [(start, source)]
    while heap:
        time_now, city = heapq.heappop(heap)
        if time_now > earliest_at[city]:
            continue
        if city == destination:
            return time_now
        for origin, next_city, departure, arrival in by_city[city]:
            ready = time_now if city == source else time_now + layover        # a layover is needed at every
transfer, not at the origin
            if departure >= ready and arrival < earliest_at.get(next_city, float("inf")):
                earliest_at[next_city] = arrival
                heapq.heappush(heap, (arrival, next_city))
    return None
```

FOLLOW-UP 2 | Fewest transfers instead of earliest arrival.

Run BFS over (city) counting trains taken, but track the earliest arrival for each count (a state is (city, number of trains)). Or Dijkstra with the pair (transfers, arrival time) compared lexicographically.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The schedule repeats daily.

Add the day to the time (departure + 24h * k) or treat times modulo 24h and carry the day count. The same Dijkstra works on absolute times.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 48

49 * Find the shortest path in a graph. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The graph is unweighted.

BFS from the source with a parent map gives a shortest path in O(V + E); Dijkstra would be correct but needlessly slow. Stop as soon as you pop the target.

FOLLOW-UP CODE

```
def shortest_path_unweighted(node_count, edges, source, target):
    graph = defaultdict(list)
    for node, neighbour in edges:
        graph[node].append(neighbour)
    parent, queue = {source: None}, deque([source])
```

```
    while queue:
        node = queue.popleft()
        if node == target:
            path = []
            while node is not None:
                path.append(node)
                node = parent[node]
            return path[::-1]
        for neighbour in graph[node]:
            if neighbour not in parent:
                parent[neighbour] = node
                queue.append(neighbour)
    return None
```

FOLLOW-UP 2 | Weights can be negative.

Dijkstra can give wrong answers. Use Bellman-Ford (O(V * E)); one more pass that still improves a distance proves a negative cycle reachable from the source, where shortest paths are undefined.

FOLLOW-UP CODE

```
def shortest_bellman_ford(node_count, edges, source, target):
    INFINITY = float("inf")
    dist = [INFINITY] * node_count
    dist[source] = 0
    for _ in range(max(0, node_count - 1)):
        changed = False
        for from_node, to_node, weight in edges:
            if dist[from_node] + weight < dist[to_node]:
                dist[to_node], changed = dist[from_node] + weight, True
        if not changed:
            break
    return None if dist[target] == INFINITY else dist[target]
```

FOLLOW-UP 3 | Weights are only 0 or 1, or the graph is huge and you know the target.

0-1 BFS uses a deque in O(V + E). With a good distance estimate use A* (Dijkstra ordered by distance plus a heuristic that never overestimates); bidirectional search roughly squares the work saved on large graphs.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 49

50 * Given a matrix with a rat, bread, cat, and obstacles, find a path from the rat to the bread that maximizes the minimum distance from cat. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Use binary search instead of a heap.

Binary search the safety threshold d and test with BFS whether the rat can reach the bread using only cells at least d from the cat. The test is monotone in d. O(m * n log(m + n)).

FOLLOW-UP CODE

```
def _cat_dist(grid):
    rows, cols = len(grid), len(grid[0])
    cat_distance = {(row, col): 0 for row in range(rows) for col in range(cols) if grid[row][col] == "C"}
    queue = deque(cat_distance)
    while queue:
        row, col = queue.popleft()
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols and grid[next_row][next_col] != "#" and (
next_row, next_col) not in cat_distance:
                cat_distance[(next_row, next_col)] = cat_distance[(row, col)] + 1
                queue.append((next_row, next_col))
    return cat_distance

def _find(grid, symbol):
    return next((row, col) for row in range(len(grid)) for col in range(len(grid[0])) if grid[row][col] ==
symbol)

def safest_path_threshold(grid):
    """Alternative: binary search the threshold d and test reachability using only cells at distance >= d."""
    rows, cols = len(grid), len(grid[0])
    cat = _cat_dist(grid)
```

```
    start, goal = _find(grid, "R"), _find(grid, "B")

    def reachable(min_cat_distance):
        if cat.get(start, 10**9) < min_cat_distance:
            return False
        seen, stack = {start}, [start]
        while stack:
            row, col = stack.pop()
            if (row, col) == goal:
                return True
            for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
                if 0 <= next_row < rows and 0 <= next_col < cols and grid[next_row][next_col] != "#" and (
next_row, next_col) not in seen and cat.get((next_row, next_col), 10**9) >= min_cat_distance:
                    seen.add((next_row, next_col))
                    stack.append((next_row, next_col))
        return False

    values = sorted({cat.get((row, col), 10**9) for row in range(rows) for col in range(cols) if grid[row][
col] != "#"})     # candidate safety values
    low, high, answer = 0, len(values) - 1, -1
    while low <= high:
        middle = (low + high) // 2
        if reachable(values[middle]):
            answer, low = values[middle], middle + 1
        else:
            high = middle - 1
    return answer
```

FOLLOW-UP 2 | Among the safest paths, return the shortest one.

First find the best safety value d*, then run an ordinary BFS restricted to cells at distance at least d*. The result is the shortest path with maximum safety.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The cat moves one step toward the rat each turn.

Then distances depend on time, and the problem becomes a game or a time-expanded graph search; ask for the cat's movement rules before designing anything.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 50

Backtracking: choose, recurse, undo

55 * Mahjong Winning Hand Detection (14 Tiles) * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | A variant needs only two melds plus a pair (8 tiles of 14).

Ask what happens to the other six tiles before coding: if they are free, check whether any 8 tiles form a pair and two melds with the same search over subsets of the 14. The rule set is intentionally vague, so confirm melds, honours and flowers first.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 55

56 * Count Prefix-Matching Simple Paths in a Labeled Graph * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The edges are undirected.

Add every edge in both directions, as above. A path is then counted once per direction unless you divide the counts for length >= 2 by 2 (a path and its reverse are different sequences only if the query is not a palindrome, so say how you count).

FOLLOW-UP CODE

```
def prefix_paths(labels, edges, query):
    neighbours = defaultdict(list)
    for node, neighbour in edges:
        neighbours[node].append(neighbour)
    counts = [0] * len(query)

    def go(node, depth, visited_mask):
        counts[depth] += 1
        if depth + 1 < len(query):
            for next_node in neighbours[node]:
                if not visited_mask >> next_node & 1 and labels[next_node] == query[depth + 1]:
                    go(next_node, depth + 1, visited_mask | 1 << next_node)

    for start_vertex in range(len(labels)):
        if query and labels[start_vertex] == query[0]:
            go(start_vertex, 0, 1 << start_vertex)
    return counts

def prefix_paths_undirected(labels, edges, query):
    both_directions = list(edges) + [(to_node, from_node) for from_node, to_node in edges]
    return prefix_paths(labels, both_directions, query)
```

FOLLOW-UP 2 | Count only the paths that spell the whole query.

Return counts[-1] instead of the list. For long queries on bigger graphs, memoise on (vertex, depth) if repeated vertices are allowed (walks, not simple paths); for simple paths the visited set prevents memoisation.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The graph has up to 100,000 vertices.

Simple paths are exponential, so ask whether walks are allowed. For walks a DP over (depth, vertex) counts them in O(L * E), the standard way to handle large graphs.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 56

57 * Determine if a 14-tile hand is winning * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Which tile would complete a 13-tile hand?

Try each of the nine values as the 14th tile and run the same check: 9 checks of a tiny search. That is how a player finds the tiles they are waiting for.

FOLLOW-UP CODE

```
def is_winning_hand(tiles):
    count_of_rank = [0] * 12

    for tile in tiles:
        count_of_rank[tile] += 1

    def melds(rank):
        while rank <= 9 and count_of_rank[rank] == 0:
            rank += 1
        if rank > 9:
            return True
        if count_of_rank[rank] >= 3:
            count_of_rank[rank] -= 3
            rest_works = melds(rank)
            count_of_rank[rank] += 3
            if rest_works:
                return True
        if rank <= 7 and count_of_rank[rank + 1] and count_of_rank[rank + 2]:
            count_of_rank[rank] -= 1; count_of_rank[rank + 1] -= 1; count_of_rank[rank + 2] -= 1
            rest_works = melds(rank)
            count_of_rank[rank] += 1; count_of_rank[rank + 1] += 1; count_of_rank[rank + 2] += 1
            if rest_works:
                return True
        return False

    for pair in range(1, 10):
        if count_of_rank[pair] >= 2:
            count_of_rank[pair] -= 2
            rest_works = melds(1)
            count_of_rank[pair] += 2
            if rest_works:
                return True
    return False

def waiting_tiles(thirteen):
    """Follow-up: which tile values complete a 13-tile hand?"""
    winning_tiles = []
    for rank in range(1, 10):
        if thirteen.count(rank) < 4 and is_winning_hand(thirteen + [rank]):
            winning_tiles.append(rank)
    return winning_tiles
```

FOLLOW-UP 2 | More suits or special hands (seven pairs).

Run the single-suit check on each suit separately (melds never cross suits) after splitting the hand; seven pairs is a separate one-line count check (seven values with count 2).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Why is the 'smallest remaining value' rule enough?

The smallest remaining value cannot be the middle or end of a sequence (the smaller values are gone), so it must be in a triplet or the start of a sequence. Trying those two options covers every legal grouping exactly once.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 57

58 * American Football Score Combinations

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | You only need how many combinations there are.

Use the unbounded coin DP: ways[0] = 1 and for each play value v, ways[amount] += ways[amount - play_points] for t from v to total. Looping over the values on the outside counts multisets. O(total * values) time and O(total) space.

FOLLOW-UP CODE

```
def count_combinations(total):
    ways = [1] + [0] * total
    for play_points in (2, 3, 6, 7, 8):
        for amount in range(play_points, total + 1):
            ways[amount] += ways[amount - play_points]
    return ways[total]
```

BACK TO PART A - PART 1: QUESTION 58

59 * Count Passcode Patterns on a Keypad

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Count patterns that use at least k keys.

Use the same DFS but add one at every depth that is at least k and keep extending instead of stopping.

FOLLOW-UP CODE

```
def count_patterns_at_least(rows, cols, min_keys):
    def dfs(row, col, keys_used, seen):
        total = 1 if keys_used >= min_keys else 0
        for row_step, col_step in ((1, 0), (-1, 0), (0, 1), (0, -1)):
            next_row, next_col = row + row_step, col + col_step
            if 0 <= next_row < rows and 0 <= next_col < cols and (next_row, next_col) not in seen:
                seen.add((next_row, next_col))
                total += dfs(next_row, next_col, keys_used + 1, seen)
                seen.remove((next_row, next_col))
        return total

    return sum(dfs(row, col, 1, {(row, col)})
               for row in range(rows) for col in range(cols))
```

BACK TO PART A - PART 1: QUESTION 59

60 * Match people to questions using tags and priorities

* MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Priorities: higher-priority people must be served first.

Process people in decreasing priority and run the same augmenting step. Matchings of this kind form a matroid, so this greedy choice gives a matched set with the maximum total priority (checked against exhaustive search).

FOLLOW-UP CODE

```
def match_by_priority(acceptable, priority):
    """Serve people in priority order (higher first); the matched set is the best possible for any priority-
consistent weights."""
    owner_of = {}

    def try_assign(person, seen):
        for question in acceptable[person]:
            if question in seen:
                continue
            seen.add(question)
            if question not in owner_of or try_assign(owner_of[question], seen):
                owner_of[question] = person
                return True
        return False

    order = sorted(range(len(acceptable)), key=lambda person: -priority[person])
```

```
    matched = [person for person in order if try_assign(person, set())]
    return sorted(matched), {question: person for question, person in owner_of.items()}
```

FOLLOW-UP 2 | Edges have weights (how good a match is).

Then it is maximum-weight bipartite matching: the Hungarian algorithm in O(n^3), or min-cost max-flow. Kuhn's algorithm only maximises the count.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | A question may take several people (capacity c).

Replicate the question c times as separate right-hand nodes, or run max flow with capacities on the question-to-sink edges.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 60

61 * Generate Parentheses * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return only the k-th string, or just the number of strings.

The count is the Catalan number C(pair_count) = (2n choose pair_count) / (pair_count + 1). For the rank-th string count the completions that start with '(' at each step and skip them if k is larger (unranking), using a small DP over (opens left, balance).

FOLLOW-UP CODE

```
def kth_parenthesization(pair_count, rank):
    """The k-th (1-based, in the order of generateParenthesis) well-formed string, without generating the
others."""
    @lru_cache(maxsize=None)
    def count(opened_left, balance):
        if opened_left == 0:
            return 1
        total = count(opened_left - 1, balance + 1)
        if balance > 0:
            total += count(opened_left, balance - 1)
        return total

    out, opened, balance = [], 0, 0
    for _ in range(2 * pair_count):
        if opened < pair_count:
            ways_if_open = count(pair_count - opened - 1, balance + 1)
            if rank <= ways_if_open:
                out.append("(")
                opened, balance = opened + 1, balance + 1
                continue
            rank -= ways_if_open
        out.append(")")
        balance -= 1
    return "".join(out)
```

FOLLOW-UP 2 | Several bracket types, such as () [] {}.

Multiply the choices: at each position you may open any of the types (until its count is used) and may close only the type on top of a stack. The pruning rule (never close more than you have opened) becomes 'close only the most recent unclosed bracket'.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Avoid recursion.

Use an explicit stack of (string so far, opens, closes) states or build the strings by length with a queue. The pruning conditions stay the same.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 61

62 * Permutations * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The input has duplicates (Permutations II).

Sort first, and skip a value that equals the previous one unless the previous one is already used in the current prefix. That forces equal values to appear in a fixed order and removes duplicates without a set.

FOLLOW-UP CODE

```
def permute_unique(nums):
    nums = sorted(nums)
    results, current, used = [], [], [False] * len(nums)

    def go():
        if len(current) == len(nums):
            results.append(current[:])
            return
        for index, value in enumerate(nums):
            if used[index] or (index and value == nums[index - 1] and not used[index - 1]):
                continue
            used[index] = True
            current.append(value)
            go()
            current.pop()
            used[index] = False

    go()
    return results
```

FOLLOW-UP 2 | Return the next lexicographic permutation in place.

Find the last position i with values[pivot] < values[pivot + 1], swap values[pivot] with the smallest larger value to its right, then reverse the suffix. O(n), O(1) space. Repeat to enumerate in order.

FOLLOW-UP CODE

```
def next_permutation(values):
    pivot = len(values) - 2
    while pivot >= 0 and values[pivot] >= values[pivot + 1]:
        pivot -= 1
    if pivot >= 0:
        successor = len(values) - 1
        while values[successor] <= values[pivot]:
            successor -= 1
        values[pivot], values[successor] = values[successor], values[pivot]
    values[pivot + 1:] = reversed(values[pivot + 1:])
    return values
```

FOLLOW-UP 3 | Return only the k-th permutation of 1..n.

Use the factorial number system: the first digit is index (rank - 1) // (count - 1)! of the remaining digits, then continue with the remainder. O(count^2), no enumeration.

FOLLOW-UP CODE

```
def kth_permutation(count, rank):
    digits, block_size, chosen = list(range(1, count + 1)), math.factorial(count - 1) if count else 1, []
    rank -= 1
    for remaining in range(count, 0, -1):
        block_index, rank = divmod(rank, block_size)
        chosen.append(digits.pop(block_index))
        if remaining > 1:
            block_size //= remaining - 1
    return "".join(map(str, chosen))
```

BACK TO PART A - PART 1: QUESTION 62

Runs and groups: extend while it fits, restart or close when it does not

66 * Longest Non-Decreasing Subarray with One Replacement * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | The run must be strictly increasing.

Use < in the run lengths, and when the replaced value must fit between the neighbours check values[index+1] - values[index-1] >= 2 (an integer strictly between them exists) instead of values[index-1] <= values[index+1]. Clarify strict versus non-strict before coding.

FOLLOW-UP CODE

```
def longest_with_one_change(values):
    length = len(values)
    if length <= 2:
        return length
    left, right = [1] * length, [1] * length
    for index in range(1, length):
        if values[index - 1] <= values[index]:
            left[index] = left[index - 1] + 1
    for index in range(length - 2, -1, -1):
        if values[index] <= values[index + 1]:
            right[index] = right[index + 1] + 1
    best = 1
    for index in range(length):
        run_before = left[index - 1] if index else 0
        run_after = right[index + 1] if index + 1 < length else 0
        best = max(best, run_before + 1 + run_after if (not run_before or not run_after or values[index - 1]
<= values[index + 1]) else max(run_before, run_after) + 1)
    return best
```

BACK TO PART A - PART 1: QUESTION 66

67 * Piano Hand Movement Grouping * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return where each group starts and where to place the hand.

Record the start index when a group opens and emit (start, end, smallest note) when it closes. The state transition is unchanged.

FOLLOW-UP CODE

```
def group_boundaries(notes, max_spread):
    out, start = [], 0
    lowest = highest = notes[0] if notes else 0
    for note_index, note in enumerate(notes):
        new_lowest, new_highest = min(lowest, note), max(highest, note)
        if new_highest - new_lowest <= max_spread:
            lowest, highest = new_lowest, new_highest
        else:
            out.append((start, note_index - 1, lowest))          # (first note index, last note index, key to
place the hand on)
            start, lowest, highest = note_index, note, note
    if notes:
        out.append((start, len(notes) - 1, lowest))
    return out
```

FOLLOW-UP 2 | Count lifts, not placements.

Lifts are placements minus one if the first placement is free; the examples differ on this, so ask and state it. In code it is a single subtraction on the group count.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The hand may slide without a lift (costing something).

Then it becomes a DP over positions with a cost per slide distance, solved with the O(n * range) DP or by greedy if the cost is monotone. Ask for the cost model.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 67

68 * Find longest increasing contiguous subarray * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | No modification allowed (LeetCode 674).

One pass keeping the length of the current strictly increasing run; reset to 1 when values[position] <= values[position - 1]. O(n), O(1).

FOLLOW-UP CODE

```
def longest_increasing_no_change(values):
    best, run = 0, 0
    for position, value in enumerate(values):
        run = run + 1 if position and value > values[position - 1] else 1
        best = max(best, run)
    return best
```

FOLLOW-UP 2 | Up to k modifications.

Use a sliding window over the array that allows at most k positions where the increasing property fails and cannot be repaired; careful with the integer room condition between neighbours. Often handled with DP over (position, changes used).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Values are real numbers, not integers.

Then there is always room between two different neighbours, so the join condition becomes a[i + 1] > a[i - 1] instead of a gap of at least 2.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 68

69 * Sum of Good Subarrays (Adjacent Diff ±1) * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Why is the singleton subtracted?

A one-element subarray [value] is both an increasing and a decreasing run of length 1, so it appears in both running totals. Subtracting x once at each index keeps it counted exactly once. A subarray of length 2 or more can only be in one total because its step is +1 or -1, not both.

FOLLOW-UP CODE

```
def good_subarray_sum(values):
    total = 0
    up_total = up_length = down_total = down_length = 0
    for index, value in enumerate(values):
        step = value - values[index - 1] if index else None
        up_total, up_length = (up_total + (up_length + 1) * value, up_length + 1) if step == 1 else (value, 1)
        down_total, down_length = (down_total + (down_length + 1) * value, down_length + 1) if step == -1
else (value, 1)
        total += up_total + down_total - value
    return total
```

BACK TO PART A - PART 1: QUESTION 69

70 * Sustained-Error Time Window

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | You get thousands of (second, interval) queries on the same data.

Compute once, for every index, the start and end of the maximal run of readings at or above the baseline (one left-to-right pass and one right-to-left pass). Each query is then a lookup plus a binary search for the index of second, O(log count).

FOLLOW-UP CODE

```
from bisect import bisect_left


def run_bounds(times, errors, baseline):
    count = len(times)
    left, right = [0] * count, [0] * count
    for index in range(count):
        left[index] = left[index - 1] if index and errors[index - 1] >= baseline and errors[index] >=
baseline else index
    for index in range(count - 1, -1, -1):
        right[index] = right[index + 1] if index + 1 < count and errors[index + 1] >= baseline and errors[
index] >= baseline else index
    return left, right


def sustained_query(times, errors, bounds, second, interval, baseline):
    at = bisect_left(times, second)
    if errors[at] < baseline:
        return False
    left, right = bounds
    return times[right[at]] - times[left[at]] >= interval
```

BACK TO PART A - PART 1: QUESTION 70

71 * Longest Non-Decreasing Subarray From Two Arrays

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | First solve it for a single array, then generalize to picking per index from two arrays.

One array: run = run + 1 if nums[i - 1] <= nums[i] else 1 and keep the maximum, O(n) time and O(1) space. The two-array version keeps one run length per choice instead of one.

FOLLOW-UP CODE

```
def longest_non_decreasing_one(nums):
    best = run = 1 if nums else 0
    for previous_value, next_value in zip(nums, nums[1:]):
        run = run + 1 if previous_value <= next_value else 1
        best = max(best, run)
    return best
```

BACK TO PART A - PART 1: QUESTION 71

72 * Count Super Streaks

* MEDIUM *

DIFFICULTY | * MEDIUM * | 2 FOLLOW-UPS

FOLLOW-UP 1 | Events carry a user_id. Return the number of super streaks for each user.

Group the events by user first and run the same pass on each group, or key the open streak state by user in one pass over the sorted events. The cost stays O(n log n).

FOLLOW-UP CODE

```
from collections import defaultdict


def super_streaks_per_user(events, min_events, max_gap, min_duration):
    by_user = defaultdict(list)
    for eid, etype, etime, user in events:
        by_user[user].append((eid, etype, etime))
```

```
    return {user: count_super_streaks(user_events, min_events, max_gap, min_duration) for user, user_events
in by_user.items()}
```

FOLLOW-UP 2 | Then answer count_streaks(user_id, time_A, time_B): how many of that user's super streaks finished entirely inside [A, B]?

Record each super streak as (start, end) when it closes, per user, sorted by start (they do not overlap). A query counts the streaks with start >= A and end <= B: bisect on the starts and scan, or keep the ends in a second sorted list. O(log s + answer) per query.

FOLLOW-UP CODE

```
from bisect import bisect_left


class StreakIndex:
    def __init__(self, events, min_events, max_gap, min_duration):
        self.spans = defaultdict(list)
        by_user = defaultdict(list)
        for eid, etype, etime, user in events:
            by_user[user].append((eid, etype, etime))
        for user, user_events in by_user.items():
            user_events.sort(key=lambda event: event[2])
            run = [user_events[0]]
            for event in user_events[1:] + [None]:
                if event and event[1] == run[-1][1] and event[2] - run[-1][2] <= max_gap:
                    run.append(event)
                    continue
                if len(run) >= min_events and run[-1][2] - run[0][2] >= min_duration:
                    self.spans[user].append((run[0][2], run[-1][2]))
                run = [event] if event else []

    def count_streaks(self, user, start_time, end_time):
        spans = self.spans.get(user, [])
        first = bisect_left(spans, (start_time, float("-inf")))
        return sum(1 for span_start, span_end in spans[first:] if span_start <= end_time and span_end <=
end_time)
```

BACK TO PART A - PART 1: QUESTION 72

Union-find: merge the pairs, then ask who is together

73 * Count Islands in a Grid and Track the Count as Land Cells Are Added * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Land can also be removed (or turned back to water).

Union-Find cannot split sets. Either recompute the affected island with a local flood fill, or process all operations offline with a segment tree over time and a rollback Union-Find.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | The grid is 10^9 x 10^9 with a few additions.

Keep the Union-Find on a dictionary keyed by coordinates, as the code does; memory depends on the number of land cells only, never on the grid size. Cost: O(a * alpha) for a additions and O(a) memory, independent of the grid size.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Report the size of the largest island after each addition.

Store a size per root in the Union-Find (union by size) and keep a running maximum, which only grows when land is added; each step stays near-constant time. Checked against recomputing the largest island from scratch. Cost: O(alpha) per addition with size tracking; the maximum is one running variable.

FOLLOW-UP CODE

```
def largest_island_after_each(grid, additions):
    parent, size, largest, largest_after_each = {}, {}, 0, []

    def find(cell):
        while parent[cell] != cell:
            parent[cell] = parent[parent[cell]]
            cell = parent[cell]
        return cell

    def add(row, col):
        nonlocal largest
        if (row, col) in parent:
            return
        parent[(row, col)], size[(row, col)] = (row, col), 1
        for neighbour in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if neighbour in parent:
                root_a, root_b = find((row, col)), find(neighbour)
                if root_a != root_b:
                    if size[root_a] < size[root_b]:
                        root_a, root_b = root_b, root_a
                    parent[root_b] = root_a
                    size[root_a] += size[root_b]
        largest = max(largest, size[find((row, col))])

    for row in range(len(grid)):
        for col in range(len(grid[0])):
            if grid[row][col] == 1:
                add(row, col)
    largest_after_each.append(largest)
    for row, col in additions:
        add(row, col)
        largest_after_each.append(largest)
    return largest_after_each
```

BACK TO PART A - PART 1: QUESTION 73

74 * 2D Points Connected Components by Radius * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Why compare squared distances?

sqrt is slow and inexact; dist <= radius is the same as dist^2 <= radius^2 for non-negative numbers, and with integer coordinates the comparison is exact. It also avoids floating-point surprises right at the boundary.

FOLLOW-UP CODE

```
def count_components(points, radius):
    cell_size = radius if radius > 0 else 1
    parent = list(range(len(points)))
    def find(node):
        while parent[node] != node:
            parent[node] = parent[parent[node]]
            node = parent[node]
        return node
    cells = defaultdict(list)
    for point_index, (point_x, point_y) in enumerate(points):
        cells[(point_x // cell_size, point_y // cell_size)].append(point_index)
    for point_index, (point_x, point_y) in enumerate(points):
        cell_x, cell_y = point_x // cell_size, point_y // cell_size
        for cell_dx in (-1, 0, 1):
            for cell_dy in (-1, 0, 1):
                for other_index in cells.get((cell_x + cell_dx, cell_y + cell_dy), ()):
                    if other_index > point_index and (point_x - points[other_index][0]) ** 2 + (point_y -
points[other_index][1]) ** 2 <= radius * radius:
                        parent[find(point_index)] = find(other_index)
    return len({find(point_index) for point_index in range(len(points))})
```

BACK TO PART A - PART 1: QUESTION 74

75 * Count Clusters of 2D Points Within a Radius * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Report the size of the largest cluster.

Count points per root after the unions (a Counter of find(first)). The unions are the same.

FOLLOW-UP CODE

```
def _find_factory(point_count):
    parent = list(range(point_count))

    def find(node):
        while parent[node] != node:
            parent[node] = parent[parent[node]]
            node = parent[node]
        return node

    return parent, find

def largest_cluster(points, radius):
    parent, find = _find_factory(len(points))
    for first in range(len(points)):
        for second in range(first + 1, len(points)):
            if (points[first][0] - points[second][0]) ** 2 + (points[first][1] - points[second][1]) ** 2 <=
radius * radius:
                parent[find(first)] = find(second)
    return max(Counter(find(point_index) for point_index in range(len(points))).values(), default=0)
```

FOLLOW-UP 2 | Points arrive one at a time.

Insert the point into its grid cell, union it with every point within r in the 9 surrounding cells, and keep the component count: components += 1 - (number of distinct roots merged).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Millions of points in 3D or higher.

The grid generalises (3^d neighbouring cells), but for high dimensions use a k-d tree or locality-sensitive hashing; or DBSCAN, which is this problem with a minimum neighbour count. Cost: grid hashing visits 3^d neighbouring cells per point, O(n * 3^d); a

k-d tree builds in O(n log n) with O(n^(1 - 1/d) + k) range queries.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 75

76 * Find A Threshold-Limited Path With Minimum Required Safety * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Use a max-heap search instead of Union-Find.

Dijkstra where the value of a path is the minimum edge so far and you maximise it: pop the best bottleneck, extend with min(value, edge). O(E log V); Union-Find is simpler when you need only the value.

FOLLOW-UP CODE

```
def best_path_safety_dijkstra(node_count, edges, source, target):
    adj = defaultdict(list)
    for from_node, to_node, safety in edges:
        adj[from_node].append((to_node, safety))
        adj[to_node].append((from_node, safety))
    best_safety = {source: float("inf")}
    heap = [(-float("inf"), source)]
    while heap:
        negated, from_node = heapq.heappop(heap)
        if -negated < best_safety.get(from_node, -1):
            continue
        if from_node == target:
            return -negated
        for to_node, safety in adj[from_node]:
            path_safety = min(-negated, safety)
            if path_safety > best_safety.get(to_node, -1):
                best_safety[to_node] = path_safety
                heapq.heappush(heap, (-path_safety, to_node))
    return -1
```

FOLLOW-UP 2 | Answer many (source, target) queries.

Build the maximum spanning tree once (Kruskal) and answer each query as the minimum edge on the tree path, with binary lifting for O(log V) per query.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Edges below a threshold are forbidden (threshold-limited).

Filter them out before the search (or stop adding edges below the threshold in the Union-Find loop); if source and target are still disconnected the answer is -1.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 76

77 * Determine Reporting Relationships * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the closest common manager of two employees.

Collect the chain of managers above one employee into a set, then climb from the other until you meet it. O(depth).

FOLLOW-UP CODE

```
class Org:
    def __init__(self):
        self.parent, self.manager_of = {}, {}

    def _find(self, person):
        self.parent.setdefault(person, person)
        while self.parent[person] != person:
            self.parent[person] = self.parent[self.parent[person]]
            person = self.parent[person]
        return person

    def add_peer(self, peer_a, peer_b):
```

```
        root_a, root_b = self._find(peer_a), self._find(peer_b)
        if root_a != root_b:
            self.parent[root_b] = root_a
            if root_b in self.manager_of and root_a not in self.manager_of:
                self.manager_of[root_a] = self.manager_of[root_b]

    def add_manager(self, manager, employee):
        self._find(manager)
        self.manager_of[self._find(employee)] = manager

    def is_manager(self, manager, employee):
        seen, current = set(), employee
        while current not in seen:
            seen.add(current)
            root = self._find(current)
            if root not in self.manager_of:
                return False
            current = self.manager_of[root]
            if current == manager:
                return True
        return False

def closest_common_manager(org, first_employee, second_employee):
    chain, current = [], first_employee
    seen = set()
    while current not in seen:
        seen.add(current)
        chain.append(current)
        root = org._find(current)
        if root not in org.manager_of:
            break
        current = org.manager_of[root]
    ancestors, current, seen = set(chain), second_employee, set()
    while current not in seen:
        if current in ancestors:
            return current
        seen.add(current)
        root = org._find(current)
        if root not in org.manager_of:
            return None
        current = org.manager_of[root]
    return None
```

FOLLOW-UP 2 | Facts can be wrong or contradictory.

When uniting groups that already have different managers, the input is inconsistent: raise, or record a conflict instead of silently overwriting. Also guard the upward walk with a visited set so a cycle cannot loop forever.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Remove a relationship.

Union-Find cannot split groups; keep the facts and rebuild lazily, or store the peers as an explicit graph and recompute connected components on demand.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 77

78 * Determine whether two nodes are related * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | 'Related' means ancestor or descendant, or sharing an ancestor.

Store the data as a parent map and walk up from one node while recording ancestors, then walk up from the other until you hit one. Keep a seen set so a corrupt cycle cannot loop forever.

FOLLOW-UP CODE

```
def share_ancestor(parent, first, second):
    chain, seen = set(), set()
    while first is not None and first not in seen:
        chain.add(first)
        seen.add(first)
        first = parent.get(first)
```

```
    seen = set()
    while second is not None and second not in seen:
        if second in chain:
            return True
        seen.add(second)
        second = parent.get(second)
    return False
```

FOLLOW-UP 2 | Relationships arrive over time and you must answer queries between updates.

Union-Find handles incremental unions in near-constant time. If relationships can also be removed, you need offline processing or a dynamic-connectivity structure.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The graph is directed and 'related' means reachability from a to b.

Union-Find no longer works (it ignores direction). Use DFS/BFS per query, or compute strongly connected components and reachability on the condensed DAG for many queries.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 78

79 * Movie Recommendations via Transitive Similarity * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Thousands of recommendation queries on the same data.

Build the groups once with union-find, sort each group by (-rating, name) once, and store the group id of every movie. A query looks up the group, takes the first how_many + 1 entries and removes the target: O(how_many) per query.

FOLLOW-UP CODE

```
from collections import defaultdict


class Recommender:
    def __init__(self, ratings, similarities):
        parent = {}

        def find(movie):
            parent.setdefault(movie, movie)
            while parent[movie] != movie:
                parent[movie] = parent[parent[movie]]
                movie = parent[movie]
            return movie

        for movie_a, movie_b in similarities:
            parent[find(movie_a)] = find(movie_b)
        groups = defaultdict(list)
        for movie in list(parent):
            groups[find(movie)].append(movie)
        self.group_of = {movie: find(movie) for movie in parent}
        self.groups = {root: sorted(movies, key=lambda movie: (-ratings[movie], movie)) for root, movies in
groups.items()}

    def recommend(self, target, how_many):
        if target not in self.group_of:
            return []
        ranked = self.groups[self.group_of[target]]
        return [movie for movie in ranked if movie != target][:how_many]
```

BACK TO PART A - PART 1: QUESTION 79

A dict of per-key state that you update, expire, or rank

82 * Active Communication Top Users * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Self communication (A-A) should now count.

Pass a flag to the class: a self pair is one edge that adds one to that user's count (and only once). The dedup set already handles repeated self pairs.

FOLLOW-UP CODE

```
def _norm(pair):
    user_a, user_b = pair.split("-")
    return (user_a, user_b) if user_a <= user_b else (user_b, user_a)

class Active:
    def __init__(self, count_self=False):
        self.edges, self.connection_count, self.buckets, self.highest_count, self.count_self = set(), Counter(
), defaultdict(set), 0, count_self

    def _bump(self, user):
        old_count = self.connection_count[user]
        if old_count:
            self.buckets[old_count].discard(user)
        self.connection_count[user] = old_count + 1
        self.buckets[old_count + 1].add(user)
        self.highest_count = max(self.highest_count, old_count + 1)

    def add(self, pair):
        user_a, user_b = _norm(pair)
        if (user_a == user_b and not self.count_self) or (user_a, user_b) in self.edges:
            return
        self.edges.add((user_a, user_b))
        self._bump(user_a)
        if user_a != user_b:
            self._bump(user_b)

    def top_k(self, how_many):
        return heapq.nsmallest(how_many, self.connection_count, key=lambda user: (-self.connection_count[
user], user))

    def most_active(self):
        return set(self.buckets[self.highest_count]) if self.highest_count else set()
```

FOLLOW-UP 2 | Top-K with a changing K, or a fixed-size priority queue.

A heap of size K can be wrong after counts change, because an element outside it can overtake one inside. Re-run heapq.nsmallest on the count dict per query (O(U log K)) or maintain a sorted structure keyed by (count, name) updated on every increment.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The space claim.

Distinct pairs can number U^2 / 2 in the worst case, so the edge set is quadratic in the users; the counts alone are linear. If duplicates are not a concern, drop the set and keep O(U).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 82

83 * Filter Repeated Robot Status Messages * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The stream never ends; memory must be bounded.

A message older than the gap can never be blocked again, so evict it. Keep a min-heap of (time shown, message) and pop entries older than the gap before each decision; delete the dictionary key only if it still points at that time.

FOLLOW-UP CODE

```
class BoundedStatusFilter:
    """Streaming version that forgets messages older than the interval, so memory stays bounded."""
    def __init__(self, gap=10):
        self.gap, self.last_shown, self.heap = gap, {}, []

    def offer(self, time, message):
        while self.heap and self.heap[0][0] + self.gap < time:            # entries older than the gap can
never block a message again
            expired_time, expired_message = heapq.heappop(self.heap)
            if self.last_shown.get(expired_message) == expired_time:
                del self.last_shown[expired_message]
        if message in self.last_shown and time - self.last_shown[message] <= self.gap:
            return False
        self.last_shown[message] = time
        heapq.heappush(self.heap, (time, message))
        return True
```

FOLLOW-UP 2 | Several robots send messages.

Use (robot id, message) as the key, or one filter per robot. The same logic applies per key.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The interval depends on the message severity.

Look up the gap from a severity table and use it in the comparison; the state (last shown time) is unchanged.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 83

84 * Next-Word Predictor * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the top K predictions.

Keep the Counter per word and use heapq.nsmallest on (-count, word): O(d log K) per query, or keep a sorted structure per word if queries dominate.

FOLLOW-UP CODE

```
class Predictor:
    def __init__(self):
        self.counts = defaultdict(Counter)
        self.best = {}

    def train(self, sentence):
        words = sentence.split()
        for word, following_word in zip(words, words[1:]):
            self.counts[word][following_word] += 1
            current_best = self.best.get(word)
            if current_best is None or self.counts[word][following_word] > self.counts[word][current_best] or
(self.counts[word][following_word] == self.counts[word][current_best] and following_word < current_best):
                self.best[word] = following_word

    def predict(self, word):
        return self.best.get(word)

class TopKPredictor(Predictor):
    def predict_k(self, word, how_many):
        following_counts = self.counts.get(word, {})
        return heapq.nsmallest(how_many, following_counts, key=lambda word: (-following_counts[word], word))
```

FOLLOW-UP 2 | Use two words of context (trigrams).

The key becomes the pair (previous two words) and everything else is identical, at the price of more memory. Back off to the bigram table when the pair was never seen.

FOLLOW-UP CODE

```
class TrigramPredictor:
    def __init__(self):
        self.counts = defaultdict(Counter)

    def train(self, sentence):
        words = sentence.split()
        for first_word, second_word, third_word in zip(words, words[1:], words[2:]):
            self.counts[(first_word, second_word)][third_word] += 1

    def predict(self, first_word, second_word):
        following_counts = self.counts.get((first_word, second_word))
        return min(following_counts, key=lambda candidate: (-following_counts[candidate], candidate)) if
following_counts else None
```

FOLLOW-UP 3 | The corpus keeps changing and old text should fade.

Use an exponentially decayed count (multiply by a factor per training step, or store a timestamp and decay on read), and recheck the best when a count drops below another.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 84

85 * Batch and Streaming Log Deduplication * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Late data: how do you decide when it is safe to emit?

Use a watermark: only release entries whose timestamp is at most (the largest timestamp seen minus the allowed lateness). get_next then refuses to return anything newer than the watermark, and anything arriving older than it is dropped or sent to a late-data side output.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Memory must stay bounded.

Evict messages older than a retention window (a second heap or the same heap keyed on timestamp) and accept that duplicates arriving after eviction are no longer detected. Cost: O(log n) per eviction with a heap; memory O(entries inside the retention window).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Keep the entry with the most occurrences instead.

Store (count, timestamp) per message and update the count on every duplicate; the heap key changes only if you rank by count, so push a new entry when the rank changes and rely on lazy deletion again.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Only release entries once they are safe (late data may still arrive).

Add a watermark: release only entries whose timestamp is at most (the largest timestamp seen minus the allowed lateness), in order; anything older than the watermark that arrives later goes to a late-data path. The lazy-deletion heap pops stale entries as it goes.

FOLLOW-UP CODE

```
class LogStream:
    def __init__(self, policy="latest"):
        self.policy, self.current, self.heap = policy, {}, []

    def add(self, timestamp, message):
        if message not in self.current or (self.policy == "latest" and timestamp > self.current[message]):
            self.current[message] = timestamp
            heapq.heappush(self.heap, (timestamp, message))

    def get_next(self):
```

```
        while self.heap:
            timestamp, message = heapq.heappop(self.heap)
            if self.current.get(message) == timestamp:
                del self.current[message]
                return timestamp, message
        return None

class WatermarkLogStream(LogStream):
    def release(self, watermark):
        out = []
        while self.heap:
            timestamp, message = self.heap[0]
            if self.current.get(message) != timestamp:
                heapq.heappop(self.heap)
                continue
            if timestamp > watermark:
                break
            heapq.heappop(self.heap)
            del self.current[message]
            out.append((timestamp, message))
        return out
```

BACK TO PART A - PART 1: QUESTION 85

86 * Design a logger that returns top-K messages * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Print a message only if it was not printed in the last 10 seconds (LeetCode 359).

Keep a dict message to the time it was last printed and compare: print if the gap is at least 10 (and update the time only when you print). O(1) per call.

FOLLOW-UP CODE

```
class RateLimitedLogger:
    """LeetCode 359: a message may be printed again only after `gap` seconds."""
    def __init__(self, gap=10):
        self.gap, self.last_printed = gap, {}

    def should_print(self, timestamp, message):
        if message in self.last_printed and timestamp - self.last_printed[message] < self.gap:
            return False
        self.last_printed[message] = timestamp
        return True
```

FOLLOW-UP 2 | Memory must stay bounded.

Expire timestamps older than the largest window you support (drop from the front of each list, delete messages with no timestamps left), or approximate counts with a count-min sketch plus a heap for the heavy hitters. Cost: O(1) amortized to drop expired timestamps; memory O(messages inside the window).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Very high top_k frequency of calls.

Maintain counts per message incrementally and a top-K structure (a heap with lazy updates or a sorted structure) so top_k costs O(k) instead of O(d log k); windowed queries still need per-message timestamps.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 86

87 * Website Activity Analytics: Unique Users + Session Time * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Users with several sessions, missing logouts or out-of-order rows.

Sort each user's events by time before pairing; a second login before a logout is ignored (or closes the old session, depending on the contract you agree). A login with no logout ends at the user's last event, or is dropped if you decide it is unreliable. Say the choice out loud: the interviewer accepted a strong discussion over perfect code.

FOLLOW-UP CODE

```
def activity_stats(records):
    from datetime import datetime
    unique, events = defaultdict(set), defaultdict(list)
    for record in records:
        unique[record["activity"]].add(record["user_id"])
        events[record["user_id"]].append((datetime.strptime(record["timestamp"], "%Y-%m-%d %H:%M:%S"), record[
"activity"]))
    total = 0
    for user_events in events.values():
        user_events.sort()
        start = None
        for event_time, activity in user_events:
            if activity == "login" and start is None:
                start = event_time
            elif activity == "logout" and start is not None:
                total += (event_time - start).total_seconds()
                start = None
        if start is not None:
            total += (user_events[-1][0] - start).total_seconds()
    return {activity: len(users) for activity, users in sorted(unique.items())}, (total / len(events) if
events else 0.0)
```

BACK TO PART A - PART 1: QUESTION 87

88 * Hit Counter * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Millions of hits per second.

Use 300 buckets (one per second) instead of one entry per hit: hit is O(1), getHits is O(300) = O(1), and memory is constant. A bucket whose stored second is older than the window is reset on reuse.

FOLLOW-UP CODE

```
class HitCounterBuckets:
    """O(1) memory: one counter per second modulo the window, reset when the slot belongs to an older second.
"""
    def __init__(self, window=300):
        self.window, self.slot_times, self.slot_hits = window, [0] * window, [0] * window

    def hit(self, time):
        slot = time % self.window
        if self.slot_times[slot] != time:
            self.slot_times[slot], self.slot_hits[slot] = time, 0
        self.slot_hits[slot] += 1

    def getHits(self, time):
        return sum(hits for slot_time, hits in zip(self.slot_times, self.slot_hits) if slot_time > time -
self.window)
```

FOLLOW-UP 2 | Timestamps can arrive out of order.

A deque no longer works. Keep a sorted structure (a sorted list with bisect or a balanced tree) and count with two binary searches, or use the buckets, which are order-insensitive within the window.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Thread safety.

Guard the queue with a lock (or use per-thread counters merged on read). The bucket version needs the compare-then-reset of a slot to be atomic. Cost: one lock per call, O(1) extra; contention rather than complexity is the concern.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 88

Binary search on a yes/no condition

90 * Cutting Ribbons (LC 1891) * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Why does the running time depend on the value range and not just on the number of ribbons?

Each probe costs O(n), and the number of probes is log2(max length) for the integer version. A huge length like 10^9 needs about 30 probes while 10 needs 4, even with the same n. For real numbers the probe count is set by the precision you promise, e.g. log2(max / epsilon).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if the candidate length is given to you?

Then it is just the feasibility check: sum(r // L for r in ribbons) >= k, in O(n). The binary search exists only to find the best L; the check is the same function you reuse inside it.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | What is a good upper bound for the search?

min(max(ribbons), sum(ribbons) // k). If the total length divided by k is smaller than the longest ribbon, no piece longer than that can ever give k pieces, which shrinks the range a little and avoids wasted probes. Cost: each probe is O(n); the tighter bound removes only a few of the log2(max) probes, so the total stays O(n log max).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Give the search a tighter upper bound.

No piece can be longer than the longest ribbon, and K pieces of length L need K * L total length, so L <= sum // K. Starting hi at min(max(ribbons), sum // K) saves a few probes and also returns 0 immediately when sum < K.

FOLLOW-UP CODE

```
def maxLength_tight(ribbons, pieces_needed):
    low_length, high_length, best_length = 1, min(max(ribbons), sum(ribbons) // pieces_needed), 0
    while low_length <= high_length:
        trial_length = (low_length + high_length) // 2
        if sum(ribbon // trial_length for ribbon in ribbons) >= pieces_needed:
            best_length, low_length = trial_length, trial_length + 1
        else:
            high_length = trial_length - 1
    return best_length
```

BACK TO PART A - PART 1: QUESTION 90

91 * First Bad Version with Parallel Search Follow-up * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | What does parallelism buy you, and what does it cost?

It cuts the number of sequential rounds from log2(n) to log_{k+1}(n), so wall-clock time drops when each probe has high latency. It increases total probes (about k per round instead of 1), so it costs more calls and money. Say which metric you are optimizing before choosing k.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if the oracle is rate limited or can fail?

Cap the pool size to the rate limit, retry transient failures with backoff (probes are idempotent), and never advance the interval from a probe that errored. Cache results by version so a retry or a later round never repeats a call.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Count how many rounds each approach needs.

Single binary search: log2(n) rounds of one probe each. k workers: about log base (k + 1) of n rounds with k probes each. For n = 10^9, one probe per round needs 30 rounds, while 7 probes per round need 10.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Show how many rounds and calls the parallel search uses.

With k workers each round shrinks the interval by about k + 1, so the number of sequential rounds is about log base (k + 1) of n and the total calls at most k times that. For version_count = 10^6 and 7 workers that is about 7 rounds and 49 calls, against 20 rounds and 20 calls for the sequential search: fewer rounds, more calls.

FOLLOW-UP CODE

```
def first_bad_version_counting(version_count, is_bad, workers=4):
    rounds = calls = 0
    low, high = 1, version_count
    while low < high:
        step = max(1, (high - low + 1) // (workers + 1))
        points = sorted(set(range(low + step - 1, high, step)))[:workers] or [low]
        results = [is_bad(version) for version in points]
        rounds += 1
        calls += len(points)
        first_true = next((index for index, result in enumerate(results) if result), None)
        if first_true is None:
            low = points[-1] + 1
        else:
            high = points[first_true]
            if first_true:
                low = points[first_true - 1] + 1
    return low, rounds, calls
```

BACK TO PART A - PART 1: QUESTION 91

92 * Solve Banana Speed and Interval Merge * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Bananas: h varies per query, or the piles change.

The check is independent of the search; for many queries keep the piles sorted with prefix sums so the hours needed at speed s can be computed in O(log n) per distinct pile size.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Intervals: insert one new interval into an already merged list (LeetCode 57).

Copy intervals ending before the new one, merge the ones that overlap it, then copy the rest. O(count), no sort needed.

FOLLOW-UP CODE

```
def insert_interval(intervals, new):
    merged, position, count = [], 0, len(intervals)
    while position < count and intervals[position][1] < new[0]:
        merged.append(intervals[position]); position += 1
    while position < count and intervals[position][0] <= new[1]:
        new = [min(new[0], intervals[position][0]), max(new[1], intervals[position][1])]; position += 1
    merged.append(new)
    merged.extend(intervals[position:])
    return merged
```

FOLLOW-UP 3 | Bananas: the guards come back in fewer hours than piles.

Then no speed works (an hour can finish at most one pile): check h >= len(piles) first and return -1 otherwise.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 92

93 * Given a list with response time for each server, calculate how long time it will take to process n requests. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Servers may fail, or there is a startup time per server.

Add the startup time to the check: server i completes max(0, (T - start_i) // r_i) requests. The check stays monotone in T, so binary search still applies.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | The inverse question: how many servers are needed to finish within a given time?

Divide: each server finishes time // response_time requests, so you need ceil(request_count / that). No search needed.

FOLLOW-UP CODE

```
def servers_needed(time_budget, response_time, request_count):
    """Different question: how many identical servers of response time r are needed to finish n requests
within time_budget."""
    per_server = time_budget // response_time
    return -(-request_count // per_server) if per_server else None
```

FOLLOW-UP 3 | Requests have different sizes.

Then it is a scheduling problem (minimise the makespan), NP-hard in general; the greedy 'longest request first to the least-loaded server' is within 4/3 of optimal.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 93

Greedy: scan once, carry the best choice so far, act only when forced

100 * Visible People in a Line (Last Person, Then Everyone) * MEDIUM *

DIFFICULTY | * MEDIUM * | 2 FOLLOW-UPS

FOLLOW-UP 1 | Part 2: return the number of people each person can see.

Run the right-looking count for every position with a monotonic stack, run it again on the reversed line for the looking-left count, and add the two. The stack makes both passes O(n), instead of the O(n^2) you would get by repeating the last-person scan for every position.

FOLLOW-UP CODE

```
def see_all(heights):
    left = see_right(heights[::-1])[::-1]
    return [seen_left + seen_right for seen_left, seen_right in zip(left, see_right(heights))]
```

FOLLOW-UP 2 | How do ties and the direction of sight change the answer?

Under the stated rule only someone strictly taller than both blocks the view, so equal heights never block: in [5, 5, 5] the first person sees both others. If the interviewer wants 'taller or equal blocks', replace the non-strict comparisons with strict ones. If a person only looks towards the front, drop the reversed pass and use see_right on the reversed line alone.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 100

101 * Bounded Knapsack Variant * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Why is greedy correct here but not for a general knapsack?

With unit weights each chosen item uses exactly one slot, so the only thing that matters is value per slot, which is just the value. If an optimal bag held a cheaper item while a dearer one was available, swapping them keeps the weight and raises the value, so the optimal bag can always be rearranged into the greedy one. With different weights a cheap item can leave room for several others, so value per weight and the leftover space interact and greedy fails.

FOLLOW-UP CODE

```
def max_value_sorted(items, capacity):
    total = 0
    for value, copies in sorted(items, reverse=True):
        take = min(copies, capacity)
        total += value * take
        capacity -= take
        if capacity == 0:
            break
    return total
```

FOLLOW-UP 2 | What if the weights are not all 1?

Then it is a real bounded knapsack and greedy no longer works; use DP over the capacity. Splitting each count into 1, 2, 4, ... copies keeps the number of 0/1 items at about log(copies_left) per type, so the cost is O(n * log(max copies_left) * capacity) instead of O(total copies * capacity).

FOLLOW-UP CODE

```
def knapsack_bounded(items, capacity):
    """Weights differ: split each count into powers of two so every count is reachable, then run a 0/1
knapsack."""
    best = [0] * (capacity + 1)
    for value, weight, copies_left in items:
        chunk = 1
        while copies_left > 0:
            take = min(chunk, copies_left)
            copies_left -= take
```

```
            cost, gain = take * weight, take * value
            for capacity_so_far in range(capacity, cost - 1, -1):
                best[capacity_so_far] = max(best[capacity_so_far], best[capacity_so_far - cost] + gain)
            chunk *= 2
    return best[capacity]
```

FOLLOW-UP 3 | W is up to 10^9. Which version still works?

Only the greedy ones: sorting is O(n log n) and quickselect O(n), and neither depends on W, while any table over the capacity needs O(W) memory. The loop simply stops when the remaining capacity reaches zero.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 101

102 * Find the Largest Monotone Increasing Number * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The number is given as a string with 100,000 digits.

The scan works directly on the digit list, so keep it as a list of characters and never convert to an integer until the end (or not at all). The algorithm is O(digits).

FOLLOW-UP CODE

```
def monotoneIncreasingDigits(number):
    digits = list(str(number))
    nines_start = len(digits)
    for position in range(len(digits) - 1, 0, -1):
        if digits[position - 1] > digits[position]:
            digits[position - 1] = str(int(digits[position - 1]) - 1)
            nines_start = position
    for position in range(nines_start, len(digits)):
        digits[position] = "9"
    return int("".join(digits))
```

FOLLOW-UP 2 | Return the smallest monotone number that is at least n instead.

Scan left to right for the first digit that is larger than its right neighbour and copy that digit over the whole suffix (332 becomes 333). The prefix before it is already monotone.

FOLLOW-UP CODE

```
def smallest_monotone_at_least(number):
    digits = list(str(number))
    for position in range(len(digits) - 1):
        if digits[position] > digits[position + 1]:
            for later_position in range(position + 1, len(digits)):
                digits[later_position] = digits[position]
            break
    return int("".join(digits))
```

FOLLOW-UP 3 | Digits must be strictly increasing.

Same idea, but the suffix to fill is the largest strictly increasing run (for example ...6789), and you must check that the prefix leaves enough room. Say the extra care and write the check as a brute-force comparison first.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 102

103 * Min boards to cover roof holes * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | You have only K boards: what is the smallest board length that covers all holes?

Binary search the length L and test with the greedy (is the number of boards at most K?); feasibility is monotone in L. O(n log range).

FOLLOW-UP CODE

```
def min_boards(holes, board_length):
    boards, covered_until = 0, float("-inf")
    for hole in sorted(set(holes)):
        if hole > covered_until:
            boards += 1
            covered_until = hole + board_length - 1
    return boards

def min_length_for_k_boards(holes, board_count):
    sorted_holes = sorted(set(holes))
    if not sorted_holes:
        return 0
    low, high = 1, sorted_holes[-1] - sorted_holes[0] + 1
    while low < high:
        length = (low + high) // 2
        if min_boards(sorted_holes, length) <= board_count:
            high = length
        else:
            low = length + 1
    return low
```

FOLLOW-UP 2 | Boards of different lengths with a cost each.

The greedy exchange argument breaks. Use DP over the sorted holes: best[i] = min over board types of best[j] + cost, where j is the first hole not covered by that board starting at hole i.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Holes are in a grid and boards are 1 x L horizontally.

Apply the greedy independently on every row. If boards may also be vertical it becomes a covering problem on a bipartite graph (König's theorem).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 103

104 * Jump Game II (Minimum Jumps) * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Return the actual indexes you jump through, not just the count.

Remember for each range the index that reached farthest. When the range ends, append that index to the path. The path still costs O(n) time.

FOLLOW-UP CODE

```
def jump_path(nums):
    path, current_end, farthest, best_start = [0], 0, 0, 0
    for position in range(len(nums) - 1):
        if position + nums[position] > farthest:
            farthest, best_start = position + nums[position], position
        if position == current_end:
            current_end = farthest
            path.append(best_start if best_start != path[-1] else position)
    return path
```

BACK TO PART A - PART 1: QUESTION 104

Sets and counters: remember the first collection, check the second against it

109 * Reconstruct original array from doubled shuffle * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the originals in the order they appear in the input.

Compute the multiset of originals, then scan the input once and keep each element while its remaining original count is positive.

FOLLOW-UP CODE

```
def findOriginalArray(changed):
    if len(changed) % 2:
        return []
    count, original = Counter(changed), []
    for value in sorted(count, key=lambda value: (abs(value), value)):
        if count[value] == 0:
            continue
        if value == 0:
            if count[0] % 2:
                return []
            original += [0] * (count[0] // 2)
            count[0] = 0
            continue
        if count[2 * value] < count[value]:
            return []
        original += [value] * count[value]
        count[2 * value] -= count[value]
        count[value] = 0
    return original

def find_original_in_input_order(changed):
    base = Counter(findOriginalArray(changed))
    if not base and changed:
        return []
    original = []
    for value in changed:
        if base[value] > 0:
            base[value] -= 1
            original.append(value)
    return original
```

FOLLOW-UP 2 | Values are small non-negative integers.

Use a counting array instead of sorting and a Counter: process values from small to large, O(n + range).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Why does processing by absolute value work with negatives?

For negatives the double has a larger absolute value, so the element with the smallest absolute value is always the original of its pair, exactly as with positives.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 109

110 * Min deletions to avoid overlap in first k

*** MEDIUM ***

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Only elements of the first part may be deleted.

Then every shared value must be removed from the first part entirely, so the cost is the number of first-part elements whose value occurs in the rest (values set lookup per element).

FOLLOW-UP CODE

```
def min_deletions_first_part_only(values, prefix_length):
    right = set(values[prefix_length:])
    return sum(1 for value in values[:prefix_length] if value in right)
```

FOLLOW-UP 2 | Elements have weights (deleting some is more costly).

Replace counts by the sum of weights per side for each value; the choice per value is still independent, so take the cheaper side. Cost: unchanged O(n); the per-value choice sums weights instead of counts.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The statement may differ from this reading.

Ask whether deletions can come from both parts, whether duplicates count separately, and whether order matters. Everything above relies on those three answers.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 110

111 * Filter Common Elements Per Window (Two Lists)

*** MEDIUM ***

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Do it in place: no new output list.

Keep a write pointer w. For each window build the set of a's window, then copy each surviving element of b to second[write_position] and advance w; at the end truncate b to length w. Time O(n), extra space O(window_size) for one window's set.

FOLLOW-UP CODE

```
def filter_windows_in_place(first, second, window_size):
    write_position = 0
    for window_start in range(0, len(first), window_size):
        seen = set(first[window_start:window_start + window_size])
        for position in range(window_start, min(window_start + window_size, len(second))):
            if second[position] not in seen:
                second[write_position] = second[position]
                write_position += 1
    del second[write_position:]
    return second
```

BACK TO PART A - PART 1: QUESTION 111

bisect on a sorted list: find a position, count a range, keep it sorted

114 * Sliding-Window Mean After Removing the Largest K Values * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Make the update O(log w) in the worst case.

A Python list insert shifts elements (fast in practice, O(w) in theory). A balanced tree or a Fenwick tree over compressed values gives O(log w) insert, delete and 'sum of the top K' via a descent.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Remove the smallest K instead, or both ends (trimmed mean).

Maintain the sum of the K smallest as well (the first K elements of the same sorted list); the mean of the middle values is (total - top - bottom) / (w - 2K).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Duplicates on the boundary.

Removing exactly K occurrences, not K distinct values, is what the positional approach does: elements equal to the boundary value are interchangeable, so which copy sits in the top K does not matter for the sum. Cost: unchanged: O(w) per update for list insertion (O(log w) with a balanced tree).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Trim the smallest K as well (a trimmed mean).

Maintain the sum of the K smallest alongside the K largest with the mirror-image boundary updates; the mean of the middle values is (window_sum - top_sum - bottom_sum) / (w - 2K). Checked against sorting every window.

FOLLOW-UP CODE

```
def trimmed_both_means(nums, window, trim_count):
    window_sorted, window_sum, top_sum, bottom_sum, means = [], 0, 0, 0, []

    def insert(value):
        nonlocal window_sum, top_sum, bottom_sum
        old_size = len(window_sorted)
        position = bisect_right(window_sorted, value)
        window_sorted.insert(position, value)
        window_sum += value
        if old_size < trim_count:
            top_sum += value; bottom_sum += value
        else:
            if position >= old_size + 1 - trim_count:
                top_sum += value - window_sorted[old_size - trim_count]
            if position < trim_count:
                bottom_sum += value - window_sorted[trim_count]

    def remove(value):
        nonlocal window_sum, top_sum, bottom_sum
        old_size = len(window_sorted)
        position = bisect_left(window_sorted, value)
        window_sum -= value
        if old_size <= trim_count:
            top_sum -= value; bottom_sum -= value
        else:
            if position >= old_size - trim_count:
                top_sum += window_sorted[old_size - trim_count - 1] - value
            if position < trim_count:
                bottom_sum += window_sorted[trim_count] - value
        del window_sorted[position]

    for index, value in enumerate(nums):
        insert(value)
        if index >= window:
            remove(nums[index - window])
        if index >= window - 1:
            kept = window - 2 * trim_count
```

```
            numerator = window_sum - top_sum - bottom_sum
            divisor = math.gcd(abs(numerator), kept)
            means.append(f"{numerator // divisor}/{kept // divisor}")
    return means
```

BACK TO PART A - PART 1: QUESTION 114

115 * Streaming Insert + Find K-th Largest * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | What if values can also be deleted?

The heap can no longer answer because removing an arbitrary element is O(k). Keep a sorted structure that supports delete: a balanced BST or sorted container with order statistics, or a Fenwick tree over compressed values (add -1 to delete). Both give O(log n) insert, delete and k-th.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What is the trade-off between the two designs?

The heap is the smallest and fastest when k is fixed and small (O(log k), O(k) memory) but cannot change k or delete. The sorted list answers any k and is simplest, at O(n) memory and a linear shift per insert. Choose from the requirements, and say which assumption you made.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Values can also be deleted.

A heap cannot delete an arbitrary value efficiently. Keep a sorted structure: a sorted list with bisect (O(n) per delete because of shifting) or a balanced tree / Fenwick tree over compressed values (O(log n) for insert, delete and rank-th).

FOLLOW-UP CODE

```
class StreamKth:
    def __init__(self):
        self.sorted_values = []

    def insert(self, value):
        insort(self.sorted_values, value)

    def findLargest(self, rank):
        return self.sorted_values[-1 - rank] if 0 <= rank < len(self.sorted_values) else None

class StreamKthWithDelete(StreamKth):
    def delete(self, value):
        position = bisect.bisect_left(self.sorted_values, value)
        if position < len(self.sorted_values) and self.sorted_values[position] == value:
            del self.sorted_values[position]
            return True
        return False
```

BACK TO PART A - PART 1: QUESTION 115

116 * Count Sorted Strings With a Given Prefix * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Thousands of different queries on the same list. Can each be O(L)?

Build a trie once where every node stores how many words pass through it. A query walks the prefix and returns the count in that node (0 if the path breaks). Building costs O(total characters); a query costs O(L), independent of the number of words.

FOLLOW-UP CODE

```
class PrefixCounter:
    def __init__(self, words):
        self.root = {}
        for word in words:
            node = self.root
            for character in word:
                node = node.setdefault(character, {})
                node["#"] = node.get("#", 0) + 1
```

```
    def count(self, prefix):
        node = self.root
        for character in prefix:
            if character not in node:
                return 0
            node = node[character]
        return node.get("#", 0) if prefix else sum(child.get("#", 0) for letter, child in self.root.items()
if letter != "#")
```

BACK TO PART A - PART 1: QUESTION 116

117 * Russian Doll Envelopes * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Why sort heights descending when widths tie?

If two envelopes have the same width they cannot nest, but an ascending height order would let the LIS pick both. Descending order makes a strictly increasing height run use at most one envelope per width.

FOLLOW-UP CODE

```
def max_envelopes(envelopes):
    tails = []
    for _, height in sorted(envelopes, key=lambda envelope: (envelope[0], -envelope[1])):
        position = bisect_left(tails, height)
        if position == len(tails):
            tails.append(height)
        else:
            tails[position] = height
    return len(tails)
```

BACK TO PART A - PART 1: QUESTION 117

118 * Given positions of N kids and M ice cream sellers on a 1D line, find the smallest distance E such that every kid lies within the range [P - E, P + E] of at least one seller. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Both lists are already sorted, or you want to avoid the binary search.

Walk the kids in order with one pointer on the sellers that only moves forward while the next seller is at least as close. O(N + M) after sorting.

FOLLOW-UP CODE

```
def smallest_radius_two_pointers(kids, sellers):
    kids, sellers = sorted(kids), sorted(sellers)
    nearest_index, worst = 0, 0
    for kid in kids:
        while nearest_index + 1 < len(sellers) and abs(sellers[nearest_index + 1] - kid) <= abs(sellers[
nearest_index] - kid):
            nearest_index += 1
        worst = max(worst, abs(sellers[nearest_index] - kid))
    return worst
```

FOLLOW-UP 2 | Each seller can serve at most C kids.

Now it is an assignment problem. Binary search the radius E and test feasibility with a greedy left-to-right matching that gives each seller up to C of the leftmost uncovered kids in its range.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Sellers can be placed anywhere (minimise the radius with S sellers).

That is the 1D k-center problem: binary search the radius and greedily place a seller at the farthest point covering the leftmost uncovered kid. O(N log range).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 118

GROUP 15 OF 43 | 5 QUESTIONS | PART A - PART 2 FOLLOW-UPS

Prefix sums, Tier 1: precompute once, answer ranges and balance points in O(1)

124 * Range Duplicate Check with Constant-Time Queries * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | The array changes between queries.

Rebuild next and reach in O(length) after a batch of changes when reads dominate. If single-element updates are frequent, keep per-value sorted index lists and a segment tree of next-occurrence minima so an update and a query cost O(log length) instead.

FOLLOW-UP CODE

```
class RangeUnique:
    def __init__(self, values):
        length, INFINITY = len(values), float("inf")
        last_index_of, next_same_index = {}, [INFINITY] * length
        for position in range(length - 1, -1, -1):
            next_same_index[position] = last_index_of.get(values[position], INFINITY)
            last_index_of[values[position]] = position
        self.reach = [INFINITY] * (length + 1)
        for position in range(length - 1, -1, -1):
            self.reach[position] = min(next_same_index[position], self.reach[position + 1])

    def unique(self, left, right):
        return self.reach[left] > right
```

BACK TO PART A - PART 1: QUESTION 124

125 * Find a Horizontal Cut That Bisects Rectangle Area * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Rectangles may overlap (LeetCode 850).

The slope is then the union width of the rectangles crossing a height, not the sum. Maintain it with a segment tree over compressed x coordinates (cover counts), O(R log R) overall.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Coordinates up to 10^9 and 200,000 rectangles.

Use integers or Fractions (areas reach 10^18, which fits in Python ints but not in 32-bit). The sweep is O(R log R); never use floats for the comparisons that decide the slab.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | A vertical cut instead, or a cut at a given ratio.

Swap the axes for a vertical cut. For a ratio p, replace half by p * total in the slab test; the interpolation formula is unchanged.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | A cut that leaves a given fraction of the area below it, not exactly half.

Replace 'half of the total' by fraction_below * total in the slab test; the interpolation formula inside the slab is the same. The code returns an exact Fraction and is checked by computing the area below the returned cut.

FOLLOW-UP CODE

```
def cut_at_ratio(rectangles, fraction_below):
    total = sum((x2 - x1) * (y2 - y1) for x1, y1, x2, y2 in rectangles)
    target = Fraction(total) * fraction_below
    events = defaultdict(int)
    for x1, y1, x2, y2 in rectangles:
        events[y1] += x2 - x1
        events[y2] -= x2 - x1
    event_heights = sorted(events)
    total_width, area_so_far = 0, Fraction(0)
```

```
    for y0, y1 in zip(event_heights, event_heights[1:]):
        total_width += events[y0]
        gain = total_width * (y1 - y0)
        if total_width and area_so_far + gain >= target:
            return y0 + (target - area_so_far) / total_width
        area_so_far += gain
    return Fraction(event_heights[-1])
```

BACK TO PART A - PART 1: QUESTION 125

126 * Check pivot after removing one element * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Without any removal (LeetCode 724).

One pass: the pivot is the index where left sum equals total minus left minus the element. O(n), O(1).

FOLLOW-UP CODE

```
def pivot_index(values):
    total, left = sum(values), 0
    for position, value in enumerate(values):
        if left == total - left - value:
            return position
        left += value
    return -1
```

FOLLOW-UP 2 | At most one removal (removal optional).

Answer true if pivot_index finds a pivot, otherwise run the removal test above.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Return which element to remove and the pivot.

Record i together with the index of the matching value (first or last occurrence) when the lookup succeeds, instead of returning True.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 126

127 * Gold Chain Split into Two Equal Halves * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | List every valid (removed segment, cut position) pair.

The answer itself can be quadratic, so a simple loop is acceptable. With positive weights prefix sums are strictly increasing, so each target has at most one cut position and the listing is O(n) after the hash lookups; with zeros or negatives a value can repeat, so keep a list of indices per prefix value.

FOLLOW-UP CODE

```
def gold_chain_all_pairs(segments):
    out = []
    for removed_index in range(len(segments)):
        remaining = segments[:removed_index] + segments[removed_index + 1:]
        remaining_total, left_total = sum(remaining), 0
        for cut_position in range(len(remaining) - 1):
            left_total += remaining[cut_position]
            if left_total * 2 == remaining_total:
                out.append((removed_index, cut_position + 1))          # remove segment r, cut before
position cut_position + 1 of the remaining chain
    return out
```

FOLLOW-UP 2 | The chain is a circle (cut anywhere).

Then a cut is two cut points on the circle and the removed segment just shortens it; use the circular window idea (duplicate the array and slide a window whose sum is half).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Remove k segments instead of one.

The remaining total depends on which k you remove, which becomes a subset-sum style search; ask for the limits before promising a fast solution.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 127

128 * Count the number of ways to split a string. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Split into m parts with equal ones.

The same argument: each of the part_count - 1 cut points sits in the gap after a multiple of k ones, so multiply the gap sizes; with no ones at all it is C(length - 1, part_count - 1).

FOLLOW-UP CODE

```
def num_ways_m_parts(bits, part_count):
    one_positions = [index for index, bit in enumerate(bits) if bit == "1"]
    length, ones_count = len(bits), len(one_positions)
    if ones_count % part_count:
        return 0
    if ones_count == 0:
        return math.comb(length - 1, part_count - 1) % MOD if length >= part_count else 0
    ones_per_part, ways = ones_count // part_count, 1
    for boundary in range(1, part_count):
        ways = ways * (one_positions[boundary * ones_per_part] - one_positions[boundary * ones_per_part - 1])
% MOD
    return ways
```

FOLLOW-UP 2 | Count ways to split into palindromes.

That is a different problem: dp[i] counts splits of the prefix, adding dp[j] for every palindromic s[j:i]; precompute the palindromic substrings with expand-around-center or a DP table in O(n^2).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The numbers get huge.

Return the answer modulo 10^9 + 7 and apply the modulus after every multiplication; in languages with fixed-width integers use 64-bit intermediates. Cost: O(n) with modular arithmetic; no big integers are needed.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 128

GROUP 16 OF 43 | 5 QUESTIONS | PART A - PART 2 FOLLOW-UPS

Sorted intervals: merge while they touch, then read the gaps or search

130 * Find Containing Range

* MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return every original range that contains x, not the merged one.

Sort by start and consider only ranges starting at or before x (one bisect); among those, keep the ones whose end is at least x. With many overlaps use an interval tree for O(log n + k) per query.

FOLLOW-UP CODE

```
def containing_original(ranges, point):
    """Return the original ranges (not merged) that contain x: sort by start and keep a running prefix max of
ends is not enough, so scan the candidates."""
    starts = sorted(ranges)
    count_started = bisect_right([start for start, _ in starts], point)
    return [(start, e) for start, e in starts[:count_started] if e >= point]
```

FOLLOW-UP 2 | Ranges are inserted and deleted over time.

Keep the disjoint ranges in a sorted container (balanced tree); an insert merges neighbours, and a query is the same predecessor search. O(log n) per operation.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Queries are given in bulk.

Sort the queries and sweep once over the merged ranges with two pointers: O((n + q) log) overall instead of a binary search per query.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 130

131 * Union of Sorted Interval Lists

* MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | What if the lists are too big for memory?

Read each list through an iterator (a file or a paged API). union_k only ever holds one interval per list plus the current merged interval, so memory is O(K) and it can emit results as it goes. Sorted order is what makes streaming possible.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if one list may overlap itself?

The two-pointer merge already works, because it folds every incoming interval into the last output interval. What breaks is only the assumption that each list is sorted by start; if that also fails, sort each list first (O(n log n)) or push everything through a heap.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Do touching intervals merge?

Decide the endpoint rule before coding. For closed intervals [1, 3] and [3, 5] share a point, so they merge (start <= last end). For half-open intervals [1, 3) and [3, 5) they do not overlap but are adjacent; whether to merge adjacent ones is a product choice, so ask.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Each list may overlap itself or be unsorted.

Normalise each list first (sort and merge it by itself, O(n log n)), after which the two-pointer union applies. Or sort everything together if memory allows.

FOLLOW-UP CODE

```
def union_sort(first, second):
```

```
    merged = []
    for start, end in sorted(first + second):
        if merged and start <= merged[-1][1]:
            merged[-1][1] = max(merged[-1][1], end)
        else:
            merged.append([start, end])
    return merged

def union_two(first, second):
    merged, first_index, second_index = [], 0, 0
    while first_index < len(first) or second_index < len(second):
        if second_index == len(second) or (first_index < len(first) and first[first_index][0] <= second[
second_index][0]):
            next_interval = first[first_index]; first_index += 1
        else:
            next_interval = second[second_index]; second_index += 1
        if merged and next_interval[0] <= merged[-1][1]:
            merged[-1][1] = max(merged[-1][1], next_interval[1])
        else:
            merged.append([next_interval[0], next_interval[1]])
    return merged

def union_unsorted_lists(first, second):
    """Follow-up: a list may overlap itself and need not be sorted: sort each list first (O(n log n)), then
merge as before."""
    return union_two(union_sort(first, []), union_sort(second, []))
```

BACK TO PART A - PART 1: QUESTION 131

132 * Range Module (Insert and Point Query) * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Add remove_range(left, right).

Find the block of intervals that overlap [left, right) with bisect, keep the part of the first one left of the removal and the part of the last one right of it, and splice the remainders in.

FOLLOW-UP CODE

```
class RemovableRangeModule(RangeModule):
    def remove_range(self, left, right):
        first_overlap = bisect_right(self.rights, left)
        after_last_overlap = bisect_left(self.lefts, right)
        if first_overlap >= after_last_overlap:
            return
        keep = []
        if self.lefts[first_overlap] < left:
            keep.append((self.lefts[first_overlap], left))
        if self.rights[after_last_overlap - 1] > right:
            keep.append((right, self.rights[after_last_overlap - 1]))
        self.lefts[first_overlap:after_last_overlap] = [left_end for left_end, _ in keep]
        self.rights[first_overlap:after_last_overlap] = [right_end for _, right_end in keep]
```

BACK TO PART A - PART 1: QUESTION 132

133 * Meetings Minus a Do-Not-Schedule Block * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | There are several do-not-schedule blocks. How do you subtract all of them?

Merge both the meetings and the blocks. Walk the merged meetings with a pointer into the blocks: keep the part of the meeting before the next block, skip past the block, and continue with what is left. O((n + b) log (n + b)).

FOLLOW-UP CODE

```
def subtract_many(meetings, blocks):
    remaining, blocks = [], merge_intervals(blocks)
    for start, end in merge_intervals(meetings):
        cursor = start
        for block_start, block_end in blocks:
            if block_end <= cursor or block_start >= end:
                continue
```

```
            if block_start > cursor:
                remaining.append((cursor, block_start))
            cursor = max(cursor, block_end)
        if cursor < end:
            remaining.append((cursor, end))
    return remaining
```

BACK TO PART A - PART 1: QUESTION 133

134 * Merge Intervals * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Insert one interval into an already merged list (LeetCode 57).

Copy the intervals that end before it, merge the ones that overlap, copy the rest: O(n) with no sort.

FOLLOW-UP CODE

```
def insert_interval(intervals, new):
    merged, position = [], 0
    while position < len(intervals) and intervals[position][1] < new[0]:
        merged.append(intervals[position]); position += 1
    while position < len(intervals) and intervals[position][0] <= new[1]:
        new = [min(new[0], intervals[position][0]), max(new[1], intervals[position][1])]; position += 1
    return merged + [new] + intervals[position:]
```

FOLLOW-UP 2 | Intervals of touching integers ([1, 2] and [3, 4]) should merge.

Change the overlap test to start <= last_end + 1. For real-valued ranges that would be wrong, so state which you assume.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Too many intervals for memory.

Sort externally (external merge sort) and stream the merge: the one-pass sweep needs only the current interval in memory. Cost: O(n log n) external sort with bounded memory runs, then an O(n) streaming merge with O(1) memory.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 134

Trie: nested dicts, one character per level

137 * URL Request Handler Router * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Different HTTP methods per route.

Make the method the first segment of the path (a tiny trick) or keep a dict method to trie. Lookup uses the method as the first step.

FOLLOW-UP CODE

```
class Router:
    def __init__(self, patterns=()):
        self.root = {"lit": {}, "wild": None, "handler": None}
        for pattern in patterns:
            self.add(pattern)

    def add(self, pattern):
        node = self.root
        for segment in _segments(pattern):
            if segment.startswith("{") and segment.endswith("}"):
                if node["wild"] is None:
                    node["wild"] = {"lit": {}, "wild": None, "handler": None}
                node = node["wild"]
            else:
                node = node["lit"].setdefault(segment, {"lit": {}, "wild": None, "handler": None})
        if node["handler"] is None:
            node["handler"] = pattern

    def match(self, url):
        url_segments = _segments(url)

        def walk(node, position):
            if position == len(url_segments):
                return node["handler"]
            child = node["lit"].get(url_segments[position])
            if child is not None:
                found = walk(child, position + 1)
                if found is not None:
                    return found
            if node["wild"] is not None:
                return walk(node["wild"], position + 1)
            return None

        return walk(self.root, 0)

def _segments(path):
    return [segment for segment in path.strip("/").split("/") if segment != ""]

class MethodRouter(Router):
    def add_route(self, method, pattern):
        self.add(f"{method}/{pattern.strip('/')}")

    def lookup(self, method, url):
        return self.match(f"{method}/{url.strip('/')}")
```

FOLLOW-UP 2 | A wildcard at the end that matches the rest of the path (/static/*).

Give each trie node a 'rest' handler. During lookup, remember the deepest node that has one and use it if the exact walk fails. Precedence: exact, then placeholder, then rest-wildcard.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Return the extracted parameters (id = 7).

Carry a list of (name, segment) while walking; append when you take a wildcard edge and drop it when you backtrack. Store the placeholder names with the pattern, not in the trie.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 137

138 * Trie Prefix Search / Autocomplete * MEDIUM *

DIFFICULTY | * MEDIUM * | 2 FOLLOW-UPS

FOLLOW-UP 1 | Return only the best K completions ranked by frequency.

Store at every node the K best (frequency, word) pairs and update them while inserting; a query then reads one node in O(L + K) and never walks the subtree. The price is more memory and slower inserts, which is the right trade when reads dominate (autocomplete).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Allow one wrong character (fuzzy match).

Walk the trie with a budget of one edit: at each level follow the matching child with the full budget and every other child with the budget reduced to 0. The work grows with the alphabet size times the prefix length, not with the dictionary size.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 138

139 * Build Prefix Lookup with a Trie * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | How many words start with this prefix?

Store a counter in each node that is incremented when a word passes through it; the answer is the counter at the end of the prefix walk, O(L) with no DFS.

FOLLOW-UP CODE

```
class CountingTrie:
    """Each node counts the words passing through it, so 'how many words start with p' needs no DFS."""
    def __init__(self):
        self.root = {"n": 0, "c": {}}

    def insert(self, word):
        node = self.root
        node["n"] += 1
        for character in word:
            node = node["c"].setdefault(character, {"n": 0, "c": {}})
            node["n"] += 1

    def count_prefix(self, prefix):
        node = self.root
        for character in prefix:
            node = node["c"].get(character)
            if node is None:
                return 0
        return node["n"]
```

FOLLOW-UP 2 | Return only the top 3 completions (by frequency or alphabetically).

Keep at each node a short list of its best K words (updated on insert), or bound the DFS: visit children in alphabetical order and stop after K words. That is the LeetCode 1268 style answer.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Memory is large (many words with long common prefixes).

A compressed (radix) trie stores whole edge labels instead of single characters; a sorted array with binary search over prefixes uses the least memory for a static dictionary. Cost: a radix trie has O(number of words) nodes instead of O(total characters); a sorted array uses O(total characters) with O(L log W) lookups.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 139

140 * Greedy Longest-Match Dictionary Tokenizer

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | The dictionary is huge.

A flat map of every key wastes memory because keys share prefixes. A compressed (radix) trie stores each shared stretch once, or shard the keys by first characters across machines. Lookup stays one pass over the text; only the node layout changes.

FOLLOW-UP CODE

```
def tokenize_trie(text, table, priority=None):
    root = {}
    for key, token_id in table.items():
        node = root
        for character in key:
            node = node.setdefault(character, {})
        node["$"] = (key, token_id)
    tokens, start = [], 0
    while start < len(text):
        node, best_match, end = root, None, start
        while end < len(text) and text[end] in node:
            node = node[text[end]]
            end += 1
            if "$" in node and (best_match is None or priority is None or priority(node["$"][0]) < priority(
best_match[0])):
                best_match = node["$"]
        if best_match is None:
            tokens.append(text[start])
            start += 1
        else:
            tokens.append(best_match[1])
            start += len(best_match[0])
    return tokens
```

BACK TO PART A - PART 1: QUESTION 140

141 * Check if all substrings are dictionary words

* MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the first failing substring.

Same loops, but return the substring instead of False, which is what a user wants to see. Check the shorter ones first so the report is the smallest counterexample.

FOLLOW-UP CODE

```
def first_failing_substring(text, words):
    for length in range(3, len(text) + 1):              # shortest first, so the report is the smallest
counterexample
        for start in range(len(text) - length + 1):
            if text[start:start + length] not in words:
                return text[start:start + length]
    return None
```

FOLLOW-UP 2 | The string is very long.

The check is inherently quadratic in the number of substrings; any substring that fails ends the search, so cheap necessary checks first help: every length-3 window must be a word (O(n) lookups), then length 4, and so on.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The dictionary is huge and sharded.

Batch the lookups by length (a set per length) or use a Bloom filter in front of the real dictionary to reject most misses cheaply. Cost: still O(n^2) lookups, spread over shards (each lookup O(L)); a Bloom filter in front removes most remote calls.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 141

142 * Return dictionary words matching a prefix * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Only the first K matches (autocomplete).

Stop the forward walk after K words, as the limit parameter does; with the sorted list the first K in alphabetical order are already the first K in the block.

FOLLOW-UP CODE

```
class PrefixIndex:
    def __init__(self, words):
        self.words = sorted(words)

    def matches(self, prefix, limit=None):
        position = bisect_left(self.words, prefix)
        matches = []
        while position < len(self.words) and self.words[position].startswith(prefix):
            matches.append(self.words[position])
            if limit is not None and len(matches) == limit:
                break
            position += 1
        return matches
```

FOLLOW-UP 2 | Words are added while queries run.

Keep the list sorted with bisect.insort (O(n) per insert, fast in practice) or use a balanced tree; a trie handles inserts in O(L).

FOLLOW-UP CODE

```
class LivePrefixIndex:
    def __init__(self):
        self.words = []

    def add(self, word):
        position = bisect_left(self.words, word)
        if position == len(self.words) or self.words[position] != word:
            self.words.insert(position, word)

    def matches(self, prefix, limit=None):
        position, matches = bisect_left(self.words, prefix), []
        while position < len(self.words) and self.words[position].startswith(prefix) and (limit is None or
len(matches) < limit):
            matches.append(self.words[position])
            position += 1
        return matches
```

FOLLOW-UP 3 | Case-insensitive matching.

Store a casefolded sort key next to each word (a tuple (key, word)) and search on the key; do not lowercase the displayed word.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 142

143 * Implement Trie Insert and Exact-Word Search * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Support '.' as a wildcard in search (LeetCode 211).

On '.', try every child at that level and recurse; other characters follow the single matching child. Worst case O(26^L) per query on adversarial patterns, so mention the bound.

FOLLOW-UP CODE

```
class Trie:
    def __init__(self):
        self.root = {"children": {}, "end": False}

    def _walk(self, text):
        node = self.root
        for character in text:
            node = node["children"].get(character)
            if node is None:
```

```
                return None
        return node

    def insert(self, word):
        node = self.root
        for character in word:
            node = node["children"].setdefault(character, {"children": {}, "end": False})
        node["end"] = True

    def search(self, word):
        node = self._walk(word)
        return node is not None and node["end"]

    def startsWith(self, prefix):
        node = self._walk(prefix)
        return node is not None and (node["end"] or bool(node["children"]))     # a node counts only if some
word passes through it

class WildcardTrie(Trie):
    """search('a.c') where '.' matches any one letter (LeetCode 211)."""
    def search(self, pattern):
        def go(node, position):
            if position == len(pattern):
                return node["end"]
            if pattern[position] == ".":
                return any(go(child, position + 1) for child in node["children"].values())
            child = node["children"].get(pattern[position])
            return child is not None and go(child, position + 1)

        return go(self.root, 0)
```

FOLLOW-UP 2 | Delete a word.

Unmark the end flag and, walking back up, remove nodes that have no children and are not word ends. Keep a path stack while walking down so you can prune on the way back. Cost: O(L) time and O(L) space for the path stack.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Unicode or large alphabets.

Store children in a dict keyed by character instead of a 26-slot array, as the code does; it handles any alphabet and wastes no space on missing letters. Cost: O(L) per operation with dictionary children; memory O(total characters).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 143

A dict for lookup plus an ordered container for who is next (lazy deletion)

146 * Design a restaurant waitlist system * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Parties that wait too long should be seated first (starvation).

Add an aging rule: if the earliest party has waited longer than a threshold and a table of at least its size appears, seat it before anyone else. The segment tree can query 'leftmost size <= c' with c = the table, so order is preserved.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Seat the best-fitting party, not the earliest.

That is a different policy: keep sizes in a sorted structure of (size, arrival) and take the largest size at most the capacity (ties by arrival). A sorted container with bisect gives O(log n) lookups.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Estimate the waiting time of a party.

Track the arrival rate and typical table turnover per capacity; the estimate is the expected time until enough tables of at least that size free up, ahead of the party in the queue.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Seat the best-fitting party (the largest that fits), not the earliest.

Group waiting parties by size in queues (arrival order inside a size) and keep the sizes in a sorted list: the best fit for a table of capacity c is the largest size at most c, found with bisect, then the oldest party of that size. O(log distinct sizes) per seating. Verified against a linear scan.

FOLLOW-UP CODE

```
class WaitlistBestFit2:
    def __init__(self):
        self.by_size = defaultdict(deque)
        self.sizes = []
        self.n = 0

    def add(self, party_id, name, size):
        if not self.by_size[size]:
            insort(self.sizes, size)
        self.by_size[size].append(party_id)

    def seat(self, capacity):
        position = bisect_right(self.sizes, capacity) - 1
        if position < 0:
            return None
        size = self.sizes[position]
        party_id = self.by_size[size].popleft()
        if not self.by_size[size]:
            del self.sizes[position]
        return party_id
```

BACK TO PART A - PART 1: QUESTION 146

147 * Implement Python's itertools.tee

* MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Memory grows when one consumer stalls. Is there a way to free items automatically?

Use one shared singly linked list with a pointer per consumer. The only references to old nodes are the consumers' pointers, so Python's garbage collector frees nodes behind the slowest consumer without any bookkeeping. This is the design CPython uses.

FOLLOW-UP CODE

```
class _Link:
    __slots__ = ("value", "next")

    def __init__(self):
        self.value = self.next = None

def tee_linked(iterable, copies=2):
    source_iterator = iter(iterable)
    head = _Link()
    state = {"tail": head, "done": False}

    def consumer(node):
        while True:
            if node.next is None:
                if state["done"]:
                    return
                try:
                    value = next(source_iterator)
                except StopIteration:
                    state["done"] = True
                    return
                link = _Link()
                link.value = value
                state["tail"].next = link
                state["tail"] = link
            node = node.next
            yield node.value

    return tuple(consumer(head) for _ in range(copies))
```

FOLLOW-UP 2 | Make it safe for several threads.

Pull from the source under one lock and append to the queues inside it. Without the lock two consumers can pull two items and append them in different orders to different queues. Cost: one lock acquisition per item pulled, O(1) extra per item; contention grows with the number of consumers.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Why not just copy the iterator?

Most iterators (files, generators, sockets) cannot be copied. tee gives independent views by buffering, which is the only thing you can do when the source cannot rewind.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 147

148 * Restaurant Waitlist: join / delete / find_first_match

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | What would you cache to speed up repeated finds?

Keep a small structure that answers 'earliest sequence among sizes <= t' directly: a segment tree or Fenwick tree over party size storing the minimum front sequence number, updated when a queue front changes. Find becomes O(log S) instead of O(S).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 148

149 * Store With Cheapest-Item Removal * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Prices are integers from 0 to 10000. Can removal be O(1) amortized?

Use an array of 10001 buckets, one per price, each a small heap by timestamp, and a pointer to the lowest price that might be non-empty. add lowers the pointer if needed; remove_cheapest moves it up past empty buckets, so the pointer only moves up between adds.

FOLLOW-UP CODE

```
class BucketStore:
    MAX_PRICE = 10000

    def __init__(self):
        self.buckets = [[] for _ in range(self.MAX_PRICE + 1)]
        self.low = self.MAX_PRICE + 1

    def add(self, value, price, timestamp):
        heapq.heappush(self.buckets[price], (timestamp, value))
        self.low = min(self.low, price)

    def remove_cheapest(self):
        while self.low <= self.MAX_PRICE and not self.buckets[self.low]:
            self.low += 1
        if self.low > self.MAX_PRICE:
            return None
        return heapq.heappop(self.buckets[self.low])[1]
```

BACK TO PART A - PART 1: QUESTION 149

150 * Round-Robin Job Scheduler * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Each job has an arrival time. How do you change it?

Sort the jobs by arrival. Before every pick admit all jobs whose arrival is at most the clock; a job preempted after its slice goes to the back after the jobs that arrived during the slice. If the queue is empty, jump the clock to the next arrival.

FOLLOW-UP CODE

```
def round_robin_arrivals(jobs, quantum):
    # jobs are (id, burst, arrival)
    pending = deque(sorted(jobs, key=lambda job: job[2]))
    queue, clock, done = deque(), 0, []
    while pending or queue:
        if not queue:
            clock = max(clock, pending[0][2])
        while pending and pending[0][2] <= clock:
            job_id, remaining_time, _ = pending.popleft()
            queue.append([job_id, remaining_time])
        job = queue.popleft()
        time_slice = min(quantum, job[1])
        clock += time_slice
        job[1] -= time_slice
        # jobs that arrived while this one ran line up before it
        while pending and pending[0][2] <= clock:
            job_id, remaining_time, _ = pending.popleft()
            queue.append([job_id, remaining_time])
        if job[1] > 0:
            queue.append(job)
        else:
            done.append((job[0], clock))
    return done
```

BACK TO PART A - PART 1: QUESTION 150

151 * Aggregate FizzBuzz Offer Matches * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | After processing, report which offers were not fully matched.

Keep the two deques as the state: whatever remains in them at the end is the unmatched quantity per offer, in priority order. Return both queues (or merge them).

FOLLOW-UP CODE

```
def unmatched_after(offers):
    queues = {"Fizz": deque(), "Buzz": deque()}
    for offer in offers:
        mine = queues[offer["type"]]
        other = queues["Buzz" if offer["type"] == "Fizz" else "Fizz"]
        quantity_left = offer["quantity"]
        while quantity_left and other:
            matched = min(quantity_left, other[0][1])
            quantity_left -= matched
            other[0][1] -= matched
            if other[0][1] == 0:
                other.popleft()
        if quantity_left:
            mine.append([offer["name"], quantity_left])
    return {side: [(name, quantity) for name, quantity in queue] for side, queue in queues.items()}
```

BACK TO PART A - PART 1: QUESTION 151

Rolling state: the next state comes only from the previous one

155 * Maximize sum without choosing adjacent elements

* MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The houses form a circle (LeetCode 213).

The first and last are adjacent, so you cannot take both: the answer is the better of 'ignore the first house' and 'ignore the last house', each solved with the linear version.

FOLLOW-UP CODE

```
def _house_dp(house_money):
    take = skip = 0
    for house_value in house_money:
        take, skip = skip + house_value, max(take, skip)
    return max(take, skip)

def _house_circular(house_money):
    if len(house_money) == 1:
        return house_money[0]
    return max(_house_dp(house_money[1:]), _house_dp(house_money[:-1]))
```

FOLLOW-UP 2 | Return which houses to rob.

Keep the table best_from[house] = best from i onward (filled from the right) and walk forward: take house i when nums[house] + best_from[house + 2] equals best_from[house], then jump two; otherwise move one.

FOLLOW-UP CODE

```
def _house_choice(nums):
    house_count = len(nums)
    best_from = [0] * (house_count + 2)
    for house in range(house_count - 1, -1, -1):
        best_from[house] = max(best_from[house + 1], nums[house] + best_from[house + 2])
    chosen, house = [], 0
    while house < house_count:
        if nums[house] + best_from[house + 2] >= best_from[house + 1] and nums[house] + best_from[house + 2]
== best_from[house]:
            chosen.append(house)
            house += 2
        else:
            house += 1
    return chosen
```

FOLLOW-UP 3 | Robbing a house also alerts the next k houses.

The recurrence becomes dp[i] = max(dp[i + 1], nums[i] + dp[i + k + 1]); the O(1) space trick needs a window of k + 1 values (a deque), or just keep the O(n) table.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 155

156 * Paint House II

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Return which colour each house gets, not just the cost.

Store the DP row for every house (O(house_count colour_count) memory), then walk back from the cheapest last cell: at house i pick the cheapest entry of the row of house i - 1 whose colour differs from the colour of house i + 1... in practice keep the argmin of the previous row and the second argmin when you store each row.

FOLLOW-UP CODE

```
def paint_plan(costs):
    house_count, colour_count = len(costs), len(costs[0])
    rows = []
```

```
    previous_totals = [0] * colour_count
    for row in costs:
        current_totals = [row[colour] + min((previous_totals[other_colour] for other_colour in range(
colour_count) if other_colour != colour), default=0) for colour in range(colour_count)]
        rows.append(current_totals)
        previous_totals = current_totals
    colour = min(range(colour_count), key=lambda colour: rows[-1][colour])
    total, plan = rows[-1][colour], [colour]
    for i in range(house_count - 1, 0, -1):
        need = rows[i][colour] - costs[i][colour]
        colour = next(other_colour for other_colour in range(colour_count) if other_colour != colour and rows[
i - 1][other_colour] == need)
        plan.append(colour)
    return total, plan[::-1]
```

BACK TO PART A - PART 1: QUESTION 156

157 * Given n houses in a line with money in each, find the maximum amount a robber can steal, without stealing from two adjacent houses. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The houses form a circle (LeetCode 213).

The first and last are adjacent, so you cannot take both: the answer is the better of 'ignore the first house' and 'ignore the last house', each solved with the linear version.

FOLLOW-UP CODE

```
def _house_dp(house_money):
    take = skip = 0
    for house_value in house_money:
        take, skip = skip + house_value, max(take, skip)
    return max(take, skip)

def _house_circular(house_money):
    if len(house_money) == 1:
        return house_money[0]
    return max(_house_dp(house_money[1:]), _house_dp(house_money[:-1]))
```

FOLLOW-UP 2 | Return which houses to rob.

Keep the table best_from[house] = best from i onward (filled from the right) and walk forward: take house i when nums[house] + best_from[house + 2] equals best_from[house], then jump two; otherwise move one.

FOLLOW-UP CODE

```
def _house_choice(nums):
    house_count = len(nums)
    best_from = [0] * (house_count + 2)
    for house in range(house_count - 1, -1, -1):
        best_from[house] = max(best_from[house + 1], nums[house] + best_from[house + 2])
    chosen, house = [], 0
    while house < house_count:
        if nums[house] + best_from[house + 2] >= best_from[house + 1] and nums[house] + best_from[house + 2]
== best_from[house]:
            chosen.append(house)
            house += 2
        else:
            house += 1
    return chosen
```

FOLLOW-UP 3 | Robbing a house also alerts the next k houses.

The recurrence becomes dp[i] = max(dp[i + 1], nums[i] + dp[i + k + 1]); the O(1) space trick needs a window of k + 1 values (a deque), or just keep the O(n) table.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 157

158 * Write SQL code to publish the Fibonacci series. * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Only the even terms, or the sum of the first n terms.

Keep the CTE and filter or aggregate in the outer query: WHERE a % 2 = 0 for the even terms, SUM(a) for the total. The recursion stays the same.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Which database?

PostgreSQL and MySQL 8 use WITH RECURSIVE; SQL Server drops the word RECURSIVE and may need a CAST so the column type is wide enough (bigint) in the anchor row; Oracle accepts WITH without RECURSIVE too. State the dialect assumption first.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Large n overflows.

Use a wider type (BIGINT, NUMERIC, or DECIMAL) in the anchor row so the column type is wide enough from the start; arbitrary-size values need a language with big integers. Cost: O(n) rows regardless of the type; a wider type only costs more bytes per row.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Use it from application code with a parameter.

Bind the limit as a query parameter (never build the SQL by string concatenation) and fetch the single column. The recursion lives inside the database, so there is one round trip regardless of n.

FOLLOW-UP CODE

```
def fibonacci_via_sql(limit):
    connection = sqlite3.connect(":memory:")
    query = """WITH RECURSIVE fib(n, a, b) AS (SELECT 1, 0, 1 UNION ALL SELECT n + 1, b, a + b FROM fib WHERE
n < :limit)
           SELECT a FROM fib ORDER BY n"""
    return [r[0] for r in connection.execute(query, {"limit": limit})]
```

BACK TO PART A - PART 1: QUESTION 158

Two pointers: move i and j toward each other (or along two lists)

164 * Find median of merged RLE arrays

*** MEDIUM ***

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | K arrays instead of two.

Merge all runs by value (heap of K heads, or sort the combined run list if memory allows) and walk to the two middle ranks. O(R log K) with a heap, R the total number of runs.

FOLLOW-UP CODE

```
def median_rle_k_arrays(arrays):
    runs = sorted((value, count) for runs_of_array in arrays for value, count in runs_of_array)
    total_count = sum(count for _, count in runs)

    def kth(rank):
        elements_passed = 0
        for value, count in runs:
            elements_passed += count
            if elements_passed > rank:
                return value

    return kth(total_count // 2) if total_count % 2 else (kth(total_count // 2 - 1) + kth(total_count // 2)) /
 2
```

FOLLOW-UP 2 | The arrays are huge and sorted, and you want O(log) time.

Binary search on the value: for a candidate value count how many elements are at most it in each array (binary search on prefix sums of counts) and find the smallest value with rank above the middle. O(log(range) * log(runs)).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Runs are not sorted by value.

Sort the runs by value first, O(R log R), then use the same walk.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 164

165 * Run-Length Vector Storage + Dot Product

*** MEDIUM ***

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | What if one vector is mostly zeros and stored sparsely, and the other as runs?

Iterate the sparse side's nonzero indices and binary-search the run list for each index to read the other value: O(s log r). Choose the smaller representation to drive the loop.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | How should the data be updated if you set element i?

Locate the run containing i (binary search over prefix run lengths), then split it into at most three runs and merge equal neighbours. That is O(log r) to find plus O(r) in the worst case to splice a list; use a balanced tree of runs for frequent updates.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | One vector is a sparse map (index to value) and the other is run-length encoded.

Iterate the smaller structure (the sparse non-zeros) and binary search the run list for each index using prefix run lengths: O(s log r) instead of expanding anything.

FOLLOW-UP CODE

```
def dot_sparse_rle(sparse, runs):
    """sparse: {index: value}; runs: RLE list. Binary search the run that contains each index."""
    import bisect
    starts, position = [], 0
```

```
    for _, run_length in runs:
        starts.append(position)
        position += run_length
    total = 0
    for index, value in sparse.items():
        run_index = bisect.bisect_right(starts, index) - 1
        if 0 <= run_index < len(runs) and index < starts[run_index] + runs[run_index][1]:
            total += value * runs[run_index][0]
    return total
```

BACK TO PART A - PART 1: QUESTION 165

166 * Count Unique Triplets That Sum to Zero * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Return the unique triplets themselves.

The same sweep, appending [nums[fixed_index], nums[left], nums[right]] on every hit instead of counting. The output size can be O(n^2), which dominates the cost.

FOLLOW-UP CODE

```
def zero_sum_triplets(nums):
    nums = sorted(nums)
    out = []
    for fixed_index in range(len(nums) - 2):
        if fixed_index and nums[fixed_index] == nums[fixed_index - 1]:
            continue
        left, right = fixed_index + 1, len(nums) - 1
        while left < right:
            total = nums[fixed_index] + nums[left] + nums[right]
            if total < 0:
                left += 1
            elif total > 0:
                right -= 1
            else:
                out.append([nums[fixed_index], nums[left], nums[right]])
                left += 1
                right -= 1
                while left < right and nums[left] == nums[left - 1]:
                    left += 1
    return out
```

BACK TO PART A - PART 1: QUESTION 166

Walk two strings at once with two indices

170 * Can board states be transformed? * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the minimum number of moves.

Because pieces never interact beyond keeping their order, each piece moves independently from its start index to its target index, so the minimum is the sum of the absolute index differences when the transformation is possible.

FOLLOW-UP CODE

```
def canTransform(start, end):
    if len(start) != len(end):
        return False
    start_index = end_index = 0
    length = len(start)
    while start_index < length or end_index < length:
        while start_index < length and start[start_index] == "X":
            start_index += 1
        while end_index < length and end[end_index] == "X":
            end_index += 1
        if start_index == length or end_index == length:
            return start_index == end_index
        if start[start_index] != end[end_index]:
            return False
        if start[start_index] == "L" and start_index < end_index:
            return False
        if start[start_index] == "R" and start_index > end_index:
            return False
        start_index, end_index = start_index + 1, end_index + 1
    return True

def min_moves(start, end):
    """If the transformation is possible, every piece moves independently, so the minimum is the sum of index
differences."""
    if not canTransform(start, end):
        return -1
    start_positions = [position for position, char in enumerate(start) if char != "X"]
    end_positions = [position for position, char in enumerate(end) if char != "X"]
    return sum(abs(start_position - end_position) for start_position, end_position in zip(start_positions,
end_positions))
```

FOLLOW-UP 2 | Add walls that cannot be crossed (next question).

A wall is just another piece that never moves: it stays in the sequence of non-blank symbols and must keep the same index in both strings. The same pointer walk handles it.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Pieces can move one step per round and several move at once.

The count of rounds is the maximum over pieces of their distance, if pieces never block each other; with blocking you simulate rounds or use the per-piece arrival times with the ordering constraint.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 170

171 * Decide string transform with directional tokens and walls * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Segments between walls are independent.

You can also split both strings at the walls (they must be at identical positions) and apply the L/R rule to each segment separately. The single pass above is equivalent.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Return the moves or the minimum number of steps.

Per piece, the steps are the index difference; list moves by letting each piece walk one cell at a time in an order that never blocks (move L pieces left to right and R pieces right to left).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The board is 2D.

Then pieces can pass each other by going around, and invariants like order no longer hold. Ask for the exact movement rules; most likely you would model it as a graph search with a heuristic.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Return the minimum number of moves.

When the transformation is possible each piece moves independently from its start index to its target index, so the minimum is the sum of the absolute index differences (checked against BFS on small strings).

FOLLOW-UP CODE

```
def can_transform_walls(start, end):
    if len(start) != len(end):
        return False
    start_index = end_index = 0
    length = len(start)
    while True:
        while start_index < length and start[start_index] == ".":
            start_index += 1
        while end_index < length and end[end_index] == ".":
            end_index += 1
        if start_index == length or end_index == length:
            return start_index == end_index
        if start[start_index] != end[end_index]:
            return False
        if start[start_index] == "X" and start_index != end_index:
            return False
        if start[start_index] == "L" and start_index < end_index:
            return False
        if start[start_index] == "R" and start_index > end_index:
            return False
        start_index, end_index = start_index + 1, end_index + 1

def min_moves_walls(start, end):
    if not can_transform_walls(start, end):
        return -1
    start_positions = [position for position, char in enumerate(start) if char != "."]
    end_positions = [position for position, char in enumerate(end) if char != "."]
    return sum(abs(start_position - end_position) for start_position, end_position in zip(start_positions,
end_positions))
```

BACK TO PART A - PART 1: QUESTION 171

172 * Determine if two sentences are similar. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Similarity is transitive (a ~ b and b ~ c means a ~ c).

Union-find over the words: union each pair, then two words are similar if they are equal or have the same root. Path compression keeps it near O(n + P).

FOLLOW-UP CODE

```
def areSentencesSimilarTwo(sentence_one, sentence_two, pairs):
    if len(sentence_one) != len(sentence_two):
        return False
    parent = {}

    def find(word):
        parent.setdefault(word, word)
        while parent[word] != word:
            parent[word] = parent[parent[word]]
            word = parent[word]
        return word

    for word_a, word_b in pairs:
        parent[find(word_a)] = find(word_b)
    return all(word_one == word_two or find(word_one) == find(word_two) for word_one, word_two in zip(
sentence_one, sentence_two))
```

FOLLOW-UP 2 | One sentence is the other with a sentence of words inserted somewhere (LeetCode 1813).

Match the longest common prefix and the longest common suffix of the word lists; the shorter list must be covered by prefix plus suffix.

FOLLOW-UP CODE

```
def areSentencesSimilar_insert(shorter, longer):
    if len(shorter) > len(longer):
        shorter, longer = longer, shorter
    prefix_match = 0
    while prefix_match < len(shorter) and shorter[prefix_match] == longer[prefix_match]:
        prefix_match += 1
    suffix_match = 0
    while suffix_match < len(shorter) - prefix_match and shorter[-1 - suffix_match] == longer[-1 -
suffix_match]:
        suffix_match += 1
    return prefix_match + suffix_match >= len(shorter)
```

FOLLOW-UP 3 | Case should not matter, or words can have synonyms with scores.

Normalise with casefold before hashing. For scores, a graph with weighted edges and a threshold query turns it into reachability along strong edges.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 172

GROUP 22 OF 43 | 5 QUESTIONS | PART A - PART 2 FOLLOW-UPS

Simulate a machine: a class with state and one branch per kind of event

175 * Tic-Tac-Toe System with K Players and 3-in-a-Row Win * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | What changes when the winning length is a parameter w?

Replace the constant 3 by w in the run check: the cost per move becomes O(w) because each of the four directions walks at most w - 1 cells each way. The board size does not matter any more, which is the point of checking locally.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 175

176 * Design Tic-Tac-Toe With K Players * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Win with m in a row, not the full line.

Counters per line no longer work, because a line can hold a run anywhere. After each move look along the four directions from that cell, counting same-player neighbours up to m in both directions. O(in_a_row) per move.

FOLLOW-UP CODE

```
class TicTacToeM:
    """Win by m in a row (not the whole board): look along 4 directions from the last move, up to m cells
each way."""
    def __init__(self, size, in_a_row):
        self.size, self.in_a_row, self.marks = size, in_a_row, {}

    def move(self, row, col, player):
        if (row, col) in self.marks:
            return -1
        self.marks[(row, col)] = player
        for row_step, col_step in ((0, 1), (1, 0), (1, 1), (1, -1)):
            line_length = 1
            for sign in (1, -1):
                distance = 1
                while distance < self.in_a_row and self.marks.get((row + sign * distance * row_step, col +
sign * distance * col_step)) == player:
                    line_length += 1
                    distance += 1
            if line_length >= self.in_a_row:
                return player
        return 0
```

FOLLOW-UP 2 | Undo the last move.

Decrement the same counters you incremented and clear the cell; remember the move history in a stack. The counters make undo as cheap as a move.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The board is huge or unbounded.

Store only the occupied cells in a dict (as TicTacToeM does) and keep the counter-per-line version only if the win condition is a complete line of a bounded board. Cost: O(1) per move with line counters on a bounded board, or O(m) per move scanning four directions for an m-in-a-row win; memory O(moves).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 176

177 * Implement iterator merging lists with filtering * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The lists are sorted and the output must stay sorted.

Replace the round robin with a k-way merge on a heap (heapq.merge), and drop a value equal to the previous output instead of keeping a seen set: memory O(k).

FOLLOW-UP CODE

```
def sorted_merge_unique(lists, blocked=()):
    blocked, last_yielded = set(blocked), object()
    for value in heapq.merge(*lists):
        if value not in blocked and value != last_yielded:
            yield value
            last_yielded = value
```

FOLLOW-UP 2 | Read the lists one after another, not interleaved.

Chain the lists with a generator; the same blocked/seen filtering applies. Generators give hasNext-style behaviour via a one-item lookahead.

FOLLOW-UP CODE

```
def concatenated_unique(lists, blocked=()):
    seen, blocked = set(), set(blocked)
    for one_list in lists:
        for value in one_list:
            if value not in blocked and value not in seen:
                seen.add(value)
                yield value
```

FOLLOW-UP 3 | The seen set grows without bound.

If the input is sorted or ids repeat only close together, remember only the last few ids; otherwise bound memory with a Bloom filter and accept rare false suppressions. Cost: O(1) per element with a Bloom filter using O(bits) memory, at the price of rare false suppressions.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 177

178 * Infinite Deck Card Duel * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Extend it to n players.

Replace the two counters with a dictionary keyed by player id. Each round every player draws a hand; the round goes to the player whose best card is strictly highest, and ties for the top give nobody a point. The champion needs a lead of at least 2 over the second best.

FOLLOW-UP CODE

```
def play_game_n(deck, players, hand_size=5, num_rounds=100):
    wins = {player: 0 for player in range(1, players + 1)}
    for _ in range(num_rounds):
        best_card_of = {player: max(deck.draw() for _ in range(hand_size)) for player in wins}
        highest_card = max(best_card_of.values())
        leaders = [player for player, card in best_card_of.items() if card == highest_card]
        if len(leaders) == 1:
            wins[leaders[0]] += 1
    order = sorted(wins.items(), key=lambda entry: -entry[1])
    if len(order) > 1 and order[0][1] - order[1][1] >= 2:
        return order[0][0]
    return 0
```

BACK TO PART A - PART 1: QUESTION 178

179 * Design an elevator control system * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Peak traffic (morning lobby rush, evening exodus).

Add modes: during the morning rush park idle cars in the lobby and let them go up with a full load; in the evening do the opposite. Zone cars (low floors, high floors) or use destination dispatch (passengers enter their floor first) to group riders by destination.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Fairness and starvation.

SCAN already guarantees every stop is reached within one sweep. For hall calls add an aging term to the cost so a long-waiting call wins over a closer new one.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Safety and fault tolerance.

Keep door and overload as explicit states; a car that fails is marked out of service and its stops are reassigned (the controller does this), and the controller should be restartable from persistent state of cars and pending calls.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Morning rush: idle cars should wait at the lobby.

When a car has no stops, give it a stop at the lobby (or a configured parking floor per car). Peak modes are just policies layered on the same step() and cost-based assignment; the elevator algorithm does not change.

FOLLOW-UP CODE

```
class Controller:
    def __init__(self, cars):
        self.cars, self.served = cars, []

    def request(self, floor, direction=0):
        call = Request(floor, direction)
        best = min(self.cars, key=lambda car: (car.cost(call), car.car_id))
        best.stops.add(floor)
        return best.car_id

    def fail(self, car_id):
        car = next(car for car in self.cars if car.car_id == car_id)
        car.in_service = False
        pending, car.stops = car.stops, set()
        for floor in pending:
            self.request(floor)

    def step(self):
        for car in self.cars:
            served_floor = car.step()
            if served_floor is not None:
                self.served.append((car.car_id, served_floor))

    def idle(self):
        return all(not car.stops for car in self.cars if car.in_service)

class LobbyController(Controller):
    def park_idle(self, lobby=0):
        for car in self.cars:
            if car.in_service and not car.stops and car.floor != lobby:
                car.stops.add(lobby)
```

BACK TO PART A - PART 1: QUESTION 179

Top-K by frequency: Counter plus nsmallest

181 * Top-K Frequent in Large Logs * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | How do you do it across many machines when the log does not fit on one?

Count per shard and send each shard's local top-K to a coordinator. That is not exact in general (a user can be 11th everywhere yet first overall), so for an exact answer use the threshold algorithm: gather the local top-K, compute a global lower bound, ask every shard for all users above bound / numShards, and merge. If an approximate answer is acceptable, merge count-min sketches (they add) and keep a heap of the K largest estimates.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if the log arrives as an unbounded stream?

Use the Misra-Gries or Space-Saving summary with m counters. Any user with more than n / (m + 1) messages is guaranteed to be in the summary, and a second pass (or a count-min sketch for the estimate) can confirm the real counts. Cost: O(1) per message with O(m) memory for m counters (Misra-Gries), instead of O(d) memory for the exact counts.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Which approach would you choose and why?

A heap is the safe default: simple, O(n + d log k) and streaming-friendly. Quickselect wins when k is a large fraction of d and you need all K but not their order. Sorting is fine only when d is small. Say that out loud, with the complexities, because the interviewer is grading the trade-off discussion.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | The log is a stream and memory is limited.

Misra-Gries keeps m counters; any user with more than n / (counter_limit + 1) messages is guaranteed to remain in the summary (checked above). A second pass, or a count-min sketch, then gives the real counts of the survivors.

FOLLOW-UP CODE

```
def misra_gries(stream, counter_limit):
    """Heavy hitters with m counters: any user with more than n / (m + 1) messages survives."""
    counters = {}
    for user in stream:
        if user in counters:
            counters[user] += 1
        elif len(counters) < counter_limit:
            counters[user] = 1
        else:
            for tracked_user in list(counters):
                counters[tracked_user] -= 1
                if counters[tracked_user] == 0:
                    del counters[tracked_user]
    return counters
```

BACK TO PART A - PART 1: QUESTION 181

182 * Rearrange Tasks With Cooldown * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Print an actual schedule, not just the time.

Use the heap-with-cooldown-queue construction: run the available task with the most copies left, park it for n slots, and emit an idle marker when nothing is available. Or rearrange a string so equal letters are at least k apart.

FOLLOW-UP CODE

```
def rearrange_k_distance(text, min_distance):
    """Return an arrangement with equal letters at least k apart, or '' if impossible (LeetCode 358)."""
    if min_distance <= 1:
```

```
        return text
    heap = [(-c, ch) for ch, c in Counter(text).items()]
    heapq.heapify(heap)
    waiting, out = deque(), []
    while heap:
        songs_left, letter = heapq.heappop(heap)
        out.append(letter)
        waiting.append((songs_left + 1, letter))
        if len(waiting) >= min_distance:
            returning_left, returning_letter = waiting.popleft()
            if returning_left:
                heapq.heappush(heap, (returning_left, returning_letter))
    return "".join(out) if len(out) == len(text) else ""
```

FOLLOW-UP 2 | Different cooldowns per task.

The closed form no longer applies. Simulate with a heap, storing for each task the time it becomes available; pick the ready task with the most remaining copies, and jump the clock to the earliest availability when nothing is ready.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Tasks have priorities or deadlines.

Then it is a scheduling problem; say what you optimise (lateness, completion time) and use a heap ordered by deadline for the ready tasks.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 182

183 * Implement frequency + distance top-K queries * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Q1 with points arriving over time.

Keep a running Counter and compute the top K on demand with a heap (O(d log how_many) per query), or maintain a sorted structure updated on every add if queries dominate.

FOLLOW-UP CODE

```
class TopPointsStream:
    """Follow-up: points arrive one at a time; top_k reads from the counts without re-counting the history."""
    def __init__(self):
        self.counts = Counter()

    def add(self, x_coord, y_coord):
        self.counts[(x_coord, y_coord)] += 1

    def top_k(self, how_many):
        return [list(point) for point in heapq.nsmallest(how_many, self.counts, key=lambda point: (-self.
counts[point], point[0] ** 2 + point[1] ** 2, point[0], point[1]))]
```

FOLLOW-UP 2 | Q2 with timestamps arriving out of order.

The append keeps the list sorted only for in-order input. For arbitrary order keep a sorted list and insert with bisect.insort (O(m) per insert) or a balanced tree (O(log m)); the query is unchanged.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Q2 returns the closest value at or before the timestamp (LeetCode 981).

Use bisect_right(timestamps, ts) - 1 and return that value or an empty string if the index is negative; there is no tie question.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 183

184 * Find top/bottom-k words in list or stream * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The stream is too big to count every distinct word.

Use the Space-Saving (or Misra-Gries) summary with m counters: a word that appears more than n / counter_limit times is guaranteed to be tracked, and its stored count is an upper bound. This works for the top K only; the least frequent words cannot be found this way.

FOLLOW-UP CODE

```
class SpaceSaving:
    """Approximate heavy hitters with m counters: any word with frequency above n / m is guaranteed to be
tracked."""
    def __init__(self, counter_limit):
        self.counter_limit, self.counts = counter_limit, {}

    def ingest(self, word):
        if word in self.counts:
            self.counts[word] += 1
        elif len(self.counts) < self.counter_limit:
            self.counts[word] = 1
        else:
            victim = min(self.counts, key=lambda word: (self.counts[word], word))
            self.counts[word] = self.counts.pop(victim) + 1

    def top(self, how_many):
        return heapq.nsmallest(how_many, self.counts, key=lambda word: (-self.counts[word], word))
```

FOLLOW-UP 2 | topK must be O(k), not O(d log k), because it is called all the time.

Keep words in buckets by count (a linked list of count buckets, like the LFU cache structure); the most frequent words are in the last bucket. An increment moves a word one bucket up in O(1).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Case and punctuation.

Normalise before counting (lowercase, strip punctuation) and say so; the counting and ranking code is unchanged.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 184

185 * Solve chat, grid paths, and car rentals * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Q1: what if the log is a stream and memory is limited?

Use a heavy-hitters sketch: Misra-Gries keeps k counters and is exact for any user with more than n / (k + 1) messages; a count-min sketch plus a small heap estimates the top K in O(1) memory per update. For exact answers across machines, count per shard, send each shard's local top-K plus a threshold, and merge.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Q2: what changes if there are obstacles or the path may also go straight up and down?

Obstacles just force ways[r] = 0 for blocked cells in the column being built. Allowing vertical moves inside a column makes each column a chain, so the transition becomes a linear recurrence solved by a prefix pass, still O(m * n). For huge n, use matrix exponentiation of the m x m transition matrix: O(m^3 log n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Q3: how do you answer 'is one more rental possible' online?

Keep the busy heap and the free-car count between calls. A new rental fits without a new car if any car's return time is <= its pickup, which is a heap peek after popping everything that returned before the pickup. That is O(log R) per query.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Q2 with obstacles.

A blocked cell contributes zero ways when its column is built; everything else is unchanged. O(rows * cols).

FOLLOW-UP CODE

```
def paths_with_obstacles(rows, cols, blocked):
    ways = [0] * rows
    if (rows - 1, 0) not in blocked:
        ways[rows - 1] = 1
    for col in range(1, cols):
        next_ways = [0] * rows
        for row in range(rows):
            if (row, col) in blocked:
                continue
            next_ways[row] = ways[row] + (ways[row - 1] if row else 0) + (ways[row + 1] if row + 1 < rows
else 0)
        ways = next_ways
    return ways[rows - 1]
```

BACK TO PART A - PART 1: QUESTION 185

186 * Top-K Chatters by Word Count

*** MEDIUM ***

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Wrap it in a ChatAnalyzer class that takes lines one at a time and answers top_k on demand.

Keep the Counter as state; add_line parses and updates it in O(words), top_k runs the selection. If top_k is called far more often than add_line, keep a sorted structure or a heap with lazy updates instead of recomputing.

FOLLOW-UP CODE

```
class ChatAnalyzer:
    def __init__(self):
        self.counts = Counter()

    def add_line(self, line):
        name, body = line.split(": ", 1)
        self.counts[name] += len(body.split())

    def top_k(self, how_many):
        return heapq.nsmallest(how_many, self.counts.items(), key=lambda entry: (-entry[1], entry[0]))
```

BACK TO PART A - PART 1: QUESTION 186

Digits and characters: peel one at a time and carry

188 * Solve several coding interview problems * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Ancestors in a tree with parent pointers, not a linked list.

The same two-pointer trick works on parent chains (LeetCode 1650); with depths known, lift the deeper node first and then move both up together.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Add two big numbers stored as linked lists.

If the lists store the least significant digit first, walk both together with the carry (LeetCode 2); if most significant first, reverse them or use stacks.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Count overlapping rectangles instead of intervals.

Sweep over x with a data structure over y (a segment tree with range add and max) to find the largest point depth in O(n log n); the pairwise count is O(n^2) unless you use a sweep with an interval tree.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Add two big numbers stored as linked lists (least significant digit first).

Walk both lists together with a carry and build the result list as you go; no reversal needed. O(max(m, n)) time. If the most significant digit comes first, reverse the lists or use stacks.

FOLLOW-UP CODE

```
class _L:
    def __init__(self, val, nxt=None):
        self.val, self.next = val, nxt

def add_reversed_lists(first_number, second_number):
    dummy = tail = _L(0)
    carry = 0
    while first_number or second_number or carry:
        total = carry + (first_number.val if first_number else 0) + (second_number.val if second_number else
0)
        tail.next = _L(total % 10)
        tail, carry = tail.next, total // 10
        first_number, second_number = (first_number.next if first_number else None), (second_number.next if
second_number else None)
    return dummy.next
```

BACK TO PART A - PART 1: QUESTION 188

189 * Check Whether Digits Are Monotone Increasing * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Strictly increasing digits.

Use >= instead of > when comparing with the previous digit, and start prev at 10 so a single digit passes.

FOLLOW-UP CODE

```
def is_strictly_monotone_digits(number):
    previous_digit = 10
    while number:
        digit = number % 10
        if digit >= previous_digit:
            return False
        previous_digit = digit
        number //= 10
    return True
```

FOLLOW-UP 2 | Negative numbers or numbers in another base.

Decide what monotone means for the sign (usually apply it to the absolute value). For base b replace 10 with b in % and //.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Find the next monotone number after n.

That is the left-to-right fix from the previous question: find the first descent and copy that digit across the rest.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 189

190 * Add two big integers from digit lists * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The digits are stored least significant first in a linked list.

Then the head is already the low end, so walk both lists together with the same carry loop and build the result list as you go. No reversal needed (LeetCode 2).

FOLLOW-UP CODE

```
class ListNode:
    def __init__(self, val=0, next=None):
        self.val, self.next = val, next

def add_two_numbers_reversed(first_list, second_list):
    """Digits stored least significant first in a linked list (LeetCode 2)."""
    dummy = tail = ListNode()
    carry = 0
    while first_list or second_list or carry:
        total = carry + (first_list.val if first_list else 0) + (second_list.val if second_list else 0)
        tail.next = ListNode(total % 10)
        tail, carry = tail.next, total // 10
        first_list, second_list = first_list.next if first_list else None, second_list.next if second_list
else None
    return dummy.next
```

FOLLOW-UP 2 | Multiply instead of add.

Schoolbook multiplication: the product of digits first_digits[first_index] and second_digits[second_index] lands in position first_index + second_index + 1, with the carry spilling to first_index + second_index. O(m * n); Karatsuba or FFT for very long numbers.

FOLLOW-UP CODE

```
def multiply_digits(first_digits, second_digits):
    out = [0] * (len(first_digits) + len(second_digits))
    for first_index in range(len(first_digits) - 1, -1, -1):
        for second_index in range(len(second_digits) - 1, -1, -1):
            out[first_index + second_index + 1] += first_digits[first_index] * second_digits[second_index]
            out[first_index + second_index] += out[first_index + second_index + 1] // 10
            out[first_index + second_index + 1] %= 10
    while len(out) > 1 and out[0] == 0:
        out.pop(0)
    return out
```

FOLLOW-UP 3 | Subtract, or numbers in another base.

Subtraction borrows instead of carrying and needs the larger number first (compare lengths, then digits). Another base replaces 10 with the base in % and //.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 190

Look at the input once and keep only a small summary (and count your calls)

193 * Streaming Points: Emit K Within Distance

* MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | What if 'within d' means every pair, and points are in 2D?

On a line pairwise distance equals the span, so nothing changes. In 2D it becomes a clique problem: candidates must be neighbours in a spatial grid of cell size d, and you test the K-subset of nearby points; that is much harder, so ask whether the 1D reading is intended.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Floating-point coordinates.

Compare with the exact subtraction hi - lo <= d (not by building a range around the target), and agree on an epsilon policy only if the interviewer wants one. Do not compare floats for equality.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Several groups qualify at once.

Agree on a rule (lowest start, as coded) so output is deterministic, and note that emitting one group can create or destroy others.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | The condition is 'every pair within d', not 'span at most d'.

On a line they are the same: the largest pairwise distance is max - min, so checking the span is enough (verified against checking all pairs). In 2D they differ and become a clique problem.

FOLLOW-UP CODE

```
def all_pairs_within(points, max_gap):
    return all(abs(first - second) <= max_gap for first, second in itertools.combinations(points, 2))
```

BACK TO PART A - PART 1: QUESTION 193

194 * Two-Day Log Intersection under Memory Limit

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Why prefer sort and two pointers to a streaming hash map?

A hash map needs memory proportional to the number of distinct objects (and clients), which is exactly what the limit forbids. An external sort needs only a chunk of memory and sequential disk passes, and the merge needs memory for one object's clients. If one object has an enormous client set, count distinct clients approximately (HyperLogLog) or stop as soon as two are seen.

FOLLOW-UP CODE

```
def log_objects_sorted(day_one_rows, day_two_rows):
    day_one_iter, day_two_iter = iter(sorted(day_one_rows, key=lambda row: row[1])), iter(sorted(day_two_rows,
 key=lambda row: row[1]))
    out = []
    def group(rows_iter, first_row):
        if first_row is None:
            return None, None
        object_id, clients, next_row = first_row[1], {first_row[2]}, None
        for row in rows_iter:
            if row[1] == object_id:
                clients.add(row[2])
            else:
                next_row = row
                break
        return (object_id, clients), next_row
    group_a, next_a = group(day_one_iter, next(day_one_iter, None))
    group_b, next_b = group(day_two_iter, next(day_two_iter, None))
```

```
    while group_a and group_b:
        if group_a[0] == group_b[0]:
            if len(group_a[1] | group_b[1]) >= 2:
                out.append(group_a[0])
            group_a, next_a = group(day_one_iter, next_a); group_b, next_b = group(day_two_iter, next_b)
        elif group_a[0] < group_b[0]:
            group_a, next_a = group(day_one_iter, next_a)
        else:
            group_b, next_b = group(day_two_iter, next_b)
    return out
```

BACK TO PART A - PART 1: QUESTION 194

195 * Find shared objects across two log files * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Both files are too big for memory.

Hash-partition both files into B buckets on disk by hash of the object id, then process bucket pairs one at a time with the in-memory set join. Choose B so that one bucket of the smaller file fits in memory. External sort and merge also works.

FOLLOW-UP CODE

```
def shared_objects_partitioned(path_a, path_b, buckets=4):
    """Neither file fits in memory: partition both by hash(id) to disk, then join bucket by bucket."""
    def partition(path, side):
        files = [open(f"{path}.{side}{i}", "w") for i in range(buckets)]
        try:
            with open(path) as file:
                for line in file:
                    if line.strip():
                        object_id = line.split()[1]
                        files[zlib.crc32(object_id.encode()) % buckets].write(object_id + "\n")
        finally:
            for bucket_file in files:
                bucket_file.close()

    partition(path_a, "a")
    partition(path_b, "b")
    out = set()
    for bucket in range(buckets):
        with open(f"{path_a}.a{bucket}") as file_a:
            bucket_ids = {line.strip() for line in file_a}
        with open(f"{path_b}.b{bucket}") as file_b:
            out |= {line.strip() for line in file_b if line.strip() in bucket_ids}
        os.remove(f"{path_a}.a{bucket}")
        os.remove(f"{path_b}.b{bucket}")
    return out
```

FOLLOW-UP 2 | Want objects seen on both days within an hour of each other.

Keep a list of timestamps per id in the set join and compare timestamps (sorted lists with two pointers), instead of just membership.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | A tiny memory budget and an approximate answer is fine.

Build a Bloom filter from one file and test the other against it, accepting a small false-positive rate; verify the few hits exactly in a second pass if needed. Cost: O(n + m) time with O(bits) memory for the Bloom filter and a small false-positive rate.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 195

196 * Minimize calls to find all bad test pairs * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | How many calls does it need, and can you prove a lower bound?

With p bad pairs among n tests about p log n calls are needed in the worst case (each pair carries log n bits of information about where it is), and the group method uses O(n * g + p log n). With no bad pairs it uses n - 1 calls, one per test after the first.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Only one bad pair exists.

Binary search for the smallest failing prefix (its last element is in the pair), then binary search among the earlier elements for the partner: about 2 log n calls.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | runTest is expensive but can run in parallel.

Test all groups for the new element in the same round, or process several elements at once with their own groups, trading a few extra calls for fewer sequential rounds.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Only one bad pair exists.

Binary search the shortest failing prefix (its last element is in the pair), then binary search the earlier tests for the partner: about 2 log2(n) calls instead of n(n - 1) / 2 (verified by counting calls).

FOLLOW-UP CODE

```
def find_single_bad_pair(tests, runTest):
    lo, hi = 1, len(tests)
    if runTest(set(tests)):
        return None
    while lo < hi:
        mid = (lo + hi) // 2
        if runTest(set(tests[:mid])):
            lo = mid + 1
        else:
            hi = mid
    last = tests[lo - 1]
    earlier = tests[:lo - 1]
    a, b = 1, len(earlier)
    while a < b:
        mid = (a + b) // 2
        if runTest(set(earlier[:mid]) | {last}):
            a = mid + 1
        else:
            b = mid
    return earlier[a - 1], last
```

BACK TO PART A - PART 1: QUESTION 196

Grid DP: each cell from its neighbours above and to the left

199 * Grid Path Count with Diagonal Moves and Checkpoints

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Two checkpoints share a column.

No path can visit both, because a path is in exactly one row per column, so the answer is 0 (the code returns it before the DP). Checkpoints at the start or end cell are fine: they only filter rows of the first or last column.

FOLLOW-UP CODE

```
def count_paths(rows, cols, checkpoints=(), ordered=False):
    checkpoint_cells = [tuple(col) for col in checkpoints]
    by_col = {}
    for row, col in checkpoint_cells:
        if by_col.setdefault(col, row) != row or len({checkpoint_col for _, checkpoint_col in
checkpoint_cells}) != len(checkpoint_cells):
            return 0
    if ordered and [col for _, col in checkpoint_cells] != sorted(col for _, col in checkpoint_cells):
        return 0
    ways = [0] * rows
    ways[rows - 1] = 1
    def keep(column_ways, col):
        if col in by_col:
            column_ways = [ways_value if row == by_col[col] else 0 for row, ways_value in enumerate(
column_ways)]
        return column_ways
    ways = keep(ways, 0)
    for col in range(1, cols):
        ways = keep([(ways[row - 1] if row else 0) + ways[row] + (ways[row + 1] if row + 1 < rows else 0) for
row in range(rows)], col)
    return ways[rows - 1]
```

BACK TO PART A - PART 1: QUESTION 199

200 * Longest Same-Color Grid Sequence

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | The line may also bend (a connected region).

Then it is a different problem: the largest connected same-colour region, found with BFS/DFS or union-find in O(m * width). The run-length DP only handles straight lines because each run depends on exactly one predecessor.

FOLLOW-UP CODE

```
def longest_same_color_line(colors):
    if not colors or not colors[0]:
        return 0
    width, best = len(colors[0]), 0
    up_run, down_left_run, down_right_run = [0] * width, [0] * width, [0] * width          # vertical, down-
left diagonal, down-right diagonal runs of the previous row
    for row_index, row in enumerate(colors):
        new_up_run, new_down_left_run, new_down_right_run = [1] * width, [1] * width, [1] * width
        left_run = 0
        for col_index, color in enumerate(row):
            left_run = left_run + 1 if col_index and row[col_index - 1] == color else 1
            if row_index:
                if colors[row_index - 1][col_index] == color:
                    new_up_run[col_index] = up_run[col_index] + 1
                if col_index and colors[row_index - 1][col_index - 1] == color:
                    new_down_right_run[col_index] = down_right_run[col_index - 1] + 1
                if col_index + 1 < width and colors[row_index - 1][col_index + 1] == color:
                    new_down_left_run[col_index] = down_left_run[col_index + 1] + 1
            best = max(best, left_run, new_up_run[col_index], new_down_left_run[col_index],
new_down_right_run[col_index])
        up_run, down_left_run, down_right_run = new_up_run, new_down_left_run, new_down_right_run
    return best
```

BACK TO PART A - PART 1: QUESTION 200

201 * Count Same-Color Squares in Matrix * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Return only the largest side instead of the count.

Keep the same dp and take the maximum instead of adding. The common bug is checking only the diagonal neighbour: the up and left neighbours must also match, otherwise a different colour can sit inside the square.

FOLLOW-UP CODE

```
def count_same_color_squares(colors):
    if not colors:
        return 0
    width, total = len(colors[0]), 0
    previous_row = [0] * width
    for row_index, row in enumerate(colors):
        current_row = [0] * width
        for col_index, color in enumerate(row):
            if row_index and col_index and colors[row_index - 1][col_index] == color and colors[row_index][
col_index - 1] == color and colors[row_index - 1][col_index - 1] == color:
                current_row[col_index] = 1 + min(previous_row[col_index], current_row[col_index - 1],
previous_row[col_index - 1])
            else:
                current_row[col_index] = 1
            total += current_row[col_index]
        previous_row = current_row
    return total
```

BACK TO PART A - PART 1: QUESTION 201

Nested structures: a stack of contexts, or recursion with a cache

204 * Decode String * MEDIUM *

DIFFICULTY | * MEDIUM * | 5 FOLLOW-UPS

FOLLOW-UP 1 | How would you do it recursively, and what is the risk?

Parse with an index: read letters and digits, on '[' recurse to parse the inner string and return where it ended, on ']' return. It reads naturally but the call depth equals the nesting depth, which can overflow the stack for adversarial inputs. The stack version trades that for explicit memory.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if the decoded string is enormous?

Do not materialize it. Return a generator, or compute the character at a given index by walking the structure with lengths (as in LeetCode 880, Decoded String at Index), using index modulo the segment length at each level.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The decoded string would be enormous.

Compute its length (or a given character) from the stack without building it: keep (length so far, repeat count) frames and combine as length_before + inner * times. The length can be computed for strings that could never fit in memory.

FOLLOW-UP CODE

```
def decoded_length(encoded):
    """Length of the decoded string without building it."""
    stack, length, pending_count = [], 0, 0
    for char in encoded:
        if char.isdigit():
            pending_count = pending_count * 10 + int(char)
        elif char == "[":
            stack.append((length, pending_count))
            length, pending_count = 0, 0
        elif char == "]":
            length_before, repeat_count = stack.pop()
            length = length_before + length * repeat_count
        else:
            length += 1
    return length
```

FOLLOW-UP 4 | Return the character at position k only.

Walk the structure with lengths: at each repeat reduce k modulo the inner length to find where in the inner string it falls (LeetCode 880 style).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 5 | Invalid input (missing bracket, count with no bracket).

Detect an unmatched ']' (empty stack), a '[' left at the end, and digits not followed by '['. Decide whether to raise or to treat a malformed part as literal text.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 204

205 * Template Variable Expansion with Follow-Ups

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Detect cycles before expanding.

Model each key as a vertex with an edge to every key its value mentions, count indegrees and run Kahn's algorithm: if fewer vertices than keys come out, some keys sit on a cycle and cannot be resolved. This reports the problem up front instead of failing halfway through a long document.

FOLLOW-UP CODE

```
def has_cycle(mapping):
    """Kahn's algorithm over the key dependency graph: True if some keys can never be resolved."""
    dependencies = {key: set(re.findall(r"(?<!%)%([A-Za-z0-9_]+)%", value.replace("%%", ""))) for key, value
in mapping.items()}
    users, indegree = defaultdict(list), {key: len(needs & mapping.keys()) for key, needs in dependencies.
items()}
    for key, needs in dependencies.items():
        for needed_key in needs & mapping.keys():
            users[needed_key].append(key)
    queue, resolved_count = deque(key for key, value in indegree.items() if value == 0), 0
    while queue:
        key = queue.popleft()
        resolved_count += 1
        for dependent in users[key]:
            indegree[dependent] -= 1
            if indegree[dependent] == 0:
                queue.append(dependent)
    return resolved_count != len(mapping)
```

BACK TO PART A - PART 1: QUESTION 205

206 * Expand Recursive Key-Value Placeholders in a Template

* MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | A literal percent sign must be possible.

Choose an escape such as %% and handle it in the same scan: when the text has %%, output one % and skip both characters; otherwise try to match a placeholder. Test percent signs next to placeholders.

FOLLOW-UP CODE

```
def expand_with_escapes(template, mapping):
    """'%%' means a literal percent sign; everything else as above."""
    cache, visiting = {}, set()

    def expand_text(text):
        pieces, position = [], 0
        while position < len(text):
            if text.startswith("%%", position):
                pieces.append("%")
                position += 2
                continue
            placeholder = _PH.match(text, position)
            if placeholder and placeholder.group(1) in mapping:
                key = placeholder.group(1)
                if key in visiting:
                    raise ValueError(f"cycle at {key}")
                if key not in cache:
                    visiting.add(key)
                    cache[key] = expand_text(mapping[key])
                    visiting.discard(key)
                pieces.append(cache[key])
                position = placeholder.end()
            else:
                pieces.append(text[position])
                position += 1
        return "".join(pieces)

    return expand_text(template)
```

FOLLOW-UP 2 | An unknown key should be an error, or replaced by an empty string.

Make the policy a parameter: raise KeyError, return the placeholder unchanged (as coded), or return ''. State your default.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Values are huge and expansion explodes.

The output can be exponentially larger than the input (a key referring to the previous key twice, repeatedly). Check the expanded length as you go and fail past a limit, or return a lazy expansion.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 206

207 * Nested List Weight Sum (LC 339) with Follow-Ups * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | When does the recursive (DFS) version become unsafe?

Python's default recursion limit is about 1000 frames, so an input nested 10,000 levels deep raises RecursionError. DFS also uses O(depth) call-stack space. The single-pass scanner uses O(1) and BFS uses O(widest level); both avoid the problem. In Java the same limit appears as StackOverflowError.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What is the inverse version (deeper integers weigh less)?

That is LeetCode 364. Do a level-by-level pass that keeps a running sum of everything seen so far and add it to the total once per level: total += running_sum on each level. Shallow integers are added many times, deep ones once, which equals weight (maxDepth + 1 - depth).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The inverse version (shallow integers weigh less, LeetCode 364).

Process level by level and keep a running sum of every integer seen so far; add the running sum to the total once per level that holds an integer or lies above one (trim trailing levels that contain only empty lists). Shallow numbers are added at every deeper level, which equals the weight maxDepth - depth + 1.

FOLLOW-UP CODE

```
def depth_sum_inverse(nested):
    """LeetCode 364: deeper integers weigh less. weight = maxDepth - depth + 1, where maxDepth is the deepest
level holding an integer."""
    sums_per_level, integers_per_level, level = [], [], list(nested)
    while level:
        level_sum = level_count = 0
        next_level = []
        for item in level:
            if isinstance(item, list):
                next_level.extend(item)
            else:
                level_sum, level_count = level_sum + item, level_count + 1
        sums_per_level.append(level_sum)
        integers_per_level.append(level_count)
        level = next_level
    while integers_per_level and integers_per_level[-1] == 0:               # levels made only of empty lists
do not count toward the maximum depth
        integers_per_level.pop()
        sums_per_level.pop()
    total = sum_so_far = 0
    for level_sum in sums_per_level:
        sum_so_far += level_sum                                  # integers on shallower levels are added again
at every deeper level
        total += sum_so_far
    return total
```

BACK TO PART A - PART 1: QUESTION 207

208 * Nested Function Expression Evaluator * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | Support variadic functions and variables.

Collect arguments until ')' instead of assuming two (as the code does). For variables, pass an environment dict into parse and treat an identifier that is not followed by '(' as a lookup. Define the error behaviour for undefined names.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | How do you make it iterative?

Use two stacks: one of pending function names and one of lists of evaluated arguments. On an identifier push the name and a new list, on a number append to the top list, on ')' pop both, apply the function, and append the result to the new top list. This removes the recursion-depth limit but is easier to get wrong off by one.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | What about division?

State the contract: integer or float, truncate toward zero or floor (Python's // floors, which differs for negatives), and divide by zero handling. The implementation above truncates toward zero.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Add variables (an identifier with no parentheses is a lookup).

In parse, check whether the identifier is followed by '('; if not, return env[name]. Decide what an undefined name does (raise, or default to 0).

FOLLOW-UP CODE

```
def evaluate_env(expr, env):
    """Variables: an identifier that is not followed by '(' is looked up in env."""
    tokens = re.findall(r"-?\d+|[A-Za-z_]\w*|[(),]", expr)

    def parse(position):
        token = tokens[position]
        if re.fullmatch(r"-?\d+", token):
            return int(token), position + 1
        if position + 1 < len(tokens) and tokens[position + 1] == "(":
            position += 2
            args = []
            while tokens[position] != ")":
                value, position = parse(position)
                args.append(value)
                if tokens[position] == ",":
                    position += 1
            return _OPS[token](*args), position + 1
        return env[token], position + 1

    return parse(0)[0]
```

BACK TO PART A - PART 1: QUESTION 208

Frontier search (BFS or DFS), Tier 2: free steps, early exit, neighbours worked out on the fly

209 * Shortest Path Nodes and Escaping a Pursuer * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | What if Bob can also meet Alice in the middle of an edge?

Then arriving at the same node at the same time is not the only way to be caught: Alice moving node -> neighbour while Bob moves neighbour -> node swaps places. Add that case by also rejecting an edge (node, neighbour) when dB[neighbour] == dA[node] and dB[node] == dA[neighbour]. Always clarify what 'caught' means first.

FOLLOW-UP CODE

```
def shortest_path_info(neighbours, alice, target, bob):
    def bfs(start):
        distance = {start: 0}
        queue = deque([start])
        while queue:
            node = queue.popleft()
            for neighbour in neighbours[node]:
                if neighbour not in distance:
                    distance[neighbour] = distance[node] + 1
                    queue.append(neighbour)
        return distance
    da, dt, db = bfs(alice), bfs(target), bfs(bob)
    if target not in da:
        return None, set(), False
    length = da[target]
    nodes = {node for node in da if node in dt and da[node] + dt[node] == length}
    safe_nodes, frontier = set(), [alice] if db.get(alice, 10 ** 9) > 0 else []
    seen = set(frontier)
    while frontier:
        next_frontier = []
        for node in frontier:
            for neighbour in neighbours[node]:
                if neighbour in nodes and da[neighbour] == da[node] + 1 and neighbour not in seen and db.get(
neighbour, 10 ** 9) > da[neighbour]:
                    seen.add(neighbour); next_frontier.append(neighbour)
        frontier = next_frontier
    return length, nodes, target in seen
```

BACK TO PART A - PART 1: QUESTION 209

210 * Radio Broadcast Reachability * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Thousands of source/target queries on the same radios.

Build the connected components once with union-find (O(radio_count^2) pair checks), then every query is two finds. Radios in one component can reach each other.

FOLLOW-UP CODE

```
class RadioNetwork:
    def __init__(self, radios, radius):
        radio_count = len(radios)
        self.parent = list(range(radio_count))
        for first in range(radio_count):
            for second in range(first + 1, radio_count):
                x_gap = radios[first][0] - radios[second][0]
                y_gap = radios[first][1] - radios[second][1]
                if x_gap * x_gap + y_gap * y_gap <= radius * radius:
                    self.parent[self.find(first)] = self.find(second)

    def find(self, radio):
        while self.parent[radio] != radio:
```

```
            self.parent[radio] = self.parent[self.parent[radio]]
            radio = self.parent[radio]
        return radio

    def can_reach(self, source, target):
        return self.find(source) == self.find(target)
```

BACK TO PART A - PART 1: QUESTION 210

211 * Minimum Direction Violations in a Directed Graph * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Why is 0-1 BFS correct, and how does it compare with Dijkstra?

The deque always holds nodes whose distances differ by at most one: front elements are at distance d, back elements at d + 1. Relaxing a zero-weight edge keeps the distance at d (front), a one-weight edge makes d + 1 (back). That is the same order a heap would produce, so it is Dijkstra without the log factor: O(n + m) versus O((n + m) log n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if edge costs can be arbitrary non-negative numbers?

Use Dijkstra with a heap on the same doubled graph (forward cost = 0 or the edge cost, backward cost = the violation cost). With negative costs use Bellman-Ford and watch for negative cycles.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Edge costs are arbitrary non-negative numbers (different for each direction).

Use Dijkstra with a heap on the same doubled graph, giving each arc its own cost. 0-1 BFS only works because the costs are 0 or 1.

FOLLOW-UP CODE

```
def min_cost_path(n, edges, start, end):
    """Edges carry costs: edges[i] = (u, v, forward_cost, backward_cost). Dijkstra with a heap."""
    import heapq
    graph = defaultdict(list)
    for node, neighbour, forward_cost, backward_cost in edges:
        graph[node].append((neighbour, forward_cost))
        graph[neighbour].append((node, backward_cost))
    dist = {start: 0}
    heap = [(0, start)]
    while heap:
        cost_so_far, node = heapq.heappop(heap)
        if cost_so_far > dist.get(node, float("inf")):
            continue
        for neighbour, cost in graph[node]:
            if cost_so_far + cost < dist.get(neighbour, float("inf")):
                dist[neighbour] = cost_so_far + cost
                heapq.heappush(heap, (cost_so_far + cost, neighbour))
    return dist.get(end, -1)
```

BACK TO PART A - PART 1: QUESTION 211

212 * Compute shortest paths with blocked nodes * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The edges have weights.

BFS is no longer correct: use Dijkstra with a heap and skip blocked nodes the same way. O((V + E) log V). Directed and undirected differ only in whether you add the reverse edge.

FOLLOW-UP CODE

```
def blocked_weighted(node_count, edges, source, blocked, directed=False):
    blocked_set = set(blocked)
    if source in blocked_set:
        return [-1] * node_count
    adj = defaultdict(list)
    for node, neighbour, weight in edges:
```

```
        adj[node].append((neighbour, weight))
        if not directed:
            adj[neighbour].append((node, weight))
    distance = {source: 0}
    heap = [(0, source)]
    while heap:
        distance_so_far, node = heapq.heappop(heap)
        if distance_so_far > distance[node]:
            continue
        for neighbour, weight in adj[node]:
            if neighbour not in blocked_set and distance_so_far + weight < distance.get(neighbour, float(
"inf")):
                distance[neighbour] = distance_so_far + weight
                heapq.heappush(heap, (distance_so_far + weight, neighbour))
    return [distance.get(node, -1) for node in range(node_count)]
```

FOLLOW-UP 2 | Directed versus undirected.

For an undirected graph add each edge in both directions; for a directed one only as given. A node may be reachable in one case and not in the other, so state which you assume.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Nodes get blocked over time and you must keep distances up to date.

Decremental shortest paths is hard in general; for few updates rerun BFS, and for many offline queries process the changes in reverse order (unblocking only improves distances).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 212

213 * Shortest Directed Cycle Through a Node * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | How do you find the shortest cycle in the whole graph?

Run the same BFS from every vertex and take the minimum: O(V * (V + E)). For undirected graphs, BFS from each node and treat any non-tree edge as closing a cycle of length dist[u] + dist[w] + 1 (the LeetCode 2608 version).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if edges have positive weights?

Replace BFS with Dijkstra from v and close the cycle at the cheapest dist[u] + weight(u, v). That is O((V + E) log V) with a heap. With negative weights use Bellman-Ford and handle negative cycles separately.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The shortest cycle anywhere in the graph (the girth).

Run the same BFS from every node and take the smallest result: O(V * (V + E)). That is the standard way and fine for graphs up to a few thousand nodes.

FOLLOW-UP CODE

```
def shortest_cycle_through(node_count, edges, start_node):
    neighbours = [[] for _ in range(node_count)]
    for from_node, to_node in edges:
        neighbours[from_node].append(to_node)
    dist = {start_node: 0}
    queue = deque([start_node])
    while queue:
        node = queue.popleft()
        for next_node in neighbours[node]:
            if next_node == start_node:
                return dist[node] + 1
            if next_node not in dist:
                dist[next_node] = dist[node] + 1
                queue.append(next_node)
    return -1

def girth_directed(node_count, edges):
    return min((cycle_length for cycle_length in (shortest_cycle_through(node_count, edges, start_node) for
start_node in range(node_count)) if cycle_length != -1), default=-1)
```

BACK TO PART A - PART 1: QUESTION 213

Topological sort (Kahn) and cycle detection

214 * Course Scheduling and Dependency Variants

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | How do you check that an order is valid?

Record each vertex's position in the order, check that every vertex appears exactly once, then check that for every edge (prerequisite, dependent) the position of u is smaller than the position of v. Accept any valid order, not one fixed answer, because independent vertices can swap.

FOLLOW-UP CODE

```
def topological_order(vertices, edges):
    dependents, indegree = defaultdict(list), {vertex: 0 for vertex in vertices}
    for prerequisite, dependent in edges:
        dependents[prerequisite].append(dependent)
        indegree[dependent] = indegree.get(dependent, 0) + 1
        indegree.setdefault(prerequisite, indegree.get(prerequisite, 0))
    queue = deque(vertex for vertex, incoming in indegree.items() if incoming == 0)
    order = []
    while queue:
        prerequisite = queue.popleft()
        order.append(prerequisite)
        for dependent in dependents[prerequisite]:
            indegree[dependent] -= 1
            if indegree[dependent] == 0:
                queue.append(dependent)
    return order if len(order) == len(indegree) else None

def can_merge_orders(first_order, second_order):
    edges = [(order[position], order[position + 1]) for order in (first_order, second_order) for position in
range(len(order) - 1)]
    return topological_order(set(first_order) | set(second_order), edges) is not None
```

BACK TO PART A - PART 1: QUESTION 214

215 * Validate Race Results (Cycle Detection)

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Recursive DFS hits Python's recursion limit on long chains. What do you do?

Either use Kahn's algorithm (already iterative) or run the three-colour DFS with an explicit stack of (node, next neighbour index) so the depth is limited by memory, not by the interpreter.

FOLLOW-UP CODE

```
def results_are_valid_dfs(results):
    graph = {}
    for winner, loser in results:
        graph.setdefault(winner, []).append(loser)
        graph.setdefault(loser, [])
    state = {}
    for start in graph:
        if start in state:
            continue
        state[start] = 1
        stack = [(start, iter(graph[start]))]
        while stack:
            racer, remaining_losers = stack[-1]
            for slower_racer in remaining_losers:
                if state.get(slower_racer) == 1:
                    return False
                if slower_racer not in state:
                    state[slower_racer] = 1
                    stack.append((slower_racer, iter(graph[slower_racer])))
                    break
            else:
```

```
                state[racer] = 2
                stack.pop()
    return True
```

BACK TO PART A - PART 1: QUESTION 215

216 * Solve tree, graph, sliding-window problems * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Course Schedule II: return an order in which to take the courses.

The same Kahn loop records the order in which courses are removed; if fewer than n courses are removed, return an empty list.

FOLLOW-UP CODE

```
def find_order(course_count, prereqs):
    dependents, indegree = defaultdict(list), [0] * course_count
    for course, prerequisite in prereqs:
        dependents[prerequisite].append(course)
        indegree[course] += 1
    queue, order = deque(course for course in range(course_count) if indegree[course] == 0), []
    while queue:
        ready_course = queue.popleft()
        order.append(ready_course)
        for dependent in dependents[ready_course]:
            indegree[dependent] -= 1
            if indegree[dependent] == 0:
                queue.append(dependent)
    return order if len(order) == course_count else []
```

FOLLOW-UP 2 | Longest substring with exactly K distinct characters.

Take atMost(K) minus atMost(K - 1) for counting substrings, or keep the window and only record lengths when the distinct count is exactly K (careful with the shrink).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Zigzag on an n-ary tree, or the right-to-left variant.

Replace the two children by the child list, and swap which levels are appended on the left.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 216

217 * Validate course catalog dependencies * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return all missing course references and one cycle.

The function above returns the sorted unknown ids when any exist and otherwise a cycle (a list of courses starting and ending at the same course), found by walking backwards along prerequisites inside the set of courses Kahn's algorithm could not remove (every such course has a leftover prerequisite, so the walk cannot dead-end).

FOLLOW-UP CODE

```
def validate_catalog(courses, prereqs):
    course_set = set(courses)
    missing = sorted({course_id for prerequisite, course in prereqs for course_id in (prerequisite, course)
if course_id not in course_set}, key=str)
    if missing:
        return False, missing, None
    dependents, indegree = defaultdict(list), {course: 0 for course in course_set}
    for prerequisite, course in prereqs:
        dependents[prerequisite].append(course)
        indegree[course] += 1
    queue = deque(sorted((course for course in course_set if indegree[course] == 0), key=str))
    order = []
    while queue:
        ready_course = queue.popleft()
        order.append(ready_course)
        for dependent in dependents[ready_course]:
```

```
            indegree[dependent] -= 1
            if indegree[dependent] == 0:
                queue.append(dependent)
    if len(order) == len(course_set):
        return True, [], order
    left = {course for course in course_set if indegree[course] > 0}
    prerequisites_of = defaultdict(list)
    for prerequisite, course in prereqs:
        if prerequisite in left and course in left:
            prerequisites_of[course].append(prerequisite)
    start = min(left, key=str)
    walk, walk_position, course_id = [], {}, start
    while course_id not in walk_position:                                    # every leftover course has a
leftover prerequisite, so this walk never stops early
        walk_position[course_id] = len(walk)
        walk.append(course_id)
        course_id = prerequisites_of[course_id][0]
    cycle = walk[walk_position[course_id]:] + [course_id]
    return False, [], cycle[::-1]                       # walked against the edges, so reverse to list
prerequisites first
```

FOLLOW-UP 2 | Group courses into semesters (the earliest round each can be taken).

Run Kahn's algorithm level by level: all courses with no remaining prerequisites form round 1, remove them, the next set forms round 2, and so on. The number of rounds is the longest dependency chain.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Find all cycles, not just one.

Compute strongly connected components (Tarjan or Kosaraju); every component with more than one course (or a self-loop) is a cycle group to report.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 217

218 * Course Schedule * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the order of courses (LeetCode 210).

The same Kahn loop records the order; if fewer than n courses are taken return an empty list. O(V + E).

FOLLOW-UP CODE

```
def find_order(course_count, prereqs):
    dependents, indegree = defaultdict(list), [0] * course_count
    for course, prerequisite in prereqs:
        dependents[prerequisite].append(course)
        indegree[course] += 1
    queue, order = deque(course for course in range(course_count) if indegree[course] == 0), []
    while queue:
        ready_course = queue.popleft()
        order.append(ready_course)
        for dependent in dependents[ready_course]:
            indegree[dependent] -= 1
            if indegree[dependent] == 0:
                queue.append(dependent)
    return order if len(order) == course_count else []
```

FOLLOW-UP 2 | The minimum number of semesters if you can take any number of courses per semester.

Run Kahn's algorithm level by level: each level is one semester. The number of levels is the longest chain of prerequisites (return -1 if there is a cycle).

FOLLOW-UP CODE

```
def semesters_needed(course_count, prereqs):
    dependents, indegree = defaultdict(list), [0] * course_count
    for course, prerequisite in prereqs:
        dependents[prerequisite].append(course)
        indegree[course] += 1
    level, taken, rounds = [c for c in range(course_count) if indegree[c] == 0], 0, 0
    while level:
        rounds += 1
```

```
        taken += len(level)
        next_level = []
        for ready_course in level:
            for dependent in dependents[ready_course]:
                indegree[dependent] -= 1
                if indegree[dependent] == 0:
                    next_level.append(dependent)
        level = next_level
    return rounds if taken == course_count else -1
```

FOLLOW-UP 3 | Report an actual cycle.

After Kahn's algorithm, the courses left over all still have an unmet prerequisite among themselves; walk backwards along prerequisites inside that set until a course repeats to extract a cycle.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 218

Counting pairs by looking up each partner in a Counter or a bucket

219 * Progressive Array Partitioning * MEDIUM *

DIFFICULTY | * MEDIUM * | 2 FOLLOW-UPS

FOLLOW-UP 1 | Stage 2 arrives. What do you do?

Do not guess. Write down what the second example pair implies, and check any candidate rule against BOTH. The common reading 'split into consecutive runs of length at least 3' gives true for [4, 5, 6, 7, 7, 8, 9], but the statement says false, so there is an extra condition (such as run lengths or a count rule) to ask about.

FOLLOW-UP CODE

```
def can_split_into_straights(nums, min_len=3):
    """One candidate reading of Stage 2 (LeetCode 659): split a sorted array into runs of consecutive values,
each at least min_len long."""
    remaining, runs_ending_at = Counter(nums), Counter()
    for value in nums:
        if remaining[value] == 0:
            continue
        remaining[value] -= 1
        if runs_ending_at[value - 1] > 0:
            runs_ending_at[value - 1] -= 1
            runs_ending_at[value] += 1
        elif remaining[value + 1] > 0 and remaining[value + 2] > 0:
            remaining[value + 1] -= 1
            remaining[value + 2] -= 1
            runs_ending_at[value + 2] += 1
        else:
            return False
    return True
```

FOLLOW-UP 2 | Keep the code adaptable between stages.

Put the counting in one helper and each stage's rule in its own small function that takes the counts. Stage 2 then replaces a function instead of rewriting the pipeline.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 219

220 * K-diff Pairs in an Array * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Do it without a hash map.

Sort, then use two pointers: advance the right pointer when the gap is too small and the left when too large; when the gap equals k count it and skip duplicate values on both sides. O(n log n) time, O(1) extra space.

FOLLOW-UP CODE

```
def count_k_diff_sorted(nums, difference):
    nums = sorted(nums)
    low_index, high_index, pairs = 0, 1, 0
    while high_index < len(nums) and low_index < len(nums):
        if low_index == high_index or nums[high_index] - nums[low_index] < difference:
            high_index += 1
        elif nums[high_index] - nums[low_index] > difference:
            low_index += 1
        else:
            pairs += 1
            low_index += 1
            while low_index < len(nums) and nums[low_index] == nums[low_index - 1]:
                low_index += 1
    return pairs
```

BACK TO PART A - PART 1: QUESTION 220

221 * Count Axis-Aligned Squares Among Grid Points

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Points may repeat. Count every distinct choice of four points.

Keep a Counter. For each unordered pair of distinct diagonal points multiply the four counts; halve the total because each square has two diagonals.

FOLLOW-UP CODE

```
from collections import Counter


def count_squares_with_duplicates(points):
    copies = Counter(points)
    keys = list(copies)
    total = 0
    for i, (x1, y1) in enumerate(keys):
        for x2, y2 in keys[i + 1:]:
            if abs(x1 - x2) == abs(y1 - y2) != 0:
                total += (copies[(x1, y1)] * copies[(x2, y2)]
                          * copies.get((x1, y2), 0) * copies.get((x2, y1), 0))
    return total // 2
```

BACK TO PART A - PART 1: QUESTION 221

222 * Sum of Two Squares in Two Ways

* MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Exactly k ways instead of two.

Change the comparison from == 2 to == ways_wanted; the counting loop is the same. For ways_wanted = 3 the smallest is 325.

FOLLOW-UP CODE

```
def two_squares_exactly_k(limit, ways_wanted):
    sum_counts = Counter()
    a = 1
    while 2 * a * a < limit:
        b = a
        while a * a + b * b < limit:
            sum_counts[a * a + b * b] += 1
            b += 1
        a += 1
    return sorted(square_sum for square_sum, ways_count in sum_counts.items() if ways_count == ways_wanted)
```

FOLLOW-UP 2 | Allow zero (0^2 + b^2).

Start a at 0; then perfect squares like 25 gain a representation, so some answers appear or disappear. Check against the examples to see which convention the question uses (here, positive only).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | n is huge (10^12) and you only need to test one number.

Factorise the number: the count of representations as a sum of two squares depends only on the primes congruent to 1 mod 4 (Jacobi's formula), so a factorisation answers it in sublinear time.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 222

Tree DP: build a parent order once, then add up from the leaves

224 * Compute File System Total Size * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The tree is queried often and files change.

Cache each directory's total and, when a file changes, clear the cache only for its ancestors (walk parent links). A query recomputes only cleared nodes. Updates cost O(depth).

FOLLOW-UP CODE

```
class File:
    def __init__(self, size):
        self.size = size

class Dir:
    def __init__(self, *children):
        self.children = list(children)

class CachedFS:
    """Directory totals cached; changing a file invalidates only its ancestors."""
    def __init__(self, root):
        self.parent, self.cache, stack = {id(root): None}, {}, [root]
        while stack:
            node = stack.pop()
            for child in getattr(node, "children", []):
                self.parent[id(child)] = node
                stack.append(child)
        self.root = root

    def total(self, node=None):
        node = node or self.root
        if id(node) in self.cache:
            return self.cache[id(node)]
        result = node.size if isinstance(node, File) else sum(self.total(child) for child in node.children)
        self.cache[id(node)] = result
        return result

    def resize(self, file, size):
        file.size = size
        current = file
        while current is not None:
            self.cache.pop(id(current), None)
            current = self.parent[id(current)]
```

FOLLOW-UP 2 | Hard links: the same file appears under two directories.

Count a file once per identity (a visited set of ids, as coded) if the question asks for disk usage, but once per path if it asks for the sum of the sizes shown. Ask which.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Find the largest directory.

Compute totals in post-order and keep the maximum; it is the same pass. For top-K directories keep a heap of size K. Cost: one post-order pass, O(n); keeping the top K directories with a heap of size K makes it O(n log K).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 224

225 * Minimum-Cost Tree Disconnection * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Weights are non-negative.

Then free(c) is always 0 and the recurrence collapses to best(v) = sum over children of (weight if the child is a leaf, else min(weight, best(child))).

FOLLOW-UP CODE

```
def disconnect_cost_nonneg(node_count, edges):
    kids = defaultdict(list)
    for parent, child, weight in edges:
        kids[parent].append((child, weight))

    def solve(node):
        return sum(weight if not kids[child] else min(weight, solve(child)) for child, weight in kids[node])

    return solve(0)
```

FOLLOW-UP 2 | The tree is a long chain and recursion overflows.

Compute nodes in reverse BFS order (children before parents) and store best and free in arrays, no recursion needed. State the space as O(n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Return which edges to cut.

Remember at each child which option was cheaper, then walk down from the root collecting the cut edges. Cut edges also include every negative edge below any cut edge.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 225

226 * Given a large n-ary tree where the root node can pass information to one child per iteration, and in subsequent iterations, nodes can pass information to one child while the root continues with others, find the minimum number of iterations to reach all nodes. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Why is it optimal to serve the child with the largest subtree time first?

Exchange argument: if a child with time a is served in round i and one with time b >= a in round i + 1, swapping them changes the finish times from max(i + a, i + 1 + b) to max(i + b, i + 1 + a), and the second is never larger because b >= a. So sorting by decreasing T minimizes the maximum. Cost: sorting the children at each node makes the whole computation O(n log n); the exchange argument adds nothing.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if each edge has a different transmission cost?

Replace the unit step by the cost: the child served in order position i can only start after the sum of the costs of the edges before it, so T(v) = max over i of (prefix cost up to child_i + T(child_i)). The best order is still decreasing (cost + T) only when costs are equal; in general it becomes a scheduling problem solved by sorting on T(child) - cost, i.e. Smith's rule.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Each call takes a different time per child.

The order still matters, and the right rule is Jackson's: serve children in decreasing order of their own finishing time T, whatever each call costs, because the finish of the child served at prefix cost P is P + T and the largest tail should start earliest. Checked against trying every order on random trees.

FOLLOW-UP CODE

```
def inform_rounds_weighted(node_count, children, cost, root=0):
    """Calling child c takes cost[c] rounds. Serve children in decreasing order of their own finishing time T
(Jackson's rule)."""
```

```
    def t(node):
        cost_and_rounds = sorted(((cost[child], t(child)) for child in children.get(node, [])), key=lambda
cost_and_time: -cost_and_time[1])
        clock = finish = 0
        for child, child_rounds in cost_and_rounds:
            clock += child
            finish = max(finish, clock + child_rounds)
        return finish

    return t(root)
```

BACK TO PART A - PART 1: QUESTION 226

Counting or reaching an amount: the knapsack loop

229 * Maximize Shelf Space (Unbounded Knapsack) * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Each box size is only available once.

That is a 0/1 knapsack (subset sum). Process the boxes one at a time and iterate the capacity downward so a box is not used twice: reachable[total] |= reachable[total - box_size].

FOLLOW-UP CODE

```
def max_shelf_space_once(capacity, sizes):
    reachable = [True] + [False] * capacity
    for box_size in sizes:
        for total in range(capacity, box_size - 1, -1):
            reachable[total] = reachable[total] or reachable[total - box_size]
    return max(total for total in range(capacity + 1) if reachable[total])
```

BACK TO PART A - PART 1: QUESTION 229

230 * Coin Change * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Count the number of combinations that make the amount (LeetCode 518).

Loop over coins in the outer loop and amounts in the inner loop, adding ways[sub_amount - coin]; the order of the loops makes combinations (not permutations) count once.

FOLLOW-UP CODE

```
def coin_change_ways(coins, amount):
    ways = [1] + [0] * amount
    for coin in coins:
        for sub_amount in range(coin, amount + 1):
            ways[sub_amount] += ways[sub_amount - coin]
    return ways[amount]
```

FOLLOW-UP 2 | Return which coins are used.

Remember the last coin chosen for each amount (sub_amount parent array) and walk back from the target.

FOLLOW-UP CODE

```
def coin_change_coins(coins, amount):
    UNREACHABLE = amount + 1
    fewest_coins, last_coin = [0] + [UNREACHABLE] * amount, [0] * (amount + 1)
    for sub_amount in range(1, amount + 1):
        for coin in coins:
            if coin <= sub_amount and fewest_coins[sub_amount - coin] + 1 < fewest_coins[sub_amount]:
                fewest_coins[sub_amount], last_coin[sub_amount] = fewest_coins[sub_amount - coin] + 1, coin
    if fewest_coins[amount] == UNREACHABLE:
        return None
    used_coins, sub_amount = [], amount
    while sub_amount:
        used_coins.append(last_coin[sub_amount]); sub_amount -= last_coin[sub_amount]
    return used_coins
```

FOLLOW-UP 3 | Does greedy work?

Only for special coin systems (like US coins). For general coins it can be wrong (coins 1, 3, 4, amount 6). Mention the counterexample, then give the DP.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 230

231 * Partition an array into two sub-arrays with equal sum. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the two subsets.

Keep, for each reachable sum, the list of indices (or a parent pointer) that first reached it; read off the subset for total / 2 and give the rest to the other side.

FOLLOW-UP CODE

```
def partition_subsets(nums):
    total = sum(nums)
    if total % 2:
        return None
    half = total // 2
    subset_at = [None] * (half + 1)
    subset_at[0] = []
    for index, number in enumerate(nums):
        for subset_sum in range(half, number - 1, -1):
            if subset_at[subset_sum] is None and subset_at[subset_sum - number] is not None:
                subset_at[subset_sum] = subset_at[subset_sum - number] + [index]
    if subset_at[half] is None:
        return None
    chosen = set(subset_at[half])
    return [nums[i] for i in sorted(chosen)], [nums[i] for i in range(len(nums)) if i not in chosen]
```

FOLLOW-UP 2 | Minimise the difference between the two parts (LeetCode 1049).

Compute all reachable subset sums with the same bitset and choose the reachable s closest to total / 2: the answer is total - 2s.

FOLLOW-UP CODE

```
def min_subset_difference(nums):
    total = sum(nums)
    reachable_sums = 1
    for number in nums:
        reachable_sums |= reachable_sums << number
    return min(abs(total - 2 * subset_sum) for subset_sum in range(total + 1) if reachable_sums >> subset_sum
& 1)
```

FOLLOW-UP 3 | The parts must be contiguous sub-arrays (a split point).

Then only one split can work: compute prefix sums and check whether some prefix equals total / 2. O(n), O(1) space; no DP is needed.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 231

232 * Explain how to find a target sum in an array. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The question was really 'find two numbers that add up to the target' (Two Sum).

One pass with a dictionary from value to index: for each x look for target - value among earlier values. O(n) time, O(n) space; with a sorted array use two pointers in O(1) space.

FOLLOW-UP CODE

```
def two_sum(nums, target):
    index_of = {}
    for index, value in enumerate(nums):
        if target - value in index_of:
            return [index_of[target - value], index]
        index_of[value] = index
    return None
```

FOLLOW-UP 2 | Only the existence of a subset with the target sum.

The same subset-sum table with booleans (or a bitset: reach |= reach << x) answers existence in O(n * sum / word size).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Negative numbers or a huge sum.

With negatives shift all values to be non-negative, or use a dictionary from reachable sum to count (memoised DFS on (index, sum)); with a huge sum the dictionary is only as large as the number of distinct reachable sums. Cost: O(n * R) with a dictionary over R reachable sums; shifting for negatives adds O(n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 232

233 * Reverse Coin Change * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The array might not come from any set of coins.

Return None (or raise) when dp[0] != 1, when an amount needs more than one extra way, or when the rebuilt table differs from the input at the end; the code does all three checks.

FOLLOW-UP CODE

```
def ways_table(coins, max_amount):
    dp = [1] + [0] * max_amount
    for coin in coins:
        for amount in range(coin, max_amount + 1):
            dp[amount] += dp[amount - coin]
    return dp

def coins_from_dp(dp):
    max_amount = len(dp) - 1
    if not dp or dp[0] != 1:
        return None
    current_table, coins = [1] + [0] * max_amount, []
    for amount in range(1, max_amount + 1):
        if dp[amount] == current_table[amount]:
            continue
        if dp[amount] != current_table[amount] + 1:
            return None
        coins.append(amount)
        for updated_amount in range(amount, max_amount + 1):
            current_table[updated_amount] += current_table[updated_amount - amount]
    return coins if current_table == list(dp) else None
```

FOLLOW-UP 2 | Order of the coins matters (the dp counts permutations, not combinations).

That is a different recurrence: dp[a] = sum of dp[a - c] over coins. Inverting it is a linear-algebra style problem: at each amount the new coins explain the excess; ask which counting is used.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Each coin may be used at most once.

The forward recurrence iterates amounts downward. The inverse still scans upward adding a coin when the target exceeds the table, but the table update must use the 0/1 form.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 233

GROUP 33 OF 43 | 4 QUESTIONS | PART A - PART 2 FOLLOW-UPS

Prefix sums, Tier 2: running total plus a dict of earlier totals

234 * Solve array-sum and city-community problems * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The numbers are all positive.

A sliding window works in O(1) memory: grow the right end, shrink the left while the sum exceeds K. With negatives the window is not monotone, which is why prefix sums are needed.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Longest subarray with sum K.

Store the first index of each prefix sum instead of a count, and take the longest distance to prefix - K.

FOLLOW-UP CODE

```
def longest_subarray_sum_k(nums, target_sum):
    first_index_of, prefix, best = {0: -1}, 0, 0
    for index, number in enumerate(nums):
        prefix += number
        if prefix - target_sum in first_index_of:
            best = max(best, index - first_index_of[prefix - target_sum])
        first_index_of.setdefault(prefix, index)
    return best
```

FOLLOW-UP 3 | Buildings are added over time and you report the number of communities after each addition.

Union-Find with a counter: +1 for the new building, -1 for each union with a different neighbouring community (LeetCode 305).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 234

235 * Maximum-Sum Subarray With Equal Endpoints * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Return only the best sum.

Drop the tie-breaking bookkeeping: the same sweep with max(prefix[j + 1] - smallest prefix of this value so far).

FOLLOW-UP CODE

```
def max_equal_ends_sum(nums):
    smallest_prefix_by_value, best, total = {}, None, 0
    for value in nums:
        smallest_prefix_by_value[value] = min(smallest_prefix_by_value.get(value, total), total)
        total += value
        candidate = total - smallest_prefix_by_value[value]
        best = candidate if best is None else max(best, candidate)
    return best
```

BACK TO PART A - PART 1: QUESTION 235

236 * Maximum Subarray Sum With a Minimum Length Constraint * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Now the length must also be at most max_len. How does it change?

The allowed start prefixes now form a sliding window of indexes [end - max_len, end - min_len]. Keep the prefix indexes in a deque ordered by increasing prefix value; pop from the back while the new value is smaller, pop from the front when an index leaves the window. The minimum is the front. O(n).

FOLLOW-UP CODE

```
from collections import deque


def max_subarray_len_range(nums, min_len, max_len):
    prefix = [0]
    for number in nums:
        prefix.append(prefix[-1] + number)
    window, best = deque(), None
    for end in range(min_len, len(nums) + 1):
        # the start prefix that just became allowed
        allowed_start = end - min_len
        while window and prefix[window[-1]] >= prefix[allowed_start]:
            window.pop()
        window.append(allowed_start)
        # drop starts that would make the subarray too long
        while window[0] < end - max_len:
            window.popleft()
        candidate = prefix[end] - prefix[window[0]]
        best = candidate if best is None else max(best, candidate)
    return best
```

BACK TO PART A - PART 1: QUESTION 236

237 * Find the maximum subarray sum. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the subarray itself.

Remember where the current run started: when starting fresh at i set current_start = index, and when current_sum beats best_sum record (current_start, index). Same single pass.

FOLLOW-UP CODE

```
def max_subarray_with_indices(nums):
    current_sum, current_start, best_sum, best_start, best_end = nums[0], 0, nums[0], 0, 0
    for index in range(1, len(nums)):
        if nums[index] > current_sum + nums[index]:
            current_sum, current_start = nums[index], index
        else:
            current_sum += nums[index]
        if current_sum > best_sum:
            best_sum, best_start, best_end = current_sum, current_start, index
    return best_sum, best_start, best_end
```

FOLLOW-UP 2 | The array is circular (LeetCode 918).

The best circular subarray is either an ordinary one or the whole array minus the worst ordinary one (the minimum subarray): max(kadane max, total - kadane min), taking care of the all-negative case.

FOLLOW-UP CODE

```
def max_subarray_circular(nums):
    total, running_max, max_sum, running_min, min_sum = sum(nums), 0, float("-inf"), 0, float("inf")
    for number in nums:
        running_max, running_min = max(running_max + number, number), min(running_min + number, number)
        max_sum, min_sum = max(max_sum, running_max), min(min_sum, running_min)
    return max_sum if max_sum < 0 else max(max_sum, total - min_sum)
```

FOLLOW-UP 3 | Divide and conquer instead of Kadane.

Split the array; the best subarray is entirely left, entirely right, or crosses the middle (best suffix of the left plus best prefix of the right). T(n) = 2T(n/2) + O(n) = O(n log n); useful when the data is split across machines.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 237

Group by a canonical key

240 * Check if all substrings are anagrams of words * MEDIUM *

DIFFICULTY | * MEDIUM * | 4 FOLLOW-UPS

FOLLOW-UP 1 | A bigger alphabet or Unicode.

Replace the fixed array by a Counter and use a frozen tuple of sorted (letter, count) pairs, or a hash of the counts, as the signature. Cost: O(n^2 * distinct letters in a substring) with a Counter signature instead of a fixed array.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Only substrings of lengths that exist in the dictionary need checking.

Record the set of word lengths; a substring whose length is not in that set can never match, so you can return False the first time such a length is reached (assuming all substrings of length >= 3 up to n must match).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Subsets of the letters (any sub-multiset of a word) instead of a full permutation.

Then the test is containment of count vectors, which is a subset-of-multiset query; use a sorted signature trie or precompute all sub-multisets for short words.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Report the first failing substring instead of True or False.

Test substrings by length (shortest first) so the report is a smallest counterexample; the signature set makes each test one lookup after sorting the substring's letters.

FOLLOW-UP CODE

```
def first_failing_anagram(text, words):
    signatures = {tuple(sorted(word)) for word in words}
    for length in range(3, len(text) + 1):
        for start in range(len(text) - length + 1):
            if tuple(sorted(text[start:start + length])) not in signatures:
                return text[start:start + length]
    return None
```

BACK TO PART A - PART 1: QUESTION 240

241 * Group anagrams * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Make the key O(L) instead of O(L log L).

Use a tuple of 26 letter counts as the key (no sort). Total O(n * L); for large alphabets use a Counter turned into a frozenset of items or a sorted tuple of (char, count).

FOLLOW-UP CODE

```
def group_anagrams_counts(words):
    groups = defaultdict(list)
    for word in words:
        letter_counts = [0] * 26
        for char in word:
            letter_counts[ord(char) - 97] += 1
        groups[tuple(letter_counts)].append(word)
    return list(groups.values())
```

FOLLOW-UP 2 | Case-insensitive, Unicode, or ignore spaces and punctuation.

Normalise first (casefold, strip non-letters, Unicode normalisation) and then compute the key. Say what counts as a letter.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The words are in many files.

Compute the key on each machine and hash-partition by key so every anagram group lands on one worker, which then groups locally (a MapReduce shuffle).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 241

242 * Given an array of strings, return groups of "buddy strings," where the strings have the same length and the distance between corresponding characters is the same. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | What if shifting is not cyclic (z does not wrap to a)?

Drop the % 26 and use the raw differences as the key. Two strings with the same raw gaps differ by one fixed integer shift, and both are already inside 'a'..'z', so that shift is automatically legal: the gap tuple alone decides buddies, with no extra range check. What changes is that 'az' and 'ba' are no longer buddies (gaps 25 and -1 instead of 25 and 25).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | How would you do this with strings that are too many for one machine?

Compute the gap key on each machine, hash-partition the strings by key (so every buddy lands on the same worker), and group locally. The key is the only thing exchanged, so shuffle cost is one pass over the data. Cost: O(total characters / machines) per machine to compute the keys, plus one shuffle of the keys; each group is then built locally in O(group size).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Shifting does not wrap around (z stays z).

Use the raw gaps (no % 26) as the key. Equal raw gaps mean one fixed integer shift, which is automatically legal because both strings are already letters, so no extra check is needed.

FOLLOW-UP CODE

```
def group_buddies_noncyclic(strings):
    groups = defaultdict(list)
    for word in strings:
        groups[tuple(ord(word[position]) - ord(word[position - 1]) for position in range(1, len(word)))].
append(word)
    return list(groups.values())
```

BACK TO PART A - PART 1: QUESTION 242

K-way merge: a heap of the next item from each source

245 * K Largest Elements Across Sorted Lists * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | The lists are not sorted, or arrive as streams. What changes?

Keep a min-heap of the k largest values seen so far: push each value and pop the smallest when the heap has more than k items. O(N log how_many) time and O(how_many) memory, in one pass.

FOLLOW-UP CODE

```
def k_largest_stream(lists, how_many):
    heap = []
    for one_list in lists:
        for value in one_list:
            if len(heap) < how_many:
                heapq.heappush(heap, value)
            elif how_many and value > heap[0]:
                heapq.heapreplace(heap, value)
    return sorted(heap)
```

BACK TO PART A - PART 1: QUESTION 245

246 * Sort a nearly sorted (or K sorted) array. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | K is not given. How do you find it?

Sort a copy with the original index as a tie-break and take the largest |original index - sorted index|; that is the smallest valid K. O(n log n), so only worth it if you will sort many similar arrays.

FOLLOW-UP CODE

```
def smallest_k(values):
    """Find the smallest K for which the array is K-sorted: the largest distance between an element's
position and its sorted position."""
    return _max_displacement(values)
```

FOLLOW-UP 2 | The data is a stream you cannot hold in memory.

The heap of K + 1 elements is already a streaming algorithm: it emits the next sorted element after each arrival once it holds K + 1 elements. Memory O(K).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | K is close to n.

Then O(n log K) approaches O(n log n) and the heap has no advantage; use the library sort. State the crossover.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 246

GROUP 36 OF 43 | 4 QUESTIONS | PART A - PART 2 FOLLOW-UPS

Tree recursion, Tier 2: pass a rule down, return a node or a yes/no up

250 * Find mode in a trinary search tree

* MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The tree may be broken (it does not respect the ordering).

In-order no longer gives sorted values, so equal values need not be adjacent. Fall back to a frequency table over a full traversal (the Counter version): O(n) time, O(distinct values) space.

FOLLOW-UP CODE

```
from collections import Counter


def modes_counter(root):
    counts = Counter()

    def dfs(node):
        if node is None:
            return
        counts[node.val] += 1
        dfs(node.left)
        dfs(node.middle)
        dfs(node.right)

    dfs(root)
    if not counts:
        return []
    best = max(counts.values())
    return sorted(value for value, count in counts.items() if count == best)
```

FOLLOW-UP 2 | Only O(1) extra space (no stack, no dictionary).

Use Morris traversal for the in-order walk (temporary threads) with the same running-count logic; it mutates the tree briefly, so restore the pointers. Adapting it to the middle child needs care.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The tree is huge and values arrive in a stream.

For a stream use a count dictionary (or a Misra-Gries summary if only frequent values matter); the sorted-order trick needs the whole tree in order. Cost: O(1) per value with a counter and O(d) memory for the distinct values (or O(1 / epsilon) with Misra-Gries).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 250

251 * Next Upcoming Birthday in a Family Tree

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Support repeated alarms: each call returns the next soonest birthday, then an empty string when all are used.

Collect the distinct (distance, month, day) values once, sort them, and keep an index: each call returns the next entry. If people can be added later, use a heap keyed by distance instead.

FOLLOW-UP CODE

```
class BirthdayAlarms:
    def __init__(self, person_me, today_month, today_day):
        today = (today_month - 1) * 31 + today_day
        seen = set()
        dates = set()

        def dfs(person):
            if person is None or person in seen:
```

```
                return
            seen.add(person)
            day = (person.month - 1) * 31 + person.day
            dates.add(((day - today) % 372, person.month, person.day))
            dfs(person.parent1)
            dfs(person.parent2)

        dfs(person_me)
        self.order = sorted(dates)
        self.index = 0

    def next(self):
        if self.index == len(self.order):
            return ""
        _, month, day = self.order[self.index]
        self.index += 1
        return month, day
```

BACK TO PART A - PART 1: QUESTION 251

252 * Lowest Common Ancestor * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The tree is n-ary.

The same recursion: collect the non-empty results from all children; if two or more children reported a hit this node is the answer, otherwise pass the single result (or None) up.

FOLLOW-UP CODE

```
class NNode:
    def __init__(self, val, children=None):
        self.val, self.children = val, children or []

def lca_nary(root, first_node, second_node):
    def dfs(node):
        if node is None or node is first_node or node is second_node:
            return node
        hits = [found for found in (dfs(child) for child in node.children) if found]
        return node if len(hits) >= 2 else (hits[0] if hits else None)

    return dfs(root)
```

FOLLOW-UP 2 | A very deep tree, or no recursion allowed.

Do one traversal that records each node's parent (a dict), then use the ancestor-set method from the parent-pointer version. O(node) time and space, no recursion.

FOLLOW-UP CODE

```
class BNode:
    def __init__(self, val, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def lca_with_parent_map(root, first_node, second_node):
    parent, stack = {root: None}, [root]
    while stack:
        node = stack.pop()
        for child in (node.left, node.right):
            if child:
                parent[child] = node
                stack.append(child)
    ancestors = set()
    while first_node:
        ancestors.add(first_node)
        first_node = parent[first_node]
    while second_node not in ancestors:
        second_node = parent[second_node]
    return second_node
```

FOLLOW-UP 3 | p or q might not be in the tree.

The recursive answer is wrong in that case (it returns the one that exists). Count how many of p and q were seen during the DFS, or first check both exist, and return None if not.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 252

253 * Determine if a given binary tree is a binary search tree (BST). * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Duplicates are allowed (all on the left, or all on the right).

Make one of the two bounds inclusive: for 'duplicates go left' every node must satisfy lo < val <= hi (the bound inherited from a left ancestor is inclusive, the one from a right ancestor is exclusive). Tested on trees built by inserting duplicates to the left.

FOLLOW-UP CODE

```
def is_bst_duplicates_left(root):
    def valid(node, low, high):
        if node is None:
            return True
        if not low < node.val <= high:
            return False
        return (valid(node.left, low, node.val)
                and valid(node.right, node.val, high))

    return valid(root, float("-inf"), float("inf"))
```

FOLLOW-UP 2 | Do it with an in-order walk that keeps only the previous value.

Walk in order with an explicit stack and compare each value with the previous one (strictly greater). O(h) space and early exit.

FOLLOW-UP CODE

```
def is_bst_inorder(root):
    previous = None

    def inorder(node):
        nonlocal previous
        if node is None:
            return True
        if not inorder(node.left):
            return False
        if previous is not None and node.val <= previous:
            return False
        previous = node.val
        return inorder(node.right)

    return inorder(root)
```

FOLLOW-UP 3 | Recover a BST where exactly two nodes were swapped (LeetCode 99).

An in-order walk shows one or two descents; the first and last nodes involved in the descents are the swapped pair. O(n) time and O(h) space (O(1) with Morris traversal).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 253

Random pick without repeats: swap the chosen item with the last one and pop

254 * Design set with O(1) random access

* MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Make it thread-safe: one lock or finer-grained locking?

One lock around every operation is simplest and correct (delete touches the array, the dict and a second element's index together, so they must change as one unit). Finer locks do not help much because every operation touches shared structure; use sharding by value hash if contention is the problem.

FOLLOW-UP CODE

```
class FancySet:
    def __init__(self):
        self.values, self.index_of = [], {}

    def add(self, value):
        if value in self.index_of:
            return False
        self.index_of[value] = len(self.values)
        self.values.append(value)
        return True

    def delete(self, value):
        if value not in self.index_of:
            return False
        slot, last_value = self.index_of[value], self.values[-1]
        self.values[slot] = last_value
        self.index_of[last_value] = slot
        self.values.pop()
        del self.index_of[value]
        return True

    def getRandom(self):
        if not self.values:
            raise IndexError("empty set")
        return self.values[random.randrange(len(self.values))]

class FancySetThreadSafe(FancySet):
    def __init__(self):
        super().__init__()
        self.lock = threading.Lock()

    def add(self, value):
        with self.lock:
            return super().add(value)

    def delete(self, value):
        with self.lock:
            return super().delete(value)

    def getRandom(self):
        with self.lock:
            return super().getRandom()
```

FOLLOW-UP 2 | What should getRandom do on an empty set?

Raise a clear error (as here) or return a sentinel such as None; whichever you pick, document it and make the caller handle it. Returning an arbitrary value hides bugs.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Duplicates allowed (a multiset, LeetCode 381).

Map each value to a set of indices in the array; delete swaps with the last element and fixes both index sets. getRandom stays a uniform pick over array positions, so values are chosen in proportion to their counts.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 254

255 * Random Generator Without Repeats

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Add reset() so all n values are available again in O(1).

Set size back to n. The array still contains every value 0..n-1 (only their order changed), and the next draws are uniform over the active region, so nothing needs to be rebuilt.

FOLLOW-UP CODE

```
class ResettableGenerator(RandomGenerator):
    def reset(self):
        self.size = self.count
```

BACK TO PART A - PART 1: QUESTION 255

256 * Random Song Shuffler With a No-Repeat Window

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Why is a plain deque or list not enough for O(1)?

A deque of the last k songs is fine for the window, but choosing a random eligible song needs a structure you can sample from and delete from in O(1). A list gives O(1) sampling; deleting from the middle is O(n) unless you swap the element with the last one and pop, which needs the index map to know where a song is. That pair is what makes play O(1).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 256

257 * Unique Random Number Generator

* MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Add a reset() so the same generator can run again.

Clear the dictionary and set the draw counter back to zero: O(draws) once, and all n values are available again.

FOLLOW-UP CODE

```
class ResettableUniqueRandom(UniqueRandom):
    def reset(self):
        self.swapped.clear()
        self.drawn = 0
```

BACK TO PART A - PART 1: QUESTION 257

Build and serialize: write a structure out, read it back with the same walk

259 * Print a Directory Tree * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Print it with box-drawing connectors like the real tree command.

Carry a prefix down the recursion. For each child choose the connector: the last child gets the corner and its subtree continues with blank space, the others get the tee and their subtree continues with a vertical bar.

FOLLOW-UP CODE

```
def tree_with_connectors(paths):
    root = {}
    for path in paths:
        node = root
        for part in path.split("/"):
            node = node.setdefault(part, {})
    lines = []
    tee = "\u251c\u2500\u2500 "
    elbow = "\u2514\u2500\u2500 "
    bar = "\u2502   "

    def walk(node, prefix):
        items = list(node.items())
        for position, (name, kids) in enumerate(items):
            last = position == len(items) - 1
            glyph = elbow if last else tee
            lines.append(prefix + glyph + name)
            walk(kids, prefix + ("    " if last else bar))

    walk(root, "")
    return "\n".join(lines)
```

BACK TO PART A - PART 1: QUESTION 259

260 * Construct a binary tree given its preorder and inorder traversals. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | You are given postorder and inorder instead.

The root is the last postorder element, and because postorder reads left, right, root, reading it backwards yields root, right, left, so build the right subtree before the left.

FOLLOW-UP CODE

```
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def buildTree_post(inorder, postorder):
    inorder_index = {value: position for position, value in enumerate(inorder)}
    next_postorder = len(postorder) - 1

    def build(low, high):
        nonlocal next_postorder
        if low >= high:
            return None
        root = TreeNode(postorder[next_postorder])
        next_postorder -= 1
        root_position = inorder_index[root.val]
        root.right = build(root_position + 1, high)           # postorder read backwards visits the right
subtree first
        root.left = build(low, root_position)
        return root

    return build(0, len(inorder))
```

FOLLOW-UP 2 | Values can repeat.

Then the tree is not unique and the inorder index of the root is ambiguous. Ask for a rule (or for a preorder with null markers). Preorder plus postorder alone also cannot determine a general binary tree.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The tree is a 100,000-node chain.

The recursion is as deep as the tree. Use an explicit stack: push nodes while walking preorder, and pop back to the parent when the next node's inorder index shows you have finished the left side.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 260

261 * Write functions to serialize and deserialize a list of strings. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Use escaping instead of a length prefix.

Escape the separator and the escape character inside each string, then terminate each item with the unescaped separator. The decoder tracks whether the previous character was an escape. It is streamable, while the length prefix can be read in O(1) jumps.

FOLLOW-UP CODE

```
def decode_escape(data):
    strings, current, position = [], [], 0
    while position < len(data):
        char = data[position]
        if char == "\\":
            current.append(data[position + 1])
            position += 2
        elif char == ";":
            strings.append("".join(current))
            current = []
            position += 1
        else:
            current.append(char)
            position += 1
    return strings

def encode_escape(strs):
    return "".join(text.replace("\\", "\\\\").replace(";", "\\;") + ";" for text in strs)
```

FOLLOW-UP 2 | The strings are long or binary.

Use bytes and a fixed-width length (for example a 4-byte big-endian integer from struct.pack). Fixed-width lengths avoid the digit-scanning loop and work for any byte value.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Just use JSON?

Yes if the format is free to choose: json.dumps handles quoting and Unicode correctly. Mention it as the practical answer, but code the length-prefix version when the interviewer wants to see you handle the delimiter problem yourself.

FOLLOW-UP CODE

```
def encode_json(strs):
    return _json.dumps(strs)
```

BACK TO PART A - PART 1: QUESTION 261

Games: win or lose with a cache, and guessing with feedback

264 * Choose optimal guesses for green-only Wordle * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Play a whole game, using the feedback after each guess.

Keep the candidates that would have produced exactly the observed green pattern (filter_candidates), then choose the best guess among the remaining ones. Each round at least removes the guess itself, so the game ends.

FOLLOW-UP CODE

```
def best_guess(words):
    word_count = len(words)
    bits = [defaultdict(int) for _ in range(len(words[0]))]
    for word_index, word in enumerate(words):
        for pos, letter in enumerate(word):
            bits[pos][letter] |= 1 << word_index
    best = None
    for guess in words:
        groups = [(1 << word_count) - 1]
        for pos, letter in enumerate(guess):
            mask = bits[pos][letter]
            next_groups = []
            for group in groups:
                green_part, other_part = group & mask, group & ~mask
                if green_part:
                    next_groups.append(green_part)
                if other_part:
                    next_groups.append(other_part)
            groups = next_groups
        score = Fraction(sum(group.bit_count() ** 2 for group in groups), word_count)
        if best is None or (score, guess) < best:
            best = (score, guess)
    return best[1]

def filter_candidates(words, guess, mask):
    return [word for word in words if _green_mask(guess, word) == tuple(mask)]

def _green_mask(guess, answer):
    return tuple(guess_letter == answer_letter for guess_letter, answer_letter in zip(guess, answer))

def play(words, answer):
    """Follow-up: play the whole game by always picking the best guess among the remaining candidates."""
    guesses, candidates = [], list(words)
    while True:
        guess = best_guess(candidates) if len(candidates) > 1 else candidates[0]
        guesses.append(guess)
        if guess == answer:
            return guesses
        candidates = filter_candidates(candidates, guess, _green_mask(guess, answer))
```

FOLLOW-UP 2 | Answers are not equally likely (weights).

Replace group sizes by group weights: score = sum(weight(group)^2) / total weight. The grouping step is unchanged; use sums of weights instead of popcounts.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Real Wordle also gives yellow feedback.

The pattern becomes three-valued per position (green, yellow, grey) with the usual duplicate-letter rules, so the pattern space is 3^L; bitsets still work if you precompute patterns, but the computation is O(W^2) pattern evaluations for a full table.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 264

Sliding window: grow on the right, shrink on the left while it is too bad

269 * Temperature Monitoring Data Structure * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The window is the last D days (elapsed time), not the last N readings.

Store (timestamp, value) pairs and evict from the front while the timestamp is at or before now minus the span, in both the window and the max deque. Everything else is unchanged.

FOLLOW-UP CODE

```
class TimeMonitor:
    """Window by elapsed time: readings older than `span` are dropped (timestamps non-decreasing)."""
    def __init__(self, span):
        self.span, self.window, self.total, self.max_queue = span, deque(), 0, deque()

    def insert(self, time, reading):
        self.window.append((time, reading))
        self.total += reading
        while self.max_queue and self.max_queue[-1][1] <= reading:
            self.max_queue.pop()
        self.max_queue.append((time, reading))
        while self.window and self.window[0][0] <= time - self.span:
            self.total -= self.window.popleft()[1]
        while self.max_queue and self.max_queue[0][0] <= time - self.span:
            self.max_queue.popleft()

    def average(self):
        return self.total / len(self.window)

    def maximum(self):
        return self.max_queue[0][1]
```

FOLLOW-UP 2 | Also support minimum, or the median.

Minimum uses a second monotonic deque with the opposite comparison. A median needs two heaps with lazy deletion or a sorted container, O(log N) per update, because it is not decomposable like sum and max.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Floating-point drift in the running sum.

Recompute the sum from the window every so often (say every N inserts), or use math.fsum or integer hundredths of a degree to avoid drift.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 269

270 * Footpath Beauty After Repainting * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Answer many queries, each with a different favourite colour (m fixed).

Group the positions of every colour once. For a colour run a window over its positions: the blocks between the first and last position of a window that are not that colour are the repaints needed; keep it at most m. The run length is the window size plus the unused repaints, capped by n. Total work is O(block_count + queries * positions of that colour).

FOLLOW-UP CODE

```
from collections import defaultdict


def beauty_queries(block_colors, queries, max_repaints):
    block_count = len(block_colors)
    positions_of = defaultdict(list)
    for index, color in enumerate(block_colors):
```

```
        positions_of[color].append(index)
    out = []
    for query_color in queries:
        positions, left, best = positions_of.get(query_color, []), 0, 0
        for right in range(len(positions)):
            while positions[right] - positions[left] - (right - left) > max_repaints:
                left += 1
            best = max(best, min(block_count, right - left + 1 + max_repaints))
        out.append(best if positions else min(block_count, max_repaints))
    return out
```

BACK TO PART A - PART 1: QUESTION 270

271 * Shortest Subarray with K Distinct Integers * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Count the subarrays with exactly k distinct values (LeetCode 992).

exactly(distinct_count) = atMost(distinct_count) - atMost(distinct_count - 1), where atMost is a sliding window that shrinks while the distinct count exceeds the limit. Both are O(n).

FOLLOW-UP CODE

```
def count_subarrays_exactly_k(nums, distinct_count):
    def at_most(max_distinct):
        count, left, total = {}, 0, 0
        for right, value in enumerate(nums):
            count[value] = count.get(value, 0) + 1
            while len(count) > max_distinct:
                leaving_value = nums[left]
                count[leaving_value] -= 1
                if count[leaving_value] == 0:
                    del count[leaving_value]
                left += 1
            total += right - left + 1
        return total

    return at_most(distinct_count) - at_most(distinct_count - 1)
```

FOLLOW-UP 2 | Longest subarray with at most k distinct values.

The same window, but keep the longest window whose distinct count never exceeds k (shrink while it does). O(n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The values are strings or large objects.

The frequency map works on any hashable value; for huge objects hash them first or map them to integer ids so the window comparisons stay cheap. Cost: O(n) plus hashing each element; mapping values to integer ids costs O(n) once.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 271

272 * Find the minimum window substring. * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The smallest window containing every distinct character of s itself.

Use required = the set of distinct characters of s and run the same window. A related problem: the longest substring with at most k distinct characters uses a shrink-when-too-many window.

FOLLOW-UP CODE

```
def minWindow(text, required):
    if not required:
        return ""
    need, missing = Counter(required), len(set(required))
    best, left = "", 0
    for right, char in enumerate(text):
        if char in need:
            need[char] -= 1
```

```
            if need[char] == 0:
                missing -= 1
        while missing == 0:
            if best == "" or right - left + 1 < len(best):
                best = text[left:right + 1]
            leaving_char = text[left]
            if leaving_char in need:
                need[leaving_char] += 1
                if need[leaving_char] > 0:
                    missing += 1
            left += 1
    return best

def smallest_window_all_distinct(text):
    return minWindow(text, "".join(sorted(set(text))))
```

FOLLOW-UP 2 | Return all minimum windows, or the number of valid substrings.

For all minimum windows record every window equal to the best length. For counting, when the window [left, right] is valid, every extension to the right is valid too, so add len(s) - right for each left.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Order matters: t must appear as a subsequence.

That is a different problem (LeetCode 727): DP over positions, or a two-pass scan (forward to match, backward to shrink) for each possible end.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 272

Parent pointers: store who is above each node and climb

275 * Employee Hierarchy Add / Delete * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Support moving an employee under another manager and reject cycles.

Walk up from the proposed new manager through manager pointers; if you meet the employee being moved, the move would put a node under its own descendant, so reject it. O(depth).

FOLLOW-UP CODE

```
class Org:
    def __init__(self, ceo):
        self.manager = {ceo: None}
        self.reports = {ceo: set()}

    def add(self, employee, boss):
        if employee in self.manager or boss not in self.manager:
            return False
        self.manager[employee] = boss
        self.reports[employee] = set()
        self.reports[boss].add(employee)
        return True

    def delete(self, employee):
        if employee not in self.manager:
            return False
        boss, direct_reports = self.manager[employee], self.reports[employee]
        if boss is None and len(direct_reports) != 1:
            return False
        for report in direct_reports:
            self.manager[report] = boss
            if boss is not None:
                self.reports[boss].add(report)
        if boss is not None:
            self.reports[boss].discard(employee)
        del self.manager[employee], self.reports[employee]
        return True

def move_employee(org, employee, new_mgr):
    """Reject a move that would create a cycle: new_mgr must not be inside emp's subtree."""
    current = new_mgr
    while current is not None:
        if current == employee:
            return False
        current = org.manager[current]
    old = org.manager[employee]
    if old is not None:
        org.reports[old].discard(employee)
    org.manager[employee] = new_mgr
    org.reports[new_mgr].add(employee)
    return True
```

FOLLOW-UP 2 | When a manager leaves, promote the most senior report instead of the grandparent.

Store a hire order (insertion counter) per employee and pick the report with the smallest one: the new manager takes over the other reports. Say how you break ties.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The CEO leaves.

Pick a policy and say it: reject, promote the most senior report (the rest report to them), or leave the company rootless and hold a set of roots. The structure above needs only a promoted root.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 275

276 * Validate parent array forms a tree * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | The graph is given as undirected edges (LeetCode 261).

A tree has exactly node_count - 1 edges and no cycle: check the edge count, then union endpoints and fail on an edge that joins two nodes already connected. O(node_count alpha(node_count)).

FOLLOW-UP CODE

```
def valid_tree_by_edges(node_count, edges):
    """Undirected form (LeetCode 261): n - 1 edges and connected."""
    if len(edges) != node_count - 1:
        return False
    parent = list(range(node_count))

    def find(node):
        while parent[node] != node:
            parent[node] = parent[parent[node]]
            node = parent[node]
        return node

    for node_a, node_b in edges:
        root_a, root_b = find(node_a), find(node_b)
        if root_a == root_b:
            return False
        parent[root_a] = root_b
    return True
```

FOLLOW-UP 2 | Report the reason it is invalid.

Return which rule failed: number of roots, an out-of-range parent, or the node where the cycle was detected (the node that was met twice on one walk).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | A binary tree given as child arrays (LeetCode 1361).

Check that exactly one node has in-degree 0, every other node has in-degree 1, and that a traversal from the root visits all n nodes.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 276

277 * Root a Tree So It Becomes a Binary Tree * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Also 2-colour the nodes so a parent and its children differ and equal depths share a colour.

Colour each node by depth % 2 from the chosen root; in a tree this satisfies both rules, so a valid colouring exists whenever a valid root does.

FOLLOW-UP CODE

```
from collections import deque


def root_and_coloring(node_count, edges):
    root = find_binary_tree_root(node_count, edges)
    if root == -1:
        return None
    adj = [[] for _ in range(node_count)]
    for node_a, node_b in edges:
        adj[node_a].append(node_b)
        adj[node_b].append(node_a)
    color = {root: 0}
    queue = deque([root])
    while queue:
        node = queue.popleft()
        for neighbour in adj[node]:
            if neighbour not in color:
                color[neighbour] = 1 - color[node]
```

```
                queue.append(neighbour)
    return root, [color[node] for node in range(node_count)]
```

BACK TO PART A - PART 1: QUESTION 277

278 * Lowest Common Ancestor of a Binary Tree III * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Do it without the swap trick (it feels like magic).

Compute the depth of each node by walking up, lift the deeper one by the difference, then step both up together until they meet. Same O(h) time and O(1) space and easier to justify.

FOLLOW-UP CODE

```
class PNode:
    def __init__(self, val):
        self.val, self.left, self.right, self.parent = val, None, None, None

def _depth(node):
    depth = 0
    while node.parent:
        node, depth = node.parent, depth + 1
    return depth

def _lca_depth(first_node, second_node):
    depth_first, depth_second = _depth(first_node), _depth(second_node)
    while depth_first > depth_second:
        first_node, depth_first = first_node.parent, depth_first - 1
    while depth_second > depth_first:
        second_node, depth_second = second_node.parent, depth_second - 1
    while first_node is not second_node:
        first_node, second_node = first_node.parent, second_node.parent
    return first_node
```

FOLLOW-UP 2 | Many LCA queries on the same tree.

Preprocess with binary lifting (up[k][v] is the 2^k-th ancestor): O(n log n) to build, O(log n) per query. Or an Euler tour with a range-minimum structure for O(1) queries.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The nodes might not be in the same tree.

Then the two pointers both end as None after two passes, and None is the correct answer. The loop terminates because both pointers reach None at the same step.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 278

Build or transform a grid with index arithmetic

280 * Connected Crop Arrangement on a Grid * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | Two adjacent square gardens (N x N next to M x M).

Use a different Hamiltonian path of the combined L-shaped area: snake down and up the columns of the first garden so that the last column ends in the top row, step into the second garden there and snake through it. Hand out the crop counts along that path exactly as before. The code checks that this path visits every cell once with each step adjacent, for every size from 1 to 6.

FOLLOW-UP CODE

```
def two_gardens_path(first_size, second_size):
    """A Hamiltonian path of an n x n garden with an m x m garden to its right (top edges aligned): cells as (
row, col)."""
    path = []
    for col in range(first_size):
        down = (col % 2 == 0) == (first_size % 2 == 0)             # the last column of the first garden must
end in row 0
        rows = range(first_size) if down else range(first_size - 1, -1, -1)
        path += [(r, col) for r in rows]
    for row in range(second_size):
        second_cols = range(first_size, first_size + second_size) if row % 2 == 0 else range(first_size +
second_size - 1, first_size - 1, -1)
        path += [(row, c) for c in second_cols]
    return path
```

BACK TO PART A - PART 1: QUESTION 280

281 * Solve matrix groups and recipe inventory * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Ingredients have quantities and a shared ingredient can only be used once.

Then availability is not enough: you also have to decide the order in which to spend scarce items. A priority order with a simulation (as coded) is a heuristic; the exact maximum is a resource-allocation problem (flow or integer programming).

FOLLOW-UP CODE

```
def recipes_with_quantities(recipes, needs, stock):
    """Follow-up: ingredients have amounts; make each recipe at most once, in the given priority order, if
the stock allows (recipes can use the output of earlier ones)."""
    stock, made = dict(stock), []
    progress = True
    while progress:
        progress = False
        for recipe, need in zip(recipes, needs):
            if recipe not in made and all(stock.get(x, 0) >= q for x, q in need.items()):
                for ingredient, amount in need.items():
                    stock[ingredient] -= amount
                stock[recipe] = stock.get(recipe, 0) + 1
                made.append(recipe)
                progress = True
    return made
```

FOLLOW-UP 2 | Blocks that do not tile the matrix exactly.

The last block row or column is smaller; the single-pass update handles that automatically because the block index is r // k and c // k.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Detect which recipes are impossible because of a cycle.

After Kahn's algorithm, recipes whose counts never reached zero are either missing an ingredient or on a cycle; a DFS with colours over just those recipes tells the two cases apart.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 281

282 * Range Sum Query 2D (Immutable Matrix) * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | The matrix can change between queries (update a cell). What do you use?

A 2D Binary Indexed Tree (Fenwick tree): update(row, col, delta) walks both indexes up by their lowest set bit, and a prefix query walks them down. Each operation is O(log R * log C), and a rectangle is four prefix queries.

FOLLOW-UP CODE

```
class MutableMatrix:
    def __init__(self, matrix):
        self.rows, self.cols = len(matrix), len(matrix[0])
        self.vals = [[0] * self.cols for _ in range(self.rows)]
        self.tree = [[0] * (self.cols + 1) for _ in range(self.rows + 1)]
        for row in range(self.rows):
            for col in range(self.cols):
                self.update(row, col, matrix[row][col])

    def update(self, row, col, value):
        delta = value - self.vals[row][col]
        self.vals[row][col] = value
        tree_row = row + 1
        while tree_row <= self.rows:
            tree_col = col + 1
            while tree_col <= self.cols:
                self.tree[tree_row][tree_col] += delta
                tree_col += tree_col & -tree_col
            tree_row += tree_row & -tree_row

    def prefix(self, row, col):
        # sum of rows 0..r - 1 and columns 0..c - 1
        total, tree_row = 0, row
        while tree_row > 0:
            tree_col = col
            while tree_col > 0:
                total += self.tree[tree_row][tree_col]
                tree_col -= tree_col & -tree_col
            tree_row -= tree_row & -tree_row
        return total

    def sum_region(self, r1, c1, r2, c2):
        return (self.prefix(r2 + 1, c2 + 1) - self.prefix(r1, c2 + 1)
                - self.prefix(r2 + 1, c1) + self.prefix(r1, c1))
```

BACK TO PART A - PART 1: QUESTION 282

283 * Rotate Any m×n 2D Matrix * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Can a non-square matrix be rotated in place?

Not in a rectangular buffer, because the dimensions swap. You can do it in place only if the matrix is stored as a flat array and you permute the elements along cycles of the index map (i, j) to (j, m - 1 - i), which is O(mn) time but needs cycle bookkeeping. In an interview, allocate the new matrix and say why.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | How would you rotate by k * 90 degrees?

Reduce k modulo 4 first. Then apply the single-turn function that many times, or write the four direct index maps so you only build the output once.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Rotate by k quarter turns.

Reduce k modulo 4 first; four turns are the identity. Apply the single-turn function that many times (at most three), or write the three index formulas directly to build the answer once.

FOLLOW-UP CODE

```
def rotate_cw(matrix):
    return [list(row) for row in zip(*matrix[::-1])]

def rotate_k(matrix, k):
    for _ in range(k % 4):
        matrix = rotate_cw(matrix)
    return matrix
```

BACK TO PART A - PART 1: QUESTION 283

Split into pieces, fix each piece, join (and pack words into lines)

285 * Snake Case to Camel Case with Repository-Wide Refactor * MEDIUM *

DIFFICULTY | * MEDIUM * | 1 FOLLOW-UPS

FOLLOW-UP 1 | How do you prove the migration is safe?

Run the full test suite before and after (green both times), compare the public API surface (exported names) before and after, and migrate in small reviewable commits with a codemod that only touches owned symbols. Anything reachable by reflection, serialisation, config files or other repositories needs an explicit alias or a deprecation period rather than a silent rename.

FOLLOW-UP CODE

```
def snake_to_camel(name):
    head, *rest = name.split("_")
    return head + "".join(part[:1].upper() + part[1:] for part in rest)

def owned_names(source, keep=()):
    names = set()
    for node in ast.walk(ast.parse(source)):
        if isinstance(node, (ast.FunctionDef, ast.AsyncFunctionDef)):
            names.add(node.name)
            names.update(argument.arg for argument in node.args.args + node.args.kwonlyargs)
        elif isinstance(node, ast.Name) and isinstance(node.ctx, ast.Store):
            names.add(node.id)
    return {name for name in names if "_" in name.strip("_") and not name.startswith("_") and name not in
keep}

def rename_identifiers(source, keep=()):
    owned = owned_names(source, keep)
    out, previous_token = [], None
    for token in tokenize.generate_tokens(io.StringIO(source).readline):
        if token.type == tokenize.NAME and token.string in owned and not (previous_token and previous_token.
string == "."):
            token = token._replace(string=snake_to_camel(token.string))
        out.append(token)
        previous_token = token
    return tokenize.untokenize(out)
```

BACK TO PART A - PART 1: QUESTION 285

286 * Google Doc Line Wrap (Word Wrap with Whitespace and Newlines) * MEDIUM *

DIFFICULTY | * MEDIUM * | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the lines, not only the count.

Keep the same loop but append the finished line to a list (the brute force above already does it). Build each line from a list of words and join once, so each word is copied once.

FOLLOW-UP CODE

```
def wrap_lines_brute(text, width):
    lines = []
    for paragraph in text.split("\n"):
        words = paragraph.split()
        current_line = ""
        for word in words:
            trial = (current_line + " " + word) if current_line else word
            if len(trial) <= width or not current_line:
                current_line = trial
            else:
                lines.append(current_line)
                current_line = word
        lines.append(current_line)
    return lines
```

FOLLOW-UP 2 | A word longer than the width must be broken.

Before placing a word, while it is longer than the width flush the current line and emit width-sized chunks of the word, then continue with the remainder as a normal word.

FOLLOW-UP CODE

```
def wrap_lines_split_long(text, width):
    lines = []
    for paragraph in text.split("\n"):
        current_line = ""
        for word in paragraph.split():
            while len(word) > width:                        # break words that can never fit
                if current_line:
                    lines.append(current_line)
                    current_line = ""
                lines.append(word[:width])
                word = word[width:]
            trial = (current_line + " " + word) if current_line else word
            if len(trial) <= width:
                current_line = trial
            else:
                lines.append(current_line)
                current_line = word
        lines.append(current_line)
    return lines
```

FOLLOW-UP 3 | Minimise raggedness instead of lines.

That is Knuth's word-wrap DP: dp[i] is the best cost of laying out words i.. with cost (spaces left)^2 on each line except the last; O(n * words per line). Greedy is not optimal for this objective.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART A - PART 1: QUESTION 286

Part B Contents - Easy and Hard Questions

91 Easy and Hard questions | question opens Part B - Part 1 | FU opens Part B - Part 2

Groups keep their original 1-43 study-path numbers; tap a question for Part 1 or FU for Part 2

1. Frontier search (BFS or DFS), Tier 1: spread, count and measure distance (3) Go >

E 1. Flood Fill — ColorStack * 2026-01-28 * [FU]
H 8. Infection Spread on a Grid (Cellular Automaton) — PracHub * 2026-06-14 * [FU]
H 9. Shortest Distance From All Buildings — ColorStack * 2025-09-22 * [FU]

2. Sweep over events in time order (3) Go >

H 16. Count Overlapping Rectangle Updates on a Grid — PracHub * 2026-07-10 * [FU]
H 17. Compute Turnstile Crossing Times — PracHub * 2026-05-04 * [FU]
H 18. Compute city skyline outline — PracHub * 2025-12-08 * [FU]

3. Tree recursion, Tier 1: one function that visits a node, asks its children, and reports upward (5) Go >

E 19. Find right-side view of binary tree — PracHub * 2025-11-01 * [FU]
E 20. Given two trees, determine if the pre-order traversals of two trees are same or not. — Exponent ND [FU]
E 21. Calculate the height of a binary tree. — Exponent ND [FU]
H 26. Repeatedly Delete Leaves Matching a Target Value From a Binary Tree — PracHub * 2026-09-16 * [FU]
H 27. Delete an Entire Binary Tree — ColorStack * 2025-09-29 * [FU]

4. A heap of what is available: pop the best, put back the changed one (1) Go >

H 36. Huffman Tree Construction — 1Point3Acres * 2026-06-01 * [FU]

5. Recurrence on a position or a range (memo or table) (3) Go >

H 43. Count Perfect Wake Numbers up to N — 1Point3Acres * 2026-05-07 * [FU]
H 44. Word Break: All Sentences — ColorStack * 2026-01-28 * [FU]
H 45. Edit distance — Exponent ND [FU]

6. Best-first search with a heap: Dijkstra and its minimax twin (4) Go >

H 51. Minimize maximum height along a grid path — PracHub * 2026-01-04 * [FU]
H 52. Find minimum threshold enabling grid path — PracHub * 2026-02-11 * [FU]
H 53. Minimize travel time with optional meeting point — PracHub * 2026-01-06 * [FU]
H 54. Compute minimax grid path and network delay — PracHub * 2026-01-22 * [FU]

7. Backtracking: choose, recurse, undo (1) Go >

H 63. Matrix Flower Placement with House Adjacency — 1Point3Acres * 2026-05-31 * [FU]

8. Runs and groups: extend while it fits, restart or close when it does not (2) Go >

E 64. First Out-of-Order Character — ColorStack * 2026-01-30 * [FU]
E 65. Group Timestamps Into Close Intervals — ColorStack * 2025-10-03 * [FU]

9. Union-find: merge the pairs, then ask who is together (1) Go >

H 80. Friend Graph with Timeline (Friend / Unfriend Events) — 1Point3Acres * 2026-09-29 * [FU]

10. A dict of per-key state that you update, expire, or rank (1) Go >

E 81. Robot Status Message Deduplication Window — 1Point3Acres * 2026-09-10 * [FU]

11. Binary search on a yes/no condition (4) Go >

E 89. First Index Reaching a Threshold (Binary Search) — ColorStack * 2025-10-11 * [FU]
H 94. Median of Two Sorted Arrays (LC 4) with a Follow-up Variant — 1Point3Acres * 2026-10-09 * [FU]
H 95. Find longest duplicated substring — PracHub * 2025-12-08 * [FU]
H 96. Find the median of two sorted arrays. — Exponent ND [FU]

12. Greedy: scan once, carry the best choice so far, act only when forced (3) Go >

E 97. Alarm for the Next Upcoming Birthday — ColorStack * 2025-10-07 * [FU]
E 98. Coins on the Table — ColorStack * 2025-09-20 * [FU]
E 99. Maximum Profit — Exponent ND [FU]

13. Sets and counters: remember the first collection, check the second against it (5) Go >

E 105. Exclude First-K-Unique From Another List — ColorStack * 2025-10-17 * [FU]
E 106. Positional Matches Between Two Lists — ColorStack * 2025-10-15 * [FU]
E 107. Contains Duplicate — ColorStack * 2025-09-16 * [FU]
E 108. Find the Duplicates — Exponent ND [FU]
H 112. Smallest Missing Positive Integer in Linear Time and Constant Extra Space — PracHub * 2026-09-09 * [FU]

14. bisect on a sorted list: find a position, count a range, keep it sorted (3) Go >

E 113. Count People in a Postal Code Range — ColorStack * 2025-10-19 * [FU]
H 119. Find Maximum Envelope Nesting — PracHub * 2026-03-18 * [FU]
H 120. Count of Smaller Numbers After Self — HackTheRounds ND [FU]

15. Prefix sums, Tier 1: precompute once, answer ranges and balance points in O(1) (3) Go >

E 121. Cake Balance Line — ColorStack * 2025-09-26 * [FU]
E 122. Find the Pivot Index (Balanced Split) in O(n) — ColorStack * 2025-09-21 * [FU]
E 123. Parity Alternating Subarray Query — HackTheRounds ND [FU]

16. Sorted intervals: merge while they touch, then read the gaps or search (3) Go >

E 129. Check an Interval Against Existing Intervals for Conflicts — PracHub * 2026-08-08 * [FU]
H 135. Find safe travel intervals between planet influences — PracHub * 2025-11-22 * [FU]
H 136. Implement a function that processes a stream of interval data. — Exponent ND [FU]

17. Trie: nested dicts, one character per level (1) Go >

H 144. Implement Longest-Match Text Replacement — PracHub * 2026-01-20 * [FU]

18. A dict for lookup plus an ordered container for who is next (lazy deletion) (2) Go >

E 145. Implement Batched Undo/Redo Layer — PracHub * 2026-02-04 * [FU]
H 152. Implement LRU Cache. — Exponent ND [FU]

19. Rolling state: the next state comes only from the previous one (3) Go >

E 153. Climbing Stairs — Exponent ND [FU]
E 154. Fibonacci Numbers — Exponent ND [FU]
H 159. Constrained Dice Roll Sequences — ColorStack * 2025-10-06 * [FU]

20. Two pointers: move i and j toward each other (or along two lists) (4) Go >

E 160. Reverse a linked list. — Exponent ND [FU]
E 161. Sort a binary array in linear time. — Exponent ND [FU]
E 162. Is this a valid palindrome? — Exponent ND [FU]
E 163. Reverse a Sentence — Exponent ND [FU]

21. Walk two strings at once with two indices (4) Go >

E 167. Find the First Occurrence of a Substring — ColorStack * 2025-09-16 * [FU]
E 168. Determine if two words are sorted in lexicographic order. — Exponent ND [FU]
E 169. Return all unique substrings of a given string. — Exponent ND [FU]
H 173. Find Dictionary Matches for a Jammed String — PracHub * 2026-04-09 * [FU]

22. Simulate a machine: a class with state and one branch per kind of event (2) Go >

E 174. Mario-Style Score Tracker (OOP) — ColorStack * 2025-10-11 * [FU]
H 180. Simulate In-Place Cellular State Updates — PracHub * 2026-05-04 * [FU]

24. Digits and characters: peel one at a time and carry (3) Go >

E 187. Caesar Cipher: Recover Key, Encrypt, and Decrypt — ColorStack * 2026-02-08 * [FU]
H 191. Check Digits and Combine Ranges — PracHub * 2026-03-29 * [FU]

H	192. Permutation Sequence	**ColorStack * 2025-10-17 * [FU]**

25. Look at the input once and keep only a small summary (and count your calls) (2) Go >

H	197. Find All Bad Pairs Using runTest Oracle	**1Point3Acres * 2025-11-27 * [FU]**
H	198. Find maximum value in huge text file	**PracHub * 2025-12-08 * [FU]**

26. Grid DP: each cell from its neighbours above and to the left (2) Go >

H	202. Find the Longest Straight Run of One Color	**PracHub * 2026-08-23 * [FU]**
H	203. Count same-color squares in a character grid	**PracHub * 2025-11-22 * [FU]**

30. Counting pairs by looking up each partner in a Counter or a bucket (1) Go >

H	223. Maximum-Area Rectangle from Planar Points	**1Point3Acres * 2026-09-29 * [FU]**

31. Tree DP: build a parent order once, then add up from the leaves (2) Go >

H	227. Compute distance-sum from every tree node	**PracHub * 2026-01-22 * [FU]**
H	228. Tree Distance Sum (Re-root DP)	**1Point3Acres * 2026-01-19 * [FU]**

33. Prefix sums, Tier 2: running total plus a dict of earlier totals (1) Go >

H	238. Given an integer array nums and an integer k, return true if nums has a subarray of at least two elements whose sum is a multiple of k.	**Exponent ND [FU]**

34. Group by a canonical key (2) Go >

E	239. Given an array of strings, return the words that can be typed using letters of the alphabet on only one row of American keyboard.	**Exponent ND [FU]**
H	243. Given two strings and two allowed operations, swapping positions of letters and converting all occurrences of one character to another, can the first string be transformed into the second?	**Exponent ND [FU]**

35. K-way merge: a heap of the next item from each source (3) Go >

E	244. Find k pairs with smallest sums	**PracHub * 2025-11-01 * [FU]**
H	247. Return top-k subset sums in descending order	**PracHub * 2026-01-07 * [FU]**
H	248. Merge k sorted linked lists.	**Exponent ND [FU]**

36. Tree recursion, Tier 2: pass a rule down, return a node or a yes/no up (1) Go >

E	249. Next Birthday (BST In-Order Successor with Wraparound)	**ColorStack * 2025-10-17 * [FU]**

37. Random pick without repeats: swap the chosen item with the last one and pop (1) Go >

H	258. Shuffle array using random integer API	**PracHub * 2025-12-08 * [FU]**

38. Build and serialize: write a structure out, read it back with the same walk (2) Go >

H	262. Serialize and Deserialize a Graph	**ColorStack * 2025-09-24 * [FU]**
H	263. Implement a decode and encode function of a general N-ary tree.	**Exponent ND [FU]**

39. Games: win or lose with a cache, and guessing with feedback (4) Go >

H	265. Determine winner in stack-merging game	**PracHub * 2026-01-06 * [FU]**
H	266. Find secret word with match-count feedback	**PracHub * 2026-02-11 * [FU]**
H	267. Wordle-Style 5-Letter Minimum-Guess Strategy	**1Point3Acres * 2025-12-15 * [FU]**
H	268. Babylon Tile-Stack Merge Game	**1Point3Acres * 2025-12-29 * [FU]**

40. Sliding window: grow on the right, shrink on the left while it is too bad (1) Go >

H	273. Necklace Cut into Two Halves with Equal D/R Counts	**1Point3Acres * 2026-05-03 * [FU]**

41. Parent pointers: store who is above each node and climb (1) Go >

E	274. Tree Node.	**Exponent ND [FU]**

42. Build or transform a grid with index arithmetic (1) Go >

E	279. Filled ASCII Diamond	**ColorStack * 2025-09-26 * [FU]**

43. Split into pieces, fix each piece, join (and pack words into lines) (3) Go >

E 284. Implement string formatting and request tracking tasks **PracHub * 2026-02-11 * [FU]**

H 287. Convert Valid Snake Case to Lower Camel Case **PracHub * 2026-04-16 * [FU]**

H 288. Implement a function to justify a list of words within a given width. **Exponent ND [FU]**

PART B - PART 1 | 91 EASY AND HARD QUESTIONS

PART B - Part 1: question to short best solution

This section contains the Easy and Hard questions. Understand the prompt, clarify the rules, state the brute-force baseline, then code the shortest complete best solution. The purple button opens the matching PART B - Part 2 follow-ups.

Frontier search (BFS or DFS), Tier 1: spread, count and measure distance

PATTERN CONTEXT

Every one of these keeps a frontier of cells or nodes still to visit, takes one out, looks at its neighbours, and adds the new ones. A dict of distances doubles as the seen set. Change what you start from, which neighbours are allowed, and what you do when a node comes out.

1 Flood Fill — EASY

DIFFICULTY | EASY

Better approach: starts = [(start_row, start_col)]; usable = same colour as the start; repaint every cell you reach; return early if old colour == new colour

QUESTION INFO

QUESTION

You are given an m x n grid image where image[r][c] is the integer color of the pixel at row r , column c . You are also given a starting pixel (sr, sc) and a new_color . Perform a flood fill: starting from (sr, sc) , repaint that pixel and every pixel connected to it 4-directionally (up, down, left, right) that shares the SAME original color as the starting pixel, changing them all to new_color . Pixels connected only diagonally are not filled. Return the modified grid.

Function signature

```
from typing import List
def flood_fill(image: List[List[int]], sr: int, sc: int, new_color: int) ->
List[List[int]]:
    ...
```

Follow-up (as reported): Implement with DFS (recursive or explicit stack) or BFS; discuss the base case that prevents re-visiting when new_color equals the original color.

EXAMPLES

Example 1	
Input	`image = [[1,1,1],[1,1,0],[1,0,1]], sr = 1, sc = 1, new_color = 2`
Output	`[[2,2,2],[2,2,0],[2,0,1]]`
Why	`Starting color is 1. All 1-pixels connected 4-directionally to (1,1) become 2. The bottom-right 1 is only diagonally reachable, so it stays.`

Example 2	
Input	`image = [[0,0,0],[0,0,0]], sr = 0, sc = 0, new_color = 5`
Output	`[[5,5,5],[5,5,5]]`
Why	`The whole grid is one connected region of color 0.`

Example 3	
Input	`image = [[0,0,0],[0,1,0]], sr = 1, sc = 1, new_color = 1`
Output	`[[0,0,0],[0,1,0]]`
Why	`The start pixel already equals new_color, so nothing changes.`

CONSTRAINTS

- 1 <= m, n <= 50
- 0 <= image[r][c], new_color < 2^16
- 0 <= sr < m , 0 <= sc < n

QUESTION

- If the starting pixel already has new_color , the grid is returned unchanged (guard against infinite recursion).

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= m, n <= 50
- 0 <= image[r][c], new_color < 2^16
- 0 <= sr < m , 0 <= sc < n
- If the starting pixel already has new_color , the grid is returned unchanged (guard against infinite recursion).

BRUTE FORCE | Repeated sweeps

BRUTE-FORCE IDEA

Mark the start with a unique temporary sentinel. Repeatedly scan the grid and mark each old-colored cell touching the marked region; when a full scan adds nothing, replace only the sentinels with the new color. The sentinel prevents growth into disconnected regions.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
old = color at the start; mark the start with a unique sentinel
repeat until a full scan adds no cells:
    scan every old-colored cell in the grid
    if it touches a sentinel cell, replace it with the sentinel
replace every sentinel with the requested new color
return the image
```

BRUTE-FORCE COMPLEXITY | Time O((m * n)^2) | Space O(1) extra

WHY THIS IS WEAKER | A single newly reached cell may force another complete m-by-n sweep, so the grid can be rescanned up to m * n times. BFS or DFS visits each cell in the connected component once.

HOW TO IMPROVE IT

starts = [(start_row, start_col)]; usable = same colour as the start; repaint every cell you reach; return early if old colour == new colour

SHORT BEST SOLUTION

Repaint the current cell, then recursively visit each 4-directional neighbour that still has the old colour. Return immediately when old and new colours match.

```python
# BASE QUESTION (optimal): Recursive DFS
# Idea: Repaint the start, then recursively spread to each
# 4-directional neighbour that still has the old colour.
def flood_fill(image, start_row, start_col, new_color):
    old_color = image[start_row][start_col]
    if old_color == new_color:
        return image

    def dfs(row, col):
        if not (0 <= row < len(image) and 0 <= col < len(image[0])):
            return
        if image[row][col] != old_color:
            return
        image[row][col] = new_color
        dfs(row + 1, col)
        dfs(row - 1, col)
        dfs(row, col + 1)
        dfs(row, col - 1)

    dfs(start_row, start_col)
    return image
```

COMPLEXITY | Time O(m * n) | Space O(m * n) recursion in the worst case

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 1

8 Infection Spread on a Grid (Cellular Automaton) HARD

DIFFICULTY | HARD

Better approach: starts = every infected cell; usable = healthy cell; the answer is the largest distance reached (number of rounds)

QUESTION INFO

QUESTION

A grid holds '.' for a healthy cell, 'I' for an infected cell and any other character (such as '#') for a barrier that is never infected. Every step, all infected cells infect their healthy neighbours (up, down, left, right) at the same time. infection_times(grid) returns the step at which each infected cell was infected (0 for the cells infected at the start); spread_steps(grid) returns the number of steps until nothing new gets infected.

Function signature

def infection_times(grid: list) -> dict

def spread_steps(grid: list) -> int

EXAMPLES

Example 1	
Input	`['I..', '...', '..#']`
Output	`3`
Why	`The infection travels from the corner: (0,1) and (1,0) at step 1, then three cells at step 2, and so on. The barrier at the bottom right is never infected.`

Example 2	
Input	`['I...']`
Output	`{(0, 0): 0, (0, 1): 1, (0, 2): 2, (0, 3): 3}`
Why	`A one-row grid: one cell per step.`

Example 3	
Input	`['I#.', 'I#.']`
Output	`0`
Why	`Both cells next to the barrier are already infected and the barrier blocks the right column, so nothing can spread: 0 steps.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Rescan the whole grid every step

BRUTE-FORCE IDEA

Build the next grid from the current one by checking every cell's four neighbors, and repeat until a step changes nothing. Easy to verify, but every step touches the whole grid even when only a thin edge is changing.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Rescan the whole grid every step
Build the next grid from the current one by checking every cell's four neighbors
repeat until a step changes nothing
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(steps * m * n) | Space O(m * n)

WHY THIS IS WEAKER | Every step scans the whole grid even when only a thin edge is changing, O(steps * m * n). Keeping a frontier of newly infected cells (BFS) touches each cell once, O(m * n).

HOW TO IMPROVE IT

starts = every infected cell; usable = healthy cell; the answer is the largest distance reached (number of rounds)

SHORT BEST SOLUTION

Put every initially infected cell in the frontier. Each step, look only at the neighbors of frontier cells, infect the healthy ones, and make them the next frontier. The loop ends when the next frontier is empty, which is the fixed point. Every cell enters the frontier at most once.

```
# BASE QUESTION (optimal): Frontier BFS (only recently
# infected cells spread)
# Idea: Put every initially infected cell in the frontier.
# Each step, look only at the neighbors of frontier cells,
# infect the healthy ones, and make them the next
# frontier. The loop ends when the next frontier is empty,
# which is the fixed point. Every cell enters the frontier
# at most once.
def spread_steps(grid):
    cells = [list(row) for row in grid]
    frontier = [(row, col) for row in range(len(cells)) for col in range(len(cells[0])) if cells[row][col] == "
I"]
    steps = 0
    while frontier:
        next_frontier = []
        for row, col in frontier:
            for row_step, col_step in ((1, 0), (-1, 0), (0, 1), (0, -1)):
                next_row, next_col = row + row_step, col + col_step
                if 0 <= next_row < len(cells) and 0 <= next_col < len(cells[0]) and cells[next_row][next_col] =
= ".":
                    cells[next_row][next_col] = "I"
                    next_frontier.append((next_row, next_col))
        if not next_frontier:
            break
        frontier, steps = next_frontier, steps + 1
    return steps
```

COMPLEXITY | Time O(m * n) | Space O(m * n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 8

9 Shortest Distance From All Buildings — HARD

DIFFICULTY | HARD

Better approach: run it once per building (one start each); add each run into a total table and count how many buildings reached each cell

QUESTION INFO

QUESTION

You are given a 2D grid where each cell is:
0 = empty land (a school can be built here), 1 = a house, 2 = a tree (an obstacle that cannot be entered or crossed). You want to build one school on an empty cell so that the total travel distance from every house to the school is minimized. Distance is measured by 4-directional steps (up/down/left/right); trees block movement. A valid school location must be reachable from every house. Return the coordinates (row, col) of the empty cell with the minimum total distance to all houses. If several cells tie on total distance, return the one that comes first in row-major order (smallest row, then smallest column). If no empty cell can reach all houses, return (-1, -1) .

Function signature

```
def shortest_distance_from_all_buildings(grid: list[list[int]]) -> tuple[int, int]:
    ...
```

QUESTION

Follow-up (as reported): (Reported) Minimize the average distance instead of the total. Since the number of houses is fixed, the minimizing cell is the same, but track the count to report the average.

EXAMPLES

Example 1	
Input	`grid = [[1, 0, 2, 0, 1], [0, 0, 0, 0, 0], [0, 0, 1, 0, 0]]`
Output	`(1, 2)`
Why	`Cell (1,2) reaches all three houses with total distance 3 + 3 + 1 = 7, the minimum over all reachable empty cells.`

Example 2	
Input	`grid = [[1, 0, 1]]`
Output	`(0, 1)`
Why	`The only empty cell is distance 1 from each house; total 2.`

Example 3	
Input	`grid = [[1, 2, 1]]`
Output	`(-1, -1)`
Why	`There is no empty cell, so no valid location exists.`

CONSTRAINTS

- 1 <= rows, cols <= 50
- Each cell is 0 , 1 , or 2 .
- There is at least one house.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= rows, cols <= 50
- Each cell is 0 , 1 , or 2 .
- There is at least one house.

BRUTE FORCE | BFS from every empty cell

BRUTE-FORCE IDEA

Start with: BFS from every empty cell. Each empty cell runs its own full search for the houses.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: BFS from every empty cell
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(cells * R * C) | Space O(R * C)

WHY THIS IS WEAKER | Each empty cell runs its own full search for the houses.

HOW TO IMPROVE IT

run it once per building (one start each); add each run into a total table and count how many buildings reached each cell

SHORT BEST SOLUTION

Every house adds its distance to the empty cells it reaches; valid cells were reached by all houses.

```
# BASE QUESTION (optimal): One BFS per house
# Idea: Every house adds its distance to the empty cells
```

```
# it reaches; valid cells were reached by all houses.
from collections import deque


def school_totals(grid):
    rows, cols = len(grid), len(grid[0])
    total = [[0] * cols for _ in range(rows)]
    reached = [[0] * cols for _ in range(rows)]
    houses = 0
    for row in range(rows):
        for col in range(cols):
            if grid[row][col] != 1:
                continue
            houses += 1
            seen, queue = {(row, col)}, deque([(row, col, 0)])
            while queue:
                cell_row, cell_col, distance = queue.popleft()
                for next_row, next_col in ((cell_row + 1, cell_col), (cell_row - 1, cell_col), (cell_row, cell_
col + 1), (cell_row, cell_col - 1)):
                    if 0 <= next_row < rows and 0 <= next_col < cols and (next_row, next_col) not in seen and g
rid[next_row][next_col] == 0:
                        seen.add((next_row, next_col))
                        total[next_row][next_col] += distance + 1
                        reached[next_row][next_col] += 1
                        queue.append((next_row, next_col, distance + 1))
    return total, reached, houses


def shortest_distance_from_all_buildings(grid):
    total, reached, houses = school_totals(grid)
    best, spot = float("inf"), (-1, -1)
    for row in range(len(grid)):
        for col in range(len(grid[0])):
            if grid[row][col] == 0 and reached[row][col] == houses and total[row][col] < best:
                best, spot = total[row][col], (row, col)
    return spot
```

COMPLEXITY | Time O(houses * R * C) | Space O(R * C)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 9

GROUP 2 OF 43 | 3 QUESTIONS | PART B - PART 1

Sweep over events in time order

PATTERN CONTEXT

Turn every item into events (a start adds one, an end removes one, an arrival joins a queue), sort the events by time, and walk them keeping a small state: a counter, a queue, a heap or a running total. What you record at each event is the only thing that changes.

16 Count Overlapping Rectangle Updates on a Grid — HARD

DIFFICULTY | HARD

Better approach: the same idea in two dimensions: a difference array (+1, -1, -1, +1 at the corners) and two running-sum passes

QUESTION INFO

QUESTION

Count Overlapping Rectangle Updates on a Grid Implement

`range_add_counts(n, rectangles)`. Start with an `n x n` matrix of zeroes. Each rectangle is represented as `[row1, col1, row2, col2]` with inclusive, zero-based coordinates. For every rectangle, add one to each matrix cell inside that rectangle. Return the final matrix, where each cell therefore contains the number of input rectangles covering it.

Example

For `n = 3` and `rectangles = [[0, 0, 1, 1], [1, 1, 2, 2]]`, return:

```
[ [1, 1, 0], [1, 2, 1], [0, 1, 1] ]
```

Constraints

- `1 <= n <= 500`
- `0 <= len(rectangles) <= 100_000`
- `0 <= row1 <= row2 < n`
- `0 <= col1 <= col2 < n`

EXAMPLES

Example 1	
Input	`3, [(0, 0, 1, 1), (1, 1, 2, 2)]`
Output	`[[1, 1, 0], [1, 2, 1], [0, 1, 1]]`
Why	`The cell (1, 1) is covered by both rectangles, so it is counted twice.`

Example 2	
Input	`2, []`
Output	`[[0, 0], [0, 0]]`
Why	`No rectangles: all zeros.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Add one to every cell of every rectangle

BRUTE-FORCE IDEA

Loop over each rectangle and increment all of its cells. Correct; with 100,000 rectangles of area up to 250,000 it is far too slow.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Add one to every cell of every rectangle
Loop over each rectangle
increment all of its cells
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(R * area) | Space O(n^2)

WHY THIS IS WEAKER | O(R * area): 100,000 rectangles of area up to 250,000 is billions of cell updates. A 2D difference array touches four corners per rectangle and one prefix-sum pass rebuilds the counts, O(R + n^2).

HOW TO IMPROVE IT

the same idea in two dimensions: a difference array (+1, -1, -1, +1 at the corners) and two running-sum passes

SHORT BEST SOLUTION

For each rectangle add +1 at (r1, c1), -1 at (r1, c2 + 1), -1 at (r2 + 1, c1) and +1 at (r2 + 1, c2 + 1) in an (size + 1) x (size + 1) array. A prefix sum over rows and then over columns turns these markers into the number of rectangles covering each cell.

```python
# BASE QUESTION (optimal): 2D difference array and prefix
# sums
# Idea: For each rectangle add +1 at (r1, c1), -1 at (r1,
# c2 + 1), -1 at (r2 + 1, c1) and +1 at (r2 + 1, c2 + 1)
# in an (size + 1) x (size + 1) array. A prefix sum over rows
# and then over columns turns these markers into the
# number of rectangles covering each cell.
def range_add_counts(size, rectangles):
    marks = [[0] * (size + 1) for _ in range(size + 1)]
    for r1, c1, r2, c2 in rectangles:
        marks[r1][c1] += 1
        marks[r1][c2 + 1] -= 1
        marks[r2 + 1][c1] -= 1
        marks[r2 + 1][c2 + 1] += 1
    for row in range(size + 1):
        for col in range(1, size + 1):
            marks[row][col] += marks[row][col - 1]
    for col in range(size + 1):
        for row in range(1, size + 1):
            marks[row][col] += marks[row - 1][col]
    return [row[:size] for row in marks[:size]]
```

COMPLEXITY | Time O(R + n^2) | Space O(n^2)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 16

17 Compute Turnstile Crossing Times HARD

DIFFICULTY | HARD

Better approach: sort people by arrival time; two queues (entering, exiting) with a rule for who goes first at each second

QUESTION INFO

QUESTION

A turnstile lets one person through per second. events[i] = (arrival time, direction) for person i, with direction 'enter' or 'exit'. When people in both directions wait in the same second: if the turnstile was not used in the previous second, people exiting go first; otherwise the direction used in the previous second goes first again. People of the same direction go in order of arrival (person id). Return the second at which each person crosses.

Function signature

def turnstile_times(events: list) -> list

EXAMPLES

QUESTION

Example 1	
Input	`[(0, 'exit'), (0, 'enter'), (0, 'enter'), (3, 'exit')]`
Output	`[0, 1, 2, 3]`
Why	The idle turnstile lets the exit (person 0) go first at second 0, then the two entering people follow at 1 and 2; person 3 arrives alone at 3.

Example 2	
Input	`[(0, 'enter'), (0, 'exit'), (0, 'exit'), (0, 'enter')]`
Output	`[2, 0, 1, 3]`
Why	Idle turnstile: the exits (persons 1 and 2) go first at seconds 0 and 1, then the entering people (persons 0 and 3) at seconds 2 and 3.

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Simulate every second

BRUTE-FORCE IDEA

Walk t = 0, 1, 2, and so on Adding arrivals to two FIFO queues and letting one person through per second by the rule. It is easy to verify, but it spends time on every empty second, so a timestamp of 10^9 is hopeless.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Simulate every second
Walk t = 0
so on Adding arrivals to two FIFO queues
letting one person through per second by the rule
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(T + n) where T is the largest timestamp | Space O(n)

WHY THIS IS WEAKER | A timestamp of 10^9 means a billion iterations even if nothing happens. Jumping to the next arrival when both queues are empty is O(n log n).

HOW TO IMPROVE IT

sort people by arrival time; two queues (entering, exiting) with a rule for who goes first at each second

SHORT BEST SOLUTION

Sort arrivals once. When both queues are empty, jump straight to the next arrival time; if the jump leaves at least one idle second, reset the previous state to 'none'. Otherwise apply the rule, record the time, and move one second ahead. Work depends on n, not on the size of the timestamps.

```python
from collections import deque


# BASE QUESTION (optimal): Event-driven simulation with
# two queues
# Idea: Sort arrivals once. When both queues are empty,
# jump straight to the next arrival time; if the jump
# leaves at least one idle second, reset the previous
# state to 'none'. Otherwise apply the rule, record the
# time, and move one second ahead. Work depends on n, not
# on the size of the timestamps.
def turnstile_times(events):
    num_people = len(events)
    people_by_arrival = sorted(range(num_people), key=lambda person: (events[person][0], person))
    enter_queue, exit_queue = deque(), deque()
    crossing_time = [0] * num_people
```

```
    time, next_arrival, last_direction, last_busy_second = 0, 0, None, -10
    while next_arrival < num_people or enter_queue or exit_queue:
        if not enter_queue and not exit_queue:
            time = max(time, events[people_by_arrival[next_arrival]][0])
            if time > last_busy_second + 1:
                last_direction = None
        while next_arrival < num_people and events[people_by_arrival[next_arrival]][0] <= time:
            (enter_queue if events[people_by_arrival[next_arrival]][1] == "enter" else exit_queue).append(peopl
e_by_arrival[next_arrival])
            next_arrival += 1
        queue = (enter_queue if last_direction == "enter" else exit_queue) if (enter_queue and exit_queue) else
 (enter_queue or exit_queue)
        person = queue.popleft()
        crossing_time[person], last_direction, last_busy_second = time, events[person][1], time
        time += 1
    return crossing_time
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 17

18 Compute city skyline outline — HARD

DIFFICULTY | HARD

Better approach: events at each building's left and right edge; a max-heap of active heights; record a point when the top height changes

QUESTION INFO

QUESTION

You are given a list of rectangular buildings in a 2D city skyline. Each building is represented by three integers **`[L, R, H]`**:

- **`L`**: the x-coordinate of the left edge of the building,
- **`R`**: the x-coordinate of the right edge of the building (**`L < R`**),
- **`H`**: the height of the building. All buildings stand on a flat horizontal line (y = 0) and extend vertically up to height **`H`**. Buildings may overlap in the x-dimension. Your task is to compute the *skyline* formed by these buildings, as seen from a distance. The skyline should be represented as a list of **key points `[x, height]`** that describe the outer contour of the union of all buildings. A **key point** is a point where the height of the skyline changes. The output list must satisfy all of the following:

1. The list is sorted by increasing **`x`**.
2. No two consecutive points have the same height (i.e., the height must change at each key point).
3. The first point has a height greater than 0.
4. The last point has height 0, marking the end of the skyline.

Input

- An integer **`n`** — the number of buildings.
- A list of **`n`** buildings, where each building is given as a triple **`[L, R, H]`** of integers. You may assume:
- **`1 <= n <= 10^4`**
- **`0 <= L < R <= 10^9`**
- **`1 <= H <= 10^9`**

Output

Return a list of key points representing the skyline. Each key point is a pair **`[x, height]`** (or a 2-element array/tuple) of integers.

Example

QUESTION

Input:

- **`n = 3`** - buildings = **`[[2, 9, 10], [3, 7, 15], [5, 12, 12]]`** One valid output:
- **`[[2, 10], [3, 15], [7, 12], [12, 0]]`** Explanation (informal):
- From x = 2 to 3, the highest building has height 10.
- From x = 3 to 7, the highest building has height 15.
- From x = 7 to 12, the remaining highest building has height 12.
- After x = 12, there are no buildings, so the height drops to 0. Design an algorithm to compute the skyline key points for the given buildings.

EXAMPLES

Example 1	
Input	**`[[2, 9, 10], [3, 7, 15], [5, 12, 12], [15, 20, 10], [19, 24, 8]]`**
Output	**`[[2, 10], [3, 15], [7, 12], [12, 0], [15, 10], [20, 8], [24, 0]]`**
Why	**`Buildings are [left, right, height]; the outline lists the points where the height changes.`**

Example 2	
Input	**`[[0, 2, 3]]`**
Output	**`[[0, 3], [2, 0]]`**
Why	**`One building: a point at its left edge and one at its right edge, where the height returns to 0.`**

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Compute the height at every edge coordinate

BRUTE-FORCE IDEA

Collect all edge x-coordinates in order. At each one the height is the maximum height among buildings with L <= x < R. Output a point where the height differs from the previous one. Correct; each coordinate scans all buildings.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compute the height at every edge coordinate
Collect all edge x-coordinates in order
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Each edge scans all buildings, O(n^2). A sweep with a max-heap of active buildings and lazy deletion is O(n log n).

HOW TO IMPROVE IT

events at each building's left and right edge; a max-heap of active heights; record a point when the top height changes

SHORT BEST SOLUTION

Make events (position, -height) for left edges and (position, 0, right) for right edges, sorted so that at the same x taller starts come first and ends come after starts. Keep a heap of (-height, right edge). Before reading the maximum, pop buildings whose right edge is at or before the current x. When the maximum height differs from the last output height, emit [position, new height].

```python
import heapq


# BASE QUESTION (optimal): Sweep line with a max-heap and
```

```python
# lazy deletion
# Idea: Make events (position, -height) for left edges and (position, 0,
# right) for right edges, sorted so that at the same x
# taller starts come first and ends come after starts.
# Keep a heap of (-height, right edge). Before reading the
# maximum, pop buildings whose right edge is at or before
# the current x. When the maximum height differs from the
# last output height, emit [position, new height].
def getSkyline(buildings):
    events = sorted([(left, -height, right) for left, right, height in buildings] + [(right, 0, 0) for _, right
, _ in buildings])
    outline, heap = [], [(0, float("inf"))]
    for position, negative_height, right_edge in events:
        while heap[0][1] <= position:
            heapq.heappop(heap)
        if negative_height:
            heapq.heappush(heap, (negative_height, right_edge))
        current_height = -heap[0][0]
        if not outline or outline[-1][1] != current_height:
            outline.append([position, current_height])
    return outline
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 18

GROUP 3 OF 43 | 5 QUESTIONS | PART B - PART 1

Tree recursion, Tier 1: one function that visits a node, asks its children, and reports upward

PATTERN CONTEXT

Write dfs(node, depth). An empty node returns a base value, a real node calls dfs on both children, updates the best answer so far, and returns one number or flag to its parent. Only three lines change: the base value, the update, and what you return.

19 Find right-side view of binary tree

EASY

DIFFICULTY | EASY

Better approach: pass depth down; the first node you reach at a new depth (visiting right before left) is the one you can see

QUESTION INFO

QUESTION

You are given the root of a binary tree. From the **right side** of the tree, at each depth you can see exactly one node: the rightmost node at that depth. Write a function that returns a list of the values of the nodes visible when looking at the tree from the right side, ordered from **top to bottom**. ---

Function Signature

(one possible form)

```
rightSideView(root: TreeNode) -> List[int]
```

Where **`TreeNode`** is defined as:

```
class TreeNode { int val; TreeNode left; TreeNode right; }
```

Example

Consider this binary tree:

```
1 / \ 2 3 \ \ 5 4
```

The right-side view is:

```
[1, 3, 4]
```

Explain the algorithm you would use, including its time and space complexity. You may use either breadth-first search (BFS) or depth-first search (DFS).

EXAMPLES

Example 1	
Input	`[1, 2, 3, null, 5, null, 4]`
Output	`[1, 3, 4]`
Why	`Looking from the right you see 1, then 3 (hiding 2), then 4 (hiding 5).`

Example 2	
Input	`[1, 2]`
Output	`[1, 2]`
Why	`A left child is visible when there is no right child at that depth.`

QUESTION

Example 3	
Input	`[]`
Output	`[]`
Why	`An empty tree shows nothing.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Collect every level, then take the last of each

BRUTE-FORCE IDEA

Build the full list of nodes per level with a queue and return the last value of each list. It stores every node of the tree when only one per level is needed.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Collect every level, then take the last of each
Build the full list of nodes per level with a queue
return the last value of each list
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | Stores every node's value when only one per level is needed, O(n) space. A depth-first walk visiting the right child first and recording the first value at each depth needs O(h).

HOW TO IMPROVE IT

pass depth down; the first node you reach at a new depth (visiting right before left) is the one you can see

SHORT BEST SOLUTION

Visit the right child first. The first node you meet at a depth that is not yet recorded is the rightmost one at that depth. Only O(height) memory is needed.

```python
# BASE QUESTION (optimal): DFS visiting right before left,
# first node at each depth
# Idea: Visit the right child first. The first node you
# meet at a depth that is not yet recorded is the
# rightmost one at that depth. Only O(height) memory is
# needed.
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def rightSideView(root):
    right_view = []

    def dfs(node, depth):
        if node is None:
            return
        if depth == len(right_view):
            right_view.append(node.val)
        dfs(node.right, depth + 1)
        dfs(node.left, depth + 1)

    dfs(root, 0)
    return right_view
```

COMPLEXITY | Time O(n) | Space O(h)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 19

20 Given two trees, determine if the pre-order traversals of two trees are same or not. EASY

DIFFICULTY | EASY

Better approach: walk both trees in preorder with an explicit stack (push right, then left) and compare value by value; stop at the first difference

QUESTION INFO

QUESTION

Given two binary trees, determine whether their pre-order traversals (node, then left subtree, then right subtree) produce the same sequence of values. The trees themselves may have different shapes.

Function signature

def same_preorder(a: TreeNode, b: TreeNode) -> bool

EXAMPLES

Example 1	
Input	`[1, 2, 3], [1, 2, 3]`
Output	`true`
Why	`Identical trees.`

Example 2	
Input	`[1, 2, null, 3], [1, null, 2, null, 3]`
Output	`true`
Why	`Different shapes (a chain going left versus a chain going right) but both read 1, 2, 3.`

Example 3	
Input	`[1, 2, 3], [1, 3, 2]`
Output	`false`
Why	`1, 2, 3 versus 1, 3, 2.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Build both pre-order lists and compare them

BRUTE-FORCE IDEA

Traverse each tree recursively into a list and compare the lists. Correct and the clearest version; it always visits both trees completely and stores them.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Build both pre-order lists and compare them
Traverse each tree recursively into a list
compare the lists
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n + m) | Space O(n + m)

WHY THIS IS WEAKER | It always visits and stores both trees completely, O(n + m) time and space. Walking both pre-orders lazily in lockstep stops at the first difference with O(h) space.

HOW TO IMPROVE IT

walk both trees in preorder with an explicit stack (push right, then left) and compare value by value; stop at the first difference

SHORT BEST SOLUTION

Yield node, left subtree and right subtree from both trees, then compare the two streams until a mismatch or the end.

```
from itertools import zip_longest


# BASE QUESTION (optimal): Recursive pre-order generators
# Idea: Yield node, left subtree, then right subtree from
# both trees and compare the streams until a mismatch.
def same_preorder(first_root, second_root):
    def preorder(node):
        if node is not None:
            yield node.val
            yield from preorder(node.left)
            yield from preorder(node.right)

    missing = object()
    return all(first_value == second_value for first_value, second_value in zip_longest(
        preorder(first_root), preorder(second_root), fillvalue=missing
    ))
```

COMPLEXITY | Time O(n) | Space O(h) recursion

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 20

21 Calculate the height of a binary tree. EASY

DIFFICULTY | EASY

Better approach: no update needed; return 1 + max(left, right) and read the value at the root

QUESTION INFO

QUESTION

Given the root of a binary tree, return its height: the number of nodes on the longest path from the root down to a leaf (an empty tree has height 0).

Function signature

def height(root: TreeNode) -> int

EXAMPLES

Example 1	
Input	[3, 9, 20, null, null, 15, 7]
Output	3
Why	Levels: [3], [9, 20], [15, 7]: height 3.

Example 2	
Input	[1, 2]
Output	2
Why	A root with one child.

Example 3	
Input	[]
Output	0
Why	An empty tree.

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Recursive height

BRUTE-FORCE IDEA

Height(node) = 1 + max(height(left), height(right)), with 0 for an empty tree. The standard answer; on a chain of a hundred thousand nodes the call stack is as deep as the tree.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Recursive height
Height(node) = 1 + max(height(left)
height(right))
with 0 for an empty tree
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(h) call stack

WHY THIS IS WEAKER | Linear time, which is optimal; the call stack is as deep as the tree, so a chain of 100,000 nodes can overflow. Counting levels with a queue avoids recursion.

HOW TO IMPROVE IT

no update needed; return 1 + max(left, right) and read the value at the root

SHORT BEST SOLUTION

An empty tree has height 0. Every other node has height 1 plus the larger height of its two children.

```python
# BASE QUESTION (optimal): Recursive height
# Idea: A node's height is one plus the larger height of
# its two children. An empty tree has height zero.
def height(root):
    if root is None:
        return 0
    return 1 + max(height(root.left), height(root.right))
```

COMPLEXITY | Time O(n) | Space O(h) recursion

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 21

26 Repeatedly Delete Leaves Matching a Target Value From a Binary Tree HARD

DIFFICULTY | HARD

Better approach: set node.left = dfs(node.left) and node.right likewise; return None when the node became a leaf with the target value

QUESTION INFO

QUESTION

Given a binary tree and an integer **`target`**, delete every leaf whose value equals **`target`**. Deleting a leaf can turn its parent into a leaf; if that parent's value also equals **`target`**, it must be deleted as well, and so on, until no leaf with value **`target`** remains. Return the resulting tree, which is empty if the root itself ends up deleted.

Function Signature

`remove_target_leaves(tree: list[int | None], target: int) -> list[int | None]`

Tree Encoding

The input and the output use the same level-order encoding:

- An empty tree is **`[]`**.
- Otherwise the first element is the root's value. Present nodes are then visited in breadth-first order, left to right, and for each one its left child and then its right child are appended, using **`None`** for an absent child. Children of absent nodes are not listed.

QUESTION

- Trailing **`None`** values are omitted. The input list is always well formed, and the output must be in exactly this canonical form.

Rules

- A leaf is a node with no children.
- A node whose value equals **`target`** but that still has at least one child after all deletions is kept.
- Kept nodes never move: every kept node keeps its original parent and its side (left or right).
- The returned tree contains no leaf whose value equals **`target`**.

Constraints

- The input tree has between 1 and 5000 nodes, and its height is at most 500.
- Node values and **`target`** are integers in **`[-1000000000, 1000000000]`**.

Examples

Input: **`tree = [5,7,7,7,3,None,7], target = 7`**
Output: **`[5,7,None,None,3]`** The leaf 7 under the root's left child is deleted, but that left child still has the child 3, so it stays. On the right side, deleting the lower 7 turns the root's right child into a leaf with value 7, so it is deleted too.
Input: **`tree = [3,3,1], target = 3`**
Output: **`[3,None,1]`** Only the left leaf is deleted. The root's value is 3, but it still has a child.
Input: **`tree = [4,4,4], target = 4`**
Output: **`[]`**

EXAMPLES

Example 1	
Input	`tree = [1, 2, 3, 2, null, 2, 4], target = 2 and tree = [1, 3, 3, 3, 2], target = 3`
Output	`values(remove_leaves(tree([1, 2, 3, 2, None, 2, 4]), 2)[0]) -> [1, null, 3, null, 4]` `values(remove_leaves(tree([1, 3, 3, 3, 2]), 3)[0]) -> [1, 3, null, null, 2]`
Why	`remove_leaves returns (the new root, how many leaves were removed). Deleting a leaf can turn its parent into a leaf, which is deleted too if it also equals the target: in the first tree the 2s go and the root's left branch disappears entirely.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Remove matching leaves, repeat until a pass changes nothing

BRUTE-FORCE IDEA

Each pass deletes only the current leaves with the target value; a parent that becomes a matching leaf waits for the next pass. On a chain of n matching nodes that is n passes of O(n) each.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Remove matching leaves, repeat until a pass changes nothing
Each pass deletes only the current leaves with the target value
a parent that becomes a matching leaf waits for the next pass
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n * h) | Space O(h)

WHY THIS IS WEAKER | A chain of matching nodes needs one pass per node, n passes of O(n). Post-order (children first) deletes the whole chain in a single O(n) pass.

HOW TO IMPROVE IT

set node.left = dfs(node.left) and node.right likewise; return None when the node became a leaf with the target value

SHORT BEST SOLUTION

Process both children first, then look at the node: if it is now a leaf and its value is the target, return None to its parent. Because children are fixed before the parent is judged, a whole matching chain collapses in one traversal.

```
# BASE QUESTION (optimal): One post-order pass
# Idea: Process both children first, then look at the
# node: if it is now a leaf and its value is the target,
# return None to its parent. Because children are fixed
# before the parent is judged, a whole matching chain
# collapses in one traversal.
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def removeLeafNodes(root, target):
    def dfs(node):
        if node is None:
            return None
        node.left = dfs(node.left)
        node.right = dfs(node.right)
        if node.left is None and node.right is None and node.val == target:
            return None
        return node

    return dfs(root)
```

COMPLEXITY | Time O(n) | Space O(h)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 26

27 Delete an Entire Binary Tree — HARD

DIFFICULTY | HARD

Better approach: post-order: free the children first, then the node itself

QUESTION INFO

QUESTION

You are given the root of a binary tree (not a BST); each node has only left and right child pointers. You must "free" every node so that after the call the tree is empty. Because a node must not be freed before its children, the correct freeing order is a post-order traversal. Implement delete_tree(root) that frees all nodes and returns the list of node values in the exact order they are freed (post-order: left subtree, right subtree, then node).

Function signature

```
from typing import Optional
class TreeNode:
    def __init__(self, val: int = 0,
                 left: "Optional[TreeNode]" = None,
                 right: "Optional[TreeNode]" = None) -> None:
        self.val = val
        self.left = left
        self.right = right
def delete_tree(root: Optional[TreeNode]) -> list[int]:
    """Free every node and return the values in the order freed (post-order)."""
    ...
```

Follow-up (as reported): Recursive DFS uses O(h) stack space. Achieve O(1) extra space by repeatedly rotating the tree (Morris-style): while the root has a left child, rotate it up; once the root has no left child, free the root and move to its right child.

EXAMPLES

QUESTION

Example 1	
Input	`root = 1 / \ 2 3`
Output	`[2, 3, 1]`
Why	`Post-order frees the left child, then the right child, then the root.`

Example 2	
Input	`root = None`
Output	`[]`
Why	`An empty tree frees nothing.`

Example 3	
Input	`root = 1 / 2 / 3`
Output	`[3, 2, 1]`
Why	`A left-leaning chain frees deepest-first.`

CONSTRAINTS

- 0 <= number of nodes <= 10^5 .
- Node values are integers.

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= number of nodes <= 10^5 .
- Node values are integers.

BRUTE FORCE | Two stacks, then reverse

BRUTE-FORCE IDEA

Start with: Two stacks, then reverse. It stores every node twice (the work stack and the output).

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Two stacks, then reverse
Start with: Two stacks
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | It stores every node twice (the work stack and the output).

HOW TO IMPROVE IT

post-order: free the children first, then the node itself

SHORT BEST SOLUTION

Free the left subtree, the right subtree and then the node.

```python
# BASE QUESTION (optimal): Recursive post-order
# Idea: Free the left subtree, the right subtree and then
# the node.
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val = val
        self.left = left
        self.right = right
```

```python
def delete_tree(root):
    freed = []

    def free(node):
        if node is None:
            return
        free(node.left)
        free(node.right)
        freed.append(node.val)
        node.left = node.right = None

    free(root)
    return freed
```

COMPLEXITY | Time O(n) | Space O(h)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 27

A heap of what is available: pop the best, put back the changed one

PATTERN CONTEXT

The heap holds the things you can choose from right now: machines and when they are free, rooms, lockers, the biggest remaining count, the two smallest weights. Each step pops the best choice, uses it, and pushes it back changed (or a combined item).

36 Huffman Tree Construction — HARD

DIFFICULTY | HARD

Better approach: pop the two smallest weights, push their sum as a new node; repeat until one node remains

QUESTION INFO

QUESTION

Given symbol frequencies, build the Huffman code: repeatedly merge the two least frequent items into one until a single tree is left; the path from the root gives each symbol its bit string. Return the code of every symbol. To make the answer unique, the codes are assigned canonically: shorter codes first, then alphabetical order.

Function signature

def huffman_codes(freq: dict) -> dict

EXAMPLES

Example 1	
Input	`{'a': 5, 'b': 9, 'c': 12, 'd': 13, 'e': 16, 'f': 45}`
Output	`{'f': '0', 'c': '100', 'd': '101', 'e': '110', 'a': '1110', 'b': '1111'}`
Why	`The frequent symbol f gets a 1-bit code; the rare ones get longer codes.`

Example 2	
Input	`{'x': 3, 'y': 8}`
Output	`{'x': '0', 'y': '1'}`
Why	`Two symbols always get one bit each.`

Example 3	
Input	`{'z': 10}`
Output	`{'z': '0'}`
Why	`A single symbol still needs a code of one bit.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan for the two smallest weights on every merge

BRUTE-FORCE IDEA

Keep a list of (weight, tree) items; each round scan for the two smallest, merge, and put the result back. The same greedy as the heap version, with a linear scan instead of a heap operation. Returns the total cost (sum of weight * code length).

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan for the two smallest weights on every merge
```

```
Keep a list of (weight
tree) items
each round scan for the two smallest
put the result back
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Finding the two smallest by scanning costs O(n) per merge and n - 1 merges, O(n^2). A min-heap does each merge in O(log n).

HOW TO IMPROVE IT

pop the two smallest weights, push their sum as a new node; repeat until one node remains

SHORT BEST SOLUTION

Put every symbol in a min-heap keyed by frequency, repeatedly pop the two lightest nodes, merge them under a new internal node with the summed weight and push it back. The last node is the root. A counter is the second heap key so ties never compare tree nodes. Codes come from a DFS: left edge '0', right edge '1'. A single-symbol alphabet needs the special case of code '0'.

```python
import heapq


# BASE QUESTION (optimal): Min-heap merge of the two
# lightest nodes
# Idea: Put every symbol in a min-heap keyed by frequency,
# repeatedly pop the two lightest nodes, merge them under
# a new internal node with the summed weight and push it
# back. The last node is the root. A counter is the second
# heap key so ties never compare tree nodes. Codes come
# from a DFS: left edge '0', right edge '1'. A single-
# symbol alphabet needs the special case of code '0'.
def huffman_codes(freq):
    heap = [(weight, insertion_order, symbol, None, None) for insertion_order, (symbol, weight) in enumerate(so
rted(freq.items()))]
    if not heap:
        return {}
    heapq.heapify(heap)
    counter = len(heap)
    while len(heap) > 1:
        lightest = heapq.heappop(heap)
        second_lightest = heapq.heappop(heap)
        heapq.heappush(heap, (lightest[0] + second_lightest[0], counter, None, lightest, second_lightest))
        counter += 1
    root = heap[0]
    codes = {}
    stack = [(root, "")]
    while stack:
        node, prefix = stack.pop()
        if node[2] is not None:
            codes[node[2]] = prefix or "0"
        else:
            stack.append((node[4], prefix + "1"))
            stack.append((node[3], prefix + "0"))
    return codes
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 36

Recurrence on a position or a range (memo or table)

PATTERN CONTEXT

Define best(i): the answer for everything from position i onward (or best(i, j) for a range, or a table over two sequences). Write what an empty or finished state returns, then combine the answers of the states you can jump to. Cache it, or fill a table in the right order.

43 Count Perfect Wake Numbers up to N — HARD

DIFFICULTY | HARD

Better approach: digit DP: cache best(remaining digits, used-digit mask, last digit, falling?) and walk along N with a tight flag

QUESTION INFO

QUESTION

A number is perfect if its digits are all different, it contains no 0, and no digit is a valley (a digit strictly smaller than both of its neighbours). Part 1: decide whether a number is perfect. Part 2: count the perfect numbers from 1 to N.

Function signature

def is_perfect(n: int) -> bool
def count_up_to(N: int) -> int

EXAMPLES

Example 1	
Input	`196`
Output	`true`
Why	`Digits 1, 9, 6 are different, no zero, and 9 is a peak, not a valley.`

Example 2	
Input	`1546`
Output	`false`
Why	`The 4 in 1546 sits between 5 and 6, both larger: a valley.`

Example 3	
Input	`99`
Output	`81`
Why	`Every number 1-9 and every two-digit number with different non-zero digits: 9 + 72 = 81.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Test every number from 1 to N

BRUTE-FORCE IDEA

Start with: Test every number from 1 to N. Apply the three rules to each number. Correct; it is hopeless for N around 10^9.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Test every number from 1 to N
process every required item or candidate directly
```

keep the result required by the question

BRUTE-FORCE COMPLEXITY | Time O(N * d) | Space O(1)

WHY THIS IS WEAKER | O(N * digits): hopeless for N around 10^9. A digit DP counts by position, used digits, last digit and direction in time independent of N's size.

HOW TO IMPROVE IT

digit DP: cache best(remaining digits, used-digit mask, last digit, falling?) and walk along N with a tight flag

SHORT BEST SOLUTION

Part 1: Reject zero, repeated digits, or an increase after the digits have started decreasing.

Part 2: Digit DP tracks used digits, the last digit, whether the sequence is falling, and the upper-bound flag.

```python
from functools import lru_cache


def is_perfect(number):
    digits = [int(digit_char) for digit_char in str(number)]
    if 0 in digits or len(set(digits)) != len(digits):
        return False
    falling = False
    for previous_digit, next_digit in zip(digits, digits[1:]):
        if falling and next_digit > previous_digit:
            return False
        falling |= next_digit < previous_digit
    return True


def count_up_to(number):
    if number < 1:
        return 0
    digits = [int(digit_char) for digit_char in str(number)]

    @lru_cache(None)
    def dp(position, used_digits, last_digit, falling, tight, started):
        if position == len(digits):
            return int(started)
        limit = digits[position] if tight else 9
        total = dp(position + 1, 0, -1, False, tight and limit == 0, False) if not started else 0
        for digit in range(1, limit + 1):
            if used_digits & (1 << digit):
                continue
            if started and falling and digit > last_digit:
                continue
            total += dp(position + 1, used_digits | (1 << digit), digit,
                        falling or (started and digit < last_digit),
                        tight and digit == limit, True)
        return total

    return dp(0, 0, -1, False, True, False)
```

COMPLEXITY | Part 1: Time O(d) | Space O(d)
Part 2: Time O(d * 2^10 * 10^2) | Space O(d * 2^10 * 10)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 43

44 Word Break: All Sentences HARD

DIFFICULTY | HARD

Better approach: best(start) = the list of sentences for the suffix; for every word that matches at start, put it in front of each sentence of best(end)

QUESTION INFO

QUESTION

You are given a string s and a dictionary word_dict of distinct words. Return ALL possible ways to segment s into a sequence of space-separated words such that every word in the sequence appears in word_dict . A dictionary word may be reused any number of times. Each returned sentence is a single string of the chosen words joined by single spaces. The sentences may be returned in any order.

Function signature

```
from typing import List
def word_break_all(s: str, word_dict: List[str]) -> List[str]:
    ...
```

Follow-up (as reported): Solve with backtracking: at each position try every dictionary word that is a prefix of the remaining string, then recurse. Add memoization keyed on the suffix (or start index) to reuse partial results and avoid exponential re-work.

EXAMPLES

Example 1	
Input	`s = "catsanddog", word_dict = ["cat","cats","and","sand","dog"]`
Output	`["cats and dog", "cat sand dog"]`
Why	`Two segmentations use only dictionary words.`

Example 2	
Input	`s = "pineapplepenapple", word_dict = ["apple","pen","applepen","pine","pineapple"]`
Output	`["pine apple pen apple", "pineapple pen apple", "pine applepen apple"]`
Why	`All three segmentations consist entirely of dictionary words.`

Example 3	
Input	`s = "catsandog", word_dict = ["cats","dog","sand","and","cat"]`
Output	`[]`
Why	`No segmentation covers the whole string with dictionary words.`

CONSTRAINTS

- 1 <= len(s) <= 20
- 1 <= len(word_dict) <= 1000
- 1 <= len(word) <= 10 for each word; words consist of lowercase letters and are distinct.
- If no valid segmentation exists, return an empty list [] .

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(s) <= 20
- 1 <= len(word_dict) <= 1000
- 1 <= len(word) <= 10 for each word; words consist of lowercase letters and are distinct.
- If no valid segmentation exists, return an empty list [] .

BRUTE FORCE | Try every set of cut positions

BRUTE-FORCE IDEA

Start with: Try every set of cut positions. It tests all 2^(n-1) cuts although most start with a non-word.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every set of cut positions
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(2^n * n) | Space O(n)

WHY THIS IS WEAKER | It tests all 2^(n-1) cuts although most start with a non-word.

HOW TO IMPROVE IT

best(start) = the list of sentences for the suffix; for every word that matches at start, put it in front of each sentence of best(end)

SHORT BEST SOLUTION

The sentences of a suffix do not depend on how you reached it, so compute them once.

```python
# BASE QUESTION (optimal): Memoized DFS on the start index
# Idea: The sentences of a suffix do not depend on how you
# reached it, so compute them once.
def word_break_all(text, word_dict):
    words = set(word_dict)
    memo = {}

    def sentences(start):
        if start == len(text):
            return [""]
        if start in memo:
            return memo[start]
        sentences_found = []
        for end in range(start + 1, len(text) + 1):
            if text[start:end] in words:
                for rest in sentences(end):
                    sentences_found.append(text[start:end] + (" " + rest if rest else ""))
        memo[start] = sentences_found
        return sentences_found

    return sentences(0)
```

COMPLEXITY | Time O(n^2 + output) | Space O(n + output)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 44

45 Edit distance HARD

DIFFICULTY | HARD

Better approach: table over two strings, two rows: equal letters copy the diagonal, otherwise 1 + the smallest of the three neighbours

QUESTION INFO

QUESTION

Given two words a and b, return the edit distance: the smallest number of single-letter operations (insert a letter, delete a letter, or replace a letter) that turn a into b.

Function signature

def edit_distance(a: str, b: str) -> (int, list)

(the distance, the whole DP

table)

EXAMPLES

QUESTION

Example 1	
Input	a = 'horse', b = 'ros'
Output	edit_distance('horse', 'ros')[0] -> 3
Why	horse -> rorse (replace h) -> rose (delete r) -> ros (delete e): 3 operations.

Example 2	
Input	a = 'intention', b = 'execution'
Output	edit_distance('intention', 'execution')[0] -> 5
Why	intention -> execution takes 5 operations.

Example 3	
Input	a = '', b = 'abc'
Output	edit_distance('', 'abc')[0] -> 3
Why	From the empty word, insert every letter.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Plain recursion on the last characters

BRUTE-FORCE IDEA

If the last characters match recurse on both prefixes; otherwise try replace, delete and insert and take the minimum. Exact; the same prefix pairs are solved exponentially often.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Plain recursion on the last characters
If the last characters match recurse on both prefixes
otherwise try replace
delete
insert
take the minimum
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(3^(m + n)) | Space O(m + n)

WHY THIS IS WEAKER | Three calls per step and the same (i, j) pair is solved many times: about 3^(m + n). A table of all (i, j) values is O(m * n), with a rolling row for O(min(m, n)) memory.

HOW TO IMPROVE IT

table over two strings, two rows: equal letters copy the diagonal, otherwise 1 + the smallest of the three neighbours

SHORT BEST SOLUTION

dp[second_length] holds the distance from the current prefix of a to the first j characters of b. Each cell is the diagonal value if the characters match, otherwise 1 + the minimum of the diagonal (replace), above (delete) and left (insert). Keeping only the previous row reduces memory to one row.

```python
# BASE QUESTION (optimal): DP table with a rolling row
# Idea: dp[second_length] holds the distance from the current prefix
# of a to the first j characters of b. Each cell is the
# diagonal value if the characters match, otherwise 1 +
# the minimum of the diagonal (replace), above (delete)
# and left (insert). Keeping only the previous row reduces
# memory to one row.
def minDistance(first, second):
    if len(first) < len(second):
```

```
        first, second = second, first
    previous_row = list(range(len(second) + 1))
    for first_length, first_char in enumerate(first, 1):
        current_row = [first_length]
        for second_length, second_char in enumerate(second, 1):
            current_row.append(previous_row[second_length - 1] if first_char == second_char else 1 + min(previo
us_row[second_length - 1], previous_row[second_length], current_row[second_length - 1]))
        previous_row = current_row
    return previous_row[-1]
```

COMPLEXITY | Time O(m * n) | Space O(min(m, n))

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 45

GROUP 6 OF 43 | 4 QUESTIONS | PART B - PART 1

Best-first search with a heap: Dijkstra and its minimax twin

PATTERN CONTEXT

Keep a heap of (cost so far, node). Pop the cheapest, skip stale entries, and push any neighbour you improved. Dijkstra adds the edge weight to the cost; the minimax questions take the maximum of the cost and the edge instead. Everything else is the same loop.

51 Minimize maximum height along a grid path — HARD

DIFFICULTY | HARD

Better approach: the same minimax loop on a height grid

QUESTION INFO

QUESTION

You are given a grid of heights and walk from the top-left cell to the bottom-right cell, moving up, down, left or right. The cost of a path is the largest height of any cell on it. Return the smallest possible cost (and, with with_path=True, a path that achieves it).

Function signature

def min_max_path(H: list, with_path=False)

EXAMPLES

Example 1	
Input	`[[1, 9], [2, 3]]`
Output	`3`
Why	`Down then right passes cells 1, 2, 3: the largest is 3; going right first would have to cross the 9.`

Example 2	
Input	`[[1, 2], [4, 3]], with_path = true`
Output	`(3, [(0, 0), (0, 1), (1, 1)])`
Why	`With with_path=True you also get the route.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every simple path

BRUTE-FORCE IDEA

Depth-first search over all simple paths from the top-left to the bottom-right, tracking the largest cell, with pruning when it cannot beat the best. Exact; exponential in the grid size.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every simple path
Depth-first search over all simple paths from the top-left to the bottom-right
tracking the largest cell
with pruning when it cannot beat the best
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(path)

WHY THIS IS WEAKER | Exponential in the grid size. Dijkstra with 'cost = largest cell so far', or binary search on the limit with a flood fill, is O(m * n log(m * n)).

HOW TO IMPROVE IT

the same minimax loop on a height grid

SHORT BEST SOLUTION

Part 1: Heap of (largest cell so far, cell). Pop the cell with the smallest value, extend to each neighbour with cost max(current, neighbour), and keep only improvements. The first time the target is popped its cost is the answer.

Part 2: Sort the distinct values; the check 'can we walk from start to goal using only cells at or below this value' is monotone in the value, so binary search the smallest value for which it is true.

```
import heapq


# BASE QUESTION (optimal): Dijkstra with cost = maximum
# cell on the path
# Idea: Heap of (largest cell so far, cell). Pop the cell
# with the smallest value, extend to each neighbour with
# cost max(current, neighbour), and keep only
# improvements. The first time the target is popped its
# cost is the answer.
def min_max_path(heights):
    rows, cols = len(heights), len(heights[0])
    best = {(0, 0): heights[0][0]}
    heap = [(heights[0][0], 0, 0)]
    while heap:
        cost, row, col = heapq.heappop(heap)
        if cost > best[(row, col)]:
            continue
        if (row, col) == (rows - 1, cols - 1):
            return cost
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols:
                new_cost = max(cost, heights[next_row][next_col])
                if new_cost < best.get((next_row, next_col), float("inf")):
                    best[(next_row, next_col)] = new_cost
                    heapq.heappush(heap, (new_cost, next_row, next_col))
```

COMPLEXITY | Part 1: Time O(m * n log(m * n)) | Space O(m * n)
Part 2: Time O(m * n log(m * n)) | Space O(m * n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 51

52 Find minimum threshold enabling grid path — HARD

DIFFICULTY | HARD

Better approach: the same minimax loop: the smallest threshold that lets you walk from corner to corner

QUESTION INFO

QUESTION

You are given a grid of heights. Find the smallest threshold T such that you can walk from the top-left cell to the bottom-right cell, moving up, down, left or right, using only cells of height at most T. Return T together with one path that achieves it.

Function signature

def min_max_path(H: list) -> (int, list)

EXAMPLES

QUESTION

Example 1	
Input	`[[1, 3], [2, 2]]`
Output	`(2, [(0, 0), (1, 0), (1, 1)])`
Why	`Going down then right uses cells 1, 2, 2: the largest is 2, and no path stays below that.`

Example 2	
Input	`[[7]]`
Output	`(7, [(0, 0)])`
Why	`A single cell: the threshold is its own height.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every simple path

BRUTE-FORCE IDEA

Depth-first search over all simple paths from the top-left to the bottom-right, tracking the largest cell, with pruning when it cannot beat the best. Exact; exponential in the grid size.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every simple path
Depth-first search over all simple paths from the top-left to the bottom-right
tracking the largest cell
with pruning when it cannot beat the best
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(path)

WHY THIS IS WEAKER | Exponential in the grid size. Dijkstra with 'cost = largest cell so far', or binary search on the limit with a flood fill, is O(m * n log(m * n)).

HOW TO IMPROVE IT

the same minimax loop: the smallest threshold that lets you walk from corner to corner

SHORT BEST SOLUTION

Part 1: Heap of (largest cell so far, cell). Pop the cell with the smallest value, extend to each neighbour with cost max(current, neighbour), and keep only improvements. The first time the target is popped its cost is the answer.

Part 2: Sort the distinct values; the check 'can we walk from start to goal using only cells at or below this value' is monotone in the value, so binary search the smallest value for which it is true.

```python
import heapq


# BASE QUESTION (optimal): Dijkstra with cost = maximum
# cell on the path
# Idea: Heap of (largest cell so far, cell). Pop the cell
# with the smallest value, extend to each neighbour with
# cost max(current, neighbour), and keep only
# improvements. The first time the target is popped its
# cost is the answer.
def min_max_path(heights):
    rows, cols = len(heights), len(heights[0])
    best = {(0, 0): heights[0][0]}
    heap = [(heights[0][0], 0, 0)]
    while heap:
```

```
        cost, row, col = heapq.heappop(heap)
        if cost > best[(row, col)]:
            continue
        if (row, col) == (rows - 1, cols - 1):
            return cost
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols:
                new_cost = max(cost, heights[next_row][next_col])
                if new_cost < best.get((next_row, next_col), float("inf")):
                    best[(next_row, next_col)] = new_cost
                    heapq.heappush(heap, (new_cost, next_row, next_col))
```

COMPLEXITY | Part 1: Time O(m * n log(m * n)) | Space O(m * n)
Part 2: Time O(m * n log(m * n)) | Space O(m * n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 52

53 Minimize travel time with optional meeting point — HARD

DIFFICULTY | HARD

Better approach: three runs: from a, from b, and from the destination on the reversed graph; answer = min over meeting points m of max(distance_from_a, distance_from_b) + distance_to_destination

QUESTION INFO

QUESTION

A weighted graph has n nodes and edges (u, v, w) (undirected unless directed=True). Two people start at nodes a and b and both must end at dest. They may meet at some node m and walk the rest of the way together, so the total time is max(time a to m, time b to m) + time m to dest. Return [the minimum total time, the meeting node] (or [-1, -1] when dest cannot be reached). Meeting at dest itself, or at a or b, is allowed.

Function signature

def meeting_point(n: int, edges: list, a: int, b: int, dest: int, directed=False) -> list

EXAMPLES

Example 1	
Input	`5, [(0, 1, 2), (3, 1, 1), (1, 4, 5)], 0, 3, 4`
Output	`[7, 1]`
Why	`They meet at node 1: a needs 2, b needs 1, then 5 more to node 4: 2 + 5 = 7. Meeting at node 4 itself also takes 7, and the smaller node number is reported.`

Example 2	
Input	`3, [(0, 1, 1)], 0, 1, 2`
Output	`[-1, -1]`
Why	`Node 2 is not connected to anything.`

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Floyd-Warshall all-pairs, then try every meeting node

BRUTE-FORCE IDEA

Compute the shortest distance between every pair of nodes, then for each node m take max(d[A][m], d[B][m]) + d[m][dest]. Simple; cubic in the number of nodes.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Floyd-Warshall all-pairs, then try every meeting node
Compute the shortest distance between every pair of nodes
for each node m take max(d[A][m]
d[B][m]) + d[m][dest]
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(V^3) | Space O(V^2)

WHY THIS IS WEAKER | All-pairs distances cost O(V^3) when only three single-source distances are needed. Dijkstra from A, from B and into the destination is O((V + E) log V).

HOW TO IMPROVE IT

three runs: from a, from b, and from the destination on the reversed graph; answer = min over meeting points m of max(distance_from_a, distance_from_b) + distance_to_destination

SHORT BEST SOLUTION

distA and distB come from forward searches. The distance from every node to the destination is a search from the destination on the reversed graph (the same graph if undirected). The best meeting node minimises max(distA, distB) + distToDest, ties to the smaller node id.

```
import heapq


def _dijkstra_all(node_count, neighbours, start):
    distance = [float("inf")] * node_count
    distance[start] = 0
    heap = [(0, start)]
    while heap:
        distance_so_far, from_node = heapq.heappop(heap)
        if distance_so_far > distance[from_node]:
            continue
        for to_node, weight in neighbours[from_node]:
            if distance_so_far + weight < distance[to_node]:
                distance[to_node] = distance_so_far + weight
                heapq.heappush(heap, (distance_so_far + weight, to_node))
    return distance


# BASE QUESTION (optimal): Three Dijkstra runs (from A,
# from B, and into the destination)
# Idea: distA and distB come from forward searches. The
# distance from every node to the destination is a search
# from the destination on the reversed graph (the same
# graph if undirected). The best meeting node minimises
# max(distA, distB) + distToDest, ties to the smaller node
# id.
def meeting_point(node_count, edges, start_a, start_b, destination, directed=False):
    forward, reverse = [[] for _ in range(node_count)], [[] for _ in range(node_count)]
    for from_node, to_node, weight in edges:
        forward[from_node].append((to_node, weight))
        reverse[to_node].append((from_node, weight))
        if not directed:
            forward[to_node].append((from_node, weight))
            reverse[from_node].append((to_node, weight))
    distance_from_a, distance_from_b, distance_to_destination = _dijkstra_all(node_count, forward, start_a), _d
ijkstra_all(node_count, forward, start_b), _dijkstra_all(node_count, reverse, destination)
    best = (float("inf"), -1)
    for meeting_node in range(node_count):
        travel_time = max(distance_from_a[meeting_node], distance_from_b[meeting_node]) + distance_to_destinati
on[meeting_node]
        if travel_time < best[0]:
            best = (travel_time, meeting_node)
    return [-1, -1] if best[0] == float("inf") else [best[0], best[1]]
```

COMPLEXITY | Time O((V + E) log V) | Space O(V + E)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 53

54 Compute minimax grid path and network delay — HARD

DIFFICULTY | HARD

Better approach: combine = max(cost, height of the next cell); the answer is the cost at the bottom-right cell

QUESTION INFO

QUESTION

Two tasks. (1) minimax_path(H): given a grid of heights, find a path from the top-left to the bottom-right cell (moving up, down, left or right) whose largest height is as small as possible, and return that largest height. (2) network_delay(n, edges, source, node_delay=None): a signal starts at source and travels along directed weighted edges (u, v, w); return the time until every node has received it, or -1 if some node never does. node_delay[u], when given, is a pause at node u before it forwards the signal.

Function signature

def minimax_path(H: list) -> int

def network_delay(n: int, edges: list, source: int, node_delay=None) -> int

EXAMPLES

Example 1	
Input	`[[1, 5, 1], [2, 2, 5]]`
Output	`5`
Why	`Both ways around pass a cell of height 5, so the answer is 5.`

Example 2	
Input	`4, [(0, 1, 2), (1, 2, 3), (0, 3, 10)], 0`
Output	`10`
Why	`The signal reaches node 1 at 2, node 2 at 5, node 3 at 10: everyone has it after 10.`

Example 3	
Input	`3, [(0, 1, 1)], 0`
Output	`-1`
Why	`Node 2 has no incoming edge: it is never reached.`

Example 4	
Input	`4, [(0, 1, 2), (1, 2, 3), (0, 3, 10)], 0, node_delay = [1, 0, 0, 0]`
Output	`11`
Why	`With a pause of 1 at node 0, every arrival is 1 later: node 3 is reached at 11.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Part 1: DFS over every simple path

BRUTE-FORCE IDEA

Try all simple paths from the top-left to the bottom-right and take the smallest maximum cell value. Exact; exponential in the grid size.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Part 1: DFS over every simple path
Try all simple paths from the top-left to the bottom-right
take the smallest maximum cell value
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(path)

WHY THIS IS WEAKER | The number of simple paths is exponential. Dijkstra where the cost of a path is its maximum cell (or a union-find over cells added in increasing order) is O(m * n log(m * n)).

BRUTE FORCE 2 | Part 2: Bellman-Ford from the source

BRUTE-FORCE IDEA

Relax every edge V - 1 times and take the largest distance. Correct and handles negative weights.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Part 2: Bellman-Ford from the source
Relax every edge V - 1 times
take the largest distance
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(V * E) | Space O(V)

WHY THIS IS WEAKER | V - 1 rounds over all edges, O(V * E). Dijkstra with a heap is O((V + E) log V) for non-negative weights.

HOW TO IMPROVE IT

combine = max(cost, height of the next cell); the answer is the cost at the bottom-right cell

SHORT BEST SOLUTION

Part 1: Heap of (best maximum so far, cell). Pop the cell with the smallest bottleneck, and for each neighbour the new bottleneck is max(current, neighbour). Improve if smaller. The first time the goal is popped, its bottleneck is optimal.

Part 2: Compute shortest distances with a heap. The signal reaches everyone at the time of the farthest node; if fewer than n nodes are reached the answer is -1.

```python
from collections import defaultdict
import heapq


# Part 1: Dijkstra where the cost of a path is its maximum
# cell
# Idea: Heap of (best maximum so far, cell). Pop the cell
# with the smallest bottleneck, and for each neighbour the
# new bottleneck is max(current, neighbour). Improve if
# smaller. The first time the goal is popped, its
# bottleneck is optimal.
def minimax_path(heights):
    rows, cols = len(heights), len(heights[0])
    best = {(0, 0): heights[0][0]}
    heap = [(heights[0][0], 0, 0)]
    while heap:
        cost, row, col = heapq.heappop(heap)
        if cost > best[(row, col)]:
            continue
        if (row, col) == (rows - 1, cols - 1):
            return cost
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols:
                new_cost = max(cost, heights[next_row][next_col])
                if new_cost < best.get((next_row, next_col), float("inf")):
                    best[(next_row, next_col)] = new_cost
                    heapq.heappush(heap, (new_cost, next_row, next_col))
    return -1


# Part 2: Dijkstra from the source, answer is the largest
```

```python
# distance
# Idea: Compute shortest distances with a heap. The signal
# reaches everyone at the time of the farthest node; if
# fewer than n nodes are reached the answer is -1.
def networkDelayTime(times, node_count, source):
    graph = defaultdict(list)
    for from_node, to_node, delay in times:
        graph[from_node].append((to_node, delay))
    arrival_time, heap = {}, [(0, source)]
    while heap:
        time, from_node = heapq.heappop(heap)
        if from_node in arrival_time:
            continue
        arrival_time[from_node] = time
        for to_node, delay in graph[from_node]:
            if to_node not in arrival_time:
                heapq.heappush(heap, (time + delay, to_node))
    return max(arrival_time.values()) if len(arrival_time) == node_count else -1
```

COMPLEXITY | Part 1: Time O(m * n log(m * n)) | Space O(m * n)
Part 2: Time O((V + E) log V) | Space O(V + E)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 54

GROUP 7 OF 43 | 1 QUESTIONS | PART B - PART 1

Backtracking: choose, recurse, undo

PATTERN CONTEXT

Build the answer one choice at a time in a list. When the list is complete, record it. Otherwise loop over the allowed choices: append one, recurse, and then remove exactly what you added so the next choice starts from the same state. Pruning is just a check that skips a choice.

63 Matrix Flower Placement with House Adjacency — HARD

DIFFICULTY | HARD

Better approach: place row by row; reject a row early when the houses two rows above are still not next to a flower

QUESTION INFO

QUESTION

A grid has houses (H) and empty cells (0). Place flowers (F) on empty cells so that every row and every column holds at most one flower (exactly one of each when exactly_one is True) and every house has a flower directly beside it (up, down, left or right). Return one valid grid as a list of strings, or None if there is none.

Function signature

def place_flowers(grid: list, exactly_one=False) -> list | None

EXAMPLES

Example 1	
Input	`['H000', 'H000', '000H', '0000']`
Output	`['HF00', 'H0F0', 'F00H', '000F']`
Why	`Flowers at (0,1), (2,0) and (1,3): one per row and column, and each house touches a flower.`

Example 2	
Input	`['H000', 'H000', '000H', '0000'], exactly_one = true`
Output	`['HF00', 'H0F0', 'F00H', '000F']`
Why	`Exactly one flower in every row and column: a fourth flower at (3,2) completes it.`

Example 3	
Input	`['H', 'H']`
Output	`null`
Why	`The two houses can only be served by the same column, so no valid placement exists.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Choose any subset of empty cells and check all the rules

BRUTE-FORCE IDEA

Enumerate every subset of empty cells, keep those with at most one flower per row and column whose every house has a flower neighbour, and count them. Exact, but exponential in the number of cells.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Choose any subset of empty cells and check all the rules
Enumerate every subset of empty cells
keep those with at most one flower per row
column whose every house has a flower neighbour
count them
```

BRUTE-FORCE COMPLEXITY | Time O(2^cells) | Space O(cells)

WHY THIS IS WEAKER | 2^cells subsets: exponential. Backtracking row by row with a used-columns set and early house checks prunes almost all of them (still exponential in the worst case).

HOW TO IMPROVE IT

place row by row; reject a row early when the houses two rows above are still not next to a flower

SHORT BEST SOLUTION

Place at most one flower per row, tracking used columns, and only on empty cells ('0'). A house at row h has neighbours in rows h - 1, h and h + 1, so its requirement is final once row h + 1 is decided; failing there prunes the whole subtree. At the last row check all remaining houses. The 'exactly one' variant just forbids skipping a row and a column. Collecting every solution instead of the first one only changes whether the recursion stops at the first success.

```python
# BASE QUESTION (optimal): Row-by-row backtracking with
# used-column set and delayed house checks
# Idea: Place at most one flower per row, tracking used
# columns, and only on empty cells ('0'). A house at row h
# has neighbours in rows h - 1, h and h + 1, so its
# requirement is final once row h + 1 is decided; failing
# there prunes the whole subtree. At the last row check
# all remaining houses. The 'exactly one' variant just
# forbids skipping a row and a column. Collecting every
# solution instead of the first one only changes whether
# the recursion stops at the first success.
def place_flowers(grid, exactly_one=False, find_all=False):
    rows, cols = len(grid), len(grid[0])
    cells = [list(row) for row in grid]
    houses = [(row, col) for row in range(rows) for col in range(cols) if cells[row][col] == "H"]
    solutions = []

    def house_ok(row, col):
        return any(0 <= row + row_step < rows and 0 <= col + col_step < cols and cells[row + row_step][col + co
l_step] == "F"
                   for row_step, col_step in ((1, 0), (-1, 0), (0, 1), (0, -1)))

    def check_row_houses(row):
        return all(house_ok(house_row, house_col) for house_row, house_col in houses if house_row == row)

    def solve(row, used_cols):
        if not find_all and solutions:
            return
        if row >= 2 and not check_row_houses(row - 2):          # rows up to row - 1 are decided, so row row -
2 houses are final
            return
        if row == rows:
            if check_row_houses(rows - 1) and (not exactly_one or len(used_cols) == cols):
                solutions.append(["".join(row) for row in cells])
            return
        for col in range(cols):
            if col not in used_cols and cells[row][col] == "0":
                cells[row][col] = "F"
                solve(row + 1, used_cols | {col})
                cells[row][col] = "0"
        if not exactly_one:
            solve(row + 1, used_cols)

    solve(0, frozenset())
    return solutions if find_all else (solutions[0] if solutions else None)
```

COMPLEXITY | Time exponential worst case, heavily pruned | Space O(rows + cols)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 63

Runs and groups: extend while it fits, restart or close when it does not

PATTERN CONTEXT

Walk the sequence once with a counter. If the next item fits the rule, the run grows; if not, the run restarts (or the group closes). Longest run, first break, number of groups, and sums over runs are all small changes to what you record.

64 First Out-of-Order Character — EASY

DIFFICULTY | EASY

Better approach: fits = text[index] >= text[index-1]; return i at the first item that does not fit

QUESTION INFO

QUESTION

You are given a string of lowercase English letters. Scanning left to right, the string is considered "in order" as long as each character is greater than or equal to the character before it (a non-decreasing sequence; equal adjacent characters are allowed). Return the index of the FIRST character that breaks this rule — that is, the first index i >= 1 where s[i] < s[i-1] . If the entire string is non-decreasing, return -1 .

Function signature

```
def first_out_of_order(s: str) -> int:
    ...
```

Follow-up (as reported): Solve in a single O(N) pass with O(1) extra space; state the time/space complexity as the interviewer expected.

EXAMPLES

Example 1	
Input	`"abcda"`
Output	`4`
Why	`a<=b<=c<=d holds, but at index 4 'a' < 'd' (s[3]), so 4 is returned.`

Example 2	
Input	`"aabbz"`
Output	`-1`
Why	`Every character is >= the previous one; fully non-decreasing.`

Example 3	
Input	`"cba"`
Output	`1`
Why	`At index 1, 'b' < 'c', the first violation.`

Example 4	
Input	`""`
Output	`-1`
Why	`An empty string is trivially in order.`

CONSTRAINTS

- 0 <= len(s) <= 10^5 s consists only of lowercase English letters.
- An empty string or a single-character string is trivially in order -> return -1 .

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(s) <= 10^5 s consists only of lowercase English letters.
- An empty string or a single-character string is trivially in order → return -1 .

BRUTE FORCE | Prefix equals its sorted copy

BRUTE-FORCE IDEA

Start with: Prefix equals its sorted copy. It sorts every prefix just to learn one comparison.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Prefix equals its sorted copy
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^2 log n) | Space O(n)

WHY THIS IS WEAKER | It sorts every prefix just to learn one comparison.

HOW TO IMPROVE IT

fits = text[index] >= text[index-1]; return i at the first item that does not fit

SHORT BEST SOLUTION

Order is broken exactly where a letter is smaller than the one before it.

```python
# BASE QUESTION (optimal): One pass, compare with the
# previous letter
# Idea: Order is broken exactly where a letter is smaller
# than the one before it.
def first_out_of_order(text):
    for index in range(1, len(text)):
        if text[index] < text[index - 1]:
            return index
    return -1
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 64

65 Group Timestamps Into Close Intervals — EASY

DIFFICULTY | EASY

Better approach: close the group when the gap to the next timestamp is at least k; otherwise extend it

QUESTION INFO

QUESTION

You are given an array of positive integer timestamps in ascending order and an integer k . Partition the timestamps into maximal runs such that within each run, the difference between every pair of adjacent values is strictly less than k . A run breaks wherever an adjacent difference is >= k . Return each run as an interval [start, end] (its first and last value), in order.

Function signature

```
def group_into_intervals(timestamps: list[int], k: int) -> list[list[int]]:
    ...
```

Follow-up (as reported): One pass, O(n) time and O(1) extra space (besides the output).

EXAMPLES

QUESTION

Example 1	
Input	`timestamps = [1,2,3,4,11,14,15,17,21,22,23], k = 2`
Output	`[[1,4],[11,11],[14,15],[17,17],[21,23]]`
Why	`Adjacent diffs of 1 stay in a run; a diff of >=2 (e.g. 4->11, 11->14, 15- >17, 17->21) starts a new run.`

Example 2	
Input	`timestamps = [5], k = 100`
Output	`[[5,5]]`
Why	`A single timestamp forms a run by itself.`

Example 3	
Input	`timestamps = [1,2,3], k = 1`
Output	`[[1,1],[2,2],[3,3]]`
Why	`A diff of 1 is not strictly less than k=1, so every element stands alone.`

CONSTRAINTS

- 1 <= len(timestamps) <= 10^5 , values strictly increasing.
- k >= 1 .

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(timestamps) <= 10^5 , values strictly increasing.
- k >= 1 .

BRUTE FORCE | Mark the cuts, then slice

BRUTE-FORCE IDEA

Start with: Mark the cuts, then slice. It builds a cut list although a single running start suffices.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Mark the cuts, then slice
Start with: Mark the cuts
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | It builds a cut list although a single running start suffices.

HOW TO IMPROVE IT

close the group when the gap to the next timestamp is at least k; otherwise extend it

SHORT BEST SOLUTION

Close the run whenever the gap to the previous value reaches k.

```python
# BASE QUESTION (optimal): One pass with a run start
# Idea: Close the run whenever the gap to the previous
# value reaches k.
def group_into_intervals(timestamps, gap_limit):
    if not timestamps:
        return []
    runs, start = [], timestamps[0]
    for previous, current in zip(timestamps, timestamps[1:]):
```

```
        if current - previous >= gap_limit:
            runs.append([start, previous])
            start = current
    runs.append([start, timestamps[-1]])
    return runs
```

COMPLEXITY | Time O(n) | Space O(1) extra

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 65

GROUP 9 OF 43 | 1 QUESTIONS | PART B - PART 1

Union-find: merge the pairs, then ask who is together

PATTERN CONTEXT

Keep a parent array. find climbs to the root (with path compression), union joins two roots. The only thing that changes between questions is where the pairs to merge come from and what you read at the end: same group, group size, number of groups, or the moment everything became one group.

80 Friend Graph with Timeline (Friend / Unfriend Events) HARD

DIFFICULTY | HARD

Better approach: pairs = friend events sorted by time; the answer is the time of the union that makes groups == 1

QUESTION INFO

QUESTION

There are n people numbered 0 to n - 1. Each event is (time, a, b, kind). A 'friend' event makes a and b friends from that time on, and two people are connected when a chain of friends links them. Return the earliest time at which all n people are connected, or -1 if that never happens.

Function signature

def earliest_all_connected(n: int, events: list) -> int

EXAMPLES

Example 1	
Input	`3, [(1, 0, 1, 'friend'), (5, 1, 2, 'friend')]`
Output	`5`
Why	`After time 1, {0, 1} are connected; at time 5 person 2 joins, so everyone is connected at 5.`

Example 2	
Input	`4, [(1, 0, 1, 'friend'), (2, 1, 2, 'friend')]`
Output	`-1`
Why	`Person 3 is never befriended, so the group never becomes one.`

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Rebuild the graph at every event time and search it

BRUTE-FORCE IDEA

For each distinct time, replay all events up to it to get the current friend set and run a DFS from person 0 to see whether everyone is reachable. Simple, correct for friend and unfriend events; it rebuilds the graph for every time.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Rebuild the graph at every event time and search it
For each distinct time
replay all events up to it to get the current friend set
run a DFS from person 0 to see whether everyone is reachable
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(T * (N + E)) for T distinct times | Space O(N + E)

WHY THIS IS WEAKER | Each time point replays all events and searches the whole graph, O(T * (N + E)). A union-find over sorted events (friend-only) is near-linear; unfriend events need offline dynamic connectivity.

HOW TO IMPROVE IT

pairs = friend events sorted by time; the answer is the time of the union that makes groups == 1

SHORT BEST SOLUTION

Sort by time and union the two people for each friend event. The first time the component count reaches 1 is the answer; -1 if it never does. Union by size with path compression keeps each operation near constant.

```
# BASE QUESTION (optimal): Base: earliest time everyone is
# connected (Union-Find on sorted events)
# Idea: Sort by time and union the two people for each
# friend event. The first time the component count reaches
# 1 is the answer; -1 if it never does. Union by size with
# path compression keeps each operation near constant.
def earliest_all_connected(people_count, events):
    parent = list(range(people_count))

    def find(person):
        while parent[person] != person:
            parent[person] = parent[parent[person]]
            person = parent[person]
        return person

    components = people_count
    if people_count == 1:
        return 0
    for time, person_a, person_b, kind in sorted(events, key=lambda event: event[0]):
        if kind != "friend":
            continue
        root_a, root_b = find(person_a), find(person_b)
        if root_a != root_b:
            parent[root_a] = root_b
            components -= 1
            if components == 1:
                return time
    return -1
```

COMPLEXITY | Time O(E log E) | Space O(N)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 80

GROUP 10 OF 43 | 1 QUESTIONS | PART B - PART 1

A dict of per-key state that you update, expire, or rank

PATTERN CONTEXT

Keep a dict from a key (message, user, word, activity) to the little bit of state you need: the last time seen, a Counter, a set, a list of timestamps. Every event updates one key's state; the rule that decides what to accept, expire or report is the only part that changes.

81 Robot Status Message Deduplication Window — EASY

DIFFICULTY | EASY

Better approach: the template as written: show a message only if it was not shown in the last 10 seconds

QUESTION INFO

QUESTION

Status events arrive from a robot as (time, message). Show a message only if the same message was not shown during the previous 10 seconds; a hidden message does not restart the timer, and exactly 10 seconds later it is shown again. show(t, msg) returns True when the message should be displayed.

Function signature

class Dedup:
    def show(self, t: int, msg: str) -> bool

EXAMPLES

Example 1

Input	`d = Dedup()`
	`d.show(0, 'low battery')`
	`d.show(4, 'low battery')`
	`d.show(6, 'door open')`
	`d.show(9, 'low battery')`
	`d.show(10, 'low battery')`

Output	`d.show(0, 'low battery') -> true`
	`d.show(4, 'low battery') -> false`
	`d.show(6, 'door open') -> true`
	`d.show(9, 'low battery') -> false`
	`d.show(10, 'low battery') -> true`

Why	The repeats at 4 and 9 are hidden and do not move the timer; at time 10 it is exactly 10 seconds after the last display, so it shows again.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Keep every displayed time and scan them

BRUTE-FORCE IDEA

Store the list of every (time, message) shown so far and scan it for the same message within the window. Correct, but the scan grows with the whole history.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Keep every displayed time and scan them
Store the list of every (time
message) shown so far
scan it for the same message within the window
```

BRUTE-FORCE COMPLEXITY | Time O(displayed) per event | Space O(displayed)

WHY THIS IS WEAKER | Each event scans the whole history. A map from message to its last displayed time answers each event in O(1).

HOW TO IMPROVE IT

the template as written: show a message only if it was not shown in the last 10 seconds

SHORT BEST SOLUTION

If the message was displayed less than `window` seconds ago hide it without changing the stored time; otherwise display it and store the new time. The comparison time - last < window means exactly 10 seconds later is allowed; change to <= for the other reading.

```
# BASE QUESTION (optimal): Hash map of last displayed time
# Idea: If the message was displayed less than `window`
# seconds ago hide it without changing the stored time;
# otherwise display it and store the new time. The
# comparison time - last < window means exactly 10 seconds
# later is allowed; change to <= for the other reading.
class Dedup:
    def __init__(self, window=10):
        self.window, self.last_shown = window, {}

    def show(self, time, message):
        if message in self.last_shown and time - self.last_shown[message] < self.window:
            return False
        self.last_shown[message] = time
        return True
```

COMPLEXITY | Time O(1) per event | Space O(distinct messages)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 81

Binary search on a yes/no condition

PATTERN CONTEXT

If answers go no, no, no, yes, yes, yes as the value grows, binary search finds the first yes. Write ok(x), set lo and hi, and halve. The questions only differ in what x is (a length, a speed, a time, an index, a version) and how ok(x) is checked.

89 First Index Reaching a Threshold (Binary Search) EASY

DIFFICULTY | EASY

Better approach: x = an index; ok(x) = sorted_values[x] >= threshold on a sorted array

QUESTION INFO

QUESTION

Given a non-decreasing (sorted ascending, duplicates allowed) array arr and a target value k , return the smallest index i such that arr[i] >= k — the first position where the array "reaches" the value k . If no element is >= k , return -1 . Solve in O(log n) time and O(1) extra space.

Function signature

```
def first_index_at_least(arr: list[int], k: int) -> int:
    ...
```

Follow-up (as reported): The interviewer probed space complexity; note the solution is already O(1) space. If arr is a huge sorted stream/file, describe reading via random access rather than loading it all.

EXAMPLES

Example 1	
Input	`arr = [1, 2, 2, 3, 5, 7], k = 2`
Output	`1`
Why	`arr[1] = 2 is the first element >= 2.`

Example 2	
Input	`arr = [1, 2, 2, 3, 5, 7], k = 4`
Output	`4`
Why	`The first element >= 4 is 5 at index 4.`

Example 3	
Input	`arr = [1, 2, 2, 3, 5, 7], k = 8`
Output	`-1`
Why	`No element is >= 8.`

CONSTRAINTS

- 0 <= len(arr) <= 10**7
- Values and k are integers.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(arr) <= 10**7

- Values and k are integers.

BRUTE FORCE | Scan from the left

BRUTE-FORCE IDEA

Start with: Scan from the left. Start with: Scan from the left. It ignores that the array is sorted.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan from the left
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(1)

WHY THIS IS WEAKER | It ignores that the array is sorted.

HOW TO IMPROVE IT

x = an index; ok(x) = sorted_values[x] >= threshold on a sorted array

SHORT BEST SOLUTION

Halve the range, keeping the boundary between values below k and values at least k.

```python
# BASE QUESTION (optimal): Binary search for the lower
# bound
# Idea: Halve the range, keeping the boundary between
# values below k and values at least k.
def first_index_at_least(sorted_values, threshold):
    low, high = 0, len(sorted_values)
    while low < high:
        middle = (low + high) // 2
        if sorted_values[middle] >= threshold:
            high = middle
        else:
            low = middle + 1
    return low if low < len(sorted_values) else -1
```

COMPLEXITY | Time O(log n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 89

94 Median of Two Sorted Arrays (LC 4) with a Follow-up Variant — HARD

DIFFICULTY | HARD

Better approach: x = how many items come from the shorter array's left side; ok(x) = the left part's largest <= the right part's smallest

QUESTION INFO

QUESTION

Given two sorted arrays of numbers, return the median of all their elements together, without merging them into one array. The median is the middle element when the total count is odd, and the average of the two middle elements when it is even. Aim for O(log(min(m, n))) time.

Function signature

def median_two(a: list, b: list) -> float

EXAMPLES

Example 1	
Input	`[1, 3], [2]`
Output	`2`
Why	`Combined [1, 2, 3]: the middle element is 2.`

QUESTION

Example 2	
Input	`[1, 2], [3, 4]`
Output	`2.5`
Why	`Combined [1, 2, 3, 4]: the two middle elements are 2 and 3, so the median is 2.5.`

Example 3	
Input	`[], [7, 9]`
Output	`8`
Why	`One array may be empty: the median is just the other array's median.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Merge everything and read the middle

BRUTE-FORCE IDEA

Concatenate, sort and take the middle one or two. It is the simplest direct approach to say first. It ignores that the inputs are sorted, and it needs the whole merged array in memory.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Merge everything and read the middle
Concatenate
sort
take the middle one or two
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O((m + n) log(m + n)) | Space O(m + n)

WHY THIS IS WEAKER | Sorting ignores that the inputs are sorted and costs O((m + n) log(m + n)) time and O(m + n) memory. A binary search for the partition reads only O(log min(m, n)) values.

HOW TO IMPROVE IT

x = how many items come from the shorter array's left side; ok(x) = the left part's largest <= the right part's smallest

SHORT BEST SOLUTION

Make a the shorter array. Binary search i in a, set second_cut = half - first_cut in b so the left side holds (first_length + second_length + 1) // 2 elements. A cut is valid when first[first_cut-1] <= second[second_cut] and second[second_cut-1] <= first[first_cut]; use minus and plus infinity at the ends. Odd total: the answer is the larger left value. Even total: the average of the larger left value and the smaller right value.

```python
INF = float("inf")


# BASE QUESTION (optimal): Binary search on the partition
# Idea: Make a the shorter array. Binary search i in a,
# set second_cut = half - first_cut in b so the left side holds (first_length + second_length + 1)
# // 2 elements. A cut is valid when first[first_cut-1] <= second[second_cut] and
# second[second_cut-1] <= first[first_cut]; use minus and plus infinity at the ends.
# Odd total: the answer is the larger left value. Even
# total: the average of the larger left value and the
# smaller right value.
def median_two(first, second):
    if len(first) > len(second):
        first, second = second, first
    first_length, second_length = len(first), len(second)
    if first_length + second_length == 0:
```

```
        raise ValueError("no elements")
    low, high, half = 0, first_length, (first_length + second_length + 1) // 2
    while low <= high:
        first_cut = (low + high) // 2
        second_cut = half - first_cut
        a_left = first[first_cut - 1] if first_cut > 0 else -INF
        a_right = first[first_cut] if first_cut < first_length else INF
        b_left = second[second_cut - 1] if second_cut > 0 else -INF
        b_right = second[second_cut] if second_cut < second_length else INF
        if a_left <= b_right and b_left <= a_right:
            if (first_length + second_length) % 2:
                return max(a_left, b_left)
            return (max(a_left, b_left) + min(a_right, b_right)) / 2
        if a_left > b_right:
            high = first_cut - 1
        else:
            low = first_cut + 1
```

COMPLEXITY | Time O(log min(m, n)) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 94

95 Find longest duplicated substring — HARD

DIFFICULTY | HARD

Better approach: x = the substring length; ok(x) = some substring of that length appears twice (rolling hash); search the largest ok

QUESTION INFO

QUESTION

You are given a string **s** consisting of lowercase English letters. A **substring** of **s** is any contiguous sequence of characters, i.e., **`s[i..j)`** for **`0 <= i < j <= len(s)`**. Design an algorithm to find **any one** of the longest substrings that appears **at least twice** in **s**. The two occurrences may overlap. If no substring appears at least twice, return the empty string.

Requirements

- Input: a string **s** of length **n** (e.g., **`1 <= n <= 2 × 10^5`**).
- Output: a string representing one of the longest duplicated substrings.
- Aim for an algorithm substantially better than the naive **`O(n^2)`** substring comparison approach.

Examples

- **`s = "banana"`**
- Possible duplicated substrings: "a", "an", "ana" - Longest length is 3 ("ana"), so acceptable outputs include **`"ana"`**.
- **`s = "abcd"`**
- No duplicated substring, so output should be **`""`** (empty string). Describe your algorithm, its time and space complexity, and how it scales for large **n**.

EXAMPLES

Example 1	
Input	`'banana'`
Output	`'ana'`
Why	`'ana' occurs twice (overlapping is allowed).`

QUESTION

Example 2	
Input	`'abcd'`
Output	`''`
Why	`No substring repeats.`

Example 3	
Input	`'aaaaa'`
Output	`'aaaa'`
Why	`'aaaa' appears at positions 0 and 1.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Check every length from longest down with a set of substrings

BRUTE-FORCE IDEA

For each length from n - 1 down, put all substrings of that length in a set until one repeats. Correct; every check builds up to n substrings of length up to n.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Check every length from longest down with a set of substrings
For each length from n - 1 down
put all substrings of that length in a set until one repeats
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^3) | Space O(n^2)

WHY THIS IS WEAKER | Each length builds up to n substrings of up to n characters: O(n^3) time and O(n^2) space. If a duplicate of length L exists then so does one of every shorter length, so binary search on L with a rolling hash is O(n log n).

HOW TO IMPROVE IT

x = the substring length; ok(x) = some substring of that length appears twice (rolling hash); search the largest ok

SHORT BEST SOLUTION

For a fixed length L compute the polynomial hash of every window in O(1) each by sliding it, store hashes with their start positions, and when a hash repeats compare the actual substrings (to rule out a collision). Binary search the largest L for which a duplicate exists. A suffix array with the LCP array gives a deterministic O(text_length log text_length) alternative.

```python
# BASE QUESTION (optimal): Binary search on the length
# with a rolling hash (Rabin-Karp)
# Idea: For a fixed length L compute the polynomial hash
# of every window in O(1) each by sliding it, store hashes
# with their start positions, and when a hash repeats
# compare the actual substrings (to rule out a collision).
# Binary search the largest L for which a duplicate
# exists. A suffix array with the LCP array gives a
# deterministic O(text_length log text_length) alternative.
def longestDupSubstring(text):
    text_length, base, modulus = len(text), 131, (1 << 61) - 1
    codes = [ord(char) for char in text]

    def find(window_length):
        window_hash, power = 0, pow(base, window_length, modulus)
        for window_start in range(window_length):
            window_hash = (window_hash * base + codes[window_start]) % modulus
        seen = {window_hash: [0]}
        for window_start in range(1, text_length - window_length + 1):
```

```
            window_hash = (window_hash * base - codes[window_start - 1] * power + codes[window_start + window_l
ength - 1]) % modulus
            for earlier_start in seen.get(window_hash, ()):
                if text[earlier_start:earlier_start + window_length] == text[window_start:window_start + window
_length]:
                    return text[window_start:window_start + window_length]
            seen.setdefault(window_hash, []).append(window_start)
        return None

    low_length, high_length, best = 1, text_length - 1, ""
    while low_length <= high_length:
        trial_length = (low_length + high_length) // 2
        duplicate = find(trial_length)
        if duplicate:
            best, low_length = duplicate, trial_length + 1
        else:
            high_length = trial_length - 1
    return best
```

COMPLEXITY | Time O(n log n) expected | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 95

96 Find the median of two sorted arrays. HARD

DIFFICULTY | HARD

Better approach: the same partition search on the shorter array

QUESTION INFO

QUESTION

Given two sorted arrays, return the median of all their elements together. The median is the middle value when the total count is odd and the average of the two middle values when it is even. The best solutions run in O(log(min(m, n))).

Function signature

def findMedianSortedArrays(nums1: list, nums2: list) -> float

EXAMPLES

Example 1	
Input	[1, 3], [2]
Output	2
Why	Combined: 1, 2, 3.

Example 2	
Input	[1, 2], [3, 4]
Output	2.5
Why	Combined: 1, 2, 3, 4: the middle two are 2 and 3.

Example 3	
Input	[], [5]
Output	5
Why	One empty array.

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Merge into one sorted list and read the middle

BRUTE-FORCE IDEA

Concatenate, sort and take the middle (or the average of the two middles). Correct and the first thing to say, but it ignores that the inputs are already sorted and costs linear memory.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Merge into one sorted list and read the middle
Concatenate
take the middle (or the average of the two middles)
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O((m + n) log(m + n)) | Space O(m + n)

WHY THIS IS WEAKER | It sorts m + n numbers from scratch, O((m + n) log(m + n)), and stores them all, even though both inputs were already sorted. The partition binary search needs only O(log min(m, n)).

HOW TO IMPROVE IT

the same partition search on the shorter array

SHORT BEST SOLUTION

Binary search how many elements of the shorter array go in the left half. A partition is right when every left element is <= every right element, which only needs the four border values to be checked.

```python
# BASE QUESTION (optimal): Binary search on the partition
# Idea: Binary search how many elements of the shorter
# array go in the left half. A partition is right when
# every left element is <= every right element, which only
# needs the four border values to be checked.
def findMedianSortedArrays(nums1, nums2):
    shorter, longer = (nums1, nums2) if len(nums1) <= len(nums2) else (nums2, nums1)
    shorter_length, longer_length = len(shorter), len(longer)
    half = (shorter_length + longer_length + 1) // 2
    low, high = 0, shorter_length
    while low <= high:
        shorter_cut = (low + high) // 2
        longer_cut = half - shorter_cut
        a_left = shorter[shorter_cut - 1] if shorter_cut > 0 else float("-inf")
        a_right = shorter[shorter_cut] if shorter_cut < shorter_length else float("inf")
        b_left = longer[longer_cut - 1] if longer_cut > 0 else float("-inf")
        b_right = longer[longer_cut] if longer_cut < longer_length else float("inf")
        if a_left <= b_right and b_left <= a_right:
            if (shorter_length + longer_length) % 2:
                return float(max(a_left, b_left))
            return (max(a_left, b_left) + min(a_right, b_right)) / 2
        if a_left > b_right:
            high = shorter_cut - 1
        else:
            low = shorter_cut + 1
    raise ValueError("inputs must be sorted")
```

COMPLEXITY | Time O(log(min(m, n))) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 96

GROUP 12 OF 43 | 3 QUESTIONS | PART B - PART 1

Greedy: scan once, carry the best choice so far, act only when forced

PATTERN CONTEXT

Decide locally and never look back: sort or scan, carry one value (how far you can reach, how far you are covered, the largest seen so far), and take an action only at the moment the current choice stops being enough.

97 Alarm for the Next Upcoming Birthday — EASY

DIFFICULTY | EASY

Better approach: minimum over people of the circular gap (their day - today) modulo 365, ties by name

QUESTION INFO

QUESTION

You maintain a list of people and their birthdays. Given today's date and a list of (name, month, day) records, return the name of the person whose birthday comes soonest starting from today. Birthdays recur every year, so if no birthday remains later this calendar year, the soonest one is the earliest birthday next year. A birthday that falls exactly on today counts as 0 days away (soonest). Ignore leap years; treat every date simply as a (month, day) pair. If two people share the soonest birthday, return the one that is alphabetically first by name.

Function signature

```
def next_birthday(today: tuple[int, int], people: list[tuple[str, int, int]]) -> str:
    ...
```

EXAMPLES

Example 1	
Input	`today = (10, 7), people = [("Alice", 10, 9), ("Bob", 1, 15), ("Carol", 12, 25)]`
Output	`"Alice"`
Why	`Alice's birthday is 2 days away; Carol's is later this year; Bob's is next January. Alice is soonest.`

Example 2	
Input	`today = (12, 31), people = [("Alice", 1, 1), ("Dan", 12, 31)]`
Output	`"Dan"`
Why	`Dan's birthday is today (0 days away), which beats Alice's tomorrow.`

Example 3	
Input	`today = (6, 1), people = [("Eve", 5, 30), ("Finn", 5, 30)]`
Output	`"Eve"`
Why	`Both birthdays already passed this year, so both are ~next-year and tie; alphabetical order picks "Eve".`

CONSTRAINTS

- 1 <= len(people) <= 10**4
- 1 <= month <= 12 , 1 <= day <= 31 (assume valid calendar dates). today is a valid (month, day) pair.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(people) <= 10**4
- 1 <= month <= 12 , 1 <= day <= 31 (assume valid calendar dates). today is a valid (month, day) pair.

BRUTE FORCE | Walk the calendar day by day

BRUTE-FORCE IDEA

Start with: Walk the calendar day by day. It checks every day of the year against every person.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Walk the calendar day by day
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(365 * p) | Space O(p)

WHY THIS IS WEAKER | It checks every day of the year against every person.

HOW TO IMPROVE IT

minimum over people of the circular gap (their day - today) modulo 365, ties by name

SHORT BEST SOLUTION

Days until the birthday is (doy - today) mod 365; take the smallest, ties by name.

```python
MONTH_DAYS = [31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31]      # a non-leap year


# BASE QUESTION (optimal): Day of the year and a cyclic
# minimum
# Idea: Days until the birthday is (doy - today) mod 365;
# take the smallest, ties by name.
def day_of_year(month, day):
    return sum(MONTH_DAYS[:month - 1]) + day


def next_birthday(today, people):
    today_number = day_of_year(*today)
    return min(people, key=lambda person: ((day_of_year(person[1], person[2]) - today_number) % 365, person[0])
)[0]
```

COMPLEXITY | Time O(p) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 97

98 Coins on the Table — EASY

DIFFICULTY | EASY

Better approach: scan the events with one counter: add on '+', subtract on '-' but never below 0

QUESTION INFO

QUESTION

There is a table that starts with 0 coins on it. You are given a string events , where each character is a single event applied left to right: '+' places one coin on the table. '-' removes one coin from the table. The table can never hold a negative number of coins: a '-' event when the table is empty has no effect (it is ignored). After processing every event, return the number of coins on the table.

Function signature

def coins_on_table(events: str) -> int:

    ...

Follow-up (as reported): Generalize so each event is an integer delta (e.g. "+3" , "-2"), still clamping the running total at 0. Hint: parse tokens instead of single characters.

EXAMPLES

QUESTION

Example 1	
Input	`events = "+++-"`
Output	`2`
Why	`+1, +1, +1, then -1 -> 2 coins remain.`

Example 2	
Input	`events = "----"`
Output	`0`
Why	`Every removal happens on an empty table and is ignored.`

Example 3	
Input	`events = "+-+-+"`
Output	`1`
Why	`1, 0, 1, 0, 1 -> 1 coin remains.`

CONSTRAINTS

- 0 <= len(events) <= 10^5
- Every character of events is either '+' or '-' .
- An empty string returns 0 .

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(events) <= 10^5
- Every character of events is either '+' or '-' .
- An empty string returns 0 .

BRUTE FORCE | A list of coins

BRUTE-FORCE IDEA

Start with: A list of coins. Start with: A list of coins. It stores every coin although only their number matters.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: A list of coins
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | It stores every coin although only their number matters.

HOW TO IMPROVE IT

scan the events with one counter: add on '+', subtract on '-' but never below 0

SHORT BEST SOLUTION

total = max(0, total + delta) for every event.

```python
# BASE QUESTION (optimal): One clamped counter
# Idea: total = max(0, total + delta) for every event.
def coins_on_table(events):
    total = 0
    for event in events:
        total = total + 1 if event == "+" else max(0, total - 1)
    return total
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 98

99 Maximum Profit — EASY

DIFFICULTY | EASY

Better approach: pick the cheapest and the dearest city with a fixed tie-break (min and max once)

QUESTION INFO

QUESTION

You and your friend have decided to start an international trading business. You want to earn a profit through some ol' fashioned arbitrage: You'll buy widgets in one country and sell them in another at a higher price. Your friend has already found the current market prices for widgets in cities around the world, and you'd like to use this data to pick which cities to buy and sell widgets in. Write a function max_profit(prices) that efficiently finds the two cities that maximize your profit (i.e. largest difference in prices).
Input: A dictionary with cities as keys and prices as values Output: An array containing the names of the cities (min, max) If the prices dictionary is empty, return None/null (depending on the programming language used). If there is 0 profit to be earned, return None/null as well. If you are using C++, return {"", ""} in such cases. Example Pythonprices = {'London': 72, 'New York': 70, 'Tokyo': 67, 'Miami': 62} max_profit(prices)

=> ['Miami', 'London']Hint 1You can think of this as two sub-problems

: Finding the lowest price, and finding the highest price.Hint 2While you could compare every pair of cities, it's more efficient to find the min and max values.Hint 3You can keep track of the lowest and highest values and update them as you iterate through the list.Challenge 1Does your solution work in O(N) time and O(1) space? If not, you can be more efficient!Challenge 2What happens if the price is the same in every city? Does your function still return

EXAMPLES

Example 1	
Input	`{'A': 3, 'B': 10, 'C': 1}`
Output	`9`
Why	`Buy in the cheapest city (C, price 1) and sell in the dearest (B, price 10): profit 9.`

Example 2	
Input	`{'A': 5}`
Output	`0`
Why	`With a single city there is nothing to trade.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every ordered pair of cities

BRUTE-FORCE IDEA

Compute price[sell] - price[buy] for every pair of different cities and keep the best. Correct, and quadratic.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every ordered pair of cities
Compute price[sell] - price[buy] for every pair of different cities
keep the best
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1)

WHY THIS IS WEAKER | N^2 pairs. Since city order does not matter, the best pair is the cheapest city and the dearest city, found in one O(n) scan.

HOW TO IMPROVE IT

pick the cheapest and the dearest city with a fixed tie-break (min and max once)

SHORT BEST SOLUTION

Find the city with the lowest price and the city with the highest price in one scan (ties by name). They are the best pair because profit is sell minus buy and the order of cities does not matter. Handle all-equal prices and fewer than two cities.

```
# BASE QUESTION (optimal): One pass for the cheapest and
# the dearest city
# Idea: Find the city with the lowest price and the city
# with the highest price in one scan (ties by name). They
# are the best pair because profit is sell minus buy and
# the order of cities does not matter. Handle all-equal
# prices and fewer than two cities.
def max_profit(prices):
    if len(prices) < 2:
        raise ValueError("need at least two cities")
    cheapest_city = min(prices, key=lambda city: (prices[city], city))
    dearest_city = min(prices, key=lambda city: (-prices[city], city))
    if cheapest_city == dearest_city:
        return sorted(prices)[:2]
    return [cheapest_city, dearest_city]
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 99

GROUP 13 OF 43 | 5 QUESTIONS | PART B - PART 1

Sets and counters: remember the first collection, check the second against it

PATTERN CONTEXT

Build a set (or a Counter) from one collection, then walk the other and ask 'is this in it?'. Membership in a set is O(1), which is the whole trick. Duplicates, intersections, filters, and missing values are the same loop with a different question.

105 Exclude First-K-Unique From Another List — EASY

DIFFICULTY | EASY

Better approach: the set holds the first k distinct values of list 1; keep list 2's values that are not in it

QUESTION INFO

QUESTION

You are given two lists list1 and list2 and an integer k . Scan list1 left to right and collect the first k distinct values it contains (in order of first appearance; if list1 has fewer than k distinct values, take all of them). Then return list2 with every occurrence of any of those collected values removed, preserving the order of the remaining elements.

Function signature

```
def exclude_first_k_unique(list1: list[int], list2: list[int], k: int) -> list[int]:
    ...
```

EXAMPLES

Example 1	
Input	`list1 = [3, 3, 1, 4, 1, 5], list2 = [1, 2, 3, 4, 3, 5], k = 2`
Output	`[2, 4, 5]`
Why	`First 2 distinct values of list1 are {3, 1}. Removing all 3s and 1s from list2 leaves [2, 4, 5].`

Example 2	
Input	`list1 = [7], list2 = [7, 8, 9], k = 3`
Output	`[8, 9]`
Why	`list1 has only one distinct value (7); remove all 7s from list2.`

CONSTRAINTS

- 0 <= len(list1), len(list2) <= 10**5 , 1 <= k .
- Values are integers.

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= len(list1), len(list2) <= 10**5 , 1 <= k .
- Values are integers.

BRUTE FORCE | Collect into a list and scan it

BRUTE-FORCE IDEA

Start with: Collect into a list and scan it. Each element of list2 searches the whole collected list linearly.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Collect into a list and scan it
Start with: Collect into a list
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n + m * k) | Space O(k)

WHY THIS IS WEAKER | Each element of list2 searches the whole collected list linearly.

HOW TO IMPROVE IT

the set holds the first k distinct values of list 1; keep list 2's values that are not in it

SHORT BEST SOLUTION

Membership in a set makes the filter linear.

```python
# BASE QUESTION (optimal): A set of the first k distinct
# values
# Idea: Membership in a set makes the filter linear.
def exclude_first_k_unique(list1, list2, distinct_count):
    blocked = set()
    for value in list1:
        if len(blocked) == distinct_count:
            break
        blocked.add(value)
    return [value for value in list2 if value not in blocked]
```

COMPLEXITY | Time O(n + m) | Space O(k)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 105

106 Positional Matches Between Two Lists — EASY

DIFFICULTY | EASY

Better approach: no set needed: zip both lists and keep a where a == b

QUESTION INFO

QUESTION

You are given two lists of equal length, first and second . Return, in order, the values at every position where the two lists agree — that is, all first[i] such that first[i] == second[i] .

Function signature

def positional_matches(first: list[int], second: list[int]) -> list[int]:

...

EXAMPLES

Example 1	
Input	`first = [3, 1, 4, 1, 5, 5, 7, 9], second = [1, 2, 3, 4, 5, 6, 7, 8]`
Output	`[5, 7]`
Why	`Positions 4 (5 == 5) and 6 (7 == 7) match. (The report described "matches" between a digits-of-pi list and an index list.)`

Example 2	
Input	`first = [1, 2, 3], second = [4, 5, 6]`
Output	`[]`
Why	`No position matches.`

CONSTRAINTS

- len(first) == len(second) , 0 <= length <= 10**6 .

QUESTION

- Values are integers.

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- len(first) == len(second) , 0 <= length <= 10**6 .
- Values are integers.

DIRECT APPROACH | Loop over the indexes

DIRECT IDEA

Start with: Loop over the indexes. Correct and linear; it is only longer than the zip form.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Loop over the indexes
process every required item or candidate directly
keep the result required by the question
```

DIRECT-APPROACH COMPLEXITY | Time O(n) | Space O(matches)

WHY THIS IS WEAKER | Correct and linear; it is only longer than the zip form.

HOW TO IMPROVE IT

no set needed: zip both lists and keep a where a == b

SHORT BEST SOLUTION

Walk both lists together and keep the agreeing values.

```python
# BASE QUESTION (optimal): zip and a filter
# Idea: Walk both lists together and keep the agreeing
# values.
def positional_matches(first, second):
    return [first_value for first_value, second_value in zip(first, second) if first_value == second_value]
```

COMPLEXITY | Time O(n) | Space O(matches)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 106

107 Contains Duplicate EASY

DIFFICULTY | EASY

Better approach: add values to a set as you go; the first value already in the set means a duplicate

QUESTION INFO

QUESTION

Given an integer array nums , return True if any value appears at least twice in the array, and False if every element is distinct.

Function signature

def contains_duplicate(nums: list[int]) -> bool:
    """Return True if any value appears more than once."""
    ...

EXAMPLES

QUESTION

Example 1	
Input	`nums = [1, 2, 3, 1]`
Output	`True`

Example 2	
Input	`nums = [1, 2, 3, 4]`
Output	`False`

Example 3	
Input	`nums = [1, 1, 1, 3, 3, 4, 3, 2, 4, 2]`
Output	`True`

CONSTRAINTS

- 1 <= len(nums) <= 10^5
- -10^9 <= nums[i] <= 10^9

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(nums) <= 10^5
- -10^9 <= nums[i] <= 10^9

BRUTE FORCE | Compare every pair

BRUTE-FORCE IDEA

Start with: Compare every pair. Start with: Compare every pair. Each element is compared with all the later ones.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compare every pair
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1)

WHY THIS IS WEAKER | Each element is compared with all the later ones.

HOW TO IMPROVE IT

add values to a set as you go; the first value already in the set means a duplicate

SHORT BEST SOLUTION

Add each value to a set; seeing one again means a duplicate.

```python
# BASE QUESTION (optimal): A set of seen values
# Idea: Add each value to a set; seeing one again means a
# duplicate.
def contains_duplicate(nums):
    seen = set()
    for value in nums:
        if value in seen:
            return True
        seen.add(value)
    return False
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 107

108 Find the Duplicates EASY

DIFFICULTY | EASY

Better approach: values that appear in both lists, each reported once (set intersection, or two pointers on sorted lists)

QUESTION INFO

QUESTION

Given two sorted arrays arr1 and arr2 of passport numbers, implement a function findDuplicates that returns an array of all passport numbers that are both arrays. Note that the output array should be sorted in ascending order. Let N and M be the lengths of arr1 and arr2, respectively. Solve for two cases and analyze the time & space complexities of your solutions: M ~ N - the array lengths are approximately the same M >> N - arr2 is much bigger than arr1. Example input: arr1 = [1, 2, 3, 5, 6, 7], arr2 = [3, 6, 7, 8, 20] output: [3, 6, 7]

since only these three values

are both in arr1 and arr2Hint 1Start with a brute force solution and optimize from there.Hint 2Note that the question asks for a sorted array. What techniques work well on sorted arrays?Hint 3If M >> N (case #2) is giving you trouble, consider how you might use the values of N, the shorter array, to increase the efficiency of your answer.Challenge 1Have you taken care to avoid array out-of-bound runtime errors?First, the brute force solution. Try two nested loops. For every value arr1[i] in the outer loop, we look it up in arr2 in the inner loop. The time complexity of this algorithm in the worst case scenario is O(N*M). Not very efficient. We can do better than that. Case 1 (M ~ N) We can use the fact the arrays are sorted to traverse both of them in an in-order manner at the same time. The general idea of the algorithm is to use two indices, i and j, for arr1 and arr2, respectively. Every time one of indices,

EXAMPLES

Example 1	
Input	`[1, 3, 5, 7], [3, 4, 5, 8]`
Output	`[3, 5]`
Why	`Two sorted lists of passport numbers: 3 and 5 appear in both.`

Example 2	
Input	`[1, 2], [3, 4]`
Output	`[]`
Why	`Nothing in common.`

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Compare every pair

BRUTE-FORCE IDEA

For each element of the first array scan the whole second array. Correct and direct, and far too slow.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compare every pair
For each element of the first array scan the whole second array
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(N * M) | Space O(1) extra

WHY THIS IS WEAKER | N * M comparisons. Two pointers on the sorted arrays are O(N + M), and binary searching the big array for each element of the small one is O(N log M).

HOW TO IMPROVE IT

values that appear in both lists, each reported once (set intersection, or two pointers on sorted lists)

SHORT BEST SOLUTION

Part 1: Walk both sorted arrays with two pointers.

Part 2: Binary-search each distinct value from the smaller list in the larger list.

```
from bisect import bisect_left
from math import log2


def common_two_pointers(first, second):
    answer, first_index, second_index = [], 0, 0
    while first_index < len(first) and second_index < len(second):
        if first[first_index] < second[second_index]:
            first_index += 1
        elif first[first_index] > second[second_index]:
            second_index += 1
        else:
            if not answer or answer[-1] != first[first_index]:
                answer.append(first[first_index])
            first_index += 1
            second_index += 1
    return answer


def common_binary_search(small, big):
    answer, start = [], 0
    for value in small:
        start = bisect_left(big, value, start)
        if start == len(big):
            break
        if big[start] == value and (not answer or answer[-1] != value):
            answer.append(value)
    return answer


def common(first, second):
    small, big = (first, second) if len(first) <= len(second) else (second, first)
    if len(small) * log2(max(2, len(big))) < len(first) + len(second):
        return common_binary_search(small, big)
    return common_two_pointers(first, second)
```

COMPLEXITY | Part 1: Time O(n + m) | Space O(1) excluding output
Part 2: Time O(s log b) | Space O(1) excluding output

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 108

112 Smallest Missing Positive Integer in Linear Time and Constant Extra Space

HARD

DIFFICULTY | HARD

Better approach: use the array itself as the set: swap each value x into slot x - 1, then the first slot with the wrong value is the answer

QUESTION INFO

QUESTION

You are given an unsorted list of integers **nums**, which may contain negative numbers, zeros, duplicates, and values far larger than the length of the list. Return the smallest positive integer, that is the smallest integer that is at least **1**, that does not appear in **nums**. In the interview, a first solution that ran in linear time but used extra memory proportional to the input was accepted as a starting point, and the interviewer then asked for the same result using only constant extra space. Aim for that tighter target.

Function Signature

QUESTION

```python
def first_missing_positive(nums: list[int]) -> int:
```

Rules

- Target complexity: **`O(n)`** time and **`O(1)`** auxiliary space, where **`n = len(nums)`**. The list may be rearranged or overwritten in place; its contents after the call do not matter.
- Return only the missing value.

Constraints

- **`1 <= len(nums) <= 10^5`**
- **`-2147483648 <= nums[i] <= 2147483647`**
- The answer is a positive integer that fits in a signed 32-bit integer.

Examples

Example 1

- Input: **`nums = [4, -2, 1, 2]`**
- Output: **`3`**
- Explanation: **`1`** and **`2`** are present and **`3`** is not.

Example 2

- Input: **`nums = [5, 6, 100, -3]`**
- Output: **`1`**
- Explanation: **`1`** does not appear, so it is the answer, however large the other values are.

Example 3

- Input: **`nums = [1, 2, 2, 3]`**
- Output: **`4`**
- Explanation: **`1`**, **`2`** and **`3`** all appear (**`2`** twice), so the smallest missing positive integer is **`4`**.

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Hash set of the values, then count up

BRUTE-FORCE IDEA

Put all numbers in a set and test 1, 2, 3, and so on until one is missing. Linear time but linear extra memory, which is the version the interviewer wants to improve.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Hash set of the values, then count up
Put all numbers in a set
test 1
so on until one is missing
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | O(n) time but O(n) extra memory, which is what the follow-up asks to remove. Placing each value v at index v - 1 inside the array itself, then finding the first mismatch, uses O(1) space.

HOW TO IMPROVE IT

use the array itself as the set: swap each value x into slot x - 1, then the first slot with the wrong value is the answer

SHORT BEST SOLUTION

For each position, while nums[position] is in 1..n and not already at its home index, swap it to index nums[position] - 1. Each swap puts one value in its final place, so the total number of swaps is at most n. Then scan for the first index whose value is not

position + 1; if none, the answer is length + 1.

```python
# BASE QUESTION (optimal): Cyclic placement: put each
# value v at index v - 1
# Idea: For each position, while nums[position] is in 1..n and
# not already at its home index, swap it to index nums[position]
# - 1. Each swap puts one value in its final place, so the
# total number of swaps is at most n. Then scan for the
# first index whose value is not position + 1; if none, the
# answer is length + 1.
def firstMissingPositive(nums):
    values, length = list(nums), len(nums)
    for position in range(length):
        while 1 <= values[position] <= length and values[values[position] - 1] != values[position]:
            home_index = values[position] - 1
            values[position], values[home_index] = values[home_index], values[position]
    for position in range(length):
        if values[position] != position + 1:
            return position + 1
    return length + 1
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 112

GROUP 14 OF 43 | 3 QUESTIONS | PART B - PART 1

bisect on a sorted list: find a position, count a range, keep it sorted

PATTERN CONTEXT

bisect_left and bisect_right give the position of a value in a sorted list in O(log n). Subtract two positions to count a range, slice between them to list a range, insort to keep inserting in sorted order, and replace an element at a position for the 'longest increasing' trick.

113 Count People in a Postal Code Range

EASY

DIFFICULTY | EASY

Better approach: sort the codes once; the answer is the template as written, never below 0

QUESTION INFO

QUESTION

You are given a list of people, each as a tuple (name, postal_code) . Build a structure that can answer many queries of the form: "how many people have a postal code within the inclusive range [low, high] ?" The interviewer explicitly wanted the optimal query implementation — not a linear scan of the people (or a scan over the range and a hash-map lookup for each value). Preprocess once, then answer each query in O(log n) . Implement a class with: a constructor that accepts the list of people, and count_in_range(low, high) returning the number of people whose postal code is in [low, high] inclusive.

Function signature

```
from typing import List, Tuple
class PostalRegistry:
    def __init__(self, people: List[Tuple[str, int]]) -> None:
        """Preprocess the (name, postal_code) pairs for fast range counts."""
        ...
    def count_in_range(self, low: int, high: int) -> int:
        """Return how many people have postal_code in [low, high] inclusive."""
        ...
```

Follow-up (as reported): Also return the list of matching names. Hint: sort by (postal_code, name) and slice between bisect_left(low) and bisect_right(high) .

EXAMPLES

Example 1	
Input	`people = [("Alice", 90210), ("Bob", 10001), ("Cara", 90211), ("Dan", 30301)]` `count_in_range(90000, 99999)`
Output	`2`
Why	`Alice (90210) and Cara (90211) fall in the range.`

Example 2	
Input	`same people; count_in_range(10001, 30301)`
Output	`2`
Why	`Bob (10001) and Dan (30301); range is inclusive on both ends.`

QUESTION

Example 3	
Input	`same people; count_in_range(50000, 60000)`
Output	`0`
Why	`No postal codes in that range.`

CONSTRAINTS

- 0 <= number of people <= 10^5 ; the same postal code may appear for multiple people (duplicates allowed).
- Postal codes are integers, 0 <= postal_code <= 10^9 .
- 0 <= low <= high <= 10^9 .
- Each count_in_range call must run in O(log n) after an O(n log n) preprocessing step.

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= number of people <= 10^5 ; the same postal code may appear for multiple people (duplicates allowed).
- Postal codes are integers, 0 <= postal_code <= 10^9 .
- 0 <= low <= high <= 10^9 .
- Each count_in_range call must run in O(log n) after an O(n log n) preprocessing step.

BRUTE FORCE | Scan the people for every query

BRUTE-FORCE IDEA

Start with: Scan the people for every query. Start with: Scan the people for every query. Every query touches every person.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan the people for every query
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n) per query | Space O(1)

WHY THIS IS WEAKER | Every query touches every person.

HOW TO IMPROVE IT

sort the codes once; the answer is the template as written, never below 0

SHORT BEST SOLUTION

The codes in the range form one block of the sorted list.

```python
# BASE QUESTION (optimal): Sorted codes and two bisects
# Idea: The codes in the range form one block of the
# sorted list.
from bisect import bisect_left, bisect_right


class PostalRegistry:
    def __init__(self, people):
        self.codes = sorted(code for _, code in people)

    def count_in_range(self, low, high):
        return max(0, bisect_right(self.codes, high) - bisect_left(self.codes, low))
```

COMPLEXITY | Time O(log n) per query | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 113

119 Find Maximum Envelope Nesting HARD

DIFFICULTY | HARD

Better approach: exactly the same sort and tails list

QUESTION INFO

QUESTION

Envelopes are given as (width, height). One envelope fits inside another only when both its width and its height are strictly smaller. Return the largest number of envelopes that can be nested one inside the next (Russian dolls).

Function signature

def max_envelopes(envelopes: list) -> int

EXAMPLES

Example 1	
Input	`[(5, 4), (6, 4), (6, 7), (2, 3)]`
Output	`3`
Why	`(2, 3) fits in (5, 4), which fits in (6, 7): 3 nested.`

Example 2	
Input	`[(1, 1), (1, 1), (1, 1)]`
Output	`1`
Why	`Equal envelopes cannot go inside each other.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | DP over pairs: longest chain ending at each envelope

BRUTE-FORCE IDEA

Sort by width, then best[i] is 1 plus the best best[j] over earlier envelopes that are strictly smaller in both dimensions. This takes quadratic time.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
sort envelopes by width
best[i] = 1 for every envelope
for each i, check every earlier j
if envelope j fits inside i: best[i] = max(best[i], best[j] + 1)
return max(best)
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Every envelope compares with all earlier ones, O(n^2). Sorting by width ascending and height descending, then a longest increasing subsequence on heights with binary search, is O(n log n).

HOW TO IMPROVE IT

exactly the same sort and tails list

SHORT BEST SOLUTION

Sorting equal widths by decreasing height means two envelopes of the same width can never both appear in a strictly increasing height sequence. The heights are then fed to the standard LIS: keep the smallest possible tail for each length and replace with bisect_left.

```python
from bisect import bisect_left
```

```python
# BASE QUESTION (optimal): Sort by width ascending and
# height descending, then LIS by binary search
# Idea: Sorting equal widths by decreasing height means
# two envelopes of the same width can never both appear in
# a strictly increasing height sequence. The heights are
# then fed to the standard LIS: keep the smallest possible
# tail for each length and replace with bisect_left.
def maxEnvelopes(envelopes):
    tails = []
    for _, height in sorted(envelopes, key=lambda envelope: (envelope[0], -envelope[1])):
        position = bisect_left(tails, height)
        if position == len(tails):
            tails.append(height)
        else:
            tails[position] = height
    return len(tails)
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 119

120 Count of Smaller Numbers After Self

HARD

DIFFICULTY | HARD

Better approach: go from the right, insort each number into a sorted list; the count is bisect_left of the number before inserting (a Fenwick tree is the fast version)

QUESTION INFO

QUESTION

Given an integer array nums, return an array counts where counts[i] is the number of elements to the right of nums[i] that are strictly smaller than nums[i].

Function signature

def count_smaller_after(nums: list) -> list

EXAMPLES

Example 1	
Input	`[5, 2, 6, 1]`
Output	`[2, 1, 1, 0]`
Why	`5 has two smaller numbers to its right (2 and 1), 2 has one (1), 6 has one (1), 1 has none.`

Example 2	
Input	`[-1]`
Output	`[0]`
Why	`A single number has nothing to its right.`

Example 3	
Input	`[-1, -1]`
Output	`[0, 0]`
Why	`Equal numbers are not smaller.`

CONSTRAINTS

- 1 <= nums.length <= 10^5
- -10^4 <= nums[i] <= 10^4

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= nums.length <= 10^5
- -10^4 <= nums[i] <= 10^4

BRUTE FORCE | For each element count smaller elements to its right

BRUTE-FORCE IDEA

Start with: For each element count smaller elements to its right. Two nested loops. Correct; too slow for n = 10^5.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: For each element count smaller elements to its right
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1) extra

WHY THIS IS WEAKER | Two nested loops, O(n^2): too slow at n = 10^5. A Fenwick tree over ranks scanned right to left (or merge sort) is O(n log n).

HOW TO IMPROVE IT

go from the right, insert each number into a sorted list; the count is bisect_left of the number before inserting (a Fenwick tree is the fast version)

SHORT BEST SOLUTION

Compress values to ranks 1..m. Scan from the right: the answer for x is the number of already inserted values with rank less than rank(value) (a prefix sum query), then insert rank(value). Both operations are O(log rank_count).

```python
# BASE QUESTION (optimal): Fenwick tree over compressed
# values, scanning right to left
# Idea: Compress values to ranks 1..m. Scan from the
# right: the answer for x is the number of already
# inserted values with rank less than rank(value) (a prefix
# sum query), then insert rank(value). Both operations are
# O(log rank_count).
def countSmaller(nums):
    ranks = {value: rank + 1 for rank, value in enumerate(sorted(set(nums)))}
    rank_count = len(ranks)
    tree = [0] * (rank_count + 1)
    out = []
    for value in reversed(nums):
        rank_index, total = ranks[value] - 1, 0
        while rank_index > 0:
            total += tree[rank_index]
            rank_index -= rank_index & -rank_index
        out.append(total)
        rank_index = ranks[value]
        while rank_index <= rank_count:
            tree[rank_index] += 1
            rank_index += rank_index & -rank_index
    return out[::-1]
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 120

GROUP 15 OF 43 | 3 QUESTIONS | PART B - PART 1

Prefix sums, Tier 1: precompute once, answer ranges and balance points in O(1)

PATTERN CONTEXT

pre[i] is the total of the first i items. Any range sum is pre[r + 1] - pre[l], and a balance point is an index where the left total equals the right total. Count something other than values (equal pairs, ones, repeats) by accumulating that instead.

121 Cake Balance Line — EASY

DIFFICULTY | EASY

Better approach: the same balance test on the cake heights

QUESTION INFO

QUESTION

A cake is described by a list heights , where heights[i] is the amount of cake sitting at position i (positions are left to right). You want to place a single vertical cut exactly at some position i so that the total amount of cake strictly to the left of the cut equals the total amount strictly to the right of the cut (the cake at position i itself is consumed by the cut and counts for neither side). Return the smallest index i where sum(heights[:i]) == sum(heights[i+1:]) , or -1 if no such position exists.

Function signature

```
def cake_balance_index(heights: list[int]) -> int:
    ...
```

Follow-up (as reported): Do it in one pass and O(1) extra space by tracking a running left sum against total - left - heights[i] .

EXAMPLES

Example 1	
Input	`heights = [1, 7, 3, 6, 5, 6]`
Output	`3`
Why	`Left of index 3 is 1+7+3=11; right of index 3 is 5+6=11.`

Example 2	
Input	`heights = [1, 2, 3]`
Output	`-1`
Why	`No cut position balances the two sides.`

Example 3	
Input	`heights = [2, 1, -1]`
Output	`0`
Why	`Left of index 0 is empty (0); right is 1+(-1)=0.`

CONSTRAINTS

- 1 <= len(heights) <= 10^5 .
- -10^4 <= heights[i] <= 10^4 (heights may be negative in this abstracted version).

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest

input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(heights) <= 10^5 .
- -10^4 <= heights[i] <= 10^4 (heights may be negative in this abstracted version).

BRUTE FORCE | Prefix and suffix arrays

BRUTE-FORCE IDEA

Start with: Prefix and suffix arrays. It stores both running sums although one number is enough.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Prefix and suffix arrays
Start with: Prefix
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | It stores both running sums although one number is enough.

HOW TO IMPROVE IT

the same balance test on the cake heights

SHORT BEST SOLUTION

The right side is total - left - heights[position].

```python
# BASE QUESTION (optimal): Running left sum
# Idea: The right side is total - left - heights[position].
def cake_balance_index(heights):
    total, left = sum(heights), 0
    for position, height in enumerate(heights):
        if left == total - left - height:
            return position
        left += height
    return -1
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 121

122 Find the Pivot Index (Balanced Split) in O(n) EASY

DIFFICULTY | EASY

Better approach: the template as written: left total equals total - left - value

QUESTION INFO

QUESTION

Given an integer array nums , return the leftmost index i such that the sum of all elements strictly to the left of i equals the sum of all elements strictly to the right of i . The element at i itself is not counted on either side. For the first index, the left sum is 0 ; for the last index, the right sum is 0 . If no such index exists, return -1 . Your solution must run in O(n) time and O(1) extra space.

Function signature

def pivot_index(nums: list[int]) -> int:
    ...

Follow-up (as reported): If several pivots exist, return them all. Hint: keep scanning after the first match, maintaining the running left sum.

EXAMPLES

QUESTION

Example 1	
Input	`nums = [1, 7, 3, 6, 5, 6]`
Output	`3`
Why	`Left of index 3: 1+7+3 = 11. Right of index 3: 5+6 = 11.`

Example 2	
Input	`nums = [1, 2, 3]`
Output	`-1`
Why	`No index balances the two sides.`

Example 3	
Input	`nums = [2, 1, -1]`
Output	`0`
Why	`Left of index 0 is 0; right is 1 + (-1) = 0.`

CONSTRAINTS

- 1 <= len(nums) <= 10^4
- -1000 <= nums[i] <= 1000

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(nums) <= 10^4
- -1000 <= nums[i] <= 1000

BRUTE FORCE | Sum both sides for every index

BRUTE-FORCE IDEA

Start with: Sum both sides for every index. Start with: Sum both sides for every index. Each index re-adds the whole array.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Sum both sides for every index
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1)

WHY THIS IS WEAKER | Each index re-adds the whole array.

HOW TO IMPROVE IT

the template as written: left total equals total - left - value

SHORT BEST SOLUTION

The right sum is total - left - nums[position], so one pass is enough.

```python
# BASE QUESTION (optimal): Running left sum
# Idea: The right sum is total - left - nums[position], so one
# pass is enough.
def pivot_index(nums):
    total, left = sum(nums), 0
    for position, value in enumerate(nums):
        if left == total - left - value:
            return position
        left += value
    return -1
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 122

123 Parity Alternating Subarray Query

EASY

DIFFICULTY | EASY

Better approach: pre counts neighbouring pairs of equal parity; a range alternates when pre[right] - pre[left] == 0

QUESTION INFO

QUESTION

Given an array nums, answer many queries alternates(l, r): do the elements nums[l], ..., nums[r] (both ends included) alternate in parity (odd, even, odd, ... or even, odd, even, ...)? Preprocess the array once so that every query takes O(1).

Function signature

class ParityQueries:
  def __init__(self, nums: list)
  def alternates(self, l: int, r: int) -> bool

EXAMPLES

Example 1

Input	`p = ParityQueries([1, 2, 3, 5, 4])`
	`p.alternates(0, 2)`
	`p.alternates(0, 3)`
	`p.alternates(3, 4)`

Output	`p.alternates(0, 2) -> true`
	`p.alternates(0, 3) -> false`
	`p.alternates(3, 4) -> true`

Why	`1, 2, 3 alternates; the neighbours 3 and 5 are both odd, so any range containing both fails; 5, 4 alternates.`

CONSTRAINTS

- 1 <= nums.length <= 10^5
- 1 <= queries.length <= 10^5
- 0 <= l <= r < nums.length

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= nums.length <= 10^5
- 1 <= queries.length <= 10^5
- 0 <= l <= r < nums.length

BRUTE FORCE | Scan the subarray for every query

BRUTE-FORCE IDEA

Check each adjacent pair in nums[l..r] for equal parity. Correct, and every query repeats work that other queries already did.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan the subarray for every query
Check each adjacent pair in nums[l..r] for equal parity
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(length) per query | Space O(1)

WHY THIS IS WEAKER | Each query scans up to n elements. A prefix count of equal-parity neighbour pairs answers every query in O(1) after one O(n) pass.

HOW TO IMPROVE IT

pre counts neighbouring pairs of equal parity; a range alternates when pre[right] - pre[left] == 0

SHORT BEST SOLUTION

bad[position] is the number of positions j < position with nums[j] and nums[j + 1] of the same parity. The pairs inside nums[left..r] are j = left .. right - 1, so the subarray alternates when bad[right] - bad[left] == 0. Use x % 2 (not x & 1 on a language where % can be negative) so negatives work.

```
# BASE QUESTION (optimal): Prefix count of equal-parity
# neighbour pairs
# Idea: bad[position] is the number of positions j < position with
# nums[j] and nums[j + 1] of the same parity. The pairs
# inside nums[left..r] are j = left .. right - 1, so the subarray
# alternates when bad[right] - bad[left] == 0. Use x % 2 (not x &
# 1 on a language where % can be negative) so negatives
# work.
class ParityQueries:
    def __init__(self, nums):
        self.bad = [0]
        for position in range(len(nums) - 1):
            self.bad.append(self.bad[-1] + (nums[position] % 2 == nums[position + 1] % 2))

    def alternates(self, left, right):
        return self.bad[right] - self.bad[left] == 0
```

COMPLEXITY | Time O(n) to build, O(1) per query | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 123

GROUP 16 OF 43 | 3 QUESTIONS | PART B - PART 1

Sorted intervals: merge while they touch, then read the gaps or search

PATTERN CONTEXT

Sort by start. Walk once: if the next interval starts before the last one ends, extend the last; otherwise start a new one. The merged list is sorted and disjoint, so gaps are the spaces between neighbours and any point can be found with bisect.

129 Check an Interval Against Existing Intervals for Conflicts — EASY

DIFFICULTY | EASY

Better approach: bisect on the starts; a new booking conflicts if the previous one ends after its start or the next one starts before its end

QUESTION INFO

QUESTION

Design a calendar of half-open intervals [start, end). book(start, end) books the interval and returns True when it does not overlap any interval booked so far, and returns False (booking nothing) when it does. Intervals that only touch, one ending exactly where the next starts, do not conflict.

Function signature

class Calendar:

    def book(self, start: int, end: int) -> bool

EXAMPLES

Example 1

Input	`c = Calendar()`
	`c.book(10, 20)`
	`c.book(15, 25)`
	`c.book(20, 30)`
	`c.book(5, 10)`
	`c.book(12, 14)`

Output	`c.book(10, 20) -> true`
	`c.book(15, 25) -> false`
	`c.book(20, 30) -> true`
	`c.book(5, 10) -> true`
	`c.book(12, 14) -> false`

Why	`[15, 25) overlaps [10, 20) and [12, 14) lies inside it, so both fail. [20, 30) and [5, 10) only touch the booked interval at an endpoint, so they succeed.`

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Scan every booked interval

BRUTE-FORCE IDEA

Compare the new interval with all existing ones using the overlap test. Correct and simple, but each booking costs time proportional to the number of bookings.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Scan every booked interval
Compare the new interval with all existing ones using the overlap test
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) per booking | Space O(n)

WHY THIS IS WEAKER | Each booking scans all earlier bookings, O(n). With the intervals sorted, binary search finds the only two neighbours that could conflict.

HOW TO IMPROVE IT

bisect on the starts; a new booking conflicts if the previous one ends after its start or the next one starts before its end

SHORT BEST SOLUTION

Keep the booked intervals sorted by start. Find where the new start would go; it can only conflict with the interval just before (if its end is past the new start) or the one just after (if its start is before the new end). A balanced tree makes the insert O(log n) as well.

```
from bisect import bisect_right


# BASE QUESTION (optimal): Sorted starts with bisect:
# check only two neighbours
# Idea: Keep the booked intervals sorted by start. Find
# where the new start would go; it can only conflict with
# the interval just before (if its end is past the new
# start) or the one just after (if its start is before the
# new end). A balanced tree makes the insert O(log n) as
# well.
class Calendar:
    def __init__(self):
        self.starts, self.ends = [], []

    def book(self, start, end):
        insert_position = bisect_right(self.starts, start)
        if insert_position > 0 and self.ends[insert_position - 1] > start:
            return False
        if insert_position < len(self.starts) and self.starts[insert_position] < end:
            return False
        self.starts.insert(insert_position, start)
        self.ends.insert(insert_position, end)
        return True
```

COMPLEXITY | Time O(log n) search, O(n) list insert | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 129

135 Find safe travel intervals between planet influences HARD

DIFFICULTY | HARD

Better approach: turn each planet into (centre - radius, centre + radius), merge, and read the gaps (including the infinite ends)

QUESTION INFO

QUESTION

You are planning a space route along a one-dimensional line (the x-axis). You are given a list of planets. Each planet is represented by an integer pair ((c, r)):

- **c** is the coordinate of the planet's center on the x-axis.
- **r** is a non-negative integer radius of its gravitational influence. A point **x** on the x-axis is **unsafe** for a space traveler if it lies within the gravitational influence of at least one planet: [|x - c_i| \le r_i \quad \text{for some planet } i] Equivalently, each planet makes the interval ([c_i - r_i,\; c_i + r_i]) unsafe. Task:
- Consider the entire real line as the possible travel path.
- Compute all **maximal continuous intervals** on this line where the traveler is **safe**, i.e., points not covered by any planet's unsafe interval.
- Return these safe intervals as a sorted list of disjoint intervals. Representation details:

QUESTION

- Represent each safe interval as a pair **`(start, end)`** with **`start < end`**.
- Intervals may be unbounded on the left or right:
- Use **`-inf`** for an interval unbounded to the left (e.g., **`(-inf, a)`**).
- Use **`+inf`** for an interval unbounded to the right (e.g., **`(b, +inf)`**). You may assume the input list of planets can be in any order and may contain overlapping or nested gravitational ranges. Return the safe intervals in increasing order of **`start`**.

You are planning a route along the real x-axis. Each planet is given as a pair (c, r), where c is its center and r is a non-negative radius of gravitational influence. A point x is unsafe if it lies inside at least one planet's influence, meaning |x - c| <= r. So each planet creates an unsafe closed interval [c - r, c + r]. Your task is to return all maximal continuous safe intervals on the real line: regions not covered by any unsafe interval. Represent each safe interval as a tuple (start, end), where the interval means start < x < end. Because JSON/Python literals do not have a direct infinity literal, use None to represent an unbounded side:

- (None, a) means (-infinity, a)
- (b, None) means (b, +infinity)
- (None, None) means the entire real line Return the safe intervals in increasing order. Overlapping, nested, and touching unsafe intervals should be treated as one merged unsafe region.

Constraints
0 <= len(planets) <= 200000-10^9 <= c <= 10^90 <= r <= 10^9

EXAMPLES

```
Input: []
Expected Output: [(None, None)]
```

Explanation: With no planets, no point is unsafe, so the entire real line is safe.

```
Input: [(3, 1), (10, 2)]
Expected Output: [(None, 2), (4, 8), (12, None)]
Explanation: The unsafe intervals are [2, 4] and [8, 12]. Their complement is (-infinity, 2), (4, 8),
    and (12, +infinity).
```

Hints

First convert every planet (c, r) into the unsafe interval [c - r, c + r].Sort the unsafe intervals and merge all overlapping or touching ones. The safe intervals are the gaps before, between, and after the merged intervals.

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Test the midpoint of every segment between critical points

BRUTE-FORCE IDEA

Collect all interval endpoints, and for every consecutive pair (and the two outer rays) test whether the midpoint is covered by some planet. Adjacent safe segments are separate because their shared endpoint is unsafe. Correct; every midpoint is tested against every planet.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Test the midpoint of every segment between critical points
Collect all interval endpoints
for every consecutive pair (
the two outer rays) test whether the midpoint is covered by some planet
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Each midpoint is tested against every planet, O(n^2). Merging the unsafe intervals after sorting and reading the gaps is O(n log n).

HOW TO IMPROVE IT

turn each planet into (centre - radius, centre + radius), merge, and read the gaps (including the infinite ends)

SHORT BEST SOLUTION

Sort the unsafe intervals by start and merge any that overlap or touch (they share a point, which is unsafe). Then the safe intervals are (-inf, first start), each gap (end of one merged interval, start of the next), and (last end, +inf). With no planets the whole line is safe.

```
INFINITY = float("inf")


# BASE QUESTION (optimal): Merge the unsafe intervals,
# then read off the gaps
# Idea: Sort the unsafe intervals by start and merge any
# that overlap or touch (they share a point, which is
# unsafe). Then the safe intervals are (-inf, first
# start), each gap (end of one merged interval, start of
# the next), and (last end, +inf). With no planets the
# whole line is safe.
def safe_intervals(planets):
    merged = []
    for start, end in sorted((centre - radius, centre + radius) for centre, radius in planets):
        if merged and start <= merged[-1][1]:
            merged[-1][1] = max(merged[-1][1], end)
        else:
            merged.append([start, end])
    if not merged:
        return [(-INFINITY, INFINITY)]
    safe = [(-INFINITY, merged[0][0])]
    for previous_interval, next_interval in zip(merged, merged[1:]):
        safe.append((previous_interval[1], next_interval[0]))
    safe.append((merged[-1][1], INFINITY))
    return safe
```

COMPLEXITY | Time O(n log n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 135

136 Implement a function that processes a stream of interval data. HARD

DIFFICULTY | HARD

Better approach: the Range Module add: bisect, merge the block, splice in one interval; intervals() returns the pairs

QUESTION INFO

QUESTION

Intervals arrive one at a time. add(start, end) receives a new interval, and intervals() returns the current list of disjoint intervals, sorted by start; intervals that overlap or touch are merged.

Function signature

class IntervalStream:
    def add(self, start: int, end: int) -> None
    def intervals(self) -> list

EXAMPLES

QUESTION

Example 1

Input	`s = IntervalStream()`
	`s.add(1, 3)`
	`s.add(6, 9)`
	`s.intervals()`
	`s.add(2, 5)`
	`s.intervals()`
	`s.add(5, 6)`
	`s.intervals()`

Output	`s.intervals() -> [(1, 3), (6, 9)]`
	`s.intervals() -> [(1, 5), (6, 9)]`
	`s.intervals() -> [(1, 9)]`

Why	`(2, 5) overlaps (1, 3) and extends it to (1, 5); (5, 6) touches (1, 5) at 5 and (6, 9) at 6, so everything merges into (1, 9).`

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Keep every interval ever added and merge on demand

BRUTE-FORCE IDEA

Append each interval to a list and, whenever the union is needed, sort and merge from scratch. It is correct; every query pays for the whole history.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Keep every interval ever added and merge on demand
Append each interval to a list
whenever the union is needed
sort
merge from scratch
```

BRUTE-FORCE COMPLEXITY | Time O(n log n) per query | Space O(n)

WHY THIS IS WEAKER | Every query re-sorts and re-merges the whole history, O(n log n) per query. Keeping the merged intervals sorted and only splicing the affected block makes an add cost O(log n) to find plus the shifted block.

HOW TO IMPROVE IT

the Range Module add: bisect, merge the block, splice in one interval; intervals() returns the pairs

SHORT BEST SOLUTION

Keep the merged intervals sorted by start. A new interval only interacts with the intervals found by two binary searches; replace that slice by the single merged interval. Touching closed intervals merge.

```python
import bisect


# BASE QUESTION (optimal): Sorted disjoint intervals with
# binary search
# Idea: Keep the merged intervals sorted by start. A new
# interval only interacts with the intervals found by two
# binary searches; replace that slice by the single merged
# interval. Touching closed intervals merge.
class IntervalStream:
    def __init__(self):
        self.starts = []
        self.ends = []

    def add(self, start, end):
```

```
        first_overlap = bisect.bisect_left(self.ends, start)           # first interval ending at or after star
t
        after_last_overlap = bisect.bisect_right(self.starts, end)          # first interval starting after end
        if first_overlap < after_last_overlap:
            start = min(start, self.starts[first_overlap])
            end = max(end, self.ends[after_last_overlap - 1])
        self.starts[first_overlap:after_last_overlap] = [start]
        self.ends[first_overlap:after_last_overlap] = [end]

    def intervals(self):
        return list(zip(self.starts, self.ends))
```

COMPLEXITY | Time O(log n) to find the spot, O(n) worst-case list shift | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 136

GROUP 17 OF 43 | 1 QUESTIONS | PART B - PART 1

Trie: nested dicts, one character per level

PATTERN CONTEXT

Store words as nested dicts, one character per level, with an end marker that holds whatever you want to remember. Insert walks and creates; search walks and fails when a character is missing; prefix queries stop in the middle; longest-match keeps the last end marker you passed.

144 Implement Longest-Match Text Replacement — HARD

DIFFICULTY | HARD

Better approach: at each text position walk the trie as far as it goes and remember the last END; replace the longest match and jump past it

QUESTION INFO

QUESTION

You are given a text string and a dictionary of replacement entries. Each dictionary entry has the form **`token:id`**, where **`token`** is a string pattern and **`id`** is the value that should be emitted when that token is matched. Implement a function that scans the text from left to right and produces an output sequence using greedy longest-prefix matching:

1. At the current text position, find all dictionary tokens that match the substring starting at that position.
2. If at least one token matches, choose the longest matching token, append its corresponding **`id`** to the output, and advance by the length of that token.
3. If no token matches, append the original character at the current position to the output and advance by one character.
4. Continue until the entire text has been processed. Example:

```
text = "abcdxyz" dictionary = { "ab": "1", "abc": "2", "xy": "3" }
```

Output:

```
["2", "d", "3", "z"]
```

Explanation: At index **`0`**, both **`"ab"`** and **`"abc"`** match, so choose the longest token **`"abc"`** and output **`"2"`**. Then **`"d"`** has no match, so output it literally. Then **`"xy"`** matches and outputs **`"3"`**. Finally, **`"z"`** is output literally. Follow-up: How would you adapt your design if the matching priority changed from longest match to a configurable priority rule, such as dictionary insertion order, highest score, or a custom comparator?

Part 1: Longest-Match Text ReplacementGiven a text string and a dictionary that maps token strings to replacement id strings, scan the text from left to right. At each position:

1. Find every token that matches the substring starting at that position.
2. If at least one token matches, choose the longest matching token, append its replacement id to the output, and advance by the token length.
3. If no token matches, append the original character at that position as a one-character string and advance by 1. Return the final output as a list of strings.

Constraints

0 <= len(text) <= 1000000 <= len(dictionary) <= 10000Each token in dictionary is non-empty.The sum of all token lengths is at most 100000.Matching is case-sensitive.

EXAMPLES

```
Input: ("abcdxyz", {"ab": "1", "abc": "2", "xy": "3"})
Expected Output: ["2", "d", "3", "z"]
```

Explanation: Both 'ab' and 'abc' match at index 0, so the longer token 'abc' is chosen.

QUESTION

```
Input: ("aaaaa", {"a": "X", "aa": "Y"})
Expected Output: ["Y", "Y", "X"]
```

Explanation: Greedy longest matching picks 'aa' twice, then the final 'a'.

Hints

A trie lets you test all token prefixes starting at one position without checking every token from scratch.While walking forward from a position, remember the last trie node that represented a complete token.

Part 2: Configurable-Priority Text ReplacementImplement the same left-to-right text replacement process, but the match-selection rule is configurable. You are given: - a text string - a list of replacement entries in insertion order - a priority rule Each replacement entry is a tuple: (token, replacement_id, score). At each text position, find all entries whose token matches the substring starting there. If one or more entries match, choose exactly one using the selected priority rule:

- 'longest': choose the longest matching token; if tied, choose the earlier entry in the list - 'insertion': choose the earliest matching entry in the input list - 'score': choose the matching entry with the highest score; if tied, choose the longer token; if still tied, choose the earlier entry in the list Append the chosen replacement_id to the output and advance by the chosen token length. If nothing matches, append the original character and advance by 1. Return the final output as a list of strings.

Constraints

0 <= len(text) <= 1000000 <= len(entries) <= 10000Each token is non-empty.Tokens in entries are unique.The sum of all token lengths i

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | At every position try every dictionary token

BRUTE-FORCE IDEA

For each position test startswith for every token and keep the longest. Correct; each position checks the entire dictionary.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: At every position try every dictionary token
For each position test startswith for every token
keep the longest
```

BRUTE-FORCE COMPLEXITY | Time O(len(text) * tokens * token length) | Space O(1) extra

WHY THIS IS WEAKER | Each position tests every token. A trie of the tokens is walked from i, remembering the last token end, so a position costs at most the longest token.

HOW TO IMPROVE IT

at each text position walk the trie as far as it goes and remember the last END; replace the longest match and jump past it

SHORT BEST SOLUTION

Build a trie of the tokens (each token's last node stores its id). At position i walk down the trie along the text; whenever the current node stores an id, record it and its length. When the walk cannot continue, use the last recorded token (advance by its length), or emit the character and advance by one.

```
# BASE QUESTION (optimal): Trie walk that remembers the
# last token end
# Idea: Build a trie of the tokens (each token's last node
# stores its id). At position i walk down the trie along
# the text; whenever the current node stores an id, record
# it and its length. When the walk cannot continue, use
# the last recorded token (advance by its length), or emit
# the character and advance by one.
```

```
def replace_longest(text, mapping):
    root = {}
    for token, token_id in mapping.items():
        node = root
        for character in token:
            node = node.setdefault(character, {})
        node["$"] = (token_id, len(token))
    output, position = [], 0
    while position < len(text):
        node, scan_position, longest_match = root, position, None
        while scan_position < len(text) and text[scan_position] in node:
            node = node[text[scan_position]]
            scan_position += 1
            if "$" in node:
                longest_match = node["$"]
        if longest_match:
            output.append(longest_match[0])
            position += longest_match[1]
        else:
            output.append(text[position])
            position += 1
    return output
```

COMPLEXITY | Time O(len(text) * longest token) | Space O(total token length)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 144

GROUP 18 OF 43 | 2 QUESTIONS | PART B - PART 1

A dict for lookup plus an ordered container for who is next (lazy deletion)

PATTERN CONTEXT

Keep a dict for O(1) lookup and a deque, list or heap for the order. When something is removed, delete it from the dict only and skip stale entries when you read from the order. LRU caches, waitlists, undo stacks and schedulers all fit.

145 Implement Batched Undo/Redo Layer — EASY

DIFFICULTY | EASY

Better approach: two stacks (undo, redo) of batches; a batch records (old value, new value) per property; undo applies the old values and moves the batch

QUESTION INFO

QUESTION

You are implementing a simplified document layer for a design tool. The layer stores properties as a **`map<string, string>`**. Implement a class that supports the following operations:

- **`beginBatch()`**: start a batch of edits.
- **`setProperty(key, value)`**: set or overwrite a property inside the current batch.
- **`endBatch()`**: commit the batch as one undoable action.
- **`undo()`**: revert the most recently committed batch.
- **`redo()`**: reapply the most recently undone batch. Requirements:

1. A batch may update the same key multiple times.
2. **`undo()`** must restore the exact document state from before that batch.
3. **`redo()`** must restore the exact document state after that batch.
4. If a property did not exist before the batch, then **`undo()`** should remove that property.
5. If **`undo()`** has been called and then a new batch is committed, the redo history must be cleared.
6. Follow-up: make the history memory-efficient. Instead of storing every intermediate edit in a batch, compact the batch so that each key appears at most once in history while still preserving enough information for correct **`undo()`** and **`redo()`**. A useful compact record for one key is:

```
Change { string key; string value; // final value after the batch string old_value; // value before the ba
tch bool has_old_value; // whether the key existed before the batch }
```

Explain how you would build and use such a compact history so that repeated updates to the same key inside one batch do not waste memory.

Part 1: Implement Batched Undo/Redo for Document PropertiesYou are given a sequence of operations for an initially empty document represented as a map<string, string>. A batch starts with ('begin',), contains zero or more ('set', key, value) operations, and ends with ('end',). Each committed batch must be undoable and redoable as one unit. Operation ('snapshot',) asks you to append the current document to the answer; snapshots may appear while a batch is open and should reflect the live document state. If a key is set multiple times in one batch, undo must restore the value from before the batch began. If a key did not exist before the batch, undo must remove it. If a new non-empty batch is committed after an undo, the redo history is cleared. If undo or redo is impossible, do nothing. If end closes an empty batch, treat it as a no-op.
Constraints
0 <= len(operations) <= 200000The total number of ('set', key, value) operations is at most 200000Keys and values are strings of length 1 to 50Batches are not nestedIf a batch contains no set operations, committing it does not create history

QUESTION

EXAMPLES

```
Input: [('begin',), ('set', 'color', 'red'), ('end',), ('snapshot',), ('undo',), ('snapshot',),
    ('redo',), ('snapshot',)]
Expected Output: [{'color': 'red'}, {}, {'color': 'red'}]
```

Explanation: A single batch is committed, undone, and redone.

```
Input: [('begin',), ('set', 'a', '1'), ('end',), ('begin',), ('set', 'a', '2'), ('set', 'a', '3'),
    ('set', 'b', '9'), ('end',), ('snapshot',), ('undo',), ('snapshot',), ('redo',), ('snapshot',)]
Expected Output: [{'a': '3', 'b': '9'}, {'a': '1'}, {'a': '3', 'b': '9'}]
```

Explanation: The second batch updates 'a' twice and adds 'b'. Undo must restore the exact pre-batch state.

Hints

For each key touched in the current batch, record its old value only the first time that key is edited in the batch.Use two stacks: one for committed batches that can be undone, and one for undone batches that can be redone. Clear the redo stack when a new non-empty batch is committed.

Part 2: Compact a Batch into Memory-Efficient Change RecordsYou are given the document state before a batch and the raw setProperty edits that happened inside that batch. Multiple edits may target the same key. Build a compact history so that each key appears at most once, while still storing enough informa

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Snapshot the whole document after each batch

BRUTE-FORCE IDEA

Push a copy of the map on every commit; undo and redo move between snapshots. Correct; it copies the whole document for a one-key change.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Snapshot the whole document after each batch
Push a copy of the map on every commit
undo
redo move between snapshots
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(size of document) per batch | Space O(size * batches)

WHY THIS IS WEAKER | A one-key change copies the whole document, O(size) per batch and O(size * batches) memory. Storing only the old and new value of each changed key makes each batch O(keys changed).

HOW TO IMPROVE IT

two stacks (undo, redo) of batches; a batch records (old value, new value) per property; undo applies the old values and moves the batch

SHORT BEST SOLUTION

While a batch is open keep {key: (old, new)} with the old value recorded only the first time a key is touched. endBatch pushes it on the undo stack and clears the redo stack. undo writes the old values back (deleting keys that did not exist before) and moves the batch to redo; redo writes the new values. The document is never copied.

```python
_MISSING = object()


# BASE QUESTION (optimal): Store only the changed keys per
# batch
# Idea: While a batch is open keep {key: (old, new)} with
# the old value recorded only the first time a key is
# touched. endBatch pushes it on the undo stack and clears
```

```
# the redo stack. undo writes the old values back
# (deleting keys that did not exist before) and moves the
# batch to redo; redo writes the new values. The document
# is never copied.
class Layer:
    def __init__(self):
        self.properties, self.undo_stack, self.redo_stack, self.open_batch = {}, [], [], None

    def beginBatch(self):
        self.open_batch = {}

    def setProperty(self, name, value):
        if self.open_batch is None:
            self.beginBatch()
            self.setProperty(name, value)
            self.endBatch()
            return
        old = self.open_batch[name][0] if name in self.open_batch else self.properties.get(name, _MISSING)
        self.open_batch[name] = (old, value)
        self.properties[name] = value

    def endBatch(self):
        if self.open_batch:
            self.undo_stack.append(self.open_batch)
            self.redo_stack.clear()
        self.open_batch = None

    def _apply(self, batch, use_new):
        for name, (old, new) in batch.items():
            value = new if use_new else old
            if value is _MISSING:
                self.properties.pop(name, None)
            else:
                self.properties[name] = value

    def undo(self):
        if self.undo_stack:
            batch = self.undo_stack.pop()
            self._apply(batch, False)
            self.redo_stack.append(batch)

    def redo(self):
        if self.redo_stack:
            batch = self.redo_stack.pop()
            self._apply(batch, True)
            self.undo_stack.append(batch)
```

COMPLEXITY | Time O(keys changed in the batch) per operation | Space O(total changed keys)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 145

152 Implement LRU Cache. HARD

DIFFICULTY | HARD

Better approach: dict key -> node plus a doubly linked list; every get/put moves the node to the front; evict from the back (or use OrderedDict.move_to_end)

QUESTION INFO

QUESTION

Your program keeps re-reading the same data from disk, so you want an in-memory cache that holds only the n most recently used items (a least-recently-used cache). Write a class LRUCache(n) with get(key), which returns the stored value or -1, and put(key, value), which stores a value and, when the cache is full, first removes the item that was used least recently. Both get and put count as a use. Both must run in O(1).

Function signature

class LRUCache:
  def __init__(self, n: int)
  def get(self, key) -> int
  def put(self, key, value) -> None

EXAMPLES

Example 1

Input	`c = LRUCache(2)`
	`c.put(1, 1)`
	`c.put(2, 2)`
	`c.get(1)`
	`c.put(3, 3)`
	`c.get(2)`
	`c.put(4, 4)`
	`c.get(1)`
	`c.get(3)`
	`c.get(4)`

Output	`c.get(1) -> 1`
	`c.get(2) -> -1`
	`c.get(1) -> -1`
	`c.get(3) -> 3`
	`c.get(4) -> 4`

Why	`get(1) makes key 1 the most recent, so put(3, 3) evicts key 2; then put(4, 4) evicts key 1 (older than 3).`

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | A list of keys ordered by recency

BRUTE-FORCE IDEA

Keep a list of (key, value) with the most recent at the end. Get and put search the list and move the entry. Correct; every operation is a linear scan.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: A list of keys ordered by recency
Keep a list of (key
value) with the most recent at the end
```

BRUTE-FORCE COMPLEXITY | Time O(n) per operation | Space O(capacity)

WHY THIS IS WEAKER | Every operation scans the list, O(capacity). A hash map to nodes of a doubly linked list gives O(1) lookup and O(1) move-to-front and eviction.

HOW TO IMPROVE IT

dict key -> node plus a doubly linked list; every get/put moves the node to the front; evict from the back (or use OrderedDict.move_to_end)

SHORT BEST SOLUTION

Part 1: Each key maps to a node of a doubly linked list ordered by recency, with sentinel head and tail nodes so no operation has to special-case an empty list. get moves the node to the front; put updates or inserts at the front and, if over capacity, removes the node just before the tail and its map entry.

Part 2: OrderedDict remembers insertion order and supports move_to_end and popitem(last=False) in O(1), so it is the hash map and the recency list in one. Good for a quick answer once you can explain what it does underneath.

```
# BASE QUESTION (optimal): Hash map plus doubly linked
# list (built by hand)
# Idea: Each key maps to a node of a doubly linked list
# ordered by recency, with sentinel head and tail nodes so
# no operation has to special-case an empty list. get
# moves the node to the front; put updates or inserts at
# the front and, if over capacity, removes the node just
# before the tail and its map entry.
class LRUCache:
    class _Node:
        __slots__ = ("key", "value", "prev", "next")

        def __init__(self, key=None, value=None):
            self.key, self.value, self.prev, self.next = key, value, None, None

    def __init__(self, capacity):
        self.max_size, self.node_by_key = capacity, {}
        self.head, self.tail = self._Node(), self._Node()
        self.head.next, self.tail.prev = self.tail, self.head

    def _remove(self, node):
        node.prev.next, node.next.prev = node.next, node.prev

    def _push_front(self, node):
        node.next, node.prev = self.head.next, self.head
        self.head.next.prev = node
        self.head.next = node

    def get(self, key):
        node = self.node_by_key.get(key)
        if node is None:
            return -1
        self._remove(node)
        self._push_front(node)
        return node.value

    def put(self, key, value):
        node = self.node_by_key.get(key)
        if node:
            node.value = value
            self._remove(node)
            self._push_front(node)
            return
        if len(self.node_by_key) == self.max_size:
            lru = self.tail.prev
            self._remove(lru)
            del self.node_by_key[lru.key]
        node = self._Node(key, value)
        self.node_by_key[key] = node
        self._push_front(node)
```

COMPLEXITY | Part 1: Time O(1) per operation | Space O(capacity)
Part 2: Time O(1) per operation | Space O(capacity)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 152

GROUP 19 OF 43 | 3 QUESTIONS | PART B - PART 1

Rolling state: the next state comes only from the previous one

PATTERN CONTEXT

You never need the whole table, only the previous state: one number, a pair, or one row. Write the state, then one line that builds the new state from the old one, and loop.

153 Climbing Stairs — EASY

DIFFICULTY | EASY

Better approach: state = the last two counts: a, current_ways = current_ways, previous_ways + current_ways, repeated step_count - 1 times

QUESTION INFO

QUESTION

Given an integer n, representing the number of steps in a staircase, write a function climbStairs that returns the number of distinct ways to climb to the top of the staircase. Each time you can either climb 1 step or 2 steps. You may assume that n is a non-negative integer. Examples Input: n = 1 Output: 1 Explanation: There is only 1 way to climb a staircase with 1 step.

Input: n = 2 Output: 2 Explanation: There are 2 ways to climb a staircase with 2 steps:

1. Climb 1 step + 1 step 2. Climb 2 steps directly Input: n = 3 Output: 3 Explanation: There are 3 ways to climb a staircase with 3 steps:
1. Climb 1 step + 1 step + 1 step 2. Climb 1 step + 2 steps 3. Climb 2 steps + 1 step Input: n = 4 Output: 5 Explanation: There are 5 ways to climb a staircase with 4 steps:
1. Climb 1 step + 1 step + 1 step + 1 step 2. Climb 1 step + 1 step + 2 steps 3. Climb 1 step + 2 steps + 1 step 4. Climb 2 steps + 1 step + 1 step 5. Climb 2 steps + 2 stepsChallenge 1Can you come up with a solution that has a O(n) time complexity?Challenge 2Can you come up with a solution that has a O(1) space complexity?Solution 1: Dynamic Programming Approach In this solution, we use dynamic programming to compute the number of distinct ways to climb n stairs. The idea is to build up the solution incrementally by solving subproblems. We initialize a list ways where ways[i] represents the number of ways to reach the ith step. We start by setting the base cases: ways[0] = 1 (1 way to stay at the ground) ways[1] =

EXAMPLES

Example 1	
Input	**2**
Output	**2**
Why	**`Two ways to climb 2 steps: 1+1 or 2.`**

Example 2	
Input	**3**
Output	**3**
Why	**`Three ways to climb 3 steps: 1+1+1, 1+2, 2+1.`**

Example 3	
Input	**5**
Output	**8**
Why	**`Eight ways for 5 steps.`**

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Plain recursion

BRUTE-FORCE IDEA

Ways(n) = ways(n - 1) + ways(n - 2) computed recursively. Correct; the same values are recomputed exponentially often.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
ways(0) = 1; ways(negative) = 0
ways(n) = ways(n - 1) + ways(n - 2)
evaluate both recursive branches every time
return ways(n)
```

BRUTE-FORCE COMPLEXITY | Time O(2^n) | Space O(n)

WHY THIS IS WEAKER | The same values are computed over and over: O(2^n). Keeping the last two values in a loop is O(n) time and O(1) space (and a matrix power gives O(log n)).

HOW TO IMPROVE IT

state = the last two counts: a, current_ways = current_ways, previous_ways + current_ways, repeated step_count - 1 times

SHORT BEST SOLUTION

Keep ways for the previous two steps and advance: (previous_ways, current_ways) = (current_ways, previous_ways + current_ways). Start with ways(0) = ways(1) = 1. No table and no recursion.

```python
# BASE QUESTION (optimal): Two running values
# Idea: Keep ways for the previous two steps and advance:
# (previous_ways, current_ways) = (current_ways, previous_ways + current_ways). Start with ways(0) = ways(1) =
1.
# No table and no recursion.
def climbStairs(step_count):
    previous_ways = current_ways = 1
    for _ in range(step_count - 1):
        previous_ways, current_ways = current_ways, previous_ways + current_ways
    return current_ways
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 153

154 Fibonacci Numbers — EASY

DIFFICULTY | EASY

Better approach: the same pair update starting from (0, 1)

QUESTION INFO

QUESTION

Write a function fib(n) that returns the nth Fibonacci number. The Fibonacci sequence is defined as F(n) = F(n-1) + F(n-2) with F(0) = 1 and F(1) = 1, which produces the following pattern: 1, 1, 2, 3, 5, 8, 13 ... Examples Pythonfib(0)

=> 1 fib(1)

=> 1 fib(2)

=> 2 fib(3)

=> 3 ... fib(10)

=> 89 fib(20)

=> 10946Hint 1The Fibonacci sequence

QUESTION

is defined recursively since it refers to itself.Hint 2We can start with a recursive implementation and optimize it.Hint 3Each call to F(n) has two subproblems: F(n-1) and F(n-2). Can we memoize these calculations?Challenge 1Does your solution work in O(n) time?This question is a classic example of how dynamic programming, or memoization, can help us solve problems faster by reducing repetitive computations. In this first version, we recursively call the fib function twice, and then we call it twice more in every subsequent recursive call, so the total number of operations expands to O(2^n)! To improve the runtime of our program, let's save our prior calculations in a dictionary called memo and pass it into our function. Now, if we've already found fib(n) before, our function will use this stored value instead of recursing again. Time Complexity: The time complexity of the memoized version is O(n). This is because each Fibonacci number from 0 to n is computed only once and then stored in the memo dictionary. The recursive calls will now simply look up the value in the dictionary if it has already been computed, reducing redundant

EXAMPLES

Example 1	
Input	`0`
Output	`1`
Why	`F(0) = 1 and F(1) = 1, as defined in this question.`

Example 2	
Input	`5`
Output	`8`
Why	`1, 1, 2, 3, 5, 8: F(5) = 8.`

Example 3	
Input	`10`
Output	`89`
Why	`F(10) = 89.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Plain recursion

BRUTE-FORCE IDEA

Fib(n) = fib(n - 1) + fib(n - 2) with the two base cases. Correct; the same values are recomputed exponentially often.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
fib(0) = 0; fib(1) = 1
fib(n) = fib(n - 1) + fib(n - 2)
evaluate both recursive branches every time
return fib(n)
```

BRUTE-FORCE COMPLEXITY | Time O(2^n) | Space O(n)

WHY THIS IS WEAKER | The call tree has about 1.6^n nodes because the same values are recomputed. A loop with two variables is O(n) time and O(1) space; fast doubling is O(log n).

HOW TO IMPROVE IT

the same pair update starting from (0, 1)

SHORT BEST SOLUTION

Part 1: Keep the previous two values and advance n times. Memoising the recursion gives the same time with O(position) memory; the loop needs no memory and no recursion.

Part 2: Use F(2k) = F(k) * (2 * F(k + 1) - F(k)) and F(2k + 1) = F(k)^2 + F(k + 1)^2. Recursing on position // 2 halves the problem each time, so only about log2(position) steps are needed.

```python
# BASE QUESTION (optimal): Iteration with two variables
# Idea: Keep the previous two values and advance n times.
# Memoising the recursion gives the same time with O(position)
# memory; the loop needs no memory and no recursion.
def fib(position):
    current, following = 0, 1
    for _ in range(position):
        current, following = following, current + following
    return current
```

COMPLEXITY | Part 1: Time O(n) | Space O(1)
Part 2: Time O(log n) big-integer operations | Space O(log n) recursion depth

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 154

159 Constrained Dice Roll Sequences — HARD

DIFFICULTY | HARD

Better approach: state = a table [value][run length]; a different value starts a run of 1, the same value extends its run (up to the limit)

QUESTION INFO

QUESTION

You roll a single 6-faced die exactly n times. You are given an array roll_max of length 6, where roll_max[i] is the maximum number of times the value i + 1 may appear consecutively. That is, you may never roll the same value more than roll_max[i] times in a row. Return the number of distinct sequences of n rolls that respect every consecutive-count limit. Because the answer can be very large, return it modulo 10**9 + 7 .

Function signature

def count_dice_sequences(n: int, roll_max: list[int]) -> int:
    ...

Follow-up (as reported): The interviewer's actual twist used two dice per turn (a pair rolled together, with the consecutive limit applied to each die independently). Extend the state to track the run lengths of both dice and count the joint sequences.

EXAMPLES

Example 1	
Input	n = 2, roll_max = [1, 1, 2, 2, 2, 3]
Output	34
Why	There are 6*6 = 36 sequences of length 2. Two of them are forbidden: rolling value 1 twice, and rolling value 2 twice (both have roll_max = 1). 36 - 2 = 34.

Example 2	
Input	n = 2, roll_max = [1, 1, 1, 1, 1, 1]
Output	30
Why	No value may repeat consecutively, so the 6 sequences "value v then v" are removed: 36 - 6 = 30.

Example 3	
Input	n = 3, roll_max = [1, 1, 1, 2, 2, 3]
Output	181
Why	Hand-verified against the memoized DP over (rolls_used, last_value, run_length).

QUESTION

CONSTRAINTS

- 1 <= n <= 5000 roll_max has exactly 6 entries, each 1 <= roll_max[i] <= 15 .
- A naive brute force over all 6**n sequences is far too slow; you must memoize over (rolls_used, last_value, consecutive_count_of_last_value) .

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= n <= 5000 roll_max has exactly 6 entries, each 1 <= roll_max[i] <= 15 .
- A naive brute force over all 6**n sequences is far too slow; you must memoize over (rolls_used, last_value, consecutive_count_of_last_value) .

BRUTE FORCE | Enumerate every sequence

BRUTE-FORCE IDEA

Start with: Enumerate every sequence. It generates all 6^n sequences and checks the runs of each.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Enumerate every sequence
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(6^n * n) | Space O(n)

WHY THIS IS WEAKER | It generates all 6^n sequences and checks the runs of each.

HOW TO IMPROVE IT

state = a table [value][run length]; a different value starts a run of 1, the same value extends its run (up to the limit)

SHORT BEST SOLUTION

dp[face][run_length] counts prefixes ending in value v repeated r times.

```python
# BASE QUESTION (optimal): DP over the last value and its
# run length
# Idea: dp[face][run_length] counts prefixes ending in value v
# repeated r times.
def count_dice_sequences(roll_count, roll_max):
    mod = 10**9 + 7
    dp = [[0] * (roll_max[face] + 1) for face in range(6)]
    for face in range(6):
        dp[face][1] = 1
    for _ in range(roll_count - 1):
        total = sum(sum(row) for row in dp) % mod
        next_dp = [[0] * (roll_max[face] + 1) for face in range(6)]
        for face in range(6):
            # a different previous value starts a run of length one
            next_dp[face][1] = (total - sum(dp[face])) % mod
            for run_length in range(1, roll_max[face]):
                next_dp[face][run_length + 1] = dp[face][run_length]
        dp = next_dp
    return sum(sum(row) for row in dp) % mod
```

COMPLEXITY | Time O(n * 6 * R) | Space O(6 * R)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 159

GROUP 20 OF 43 | 4 QUESTIONS | PART B - PART 1

Two pointers: move i and j toward each other (or along two lists)

PATTERN CONTEXT

Put one pointer at each end (or one at the start of each of two sorted lists). Look at the two items, decide which pointer moves, and stop when they meet. Most of these are O(n) because each pointer only moves one way.

160 Reverse a linked list. EASY

DIFFICULTY | EASY

Better approach: three pointers previous_node, current, next: point current.next back at previous_node, then move all three one step

QUESTION INFO

QUESTION

Given the head of a singly linked list, reverse the list and return its head. Implement a function reverseList that takes the head of the linked list as input and returns the new head of the reversed list. Refer to the ListNode class/struct and do not comment it out. Example Input: 1 -> 2 -> 3 -> 4 Output: 4 -> 3 -> 2 -> 1Challenge 1Can you come up with a solution with a time complexity of O(n)?Challenge 2Can you come up with a solution with a space complexity of O(1)?

EXAMPLES

Example 1	
Input	`head = 1 -> 2 -> 3`
Output	`vals(reverse_list(head)) -> [3, 2, 1]`
Why	`1 -> 2 -> 3 becomes 3 -> 2 -> 1.`

CLARIFY BEFORE CODING

May I relink the original nodes in place? Can the list be empty or cyclic, and must relative order or node identity be preserved? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Copy the values to an array and rebuild

BRUTE-FORCE IDEA

Read all values, then build a new list in reverse order. Correct, but it allocates a whole new list instead of changing the links.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Copy the values to an array and rebuild
Read all values
build a new list in reverse order
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | Time is O(n), but it allocates a whole new list, O(n) extra space. Flipping each node's next pointer in place with three variables needs O(1) space.

HOW TO IMPROVE IT

three pointers previous_node, current, next: point current.next back at previous_node, then move all three one step

SHORT BEST SOLUTION

Walk with previous_node and cur. Save cur.next, point cur at previous_node, then move both forward. When cur is None, previous_node is the new head. Saving next before the overwrite is the step people forget.

```
# BASE QUESTION (optimal): Flip the pointers in one pass
# Idea: Walk with previous_node and cur. Save cur.next, point cur
# at previous_node, then move both forward. When cur is None, previous_node
```

```
# is the new head. Saving next before the overwrite is the
# step people forget.
class ListNode:
    def __init__(self, val=0, next=None):
        self.val, self.next = val, next

def reverseList(head):
    previous_node = None
    while head:
        next_node = head.next
        head.next = previous_node
        previous_node, head = head, next_node
    return previous_node
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 160

161 Sort a binary array in linear time. EASY

DIFFICULTY | EASY

Better approach: left skips 0s, right skips 1s; when left sits on a 1 and right on a 0, swap them

QUESTION INFO

QUESTION

Sort an array that contains only 0s and 1s in place, in one pass over the array (O(n) time) and with O(1) extra space.

Function signature

def sort_binary(arr: list) -> list

EXAMPLES

Example 1	
Input	`[1, 0, 1, 0, 0, 1]`
Output	`[0, 0, 0, 1, 1, 1]`
Why	`All the 0s move to the front.`

Example 2	
Input	`[0, 0, 1, 1]`
Output	`[0, 0, 1, 1]`
Why	`Already sorted arrays stay as they are.`

Example 3	
Input	`[1]`
Output	`[1]`
Why	`A single value.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Use the library sort

BRUTE-FORCE IDEA

Sorted(arr) is correct, but it is a comparison sort and ignores that the values are only 0 and 1.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Use the library sort
Sorted(arr) is correct
but it is a comparison sort
ignores that the values are only 0
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n log n) | Space O(n)

WHY THIS IS WEAKER | A comparison sort is O(n log n) and needs a copy; with only two values two pointers (swap a 1 on the left with a 0 on the right) finish in one O(n) pass in place.

HOW TO IMPROVE IT

left skips 0s, right skips 1s; when left sits on a 1 and right on a 0, swap them

SHORT BEST SOLUTION

Part 1: Keep zero_slot at the next position for a 0 and one_slot at the next position for a 1. Advance zero_slot past zeros, retreat one_slot past ones, and when both are stuck swap them. One pass, in place.

Part 2: Maintain three regions: [0, low) zeros, [low, mid) ones, (high, end] twos. Examine bits[mid]: swap a 0 to low, skip a 1, swap a 2 to high without advancing mid (the swapped-in value has not been inspected).

```python
# BASE QUESTION (optimal): Two pointers swapping from both
# ends
# Idea: Keep zero_slot at the next position for a 0 and one_slot at the
# next position for a 1. Advance zero_slot past zeros, retreat one_slot
# past ones, and when both are stuck swap them. One pass,
# in place.
def sort_binary(bits):
    zero_slot, one_slot = 0, len(bits) - 1
    while zero_slot < one_slot:
        if bits[zero_slot] == 0:
            zero_slot += 1
        elif bits[one_slot] == 1:
            one_slot -= 1
        else:
            bits[zero_slot], bits[one_slot] = bits[one_slot], bits[zero_slot]
            zero_slot += 1
            one_slot -= 1
    return bits
```

COMPLEXITY | Part 1: Time O(n) | Space O(1)
Part 2: Time O(n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 161

162 Is this a valid palindrome? EASY

DIFFICULTY | EASY

Better approach: compare the two ends; skip characters that are not letters or digits; stop when the pointers cross

QUESTION INFO

QUESTION

Given a string s, return true if it is a palindrome, and false otherwise. A string is considered a palindrome if, after converting all uppercase letters into lowercase letters and removing all non-alphanumeric characters, it reads the same forward and backward. Write a function isPalindrome that implements this logic. Examples Input: s = "H123!@321h" Output: true Input: s = "race a car" Output: false Constraints 1 <= s.length <= 2 * 10^5 The input string s can include any printable ASCII characters. Challenge 1Can you come up with a solution that achieves O(n) time complexity?Challenge 2Can you come up with a solution that achieves O(1) space complexity?Solution 1: Filtering and Reversing Approach This solution processes the input string by first filtering out non-alphanumeric characters and converting all remaining characters to lowercase. It then checks if this filtered string reads the same forward and backward. Filtering: Iterate through the string and collect alphanumeric characters while converting them to lowercase. Reversing and Comparing: Create a reversed version of the filtered string and compare it with the original filtered string to check for palindrome properties. Time Complexity: The time complexity is O(n), where n is the length of the input string. This is because filtering the characters and creating the reversed string both require iterating through the input once. Space Complexity: The space complexity is O(n) for storing the filtered characters and their

EXAMPLES

Example 1	
Input	`'A man, a plan, a canal: Panama'`
Output	`true`
Why	`Ignoring punctuation, spaces and case it reads 'amanaplanacanalpanama'.`

Example 2	
Input	`'race a car'`
Output	`false`
Why	`'raceacar' is not a palindrome.`

Example 3	
Input	`' '`
Output	`true`
Why	`Nothing but a space: nothing to compare, so it is a palindrome.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Clean the string and compare with its reverse

BRUTE-FORCE IDEA

Keep only letters and digits in lowercase, and compare with the reverse. Short and correct, with two extra copies of the string.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Clean the string and compare with its reverse
Keep only letters
digits in lowercase
compare with the reverse
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | O(n) time but two extra copies of the string, O(n) space. Two pointers moving inward, skipping punctuation, need O(1) space and stop at the first mismatch.

HOW TO IMPROVE IT

compare the two ends; skip characters that are not letters or digits; stop when the pointers cross

SHORT BEST SOLUTION

Move left and right toward each other, skipping characters that are not letters or digits, and compare the lowercase forms. The first mismatch ends it. Nothing is copied.

```
# BASE QUESTION (optimal): Two pointers that skip ignored
# characters
# Idea: Move left and right toward each other, skipping
# characters that are not letters or digits, and compare
# the lowercase forms. The first mismatch ends it. Nothing
# is copied.
def isPalindrome(text):
    left, right = 0, len(text) - 1
    while left < right:
        while left < right and not text[left].isalnum():
            left += 1
        while left < right and not text[right].isalnum():
            right -= 1
        if text[left].lower() != text[right].lower():
            return False
        left, right = left + 1, right - 1
    return True
```

COMPLEXITY | Time O(n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 162

163 Reverse a Sentence — EASY

DIFFICULTY | EASY

Better approach: swap from both ends to reverse the whole array, then reverse every word the same way

QUESTION INFO

QUESTION

You are given an array of characters arrthat consists of sequences of characters separated by space characters. Each space-delimited sequence of characters defines a word. Implement a function reverseWords that reverses the order of the words in the array in the most efficient manner. Explain your solution and analyze its time and space complexities. Example input: arr = ['p', 'e', 'r', 'f', 'e', 'c', 't', ' ', 'm', 'a', 'k', 'e', 's', ' ', 'p', 'r', 'a', 'c', 't', 'i', 'c', 'e'] output: ['p', 'r', 'a', 'c', 't', 'i', 'c', 'e', ' ', 'm', 'a', 'k', 'e', 's', ' ', 'p', 'e', 'r', 'f', 'e', 'c', 't']Hint 1Even if your language of choice enables this, don't be tempted to use standard functions. Consider how inefficient it is to separately create a string, split the words, and reverse the split.Challenge 1As a general rule, if you're tempted to use a standard or library function, be sure to explain that function's time and space complexity before committing to that solution. You may find a more efficient way to solve the problem yourself. Oftentimes in coding interviews, this is the whole point of the exercise.Challenge 2How have you handled edge cases? For example, empty arrays, arrays with nothing but spaces, arrays with single words only, or with multiple spaces between words, etc.Challenge 3Were you able to achieve a linear time complexity and a constant space complexity?Challenge 4Bonus: how might the mirrorReverse function be implemented with a single index?One possible s

EXAMPLES

Example 1	
Input	`arr = ['t','h','e',' ','s','k','y',' ','i','s',' ','b','l','u','e']`
Output	`''.join(reverse_words(list('the sky is blue'))) -> 'blue is sky the'`
Why	`The words come out in reverse order while each word keeps its letters in order (the array is reversed in place).`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Split into words and join them backwards

BRUTE-FORCE IDEA

Split on whitespace, reverse the list, join with single spaces. Short and clear; it uses extra space for the list of words and also collapses repeated spaces.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Split into words and join them backwards
Split on whitespace
reverse the list
join with single spaces
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | O(n) time but extra space for the word list, and it collapses repeated spaces. Reversing the whole character array and then each word in place needs O(1) extra space.

HOW TO IMPROVE IT

swap from both ends to reverse the whole array, then reverse every word the same way

SHORT BEST SOLUTION

Reverse every character; each word is now backwards but in the right place. Walk the array and reverse each maximal run of non-space characters. Everything happens inside the same array.

```python
# BASE QUESTION (optimal): Reverse the whole array, then
# reverse each word in place
# Idea: Reverse every character; each word is now
# backwards but in the right place. Walk the array and
# reverse each maximal run of non-space characters.
# Everything happens inside the same array.
def reverse_sentence(chars):
    def rev(left, right):
        while left < right:
            chars[left], chars[right] = chars[right], chars[left]
            left, right = left + 1, right - 1

    rev(0, len(chars) - 1)
    start = 0
    for position in range(len(chars) + 1):
        if position == len(chars) or chars[position] == " ":
            rev(start, position - 1)
            start = position + 1
    return chars
```

COMPLEXITY | Time O(n) | Space O(1) extra on the character array

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 163

Walk two strings at once with two indices

PATTERN CONTEXT

Keep i on one string and j on the other. Skip characters that do not matter, compare the rest, and decide what to do at the first difference. Substring search, ordering, similarity and move-based transforms are all this loop.

167 Find the First Occurrence of a Substring — EASY

DIFFICULTY | EASY

Better approach: try every start i; inside, matched walks the needle; a full match returns i

QUESTION INFO

QUESTION

Reconstructing the "find substring in array" item as substring search: given two strings haystack and needle , return the index of the first occurrence of needle in haystack , or -1 if needle is not part of haystack .

Function signature

```
def str_str(haystack: str, needle: str) -> int:
    """Return the index of the first occurrence of needle in haystack, or -1."""
    ...
```

Follow-up (as reported): Achieve O(n + m) worst-case time with the Knuth-Morris-Pratt (KMP) failure-function approach instead of the naive O(n * m) scan.

EXAMPLES

Example 1	
Input	`haystack = "sadbutsad", needle = "sad"`
Output	`0`
Why	`"sad" occurs at indices 0 and 6; the first is index 0.`

Example 2	
Input	`haystack = "leetcode", needle = "leeto"`
Output	`-1`
Why	`"leeto" does not occur in "leetcode".`

Example 3	
Input	`haystack = "abc", needle = "c"`
Output	`2`

CONSTRAINTS

- 1 <= len(haystack), len(needle) <= 10^4 haystack and needle consist of only lowercase English letters.
- If needle is the empty string it is considered to occur at index 0 (though the constraints above make it non-empty).

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(haystack), len(needle) <= 10^4 haystack and needle consist of only lowercase English letters.
- If needle is the empty string it is considered to occur at index 0 (though the constraints above make it non-empty).

BRUTE FORCE | Compare at every start

BRUTE-FORCE IDEA

Start with: Compare at every start. Each of the n - m + 1 starts may compare up to m characters.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compare at every start
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n * m) | Space O(1)

WHY THIS IS WEAKER | Each of the n - m + 1 starts may compare up to m characters.

HOW TO IMPROVE IT

try every start i; inside, matched walks the needle; a full match returns i

SHORT BEST SOLUTION

Compare character by character and stop at the first mismatch.

```python
# BASE QUESTION (optimal): Naive scan with early exit
# Idea: Compare character by character and stop at the
# first mismatch.
def str_str(haystack, needle):
    haystack_length, needle_length = len(haystack), len(needle)
    for start in range(haystack_length - needle_length + 1):
        matched = 0
        while matched < needle_length and haystack[start + matched] == needle[matched]:
            matched += 1
        if matched == needle_length:
            return start
    return -1
```

COMPLEXITY | Time O(n * m) worst case | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 167

168 Determine if two words are sorted in lexicographic order. EASY

DIFFICULTY | EASY

Better approach: walk both words; at the first differing letter compare the letters' ranks in the given alphabet; if one word ends first, it must be the shorter

QUESTION INFO

QUESTION

Words are ordered using a custom alphabet: rank maps each letter to its position in that alphabet. Return whether word a comes before word b (or is equal to it). The first letter where the words differ decides; if one word is a prefix of the other, the shorter one comes first.

Function signature

def in_order(a: str, b: str, rank: dict) -> bool

EXAMPLES

QUESTION

Example 1

Input	`alphabet = 'hlabcdefgijkmnopqrstuvwxyz' # 'h' comes first, then 'l', then a, b, c ...`
	`rank = {c: i for i, c in enumerate(alphabet)}`
	`in_order('hello', 'leetcode', rank)`
	`in_order('word', 'world', rank)`
	`in_order('app', 'apple', rank)`

Output	`in_order('hello', 'leetcode', rank) -> true`
	`in_order('word', 'world', rank) -> false`
	`in_order('app', 'apple', rank) -> true`

Why	`'h' ranks before 'l', so hello comes first. word vs world differ at d versus l, and d ranks after l, so the order is wrong. A prefix always comes first.`

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Translate both words into ranks and compare the lists

BRUTE-FORCE IDEA

Map every letter to its position in the alphabet and compare the two rank lists with Python's list comparison. Correct; it builds both lists even when the first letters already differ.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Translate both words into ranks and compare the lists
Map every letter to its position in the alphabet
compare the two rank lists with Python's list comparison
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(L) | Space O(L)

WHY THIS IS WEAKER | It builds both full rank lists even when the first letters already decide the answer. Comparing character by character and stopping at the first difference is O(min(len a, len b)) with O(1) extra space.

HOW TO IMPROVE IT

walk both words; at the first differing letter compare the letters' ranks in the given alphabet; if one word ends first, it must be the shorter

SHORT BEST SOLUTION

Find the first position where the words differ and compare the two characters' ranks. If one word is a prefix of the other, the shorter one comes first, so 'app' is before 'apple' but not the reverse. Using a rank dict makes the same code work for any alphabet.

```python
# BASE QUESTION (optimal): Compare character by character,
# optionally with a custom alphabet
# Idea: Find the first position where the words differ and
# compare the two characters' ranks. If one word is a
# prefix of the other, the shorter one comes first, so
# 'app' is before 'apple' but not the reverse. Using a
# rank dict makes the same code work for any alphabet.
def is_sorted_pair(first, second, alphabet="abcdefghijklmnopqrstuvwxyz"):
    rank = {letter: position for position, letter in enumerate(alphabet)}
    for first_letter, second_letter in zip(first, second):
        if first_letter != second_letter:
            return rank[first_letter] < rank[second_letter]
    return len(first) <= len(second)
```

COMPLEXITY | Time O(min(len a, len b)) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 168

169 Return all unique substrings of a given string. EASY

DIFFICULTY | EASY

Better approach: sort all suffixes; the unique substrings are all prefixes minus the shared prefixes of neighbouring suffixes

QUESTION INFO

QUESTION

Return every distinct substring of a string, sorted alphabetically (a substring is a contiguous piece, the whole string and single letters included).

Function signature

def unique_substrings(s: str) -> list

EXAMPLES

Example 1	
Input	`'abc'`
Output	`['a', 'ab', 'abc', 'b', 'bc', 'c']`
Why	`Six different substrings of 'abc'.`

Example 2	
Input	`'aaa'`
Output	`['a', 'aa', 'aaa']`
Why	`'a' appears three times but is listed once.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Set of every slice

BRUTE-FORCE IDEA

Add s[i:j] for every i < j to a set. There are n(n + 1) / 2 slices, each up to n characters, so the work and memory are cubic in characters. Fine for short strings.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Set of every slice
Add s[i:j] for every i < j to a set
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^3) | Space O(n^3)

WHY THIS IS WEAKER | There are about n^2 / 2 substrings and each slice copies up to n characters: O(n^3) time and space. If only the count is needed, a suffix array with LCP counts them in O(n log n) without storing any.

HOW TO IMPROVE IT

sort all suffixes; the unique substrings are all prefixes minus the shared prefixes of neighbouring suffixes

SHORT BEST SOLUTION

Sort the suffixes; the number of distinct substrings is sum over suffixes of (length) minus sum of the LCP of neighbours in sorted order, because each new suffix adds only the prefixes that the previous one did not already contribute. This counts without storing a single substring.

```python
# BASE QUESTION (optimal): Count with a suffix array and
# LCP
# Idea: Sort the suffixes; the number of distinct
# substrings is sum over suffixes of (length) minus sum of
# the LCP of neighbours in sorted order, because each new
# suffix adds only the prefixes that the previous one did
# not already contribute. This counts without storing a
# single substring.
def count_unique_substrings(text):
    length = len(text)
    suffixes = sorted(text[start:] for start in range(length))
    total = length * (length + 1) // 2
    for earlier_suffix, later_suffix in zip(suffixes, suffixes[1:]):
        common_length = 0
        while common_length < min(len(earlier_suffix), len(later_suffix)) and earlier_suffix[common_length] ==
later_suffix[common_length]:
            common_length += 1
        total -= common_length
    return total
```

COMPLEXITY | Time O(n^2 log n) with naive sorting (O(n log n) with a real suffix array) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 169

173 Find Dictionary Matches for a Jammed String — HARD

DIFFICULTY | HARD

Better approach: compress every word to (letter, run length); a query matches when the letters agree and each run is equal or stretched to at least 3

QUESTION INFO

QUESTION

A word was typed on a keyboard where some keys got stuck, so a letter may be repeated. Given a dictionary of words, matches(query) returns the dictionary words the query could be a jammed version of: the runs of equal letters line up with the word's runs, and each run of the query is either exactly as long as in the word or longer and at least min_stretch (default 3) letters long.

Function signature

class JammedIndex:
    def __init__(self, words: list)
    def matches(self, query: str, min_stretch: int = 3) -> list

EXAMPLES

Example 1

Input	`idx = JammedIndex(['hello', 'help', 'hel'])`
	`idx.matches('heeello')`
	`idx.matches('hello')`
	`idx.matches('heello')`
	`idx.matches('helllo')`

Output	`idx.matches('heeello') -> ['hello']`
	`idx.matches('hello') -> ['hello']`
	`idx.matches('heello') -> []`
	`idx.matches('helllo') -> ['hello']`

Why	`'heeello' stretches the e to 3 letters, which is allowed. 'heello' only stretches it to 2 (shorter than min_stretch), and 'helllo' stretches the l run from 2 to 3 letters, which is allowed.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Compare the query with every dictionary word

BRUTE-FORCE IDEA

Encode the query once and compare it with each word's runs. It is correct; the cost is that every query touches every word, even words whose letters cannot match.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Compare the query with every dictionary word
Encode the query once
compare it with each word's runs
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(W * L) per query | Space O(L)

WHY THIS IS WEAKER | Every query touches every word, O(W * L), even words whose letters cannot match. Grouping words by their run letters in a hash map checks only the right bucket.

HOW TO IMPROVE IT

compress every word to (letter, run length); a query matches when the letters agree and each run is equal or stretched to at least 3

SHORT BEST SOLUTION

Group words by the string of their distinct run letters ('hello' -> 'helo'). A query can only match words in its own group, so compare run counts only there. Build the index once and reuse it for many queries.

```
def _runs(text):
    runs = []
    for letter in text:
        if runs and runs[-1][0] == letter:
            runs[-1][1] += 1
        else:
            runs.append([letter, 1])
    return runs


# BASE QUESTION (optimal): Index the dictionary by its run
# letters
# Idea: Group words by the string of their distinct run
# letters ('hello' -> 'helo'). A query can only match
# words in its own group, so compare run counts only
# there. Build the index once and reuse it for many
# queries.
class JammedIndex:
    def __init__(self, words):
        self.groups = {}
        for word in words:
            runs = _runs(word)
            self.groups.setdefault("".join(letter for letter, _ in runs), []).append((word, [run_length for _,
run_length in runs]))

    def matches(self, query, min_stretch=3):
        runs = _runs(query)
        key = "".join(letter for letter, _ in runs)
        query_run_lengths = [run_length for _, run_length in runs]
        return [word for word, word_run_lengths in self.groups.get(key, [])
                if all(query_run_length == run_length or (query_run_length > run_length and query_run_length >=
 min_stretch) for query_run_length, run_length in zip(query_run_lengths, word_run_lengths))]
```

COMPLEXITY | Time O(L) per query plus the size of the matching bucket | Space O(total dictionary length)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 173

GROUP 22 OF 43 | 2 QUESTIONS | PART B - PART 1

Simulate a machine: a class with state and one branch per kind of event

PATTERN CONTEXT

Put the state in __init__, give it one method that takes an event, and write one branch per event kind. Board games keep a grid or counters and check for a win after each move; game-of-life style updates compute every cell from the old state before changing it.

174 Mario-Style Score Tracker (OOP) EASY

DIFFICULTY | EASY

Better approach: the template as written: an event name decides how score, lives and size change

QUESTION INFO

QUESTION

Design a class that tracks the state of a platformer character across a sequence of events. The character has a score (starts at 0), a lives count (starts at 3), and a big flag (starts False). Process events with these rules: "coin" : add 100 to the score. "mushroom" : add 1000 to the score and set big = True . "enemy" : add 200 to the score (the character stomps an enemy). "damage" : if big is True , set big = False (no life lost); otherwise decrease lives by 1. Score is unchanged. The character is alive while lives > 0 . Once lives reaches 0, the game is over and any further events are ignored. Implement the class and a method to run a batch of events.

Function signature

```
class MarioGame:
    def __init__(self) -> None:
        """score=0, lives=3, big=False."""
        ...
    def process(self, event: str) -> None:
        """Apply one event, ignoring it if the game is already over."""
        ...
    def run(self, events: list[str]) -> None:
        """Apply a sequence of events in order."""
        ...
    def get_score(self) -> int:
        ...
    def is_alive(self) -> bool:
        ...
```

EXAMPLES

Example 1	
Input	`g = MarioGame(); g.run(["coin", "mushroom", "enemy", "damage", "damage"]); (g.get_score(), g.is_alive())`
Output	`(1300, True)`
Why	`coin -> 100; mushroom -> 1100, big=True; enemy -> 1300; damage -> big=False (no life lost); damage -> small, lives 3->2. Score 1300, still alive.`

QUESTION

Example 2	
Input	`g = MarioGame(); g.run(["damage", "damage", "damage", "coin"]); (g.get_score(), g.is_alive())`
Output	`(0, False)`
Why	`Three small-state hits drop lives 3->0 (game over). The trailing "coin" is ignored.`

CONSTRAINTS

- Event strings are always one of the four listed above.

CLARIFY BEFORE CODING

Are intervals closed or half-open, does touching count as overlap, and is the input already sorted? What is the tie rule when a start and end happen together? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- Event strings are always one of the four listed above.

BRUTE FORCE | Fold the events over a tuple

BRUTE-FORCE IDEA

Start with: Fold the events over a tuple. It uses the same rules as the class, written as a pure function.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Fold the events over a tuple
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(events) | Space O(1)

WHY THIS IS WEAKER | It uses the same rules as the class, written as a pure function.

HOW TO IMPROVE IT

the template as written: an event name decides how score, lives and size change

SHORT BEST SOLUTION

One method owns the rules; the game ends when lives hit zero.

```python
# BASE QUESTION (optimal): A small state class
# Idea: One method owns the rules; the game ends when
# lives hit zero.
class MarioGame:
    def __init__(self):
        self.score = 0
        self.lives = 3
        self.big = False

    def process(self, event):
        if self.lives == 0:
            return
        if event == "coin":
            self.score += 100
        elif event == "mushroom":
            self.score += 1000
            self.big = True
        elif event == "enemy":
            self.score += 200
        elif event == "damage":
            if self.big:
                self.big = False
            else:
                self.lives -= 1
```

```
    def run(self, events):
        for event in events:
            self.process(event)

    def get_score(self):
        return self.score

    def is_alive(self):
        return self.lives > 0
```

COMPLEXITY | Time O(events) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 174

180 Simulate In-Place Cellular State Updates — HARD

DIFFICULTY | HARD

Better approach: encode old and new state in two bits of each cell so every cell is computed from the old state; shift at the end

QUESTION INFO

QUESTION

Conway's Game of Life on a grid of 0 (dead) and 1 (alive) cells. A live cell with 2 or 3 live neighbours (of its 8 neighbours) stays alive, a dead cell with exactly 3 live neighbours becomes alive, and every other cell is dead in the next generation. All cells change at the same time. Do it in place with O(1) extra space; with wrap=True the grid wraps around its edges.

Function signature

def step_in_place(board: list, wrap=False) -> list

EXAMPLES

Example 1	
Input	`[[0, 1, 0], [0, 1, 0], [0, 1, 0]]`
Output	`[[0, 0, 0], [1, 1, 1], [0, 0, 0]]`
Why	`The vertical blinker turns into a horizontal one.`

Example 2	
Input	`[[1, 1], [1, 1]]`
Output	`[[1, 1], [1, 1]]`
Why	`A 2x2 block is stable: every cell has 3 live neighbours.`

Example 3	
Input	`[[0, 0, 0], [0, 1, 0], [0, 0, 0]]`
Output	`[[0, 0, 0], [0, 0, 0], [0, 0, 0]]`
Why	`A lonely cell dies of isolation.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Copy the board, read from the copy

BRUTE-FORCE IDEA

Compute the next generation from an untouched copy of the board. It is the simplest correct answer, and the extra copy is exactly what the in-place version removes.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Copy the board, read from the copy
Compute the next generation from an untouched copy of the board
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(m * n) | Space O(m * n)

WHY THIS IS WEAKER | Time O(m * n) is already optimal; the cost is a second full board, O(m * n) extra space. Storing the next state in a spare bit of each cell removes the copy.

HOW TO IMPROVE IT

encode old and new state in two bits of each cell so every cell is computed from the old state; shift at the end

SHORT BEST SOLUTION

Bit 0 holds the current state and bit 1 holds the next state. First pass: count neighbors using only bit 0 (value & 1) and set bit 1 by the rules. Second pass: shift every cell right by one to expose the new state. Reading with & 1 means neighbors that were already updated still show their old state.

```
def _life_neighbors(board, row, col, is_live):
    return sum(is_live(board[neighbour_row][neighbour_col]) for neighbour_row in range(row - 1, row + 2) for ne
ighbour_col in range(col - 1, col + 2)
               if (neighbour_row, neighbour_col) != (row, col) and 0 <= neighbour_row < len(board) and 0 <= nei
ghbour_col < len(board[0]))


# BASE QUESTION (optimal): Two-bit encoding in the same
# board
# Idea: Bit 0 holds the current state and bit 1 holds the
# next state. First pass: count neighbors using only bit 0
# (value & 1) and set bit 1 by the rules. Second pass:
# shift every cell right by one to expose the new state.
# Reading with & 1 means neighbors that were already
# updated still show their old state.
def gameOfLife(board):
    rows, cols = len(board), len(board[0])
    for row in range(rows):
        for col in range(cols):
            live_neighbours = _life_neighbors(board, row, col, lambda cell: cell & 1)
            if (board[row][col] & 1 and live_neighbours in (2, 3)) or (not board[row][col] & 1 and live_neighbo
urs == 3):
                board[row][col] |= 2
    for row in range(rows):
        for col in range(cols):
            board[row][col] >>= 1
    return board
```

COMPLEXITY | Time O(m * n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 180

Digits and characters: peel one at a time and carry

PATTERN CONTEXT

Work one digit or character at a time: n % 10 peels the last digit, n // 10 removes it, a carry moves overflow to the next place, and (ord(c) - 97 + k) % 26 shifts a letter. Build the answer in a list and reverse or join at the end.

187 Caesar Cipher: Recover Key, Encrypt, and Decrypt — EASY

DIFFICULTY | EASY

Better approach: per letter: (ord(letter) - 97 + shift) % 26 + 97; the key comes from one letter pair; decrypt uses -shift

QUESTION INFO

QUESTION

A message was scrambled with a Caesar cipher: every lowercase letter was shifted forward by the same secret amount k (0 <= k < 26) with wraparound (z +1 -> a); all non-lowercase characters were left unchanged. You are given one example pair — a known plaintext and its corresponding ciphertext — from which you must recover the secret shift, and then implement the cipher in both directions. Specifically implement: find_shift(plaintext, ciphertext) -> int : return the shift k (0..25) that maps plaintext to ciphertext . The two strings have equal length and contain at least one lowercase letter that pins down k . encrypt(text, k) -> str : shift each lowercase letter forward by k with wraparound; leave other characters unchanged. decrypt(text, k) -> str : the inverse of encrypt — shift each lowercase letter backward by k with wraparound.

Function signature

```
def find_shift(plaintext: str, ciphertext: str) -> int:
    ...
def encrypt(text: str, k: int) -> str:
    ...
def decrypt(text: str, k: int) -> str:
    ...
```

Follow-up (as reported): Given only a ciphertext (no key and no plaintext), how would you break the cipher? Discuss trying all 26 shifts (brute force) or English letter-frequency analysis.

EXAMPLES

Example 1	
Input	`find_shift("abc", "def")`
Output	`3`
Why	`'a'->'d', 'b'->'e', 'c'->'f' is a uniform forward shift of 3.`

Example 2	
Input	`encrypt("hello world", 3)`
Output	`"khoor zruog"`
Why	`Each letter moves forward 3 ('h'->'k', 'o'->'r', 'w'->'z'); the space is unchanged.`

Example 3	
Input	`decrypt("khoor zruog", 3)`
Output	`"hello world"`
Why	`decrypt inverts encrypt, so the original message is recovered.`

QUESTION

Example 4	
Input	`encrypt("xyz", 3)`
Output	`"abc"`
Why	`Wraparound: 'x'->'a', 'y'->'b', 'z'->'c'.`

CONSTRAINTS

- 1 <= len(text) <= 10^5 . text may contain lowercase letters, spaces, digits, and punctuation; only lowercase letters a – z are shifted.
- For find_shift , len(plaintext) == len(ciphertext) and they align character by character.

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= len(text) <= 10^5 . text may contain lowercase letters, spaces, digits, and punctuation; only lowercase letters a - z are shifted.
- For find_shift , len(plaintext) == len(ciphertext) and they align character by character.

BRUTE FORCE | Try all 26 shifts

BRUTE-FORCE IDEA

Start with: Try all 26 shifts. It encrypts the whole text for each guess although one letter pair fixes the key.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try all 26 shifts
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(26 n) | Space O(n)

WHY THIS IS WEAKER | It encrypts the whole text for each guess although one letter pair fixes the key.

HOW TO IMPROVE IT

per letter: (ord(letter) - 97 + shift) % 26 + 97; the key comes from one letter pair; decrypt uses -shift

SHORT BEST SOLUTION

The difference of one letter pair is the key; the same helper shifts forward or backward.

```python
# BASE QUESTION (optimal): Key from one letter pair,
# shared shift helper
# Idea: The difference of one letter pair is the key; the
# same helper shifts forward or backward.
def find_shift(plaintext, ciphertext):
    for plain_letter, cipher_letter in zip(plaintext, ciphertext):
        if "a" <= plain_letter <= "z":
            return (ord(cipher_letter) - ord(plain_letter)) % 26
    return 0


def shift_text(text, shift):
    shifted = []
    for letter in text:
        if "a" <= letter <= "z":
            letter = chr((ord(letter) - 97 + shift) % 26 + 97)
        shifted.append(letter)
    return "".join(shifted)


def encrypt(text, shift):
    return shift_text(text, shift)
```

```
def decrypt(text, shift):
    return shift_text(text, -shift)
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 187

191 Check Digits and Combine Ranges HARD

DIFFICULTY | HARD

Better approach: palindrome number: reverse only the second half with n % 10 and n // 10 and compare with the first half

QUESTION INFO

QUESTION

You are asked to solve two independent coding tasks.

Task 1:

Check Whether an Integer Reads the Same Backward Given an integer **`x`**, return **`true`** if its decimal representation reads the same from left to right and from right to left; otherwise return **`false`**. Requirements and edge cases:

- Negative integers should return **`false`** because of the leading minus sign.
- You may assume **`x`** fits in a signed 32-bit integer.
- Try to solve it without converting the integer to a string. Example:

```
Input: x = 121
Output: true
```

```
Input: x = -121
Output: false
```

```
Input: x = 10
Output: false
```

Task 2:

Combine Overlapping Ranges Given an array of intervals **`intervals`**, where each interval is represented as **`[start, end]`**, combine all overlapping intervals and return an array of non-overlapping intervals that covers exactly the same ranges. Requirements and edge cases:

- Intervals may be provided in any order.
- Two intervals overlap if they share at least one point. For example, **`[1, 4]`** and **`[4, 5]`** should be combined into **`[1, 5]`**.
- Return the intervals in increasing order by start value. Example:

```
Input: intervals = [[1,3],[2,6],[8,10],[15,18]]
Output: [[1,6],[8,10],[15,18]]
```

```
Input: intervals = [[1,4],[4,5]]
Output: [[1,5]]
```

Part 1: Check Whether an Integer Reads the Same BackwardGiven an integer x, return True if its decimal representation reads the same from left to right and from right to left. Otherwise, return False. Negative integers are not palindromes because of the leading minus sign. Try to solve the problem using arithmetic operations instead of converting the integer to a string.
Constraints-2^31 <= x <= 2^31 - 1Negative integers must return FalseAn arithmetic-based solution is preferred over string conversion

QUESTION

EXAMPLES

```
Input: 121
Expected Output: True
```

Explanation: 121 reads the same forward and backward.

```
Input: -121
Expected Output: False
```

Explanation: Negative numbers are not palindromes.

Hints

A negative number can never be a palindrome because the minus sign only appears on the left.Try reversing only the second half of the digits instead of the whole number.

Part 2: Combine Overlapping RangesGiven a list of intervals, where each interval is represented as [start, end], merge all overlapping intervals and return a new list of non-overlapping intervals that covers exactly the same ranges. Intervals may appear in any order. Two intervals overlap if they share at least one point, so [1, 4] and [4, 5] should be merged into [1, 5]. Return the result sorted by starting value.

Constraints

0 <= len(intervals) <= 10^4-10^9 <= start <= end <= 10^9Intervals may be unsorted

Examples

Input: [[1, 3], [2, 6], [8, 10], [15, 18]]

Expected Output: [[1, 6], [8, 10], [15, 18]]

Explanation: The first two intervals overlap and merge into [1, 6].

Input: [[1, 4], [4, 5]]

Expected Output: [[1, 5]]

Explanation: The intervals share the point 4, so they overlap.

Hints

Sort the intervals by their start value before trying to merge them.Keep track of the last interval in the output. If the next interval overlaps, extend the end; otherwise, start a new interval.

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Task 1: convert to a string and compare with its reverse

BRUTE-FORCE IDEA

Str(x) == str(x)[::-1] after rejecting negatives. Correct and short; the question asks to avoid the string.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Task 1: convert to a string and compare with its reverse
Str(x) == str(x)[::-1] after rejecting negatives
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(d) | Space O(d)

WHY THIS IS WEAKER | O(d) time but it builds strings, which the question forbids. Reversing only the second half of the digits arithmetically (x % 10, x // 10) uses O(1) space.

BRUTE FORCE 2 | Task 2: merge any two overlapping ranges until none overlap

BRUTE-FORCE IDEA

Repeatedly look for a pair of ranges that share a point and replace them with their union. Correct, and each pass is quadratic with many passes possible.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Task 2: merge any two overlapping ranges until none overlap
Repeatedly look for a pair of ranges that share a point
replace them with their union
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | Each pass compares all pairs, O(n^2), and several passes may be needed. Sorting by start and sweeping once is O(n log n).

HOW TO IMPROVE IT

palindrome number: reverse only the second half with n % 10 and n // 10 and compare with the first half

SHORT BEST SOLUTION

Part 1: Reject negatives and positive numbers ending in 0. Move digits from x to rev (reversed_half = reversed_half * 10 + number % 10, number //= 10) until reversed_half >= number; then x equals rev (even number of digits) or rev // 10 (odd, middle digit dropped). Only half the digits are processed, so rev cannot overflow.

Part 2: Sort by start. For each range, if it starts at or before the end of the last merged range extend that range's end with max(); otherwise append it as a new range. Sorting guarantees nothing earlier can overlap the new range.

```python
# Task 1: reverse only the second half arithmetically
# Idea: Reject negatives and positive numbers ending in 0.
# Move digits from x to rev (reversed_half = reversed_half * 10 + number % 10, number
# //= 10) until reversed_half >= number; then x equals rev (even number
# of digits) or rev // 10 (odd, middle digit dropped).
# Only half the digits are processed, so rev cannot
# overflow.
def isPalindrome(number):
    if number < 0 or (number % 10 == 0 and number != 0):
        return False
    reversed_half = 0
    while number > reversed_half:
        reversed_half = reversed_half * 10 + number % 10
        number //= 10
    return number == reversed_half or number == reversed_half // 10


# Task 2: sort by start, then one sweep
# Idea: Sort by start. For each range, if it starts at or
# before the end of the last merged range extend that
# range's end with max(); otherwise append it as a new
# range. Sorting guarantees nothing earlier can overlap
# the new range.
def merge(ranges):
    merged = []
    for start, end in sorted(ranges):
        if merged and start <= merged[-1][1]:
            merged[-1][1] = max(merged[-1][1], end)
        else:
            merged.append([start, end])
    return merged
```

COMPLEXITY | Part 1: Time O(d) | Space O(1)
Part 2: Time O(n log n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 191

192 Permutation Sequence HARD

DIFFICULTY | HARD

Better approach: factorial number system: divide rank - 1 by (count - 1)!, (count - 2)!, ... to pick each digit from the list of unused digits

QUESTION INFO

QUESTION

The set [1, 2, ..., n] has n! distinct permutations. Listing them in strictly increasing (lexicographic) order and numbering them from 1 , return the k -th permutation as a string. Aim for an O(n) -ish construction using the factorial number system rather than generating all permutations.

Function signature

```
def get_permutation(n: int, k: int) -> str:
    ...
```

Follow-up (as reported): A classmate asked whether a brute-force "generate all and index" is acceptable; discuss why the factorial-base method avoids O(n!) .

EXAMPLES

Example 1	
Input	`n = 3, k = 3`
Output	`"213"`
Why	`Order is 123, 132, 213, 231, 312, 321; the 3rd is "213".`

Example 2	
Input	`n = 4, k = 9`
Output	`"2314"`
Why	`The 9th permutation of 1234 in lexicographic order is "2314".`

Example 3	
Input	`n = 3, k = 1`
Output	`"123"`
Why	`The first permutation is the sorted sequence.`

CONSTRAINTS

- 1 <= n <= 9
- 1 <= k <= n!

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 1 <= n <= 9
- 1 <= k <= n!

BRUTE FORCE | Generate every permutation

BRUTE-FORCE IDEA

Start with: Generate every permutation. It builds permutations one by one until it reaches the k-th.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Generate every permutation
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n! * n) | Space O(n)

WHY THIS IS WEAKER | It builds permutations one by one until it reaches the k-th.

HOW TO IMPROVE IT

factorial number system: divide rank - 1 by (count - 1)!, (count - 2)!, ... to pick each digit from the list of unused digits

SHORT BEST SOLUTION

Each digit is chosen by dividing the remaining index by the size of a block of permutations.

```python
# BASE QUESTION (optimal): Factorial number system
# Idea: Each digit is chosen by dividing the remaining
# index by the size of a block of permutations.
from math import factorial


def get_permutation(count, rank):
    digits = list(range(1, count + 1))
    rank -= 1
    chosen = []
    for remaining in range(count, 0, -1):
        block = factorial(remaining - 1)
        chosen.append(digits.pop(rank // block))
        rank %= block
    return "".join(map(str, chosen))
```

COMPLEXITY | Time O(n^2) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 192

GROUP 25 OF 43 | 2 QUESTIONS | PART B - PART 1

Look at the input once and keep only a small summary (and count your calls)

PATTERN CONTEXT

When the input is too big to hold, read it one item at a time and keep a tiny summary: a maximum, a set from the smaller file, a sorted window. When asking a question is expensive (an oracle), use binary search on how many items you include so the number of calls stays small.

197 Find All Bad Pairs Using runTest Oracle — HARD

DIFFICULTY | HARD

Better approach: for each test, binary search the prefix of earlier tests where runTest first fails; that test is a bad partner; repeat until none is left

QUESTION INFO

QUESTION

PhD intern coding round 2: given a black-box oracle **`runTest(S)`** that returns **`true`** iff a subset **`S`** contains no bad pair, identify all bad pairs across N test items with as few oracle calls as possible.

Items t_1, ..., t_N are given.
Oracle: runTest(S: Set[Item]) -> bool returns true iff S contains zero bad pairs; otherwise false.
Output: every bad pair (t_i, t_j).
Minimize total oracle calls.

EXAMPLES

```
N=4, bad pairs = {(1,2)}.
runTest({1,2,3,4}) -> false -> at least one bad pair somewhere.
runTest({1,3,4})   -> true  -> 2 is in every bad pair containing items from this set.
... binary search to isolate.
```

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Call runTest on every pair

BRUTE-FORCE IDEA

Start with: Call runTest on every pair. Test every pair of items with runTest and keep the failing pairs. Always N(N-1)/2 calls.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Call runTest on every pair
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(N^2) calls | Space O(1)

WHY THIS IS WEAKER | Always N(N - 1) / 2 calls. Testing whole groups and binary searching a failing group finds each bad pair in about log N calls.

HOW TO IMPROVE IT

for each test, binary search the prefix of earlier tests where runTest first fails; that test is a bad partner; repeat until none is left

SHORT BEST SOLUTION

Keep groups of tests that pass together. For a new test test and a group G: if runTest(G + {test}) passes there is no partner in G; otherwise binary search for the smallest prefix of G that fails with test, record that partner, remove it and repeat. Put test into a group with no partner (or a new group). Pairs inside a group cannot exist and pairs across groups are found when the later test

arrives.

```
# BASE QUESTION (optimal): Group the tests and binary
# search for partners
# Idea: Keep groups of tests that pass together. For a new
# test test and a group G: if runTest(G + {test}) passes there
# is no partner in G; otherwise binary search for the
# smallest prefix of G that fails with test, record that
# partner, remove it and repeat. Put test into a group with
# no partner (or a new group). Pairs inside a group cannot
# exist and pairs across groups are found when the later
# test arrives.
def find_bad_pairs(tests, runTest):
    groups, pairs = [], set()

    def partners(group, test):
        found, rest = [], list(group)
        while rest and not runTest(set(rest) | {test}):
            low, high = 1, len(rest)
            while low < high:
                middle = (low + high) // 2
                if runTest(set(rest[:middle]) | {test}):
                    low = middle + 1
                else:
                    high = middle
            found.append(rest[low - 1])
            rest = rest[:low - 1] + rest[low:]
        return found

    for test in tests:
        placed = False
        for group in groups:
            found_partners = partners(group, test)
            for partner in found_partners:
                pairs.add((min(partner, test), max(partner, test)))
            if not found_partners and not placed:
                group.append(test)
                placed = True
        if not placed:
            groups.append([test])
    return pairs
```

COMPLEXITY | Time about O(K log N + N * G) calls (K bad pairs, G groups) | Space O(N)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 197

198 Find maximum value in huge text file — HARD

DIFFICULTY | HARD

Better approach: the template as written; the file is never loaded as a whole

QUESTION INFO

QUESTION

A text file is too large to load into memory (tens or hundreds of gigabytes). Each line holds one signed 64-bit integer. Return the largest value by reading the file sequentially, holding only a small amount in memory.

Function signature

def max_in_file(path: str) -> int

EXAMPLES

QUESTION

Example 1

Input	**nums.txt contains the lines:** **5** **-3** **12** **7**
Output	**max_in_file(path) -> 12**
Why	**Only the current line and the best value so far are in memory, so the file size does not matter.**

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Load the whole file and take max

BRUTE-FORCE IDEA

Read every number into a list and call max. Correct for small files; it needs memory proportional to the file, which is the thing the problem rules out.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Load the whole file and take max
Read every number into a list
call max
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | Needs memory for the whole file, which is exactly what is not available at tens of gigabytes. Streaming with one running maximum uses O(1) memory.

HOW TO IMPROVE IT

the template as written; the file is never loaded as a whole

SHORT BEST SOLUTION

Part 1: Never load the file. Read it through a buffered iterator and keep one variable, the maximum so far, started at None so that an all-negative file or a one-line file is handled correctly (do not start at 0). Python ints are unbounded, but in a fixed-width language parse into 64 bits and treat overflow as bad input. Disk read speed is the bottleneck, not the comparison.

Part 2: Divide the file into roughly equal byte ranges. Each worker seeks to its start and, unless it starts at byte 0, discards the partial line it landed in (the previous worker finishes that line). It reads lines until the line it just read began past its end offset. Every line is therefore processed exactly once. The reducer takes the max of the partial results and ignores workers whose range was empty.

```python
# BASE QUESTION (optimal): Single machine: stream the
# file, keep one running maximum
# Idea: Never load the file. Read it through a buffered
# iterator and keep one variable, the maximum so far,
# started at None so that an all-negative file or a one-
# line file is handled correctly (do not start at 0).
# Python ints are unbounded, but in a fixed-width language
# parse into 64 bits and treat overflow as bad input. Disk
# read speed is the bottleneck, not the comparison.
def max_in_file(path):
    best = None
    with open(path, "r", buffering=1 << 20) as file:
        for line in file:
            line = line.strip()
            if not line:
```

```
            continue
        value = int(line)
        if best is None or value > best:
            best = value
    return best
```

COMPLEXITY | Part 1: Time O(n) | Space O(1) plus the read buffer
Part 2: Time O(n / workers) plus I/O limits | Space O(workers)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 198

GROUP 26 OF 43 | 2 QUESTIONS | PART B - PART 1

Grid DP: each cell from its neighbours above and to the left

PATTERN CONTEXT

Walk the grid row by row and keep only the previous row. A cell's value depends on the cell above, the cell to the left, and the diagonal. Change the condition for a cell to count and the formula that combines the three neighbours.

202 Find the Longest Straight Run of One Color — HARD

DIFFICULTY | HARD

Better approach: the same four run lengths; extend a run when the neighbour in that direction has the same colour

QUESTION INFO

QUESTION

Given a rectangular grid of colors, return the length of the longest consecutive sequence of cells with the same color along a horizontal, vertical, or diagonal straight line. A sequence cannot change direction or skip a cell. Both diagonal orientations count. The length is the number of cells in the sequence.

Input

- **`grid`**: an array of strings of equal length. Each character represents one cell's color.

Output

Return the maximum length as an integer.

Constraints

and Edge Cases - For this practice version, the grid contains at least one row and one column, with at most **`1000000`** cells in total.

- Colors are uppercase English letters; equal letters mean equal colors.
- A one-cell grid has answer **`1`**.
- Diagonal lines may run down-right or down-left. Reading the same line in reverse does not change its length.
- A connected region that bends is not necessarily one valid sequence.

Example

1

```
grid = ["RRRG", "RRBB", "GRRR", "BGBR"] output = 4
```

The main diagonal contains four consecutive **`R`** cells.

Example

2

```
grid = ["RBG", "BGR", "GRB"] output = 3
```

The down-left diagonal from the top-right cell contains three **`G`** cells. No horizontal or vertical sequence has that length.

EXAMPLES

Example 1	
Input	`['RRRG', 'RRBB', 'GRRR', 'BGBR']`
Output	`4`
Why	`Four R cells lie on the main diagonal.`

QUESTION

Example 2	
Input	`['RGB', 'BGR', 'RGB']`
Output	**3**
Why	**Three G cells in the middle column.**

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | From every cell, walk in each of four directions

BRUTE-FORCE IDEA

For each start cell and direction, step while the color matches. Correct and short, but cells in a long run are re-walked from every start.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: From every cell, walk in each of four directions
For each start cell
step while the color matches
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(m * n * min(m, n)) | Space O(1)

WHY THIS IS WEAKER | A long run is re-walked from every cell inside it, O(m * n * min(m, n)). A table of the run length ending at each cell per direction is O(m * n).

HOW TO IMPROVE IT

the same four run lengths; extend a run when the neighbour in that direction has the same colour

SHORT BEST SOLUTION

run[d][i][j] is the length of the same-color run ending at (i, j) in direction d. It is the previous cell's run plus one when the colors match, else 1. The previous cell is left, up, up-left and up-right for the four directions, so one previous row of each direction is enough.

```python
# BASE QUESTION (optimal): DP per direction with a rolling
# row
# Idea: run[d][i][j] is the length of the same-color run
# ending at (i, j) in direction d. It is the previous
# cell's run plus one when the colors match, else 1. The
# previous cell is left, up, up-left and up-right for the
# four directions, so one previous row of each direction
# is enough.
def longest_run(grid):
    if not grid:
        return 0
    width = len(grid[0])
    horiz = [0] * width
    vert = [0] * width
    diag = [0] * width
    anti = [0] * width
    best = 0
    previous_row = None
    for row_index, row in enumerate(grid):
        new_horiz, new_vert, new_diag, new_anti = [1] * width, [1] * width, [1] * width, [1] * width
        for col in range(width):
            if col and row[col] == row[col - 1]:
                new_horiz[col] = new_horiz[col - 1] + 1
            if previous_row is not None:
                if row[col] == previous_row[col]:
                    new_vert[col] = vert[col] + 1
```

```
            if col and row[col] == previous_row[col - 1]:
                new_diag[col] = diag[col - 1] + 1
            if col + 1 < width and row[col] == previous_row[col + 1]:
                new_anti[col] = anti[col + 1] + 1
            best = max(best, new_horiz[col], new_vert[col], new_diag[col], new_anti[col])
        horiz, vert, diag, anti, previous_row = new_horiz, new_vert, new_diag, new_anti, row
    return best
```

COMPLEXITY | Time O(m * n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 202

203 Count same-color squares in a character grid — HARD

DIFFICULTY | HARD

Better approach: exactly the same recurrence on characters

QUESTION INFO

QUESTION

You are given a 2D grid (matrix) of characters. Each character represents a color: cells with the same character are considered the same color. Formally:

- The grid has `m` rows and `n` columns.
- `grid[i][j]` is an alphabetic character (e.g., `'a'`–`'z'` or `'A'`–`'Z'`). A **square** in this grid is defined as a contiguous `k × k` submatrix aligned with the grid axes, for some integer `k >= 1`. A square is **monochromatic** if **all** of its cells contain the **same** character (i.e., the same color). Task:
- Count and return the total number of monochromatic squares in the grid.
- Count all sizes of squares (1×1, 2×2, ..., up to the largest possible that fits in the grid). Example (just for clarity, not necessarily exhaustive): If the grid is:

```
a a b a a b b b
```

Then some monochromatic squares include:

- All 1×1 cells (each single cell is a monochromatic square).
- The 2×2 square in the top-left corner formed by `'a'`. Your function should return the total count of such monochromatic squares for a given input grid.

You are given a 2D grid of characters, where each character represents a color. A square is a contiguous k × k submatrix aligned with the grid axes, for some k >= 1. A square is monochromatic if all of its cells contain the same character. Return the total number of monochromatic squares of all possible sizes and positions in the grid. Every 1 × 1 cell is considered a monochromatic square.
Constraints
0 <= len(grid) <= 1000If grid is non-empty, 0 <= len(grid[0]) <= 1000All rows in grid have the same lengthgrid[i][j] is an alphabetic characterThe total number of cells is at most 1,000,000

EXAMPLES

```
Input: ([])
Expected Output: 0
```

Explanation: An empty grid contains no squares.

```
Input: (["aab", "aab", "bbb"])
Expected Output: 10
```

Explanation: There are 9 single-cell squares and one 2x2 monochromatic square of 'a' in the top-left corner.

QUESTION

Hints

Try defining dp[i][j] as the side length of the largest monochromatic square whose bottom-right corner is cell (i, j).A square larger than 1 can end at (i, j) only if the current cell matches its top, left, and top-left neighboring cells.

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Check every square from every corner

BRUTE-FORCE IDEA

For each top-left corner and each size k, verify that all k * k cells are equal. Correct; the cost is huge because the same cells are rechecked for every size.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Check every square from every corner
For each top-left corner
each size k
verify that all k * k cells are equal
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(m * n * min(m, n)^3) | Space O(1)

WHY THIS IS WEAKER | The same cells are rechecked for every size: O(m * n * min(m, n)^3). A table of the largest same-colour square ending at each cell, summed over all cells, is O(m * n).

HOW TO IMPROVE IT

exactly the same recurrence on characters

SHORT BEST SOLUTION

If the cell matches its top, left and top-left neighbours then dp = 1 + min(dp top, dp left, dp top-left); otherwise 1. Summing dp over all cells counts every square exactly once, at its bottom-right corner. Only the previous row is needed.

```python
# BASE QUESTION (optimal): DP: side of the largest square
# ending at each cell, summed
# Idea: If the cell matches its top, left and top-left
# neighbours then dp = 1 + min(dp top, dp left, dp top-
# left); otherwise 1. Summing dp over all cells counts
# every square exactly once, at its bottom-right corner.
# Only the previous row is needed.
def count_same_color_squares(grid):
    rows, cols = len(grid), len(grid[0])
    previous_row, total = [0] * cols, 0
    for row in range(rows):
        current_row = [1] * cols
        for col in range(cols):
            if row and col and grid[row][col] == grid[row - 1][col] == grid[row][col - 1] == grid[row - 1][col
- 1]:
                current_row[col] = 1 + min(previous_row[col], current_row[col - 1], previous_row[col - 1])
        total += sum(current_row)
        previous_row = current_row
    return total
```

COMPLEXITY | Time O(m * n) | Space O(n) with a rolling row

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 203

Counting pairs by looking up each partner in a Counter or a bucket

PATTERN CONTEXT

For each value, compute the partner it would need (v + k, 2v, the other corner of a square) and look it up in a Counter or a set. When the partner depends on two things, enumerate the pairs once and bucket them by a shared key.

223 Maximum-Area Rectangle from Planar Points — HARD

DIFFICULTY | HARD

Better approach: bucket every pair of points by (sum of coordinates, squared length); two pairs in one bucket are the diagonals of a rectangle

QUESTION INFO

QUESTION

Given 2D points, find the maximum area of a rectangle whose four corners are input points; sides need not be parallel to the axes. Candidates compare it to LC 963 (Minimum Area Rectangle II) but maximizing. Interviewers ask to improve beyond an O(N^3) approach.

Input: a list of 2D coordinate points.
Return the maximum area of any rectangle whose four vertices are all input points.
Rectangle sides do not need to be parallel to the coordinate axes.
Follow-up pressure: start from an O(N^3) idea, then optimize toward a better complexity.

EXAMPLES

Example 1	
Input	`[(1, 1), (1, 3), (3, 1), (3, 3)]`
Output	`4`
Why	`The four points form a 2 x 2 axis-aligned square.`

Example 2	
Input	`[(0, 1), (1, 0), (2, 1), (1, 2)]`
Output	`2`
Why	`A tilted rectangle: a square with diagonal 2 has area 2.`

Example 3	
Input	`[(0, 0), (1, 0), (0, 1)]`
Output	`0`
Why	`Three points never make a rectangle.`

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every triple with a right angle

BRUTE-FORCE IDEA

Choose a corner p and two other points q and r. If dot(q - p, r - p) is zero, they form a right angle; check whether q + r - p exists as the fourth corner and update the maximum area. This checks all triples, so it is cubic.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
for each point p as a possible corner:
    try every pair of other points q and r
    require dot(q - p, r - p) == 0
    require the fourth corner q + r - p to exist
    update the maximum rectangle area
```

BRUTE-FORCE COMPLEXITY | Time O(n^3) | Space O(n)

WHY THIS IS WEAKER | Three nested loops over the points, O(n^3). Bucketing pairs by their diagonal midpoint and length finds rectangles from O(n^2) pairs.

HOW TO IMPROVE IT

bucket every pair of points by (sum of coordinates, squared length); two pairs in one bucket are the diagonals of a rectangle

SHORT BEST SOLUTION

For every pair (p, q) use the key (p.x + q.x, p.y + q.y, squared length). All pairs with the same key are diagonals of one circle around the same centre, so any two of them form a rectangle. For two diagonals (p, q) and (r, s) the sides from p are r - p and s - p, and the area is the absolute cross product. Keep the maximum; return 0 when no bucket has two pairs.

```python
from collections import defaultdict
import itertools


# BASE QUESTION (optimal): Bucket pairs by diagonal
# midpoint and length
# Idea: For every pair (p, q) use the key (p.x + q.x, p.y
# + q.y, squared length). All pairs with the same key are
# diagonals of one circle around the same centre, so any
# two of them form a rectangle. For two diagonals (p, q)
# and (r, s) the sides from p are r - p and s - p, and the
# area is the absolute cross product. Keep the maximum;
# return 0 when no bucket has two pairs.
def max_rect_area(points):
    unique_points = list(set(map(tuple, points)))
    diagonals_by_key = defaultdict(list)
    for first_index in range(len(unique_points)):
        for second_index in range(first_index + 1, len(unique_points)):
            p, q = unique_points[first_index], unique_points[second_index]
            diagonals_by_key[(p[0] + q[0], p[1] + q[1], (p[0] - q[0]) ** 2 + (p[1] - q[1]) ** 2)].append((p, q)
)
    best = 0
    for pairs in diagonals_by_key.values():
        for (p, q), (r, s) in itertools.combinations(pairs, 2):
            best = max(best, abs((r[0] - p[0]) * (s[1] - p[1]) - (r[1] - p[1]) * (s[0] - p[0])))
    return best
```

COMPLEXITY | Time O(n^2) pairs (worst case O(n^3) if every pair shares a bucket) | Space O(n^2)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 223

GROUP 31 OF 43 | 2 QUESTIONS | PART B - PART 1

Tree DP: build a parent order once, then add up from the leaves

PATTERN CONTEXT

Get an order where every parent comes before its children (a BFS order from the root), then sweep it backwards so every child is finished before its parent. Subtree sizes and sums fall out of that one backwards sweep; the re-rooting questions add a second forward sweep.

227 Compute distance-sum from every tree node — HARD

DIFFICULTY | HARD

Better approach: down[neighbour] = sum of distances to v's subtree; then a forward sweep: distance_sum[child] = distance_sum[parent] - size[child] + (node_count - size[child])

QUESTION INFO

QUESTION

A tree has n nodes numbered 0 to n - 1 and n - 1 edges (u, v). For every node, return the sum of the distances from that node to all the other nodes.

Function signature

def distance_sums(n: int, edges: list) -> list

EXAMPLES

Example 1	
Input	`4, [(0, 1), (0, 2), (2, 3)]`
Output	`[4, 6, 4, 6]`
Why	`Node 0: 1 + 1 + 2 = 4. Node 1 is a leaf: 1 + 2 + 3 = 6. Node 2: 1 + 2 + 1 = 4. Node 3: 1 + 2 + 3 = 6.`

Example 2	
Input	`3, [(0, 1), (1, 2)]`
Output	`[3, 2, 3]`
Why	`A path of three nodes: the middle one is closest to the others.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | BFS from every node

BRUTE-FORCE IDEA

Start with: BFS from every node. For each node run a BFS and add up the distances. Correct; n searches of O(n) each.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: BFS from every node
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | N searches of O(n) each, O(n^2). Compute the answer for the root, then move the root edge by edge using subtree sizes (re-rooting), O(n).

HOW TO IMPROVE IT

down[neighbour] = sum of distances to v's subtree; then a forward sweep: distance_sum[child] = distance_sum[parent] - size[child] + (node_count - size[child])

SHORT BEST SOLUTION

Root the tree at 0 and compute size[neighbour] (subtree sizes) and distance_sum[0] = the sum of depths. For a child c of v: distance_sum[c] = distance_sum[neighbour] - size[c] + (node_count - size[c]), because the size[c] nodes below c get one closer and the other node_count - size[c] nodes get one farther. Done iteratively so deep trees are safe.

```
from collections import defaultdict


# BASE QUESTION (optimal): Re-rooting with subtree sizes
# Idea: Root the tree at 0 and compute size[neighbour] (subtree
# sizes) and distance_sum[0] = the sum of depths. For a child c of
# v: distance_sum[c] = distance_sum[neighbour] - size[c] + (node_count - size[c]), because
# the size[c] nodes below c get one closer and the other n
# - size[c] nodes get one farther. Done iteratively so
# deep trees are safe.
def sumOfDistancesInTree(node_count, edges):
    adj = defaultdict(list)
    for node, neighbour in edges:
        adj[node].append(neighbour); adj[neighbour].append(node)
    order, parent, depth = [], {0: -1}, {0: 0}
    stack = [0]
    while stack:
        node = stack.pop()
        order.append(node)
        for neighbour in adj[node]:
            if neighbour not in parent:
                parent[neighbour], depth[neighbour] = node, depth[node] + 1
                stack.append(neighbour)
    size = [1] * node_count
    for node in reversed(order):
        if parent[node] != -1:
            size[parent[node]] += size[node]
    distance_sum = [0] * node_count
    distance_sum[0] = sum(depth.values())
    for node in order[1:]:
        distance_sum[node] = distance_sum[parent[node]] - size[node] + (node_count - size[node])
    return distance_sum
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 227

228 Tree Distance Sum (Re-root DP) HARD

DIFFICULTY | HARD

Better approach: exactly the same two sweeps (this is the re-rooting version of the question)

QUESTION INFO

QUESTION

Onsite coding round 1: given an undirected tree on N nodes, return for every node the sum of distances to all other nodes. Equivalent to LC 834. Interviewer expects baseline O(n²) DFS followed by the O(n) re-root DP.

Input: undirected tree with N nodes and N-1 edges.
Output: array ans where ans[u] is the sum of shortest-path distances from u to every other node.
Walk through baseline (BFS/DFS from every node, O(n²)) before optimizing.
Final solution should be O(n) via two-pass re-root DP.

QUESTION

EXAMPLES

```
Edges: [[0,1],[0,2],[2,3],[2,4],[2,5]]
ans:   [8, 12, 6, 10, 10, 10]
```

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | BFS from every node

BRUTE-FORCE IDEA

Run a BFS from each node and add up the distances. Correct and the right direct approach to state in the first minutes; quadratic is too slow for 10^5 nodes.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: BFS from every node
Run a BFS from each node
add up the distances
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(n)

WHY THIS IS WEAKER | N BFS runs of O(n) each, O(n^2). Moving the root across one edge changes every distance by exactly 1, so two passes over the tree give all answers in O(n).

HOW TO IMPROVE IT

exactly the same two sweeps (this is the re-rooting version of the question)

SHORT BEST SOLUTION

Root the tree at 0 and get a BFS order. In reverse order compute size[node] and the sum of distances inside u's subtree. For the root that sum is already the answer. In forward order, distance_sum[neighbour] = distance_sum[parent] - size[neighbour] + (node_count - size[neighbour]): the size[neighbour] nodes inside v's subtree are one step closer and the other node_count - size[neighbour] nodes one step farther. Everything is iterative, so a chain of 10^5 nodes cannot overflow the stack.

```python
# BASE QUESTION (optimal): Two passes: subtree sizes, then
# re-root
# Idea: Root the tree at 0 and get a BFS order. In reverse
# order compute size[node] and the sum of distances inside
# u's subtree. For the root that sum is already the
# answer. In forward order, distance_sum[neighbour] = distance_sum[parent] - size[neighbour]
# + (node_count - size[neighbour]): the size[neighbour] nodes inside v's subtree
# are one step closer and the other node_count - size[neighbour] nodes one
# step farther. Everything is iterative, so a chain of
# 10^5 nodes cannot overflow the stack.
def tree_distance_sums(node_count, edges):
    if node_count == 0:
        return []
    adj = [[] for _ in range(node_count)]
    for node_a, node_b in edges:
        adj[node_a].append(node_b); adj[node_b].append(node_a)
    parent, order = [-1] * node_count, [0]
    for node in order:
        for neighbour in adj[node]:
            if neighbour != parent[node]:
                parent[neighbour] = node; order.append(neighbour)
    size, distance_inside = [1] * node_count, [0] * node_count
    for node in reversed(order):
        if parent[node] >= 0:
            size[parent[node]] += size[node]
            distance_inside[parent[node]] += distance_inside[node] + size[node]
    distance_sum = [0] * node_count
```

```
    distance_sum[0] = distance_inside[0]
    for node in order[1:]:
        distance_sum[node] = distance_sum[parent[node]] + node_count - 2 * size[node]
    return distance_sum
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 228

Prefix sums, Tier 2: running total plus a dict of earlier totals

PATTERN CONTEXT

Keep a running total. At each step ask the dict about an earlier total: total - k (subarray sums to k), total % k (sum divisible by k), or the smallest earlier total (best subarray ending here). Store the first index, a count, or the minimum.

238 Given an integer array nums and an integer k, return true if nums has a subarray of at least two elements whose sum is a multiple of k. HARD

DIFFICULTY | HARD

Better approach: remainder = total % divisor; store the first index; a hit at least 2 positions back means a subarray whose sum is a multiple of k

QUESTION INFO

QUESTION

Determine if a given array contains a subarray of at least two elements whose sum is a multiple of a specified number k. An array is considered to have a "good subarray" if there exists at least one subarray (consisting of two or more elements) such that the sum of the elements in this subarray is a multiple of k. For example, the array [23, 2, 4, 7] with k = 6 has a "good subarray" ([2, 4]), as the sum 6 is a multiple of k = 6. The array [5, 0, 0, 0] with k = 3 also has a "good subarray" ([0, 0]), since the sum is 0, and 0 is a multiple of 3. Examples nums = [23, 2, 4, 7], k = 6 output: true nums = [5, 0, 0, 0], k = 3 output: true nums = null, k = 1 output: false In these examples, the output is true if there exists a "good subarray" in the given array nums for the specified k, and false otherwise.Challenge 1Can you come up with a solution with a time complexity of O(n)?Challenge 2Can you come up with a solution with a space complexity of O(n)?

EXAMPLES

Example 1	
Input	`[23, 2, 4, 6, 7], 6`
Output	`true`
Why	`[2, 4] adds up to 6, a multiple of 6.`

Example 2	
Input	`[23, 2, 6, 4, 7], 13`
Output	`false`
Why	`No subarray of at least two elements adds up to a multiple of 13.`

Example 3	
Input	`[23, 2, 6, 4, 7], 6`
Output	`true`
Why	`[23, 2, 6, 4, 7] adds up to 42, a multiple of 6.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Check every subarray of length at least two

BRUTE-FORCE IDEA

Extend each start with a running sum and test every end at least one element later. This takes quadratic time.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Check every subarray of length at least two
Extend each start with a running sum
test every end at least one element later
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) | Space O(1)

WHY THIS IS WEAKER | O(n^2) pairs. Two prefix sums with the same remainder mod k bound a subarray whose sum is a multiple of k, so a dictionary of first-seen remainders is O(n).

HOW TO IMPROVE IT

remainder = total % divisor; store the first index; a hit at least 2 positions back means a subarray whose sum is a multiple of k

SHORT BEST SOLUTION

Walk the array keeping the running sum modulo k. If this remainder was seen before at index j and the current index i satisfies index - j >= 2, the elements between form a good subarray. Store only the first index of each remainder (a later one could only give a shorter subarray). For divisor = 0 compare the raw prefix sums instead.

```python
# BASE QUESTION (optimal): Prefix sums mod k with the
# first index of each remainder
# Idea: Walk the array keeping the running sum modulo k.
# If this remainder was seen before at index j and the
# current index i satisfies index - j >= 2, the elements
# between form a good subarray. Store only the first index
# of each remainder (a later one could only give a shorter
# subarray). For divisor = 0 compare the raw prefix sums
# instead.
def checkSubarraySum(nums, divisor):
    if not nums or len(nums) < 2:
        return False
    first_index_of, running_sum = {0: -1}, 0
    for index, number in enumerate(nums):
        running_sum += number
        remainder = running_sum if divisor == 0 else running_sum % divisor
        if remainder in first_index_of:
            if index - first_index_of[remainder] >= 2:
                return True
        else:
            first_index_of[remainder] = index
    return False
```

COMPLEXITY | Time O(n) | Space O(min(n, k))

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 238

Group by a canonical key

PATTERN CONTEXT

Find a key that is the same for everything that should be grouped (sorted letters, letter counts, shifts between letters, the keyboard row) and put the items into a dict under that key. Equal key means same group.

239 Given an array of strings, return the words that can be typed using letters of the alphabet on only one row of American keyboard. EASY

DIFFICULTY | EASY

Better approach: key = the keyboard row of each letter; a word qualifies when its letters map to one row

QUESTION INFO

QUESTION

Given an array of words, return those that can be typed using letters from a single row of an American keyboard (the rows are qwertyuiop, asdfghjkl and zxcvbnm). Ignore upper and lower case and keep the input order.

Function signature

def findWords(words: list) -> list

EXAMPLES

Example 1	
Input	`['Hello', 'Alaska', 'Dad', 'Peace']`
Output	`['Alaska', 'Dad']`
Why	`'Alaska' and 'Dad' use only the middle row; 'Hello' and 'Peace' mix rows.`

Example 2	
Input	`['omk']`
Output	`[]`
Why	`'omk' mixes the top and bottom rows.`

Example 3	
Input	`['adsdf', 'sfd']`
Output	`['adsdf', 'sfd']`
Why	`Both words stay on one row.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | For each word test whether it fits each row

BRUTE-FORCE IDEA

For every word, check whether all of its lowercase letters are in the first row, or all in the second, or all in the third. Correct; it rescans each word up to three times.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: For each word test whether it fits each row
For every word
check whether all of its lowercase letters are in the first row
```

```
or all in the second
or all in the third
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(W * L * 3) | Space O(1)

WHY THIS IS WEAKER | Up to three scans per word; building one letter-to-row table makes each word a single scan, O(total characters).

HOW TO IMPROVE IT

key = the keyboard row of each letter; a word qualifies when its letters map to one row

SHORT BEST SOLUTION

Build a dict from letter to row index (26 entries). A word qualifies if all of its lowercased letters map to the same row, i.e. the set of rows has size one. Lowercase first so 'Alaska' works.

```python
# BASE QUESTION (optimal): Map each letter to its row and
# compare sets
# Idea: Build a dict from letter to row index (26
# entries). A word qualifies if all of its lowercased
# letters map to the same row, i.e. the set of rows has
# size one. Lowercase first so 'Alaska' works.
def findWords(words):
    rows = ["qwertyuiop", "asdfghjkl", "zxcvbnm"]
    row_of = {letter: row_number for row_number, row in enumerate(rows) for letter in row}
    return [word for word in words if len({row_of[letter] for letter in word.lower()}) <= 1]
```

COMPLEXITY | Time O(total characters) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 239

243 Given two strings and two allowed operations, swapping positions of letters and converting all occurrences of one character to another, can the first string be transformed into the second? HARD

DIFFICULTY | HARD

Better approach: same letters used, and the sorted list of letter frequencies must match: compare set(first) == set(second) and the sorted Counter values

QUESTION INFO

QUESTION

Two strings are 'close' if you can turn one into the other with these operations, used any number of times: (1) swap the positions of any two letters; (2) take two letters that occur in the string and swap all of their occurrences (every a becomes b and every b becomes a). Return whether the first string can be transformed into the second.

Function signature

def close_strings(s: str, t: str) -> bool

EXAMPLES

Example 1	
Input	'abc', 'bca'
Output	true
Why	Swapping positions turns 'abc' into 'bca'.

Example 2	
Input	'a', 'aa'
Output	false
Why	The lengths differ.

QUESTION

Example 3	
Input	`'cabbba', 'abbccc'`
Output	`true`
Why	`Use operation 1 to reorder, then operation 2 to swap the letters a and b, and c and ... until the counts match.`

Example 4	
Input	`'uau', 'ssx'`
Output	`false`
Why	`'u' appears in the first string but not the second, so they cannot be close.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | BFS over every reachable string (Close Strings reading)

BRUTE-FORCE IDEA

From the first string apply every swap of two positions and every swap of two letters' identities, searching for the second. Exact for the frequency test, but the state space explodes.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: BFS over every reachable string (Close Strings reading)
From the first string apply every swap of two positions
every swap of two letters' identities
searching for the second
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space exponential

WHY THIS IS WEAKER | The number of reachable strings grows exponentially with the length. Because swaps make order irrelevant, comparing the sets of letters and their sorted counts decides it in O(n).

HOW TO IMPROVE IT

same letters used, and the sorted list of letter frequencies must match: compare set(first) == set(second) and the sorted Counter values

SHORT BEST SOLUTION

Part 1: Operation 1 reorders positions freely, so only the multiset of letters matters. Operation 2 exchanges the identities of two letters that exist, which permutes the frequencies among existing letters. Therefore first can become second exactly when both use the same set of letters (a letter cannot be created or destroyed) and their sorted frequency lists are equal.

Part 2: Each source letter must map to exactly one target letter (a function), otherwise it is impossible. If first != second and second already uses all 26 letters, no spare letter exists to break mapping cycles, so it is impossible. Otherwise convert in an order that resolves chains, using a spare letter for any cycle.

```python
from collections import Counter


# BASE QUESTION (optimal): Reading 1 (LeetCode 1657): same
# letter set and same multiset of frequencies
# Idea: Operation 1 reorders positions freely, so only the
# multiset of letters matters. Operation 2 exchanges the
# identities of two letters that exist, which permutes the
# frequencies among existing letters. Therefore first can
# become second exactly when both use the same set of letters
# (a letter cannot be created or destroyed) and their
```

```
# sorted frequency lists are equal.
def close_strings(first, second):
    return set(first) == set(second) and sorted(Counter(first).values()) == sorted(Counter(second).values())
```

COMPLEXITY | Part 1: Time O(n) | Space O(1) for a fixed alphabet
Part 2: Time O(n) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 243

K-way merge: a heap of the next item from each source

PATTERN CONTEXT

Put the first item of every list (with its list number and position) on a heap. Pop the smallest, record it, and push the next item from the same list. After k pops you have the k best without sorting everything.

244 Find k pairs with smallest sums — EASY

DIFFICULTY | EASY

Better approach: seed with (first[first_index] + second[0], first_index, 0); the successor of pair (first_index, second_index) is (first_index, second_index + 1)

QUESTION INFO

QUESTION

You are given two **sorted** integer arrays **`arr1`** and **`arr2`** (each sorted in non-decreasing order), and an integer **`k`**. Consider all possible pairs formed by taking one element from **`arr1`** and one element from **`arr2`**. For indices **`i`** and **`j`**, the pair is **`(arr1[i], arr2[j])`** with sum **`arr1[i] + arr2[j]`**. Your task is to return the **k pairs `(arr1[i], arr2[j])`** with the smallest sums. If there are fewer than **`k`** total possible pairs, return all possible pairs. You may return the pairs in any order. ---

Function Signature

(one possible form)

```
kSmallestPairs(arr1: List[int], arr2: List[int], k: int) -> List[Tuple[int, int]]
```

Example

```
arr1 = [1, 7, 11] arr2 = [2, 4, 6] k = 3
```

All possible pairs and their sums are:

- (1, 2) -> 3 - (1, 4) -> 5 - (1, 6) -> 7 - (7, 2) -> 9 - (7, 4) -> 11 - (7, 6) -> 13 - (11, 2) -> 13 - (11, 4) -> 15 - (11, 6) -> 17
  The 3 pairs with the smallest sums are:

```
[(1, 2), (1, 4), (1, 6)]
```

(The order among these returned pairs does not matter.) Explain an efficient algorithm to solve this problem and analyze its time and space complexity. You may assume the arrays are large, so a solution that explicitly generates and sorts all **`len(arr1) * len(arr2)`** pairs is too slow.

EXAMPLES

Example 1	
Input	`[1, 7, 11], [2, 4, 6], 3`
Output	`[(1, 2), (1, 4), (1, 6)]`
Why	`Pairs (one from each array) in order of their sum.`

Example 2	
Input	`[1, 1, 2], [1, 2, 3], 2`
Output	`[(1, 1), (1, 1)]`
Why	`Equal numbers give equal pairs.`

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Generate all pairs and sort

BRUTE-FORCE IDEA

Build all m * n pairs, sort by sum and take k. Correct; with 10^5 elements in each array it would create 10^10 pairs.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Generate all pairs and sort
Build all m * n pairs
sort by sum
take k
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(m * n log(m * n)) | Space O(m * n)

WHY THIS IS WEAKER | M * n pairs sorted: with 10^5 elements in each array that is 10^10 pairs. Because both arrays are sorted, a min-heap that starts from each x's first pair and expands only the next pair in that row is O(k log k).

HOW TO IMPROVE IT

seed with (first[first_index] + second[0], first_index, 0); the successor of pair (first_index, second_index) is (first_index, second_index + 1)

SHORT BEST SOLUTION

Push (first[first_index] + second[0], first_index, 0) for the first min(pair_count, m) values of a. Pop the smallest, record the pair, and push the next pair in the same row, (first[first_index] + second[second_index + 1], first_index, second_index + 1). After k pops you have the k smallest sums. Nothing larger than needed is ever generated.

```python
import heapq


# BASE QUESTION (optimal): Min-heap over the sorted rows
# Idea: Push (first[first_index] + second[0], first_index, 0) for the first min(pair_count, m)
# values of a. Pop the smallest, record the pair, and push
# the next pair in the same row, (first[first_index] + second[second_index + 1], first_index, second_index +
# 1). After k pops you have the k smallest sums. Nothing
# larger than needed is ever generated.
def kSmallestPairs(first, second, pair_count):
    if not first or not second or pair_count <= 0:
        return []
    heap = [(first[first_index] + second[0], first_index, 0) for first_index in range(min(pair_count, len(first
))))]
    heapq.heapify(heap)
    pairs = []
    while heap and len(pairs) < pair_count:
        _, first_index, second_index = heapq.heappop(heap)
        pairs.append((first[first_index], second[second_index]))
        if second_index + 1 < len(second):
            heapq.heappush(heap, (first[first_index] + second[second_index + 1], first_index, second_index + 1)
)
    return pairs
```

COMPLEXITY | Time O(k log min(k, m)) | Space O(min(k, m))

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 244

247 Return top-k subset sums in descending order HARD

DIFFICULTY | HARD

Better approach: heap of (sum, index); each popped state pushes two successors: add the next item, or swap the last item for the next

QUESTION INFO

QUESTION

Given a list of integers (possibly negative), return the k largest subset sums in descending order. The empty subset counts and has sum 0; different subsets with the same sum are listed separately.

Function signature

def top_k_sums(nums: list, k: int) -> list

EXAMPLES

Example 1	
Input	`[1, 2, 3], 4`
Output	`[6, 5, 4, 3]`
Why	`The eight subset sums are 6, 5, 4, 3, 3, 2, 1, 0; the top four are 6, 5, 4, 3.`

Example 2	
Input	`[3, -2, 1], 3`
Output	`[4, 3, 2]`
Why	`With a negative number: 3 + 1 = 4, then 3, then 3 + (-2) + 1 = 2.`

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Enumerate all 2^n subsets and sort

BRUTE-FORCE IDEA

Compute every subset sum, sort in descending order and take k. Fine for n up to about 20, hopeless beyond that.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Enumerate all 2^n subsets and sort
Compute every subset sum
sort in descending order
take k
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(2^n * n) | Space O(2^n)

WHY THIS IS WEAKER | 2^n subsets and a sort of them: O(2^n * n) time and O(2^n) space, hopeless past about n = 20. A heap that grows the k best sums step by step is O(n log n + k log k).

HOW TO IMPROVE IT

heap of (sum, index); each popped state pushes two successors: add the next item, or swap the last item for the next

SHORT BEST SOLUTION

Every subset sum equals (sum of positives) minus a subset sum of the absolute values, so the k largest sums are the sum of positives minus the k smallest subset sums of the sorted absolute values. Generate those with a heap of (sum, last index): popping a state gives the next smallest sum and pushes at most two successors, 'add the next element' and 'replace the last element with the next'. Each subset is produced once, in nondecreasing order.

```python
import heapq
```

```python
# BASE QUESTION (optimal): Heap of the k smallest sums of
# absolute values
# Idea: Every subset sum equals (sum of positives) minus a
# subset sum of the absolute values, so the k largest sums
# are the sum of positives minus the k smallest subset
# sums of the sorted absolute values. Generate those with
# a heap of (sum, last index): popping a state gives the
# next smallest sum and pushes at most two successors,
# 'add the next element' and 'replace the last element
# with the next'. Each subset is produced once, in
# nondecreasing order.
def top_k_sums(nums, how_many):
    base = sum(number for number in nums if number > 0)
    absolute_values = sorted(abs(number) for number in nums)
    sums = [0]
    heap = [(absolute_values[0], 0)] if absolute_values else []
    while heap and len(sums) < how_many:
        subset_sum, last_index = heapq.heappop(heap)
        sums.append(subset_sum)
        if last_index + 1 < len(absolute_values):
            heapq.heappush(heap, (subset_sum + absolute_values[last_index + 1], last_index + 1))
            heapq.heappush(heap, (subset_sum - absolute_values[last_index] + absolute_values[last_index + 1], l
ast_index + 1))
    return [base - number for number in sums[:how_many]]
```

COMPLEXITY | Time O(n log n + k log k) | Space O(n + k)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 247

248 Merge k sorted linked lists. HARD

DIFFICULTY | HARD

Better approach: seed with each list head (value, list number, node); the successor of a node is node.next

QUESTION INFO

QUESTION

You are given k linked lists, each sorted in ascending order. Merge them into one sorted linked list and return its head. (In the examples the linked lists are written as arrays.)

Function signature

def merge_k_heap(lists: list) -> ListNode

EXAMPLES

Example 1	
Input	`lists = [[1, 4, 5], [1, 3, 4], [2, 6]]`
Output	`_to_list(merge_k_heap([mk([1, 4, 5]), mk([1, 3, 4]), mk([2, 6])])) -> [1, 1, 2, 3, 4, 4, 5, 6]`
Why	The three sorted lists merge into one sorted list.

Example 2	
Input	`lists = []`
Output	`_to_list(merge_k_heap([])) -> []`
Why	No lists: an empty result.

CLARIFY BEFORE CODING

Is k guaranteed valid, are results over distinct values or all occurrences, and what deterministic ordering should break ties? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Collect all values, sort them, rebuild a list

BRUTE-FORCE IDEA

Read every node into one array, sort it and build a new list. Correct; it ignores that each input is already sorted and uses O(N) extra memory.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Collect all values, sort them, rebuild a list
Read every node into one array
sort it
build a new list
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(N log N) | Space O(N)

WHY THIS IS WEAKER | O(N log N) time and O(N) extra memory, ignoring that each list is already sorted. A min-heap of the k current heads is O(N log k) with O(k) memory.

HOW TO IMPROVE IT

seed with each list head (value, list number, node); the successor of a node is node.next

SHORT BEST SOLUTION

Push the head of each non-empty list as (value, list index, node). Pop the smallest, append it to the output, and push its successor. The list index is a tie-breaker so nodes are never compared. The heap never holds more than k items.

```python
import heapq


# BASE QUESTION (optimal): Min-heap of the current heads
# Idea: Push the head of each non-empty list as (value,
# list index, node). Pop the smallest, append it to the
# output, and push its successor. The list index is a tie-
# breaker so nodes are never compared. The heap never
# holds more than k items.
class ListNode:
    def __init__(self, val=0, next=None):
        self.val, self.next = val, next

def mergeKLists(lists):
    heap = [(first_node.val, list_index, first_node) for list_index, first_node in enumerate(lists) if first_no
de]
    heapq.heapify(heap)
    dummy = tail = ListNode()
    while heap:
        _, list_index, node = heapq.heappop(heap)
        tail.next = node
        tail = node
        if node.next:
            heapq.heappush(heap, (node.next.val, list_index, node.next))
    return dummy.next
```

COMPLEXITY | Time O(N log k) | Space O(k)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 248

GROUP 36 OF 43 | 1 QUESTIONS | PART B - PART 1

Tree recursion, Tier 2: pass a rule down, return a node or a yes/no up

PATTERN CONTEXT

Same recursive walk as Tier 1, but now the information also travels down: lower and upper bounds for a binary search tree, or the target nodes you are looking for. The BST rule (left < node < right) lets you go left or right instead of visiting everything.

249 Next Birthday (BST In-Order Successor with Wraparound) EASY

DIFFICULTY | EASY

Better approach: descend(): go left after a hit (a closer candidate may exist), right otherwise; if no candidate exists wrap around to the leftmost node

QUESTION INFO

QUESTION

This is the "alarm / next-birthday" problem: from a set of birthdays stored in a binary search tree, find the next one that will occur. Each birthday is stored as an integer day-of-year in [1, 365] , and the tree is a valid BST keyed on that value. Given the root of the BST and an integer today , return the next upcoming birthday: the smallest key that is strictly greater than today . If no key is strictly greater than today , the calendar wraps around, so return the smallest key in the whole tree. If the tree is empty, return None .

Function signature

```
from typing import Optional
class TreeNode:
    def __init__(self, val: int, left: "Optional[TreeNode]" = None,
                 right: "Optional[TreeNode]" = None) -> None:
        self.val = val
        self.left = left
        self.right = right
def next_birthday(root: Optional[TreeNode], today: int) -> Optional[int]:
    """Return the smallest key strictly greater than today, wrapping around
    to the smallest key if none is greater. Return None if the tree is empty."""
    ...
```

Follow-up (as reported): If the input were an unsorted binary tree (not a BST), extract all values, sort them, and binary-search for the successor. Hint: bisect_right .

EXAMPLES

Example 1	
Input	`root = above, today = 25`
Output	`30`
Why	`Smallest key strictly greater than 25 is 30.`

Example 2	
Input	`root = above, today = 40`
Output	`10`
Why	`No key > 40, so wrap around to the smallest key, 10.`

QUESTION

Example 3	
Input	`root = above, today = 5`
Output	`10`
Why	`Smallest key greater than 5 is 10.`

Example 4	
Input	`root = None, today = 100`
Output	`None`

CONSTRAINTS

- 0 <= number of nodes <= 10^5 ; all node values are distinct and in [1, 365] .
- 1 <= today <= 365 .
- Aim for O(h) time where h is the tree height (do not sort all nodes if you can avoid it).

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= number of nodes <= 10^5 ; all node values are distinct and in [1, 365] .
- 1 <= today <= 365 .
- Aim for O(h) time where h is the tree height (do not sort all nodes if you can avoid it).

BRUTE FORCE | In-order list and a scan

BRUTE-FORCE IDEA

Start with: In-order list and a scan. It visits every node although the BST order points to the answer.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: In-order list and a scan
Start with: In-order list
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | It visits every node although the BST order points to the answer.

HOW TO IMPROVE IT

descend(): go left after a hit (a closer candidate may exist), right otherwise; if no candidate exists wrap around to the leftmost node

SHORT BEST SOLUTION

A node above today is a candidate; its left subtree may hold a closer one.

```python
# BASE QUESTION (optimal): Descend the BST once
# Idea: A node above today is a candidate; its left
# subtree may hold a closer one.
def next_birthday(root, today):
    if root is None:
        return None
    best, node = None, root
    while node:
        if node.val > today:
            best, node = node.val, node.left
        else:
            node = node.right
    if best is not None:
        return best
```

```
    node = root
    while node.left:
        node = node.left
    return node.val
```

COMPLEXITY | Time O(h) | Space O(1)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 249

GROUP 37 OF 43 | 1 QUESTIONS | PART B - PART 1

Random pick without repeats: swap the chosen item with the last one and pop

PATTERN CONTEXT

To pick a random item and never pick it again in O(1), keep the items in an array. Choose a random index, swap that item with the last one, pop the last, and fix its position in the index dict. The same trick gives O(1) delete and O(1) random access.

258 Shuffle array using random integer API — HARD

DIFFICULTY | HARD

Better approach: Fisher-Yates: for i from the end, pick = rand_int(0, last_slot), swap a[last_slot] and a[pick]

QUESTION INFO

QUESTION

You have an array nums and no built-in shuffle, only rand_int(l, r), which returns a uniformly random integer in [l, r] (both ends included). Shuffle nums in place so that every one of the n! orders is equally likely.

Function signature

def shuffle(nums: list, rand_int) -> list

EXAMPLES

Example 1

Input	`rand_int = lambda l, r: l # a fake 'random' source that always returns the lowest allowed index`
	`nums = [1, 2, 3, 4]`
	`shuffle(nums, rand_int)`

Output	`shuffle(nums, rand_int) -> [2, 3, 4, 1]`
Why	`Fisher-Yates walks i from the end and swaps nums[i] with nums[rand_int(0, i)]. With the fake source the swaps are (3,0), (2,0), (1,0), giving a predictable result; with a real source each of the 24 orders has probability 1/24.`

CLARIFY BEFORE CODING

Can values be negative or duplicated, is order significant, and is the range or subarray required to be contiguous? Do we return one answer, all answers, or only a count? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Give each element a random key and sort

BRUTE-FORCE IDEA

Pair each element with a random number and sort by it. With keys from a large range it is practically uniform, but it spends O(n log n) and needs extra space; if two keys collide the result is slightly biased.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Give each element a random key and sort
Pair each element with a random number
sort by it
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n log n) | Space O(n)

WHY THIS IS WEAKER | O(n log n) time and O(n) extra space, and equal random keys make the result very slightly non-uniform. Fisher-Yates (swap position i with a random position in 0..i) is O(n), in place and exactly uniform.

HOW TO IMPROVE IT

Fisher-Yates: for i from the end, pick = rand_int(0, last_slot), swap a[last_slot] and a[pick]

SHORT BEST SOLUTION

Walk i from n - 1 down to 1, pick j uniformly from 0..i (inclusive) and swap nums[last_slot] with nums[pick]. Position i is filled by one of the last_slot + 1 elements still unplaced, each equally likely, so every permutation has probability 1 / n!. Picking j from 0..n-1 every time is the common biased mistake.

```
import random


def rand_int(low, high):
    return random.randint(low, high)


# BASE QUESTION (optimal): Fisher-Yates
# Idea: Walk i from n - 1 down to 1, pick j uniformly from
# 0..i (inclusive) and swap nums[last_slot] with nums[pick]. Position
# i is filled by one of the last_slot + 1 elements still unplaced,
# each equally likely, so every permutation has
# probability 1 / n!. Picking j from 0..n-1 every time is
# the common biased mistake.
def shuffle(nums):
    for last_slot in range(len(nums) - 1, 0, -1):
        pick = rand_int(0, last_slot)
        nums[last_slot], nums[pick] = nums[pick], nums[last_slot]
    return nums
```

COMPLEXITY | Time O(n) | Space O(1) extra

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 258

Build and serialize: write a structure out, read it back with the same walk

PATTERN CONTEXT

Encoding is one walk that writes tokens; decoding is the same walk reading tokens back (an iterator or an index). Put a length or a child count before the data so the reader always knows how much to take.

262 Serialize and Deserialize a Graph — HARD

DIFFICULTY | HARD

Better approach: BFS with a seen set; write one line per node: value, then the values of its neighbours; rebuild by creating all nodes first, then linking

QUESTION INFO

QUESTION

Design a codec that can serialize a connected, undirected graph to a string and deserialize that string back into an identical graph. Each node has a unique integer val and a list of neighbors . The graph may contain cycles. After a round trip, the reconstructed graph must have the same node values and the same adjacency relationships as the original. You are given the reference node class below. Implement serialize (node -> string) and deserialize (string -> node). serialize(None) must round-trip to None via deserialize .

Function signature

```
class Node:
    def __init__(self, val: int = 0, neighbors: list["Node"] | None = None):
        self.val = val
        self.neighbors = neighbors if neighbors is not None else []
class Codec:
    def serialize(self, node: "Node | None") -> str:
        """Encode the whole graph reachable from node into a string."""
        ...
    def deserialize(self, data: str) -> "Node | None":
        """Rebuild the graph from the string produced by serialize."""
        ...
```

Follow-up (as reported): Support directed graphs (asymmetric adjacency). Hint: record each directed edge (u -> v) rather than assuming both directions during deserialization.

EXAMPLES

Example 1	
Input	`adjList = [[]] # a single node with no neighbors`
Output	`A single node with val 1 and an empty neighbor list.`

Example 2	
Input	`node = None # empty graph`
Output	`codec.deserialize(codec.serialize(None)) is None`

CONSTRAINTS

- 0 <= number of nodes <= 100
- Node values are unique.
- The graph is connected when non-empty; you receive any one node as the entry point.

CLARIFY BEFORE CODING

Is the graph directed or undirected and weighted or unweighted? Can it contain cycles, duplicate edges, or disconnected nodes, and what should I return when no path exists? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

Known bounds to confirm:
- 0 <= number of nodes <= 100
- Node values are unique.
- The graph is connected when non-empty; you receive any one node as the entry point.

BRUTE FORCE | Naive nesting (the idea that fails on cycles)

BRUTE-FORCE IDEA

Without remembering visited nodes a cycle is expanded forever; a visited set is the minimum fix.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Naive nesting (the idea that fails on cycles)
Without remembering visited nodes a cycle is expanded forever
a visited set is the minimum fix
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time does not finish on a cycle | Space -

WHY THIS IS WEAKER | Without remembering visited nodes a cycle is expanded forever; a visited set is the minimum fix.

HOW TO IMPROVE IT

BFS with a seen set; write one line per node: value, then the values of its neighbours; rebuild by creating all nodes first, then linking

SHORT BEST SOLUTION

One token per node with its neighbour values; rebuild by creating the nodes first and linking afterwards.

```python
# BASE QUESTION (optimal): BFS to a text, two passes back
# Idea: One token per node with its neighbour values;
# rebuild by creating the nodes first and linking
# afterwards.
from collections import deque


class Node:
    def __init__(self, val=0, neighbors=None):
        self.val = val
        self.neighbors = neighbors if neighbors is not None else []


class Codec:
    def serialize(self, node):
        if node is None:
            return ""
        seen, queue, parts = {node.val}, deque([node]), []
        while queue:
            current = queue.popleft()
            parts.append(f"{current.val}:" + ",".join(str(neighbour.val) for neighbour in current.neighbors))
            for neighbour in current.neighbors:
                if neighbour.val not in seen:
                    seen.add(neighbour.val)
                    queue.append(neighbour)
        return ";".join(parts)

    def deserialize(self, data):
        if not data:
            return None
        rows = [part.split(":") for part in data.split(";")]
        nodes = {int(node_value): Node(int(node_value)) for node_value, _ in rows}
        for node_value, neighbour_text in rows:
            nodes[int(node_value)].neighbors = [nodes[int(neighbour_value)] for neighbour_value in neighbour_te
xt.split(",") if neighbour_value]
        return nodes[int(rows[0][0])]
```

COMPLEXITY | Time O(V + E) | Space O(V + E)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 262

263 Implement a decode and encode function of a general N-ary tree. HARD

DIFFICULTY | HARD

Better approach: write value and child count for each node (preorder); read the count to know how many children to build

QUESTION INFO

QUESTION

A general tree has a value in every node and any number of children. Write encode(root), which turns the tree into a string, and decode(text), which rebuilds the identical tree from that string.

Function signature

def encode(root: Node) -> str

def decode(text: str) -> Node

EXAMPLES

Example 1

Input	`root = Node(1, [Node(3, [Node(5), Node(6)]), Node(2), Node(4)])`
	`text = encode(root)`
	`text`
	`encode(decode(text)) == text`

Output	`text -> '[1, 3, 3, 2, 5, 0, 6, 0, 2, 0, 4, 0]'` `encode(decode(text)) == text -> true`
Why	`Each node is written as its value followed by its number of children, in pre-order: 1 has 3 children, 3 has 2, and so on. Decoding reads exactly that many children for each node, so encoding the decoded tree gives the same text.`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Nested brackets with a recursive parser

BRUTE-FORCE IDEA

Write each node as value[child,child, and so on]. It is the first format most people think of and it works, but encoding by joining child strings copies text at every level, and the recursive parser can overflow the stack on a very deep tree. Values that contain brackets or commas break it.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Nested brackets with a recursive parser
Write each node as value[child,child
so on]
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n * h) for building strings (O(n) with a list buffer) | Space O(n)

WHY THIS IS WEAKER | Joining child strings copies text at every level (O(n * h) in the worst case) and the recursive parser can overflow on a deep tree; values containing brackets or commas break it. Pre-order with child counts is O(n) and needs no markers.

HOW TO IMPROVE IT

write value and child count for each node (preorder); read the count to know how many children to build

SHORT BEST SOLUTION

Write each node before its children with its child count. The decoder recursively reads exactly that many children.

```python
# BASE QUESTION (optimal): Recursive pre-order with child counts
# Idea: Write each node before its children with its child
# count. The decoder recursively reads exactly that many children.
class Node:
    def __init__(self, val=None, children=None):
        self.val = val
        self.children = children or []


def encode(root):
    tokens = []

    def dfs(node):
        tokens.extend((str(node.val), str(len(node.children))))
        for child in node.children:
            dfs(child)

    if root is not None:
        dfs(root)
    return " ".join(tokens)


def decode(text):
    if not text:
        return None
    tokens = iter(text.split())

    def dfs():
        node = Node(next(tokens))
        child_count = int(next(tokens))
        node.children = [dfs() for _ in range(child_count)]
        return node

    return dfs()
```

COMPLEXITY | Time O(n) | Space O(h) recursion plus output

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 263

GROUP 39 OF 43 | 4 QUESTIONS | PART B - PART 1

Games: win or lose with a cache, and guessing with feedback

PATTERN CONTEXT

For two-player games, a state is winning if some move leads to a state where the opponent loses; cache every state. For guessing games, pick the guess that shrinks the worst remaining group of candidates the most, then keep only the candidates that agree with the feedback.

265 Determine winner in stack-merging game — HARD

DIFFICULTY | HARD

Better approach: state = the sorted tuple of stack sizes; a move merges two whose sum stays within the cap

QUESTION INFO

QUESTION

A game starts with stacks of the given sizes. A move merges two stacks into one if their combined size is at most cap. The player who cannot move loses (in the misere variant, the player who cannot move wins). Return whether the first player wins with best play.

Function signature

def first_player_wins(stacks: list, cap: int, misere=False) -> bool

EXAMPLES

Example 1	
Input	`[1, 1], 2`
Output	`true`
Why	`Merge the two stacks of 1 into 2; the opponent has a single stack and cannot move.`

Example 2	
Input	`[1, 1, 1], 2`
Output	`true`
Why	`Any merge makes a stack of 2 next to a stack of 1, whose sum 3 exceeds the cap: the opponent is stuck.`

Example 3	
Input	`[3], 5`
Output	`false`
Why	`A single stack: the first player cannot move and loses.`

Example 4	
Input	`[3], 5, misere = true`
Output	`true`
Why	`The same single stack in the misere variant: being unable to move wins.`

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Plain recursion over every move

BRUTE-FORCE IDEA

Try every merge, and the current player wins if some merge leaves the opponent in a losing position. Correct, but the same positions are explored again and again through different move orders.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Plain recursion over every move
Try every merge
the current player wins if some merge leaves the opponent in a losing position
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential in the number of stacks | Space O(depth)

WHY THIS IS WEAKER | The same position is reached through many different move orders, so the work is exponential. Remembering the result per position (sorted state as key) solves each position once.

HOW TO IMPROVE IT

state = the sorted tuple of stack sizes; a move merges two whose sum stays within the cap

SHORT BEST SOLUTION

Merging does not depend on order, so a sorted tuple identifies the position. Each state is solved once and then looked up. This is the standard fix for 'many move orders reach the same position'.

```python
# BASE QUESTION (optimal): Memoize on the sorted state
# Idea: Merging does not depend on order, so a sorted
# tuple identifies the position. Each state is solved once
# and then looked up. This is the standard fix for 'many
# move orders reach the same position'.
def merge_game(stacks, height_limit):
    memo = {}

    def wins(state):
        if state in memo:
            return memo[state]
        result = False
        for first in range(len(state)):
            for second in range(first + 1, len(state)):
                if state[first] + state[second] <= height_limit:
                    rest = state[:first] + state[first + 1:second] + state[second + 1:] + (state[first] + state
[second],)
                    if not wins(tuple(sorted(rest))):
                        result = True
                        break
            if result:
                break
        memo[state] = result
        return result

    return wins(tuple(sorted(stacks)))
```

COMPLEXITY | Time O(number of distinct sorted states * k^2) | Space O(number of states)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 265

266 Find secret word with match-count feedback HARD

DIFFICULTY | HARD

Better approach: guess the word that minimises the size of the biggest feedback group; keep the words with the same match count

QUESTION INFO

QUESTION

Problem

You are given a list of **unique** lowercase words, all of the same length (e.g., length = 6). One of these words is a **secret** word. You can interact with a black-box API:

- **`matchCount(guess) -> int`**: when you submit a word **`guess`** from the list, the API returns the number of positions **`i`** such that **`guess[i] == secret[i]`** (i.e., exact character match at the same index). You are allowed to call **`matchCount`**

at most 10 times.

Task

Design/implement a function (e.g., **`findSecretWord(words)`**) that uses the API to identify the secret word within the guess limit.

Notes / Constraints

- **`words`** contains between 1 and 100 words.
- All words have equal length **`L`** (commonly **`L = 6`**).
- Every guess must be a word from **`words`**.
- After each API response, you may adapt your next guess.

What to return

Depending on the interface, you may either: - return the identified secret word, or - stop once **`matchCount(guess) == L`** (found it). (Your interviewer may treat this as an interactive problem; focus on strategy + correctness under the 10-guess constraint.)

You are given a list of unique lowercase words, and every word has the same length **`L`**. Exactly one word in the list is the secret word. In the original interview version, you interact with a black-box API:

- **`matchCount(guess) -> int`**
- It returns how many positions **`i`** satisfy **`guess[i] == secret[i]`** You may call this API at most 10 times, and every guess must be a word from the given list. In this non-interactive version, implement **`solution(words, secret)`** and simulate the API inside your function. The parameter **`secret`** is provided only so the judge can verify your strategy. Your function should repeatedly make guesses, use the returned match counts to eliminate impossible candidates, and return the discovered secret word. If your strategy does not find the secret within 10 guesses, return an empty string **`""`**.

Constraints**`1 <= len(words) <= 1001 <= len(words[i]) <= 10`**All words are unique, lowercase, and have the same length **`Lsecret`** is guaranteed to be one of the words in **`words`**Your algorithm may use at most 10 simulated **`matchCount`** callsTest data is designed so that a correct elimination strategy can identify the secret within the limit

EXAMPLES

```
Input: (["acckzz", "ccbazz", "eiowzz", "abcczz"], "acckzz")
```

Expected Output: 'acckzz'Explanation: The secret is already one of the candidates. Using match-count feedback, the candidate set shrinks until **`acckzz`** is identified.

```
Input: (["secret"], "secret")
```

Expected Output: 'secret'Explanation: With only one word in the list, the first allowed guess must be the secret.

QUESTION

Hints

After a guess returns **k**, any valid remaining candidate must also have exactly **k** matching positions with that guess.A strong strategy is to choose the next guess that splits the remaining candidates as evenly as possible, minimizing the size of the largest feedback bucket.

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Guess the first candidate each time, then filter

BRUTE-FORCE IDEA

Pick any remaining word, call the API, and keep only the words that have the same match count with it. Correct but there is no guarantee of finishing in 10 guesses on adversarial lists.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Guess the first candidate each time, then filter
Pick any remaining word
call the API
keep only the words that have the same match count with it
```

BRUTE-FORCE COMPLEXITY | Time O(guesses * W * L) | Space O(W)

WHY THIS IS WEAKER | A poor guess may remove very few candidates, so there is no guarantee of finishing in 10 guesses. Guessing the word whose largest match-count group is smallest (minimax) keeps the list shrinking fast.

HOW TO IMPROVE IT

guess the word that minimises the size of the biggest feedback group; keep the words with the same match count

SHORT BEST SOLUTION

For each candidate guess count how many other candidates would fall into each match-count group, and pick the guess whose biggest group is smallest (first minimax choice). Query it and keep the words consistent with the answer. With up to 100 words of length 6 this finishes within 10 guesses in practice.

```python
from collections import Counter


def _match(first, second):
    return sum(letter_a == letter_b for letter_a, letter_b in zip(first, second))


# BASE QUESTION (optimal): Guess the word that minimises
# the size of its largest match-count group
# Idea: For each candidate guess count how many other
# candidates would fall into each match-count group, and
# pick the guess whose biggest group is smallest (first
# minimax choice). Query it and keep the words consistent
# with the answer. With up to 100 words of length 6 this
# finishes within 10 guesses in practice.
def find_secret_minimax(words, secret):
    candidates, guesses = list(words), 0
    while True:
        best_guess = min(candidates, key=lambda guess: (max(Counter(_match(guess, word) for word in candidates)
.values()), guess))
        guesses += 1
        match_count = _match(best_guess, secret)
        if match_count == len(secret):
            return guesses
        candidates = [word for word in candidates if _match(best_guess, word) == match_count]
```

COMPLEXITY | Time O(guesses * W^2 * L) | Space O(W)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 266

267 Wordle-Style 5-Letter Minimum-Guess Strategy HARD

DIFFICULTY | HARD

Better approach: ask about the letter at a position that splits the remaining words most evenly (min of yes-count and no-count is largest)

QUESTION INFO

QUESTION

A hidden word has 5 lowercase letters. The only thing you can ask is the oracle query(pos, letter), which says whether the letter is at that position (positions 0 to 4). Find the word with as few oracle calls as possible. by_frequency tries letters in order of English frequency; with_dictionary also knows a list of candidate words and always asks the question that splits the candidates most evenly. Both return (word, number of calls).

Function signature

def by_frequency(query) -> (str, int)
def with_dictionary(query, dictionary: list) -> (str, int)

EXAMPLES

Example 1	

Input	`secret = 'crane'`
	`query = lambda pos, letter: secret[pos] == letter`
	`by_frequency(query)`
	`with_dictionary(query, ['crane', 'crate', 'slate', 'trace', 'grape'])`

Output	`by_frequency(query) -> ('crane', 31)` `with_dictionary(query, ['crane', 'crate', 'slate', 'trace', 'grape']) -> ('crane', 2)`
Why	`Without a dictionary every position is searched letter by letter; with five candidate words a handful of well-chosen questions are enough.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Ask every position and letter

BRUTE-FORCE IDEA

Query all 5 x 26 pairs and read the word off the answers. Always 130 calls, no matter how lucky the word is.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Ask every position and letter
Query all 5 x 26 pairs
read the word off the answers
```

BRUTE-FORCE COMPLEXITY | Time 130 calls | Space O(1)

WHY THIS IS WEAKER | Always 130 calls. Asking likely letters first, skipping the implied last letter, or splitting a dictionary of candidates in half needs far fewer.

HOW TO IMPROVE IT

ask about the letter at a position that splits the remaining words most evenly (min of yes-count and no-count is largest)

SHORT BEST SOLUTION

Without a dictionary: for each position ask letters from most to least common and stop at the first yes; if 25 letters said no the 26th is implied and needs no call. With a dictionary: keep the words consistent with every answer so far. Each round pick the (position, letter) pair whose yes-count is closest to half of the candidates, ask it, and keep only the words that agree. The loop ends when one word is left.

```
LETTERS_BY_FREQUENCY = "etaoinshrdlcumwfgypbvkjxqz"


# BASE QUESTION (optimal): Frequency order, or entropy
# split when a dictionary is known
# Idea: Without a dictionary: for each position ask
# letters from most to least common and stop at the first
# yes; if 25 letters said no the 26th is implied and needs
# no call. With a dictionary: keep the words consistent
# with every answer so far. Each round pick the (position,
# letter) pair whose yes-count is closest to half of the
# candidates, ask it, and keep only the words that agree.
# The loop ends when one word is left.
def wordle_guess(query, dictionary=None):
    calls = 0
    def ask(position, letter):
        nonlocal calls
        calls += 1
        return query(position, letter)
    if dictionary:
        words = list(dictionary)
        while len(words) > 1:
            best_probe = max(((position, letter) for position in range(5) for letter in set(word[position] for
word in words)),
                             key=lambda probe: min(sum(word[probe[0]] == probe[1] for word in words), sum(word[probe[
0]] != probe[1] for word in words)))
            answer = ask(*best_probe)
            words = [word for word in words if (word[best_probe[0]] == best_probe[1]) == answer]
        return words[0], calls
    letters = []
    for position in range(5):
        for rank, letter in enumerate(LETTERS_BY_FREQUENCY):
            if rank == 25 or ask(position, letter):
                letters.append(letter)
                break
    return "".join(letters), calls
```

COMPLEXITY | Time at most 5 x 25 calls without a dictionary; about log2(words) with one | Space O(words)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 267

268 Babylon Tile-Stack Merge Game — HARD

DIFFICULTY | HARD

Better approach: state = the sorted tuple of stacks; a move merges two stacks that share a height or a colour; you win if some merge leaves the opponent without a winning reply

QUESTION INFO

QUESTION

A two-player game on stacks of tiles. Each stack has a height and a colour (the colour of its top tile). A move places one stack on top of another when the two stacks have the same height or the same colour; the new stack has the combined height and the colour of the stack that was placed on top. The player who cannot move loses. Given the stacks as (height, colour) pairs, return whether the player to move wins with best play.

Function signature

def babylon_wins(stacks: list) -> bool

EXAMPLES

QUESTION

Example 1	
Input	`[(1, 'Y'), (1, 'W')]`
Output	`true`
Why	`Two stacks of height 1: the mover merges them, and the opponent has nothing left to move.`

Example 2	
Input	`[(1, 'Y')]`
Output	`false`
Why	`A single stack: the mover has no move and loses.`

Example 3	
Input	`[(1, 'Y'), (2, 'W')]`
Output	`false`
Why	`Heights differ and colours differ: no move is possible, so the mover loses.`

Example 4	
Input	`[(1, 'Y'), (1, 'Y'), (1, 'W'), (1, 'W')]`
Output	`true`
Why	`Four height-1 stacks of two colours: the mover has a winning plan.`

CLARIFY BEFORE CODING

Please confirm the required API, behavior for invalid or empty operations, ordering guarantees, capacity limits, and whether concurrent access matters. Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Plain recursion over every move

BRUTE-FORCE IDEA

Try every legal move recursively with no caching. Correct and fine for a few stacks; it recomputes the same positions over and over.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Plain recursion over every move
Try every legal move recursively with no caching
process every required item or candidate directly
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time exponential | Space O(depth)

WHY THIS IS WEAKER | Explores the same positions many times, exponential. Sorting the stacks and caching each position's result visits every distinct position once.

HOW TO IMPROVE IT

state = the sorted tuple of stacks; a move merges two stacks that share a height or a colour; you win if some merge leaves the opponent without a winning reply

SHORT BEST SOLUTION

legal_moves lists the ordered pairs that may merge, apply builds the next state, and wins(state) is true when some move reaches a state where the opponent loses. Sorting the stacks makes equal positions share one cache entry. Playing the game out is then just: pick any move that leads to a state where wins() is False.

```python
import itertools
```

```
# BASE QUESTION (optimal): Negamax with memoisation on the
# sorted stack tuple
# Idea: legal_moves lists the ordered pairs that may
# merge, apply builds the next state, and wins(state) is
# true when some move reaches a state where the opponent
# loses. Sorting the stacks makes equal positions share
# one cache entry. Playing the game out is then just: pick
# any move that leads to a state where wins() is False.
def babylon_wins(stacks):
    cache = {}
    def wins(state):
        if state in cache:
            return cache[state]
        can_win = False
        for first, second in itertools.permutations(range(len(state)), 2):
            (height_a, colour_a), (height_b, colour_b) = state[first], state[second]
            if height_a == height_b or colour_a == colour_b:
                next_state = tuple(sorted([stack for position, stack in enumerate(state) if position not in (fi
rst, second)] + [(height_a + height_b, colour_a)]))
                if not wins(next_state):
                    can_win = True
                    break
        cache[state] = can_win
        return can_win
    return wins(tuple(sorted(stacks)))
```

COMPLEXITY | Time O(states * moves^2) | Space O(states)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 268

GROUP 40 OF 43 | 1 QUESTIONS | PART B - PART 1

Sliding window: grow on the right, shrink on the left while it is too bad

PATTERN CONTEXT

Move the right end one step at a time and add that item to your state. While the window breaks the rule, remove items from the left end. After each step the window is valid, so record its length (or the shortest valid one).

273 Necklace Cut into Two Halves with Equal D/R Counts — HARD

DIFFICULTY | HARD

Better approach: a fixed-size window of half the length slides around the doubled string, keeping a count of D

QUESTION INFO

QUESTION

A necklace is a circular string of beads D and R. Cut it in two places into two arcs of equal length that contain the same number of D beads (and so the same number of R beads). Return the two cut positions and the two arcs, or None if no such cut exists.

Function signature

def necklace_two_cuts(s: str) -> tuple | None

(cut1, cut2, arc1, arc2)

EXAMPLES

Example 1	
Input	`'DRDR'`
Output	`(0, 2, 'DR', 'DR')`
Why	`Cut at 0 and 2: both arcs are DR.`

Example 2	
Input	`'DDRR'`
Output	`(1, 3, 'DR', 'RD')`
Why	`Linear halves DD and RR are unbalanced, but rotating the cuts by one gives DR and RD.`

Example 3	
Input	`'DDR'`
Output	`null`
Why	`An odd number of beads cannot be split into equal arcs.`

CLARIFY BEFORE CODING

Can the input be empty, contain duplicates or negative values, and are there any ordering or stability requirements not shown in the example? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Try every pair of cut points

BRUTE-FORCE IDEA

Choose two cut positions, form the two arcs, and compare their letter counts. Correct; it checks every pair of cuts and recounts letters each time.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Try every pair of cut points
```

```
Choose two cut positions
form the two arcs
compare their letter counts
```

BRUTE-FORCE COMPLEXITY | Time O(n^3) | Space O(n)

WHY THIS IS WEAKER | O(n^2) pairs and each recount costs O(n), so O(n^3). Because the arcs must both have length n / 2, one circular window of that length slid around the string finds it in O(n).

HOW TO IMPROVE IT

a fixed-size window of half the length slides around the doubled string, keeping a count of D

SHORT BEST SOLUTION

If each arc has half of the D's and half of the R's, each arc has length length/2, and the total D and R counts must both be even. Slide a window of length length/2 around the circle (use necklace + necklace), tracking its D count. The window starting at i and the one starting at start + length/2 are complementary arcs, so their D counts sum to the total. Counts change by at most one per step, so if one is below half and the other above half the count must hit exactly half somewhere. Hence a valid pair of cuts always exists when parity allows.

```python
# BASE QUESTION (optimal): Fixed-length circular window
# with the discrete intermediate value argument
# Idea: If each arc has half of the D's and half of the
# R's, each arc has length length/2, and the total D and R
# counts must both be even. Slide a window of length length/2
# around the circle (use necklace + necklace), tracking its D count. The
# window starting at i and the one starting at start + length/2 are
# complementary arcs, so their D counts sum to the total.
# Counts change by at most one per step, so if one is
# below half and the other above half the count must hit
# exactly half somewhere. Hence a valid pair of cuts
# always exists when parity allows.
def necklace_two_cuts(necklace):
    length = len(necklace)
    d_total, r_total = necklace.count("D"), necklace.count("R")
    if length % 2 or d_total % 2 or r_total % 2:
        return None
    half, target = length // 2, d_total // 2
    doubled = necklace + necklace
    window = doubled[:half].count("D")
    for start in range(length):
        if window == target:
            return start, (start + half) % length, doubled[start:start + half], doubled[start + half:start + le
ngth]
        window += (doubled[start + half] == "D") - (doubled[start] == "D")
    return None
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 273

GROUP 41 OF 43 | 1 QUESTIONS | PART B - PART 1

Parent pointers: store who is above each node and climb

PATTERN CONTEXT

Keep parent[x] (or manager[x]). Climbing is a while loop that follows parent until it runs out. Questions about common ancestors, cycles and who reports to whom are all small variations of that climb.

274 Tree Node. EASY

DIFFICULTY | EASY

Better approach: the SQL version of the same idea: p_id is the parent pointer, NULL means root, being someone's parent means inner, else leaf

QUESTION INFO

QUESTION

A binary tree is stored in a table tree_node_table(id, p_id), where p_id is the id of the parent node (NULL for the root). Write a query that labels every node: 'Root' if it has no parent, 'Inner' if it has a parent and at least one child, and 'Leaf' if it has a parent and no children. The output has the columns id and node_types, ordered by id.

Function signature

-- columns: id, node_types

EXAMPLES

Example 1

Input	`CREATE TABLE tree_node_table (id INTEGER, p_id INTEGER);` `INSERT INTO tree_node_table VALUES (1, NULL), (2, 1), (3, 1), (4, 2), (5, 2);`

Output	`1, Root`
	`2, Inner`
	`3, Leaf`
	`4, Leaf`
	`5, Leaf`

Why	`Node 1 has no parent, node 2 has a parent and children 4 and 5, and nodes 3, 4 and 5 have no children.`

Example 2

Input	`CREATE TABLE tree_node_table (id INTEGER, p_id INTEGER);` `INSERT INTO tree_node_table VALUES (1, NULL);`
Output	`1, Root`
Why	`A tree with one node: it has no parent, so it is the Root (not a Leaf).`

CLARIFY BEFORE CODING

Can the tree be empty, can values repeat, and should the answer contain node objects, values, or a count? Is recursion depth safe for the maximum tree height? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Three queries combined with UNION and a correlated subquery

BRUTE-FORCE IDEA

One query per node type, each using EXISTS to ask whether the node has a child. Correct and readable; the correlated subquery runs once per row, and the table is scanned three times.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Three queries combined with UNION and a correlated subquery
One query per node type
```

```
each using EXISTS to ask whether the node has a child
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n^2) without an index on p_id | Space O(n)

WHY THIS IS WEAKER | The correlated subquery runs once per row and the table is scanned three times, O(n^2) without an index. One CASE expression with a single set of parent ids scans once.

HOW TO IMPROVE IT

the SQL version of the same idea: p_id is the parent pointer, NULL means root, being someone's parent means inner, else leaf

SHORT BEST SOLUTION

CASE tests in order: no parent means Root, an id that appears as someone's parent means Inner, anything else is a Leaf. The subquery filters out NULL so the IN test is safe (NOT IN with a NULL would return no rows). The table is scanned once and the parent id set is built once.

```sql
-- BASE QUESTION (optimal): One CASE expression
-- Idea: CASE tests in order: no parent means Root, an id
-- that appears as someone's parent means Inner, anything
-- else is a Leaf. The subquery filters out NULL so the IN
-- test is safe (NOT IN with a NULL would return no rows).
-- The table is scanned once and the parent id set is
-- built once.
SELECT id,
       CASE
           WHEN p_id IS NULL THEN 'Root'
           WHEN id IN (SELECT p_id FROM tree WHERE p_id IS NOT NULL) THEN 'Inner'
           ELSE 'Leaf'
       END AS node_types
FROM tree
ORDER BY id;
```

COMPLEXITY | Time O(n) with the set of parent ids built once | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 274

Build or transform a grid with index arithmetic

PATTERN CONTEXT

Decide where each output cell comes from with a formula in (r, c), then build the whole grid with one comprehension. Rotation, diamonds, block summaries and prefix tables are all 'cell(r, c) = something computed from r and c'.

279 Filled ASCII Diamond — EASY

DIFFICULTY | EASY

Better approach: row i has side_underscores = |row_number - radius| underscores on each side and 2 * (radius - side_underscores) + 1 x's in the middle

QUESTION INFO

QUESTION

Given a non-negative integer radius R , render a filled diamond on a square grid of side 2*R + 1 . The grid is indexed by (i, j) with 0 <= i, j <= 2*R , and its center is (R, R) . A cell belongs to the diamond (draw 'x') if its Manhattan distance from the center is at most R , i.e. abs(i - R) + abs(j - R) <= R . Every other cell is background (draw '_'). Return the picture as a list of 2*R + 1 strings, one per row (top to bottom), each of length 2*R + 1 .

Function signature

```
def render_diamond(R: int) -> list[str]:
    ...
```

Follow-up (as reported): The naive fill is O(R^2). Can you produce each row directly from its width instead of testing every cell? (Row i is 'x' repeated 2*(R - abs(i-R)) + 1 times, padded with '_' .)

EXAMPLES

Example 1

Input	`R = 1`
Output	`["_x_", "xxx", "_x_"]`
Why	`Center (1,1). Row 0 has distance 1 from center vertically, so only column 1 (distance 0) is inside.`

Example 2

Input	`R = 2`
Output	`["__x__", "_xxx_", "xxxxx", "_xxx_", "__x__"]`
Why	`Each row i allows columns within (R - abs(i-R)) of the center.`

Example 3

Input	`R = 0`
Output	`["x"]`
Why	`A 1x1 grid whose only cell is the center.`

CONSTRAINTS

- 0 <= R <= 500 .
- For R = 0 the result is the single cell ["x"] .

CLARIFY BEFORE CODING

Should adjacency be four-directional or include diagonals? Can the grid be empty or ragged, and are start/end cells guaranteed valid? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target

the expected complexity.

Known bounds to confirm:
- 0 <= R <= 500 .
- For R = 0 the result is the single cell ["x"] .

BRUTE FORCE | Test every cell

BRUTE-FORCE IDEA

Start with: Test every cell. Start with: Test every cell. Each of the (2R + 1)^2 cells runs the distance test.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Test every cell
process every required item or candidate directly
keep the result required by the question
```

BRUTE-FORCE COMPLEXITY | Time O(R^2) | Space O(R^2)

WHY THIS IS WEAKER | Each of the (2R + 1)^2 cells runs the distance test.

HOW TO IMPROVE IT

row i has side_underscores = |row_number - radius| underscores on each side and 2 * (radius - side_underscores) + 1 x's in the middle

SHORT BEST SOLUTION

Row i has 2 * (radius - |row_number - radius|) + 1 x's with |row_number - radius| underscores on each side.

```python
# BASE QUESTION (optimal): Build each row from its width
# Idea: Row i has 2 * (radius - |row_number - radius|) + 1 x's with |row_number - radius|
# underscores on each side.
def render_diamond(radius):
    rows = []
    for row_number in range(2 * radius + 1):
        side_underscores = abs(row_number - radius)
        rows.append("_" * side_underscores + "x" * (2 * (radius - side_underscores) + 1) + "_" * side_underscor
es)
    return rows
```

COMPLEXITY | Time O(R^2) characters | Space O(R^2)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 279

GROUP 43 OF 43 | 3 QUESTIONS | PART B - PART 1

Split into pieces, fix each piece, join (and pack words into lines)

PATTERN CONTEXT

Parsing questions split the text on a separator, transform every piece with one small rule, and join again. Layout questions go word by word and decide whether the next word still fits on the current line.

284 Implement string formatting and request tracking tasks — EASY

DIFFICULTY | EASY

Better approach: split, fix each piece, join; the request-tracking part is a small dict

QUESTION INFO

QUESTION

Four small tasks. (A) snake_to_camel(s): convert snake_case to camelCase, ignoring leading, trailing and repeated underscores. (B) format_thousands(n): write an integer with a comma between every group of three digits (keep the minus sign). (C) ocr_valid(t, rot): an OCR scanner may read a label upside down; given the rotation table rot (character -> character seen after turning 180 degrees), return true if t, or t turned 180 degrees (reversed and mapped through rot), is an integer from 1 to 200 without leading zeros. (D) Server: begin(rid, ts) and end(rid, ts) arrive in any order for each request id; drain(watermark) returns the completed requests as 'id begin=b end=e' lines in order of end time, only those with end at most the watermark (all of them when no watermark is given).

Function signature

```
def snake_to_camel(s: str) -> str
def format_thousands(n: int) -> str
def ocr_valid(t: str, rot: dict) -> bool
class Server:
    def begin(self, rid, ts) -> None
    def end(self, rid, ts) -> None
    def drain(self, watermark=None) -> list
```

EXAMPLES

Example 1	
Input	`'make__it_camel_'`
Output	`'makeItCamel'`
Why	`Repeated and trailing underscores are ignored.`

Example 2	
Input	`1234567`
Output	`'1,234,567'`
Why	`A comma before every third digit from the right.`

Example 3	
Input	`-1000`
Output	`'-1,000'`
Why	`The sign stays at the front.`

QUESTION

Example 4

Input	`rot = {'0': '0', '1': '1', '2': '2', '5': '5', '6': '9', '8': '8', '9': '6'}`
	`ocr_valid('186', rot)`
	`ocr_valid('981', rot)`
	`ocr_valid('500', rot)`

Output	`ocr_valid('186', rot) -> true`
	`ocr_valid('981', rot) -> true`
	`ocr_valid('500', rot) -> false`

Why	`981 is above 200, but turned upside down it reads 186, which is valid. 500 is too large and its rotation 005 has leading zeros.`

Example 5

Input	`sv = Server()`
	`sv.begin('r2', 5)`
	`sv.end('r1', 4)`
	`sv.begin('r1', 1)`
	`sv.end('r2', 9)`
	`sv.drain(8)`
	`sv.drain()`

Output	`sv.drain(8) -> ['r1 begin=1 end=4']` `sv.drain() -> ['r2 begin=5 end=9']`
Why	`r1 finished at 4 and r2 at 9. Draining with watermark 8 prints only r1; draining everything then prints r2.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE 1 | Task A by walking characters

BRUTE-FORCE IDEA

Walk the identifier, set a flag after each underscore, and uppercase the next letter. It ignores leading underscores by construction when the flag is applied to the first output letter, so handle that case explicitly.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Task A by walking characters
Walk the identifier
set a flag after each underscore
uppercase the next letter
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | Linear and fine; the split-based version is shorter. The only care is leading and repeated underscores.

BRUTE FORCE 2 | Task B with the language's format specifier

BRUTE-FORCE IDEA

Python's format(n, ',') inserts the separators. Correct for every integer including negatives and the 64-bit minimum; in an interview you are usually asked to write the grouping yourself, as the optimal version does.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Task B with the language's format specifier
Python's format(n
',') inserts the separators
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(d) | Space O(d)

WHY THIS IS WEAKER | Correct and O(d), but it hides the algorithm; interviews usually ask for the grouping by hand: peel three digits at a time from the right.

BRUTE FORCE 3 | Task C by checking all three readings with a straightforward helper

BRUTE-FORCE IDEA

Try the string as-is, then the rotated string, and test each for a base-10 integer in 1..200. This is the same logic as the optimal version without clever shortcuts; its only risk is forgetting that unmapped characters make the rotation invalid.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Task C by checking all three readings with a straightforward helper
Try the string as-is
the rotated string
test each for a base-10 integer in 1..200
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(L) | Space O(L)

WHY THIS IS WEAKER | It is already linear; there is little to speed up. The risk is correctness: unmapped characters make the rotation invalid, and leading zeros are not valid numbers.

BRUTE FORCE 4 | Task D by storing all events and sorting when asked

BRUTE-FORCE IDEA

Record every begin and end, and on a print request pair them up and sort by (end, id). Correct for the heap version; each print re-sorts everything.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
store every begin and end event
when asked, pair each request's begin and end
sort completed requests by (end time, request id)
return the sorted requests
```

BRUTE-FORCE COMPLEXITY | Time O(n log n) per print | Space O(n)

WHY THIS IS WEAKER | Each print sorts everything pending, O(n log n); a min-heap of completed requests prints in order with O(log n) per event, releasing only entries below a safe watermark.

HOW TO IMPROVE IT

split, fix each piece, join; the request-tracking part is a small dict

SHORT BEST SOLUTION

Part 1: Split on underscores, skip empty pieces, and capitalize every piece after the first.

Part 2: Take groups of three digits from the right.

Part 3: Validate the text directly or reverse it through the rotation map and validate again.

Part 4: Join begin and end events by ID, then heap completed requests by end time.

```python
import heapq


def snake_to_camel(text):
    words = [word for word in text.split("_") if word]
    return "" if not words else words[0] + "".join(word.capitalize() for word in words[1:])


def format_thousands(number):
    sign, digits = ("-", str(-number)) if number < 0 else ("", str(number))
    groups = []
    while digits:
        groups.append(digits[-3:])
        digits = digits[:-3]
    return sign + ",".join(reversed(groups))


def ocr_valid(text, rotations):
    def valid(value):
        return value.isdigit() and value[0] != "0" and 1 <= int(value) <= 200
    if valid(text):
```

```
        return True
    try:
        turned = "".join(rotations[ch] for ch in reversed(text))
    except KeyError:
        return False
    return valid(turned)


class Server:
    def __init__(self):
        self.pending = {}
        self.done = []

    def _add(self, request_id, timestamp, side):
        times = self.pending.setdefault(request_id, [None, None])
        if times[side] is not None:
            raise ValueError("duplicate event")
        times[side] = timestamp
        if None not in times:
            begin, end = self.pending.pop(request_id)
            if end < begin:
                raise ValueError("ends before it begins")
            heapq.heappush(self.done, (end, request_id, begin))

    def begin(self, request_id, timestamp):
        self._add(request_id, timestamp, 0)

    def end(self, request_id, timestamp):
        self._add(request_id, timestamp, 1)

    def drain(self, watermark=None):
        lines = []
        while self.done and (watermark is None or self.done[0][0] <= watermark):
            end, request_id, begin = heapq.heappop(self.done)
            lines.append(f"{request_id} begin={begin} end={end}")
        return lines
```

COMPLEXITY | Part 1: Time O(n) | Space O(n)
Part 2: Time O(d) | Space O(d)
Part 3: Time O(n) | Space O(n)
Part 4: Time O(log n) per completed request | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 284

287 Convert Valid Snake Case to Lower Camel Case — HARD

DIFFICULTY | HARD

Better approach: reformat(): split on '_', keep the first piece, capitalise the others

QUESTION INFO

QUESTION

Given a valid snake_case identifier (one or more lowercase words separated by single underscores, with no leading, trailing or doubled underscores), convert it to lowerCamelCase: keep the first word as it is, remove the underscores and capitalise the first letter of every later word.

Function signature

def snake_to_camel(s: str) -> str

EXAMPLES

QUESTION

Example 1	
Input	`'get_http_response_code'`
Output	`'getHttpResponseCode'`
Why	Words are joined and every word after the first is capitalised.

Example 2	
Input	`'user_id'`
Output	`'userId'`
Why	Two words.

Example 3	
Input	`'name'`
Output	`'name'`
Why	A single word has nothing to change.

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

DIRECT APPROACH | Walk the characters and capitalize after each underscore

DIRECT IDEA

Keep a flag that is set after an underscore; when set, append the uppercase of the next letter. Linear and fine, just longer than the split-based answer.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Walk the characters and capitalize after each underscore
Keep a flag that is set after an underscore
when set
append the uppercase of the next letter
```

DIRECT-APPROACH COMPLEXITY | Time O(n) | Space O(n)

WHY THIS IS WEAKER | Already O(n); it is just longer than needed. Splitting on '_' and capitalising each word after the first is the short form.

HOW TO IMPROVE IT

reformat(): split on '_', keep the first piece, capitalise the others

SHORT BEST SOLUTION

Split on '_', keep the first word, and capitalize the first letter of each other word with a join. For valid input there are no empty parts.

```python
# BASE QUESTION (optimal): Split on underscores and
# capitalize
# Idea: Split on '_', keep the first word, and capitalize
# the first letter of each other word with a join. For
# valid input there are no empty parts.
def snake_to_camel(snake_name):
    first, *rest = snake_name.split("_")
    return first + "".join(word[:1].upper() + word[1:] for word in rest)
```

COMPLEXITY | Time O(n) | Space O(n)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 287

288 Implement a function to justify a list of words within a given width. HARD

DIFFICULTY | HARD

Better approach: pack_lines for the words, then spread the extra spaces over the gaps (the left gaps get the remainder); the last line is left-aligned

QUESTION INFO

QUESTION

Given a list of words and a line width, pack the words into lines of exactly that width (full justification): put as many words on a line as fit (with at least one space between them), then spread the extra spaces between the words as evenly as possible, giving the leftmost gaps the extra ones. The last line, and any line with a single word, is left-justified and padded with spaces on the right. Return the list of lines.

Function signature

def fullJustify(words: list, max_width: int) -> list

EXAMPLES

Example 1	
Input	`['This', 'is', 'an', 'example', 'of', 'text', 'justification.'], 16`
Output	`['This is an', 'example of text', 'justification. ']`
Why	`Three lines of width 16: the first two are spread out, the last is left-aligned and padded.`

Example 2	
Input	`['a', 'b'], 5`
Output	`['a b ']`
Why	`Both words fit on one line, which is the last line: left-aligned with single spaces and padded on the right.`

CLARIFY BEFORE CODING

Is matching case-sensitive, what alphabet or character set is allowed, and how should spaces, punctuation, duplicates, and the empty string be handled? Confirm the exact return value, whether I may modify the input, how ties are broken, and the largest input size so I target the expected complexity.

BRUTE FORCE | Build each line by trying to add one more word with join

BRUTE-FORCE IDEA

Grow a line word by word, testing the joined length each time, then distribute the spaces. It does the same packing as the optimal version but recomputes lengths from scratch, which costs extra time.

SHORT PSEUDOCODE (CODE ONLY IF ASKED)

```
method: Build each line by trying to add one more word with join
Grow a line word by word
testing the joined length each time
distribute the spaces
return the requested result
```

BRUTE-FORCE COMPLEXITY | Time O(total characters * words per line) | Space O(width)

WHY THIS IS WEAKER | Every attempt to add a word re-joins the whole line to measure it, so each line costs about (words per line) times its length. Tracking only a running length makes it linear in the total characters.

HOW TO IMPROVE IT

pack_lines for the words, then spread the extra spaces over the gaps (the left gaps get the remainder); the last line is left-aligned

SHORT BEST SOLUTION

Pack as many words as fit on a line (counting one space between words). Spread the spare spaces over the gaps, giving the extra ones to the leftmost gaps. The last line and any single-word line are left-justified.

```python
# BASE QUESTION (optimal): Greedy line packing
# Idea: Pack as many words as fit on a line (counting one
# space between words). Spread the spare spaces over the
# gaps, giving the extra ones to the leftmost gaps. The
# last line and any single-word line are left-justified.
def fullJustify(words, max_width):
    lines, first = [], 0
    while first < len(words):
        after_last, length = first, 0
        while after_last < len(words) and length + len(words[after_last]) + (after_last - first) <= max_width:
            length += len(words[after_last])
            after_last += 1
        gaps = after_last - first - 1
        if after_last == len(words) or gaps == 0:
            line = " ".join(words[first:after_last])
            line += " " * (max_width - len(line))
        else:
            spaces, extra = divmod(max_width - length, gaps)
            line = ""
            for word_index in range(first, after_last - 1):
                line += words[word_index] + " " * (spaces + (1 if word_index - first < extra else 0))
            line += words[after_last - 1]
        lines.append(line)
        first = after_last
    return lines
```

COMPLEXITY | Time O(total characters) | Space O(total characters)

GO TO PART B - PART 2: FOLLOW-UPS FOR QUESTION 288

PART B - PART 2 FOLLOW-UPS

PART B - Part 2: follow-up information and code

These are all 231 follow-ups for the 91 Easy and Hard questions in PART B. When implementation is useful, the code is included. The blue button returns to the matching PART B - Part 1 solution.

Frontier search (BFS or DFS), Tier 1: spread, count and measure distance

1 Flood Fill — EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Do it with BFS. Why is the early return needed?

A deque gives BFS with the same cost. If the new colour equals the old one, repainting never changes the test 'pixel has the old colour', so pixels would be pushed again forever; the early return (or a visited set) stops that.

FOLLOW-UP CODE

```
from collections import deque


def flood_fill_bfs(image, start_row, start_col, new_color):
    old_color = image[start_row][start_col]
    if old_color == new_color:
        return image
    image[start_row][start_col] = new_color
    queue = deque([(start_row, start_col)])
    while queue:
        row, col = queue.popleft()
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if (0 <= next_row < len(image) and 0 <= next_col < len(image[0])
                    and image[next_row][next_col] == old_color):
                image[next_row][next_col] = new_color
                queue.append((next_row, next_col))
    return image
```

BACK TO PART B - PART 1: QUESTION 1

8 Infection Spread on a Grid (Cellular Automaton) — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Add a recovery delay: an infected cell recovers after d steps and cannot spread afterwards.

A cell infects every healthy neighbor on the very next step, so recovery does not change how long spreading takes; it only changes what the grid looks like at time t. Store the step at which each cell was infected (the BFS level) and derive the state at any time: healthy, infected, or recovered when t minus its infection step is at least d.

FOLLOW-UP CODE

```
def state_after(grid, steps, delay=None):
    """Grid after t steps; with a recovery delay a cell infected at step s becomes 'R' at step s + delay."""
    cells = [list(row) for row in grid]
    infected_at = {(row, col): 0 for row in range(len(cells)) for col in range(len(cells[0])) if cells[row][
col] == "I"}
    frontier, step = list(infected_at), 0
    while frontier and step < steps:
        step += 1
        next_frontier = []
        for row, col in frontier:
            for row_step, col_step in ((1, 0), (-1, 0), (0, 1), (0, -1)):
                next_row, next_col = row + row_step, col + col_step
                if 0 <= next_row < len(cells) and 0 <= next_col < len(cells[0]) and cells[next_row][next_col]
== "." and (next_row, next_col) not in infected_at:
                    infected_at[(next_row, next_col)] = step
                    next_frontier.append((next_row, next_col))
        frontier = next_frontier
    for (row, col), infected_step in infected_at.items():
        cells[row][col] = "R" if delay is not None and steps - infected_step >= delay else "I"
    return ["".join(row) for row in cells]
```

FOLLOW-UP 2 | Why can you stop early?

When the next frontier is empty no cell can change in any later step (the update only depends on the frontier), so the grid is at a fixed point. In a variant with pending recoveries keep going until those are done too.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Make the spread probabilistic.

A frontier cell may fail to infect a neighbor this step, so it stays in the frontier. Keep frontier cells until all their healthy neighbors are infected, and draw a random number per (cell, neighbor, step). The expected number of steps now needs simulation.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 8

9 Shortest Distance From All Buildings HARD

DIFFICULTY | HARD | 1 FOLLOW-UPS

FOLLOW-UP 1 | Minimize the average distance instead of the total.

With a fixed number of houses the average is total / houses, so the best cell is the same; compute it once at the end to report the average.

FOLLOW-UP CODE

```
def best_school_average(grid):
    spot = shortest_distance_from_all_buildings(grid)
    if spot == (-1, -1):
        return spot, None
    total, _, houses = school_totals(grid)
    return spot, total[spot[0]][spot[1]] / houses
```

BACK TO PART B - PART 1: QUESTION 9

Sweep over events in time order

16 Count Overlapping Rectangle Updates on a Grid — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | How many cells are covered by at least k rectangles?

Build the count matrix with the difference array and count the cells with value at least k. O(R + size^2).

FOLLOW-UP CODE

```
def range_add_counts(size, rectangles):
    marks = [[0] * (size + 1) for _ in range(size + 1)]
    for r1, c1, r2, c2 in rectangles:
        marks[r1][c1] += 1
        marks[r1][c2 + 1] -= 1
        marks[r2 + 1][c1] -= 1
        marks[r2 + 1][c2 + 1] += 1
    for row in range(size + 1):
        for col in range(1, size + 1):
            marks[row][col] += marks[row][col - 1]
    for col in range(size + 1):
        for row in range(1, size + 1):
            marks[row][col] += marks[row - 1][col]
    return [row[:size] for row in marks[:size]]

def count_cells_covered_at_least(size, rectangles, min_rectangles):
    counts = range_add_counts(size, rectangles)
    return sum(1 for row in counts for cover in row if cover >= min_rectangles)
```

FOLLOW-UP 2 | The grid is huge (10^9 x 10^9) but there are few rectangles.

Coordinate-compress the corner coordinates and apply the same difference array on the compressed grid (at most (2R)^2 cells), or use a sweep line with a segment tree over columns. Cost: O(R log R) to sort the corner coordinates plus up to O(R^2) compressed cells, instead of 10^18 cells.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Rectangles are added and the count of a cell is queried online.

Use a 2D Fenwick tree with range update and point query (the difference-array idea turned into a BIT): O(log^2 n) per operation.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 16

17 Compute Turnstile Crossing Times — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | The priority when the turnstile was idle is different in this version of the question.

Keep the same loop and make the idle-state winner a parameter. Only the line that picks the queue changes, which is why the state machine should be written first.

FOLLOW-UP CODE

```
def turnstile_times_priority(events, idle_first="exit"):
    """Same simulation with a configurable idle priority (some versions let entering go first when the door
was idle)."""
    num_people = len(events)
    people_by_arrival = sorted(range(num_people), key=lambda person: (events[person][0], person))
    enter_queue, exit_queue, pass_time = deque(), deque(), [0] * num_people
    time, next_arrival, last_direction, last_busy_second = 0, 0, None, -10
    while next_arrival < num_people or enter_queue or exit_queue:
        if not enter_queue and not exit_queue:
            time = max(time, events[people_by_arrival[next_arrival]][0])
            if time > last_busy_second + 1:
                last_direction = None
        while next_arrival < num_people and events[people_by_arrival[next_arrival]][0] <= time:
```

```
            (enter_queue if events[people_by_arrival[next_arrival]][1] == "enter" else exit_queue).append(
people_by_arrival[next_arrival])
            next_arrival += 1
        if enter_queue and exit_queue:
            direction = last_direction if last_direction is not None else idle_first
            queue = enter_queue if direction == "enter" else exit_queue
        else:
            queue = enter_queue or exit_queue
        person = queue.popleft()
        pass_time[person], last_direction, last_busy_second = time, events[person][1], time
        time += 1
    return pass_time
```

FOLLOW-UP 2 | Timestamps are up to 10^9.

That is exactly why the simulation jumps between events: the loop runs once per person, so the answer is O(n log n) no matter how large the times are.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Two turnstiles or a capacity of two people per second.

Run the same pick-a-queue step twice per tick (or per turnstile) and carry a previous-state variable per turnstile. Decide whether the two turnstiles share the queues.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 17

18 Compute city skyline outline — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Divide and conquer instead of a heap.

Build the skyline of each half and merge the two by sweeping both lists with the current height of each; the merged height is the max of the two. T(n) = 2T(n/2) + O(n) = O(n log n), same as the heap.

FOLLOW-UP CODE

```
def skyline_divide_conquer(buildings):
    def merge(left_skyline, right_skyline):
        merged, left_index, right_index, left_height, right_height = [], 0, 0, 0, 0
        while left_index < len(left_skyline) or right_index < len(right_skyline):
            if right_index >= len(right_skyline) or (left_index < len(left_skyline) and left_skyline[
left_index][0] < right_skyline[right_index][0]):
                position, left_height = left_skyline[left_index]; left_index += 1
            elif left_index >= len(left_skyline) or right_skyline[right_index][0] < left_skyline[left_index][
0]:
                position, right_height = right_skyline[right_index]; right_index += 1
            else:
                position, left_height = left_skyline[left_index]; right_height = right_skyline[right_index][
1]; left_index += 1; right_index += 1
            height = max(left_height, right_height)
            if not merged or merged[-1][1] != height:
                merged.append([position, height])
        return merged

    def solve(start, end):
        if end - start == 0:
            return []
        if end - start == 1:
            left, right, height = buildings[start]
            return [[left, height], [right, 0]]
        middle = (start + end) // 2
        return merge(solve(start, middle), solve(middle, end))

    return solve(0, len(buildings))
```

FOLLOW-UP 2 | Coordinates up to 10^9 or buildings of a very large count.

Only the edge coordinates matter, so nothing depends on the coordinate range. The sweep is O(n log n) in the number of buildings; use a balanced tree or a heap with lazy deletion for the active set.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Output the area under the skyline.

Sum height * (next x - x) over consecutive key points: one extra pass over the output.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 18

GROUP 3 OF 43 | 5 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Tree recursion, Tier 1: one function that visits a node, asks its children, and reports upward

19 Find right-side view of binary tree

EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | Left side view, or both.

Swap the order of the two recursive calls. For both views record the first and last node per depth in one pass.

FOLLOW-UP CODE

```
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def leftSideView(root):
    left_view = []

    def dfs(node, depth):
        if node is None:
            return
        if depth == len(left_view):
            left_view.append(node.val)
        dfs(node.left, depth + 1)
        dfs(node.right, depth + 1)

    dfs(root, 0)
    return left_view
```

FOLLOW-UP 2 | Avoid recursion.

Use the BFS version and keep only the last node per level (no per-level list), or an explicit stack of (node, depth) pushing left before right so that right is popped first.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Top view or bottom view.

Track a horizontal distance per node (left minus one, right plus one) and keep the first (top) or last (bottom) node seen per distance in BFS order.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 19

20 Given two trees, determine if the pre-order traversals of two trees are same or not.

EASY

DIFFICULTY | EASY | 4 FOLLOW-UPS

FOLLOW-UP 1 | What if the interviewer meant the trees must be identical?

Then compare structure as well as values: same_tree above walks both trees together and treats two missing nodes as equal. Equivalently, serialize pre-order with null markers and compare the strings, since pre-order with nulls is unique. Ask which definition they mean before coding.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Do it with O(1) extra space.

Morris traversal visits pre-order using temporary threads on right pointers, giving O(1) space, but it mutates the tree while running (restoring it after), so it is not safe for concurrent readers. Mention that trade-off.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | They want identical trees, not just sequences.

Compare structure as well: walk both trees together and treat a missing child on one side only as a mismatch, or serialise pre-order with null markers and compare the strings.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | They want identical trees (structure and values), not just the same pre-order values.

Compare the two trees together: a node must match in value and both children must match; two missing children match. Equivalently compare the pre-order serialisation with null markers (which is unique). O(min(n, m)) with early exit.

FOLLOW-UP CODE

```
def same_tree(first, second):
    if first is None or second is None:
        return first is second
    return (first.val == second.val
            and same_tree(first.left, second.left)
            and same_tree(first.right, second.right))
```

BACK TO PART B - PART 1: QUESTION 20

21 Calculate the height of a binary tree. EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | Nodes or edges?

Height in nodes is what the code returns; in edges subtract one (an empty tree is -1 and a single node is 0). Say which you use before you code.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Diameter (the longest path between any two nodes).

Return the height from the recursion and update a global best with left + right at each node: one pass, O(node).

FOLLOW-UP CODE

```
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def diameter(root):
    best = 0

    def depth(node):
        nonlocal best
        if node is None:
            return 0
        left_depth, right_depth = depth(node.left), depth(node.right)
        best = max(best, left_depth + right_depth)
        return 1 + max(left_depth, right_depth)

    depth(root)
    return best
```

FOLLOW-UP 3 | Is the tree balanced?

Return the height, or -1 as soon as a subtree is unbalanced or the heights differ by more than one, so the whole check is one O(node) pass.

FOLLOW-UP CODE

```
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def is_balanced(root):
    def check(node):
        if node is None:
            return 0
        left_height = check(node.left)
        if left_height < 0:
            return -1
        right_height = check(node.right)
        if right_height < 0 or abs(left_height - right_height) > 1:
```

```
            return -1
        return 1 + max(left_height, right_height)

    return check(root) >= 0
```

BACK TO PART B - PART 1: QUESTION 21

26 Repeatedly Delete Leaves Matching a Target Value From a Binary Tree — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | The tree is a 100,000-node chain. How do you avoid a stack overflow?

Collect nodes in pre-order with an explicit stack and remember each node's parent, then walk that list in reverse (children before parents) and unlink matching leaves. This keeps the post-order guarantee without recursion.

FOLLOW-UP CODE

```
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def removeLeafNodes_iterative(root, target):
    if root is None:
        return None
    parent = {id(root): None}
    order, stack = [], [root]
    while stack:
        node = stack.pop()
        order.append(node)
        for child in (node.left, node.right):
            if child:
                parent[id(child)] = node
                stack.append(child)
    for node in reversed(order):                        # children always come before their parent
        if node.left is None and node.right is None and node.val == target:
            parent_node = parent[id(node)]
            if parent_node is None:
                return None
            if parent_node.left is node:
                parent_node.left = None
            else:
                parent_node.right = None
    return root
```

FOLLOW-UP 2 | Delete any node whose whole subtree sums to zero, or whose value matches a predicate.

Same shape of solution: the decision for a node depends on information returned from its children. Return (new subtree, subtree sum) from the helper and drop the node when the rule holds. The key is to define what a child reports upward. Cost: one post-order pass, O(n) time and O(h) space.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Return how many nodes were deleted.

Count inside the same post-order helper (a nonlocal counter, or return it with the subtree). Never do a second traversal to count. Cost: O(1) extra, counted inside the same O(n) pass.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 26

27 Delete an Entire Binary Tree HARD

DIFFICULTY | HARD | 1 FOLLOW-UPS

FOLLOW-UP 1 | Do it with O(1) extra space.

Rotate instead of recursing: while the root has a left child, rotate it up (the left child becomes the root and the old root becomes its right child). When the root has no left child, free it and move to its right child. Every node is freed once and no stack is used; the order is no longer post-order.

FOLLOW-UP CODE

```
def delete_tree_constant_space(root):
    freed = []
    while root:
        if root.left:
            left = root.left
            root.left = left.right
            left.right = root
            root = left
        else:
            next_node = root.right
            freed.append(root.val)
            root.right = None
            root = next_node
    return freed
```

BACK TO PART B - PART 1: QUESTION 27

A heap of what is available: pop the best, put back the changed one

36 Huffman Tree Construction HARD

DIFFICULTY | HARD | 4 FOLLOW-UPS

FOLLOW-UP 1 | Why is the greedy choice optimal?

Exchange argument: in some optimal tree the two least frequent symbols are siblings at the deepest level (swapping a shallower, more frequent symbol with a deeper, less frequent one never increases cost). Merging them into one symbol reduces the problem to n - 1 symbols with the same optimal-substructure property, so induction closes it.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | How do you make the code canonical, and why would you?

Keep only each symbol's code length from the tree, sort symbols by (length, symbol), and assign consecutive binary codes. A decoder then needs just the lengths rather than the whole tree, which is how DEFLATE stores its tables.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Time complexity if symbols arrive already sorted?

Two-queue method: one queue holds the sorted leaves, the other the merged nodes (which are produced in nondecreasing order). Always pop the smaller front of the two queues. That is O(n) after sorting.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Make the codes canonical so only the code lengths need to be stored.

Sort symbols by (next_code length, symbol) and assign consecutive binary numbers, shifting left when the length grows. A decoder rebuilds the codes from the lengths alone (this is how DEFLATE stores its tables).

FOLLOW-UP CODE

```
def huffman_codes(freq):
    heap = [(weight, insertion_order, symbol, None, None) for insertion_order, (symbol, weight) in enumerate(
sorted(freq.items()))]
    if not heap:
        return {}
    heapq.heapify(heap)
    counter = len(heap)
    while len(heap) > 1:
        lightest = heapq.heappop(heap)
        second_lightest = heapq.heappop(heap)
        heapq.heappush(heap, (lightest[0] + second_lightest[0], counter, None, lightest, second_lightest))
        counter += 1
    root = heap[0]
    codes = {}
    stack = [(root, "")]
    while stack:
        node, prefix = stack.pop()
        if node[2] is not None:
            codes[node[2]] = prefix or "0"
        else:
            stack.append((node[4], prefix + "1"))
            stack.append((node[3], prefix + "0"))
    return codes

def canonical_codes(freq):
    lengths = {symbol: len(code_text) for symbol, code_text in huffman_codes(freq).items()}
    canonical, next_code, previous_length = {}, 0, 0
    for symbol in sorted(lengths, key=lambda symbol: (lengths[symbol], symbol)):
        next_code <<= lengths[symbol] - previous_length
        canonical[symbol] = format(next_code, f"0{lengths[symbol]}b")
        next_code, previous_length = next_code + 1, lengths[symbol]
    return canonical
```

BACK TO PART B - PART 1: QUESTION 36

GROUP 5 OF 43 | 3 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Recurrence on a position or a range (memo or table)

43 Count Perfect Wake Numbers up to N — HARD

DIFFICULTY | HARD | 5 FOLLOW-UPS

FOLLOW-UP 1 | Why does the DP only need a 'falling' flag instead of the full history?

A valley needs a digit that is lower than the digit before it and then higher than the one after it. Once any digit has decreased, a later increase would create exactly that valley, so the only information to carry is whether we have already gone down. The used mask and last digit cover the other two rules.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | How do you count the numbers in a range [A, B]?

Use inclusion-exclusion: count(B) - count(A - 1). The DP is a function of the upper bound only, and it is cheap enough (a few thousand states) to call twice.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | What is the closed form without a DP?

Since digits are distinct and unimodal, pick which k digits (from 1..9) appear and the peak is the largest; each of the other k - 1 digits goes either left or right of the peak: 2^(k-1) arrangements per set. So the count of perfect numbers with k digits is C(9, k) * 2^(k-1). Only the 'at most N' bound needs the DP.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Is there a closed form for k-digit numbers?

Pick the k distinct digits from 1..9 (C(9, digit_count) ways); the largest must be the peak, and each of the other digit_count - 1 digits goes either left or right of it in sorted order, so 2^(digit_count-1) arrangements: C(9, digit_count) * 2^(digit_count-1). Only the bound N needs the DP.

FOLLOW-UP CODE

```
def perfect_with_k_digits(digit_count):
    return math.comb(9, digit_count) * 2 ** (digit_count - 1) if 1 <= digit_count <= 9 else 0
```

FOLLOW-UP 5 | Count perfect numbers in a range [A, B].

Compute count(B) - count(A - 1); the DP depends on the upper bound only.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 43

44 Word Break: All Sentences — HARD

DIFFICULTY | HARD | 1 FOLLOW-UPS

FOLLOW-UP 1 | Only count the segmentations.

A DP over suffixes: ways[start] is the sum of ways[j] over words text[start:j]. No strings are built, so it is O(n * L) with a set of words.

FOLLOW-UP CODE

```
def count_sentences(text, word_dict):
    words = set(word_dict)
    ways = [0] * (len(text) + 1)
    ways[len(text)] = 1
    for start in range(len(text) - 1, -1, -1):
        ways[start] = sum(ways[end] for end in range(start + 1, len(text) + 1)
                          if text[start:end] in words)
    return ways[0]
```

BACK TO PART B - PART 1: QUESTION 44

45 Edit distance HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the actual operations.

Keep the full table and walk back from dp[first_length][second_length]: if the characters match step diagonally; otherwise take the neighbour whose value is one smaller and record replace, delete or insert. O(first_length * second_length) memory is needed for the walk back.

FOLLOW-UP CODE

```
def edit_operations(first, second):
    first_length, second_length = len(first), len(second)
    dp = [[0] * (second_length + 1) for _ in range(first_length + 1)]
    for first_used in range(first_length + 1):
        dp[first_used][0] = first_used
    for second_used in range(second_length + 1):
        dp[0][second_used] = second_used
    for first_used in range(1, first_length + 1):
        for second_used in range(1, second_length + 1):
            dp[first_used][second_used] = dp[first_used - 1][second_used - 1] if first[first_used - 1] ==
second[second_used - 1] else 1 + min(dp[first_used - 1][second_used - 1], dp[first_used - 1][second_used], dp[
first_used][second_used - 1])
    ops, first_used, second_used = [], first_length, second_length
    while first_used or second_used:
        if first_used and second_used and first[first_used - 1] == second[second_used - 1]:
            first_used, second_used = first_used - 1, second_used - 1
        elif first_used and second_used and dp[first_used][second_used] == dp[first_used - 1][second_used -
1] + 1:
            ops.append(f"replace {first[first_used - 1]} with {second[second_used - 1]}"); first_used,
second_used = first_used - 1, second_used - 1
        elif first_used and dp[first_used][second_used] == dp[first_used - 1][second_used] + 1:
            ops.append(f"delete {first[first_used - 1]}"); first_used -= 1
        else:
            ops.append(f"insert {second[second_used - 1]}"); second_used -= 1
    return ops[::-1]
```

FOLLOW-UP 2 | Are the strings exactly one edit apart (LeetCode 161)?

No table needed: if the lengths differ by more than one answer false; scan to the first mismatch and compare the remainders (equal lengths: skip one char in each; otherwise skip one in the longer). O(n), O(1).

FOLLOW-UP CODE

```
def one_edit_apart(first, second):
    if abs(len(first) - len(second)) > 1:
        return False
    if len(first) > len(second):
        first, second = second, first
    for position in range(len(first)):
        if first[position] != second[position]:
            return first[position + 1:] == second[position + 1:] if len(first) == len(second) else first[
position:] == second[position + 1:]
    return len(first) != len(second) or False
```

FOLLOW-UP 3 | Operations have different costs, or only a small edit distance is of interest (at most k).

Weighted costs just change the numbers added in the recurrence. For 'at most k' only compute a band of width 2k + 1 around the diagonal: O(k * n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 45

Best-first search with a heap: Dijkstra and its minimax twin

51 Minimize maximum height along a grid path — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | The cost is the largest difference between adjacent cells on the path (LeetCode 1631).

Same Dijkstra with max, but the step cost is abs(height difference) instead of the cell value. The binary-search version also works: the check allows an edge only if the difference is at most the limit.

FOLLOW-UP CODE

```
def min_effort_path(heights):
    """LeetCode 1631: minimise the largest absolute difference between consecutive cells on the path."""
    rows, cols = len(heights), len(heights[0])
    best = {(0, 0): 0}
    heap = [(0, 0, 0)]
    while heap:
        cost, row, col = heapq.heappop(heap)
        if cost > best[(row, col)]:
            continue
        if (row, col) == (rows - 1, cols - 1):
            return cost
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols:
                new_effort = max(cost, abs(heights[next_row][next_col] - heights[row][col]))
                if new_effort < best.get((next_row, next_col), float("inf")):
                    best[(next_row, next_col)] = new_effort
                    heapq.heappush(heap, (new_effort, next_row, next_col))
    return 0
```

FOLLOW-UP 2 | Water rises over time (the cell is usable from time H[r][c]).

That is exactly the maximum-cell cost above; the answer is the time at which the corner becomes reachable.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Return the path, not only the value.

Remember each cell's parent at its last improvement and rebuild the path from the target; for the binary-search version run the flood fill once more at the answer limit with parent pointers.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 51

52 Find minimum threshold enabling grid path — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | The cost is the largest difference between adjacent cells on the path (LeetCode 1631).

Same Dijkstra with max, but the step cost is abs(height difference) instead of the cell value. The binary-search version also works: the check allows an edge only if the difference is at most the limit.

FOLLOW-UP CODE

```
def min_effort_path(heights):
    """LeetCode 1631: minimise the largest absolute difference between consecutive cells on the path."""
    rows, cols = len(heights), len(heights[0])
    best = {(0, 0): 0}
    heap = [(0, 0, 0)]
    while heap:
        cost, row, col = heapq.heappop(heap)
        if cost > best[(row, col)]:
            continue
        if (row, col) == (rows - 1, cols - 1):
            return cost
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if 0 <= next_row < rows and 0 <= next_col < cols:
```

```
                new_effort = max(cost, abs(heights[next_row][next_col] - heights[row][col]))
                if new_effort < best.get((next_row, next_col), float("inf")):
                    best[(next_row, next_col)] = new_effort
                    heapq.heappush(heap, (new_effort, next_row, next_col))
    return 0
```

FOLLOW-UP 2 | Water rises over time (the cell is usable from time H[r][c]).

That is exactly the maximum-cell cost above; the answer is the time at which the corner becomes reachable.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Return the path, not only the value.

Remember each cell's parent at its last improvement and rebuild the path from the target; for the binary-search version run the flood fill once more at the answer limit with parent pointers.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 52

53 Minimize travel time with optional meeting point — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | More than two travellers.

Run one Dijkstra per traveller and one from the destination, then for each node take the maximum arrival time (if they wait for each other) or the sum of distances (if the cost is total distance) plus the shared leg. K + 1 searches.

FOLLOW-UP CODE

```
def _dijkstra_all(node_count, neighbours, start):
    distance = [float("inf")] * node_count
    distance[start] = 0
    heap = [(0, start)]
    while heap:
        distance_so_far, from_node = heapq.heappop(heap)
        if distance_so_far > distance[from_node]:
            continue
        for to_node, weight in neighbours[from_node]:
            if distance_so_far + weight < distance[to_node]:
                distance[to_node] = distance_so_far + weight
                heapq.heappush(heap, (distance_so_far + weight, to_node))
    return distance

def meeting_point_total_distance(node_count, edges, travelers, dest):
    """Follow-up: K travellers who each walk to the meeting node, then everyone goes on together; minimise
the sum of the walks plus the shared leg (undirected)."""
    neighbours = [[] for _ in range(node_count)]
    for from_node, to_node, weight in edges:
        neighbours[from_node].append((to_node, weight)); neighbours[to_node].append((from_node, weight))
    distances_from_travelers = [_dijkstra_all(node_count, neighbours, traveler) for traveler in travelers]
    distance_to_destination = _dijkstra_all(node_count, neighbours, dest)
    return min(sum(traveler_distances[meeting_node] for traveler_distances in distances_from_travelers) +
distance_to_destination[meeting_node] for meeting_node in range(node_count))
```

FOLLOW-UP 2 | Meeting is optional.

Also compute the cost with no meeting (each goes straight to the destination: max(distA[dest], distB[dest])) and take the smaller. Meeting at the destination itself already covers it when the cost is the later arrival.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Edge weights change over time (traffic).

Dijkstra still works if the graph is FIFO (leaving later never arrives earlier); each edge's cost becomes a function of the departure time, evaluated when you relax it.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 53

54 Compute minimax grid path and network delay HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Part 1 with a union-find (sort cells, add them one by one).

Add cells in increasing order of height, union neighbours already added, and stop when the start and the goal are connected: that height is the answer. O(mn log mn) for the sort.

FOLLOW-UP CODE

```
def minimax_path_union_find(heights):
    """Alternative: add cells in increasing order of value and stop when start and goal are connected."""
    rows, cols = len(heights), len(heights[0])
    parent = list(range(rows * cols))

    def find(cell):
        while parent[cell] != cell:
            parent[cell] = parent[parent[cell]]
            cell = parent[cell]
        return cell

    opened = set()
    for value, row, col in sorted((heights[row][col], row, col) for row in range(rows) for col in range(
cols)):
        opened.add((row, col))
        for next_row, next_col in ((row + 1, col), (row - 1, col), (row, col + 1), (row, col - 1)):
            if (next_row, next_col) in opened:
                parent[find(row * cols + col)] = find(next_row * cols + next_col)
        if (0, 0) in opened and (rows - 1, cols - 1) in opened and find(0) == find((rows - 1) * cols + cols -
1):
            return value
    return -1
```

FOLLOW-UP 2 | Part 2: weights are huge.

Use 64-bit integers (Python is fine); keep sums in a wide type, and with weights up to 10^9 and 10^5 edges a path can reach 10^14. Cost: unchanged O((V + E) log V); sums need 64-bit (or Python) integers.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Part 2: the signal can be blocked at some nodes, or each node adds a processing delay.

A node delay is added to the edge cost leaving the node (or modelled by splitting each node into in and out with an edge of the delay). The same Dijkstra then applies.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 54

Backtracking: choose, recurse, undo

63 Matrix Flower Placement with House Adjacency — HARD

DIFFICULTY | HARD | 4 FOLLOW-UPS

FOLLOW-UP 1 | What changes between 'at most one' and 'exactly one' flower per row and column?

With 'exactly one' the number of flowers is forced (rows must equal columns) and a row cannot be skipped, which prunes much harder, like N-Queens without the diagonal rule. With 'at most one' you also branch on skipping a row. State which reading you are using before writing code.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | How would you speed up the search?

Prune earlier by checking each house as soon as its last neighbour row is decided (done above), order houses by how few candidate cells they have (most-constrained first), and keep bitmasks for used columns. For large grids this becomes a constraint satisfaction or exact-cover problem, so a SAT solver or DLX is the next step. Cost: the worst case stays exponential; most-constrained-first ordering and early house checks cut the explored states by orders of magnitude on typical grids.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Exactly one flower per row and column.

Add a flag that forbids skipping a row and requires as many columns as rows (otherwise no solution). The pruning is stronger, since every row must find a column.

FOLLOW-UP CODE

```
def place_flowers(grid, exactly_one=False, find_all=False):
    rows, cols = len(grid), len(grid[0])
    cells = [list(row) for row in grid]
    houses = [(row, col) for row in range(rows) for col in range(cols) if cells[row][col] == "H"]
    solutions = []

    def house_ok(row, col):
        return any(0 <= row + row_step < rows and 0 <= col + col_step < cols and cells[row + row_step][col +
col_step] == "F"
                   for row_step, col_step in ((1, 0), (-1, 0), (0, 1), (0, -1)))

    def check_row_houses(row):
        return all(house_ok(house_row, house_col) for house_row, house_col in houses if house_row == row)

    def solve(row, used_cols):
        if not find_all and solutions:
            return
        if row >= 2 and not check_row_houses(row - 2):          # rows up to row - 1 are decided, so row row -
 2 houses are final
            return
        if row == rows:
            if check_row_houses(rows - 1) and (not exactly_one or len(used_cols) == cols):
                solutions.append(["".join(row) for row in cells])
            return
        for col in range(cols):
            if col not in used_cols and cells[row][col] == "0":
                cells[row][col] = "F"
                solve(row + 1, used_cols | {col})
                cells[row][col] = "0"
        if not exactly_one:
            solve(row + 1, used_cols)

    solve(0, frozenset())
    return solutions if find_all else (solutions[0] if solutions else None)
```

FOLLOW-UP 4 | Count the solutions instead of listing them.

Keep the same recursion but add one to a counter at the leaf instead of copying the grid; with a bitmask of used columns and memoisation on (row, mask, pending-house state) it can become a DP.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 63

GROUP 8 OF 43 | 2 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Runs and groups: extend while it fits, restart or close when it does not

64 First Out-of-Order Character EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Report every break and the longest in-order run.

Collect each index where text[i] < text[i - 1]; the runs are the gaps between breaks, so the longest run is the largest difference between consecutive break positions (with the ends included).

FOLLOW-UP CODE

```
def order_breaks(text):
    return [position for position in range(1, len(text)) if text[position] < text[position - 1]]


def longest_in_order_run(text):
    boundaries = [0] + order_breaks(text) + [len(text)]
    return max((end - start for start, end in zip(boundaries, boundaries[1:])), default=0)
```

BACK TO PART B - PART 1: QUESTION 64

65 Group Timestamps Into Close Intervals EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | A run must also never last k seconds or more from its first value (a session limit).

Compare against the run's start: close the run when current - start >= gap_limit (the neighbour gap rule is then implied). One pass again.

FOLLOW-UP CODE

```
def group_by_span(timestamps, gap_limit):
    if not timestamps:
        return []
    runs, start, previous = [], timestamps[0], timestamps[0]
    for current in timestamps[1:]:
        if current - start >= gap_limit:
            runs.append([start, previous])
            start = current
        previous = current
    runs.append([start, previous])
    return runs
```

BACK TO PART B - PART 1: QUESTION 65

Union-find: merge the pairs, then ask who is together

80 Friend Graph with Timeline (Friend / Unfriend Events) HARD

DIFFICULTY | HARD | 4 FOLLOW-UPS

FOLLOW-UP 1 | Why does unfriending break plain Union-Find?

A union merges two sets and forgets which edge caused it, so removing that edge cannot tell which members should separate. Online, fully dynamic connectivity needs heavy structures (Holm-de Lichtenberg-Thorup or link-cut trees, polylogarithmic time). When all events are known in advance, the offline segment-tree-over-time plus rollback DSU is the practical answer and the one to code in an interview.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What is the simple answer if the interviewer wants something quick?

Rebuild the graph from the events up to time t and run BFS or DFS for each query: O(E + N) per query. State the cost and then explain how the offline method amortizes it. A correct simple answer is accepted; naming the optimal one shows depth.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Report how many friend groups exist after each event (friends only).

Union-find with a component counter: decrement when a union merges two different roots; record the counter after each event. O(E alpha(N)).

FOLLOW-UP CODE

```
def components_over_time(people_count, events):
    """Insert-only events: the number of components after each event time."""
    parent = list(range(people_count))

    def find(person):
        while parent[person] != person:
            parent[person] = parent[parent[person]]
            person = parent[person]
        return person

    group_count, history = people_count, []
    for time, person_a, person_b, kind in sorted(events, key=lambda event: event[0]):
        root_a, root_b = find(person_a), find(person_b)
        if root_a != root_b:
            parent[root_a] = root_b
            group_count -= 1
        history.append((time, group_count))
    return history
```

FOLLOW-UP 4 | Queries ask 'were A and B connected at time t?' with no unfriending.

Union by size without path compression, recording the time of each union on the edge it creates; two people are connected at t if the largest time on the path between them is at most t.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 80

GROUP 10 OF 43 | 1 QUESTIONS | PART B - PART 2 FOLLOW-UPS

A dict of per-key state that you update, expire, or rank

81 Robot Status Message Deduplication Window EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Events can arrive out of order.

Decide the contract first: either a late event is judged against the stored time with abs(t - last), or you keep a sorted history per message and compare with the nearest displayed neighbours on both sides. The simple map is only correct for a chronological stream.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 81

Binary search on a yes/no condition

89 First Index Reaching a Threshold (Binary Search) EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | The array is a huge sorted file with random access and an unknown length.

Probe positions 1, 2, 4, 8, ... until the value there is at least k or the position is past the end, then binary search inside the last doubled range. It needs O(log position) reads and no loading of the data.

FOLLOW-UP CODE

```
def first_at_least_unbounded(get, threshold):
    # get(position) returns the position-th value or raises IndexError past the end
    def value(position):
        try:
            return get(position)
        except IndexError:
            return float("inf")

    high = 1
    while value(high - 1) < threshold:
        high *= 2
        if high > 1 << 40:
            return -1
    low = high // 2
    while low < high:
        middle = (low + high) // 2
        if value(middle) >= threshold:
            high = middle
        else:
            low = middle + 1
    return low if value(low) != float("inf") else -1
```

BACK TO PART B - PART 1: QUESTION 89

94 Median of Two Sorted Arrays (LC 4) with a Follow-up Variant HARD

DIFFICULTY | HARD | 2 FOLLOW-UPS

FOLLOW-UP 1 | Now there are k sorted arrays instead of two. How do you find the median?

Binary search on the answer value. For a candidate x, count the elements at most x with one bisect per array (O(k log n)); the median is the smallest x whose count reaches the middle rank. With integers the search range is the global min to max, so the cost is O(k log n log(range)). The heap alternative pops the smallest element from a k-way merge until it passes the middle, O((N / 2) log k), which is simpler but linear in the total size.

FOLLOW-UP CODE

```
def median_k_value(arrays):
    arrays = [sorted_array for sorted_array in arrays if sorted_array]
    total = sum(len(sorted_array) for sorted_array in arrays)
    if total == 0:
        raise ValueError("no elements")

    def kth(rank):
        # smallest value x with at least rank elements <= x (rank counts from 1)
        low = min(sorted_array[0] for sorted_array in arrays)
        high = max(sorted_array[-1] for sorted_array in arrays)
        while low < high:
            middle = (low + high) // 2
            if sum(bisect.bisect_right(sorted_array, middle) for sorted_array in arrays) >= rank:
                high = middle
            else:
                low = middle + 1
        return low

    if total % 2:
        return kth(total // 2 + 1)
```

```
    return (kth(total // 2) + kth(total // 2 + 1)) / 2
```

FOLLOW-UP 2 | What is the heap version and when would you choose it?

A min-heap holds the front of every array. Pop (N + 1) / 2 times, remembering the one or two middle values. It is O(N log k) at worst, trivial to get right and needs no bound on the values, so choose it when the arrays are small or the values are not integers; choose the value search when N is huge and k is small.

FOLLOW-UP CODE

```
def median_k_heap(arrays):
    heap = [(sorted_array[0], array_index, 0) for array_index, sorted_array in enumerate(arrays) if
sorted_array]
    heapq.heapify(heap)
    total = sum(len(sorted_array) for sorted_array in arrays)
    if total == 0:
        raise ValueError("no elements")
    low_rank, high_rank = (total - 1) // 2, total // 2
    seen, middle = 0, []
    while heap:
        value, array_index, position = heapq.heappop(heap)
        if seen >= low_rank:
            middle.append(value)
        if seen >= high_rank:
            break
        seen += 1
        if position + 1 < len(arrays[array_index]):
            heapq.heappush(heap, (arrays[array_index][position + 1], array_index, position + 1))
    return middle[0] if total % 2 else (middle[0] + middle[1]) / 2
```

BACK TO PART B - PART 1: QUESTION 94

95 Find longest duplicated substring — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Avoid hash collisions entirely.

Use a suffix array with the LCP array: the longest duplicate is the largest LCP between neighbours in suffix order. Naive construction (sorting suffixes) is O(n^2 log n); real suffix array algorithms are O(n log n) or O(n).

FOLLOW-UP CODE

```
def longest_dup_suffix_array(text):
    suffix_array = sorted(range(len(text)), key=lambda start: text[start:])
    best = ""
    for earlier, later in zip(suffix_array, suffix_array[1:]):
        common_length = 0
        while earlier + common_length < len(text) and later + common_length < len(text) and text[earlier +
common_length] == text[later + common_length]:
            common_length += 1
        if common_length > len(best):
            best = text[earlier:earlier + common_length]
    return best
```

FOLLOW-UP 2 | The substring must appear at least k times, or must not overlap.

At least k times: count equal hashes in the check (need k of them). Non-overlapping: when two windows match, require their start positions to differ by at least L.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The string is huge and does not fit in memory.

Hash windows on disk in passes, or use a suffix automaton built online; for approximate answers use minhash sketches. State that exact answers need memory proportional to the number of distinct windows. Cost: O(n log n) time and O(n) memory for the hashes; on disk the hashing passes multiply the I/O by about log n.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 95

96 Find the median of two sorted arrays. HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | What if there are k sorted arrays instead of two?

Binary search on the answer value instead of on a position. For a candidate value v, count how many elements are <= v in each array with bisect; the median is the smallest v whose total count reaches the middle rank. That costs O(rank log n log(range)) time and O(1) extra space and needs no merging.

FOLLOW-UP CODE

```
def _kth_across(arrays, rank):
    low = min(sorted_array[0] for sorted_array in arrays if sorted_array)
    high = max(sorted_array[-1] for sorted_array in arrays if sorted_array)
    while low < high:
        middle = (low + high) // 2
        if sum(bisect.bisect_right(sorted_array, middle) for sorted_array in arrays) >= rank:
            high = middle
        else:
            low = middle + 1
    return low

def find_median_of_k_sorted_arrays(arrays):
    total = sum(len(sorted_array) for sorted_array in arrays)
    if total % 2:
        return float(_kth_across(arrays, total // 2 + 1))
    return (_kth_across(arrays, total // 2) + _kth_across(arrays, total // 2 + 1)) / 2
```

FOLLOW-UP 2 | What if the arrays are too large for memory?

The two-array binary search only touches O(log n) positions, so with random access (an indexed file or a remote array) it reads almost nothing. If only sequential access is possible, do a streaming selection: a two-pointer merge that stops after (m + n) / 2 elements, using O(1) memory and O(m + n) time.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Find the k-th smallest element, not just the median.

The same partition search: choose i elements from the shorter array and rank - from_shorter from the other, and move i until the borders are in order. The median is just rank = (m + n + 1) // 2.

FOLLOW-UP CODE

```
def kth_of_two(shorter, longer, rank):
    """k-th smallest (1-based) of two sorted arrays in O(log min(m, n))."""
    if len(shorter) > len(longer):
        shorter, longer = longer, shorter
    low, high = max(0, rank - len(longer)), min(rank, len(shorter))
    while low < high:
        from_shorter = (low + high) // 2
        if shorter[from_shorter] < longer[rank - from_shorter - 1]:
            low = from_shorter + 1
        else:
            high = from_shorter
    from_shorter = low
    return max(([shorter[from_shorter - 1]] if from_shorter else []) + ([longer[rank - from_shorter - 1]] if
rank - from_shorter else []))
```

BACK TO PART B - PART 1: QUESTION 96

Greedy: scan once, carry the best choice so far, act only when forced

97 Alarm for the Next Upcoming Birthday

EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Many queries with different dates. How do you make each one fast?

Sort the birthdays by (day of year, name) once. For a query binary search the first birthday at or after today's day; if there is none, wrap around to the first birthday in the list. O(log p) per query after an O(p log p) setup.

FOLLOW-UP CODE

```
from bisect import bisect_left


class BirthdayBook:
    def __init__(self, people):
        self.entries = sorted((day_of_year(month, day), name) for name, month, day in people)
        self.day_numbers = [day for day, _ in self.entries]

    def next_birthday(self, today):
        insert_position = bisect_left(self.day_numbers, day_of_year(*today))
        return self.entries[insert_position % len(self.entries)][1]
```

BACK TO PART B - PART 1: QUESTION 97

98 Coins on the Table

EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Each event is an integer delta like +3 or -2, still never below zero.

Split the string into tokens, convert each with int(), and apply total = max(0, total + delta). The clamp must happen after every event, not once at the end.

FOLLOW-UP CODE

```
def coins_with_deltas(events):
    total = 0
    for token in events.split():
        total = max(0, total + int(token))
    return total
```

BACK TO PART B - PART 1: QUESTION 98

99 Maximum Profit

EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | The prices are a time series: you must buy before you sell (LeetCode 121).

Keep the minimum price seen so far and the best difference: O(n), O(1). The dictionary version ignores order, so the two extremes are enough.

FOLLOW-UP CODE

```
def max_profit_series(prices):
    """The ordered version (LeetCode 121): buy before sell, one pass with a running minimum."""
    best_profit, lowest_price = 0, float("inf")
    for price in prices:
        lowest_price = min(lowest_price, price)
        best_profit = max(best_profit, price - lowest_price)
    return best_profit
```

FOLLOW-UP 2 | Allow many trades (LeetCode 122) or a limit of k trades.

Many trades: add every positive day-to-day rise. At most k trades: DP over (trades used, holding or not), O(n * k).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Transaction fee or cooldown.

A two-state DP (holding, not holding) per day: hold = max(hold, free - price), free = max(free, hold + price - fee).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 99

GROUP 13 OF 43 | 5 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Sets and counters: remember the first collection, check the second against it

105 Exclude First-K-Unique From Another List — EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | list2 is a huge stream. How do you avoid loading it?

Build the blocked set from list1 first (O(distinct_count) memory), then yield the stream's values lazily, skipping the blocked ones. The output is a generator.

FOLLOW-UP CODE

```
def exclude_stream(list1, stream, distinct_count):
    blocked = set()
    for value in list1:
        if len(blocked) == distinct_count:
            break
        blocked.add(value)
    return (value for value in stream if value not in blocked)
```

BACK TO PART B - PART 1: QUESTION 105

106 Positional Matches Between Two Lists — EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Return the positions instead of the values, lazily for a huge input.

Use enumerate over zip inside a generator: nothing is stored, and the caller can stop early. Memory is O(1).

FOLLOW-UP CODE

```
def matching_positions(first, second):
    for position, (first_value, second_value) in enumerate(zip(first, second)):
        if first_value == second_value:
            yield position
```

BACK TO PART B - PART 1: QUESTION 106

107 Contains Duplicate — EASY

DIFFICULTY | EASY | 2 FOLLOW-UPS

FOLLOW-UP 1 | What if you cannot use extra memory?

Sort in place and compare neighbours: duplicates become adjacent. O(n log n) time and O(1) extra space (not counting the sort's own stack).

FOLLOW-UP CODE

```
def contains_duplicate_sorted(nums):
    ordered = sorted(nums)
    return any(earlier == later for earlier, later in zip(ordered, ordered[1:]))
```

FOLLOW-UP 2 | Only count it if the two equal values are at most k positions apart.

Keep a set of the last k values (a sliding window): add each new value, drop the one that left the window, and check membership before adding. O(n).

FOLLOW-UP CODE

```
def duplicate_within_k(nums, max_distance):
    window = set()
    for position, value in enumerate(nums):
        if value in window:
```

```
            return True
        window.add(value)
        if len(window) > max_distance:
            window.remove(nums[position - max_distance])
    return False
```

BACK TO PART B - PART 1: QUESTION 107

108 Find the Duplicates — EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | Three or more arrays.

Intersect pairwise, smallest arrays first so the running result shrinks quickly: O(total size). With K arrays you can also keep K pointers and advance the smallest.

FOLLOW-UP CODE

```
def duplicates_two_pointers(first, second):
    first_index = second_index = 0
    out = []
    while first_index < len(first) and second_index < len(second):
        if first[first_index] < second[second_index]:
            first_index += 1
        elif first[first_index] > second[second_index]:
            second_index += 1
        else:
            if not out or out[-1] != first[first_index]:
                out.append(first[first_index])
            first_index, second_index = first_index + 1, second_index + 1
    return out

def duplicates_in_all(arrays):
    out = arrays[0] if arrays else []
    for next_array in arrays[1:]:
        out = duplicates_two_pointers(out, next_array)
    return out
```

FOLLOW-UP 2 | The arrays are too big for memory.

Read both as streams (two pointers need only the current element of each), or partition both by value range into files that fit in memory and intersect partition by partition. Cost: O(N + M) time and O(1) memory with two streaming pointers; partitioning by value range multiplies the disk passes by the number of partitions.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Which one do you choose when N = 1,000 and M = 10^9?

Binary search: N log M is about 30,000 steps while two pointers need 10^9. The break-even is when N log M is around M.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 108

112 Smallest Missing Positive Integer in Linear Time and Constant Extra Space — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Do it without swapping (sign marking).

Replace out-of-range values by length + 1, then for every value v in 1..n make the entry at index value - 1 negative as a 'seen' flag; the first positive entry marks the missing number. Same O(length) time and O(1) space.

FOLLOW-UP CODE

```
def first_missing_sign_marking(nums):
    values, length = list(nums), len(nums)
    for position in range(length):
        if values[position] <= 0 or values[position] > length:
            values[position] = length + 1
    for position in range(length):
        value = abs(values[position])
        if value <= length:
```

```
        values[value - 1] = -abs(values[value - 1])
    for position in range(length):
        if values[position] > 0:
            return position + 1
    return length + 1
```

FOLLOW-UP 2 | The array must not be modified.

Then O(1) space with O(n) time is not possible in general; copy the array (O(n) space) or accept O(n log n) time with a sort of a copy. State the trade-off.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Smallest missing non-negative integer, or the k-th missing positive.

Shift the target range by one for non-negative. For the k-th missing positive in a sorted array use binary search on 'how many numbers are missing before index i' = a[i] - (i + 1).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 112

bisect on a sorted list: find a position, count a range, keep it sorted

113 Count People in a Postal Code Range — EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Also return the names of the people in the range.

Store the (code, name) pairs sorted. Find the two positions with bisect (using (low, '') and (high, a sentinel after every name)) and slice: the names come out ordered by code, then name. O(log n + answer size).

FOLLOW-UP CODE

```
class NamedPostalRegistry:
    def __init__(self, people):
        self.entries = sorted((code, name) for name, code in people)

    def names_in_range(self, low, high):
        first_index = bisect_left(self.entries, (low, ""))
        after_last_index = bisect_left(self.entries, (high + 1, ""))
        return [name for _, name in self.entries[first_index:after_last_index]]
```

BACK TO PART B - PART 1: QUESTION 113

119 Find Maximum Envelope Nesting — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the actual chain of envelopes.

Remember, for each element, the index of the tail it extended (parent pointers), and rebuild from the last tail of the longest length. The sorted order and binary search are unchanged.

FOLLOW-UP CODE

```
def nesting_chain(envelopes):
    sorted_envelopes = sorted(envelopes, key=lambda envelope: (envelope[0], -envelope[1]))
    tails, tail_indexes, parent = [], [], [-1] * len(sorted_envelopes)
    for index, (_, height) in enumerate(sorted_envelopes):
        position = bisect_left(tails, height)
        if position == len(tails):
            tails.append(height); tail_indexes.append(index)
        else:
            tails[position], tail_indexes[position] = height, index
        parent[index] = tail_indexes[position - 1] if position else -1
    chain, index = [], tail_indexes[-1] if tail_indexes else -1
    while index != -1:
        chain.append(sorted_envelopes[index])
        index = parent[index]
    return chain[::-1]
```

FOLLOW-UP 2 | Three dimensions (boxes).

Sorting plus LIS no longer reduces the problem; use an O(n^2) DP, or CDQ divide and conquer with a Fenwick tree for O(n log^2 n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Non-strict nesting (equal sizes allowed).

Sort by width ascending and height ascending, and use bisect_right (longest non-decreasing subsequence) instead of bisect_left.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 119

120 Count of Smaller Numbers After Self — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Do it with merge sort instead of a Fenwick tree.

Sort indices by value with merge sort; whenever you place an element from the left half, add the number of right-half elements already placed (they are smaller). Same O(length log length), and it avoids coordinate compression.

FOLLOW-UP CODE

```
def count_smaller_merge_sort(nums):
    length = len(nums)
    counts = [0] * length
    order = list(range(length))

    def sort(start, end):
        if end - start <= 1:
            return
        middle = (start + end) // 2
        sort(start, middle)
        sort(middle, end)
        merged, left_index, right_index, smaller_right = [], start, middle, 0
        while left_index < middle or right_index < end:
            if right_index >= end or (left_index < middle and nums[order[left_index]] <= nums[order[
right_index]]):
                counts[order[left_index]] += smaller_right          # right elements already placed are
smaller than this left element
                merged.append(order[left_index])
                left_index += 1
            else:
                smaller_right += 1
                merged.append(order[right_index])
                right_index += 1
        order[start:end] = merged

    sort(0, length)
    return counts
```

FOLLOW-UP 2 | Count larger numbers after self, or smaller numbers before self.

Larger after: query the suffix instead of the prefix of the Fenwick tree. Smaller before: scan left to right instead of right to left. Cost: still O(n log n) with the Fenwick tree.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Count inversions in an array.

The sum of the answers above is the number of inversions; merge sort counts them directly in O(n log n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 120

GROUP 15 OF 43 | 3 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Prefix sums, Tier 1: precompute once, answer ranges and balance points in O(1)

121 Cake Balance Line — EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Which cut makes the two sides as equal as possible?

Same scan, but track the smallest absolute difference |left - (total - left - height)| and the first index that reaches it. If a perfectly balanced cut exists it wins with difference 0.

FOLLOW-UP CODE

```
def closest_cut(heights):
    total, left = sum(heights), 0
    best, best_position = float("inf"), -1
    for position, height in enumerate(heights):
        imbalance = abs(left - (total - left - height))
        if imbalance < best:
            best, best_position = imbalance, position
        left += height
    return best_position
```

BACK TO PART B - PART 1: QUESTION 121

122 Find the Pivot Index (Balanced Split) in O(n) — EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Several pivots can exist: return all of them.

Do not return at the first match; collect every index where the left sum equals the right sum while the scan continues. Still O(n).

FOLLOW-UP CODE

```
def all_pivots(nums):
    total, left, pivots = sum(nums), 0, []
    for position, value in enumerate(nums):
        if left == total - left - value:
            pivots.append(position)
        left += value
    return pivots
```

BACK TO PART B - PART 1: QUESTION 122

123 Parity Alternating Subarray Query — EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | The array also changes (point updates).

Keep the bad-pair indicators in a Fenwick tree: an update changes at most two indicators, each an O(log n) update, and a query is two prefix sums.

FOLLOW-UP CODE

```
class FenwickParity:
    """Point updates: set nums[i] = v; the bad-pair prefix counts live in a Fenwick tree."""
    def __init__(self, nums):
        self.nums, self.length = list(nums), len(nums)
        self.tree = [0] * (self.length + 1)
        for position in range(self.length - 1):
            self._add(position, self._bad(position))

    def _bad(self, position):
        return int(self.nums[position] % 2 == self.nums[position + 1] % 2)
```

```
    def _add(self, position, change):
        position += 1
        while position <= self.length:
            self.tree[position] += change
            position += position & -position

    def _prefix(self, position):
        total = 0
        while position > 0:
            total += self.tree[position]
            position -= position & -position
        return total

    def update(self, position, value):
        for pair_index in (position - 1, position):
            if 0 <= pair_index < self.length - 1:
                self._add(pair_index, -self._bad(pair_index))
        self.nums[position] = value
        for pair_index in (position - 1, position):
            if 0 <= pair_index < self.length - 1:
                self._add(pair_index, self._bad(pair_index))

    def alternates(self, left, right):
        return self._prefix(right) - self._prefix(left) == 0
```

FOLLOW-UP 2 | Find the longest alternating subarray.

One pass with a running length that resets to 1 on a bad pair. O(n), O(1).

FOLLOW-UP CODE

```
def longest_alternating(nums):
    best = run = 1 if nums else 0
    for position in range(1, len(nums)):
        run = run + 1 if nums[position] % 2 != nums[position - 1] % 2 else 1
        best = max(best, run)
    return best
```

FOLLOW-UP 3 | Offline queries, different property (for example all elements distinct).

The prefix-sum trick applies whenever the property is 'no violating neighbour pair'. Properties about distinctness need Mo's algorithm or an offline sweep with last-occurrence tracking. Cost: Mo's algorithm is O((n + q) sqrt(n)); the prefix trick for 'no violating neighbour pair' stays O(n + q).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 123

GROUP 16 OF 43 | 3 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Sorted intervals: merge while they touch, then read the gaps or search

129 Check an Interval Against Existing Intervals for Conflicts — EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | Allow double booking but never triple booking (LeetCode 731).

Keep a difference map (+1 at start, -1 at end) and sweep it in sorted order; if the running count ever exceeds 2, undo the change and reject. O(n log n) per booking, or use a segment tree for faster.

FOLLOW-UP CODE

```
class CalendarTwo:
    """At most two overlapping bookings at any time (LeetCode 731)."""
    def __init__(self):
        self.delta = defaultdict(int)

    def book(self, start, end):
        self.delta[start] += 1
        self.delta[end] -= 1
        running = 0
        for time in sorted(self.delta):
            running += self.delta[time]
            if running > 2:
                self.delta[start] -= 1
                self.delta[end] += 1
                return False
        return True
```

FOLLOW-UP 2 | Report the maximum number of overlapping bookings after each booking (LeetCode 732).

The same difference map, taking the maximum running count in the sweep. A segment tree with lazy range increments gives O(log n) per booking.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Closed intervals instead of half-open.

Then touching endpoints conflict, so the tests become <= instead of <. Say it, and keep the choice in one comparison.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 129

135 Find safe travel intervals between planet influences — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Only the part between two given positions matters (a route from A to B).

Intersect each safe interval with [A, B] and keep the non-empty ones. Or clip the planets first; the merge-and-gap logic is the same.

FOLLOW-UP CODE

```
def safe_intervals(planets):
    merged = []
    for start, end in sorted((centre - radius, centre + radius) for centre, radius in planets):
        if merged and start <= merged[-1][1]:
            merged[-1][1] = max(merged[-1][1], end)
        else:
            merged.append([start, end])
    if not merged:
        return [(-INF, INF)]
    safe = [(-INF, merged[0][0])]
    for previous_interval, next_interval in zip(merged, merged[1:]):
        safe.append((previous_interval[1], next_interval[0]))
    safe.append((merged[-1][1], INF))
    return safe
```

```
def safe_within_range(planets, range_start, range_end):
    safe = []
    for start, end in safe_intervals(planets):
        clipped_start, clipped_end = max(start, range_start), min(end, range_end)
        if clipped_start < clipped_end:
            safe.append((clipped_start, clipped_end))
    return safe
```

FOLLOW-UP 2 | Are the endpoints safe?

The statement uses <=, so the endpoints of an unsafe interval are unsafe and the safe intervals are open. If the influence were strict (<), touching intervals would leave a single safe point between them and the merge test would use < instead of <=.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Planets arrive over time.

Keep the merged unsafe intervals in a sorted structure; an insert merges neighbours (like the stream-of-intervals problem) and the safe intervals are always the gaps.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 135

136 Implement a function that processes a stream of interval data. HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | How do you keep adds fast for a very large number of intervals?

Use an ordered map (TreeMap, or a balanced tree / skip list in Python via the sortedcontainers package) keyed by start so a merge removes the swallowed intervals in amortized O(log n) instead of shifting an array. Each interval is inserted once and removed at most once, so n adds cost O(n log n) overall.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Should touching intervals such as [1, 3] and [4, 5] merge?

Agree on the semantics first. For integer points, [1, 3] and [4, 5] cover every point from 1 to 5, so they merge; use bisect_left(self.ends, start - 1). For real-valued closed intervals only overlap or a shared endpoint merges, which is what the code above does. For half-open intervals [a, b), a shared endpoint merges but nothing else does.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Answer 'is this point covered?' and 'how much is covered in total?'

The sorted disjoint intervals already support both: a point is covered if the interval with the largest start at or before it ends at or after it (one bisect), and the total length is the sum of the interval lengths (or keep it updated on each add).

FOLLOW-UP CODE

```
class IntervalStream:
    def __init__(self):
        self.starts = []
        self.ends = []

    def add(self, start, end):
        first_overlap = bisect.bisect_left(self.ends, start)           # first interval ending at or after
start
        after_last_overlap = bisect.bisect_right(self.starts, end)          # first interval starting after
end
        if first_overlap < after_last_overlap:
            start = min(start, self.starts[first_overlap])
            end = max(end, self.ends[after_last_overlap - 1])
        self.starts[first_overlap:after_last_overlap] = [start]
        self.ends[first_overlap:after_last_overlap] = [end]

    def intervals(self):
        return list(zip(self.starts, self.ends))

class CoverageStream(IntervalStream):
    """Adds point queries and total covered length to the same structure."""
    def covers(self, point):
        import bisect
        position = bisect.bisect_right(self.starts, point) - 1
```

```
        return position >= 0 and self.ends[position] >= point

    def total_length(self):
        return sum(end - start for start, end in self.intervals())
```

BACK TO PART B - PART 1: QUESTION 136

Trie: nested dicts, one character per level

144 Implement Longest-Match Text Replacement HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | The priority is not longest match: insertion order, a score, or a custom comparator.

Collect every token that matches at the position (walk the trie and gather all token ends along the path, not just the last) and choose with the comparator. The trie gives the candidates; the policy is separate.

FOLLOW-UP CODE

```
def replace_with_priority(text, entries, key):
    """Follow-up: the winner among matching tokens is chosen by an arbitrary comparator key(token, id, order).
"""
    output, position = [], 0
    while position < len(text):
        matches = [(tok, ident, order) for order, (tok, ident) in enumerate(entries) if text.startswith(tok,
position)]
        if matches:
            token, token_id, _ = max(matches, key=lambda candidate: key(*candidate))
            output.append(token_id)
            position += len(token)
        else:
            output.append(text[position])
            position += 1
    return output
```

FOLLOW-UP 2 | The text is huge or the dictionary changes often.

Use Aho-Corasick if you need all matches in one pass over the text, or a trie that supports inserts and deletes; a longest-match scan is then linear in the text length times the longest token.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Overlapping matches or word boundaries.

Require a boundary before and after a token by checking the neighbouring characters when you record a candidate end, or tokenise on word boundaries first and look up whole words. Cost: still O(len(text) * longest token) with the trie walk; a boundary check is O(1) per candidate.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 144

A dict for lookup plus an ordered container for who is next (lazy deletion)

145 Implement Batched Undo/Redo Layer — EASY

DIFFICULTY | EASY | 4 FOLLOW-UPS

FOLLOW-UP 1 | Limit the history to the last N batches.

Keep the undo stack in a deque with maxlen N so the oldest batch is dropped automatically. Redo is unaffected.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Nested batches.

Keep a stack of open batches and merge a child into its parent on endBatch (keep the parent's old value if the key was already touched). Only the outermost endBatch commits an undoable action.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Undo of individual setProperty calls, not whole batches.

Treat every setProperty outside a batch as a one-key batch (the code does), so undo granularity equals commit granularity.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Limit the history to the last N batches.

Cap the undo stack: when it grows beyond N drop the oldest batch (use a deque with maxlen, or pop(0) as here). The redo stack is unaffected. Memory is then O(N batches).

FOLLOW-UP CODE

```
class Layer:
    def __init__(self):
        self.props, self.undo_s, self.redo_s, self.open = {}, [], [], None

    def beginBatch(self):
        self.open = {}

    def setProperty(self, k, v):
        if self.open is None:
            self.beginBatch()
            self.setProperty(k, v)
            self.endBatch()
            return
        old = self.open[k][0] if k in self.open else self.props.get(k, _MISSING)
        self.open[k] = (old, v)
        self.props[k] = v

    def endBatch(self):
        if self.open:
            self.undo_s.append(self.open)
            self.redo_s.clear()
        self.open = None

    def _apply(self, batch, use_new):
        for k, (old, new) in batch.items():
            value = new if use_new else old
            if value is _MISSING:
                self.props.pop(k, None)
            else:
                self.props[k] = value

    def undo(self):
        if self.undo_s:
            batch = self.undo_s.pop()
            self._apply(batch, False)
            self.redo_s.append(batch)

    def redo(self):
        if self.redo_s:
            batch = self.redo_s.pop()
            self._apply(batch, True)
            self.undo_s.append(batch)
```

```
class LayerBounded(Layer):
    def __init__(self, limit):
        super().__init__()
        self.limit = limit

    def endBatch(self):
        super().endBatch()
        if len(self.undo_s) > self.limit:
            self.undo_s.pop(0)
```

BACK TO PART B - PART 1: QUESTION 145

152 Implement LRU Cache. HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Many threads share the cache.

Wrap get and put in one lock (both reorder the list, so even get is a write). For more throughput shard the cache by key hash, each shard with its own lock and capacity.

FOLLOW-UP CODE

```
class LRUCacheOrdered:
    def __init__(self, capacity):
        self.max_size, self.entries = capacity, OrderedDict()

    def get(self, key):
        if key not in self.entries:
            return -1
        self.entries.move_to_end(key)
        return self.entries[key]

    def put(self, key, value):
        if key in self.entries:
            self.entries.move_to_end(key)
        self.entries[key] = value
        if len(self.entries) > self.max_size:
            self.entries.popitem(last=False)

class LRUCacheThreadSafe(LRUCacheOrdered):
    def __init__(self, capacity):
        super().__init__(capacity)
        self.lock = threading.Lock()

    def get(self, key):
        with self.lock:
            return super().get(key)

    def put(self, key, value):
        with self.lock:
            super().put(key, value)
```

FOLLOW-UP 2 | Entries expire after a time to live (TTL).

Store an expiry time with each entry, check it on get (a stale entry is treated as missing and removed), and optionally sweep from the old end of the list on put. The recency order and the O(1) operations are unchanged.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Least frequently used (LFU) instead of least recently used.

Keep a frequency per key and buckets of keys per frequency (each bucket an ordered dict) with a pointer to the minimum frequency; get and put move a key to the next bucket in O(1) (LeetCode 460).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 152

Rolling state: the next state comes only from the previous one

153 Climbing Stairs — EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | n is huge (10^18).

Use matrix exponentiation of [[1, 1], [1, 0]] (or fast doubling) for O(log stairs) multiplications; take the result modulo a prime if asked.

FOLLOW-UP CODE

```
def climb_matrix(stairs):
    def mul(left_matrix, right_matrix):
        return [[left_matrix[0][0] * right_matrix[0][0] + left_matrix[0][1] * right_matrix[1][0], left_matrix[
0][0] * right_matrix[0][1] + left_matrix[0][1] * right_matrix[1][1]],
                [left_matrix[1][0] * right_matrix[0][0] + left_matrix[1][1] * right_matrix[1][0], left_matrix[
1][0] * right_matrix[0][1] + left_matrix[1][1] * right_matrix[1][1]]]

    result, base, remaining_bits = [[1, 0], [0, 1]], [[1, 1], [1, 0]], stairs
    while remaining_bits:
        if remaining_bits & 1:
            result = mul(result, base)
        base = mul(base, base)
        remaining_bits >>= 1
    return result[0][0]
```

FOLLOW-UP 2 | You can take 1 to k steps (or a given set of step sizes).

ways[stair] = sum of ways[stair - s] over allowed step sizes s; for 1..k keep a sliding-window sum for O(stairs).

FOLLOW-UP CODE

```
def climb_k_steps(stairs, max_step):
    ways = [1] + [0] * stairs
    for stair in range(1, stairs + 1):
        ways[stair] = sum(ways[stair - step] for step in range(1, min(max_step, stair) + 1))
    return ways[stairs]
```

FOLLOW-UP 3 | Some steps are broken (cannot be stood on).

Set ways[i] = 0 for broken steps while filling the table; everything else is unchanged.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 153

154 Fibonacci Numbers — EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | n is up to 10^18 and the answer is needed modulo a prime.

Fast doubling (or matrix power) with every multiplication reduced modulo the prime: O(log position) steps with small numbers.

FOLLOW-UP CODE

```
def fib_mod(position, modulus):
    def go(index):
        if index == 0:
            return 0, 1
        fib_k, fib_k_plus_one = go(index // 2)
        fib_doubled = fib_k * (2 * fib_k_plus_one - fib_k) % modulus
        fib_doubled_plus_one = (fib_k * fib_k + fib_k_plus_one * fib_k_plus_one) % modulus
        return (fib_doubled_plus_one, (fib_doubled + fib_doubled_plus_one) % modulus) if index % 2 else (
fib_doubled, fib_doubled_plus_one)

    return go(position)[0]
```

FOLLOW-UP 2 | Why is the recursive version slow, and how do you fix it without changing the structure?

It recomputes the same values exponentially often (the call tree has about phi^n nodes). Memoisation (lru_cache) makes each value computed once, so O(position) time with O(position) memory; bottom-up with two variables drops the memory to O(1).

FOLLOW-UP CODE

```
def fib_memo(position):
    return position if position < 2 else fib_memo(position - 1) + fib_memo(position - 2)
```

FOLLOW-UP 3 | Closed form (Binet's formula).

F(n) = round(phi^n / sqrt(5)) is exact only while floating point has enough precision (up to about n = 70); beyond that it gives wrong digits, so use the integer methods.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 154

159 Constrained Dice Roll Sequences

HARD

DIFFICULTY | HARD | 1 FOLLOW-UPS

FOLLOW-UP 1 | Each turn rolls a pair of dice together, and the limit applies to each die independently. How many joint sequences are there?

The two dice never influence each other, so the joint sequences are the pairs of one valid sequence per die: the answer is the single-die count squared (mod 10^9 + 7). Tracking both run lengths in one DP gives the same number with a much bigger state.

FOLLOW-UP CODE

```
def count_two_dice(n, roll_max):
    one = count_dice_sequences(n, roll_max)
    return one * one % (10**9 + 7)
```

BACK TO PART B - PART 1: QUESTION 159

Two pointers: move i and j toward each other (or along two lists)

160 Reverse a linked list. EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | Do it recursively.

Reverse the rest first, then make the next node point back at this one and cut this node's own pointer. An inner helper that carries prev (as coded) keeps it tail-recursive in spirit; the call stack is O(n), so mention the depth limit.

FOLLOW-UP CODE

```
class ListNode:
    def __init__(self, val=0, next=None):
        self.val, self.next = val, next

def reverse_recursive(head):
    def go(node, reversed_so_far):
        if node is None:
            return reversed_so_far
        next_node = node.next
        node.next = reversed_so_far
        return go(next_node, node)

    return go(head, None)
```

FOLLOW-UP 2 | Reverse only positions left to right (LeetCode 92).

Walk to the node before the segment, reverse right - left + 1 nodes with the same loop, then reconnect: the old first node of the segment must point to what followed the segment, and the node before it points to the new first.

FOLLOW-UP CODE

```
class ListNode:
    def __init__(self, val=0, next=None):
        self.val, self.next = val, next

def reverse_between(head, left, right):
    dummy = ListNode(0, head)
    before = dummy
    for _ in range(left - 1):
        before = before.next
    reversed_part, current = None, before.next
    for _ in range(right - left + 1):
        next_node = current.next
        current.next = reversed_part
        reversed_part, current = current, next_node
    before.next.next = current                  # the old first node of the segment now ends it
    before.next = reversed_part
    return dummy.next
```

FOLLOW-UP 3 | Reverse in groups of k and leave a short last group alone (LeetCode 25).

Check that k nodes exist before reversing a group, reverse them with the same loop, reconnect both ends, and move the group pointer to the old first node of the group.

FOLLOW-UP CODE

```
class ListNode:
    def __init__(self, val=0, next=None):
        self.val, self.next = val, next

def reverse_k_group(head, k):
    dummy = ListNode(0, head)
    group_prev = dummy
    while True:
        kth = group_prev
        for _ in range(k):
            kth = kth.next
            if kth is None:
                return dummy.next
        group_next = kth.next
```

```
        reversed_part, current = group_next, group_prev.next
        while current is not group_next:
            next_node = current.next
            current.next = reversed_part
            reversed_part, current = current, next_node
        old_first = group_prev.next
        group_prev.next = kth
        group_prev = old_first
```

BACK TO PART B - PART 1: QUESTION 160

161 Sort a binary array in linear time. EASY

DIFFICULTY | EASY | 2 FOLLOW-UPS

FOLLOW-UP 1 | Why not just count zeros and fill?

Counting is also O(n) time and O(1) space and is fine for bare numbers. It needs two passes, and it cannot sort records that merely have a binary key, because it loses the objects. The swap version sorts the records themselves.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Count the zeros and rebuild (also O(n)).

Fine for bare numbers: one pass to count, one to fill. It cannot sort records that merely have a 0 or 1 key, because it loses the objects; the swapping version can.

FOLLOW-UP CODE

```
def sort_by_counting(bits):
    zeros = bits.count(0)
    return [0] * zeros + [1] * (len(bits) - zeros)
```

BACK TO PART B - PART 1: QUESTION 161

162 Is this a valid palindrome? EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | You may delete at most one character (LeetCode 680).

Move inward until the first mismatch, then the answer is whether skipping the left character or skipping the right character leaves a palindrome. That is two more linear checks, so O(n) overall.

FOLLOW-UP CODE

```
def valid_palindrome_one_deletion(text):
    def pal(left, right):
        while left < right:
            if text[left] != text[right]:
                return False
            left, right = left + 1, right - 1
        return True

    left, right = 0, len(text) - 1
    while left < right:
        if text[left] != text[right]:
            return pal(left + 1, right) or pal(left, right - 1)
        left, right = left + 1, right - 1
    return True
```

FOLLOW-UP 2 | Unicode, accents, or locale-specific case.

Normalise first (unicodedata.normalize and casefold) and decide what counts as a letter. str.isalnum handles many scripts, but combining marks need normalisation.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Longest palindromic substring.

Expand around each of the 2n - 1 centers in O(n^2), or use Manacher's algorithm in O(n). The two-pointer comparison is the building block.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 162

163 Reverse a Sentence

EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | The input may have leading, trailing or repeated spaces.

The split-based version already normalises them; the in-place version keeps them, so say which you want. To normalise in place add a two-pointer pass that writes single spaces.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Reverse the letters of each word but keep the word order (LeetCode 557).

Split on single spaces and reverse each word, or reverse each run in place without the first full reversal.

FOLLOW-UP CODE

```
def reverse_each_word(sentence):
    return " ".join(word[::-1] for word in sentence.split(" "))
```

FOLLOW-UP 3 | The sentence is a linked list of characters.

Reverse each word segment, then reverse the whole list, using the pointer-flip routine for linked lists.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 163

Walk two strings at once with two indices

167 Find the First Occurrence of a Substring — EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Make it O(n + m) in the worst case.

Knuth-Morris-Pratt: failure_table[position] is the length of the longest proper prefix of needle[:position + 1] that is also a suffix. While scanning the haystack, on a mismatch fall back to failure_table[matched - 1] instead of restarting, so the haystack pointer only moves forward.

FOLLOW-UP CODE

```
def str_str_kmp(haystack, needle):
    if not needle:
        return 0
    failure_table, matched = [0] * len(needle), 0
    for position in range(1, len(needle)):
        while matched and needle[position] != needle[matched]:
            matched = failure_table[matched - 1]
        if needle[position] == needle[matched]:
            matched += 1
        failure_table[position] = matched
    matched = 0
    for position, char in enumerate(haystack):
        while matched and char != needle[matched]:
            matched = failure_table[matched - 1]
        if char == needle[matched]:
            matched += 1
        if matched == len(needle):
            return position - matched + 1
    return -1
```

BACK TO PART B - PART 1: QUESTION 167

168 Determine if two words are sorted in lexicographic order. — EASY

DIFFICULTY | EASY | 4 FOLLOW-UPS

FOLLOW-UP 1 | What about uppercase, accents, or case-insensitive order?

Normalize first (casefold, unicode normalization) or give the rank map entries for the extra characters. Python's own string comparison orders by code point, which is not locale order, so use locale.strxfrm or a collation library when real language order matters.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | A whole list of words in an alien alphabet (LeetCode 953).

Check each adjacent pair; sortedness of a list is the AND of its adjacent pairs. Build the rank table once.

FOLLOW-UP CODE

```
def is_sorted_list(words, order):
    rank = {letter: position for position, letter in enumerate(order)}
    return all([rank[letter] for letter in earlier] <= [rank[letter] for letter in later] for earlier, later
in zip(words, words[1:]))
```

FOLLOW-UP 3 | Case-insensitive or accented comparison.

Normalise (casefold, unicode normalisation) before building ranks, or use locale-aware collation keys; plain code-point order is not language order.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Prefix rule.

'app' comes before 'apple' but 'apple' does not come before 'app': compare lengths only after all shared positions are equal.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 168

169 Return all unique substrings of a given string. EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | List them without building duplicates first.

Insert every suffix into a trie; a substring is new exactly when its trie node is created. O(n^2) time and nodes, and the substrings appear without duplicates.

FOLLOW-UP CODE

```
def unique_substrings_trie(text):
    trie, substrings = {}, []
    for start in range(len(text)):
        node, letters = trie, []
        for char in text[start:]:
            letters.append(char)
            if char not in node:
                node[char] = {}
                substrings.append("".join(letters))
            node = node[char]
    return substrings
```

FOLLOW-UP 2 | Return the k-th lexicographic distinct substring, or the longest repeated substring.

Both use the suffix array with LCP: the longest repeated substring is the maximum LCP value, and the k-th substring is found by walking the sorted suffixes and subtracting the new prefixes each one contributes.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The string has a million characters.

Quadratic is out. Build a suffix automaton (O(n)) and count distinct substrings as the sum of len(v) - len(link(v)) over states, or use a suffix array with O(n log n) construction.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 169

173 Find Dictionary Matches for a Jammed String HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | A different stretch rule: any run may be longer, no minimum.

Only the comparison changes. The index (grouping by run letters) is the same, so make the minimum stretch a parameter, as the code does with min_stretch.

FOLLOW-UP CODE

```
class JammedIndex:
    def __init__(self, words):
        self.groups = {}
        for word in words:
            runs = _runs(word)
            self.groups.setdefault("".join(letter for letter, _ in runs), []).append((word, [run_length for _,
 run_length in runs]))

    def matches(self, query, min_stretch=3):
        runs = _runs(query)
        key = "".join(letter for letter, _ in runs)
        query_run_lengths = [run_length for _, run_length in runs]
        return [word for word, word_run_lengths in self.groups.get(key, [])
                if all(query_run_length == run_length or (query_run_length > run_length and query_run_length
>= min_stretch) for query_run_length, run_length in zip(query_run_lengths, word_run_lengths))]

def _runs(text):
    runs_list = []
    for letter in text:
        if runs_list and runs_list[-1][0] == letter:
```

```
            runs_list[-1][1] += 1
        else:
            runs_list.append([letter, 1])
    return runs_list

def jammed_any_stretch(words, query):
    return JammedIndex(words).matches(query, min_stretch=2)
```

FOLLOW-UP 2 | Millions of dictionary words and many queries.

Keep the group index in memory and put a trie over the run-letter strings if the keys share long prefixes. For counts compare only the first words in a bucket that survive a cheap check such as total length. Cost: O(L) per query to build the run key plus the size of the bucket scanned; building the index is O(total dictionary length) once.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The jammed string may also be missing letters.

That becomes an edit-distance style problem on run encodings, usually solved with DP over (query run, word run). Ask for the exact rule before coding it.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 173

GROUP 22 OF 43 | 2 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Simulate a machine: a class with state and one branch per kind of event

174 Mario-Style Score Tracker (OOP) EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Add undo() to take back the last event.

Before applying an event push a snapshot (score, lives, big) on a stack; undo pops it and restores the three fields. Events ignored after game over should not push a snapshot.

FOLLOW-UP CODE

```
class UndoMarioGame(MarioGame):
    def __init__(self):
        super().__init__()
        self.history = []

    def process(self, event):
        if self.lives == 0:
            return
        self.history.append((self.score, self.lives, self.big))
        super().process(event)

    def undo(self):
        if self.history:
            self.score, self.lives, self.big = self.history.pop()
```

BACK TO PART B - PART 1: QUESTION 174

180 Simulate In-Place Cellular State Updates HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | The board is infinite, or huge and mostly empty.

Store only the live cells in a set. A cell's next state depends on its live neighbor count, so count how many live neighbors every candidate cell has with a Counter and keep cells with 3, or with 2 if already live. Work is proportional to the number of live cells.

FOLLOW-UP CODE

```
def life_infinite(live_cells):
    """One generation on an unbounded board, storing only the live cells."""
    from collections import Counter
    counts = Counter((row + row_step, col + col_step) for row, col in live_cells for row_step in (-1, 0, 1)
for col_step in (-1, 0, 1) if (row_step, col_step) != (0, 0))
    return {cell for cell, neighbour_count in counts.items() if neighbour_count == 3 or (neighbour_count == 2
and cell in live_cells)}
```

FOLLOW-UP 2 | Why is bit 0 used when counting neighbors?

Neighbors that were already processed in this pass carry their next state in bit 1. Masking with & 1 always reads the original generation, which is the whole point of the encoding.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Cells wrap around the edges (a torus).

Replace the bounds check with modular indexing: (r + dr) % m and (c + dc) % n. Be careful on very small boards where the same neighbor can be reached through two offsets.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 180

Digits and characters: peel one at a time and carry

187 Caesar Cipher: Recover Key, Encrypt, and Decrypt — EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | You only have a ciphertext. How do you break it?

Try all 26 shifts and score each candidate by how English it looks: add a weight for every common letter (e, t, a, o, i, n, s, h, r, d, l, u) and keep the shift with the best score. Longer texts make the frequency signal reliable.

FOLLOW-UP CODE

```
def break_caesar(ciphertext):
    common = "etaoinshrdlu"
    weight = {letter: 12 - i for i, letter in enumerate(common)}

    def score(shift):
        text = decrypt(ciphertext, shift)
        return sum(weight.get(letter, 0) for letter in text)

    return max(range(26), key=score)
```

BACK TO PART B - PART 1: QUESTION 187

191 Check Digits and Combine Ranges — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Adjacent integer ranges ([1, 2] and [3, 4]) should merge too.

Change the overlap test to s <= out[-1][1] + 1 (integers only). For real-valued ranges that is wrong, so say which you assume.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Summarise a sorted list of integers as ranges (LeetCode 228).

One pass keeping the current range and extending it while the next number is exactly one more; the same 'extend or start new' structure as the merge.

FOLLOW-UP CODE

```
def summary_ranges(nums):
    out = []
    for number in nums:
        if out and number == out[-1][1] + 1:
            out[-1][1] = number
        else:
            out.append([number, number])
    return [f"{start}->{end}" if start != end else str(start) for start, end in out]
```

FOLLOW-UP 3 | Palindrome in another base, or as a linked list.

Another base replaces 10 by the base in % and //. For a linked list find the middle with slow and fast pointers, reverse the second half in place and compare.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 191

192 Permutation Sequence HARD

DIFFICULTY | HARD | 1 FOLLOW-UPS

FOLLOW-UP 1 | Given a permutation, what is its position k in the list?

Run the idea backwards: for every position count how many unused digits are smaller than the current one and multiply by the factorial of the digits that remain; add one at the end.

FOLLOW-UP CODE

```
def permutation_rank(perm):
    digits = sorted(int(c) for c in perm)
    rank = 0
    for position, digit_char in enumerate(perm):
        smaller = digits.index(int(digit_char))
        rank += smaller * factorial(len(perm) - position - 1)
        digits.pop(smaller)
    return rank + 1
```

BACK TO PART B - PART 1: QUESTION 192

Look at the input once and keep only a small summary (and count your calls)

197 Find All Bad Pairs Using runTest Oracle — HARD

DIFFICULTY | HARD | 1 FOLLOW-UPS

FOLLOW-UP 1 | How many calls are needed in the best case and the worst case?

With no bad pairs it takes N - 1 calls, one per test after the first. With K bad pairs each one costs about log N calls to locate, so roughly K log N plus the cost of building clean groups. The brute force is always N(N-1)/2, which is far larger as soon as bad pairs are rare.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 197

198 Find maximum value in huge text file — HARD

DIFFICULTY | HARD | 5 FOLLOW-UPS

FOLLOW-UP 1 | Edge cases to mention.

Empty file (return None or raise, by contract); a final line without a newline; blank lines; malformed lines; values at the 64-bit extremes; an all-negative file (do not initialize the max to 0); and Windows line endings. Decide whether a bad line fails the job or is skipped and counted.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | How do you scale to many machines and handle failures?

Assign byte ranges (map tasks) to workers through a coordinator, and make each task idempotent (it only returns its maximum), so a failed or slow worker is simply retried or speculatively re-executed elsewhere. The reduce step is a max over a tiny list. On a distributed filesystem, align ranges to its block size for locality. Cost: O(n / workers) wall-clock time plus O(workers) to combine; total work stays O(n).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Is it I/O-bound or CPU-bound, and why does that matter?

Parsing integers is cheap compared with disk throughput, so one sequential reader often saturates the disk and extra threads help only if the data sits on several disks or on fast SSD or network storage. Measure first; the parallel version pays off mainly across machines or storage devices. Cost: reading takes about n / disk throughput and parsing is O(n) as well, so extra threads help only beyond the disk's speed.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | The file has billions of lines and one machine is too slow.

Split by byte offsets across machines (each skips to the next newline), compute local maxima, and take the maximum of those few numbers. Tasks are idempotent, so a failed worker can simply be retried. Cost: O(n / machines) per machine and O(machines) to combine; a retry re-runs only the failed byte range.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 5 | Now return the K largest values instead of only the maximum.

Keep a min-heap of size K while streaming: push a value if the heap is not full, or replace the root when the value is larger. Memory O(K), time O(n log K); merge the per-worker heaps for a parallel version.

FOLLOW-UP CODE

```
def top_k_in_file(path, how_many):
    heap = []
    with open(path) as file:
        for line in file:
```

```
            if line.strip():
                value = int(line)
                if len(heap) < how_many:
                    heapq.heappush(heap, value)
                elif value > heap[0]:
                    heapq.heapreplace(heap, value)
    return sorted(heap, reverse=True)
```

BACK TO PART B - PART 1: QUESTION 198

Grid DP: each cell from its neighbours above and to the left

202 Find the Longest Straight Run of One Color — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return where the run is, not only its length.

Only start a walk at cells that begin a run (the previous cell in that direction has a different color); that keeps the total work linear, and you record the start cell and direction of the best run. Equivalent: keep the DP table and read the end cell of the maximum, then step back by its length.

FOLLOW-UP CODE

```
def longest_run_cells(grid):
    best = (0, None, None)
    for row in range(len(grid)):
        for col in range(len(grid[0])):
            for row_step, col_step in DIRS4:
                if 0 <= row - row_step < len(grid) and 0 <= col - col_step < len(grid[0]) and grid[row -
row_step][col - col_step] == grid[row][col]:
                    continue                                    # not the start of a run in this direction
                run_length = 0
                while 0 <= row + row_step * run_length < len(grid) and 0 <= col + col_step * run_length < len(
grid[0]) and grid[row + row_step * run_length][col + col_step * run_length] == grid[row][col]:
                    run_length += 1
                if run_length > best[0]:
                    best = (run_length, (row, col), (row_step, col_step))
    return best
```

FOLLOW-UP 2 | Require the run to be at least K and count how many such runs exist.

Count a run once, at its start, when its length is at least K (use the begins-a-run test above). Counting at every cell would count a long run many times.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The grid is huge and arrives row by row.

The rolling-row version already needs only the previous row, so it works on a stream of rows with O(n) memory.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 202

203 Count same-color squares in a character grid — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Only the largest square (LeetCode 221).

Keep the maximum of the same dp table instead of the sum.

FOLLOW-UP CODE

```
def largest_same_color_square(grid):
    rows, cols, best = len(grid), len(grid[0]), 1
    dp = [[1] * cols for _ in range(rows)]
    for row in range(1, rows):
        for col in range(1, cols):
            if grid[row][col] == grid[row - 1][col] == grid[row][col - 1] == grid[row - 1][col - 1]:
                dp[row][col] = 1 + min(dp[row - 1][col], dp[row][col - 1], dp[row - 1][col - 1])
                best = max(best, dp[row][col])
    return best
```

FOLLOW-UP 2 | Allow at most one different cell inside a square.

The simple min-of-three recurrence no longer works. Use 2D prefix counts per colour and test each (corner, size) with binary search on the size, or DP with a second layer 'one defect used'.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Count only squares of colour c, or only side >= 2.

Filter while summing: add dp only for cells of colour c, or add max(0, dp - 1) for sizes at least 2.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 203

GROUP 30 OF 43 | 1 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Counting pairs by looking up each partner in a Counter or a bucket

223 Maximum-Area Rectangle from Planar Points HARD

DIFFICULTY | HARD | 1 FOLLOW-UPS

FOLLOW-UP 1 | Can you do better than O(n^2) pairs?

Not in general: the answer can depend on all pairs, and even deciding whether any rectangle exists is believed to need about n^2 work. The practical gain is in the buckets: most buckets hold one pair, so the work after bucketing is small. State the worst case honestly (points on a circle put many pairs in few diagonals_by_key).

FOLLOW-UP CODE

```
def max_rect_area(points):
    unique_points = list(set(map(tuple, points)))
    diagonals_by_key = defaultdict(list)
    for first_index in range(len(unique_points)):
        for second_index in range(first_index + 1, len(unique_points)):
            p, q = unique_points[first_index], unique_points[second_index]
            diagonals_by_key[(p[0] + q[0], p[1] + q[1], (p[0] - q[0]) ** 2 + (p[1] - q[1]) ** 2)].append((p,
q))
    best = 0
    for pairs in diagonals_by_key.values():
        for (p, q), (r, s) in itertools.combinations(pairs, 2):
            best = max(best, abs((r[0] - p[0]) * (s[1] - p[1]) - (r[1] - p[1]) * (s[0] - p[0])))
    return best
```

BACK TO PART B - PART 1: QUESTION 223

Tree DP: build a parent order once, then add up from the leaves

227 Compute distance-sum from every tree node — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Find the node with the smallest distance sum (the tree's centroid).

The answer array above gives it directly: argmin of ans. Equivalent: the centroid, found by walking toward the heavier subtree.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | The tree has weighted edges.

Re-rooting still works, but the change when moving the root across an edge of weight w is w * (n - 2 * size[c]) instead of n - 2 * size[c]. Sizes and the first pass are the same.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Maximum distance (eccentricity) from each node instead of the sum.

The farthest node from any node is one end of a diameter; compute distances from both diameter ends and take the maximum per node. O(node_count).

FOLLOW-UP CODE

```
def tree_eccentricities(node_count, edges):
    """Maximum distance from every node: the farthest node from any node is one end of a diameter."""
    adj = defaultdict(list)
    for node, neighbour in edges:
        adj[node].append(neighbour); adj[neighbour].append(node)

    def bfs(start):
        distance, queue = {start: 0}, deque([start])
        last = start
        while queue:
            node = queue.popleft()
            last = node
            for neighbour in adj[node]:
                if neighbour not in distance:
                    distance[neighbour] = distance[node] + 1
                    queue.append(neighbour)
        return last, distance

    first_end, _ = bfs(0)
    second_end, distance_from_first = bfs(first_end)
    _, distance_from_second = bfs(second_end)
    return [max(distance_from_first[node], distance_from_second[node]) for node in range(node_count)]
```

BACK TO PART B - PART 1: QUESTION 227

228 Tree Distance Sum (Re-root DP) — HARD

DIFFICULTY | HARD | 1 FOLLOW-UPS

FOLLOW-UP 1 | Why is the re-root formula correct?

Compare the distance from every node w to p and to its child v. If w is inside v's subtree the path to p goes through v, so dist(w, p) = dist(w, neighbour) + 1. If w is outside it, dist(w, neighbour) = dist(w, p) + 1. There are size[neighbour] nodes of the first kind and node_count - size[neighbour] of the second, so distance_sum[neighbour] = distance_sum[p] - size[neighbour] + (node_count - size[neighbour]).

FOLLOW-UP CODE

```
def tree_distance_sums(node_count, edges):
    if node_count == 0:
        return []
    adj = [[] for _ in range(node_count)]
    for node_a, node_b in edges:
        adj[node_a].append(node_b); adj[node_b].append(node_a)
```

```
    parent, order = [-1] * node_count, [0]
    for node in order:
        for neighbour in adj[node]:
            if neighbour != parent[node]:
                parent[neighbour] = node; order.append(neighbour)
    size, distance_inside = [1] * node_count, [0] * node_count
    for node in reversed(order):
        if parent[node] >= 0:
            size[parent[node]] += size[node]
            distance_inside[parent[node]] += distance_inside[node] + size[node]
    distance_sum = [0] * node_count
    distance_sum[0] = distance_inside[0]
    for node in order[1:]:
        distance_sum[node] = distance_sum[parent[node]] + node_count - 2 * size[node]
    return distance_sum
```

BACK TO PART B - PART 1: QUESTION 228

GROUP 33 OF 43 | 1 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Prefix sums, Tier 2: running total plus a dict of earlier totals

238 **Given an integer array nums and an integer k, return true if nums has a subarray of at least two elements whose sum is a multiple of k.** **HARD**

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Count the subarrays whose sum is a multiple of k (any length, LeetCode 974).

Count how many times each remainder has occurred and add that count each time the remainder repeats (Python's % is already non-negative for negative sums). O(n).

FOLLOW-UP CODE

```
def count_good_subarrays(nums, divisor):
    remainder_counts, prefix_sum, count = Counter({0: 1}), 0, 0
    for number in nums:
        prefix_sum += number
        remainder = prefix_sum % divisor
        count += remainder_counts[remainder]
        remainder_counts[remainder] += 1
    return count
```

FOLLOW-UP 2 | Why store only the first index of each remainder?

A later occurrence of the same remainder would make the subarray shorter, never longer, so the first index is the best partner for the length-at-least-two condition.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The values can be negative or k huge.

Python's modulo gives a non-negative remainder; in languages where % can be negative normalise with ((s % k) + k) % k. A huge k just means the dictionary holds up to n entries. Cost: O(n) time with up to min(n, k) dictionary entries.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 238

Group by a canonical key

239 Given an array of strings, return the words that can be typed using letters of the alphabet on only one row of American keyboard. EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | What if the keyboard layout is a parameter (Dvorak, AZERTY, a phone keypad)?

Nothing changes except the row table, which is built from the layout argument. Build the letter-to-row map once and reuse it across queries.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Other layouts (Dvorak, AZERTY) or a phone keypad.

Pass the rows as a parameter and build the letter-to-row dict from it. Nothing else changes.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | A different keyboard layout.

Pass the rows as a parameter and build the letter-to-row map from it (a phone keypad or Dvorak works the same way). Letters that are on no row make the word invalid, which the code checks.

FOLLOW-UP CODE

```
def find_words_layout(words, rows):
    row_of = {letter: row_number for row_number, row in enumerate(rows) for letter in row}
    return [word for word in words if len({row_of.get(letter) for letter in word.lower()}) <= 1 and all(
letter in row_of for letter in word.lower())]
```

BACK TO PART B - PART 1: QUESTION 239

243 Given two strings and two allowed operations, swapping positions of letters and converting all occurrences of one character to another, can the first string be transformed into the second? HARD

DIFFICULTY | HARD | 5 FOLLOW-UPS

FOLLOW-UP 1 | What if the two operations are combined (one-directional conversion plus swaps)?

Swaps make positions irrelevant, so only letter counts matter. Conversions can only merge letter classes, never split one. So t is reachable only if the counts of t can be formed by grouping the counts of s into the t letters (each s letter's count goes wholly to one t letter) and, unless the relabeling is an identity or chain, there is a spare letter. With at most 26 classes the grouping is a small bin-packing search.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | Why does a full 26-letter target make the one-directional version impossible?

Every conversion rewrites a letter into another letter, and if a mapping contains a cycle (a to b and b to a) you need a temporary letter not currently in use to break it. If t uses all 26 letters then s must use all 26 as well (the mapping is a function onto 26 distinct letters), so the mapping is a permutation and any non-identity permutation has a cycle. The only possible case is therefore s equal to t.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Only position swaps are allowed.

Then first can become second exactly when they are anagrams: compare the letter counts. O(n).

FOLLOW-UP CODE

```
def only_swaps_possible(first, second):
    return Counter(first) == Counter(second)
```

FOLLOW-UP 4 | Conversion is one-directional (every a becomes b) and there are no position swaps (LeetCode 1153).

Each source letter must map to one target letter, and if the strings differ and the target already uses all 26 letters there is no spare letter to break mapping cycles, so it is impossible. Verified against brute force on a small alphabet.

FOLLOW-UP CODE

```
def can_convert(source, target, alphabet_size=26):
    if source == target:
        return True
    if len(source) != len(target):
        return False
    mapping = {}
    for source_letter, target_letter in zip(source, target):
        if mapping.setdefault(source_letter, target_letter) != target_letter:
            return False
    return len(set(target)) < alphabet_size
```

FOLLOW-UP 5 | Return the sequence of operations, or the minimum number.

Minimum operations for the swap-only case is n minus the number of cycles in the permutation that sorts s into t; for letter-identity swaps it is the number of cycles in the letter mapping minus one per cycle of length greater than one.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 243

K-way merge: a heap of the next item from each source

244 Find k pairs with smallest sums — EASY

DIFFICULTY | EASY | 3 FOLLOW-UPS

FOLLOW-UP 1 | The arrays are not sorted.

Sort both first (O(m log m + n log n)); the heap method needs sorted rows. For one huge array you may instead use a heap of size k with a selection of candidates.

FOLLOW-UP CODE

```
def kSmallestPairs(first, second, pair_count):
    if not first or not second or pair_count <= 0:
        return []
    heap = [(first[first_index] + second[0], first_index, 0) for first_index in range(min(pair_count, len(
first)))]
    heapq.heapify(heap)
    pairs = []
    while heap and len(pairs) < pair_count:
        _, first_index, second_index = heapq.heappop(heap)
        pairs.append((first[first_index], second[second_index]))
        if second_index + 1 < len(second):
            heapq.heappush(heap, (first[first_index] + second[second_index + 1], first_index, second_index +
1))
    return pairs

def k_smallest_unsorted(first, second, pair_count):
    return kSmallestPairs(sorted(first), sorted(second), pair_count)
```

FOLLOW-UP 2 | Only the k-th smallest sum, with huge arrays.

Binary search the sum value and count how many pairs have sum at most it with two pointers (O(m + n) per check): O((m + n) log(range)) with O(1) space, no heap.

FOLLOW-UP CODE

```
def kSmallestPairs(first, second, pair_count):
    if not first or not second or pair_count <= 0:
        return []
    heap = [(first[first_index] + second[0], first_index, 0) for first_index in range(min(pair_count, len(
first)))]
    heapq.heapify(heap)
    pairs = []
    while heap and len(pairs) < pair_count:
        _, first_index, second_index = heapq.heappop(heap)
        pairs.append((first[first_index], second[second_index]))
        if second_index + 1 < len(second):
            heapq.heappush(heap, (first[first_index] + second[second_index + 1], first_index, second_index +
1))
    return pairs

def kth_smallest_pair_sum(first, second, pair_count):
    pairs = kSmallestPairs(first, second, pair_count)
    return None if len(pairs) < pair_count else sum(pairs[-1])
```

FOLLOW-UP 3 | Three arrays, or k pairs with the largest sums.

For largest sums run the same algorithm on the reversed arrays (or negate the values). For three arrays use a heap of triples (i, j, l), pushing the three neighbours with a visited set. Cost: O(k log k) with the heap of triples; reversing the arrays gives the largest sums at the same cost.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 244

247 Return top-k subset sums in descending order — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return the subsets themselves, not only the sums.

Store each heap state as a node with (last index, parent node) instead of copying lists, and rebuild a subset by walking parents only for the k answers. That keeps pushes O(1).

FOLLOW-UP CODE

```
def top_k_subsets(nums, how_many):
    """Non-negative inputs: also return the subsets, using parent pointers instead of copying lists."""
    sorted_values = sorted(nums)
    picks = [(None, None)]                          # node id -> (index added, parent id)
    heap = [(sorted_values[0], 0, 1)] if sorted_values else []
    picks.append((0, 0))
    out = [(0, [])]
    while heap and len(out) < how_many:
        subset_sum, last_index, pick_id = heapq.heappop(heap)
        chosen, cur = [], pick_id
        while cur:
            added_index, cur = picks[cur]
            chosen.append(sorted_values[added_index])
        out.append((subset_sum, chosen[::-1]))
        if last_index + 1 < len(sorted_values):
            picks.append((last_index + 1, pick_id))
            heapq.heappush(heap, (subset_sum + sorted_values[last_index + 1], last_index + 1, len(picks) - 1))
            picks.append((last_index + 1, picks[pick_id][1]))
            heapq.heappush(heap, (subset_sum - sorted_values[last_index] + sorted_values[last_index + 1],
last_index + 1, len(picks) - 1))
    return out[:how_many]
```

FOLLOW-UP 2 | k can be as large as 2^n.

Then the output itself is that large; the heap approach is still O(k log k) and stops when the heap empties. Check k against 2^n first and fall back to enumeration for tiny n.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Limit subsets to at most m elements.

Add the element count to each heap state and refuse to push successors that exceed m. The ordering argument is unchanged because both successors never make the sum smaller.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 247

248 Merge k sorted linked lists. — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Merge the lists in pairs instead of using a heap.

Repeatedly merge neighbouring pairs with the two-list merge; there are log k rounds, each touching all N nodes: O(N log k) time and O(1) extra space (no heap).

FOLLOW-UP CODE

```
class ListNode:
    def __init__(self, val=0, next=None):
        self.val, self.next = val, next

def merge_two(first, second):
    dummy = tail = ListNode()
    while first and second:
        if first.val <= second.val:
            tail.next, first = first, first.next
        else:
            tail.next, second = second, second.next
        tail = tail.next
    tail.next = first or second
    return dummy.next

def merge_k_divide_and_conquer(lists):
```

```
    lists = [list_head for list_head in lists if list_head]
    while len(lists) > 1:
        lists = [merge_two(lists[position], lists[position + 1]) if position + 1 < len(lists) else lists[
position] for position in range(0, len(lists), 2)]
    return lists[0] if lists else None
```

FOLLOW-UP 2 | The lists are too large for memory (files or streams).

The heap approach already works on streams: it holds one item per list. Read each input through an iterator and write the output as you go; this is the merge phase of an external sort (heapq.merge does it). Cost: O(N log k) time and O(k) memory; one sequential read per list.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Why not merge the lists one by one into a running result?

Merging list i into a result of size about i * (N / k) costs O(i * N / k), so the total is O(k * N), far worse than O(N log k).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 248

GROUP 36 OF 43 | 1 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Tree recursion, Tier 2: pass a rule down, return a node or a yes/no up

249 Next Birthday (BST In-Order Successor with Wraparound) EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | The tree is not a BST. What now?

Collect every value, sort once, then binary-search with bisect_right for the first value above today and wrap to index 0. O(n log n) once, O(log n) per query.

FOLLOW-UP CODE

```
from bisect import bisect_right


def next_birthday_unsorted(root, today):
    dates = []

    def dfs(node):
        if node is None:
            return
        dates.append(node.val)
        dfs(node.left)
        dfs(node.right)

    dfs(root)
    if not dates:
        return None
    dates.sort()
    position = bisect_right(dates, today)
    return dates[position % len(dates)]
```

BACK TO PART B - PART 1: QUESTION 249

GROUP 37 OF 43 | 1 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Random pick without repeats: swap the chosen item with the last one and pop

258 Shuffle array using random integer API HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Return only k random elements.

Run Fisher-Yates for k steps from the front (swap position i with a random index in i..n-1) and return the first k. O(count) after copying, still uniform.

FOLLOW-UP CODE

```
def sample(nums, count):
    """k uniformly random distinct elements: stop Fisher-Yates after k steps."""
    shuffled = list(nums)
    for position in range(count):
        pick = rand_int(position, len(shuffled) - 1)
        shuffled[position], shuffled[pick] = shuffled[pick], shuffled[position]
    return shuffled[:count]
```

FOLLOW-UP 2 | Only a coin-flip API exists, no rand_int.

Build a uniform integer in [0, m) by rejection sampling: draw ceil(log2 m) random bits and retry if the number is at least m. That keeps every value equally likely, unlike taking a remainder.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Why is the biased version biased?

It makes n choices of n options, n^n equally likely paths, and n! does not divide n^n for n >= 3, so some permutations must appear more often than others.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 258

GROUP 38 OF 43 | 2 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Build and serialize: write a structure out, read it back with the same walk

262 Serialize and Deserialize a Graph HARD

DIFFICULTY | HARD | 1 FOLLOW-UPS

FOLLOW-UP 1 | Support directed graphs, where the adjacency is not symmetric.

The format already stores each node's own neighbour list, so a directed edge u -> v is kept as given and nothing assumes the reverse edge exists. Only nodes reachable from the starting node are encoded, which is the same as for the undirected case.

FOLLOW-UP CODE

```
def build_graph(neighbour_lists):
    nodes = [Node(index + 1) for index in range(len(neighbour_lists))]
    for node, neighbour_numbers in zip(nodes, neighbour_lists):
        node.neighbors = [nodes[neighbour_number - 1] for neighbour_number in neighbour_numbers]
    return nodes[0] if nodes else None
```

BACK TO PART B - PART 1: QUESTION 262

263 Implement a decode and encode function of a general N-ary tree. HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | The values are arbitrary strings that may contain spaces, commas or brackets.

Do not invent delimiters. Put the values in a real container: a JSON list of alternating value and child count, which handles quoting for you, or length-prefix every value (5#hello). The structure part (the counts) stays the same.

FOLLOW-UP CODE

```
import json


def encode_any(root):
    def pack(node):
        return [node.val, [pack(child) for child in node.children]]
    return json.dumps(None if root is None else pack(root))


def decode_any(text):
    def unpack(data):
        if data is None:
            return None
        return Node(data[0], [unpack(child) for child in data[1]])
    return unpack(json.loads(text))
```

FOLLOW-UP 2 | The tree is huge and you want a compact binary encoding.

Counts and node values can be packed with variable-length integers (varint), which makes small values one byte. Keep the same pre-order count layout so decoding is still one pass. Add a version byte at the front so the format can change later. Cost: still O(n) time and space; variable-length integers cost about one byte per small value.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Encode in level order instead.

Write each node's value and child count in BFS order and decode with a queue: pop a parent, read its count, attach that many new nodes, and push them. BFS order is handy when the consumer wants to stream levels.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 263

GROUP 39 OF 43 | 4 QUESTIONS | PART B - PART 2 FOLLOW-UPS

Games: win or lose with a cache, and guessing with feedback

265 Determine winner in stack-merging game — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | The game is a sum of independent games (several separate boards or piles).

Compute each part's Grundy number (the mex of the Grundy numbers of its moves) and XOR them. The first player wins exactly when the XOR is not zero. This turns an exponential search over the product of states into a sum of small searches.

FOLLOW-UP CODE

```
def grundy(state, moves, memo=None):
    memo = {} if memo is None else memo
    if state in memo:
        return memo[state]
    reachable_values = {grundy(next_state, moves, memo) for next_state in moves(state)}
    mex = 0
    while mex in reachable_values:
        mex += 1
    memo[state] = mex
    return mex
```

FOLLOW-UP 2 | Misere play (the player who cannot move wins).

Flip the base case in the memoized search: no moves means the player to move wins. Grundy numbers no longer compose by XOR in general, so search the whole state or use the known special-case theory.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Count the number of winning first moves.

Keep the same memoized win/lose function and count the moves whose resulting state is a loss for the opponent.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 265

266 Find secret word with match-count feedback — HARD

DIFFICULTY | HARD | 3 FOLLOW-UPS

FOLLOW-UP 1 | Show how the candidate list shrinks.

After a guess with answer m, only words with the same match count against the guess survive (the secret and the candidate agree on every comparison). Printing the sizes round by round shows why a balanced guess finishes in a few rounds.

FOLLOW-UP CODE

```
def candidates_after(words, guess, count):
    return [word for word in words if _match(guess, word) == count]
```

FOLLOW-UP 2 | Guess at random instead of the minimax choice.

Expected to finish in far fewer than W guesses, but not guaranteed within 10 on a hard list. The minimax choice is the safe answer; random with a 'zero-match' preference (guess words that share no letters with earlier guesses) is a cheap improvement.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The word list is much larger.

Computing all pairwise match counts is O(W^2 L). Sample candidates when choosing a guess, or precompute buckets by letter at each position to count matches faster.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 266

267 Wordle-Style 5-Letter Minimum-Guess Strategy HARD

DIFFICULTY | HARD | 1 FOLLOW-UPS

FOLLOW-UP 1 | What is the lower bound and when is it reached?

Every call returns at most one bit, so at least ceil(log2(candidates)) calls are needed. The entropy split reaches it only when a question can cut the candidates exactly in half; real dictionaries usually need a few more. A greedy scoring rule (closest to half) is enough to state; the interviewer mostly wants the entropy idea.

FOLLOW-UP CODE

```
def wordle_guess(query, dictionary=None):
    calls = 0
    def ask(position, letter):
        nonlocal calls
        calls += 1
        return query(position, letter)
    if dictionary:
        words = list(dictionary)
        while len(words) > 1:
            best_probe = max(((position, letter) for position in range(5) for letter in set(word[position]
for word in words)),
                             key=lambda probe: min(sum(word[probe[0]] == probe[1] for word in words), sum(word[
probe[0]] != probe[1] for word in words)))
            answer = ask(*best_probe)
            words = [word for word in words if (word[best_probe[0]] == best_probe[1]) == answer]
        return words[0], calls
    letters = []
    for position in range(5):
        for rank, letter in enumerate(LETTERS_BY_FREQUENCY):
            if rank == 25 or ask(position, letter):
                letters.append(letter)
                break
    return "".join(letters), calls
```

BACK TO PART B - PART 1: QUESTION 267

268 Babylon Tile-Stack Merge Game HARD

DIFFICULTY | HARD | 1 FOLLOW-UPS

FOLLOW-UP 1 | How big is the real state space?

With 12 starting stacks the first position has at most 66 unordered pairs and the number of reachable multisets of (height, colour) stacks shrinks quickly because every merge removes a stack, so memoised search is feasible. In the interview build the clean state class, legal moves and play(move) first and add the solver only if there is time.

FOLLOW-UP CODE

```
def babylon_wins(stacks):
    cache = {}
    def wins(state):
        if state in cache:
            return cache[state]
        can_win = False
        for first, second in itertools.permutations(range(len(state)), 2):
            (height_a, colour_a), (height_b, colour_b) = state[first], state[second]
            if height_a == height_b or colour_a == colour_b:
                next_state = tuple(sorted([stack for position, stack in enumerate(state) if position not in (
first, second)] + [(height_a + height_b, colour_a)]))
                if not wins(next_state):
                    can_win = True
                    break
        cache[state] = can_win
        return can_win
    return wins(tuple(sorted(stacks)))
```

BACK TO PART B - PART 1: QUESTION 268

Sliding window: grow on the right, shrink on the left while it is too bad

273 Necklace Cut into Two Halves with Equal D/R Counts

HARD

DIFFICULTY | HARD | 4 FOLLOW-UPS

FOLLOW-UP 1 | When is a single cut enough?

A single linear cut works when some prefix has exactly half the D's and half the R's, i.e. a prefix of length n/2 whose D count is d_total / 2. Otherwise rotate: the circular window above finds the start, which is equivalent to two cuts on the circle.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if the two pieces must be equal strings, not just equal counts?

That is a different question: the necklace must be a square rotation, i.e. some rotation of s is t + t. Check whether s appears inside (s + s)[1:-1] at index n/2, or compare s[i:i + n/2] with the complementary arc using a rolling hash or KMP over s + s. It is a string matching problem, not a counting one.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | The two pieces must be equal strings, not just equal counts.

Then the necklace must be a square in some rotation: some rotation of s equals t + t. Test with s in (s + s)[1:-1] restricted to an even-length split, or compare s[i:i + n/2] with s[i + n/2:i + n] using a rolling hash or KMP.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | The two pieces must be equal strings, not just have equal counts.

A rotation of the necklace must be of the form t + t. Try every rotation and compare its two halves (O(length^2), or O(length) by searching s in (necklace + necklace)[1:-1] and checking the match at an offset of length / 2).

FOLLOW-UP CODE

```
def necklace_equal_strings(necklace):
    length = len(necklace)
    if length % 2:
        return None
    half = length // 2
    for start in range(length):
        rotation = necklace[start:] + necklace[:start]
        if rotation[:half] == rotation[half:]:
            return start, rotation[:half]
    return None
```

BACK TO PART B - PART 1: QUESTION 273

Parent pointers: store who is above each node and climb

274 Tree Node. EASY

DIFFICULTY | EASY | 4 FOLLOW-UPS

FOLLOW-UP 1 | Why not use NOT IN for the leaves?

If the subquery returns a NULL (the root's p_id), id NOT IN (...) is never true, so you get no rows. Filter NULLs out of the subquery or use NOT EXISTS.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | The table has no row for the root's parent and p_id can point to missing ids.

Validate with a LEFT JOIN to find orphans. A node whose p_id is not NULL but missing from the table is a data error, not a root; decide whether to report it.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Add the depth of each node.

A recursive CTE starting from the root rows and joining children on p_id, carrying depth + 1, gives every node's depth in one query.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | Add the depth of each node.

A recursive CTE starts from the root rows (depth 0) and joins the table to itself on p_id, adding 1 at every step; each node appears once because every node has one parent. Runs in one query, O(n) with an index on p_id.

FOLLOW-UP CODE

```
def node_depths(rows):
    conn = sqlite3.connect(":memory:")
    conn.execute("CREATE TABLE tree(id INTEGER PRIMARY KEY, p_id INTEGER)")
    conn.executemany("INSERT INTO tree VALUES (?, ?)", rows)
    q = """WITH RECURSIVE d(id, depth) AS (SELECT id, 0 FROM tree WHERE p_id IS NULL
                                           UNION ALL SELECT t.id, d.depth + 1 FROM tree t JOIN d ON t.p_id =
d.id)
           SELECT id, depth FROM d ORDER BY id"""
    return conn.execute(q).fetchall()
```

BACK TO PART B - PART 1: QUESTION 274

Build or transform a grid with index arithmetic

279 Filled ASCII Diamond EASY

DIFFICULTY | EASY | 1 FOLLOW-UPS

FOLLOW-UP 1 | Draw only the outline of the diamond.

Keep the same loop but put 'x' only where the distance is exactly R; inside cells become background. In the row form: the two end cells of each row are 'x' (one cell at the tips).

FOLLOW-UP CODE

```
def render_hollow_diamond(radius):
    rows = []
    for row_number in range(2 * radius + 1):
        side_underscores = abs(row_number - radius)
        row = ["_"] * (2 * radius + 1)
        row[side_underscores] = row[2 * radius - side_underscores] = "x"
        rows.append("".join(row))
    return rows
```

BACK TO PART B - PART 1: QUESTION 279

Split into pieces, fix each piece, join (and pack words into lines)

284 Implement string formatting and request tracking tasks EASY

DIFFICULTY | EASY | 5 FOLLOW-UPS

FOLLOW-UP 1 | Why not print as soon as a request completes in the follow-up?

Because another request that completes later can still have an earlier end time. An ordered printer needs a contract for when it is safe to emit: either a watermark (everything with end <= W is final because ends arrive within some known lateness), or a flush at the end of input. State that contract, then the heap pops only below the watermark.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | How do you handle bad input (duplicate begin, end before begin, never-completed requests)?

Duplicate begins overwrite or are rejected by policy; an end timestamp earlier than its begin is flagged as invalid rather than printed. Requests that stay in the pending map are the leak to watch: evict by age with a timer or a second heap keyed on first-seen time, and report them as incomplete.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Rotation map provided at runtime with extra pairs.

Nothing changes: the map is a parameter. Validate that it is consistent (rot of rot is the original) if you control it, and treat characters that are not in it as invalid.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 4 | The tracker must bound its memory.

Evict requests whose begin or end has been waiting longer than a retention window (a second heap keyed by first-seen time) and report them as incomplete. Cost: O(log n) per event for the eviction heap; memory O(requests inside the retention window).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 5 | Another locale: a different separator or grouping (Indian style 12,34,567).

Parameterise the separator and the group size: the lowest group may differ from the others (3 digits, then groups of 2 in the Indian system). The loop peels a group from the right each time.

FOLLOW-UP CODE

```
def format_grouped(n, sep=",", group=3, first_group=None):
    sign = "-" if n < 0 else ""
    digits = str(abs(n))
    parts = []
    size = first_group or group
    while len(digits) > size:
        parts.append(digits[-size:])
        digits, size = digits[:-size], group
    parts.append(digits)
    return sign + sep.join(reversed(parts))
```

BACK TO PART B - PART 1: QUESTION 284

287 Convert Valid Snake Case to Lower Camel Case

HARD

DIFFICULTY | HARD | 2 FOLLOW-UPS

FOLLOW-UP 1 | The input may have leading, trailing or repeated underscores, or mixed case.

Drop empty parts after the split, lowercase the first word, capitalize the rest. Say what you do with digits and non-letters, and keep the rule in one place.

FOLLOW-UP CODE

```
def snake_to_camel_robust(snake_name):
    words = [word for word in snake_name.split("_") if word]
    return (words[0].lower() + "".join(word.capitalize() for word in words[1:])) if words else ""
```

FOLLOW-UP 2 | Convert back from camel case to snake case.

Insert an underscore before each uppercase letter and lowercase it. Acronyms such as HTTPServer need a smarter rule (split between a lowercase-to-uppercase edge and between an acronym and the next word).

FOLLOW-UP CODE

```
def camel_to_snake(camel_name):
    pieces = []
    for char in camel_name:
        if char.isupper():
            pieces.append("_")
        pieces.append(char.lower())
    return "".join(pieces)
```

BACK TO PART B - PART 1: QUESTION 287

288 Implement a function to justify a list of words within a given width.

HARD

DIFFICULTY | HARD | 5 FOLLOW-UPS

FOLLOW-UP 1 | How would you change it to fully justify the last line too, or to center the text?

Only the final-line branch changes. For full justification drop the special case for the last line (it would be spread like every other line). For centering, split the leftover spaces between the two sides, giving the extra space to the right. The greedy packing step stays identical.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 2 | What if the words contain wide characters or different pixel widths?

Replace len(word) by a measure(word) function (for example font metrics or East Asian width) and compare against the pixel width. Packing and distribution logic stay the same; spaces are measured in the same unit, so the spare width may not divide evenly and the remainder must be assigned to gaps in whole space units.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 3 | Fully justify the last line too.

Drop the special case for the last line (a line with more than one word is spread like the others). Single-word lines stay left-aligned.

FOLLOW-UP CODE

```
def fullJustify(words, max_width):
    lines, first = [], 0
    while first < len(words):
        after_last, length = first, 0
        while after_last < len(words) and length + len(words[after_last]) + (after_last - first) <= max_width:
            length += len(words[after_last])
            after_last += 1
        gaps = after_last - first - 1
        if after_last == len(words) or gaps == 0:
            line = " ".join(words[first:after_last])
            line += " " * (max_width - len(line))
        else:
            spaces, extra = divmod(max_width - length, gaps)
            line = ""
            for word_index in range(first, after_last - 1):
```

```
                line += words[word_index] + " " * (spaces + (1 if word_index - first < extra else 0))
            line += words[after_last - 1]
        lines.append(line)
        first = after_last
    return lines

def justify_all_lines(words, max_width):
    lines = fullJustify(words, max_width)
    if not lines:
        return lines
    # recompute which words are on the last line and spread them like the others
    after_last, rows = 0, []
    while after_last < len(words):
        word_index, length = after_last, 0
        while word_index < len(words) and length + len(words[word_index]) + (word_index - after_last) <=
max_width:
            length += len(words[word_index])
            word_index += 1
        rows.append(words[after_last:word_index])
        after_last = word_index
    last = rows[-1]
    if len(last) > 1:
        letters = sum(len(word) for word in last)
        gap_spaces, extra = divmod(max_width - letters, len(last) - 1)
        lines[-1] = "".join(word + " " * (gap_spaces + (1 if index < extra else 0)) for index, word in
enumerate(last[:-1])) + last[-1]
    return lines
```

FOLLOW-UP 4 | Words have pixel widths (a proportional font).

Replace len(word) with a measure function and compare against the pixel width; the extra space is distributed in whole space units, so the remainder rule stays the same.

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

FOLLOW-UP 5 | Minimise raggedness instead of packing greedily.

That is the Knuth-Plass style DP: cost of a line = (spare space)^2, minimise the total over all ways to break the text. O(n * words per line).

DISCUSSION / MODIFY THE MAIN SOLUTION | No separate implementation is needed unless the interviewer asks for one.

BACK TO PART B - PART 1: QUESTION 288